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(54) **COMPOSITIONS AND METHODS FOR
PRODUCING PROTEIN**

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(71) Applicant: **Terraferma Foods, Inc.**, San Francisco,
CA (US)

(72) Inventors: **Ruby YU**, San Francisco, CA (US);
Victoria MO, San Francisco, CA (US)

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(57)

ABSTRACT

Disclosed herein is a food product comprising an animal-
free composition comprising a meat protein.

Specification includes a Sequence Listing.

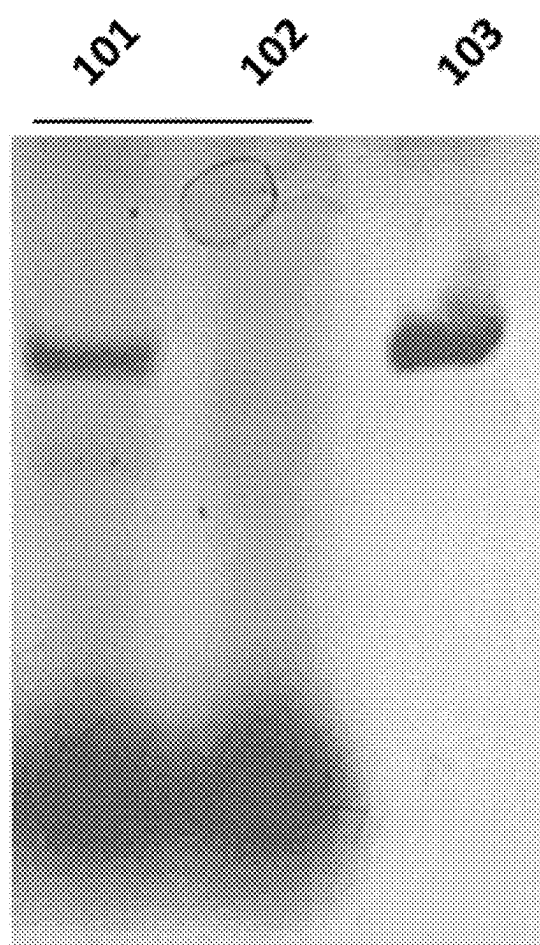


FIG. 1

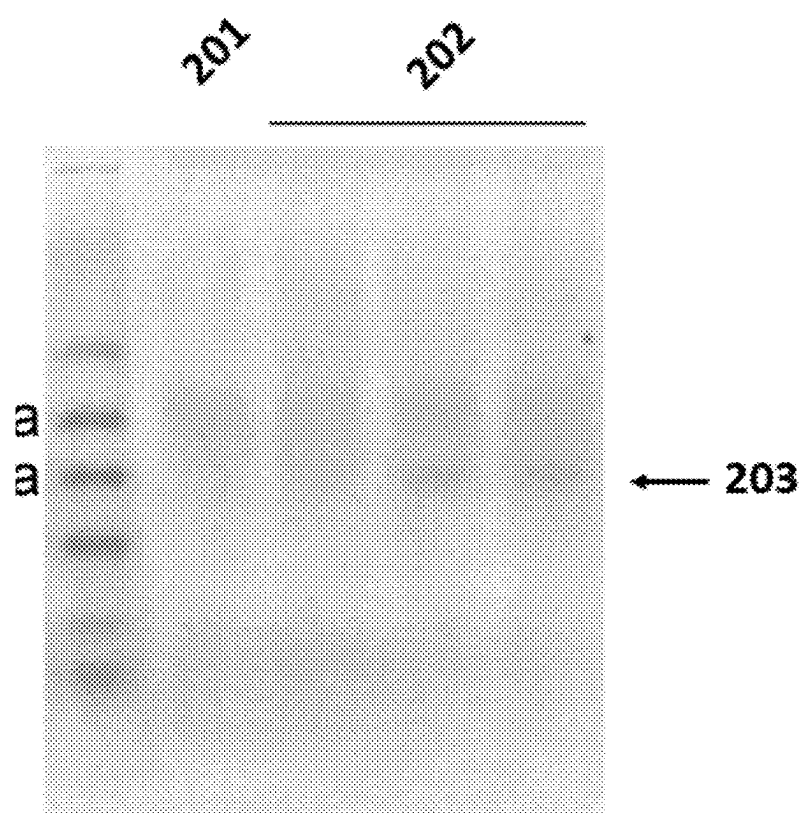
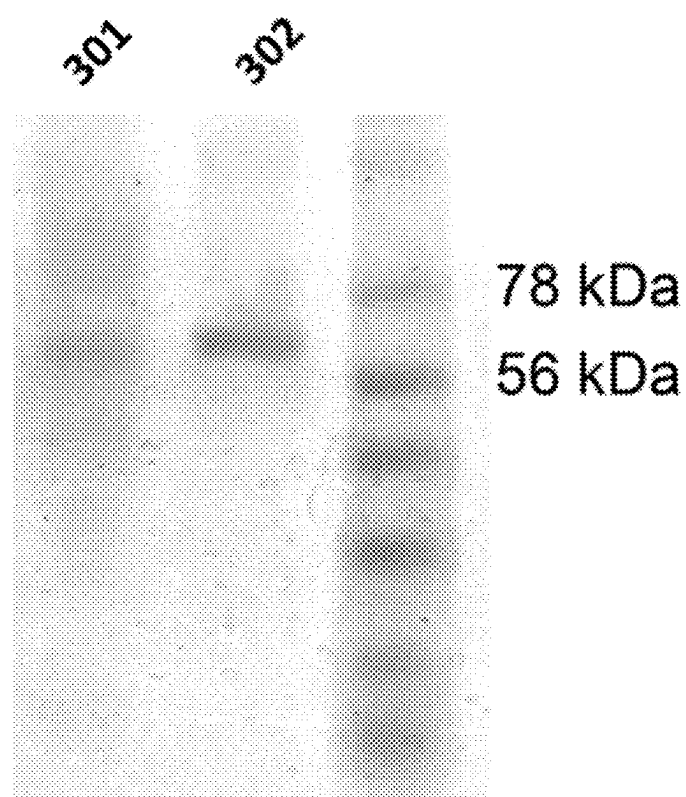


FIG. 2

***FIG. 3***

COMPOSITIONS AND METHODS FOR PRODUCING PROTEIN

CROSS-REFERENCE

[0001] This application is a continuation of International Application No. PCT/US2023/028715, filed Jul. 26, 2023, which claims the benefit of U.S. Provisional Application No. 63/392,613, filed Jul. 27, 2022, and U.S. Provisional Application No. 63/402,544, filed Aug. 31, 2022, each of which is entirely incorporated herein by reference.

SEQUENCE LISTING

[0002] The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jan. 21, 2025, is named 63116-701_601_SL.xml and is 458,343 bytes in size.

BACKGROUND

[0003] Livestock is the world's largest user of land resources on Earth. Land dedicated to the production of meat and dairy, which includes animal feed production, encompasses nearly 80% of global agricultural land. One-fifth of the world's meat production is consumed by companion animals (e.g., pets). Recombinant protein production can be used to supplement or offset meat production.

SUMMARY

[0004] An aspect of the present disclosure provides a method for producing a food product, the method comprising: a) contacting a plurality of *Pichia* cells with a recombinant nucleic acid; b) expressing a protein encoded by the recombinant nucleic acid; and c) isolating the protein from the plurality of *Pichia* cells, wherein the protein comprises an amino acid sequence that is at least 80% identical to any one of SEQ ID NOs: 54-232.

[0005] In some embodiments, the food product is a cat food product. In some embodiments, the protein is an albumin protein. In some embodiments, the albumin protein is a rabbit albumin protein. In some embodiments, the albumin protein is a duck albumin protein. In some embodiments, the duck albumin protein is a mallard duck albumin protein. In some embodiments, the duck albumin protein is a Muscovy duck albumin protein. In some embodiments, the albumin protein is a cow albumin protein. In some embodiments, the albumin protein is a sheep albumin protein. In some embodiments, the albumin protein is a pig albumin protein. In some embodiments, the albumin protein is a bison albumin protein. In some embodiments, the albumin protein is a deer albumin protein. In some embodiments, the albumin protein is a chicken albumin protein. In some embodiments, the albumin protein is a turkey albumin protein. In some embodiments, the albumin protein is a salmon albumin protein. In some embodiments, the albumin protein is a cod albumin protein. In some embodiments, the albumin protein is a tuna albumin protein.

[0006] In some embodiments, the *Pichia* cells are *Pichia pastoris* cells. In some embodiments, the protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the protein comprises an amino acid sequence that is at least 95% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is substantially animal-free.

[0007] Another aspect of the present disclosure provides a food product comprising a recombinant protein, wherein the recombinant protein comprises an amino acid sequence that is at least 80% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is substantially animal-free. In some embodiments, the recombinant protein comprises an amino acid sequence that is at least 90% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the recombinant protein comprises an amino acid sequence that is at least 95% identical to any one of SEQ ID NOs: 54-232. In some embodiments, the food product is a cat food product.

[0008] In some embodiments, the recombinant protein is a recombinant albumin protein. In some embodiments, the recombinant albumin protein is a recombinant rabbit albumin protein. In some embodiments, the albumin protein is a recombinant duck albumin protein. In some embodiments, the recombinant duck albumin protein is a recombinant mallard duck albumin protein. In some embodiments, the recombinant duck albumin protein is a recombinant Muscovy duck albumin protein. In some embodiments, the recombinant albumin protein is a recombinant cow albumin protein. In some embodiments, the recombinant albumin protein is a recombinant sheep albumin protein. In some embodiments, the recombinant albumin protein is a recombinant pig albumin protein. In some embodiments, the recombinant albumin protein is a recombinant bison albumin protein. In some embodiments, the recombinant albumin protein is a recombinant deer albumin protein. In some embodiments, the recombinant albumin protein is a recombinant chicken albumin protein. In some embodiments, the recombinant albumin protein is a recombinant turkey albumin protein. In some embodiments, the recombinant albumin protein is a recombinant salmon albumin protein. In some embodiments, the recombinant albumin protein is a recombinant cod albumin protein. In some embodiments, the recombinant albumin protein is a recombinant tuna albumin protein.

[0009] Additional aspects and advantages of the present disclosure will become readily apparent to those skilled in this art from the following detailed description, wherein only illustrative embodiments of the present disclosure are shown and described. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

INCORPORATION BY REFERENCE

[0010] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference. To the extent publications and patents or patent applications incorporated by reference contradict the disclosure contained in the specification, the specification is intended to supersede and/or take precedence over any such contradictory material.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The novel features of the invention are set forth with particularity in the appended claims. A better under-

standing of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings (also “figure” and “FIG.” herein), of which:

[0012] FIG. 1 shows a Coomassie stained sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit albumin or wild-type cells.

[0013] FIG. 2 shows a Coomassie stained SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit Tropomyosin alpha-1 chain or wildtype cells.

[0014] FIG. 3 shows a Coomassie stained SDS-PAGE gel of an example recombinant rabbit albumin.

DETAILED DESCRIPTION

[0015] While various embodiments of the invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions may occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed.

[0016] Generation of animal-free protein is a way to reduce global meat production and offset the agricultural and environmental consequences of animal farming. Animal free protein can be made using microorganisms designed to produce meat proteins using recombinant protein expression. Foods, such as pet food (e.g., cat food, dog food, or food for other companion animals) or foods for human consumption can be developed from animal-free meat proteins as an alternative or supplement to traditional meat-based foods.

[0017] Using recombinant protein expression methods, microorganisms (e.g., a yeast or bacteria) can be contacted with a DNA expression cassette. This DNA sequence can contain a gene or genes of interest, as well as regulatory sequences which will help in the production of the gene or genes of interest. Regulatory sequences (e.g., UTRs, promoters, signal sequences, etc.) can aid in, for example, transcription or translation efficiency, protein product yield, protein expression, or protein secretion.

[0018] Microorganisms can be induced to express and secrete meat proteins of specific animals (e.g., a rabbit protein or a duck protein). Microorganisms can alternatively or additionally be induced to express and secrete specific types of meat proteins (e.g., an albumin protein). Animal-free meat proteins can be purified and formulated into food products (e.g., a pet food or food for human consumption).

Definitions

[0019] Whenever the term “at least,” “greater than,” or “greater than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “at least,” “greater than” or “greater than or equal to” applies to each of the numerical values in that series of numerical values. For example, greater than or equal to 1, 2, or 3 is equivalent to greater than or equal to 1, greater than or equal to 2, or greater than or equal to 3.

[0020] Whenever the term “no more than,” “less than,” or “less than or equal to” precedes the first numerical value in a series of two or more numerical values, the term “no more than,” “less than,” or “less than or equal to” applies to each of the numerical values in that series of numerical values. For example, less than or equal to 3, 2, or 1 is equivalent to less than or equal to 3, less than or equal to 2, or less than or equal to 1.

[0021] The terms “companion animal” and “pet” are used interchangeably herein. A pet is a domestic or tamed animal that is kept for companionship. A pet can be, but is not limited to, a dog, a cat, a pig, a rat, a mouse, a snake, a turtle, a cow, a sheep, a llama, a donkey, a goat, a horse, a chicken, a parrot, a ferret, a guinea pig, a hamster, a rabbit, a fish, a frog, a toad, a salamander, an iguana, a lizard, a duck, a goose, or an insect.

[0022] The term “meat” can be defined as skeletal muscle and its associated tissues, including blood, derived from mammalian, avian, reptilian, amphibian, and aquatic species harvested for consumption. Edible offal comprising organs and non-skeletal muscle tissue can also be considered meat.

[0023] The term “animal-free” can be defined as something that is not harvested directly from an animal, living or dead (e.g., meat harvested from a dead animal).

Food Products

[0024] Described herein are food products comprising animal-free proteins. An animal-free protein can be a protein or a fragment or truncation of a protein that is not obtained from an animal (e.g., by harvesting meat from an animal). An animal-free protein can be obtained through in vitro or ex vivo methods such as cell-free protein synthesis (CFPS). Alternatively, an animal-free protein can be obtained through in vivo methods such as genetic engineering of cells for recombinant protein expression. Genetically engineered cells used for recombinant protein expression can be, but are not limited to, microorganism cells (e.g., bacteria, yeast), plant cells, insect cells, or animal cells.

[0025] In recombinant protein expression, a cell is contacted with a DNA sequence. The DNA sequence will be incorporated by the cell, and the cell’s internal protein expression machinery will transcribe and translate the DNA sequence into a peptide.

[0026] A cell used in recombinant protein expression can be a microorganism cell. Microorganisms are microscopic organisms such as bacterium, viruses, or fungi (e.g., yeast). A microorganism cell used in recombinant protein expression can be a bacterial cell. A bacterial cell can be, but is not limited to, a cell from *Escherichia coli*, *Bacillus licheniformis*, *Bacillus subtilis*, or *Lactococcus lactis*. A microorganism cell used in recombinant protein expression can be a yeast cell. A yeast cell can be, but is not limited to, a cell from *Schizosaccharomyces pombe*, *Saccharomyces cerevisiae*, *Hansenula polymorpha* (also known as *Ogataea polymorpha*), *Pichia farinosa*, *Pichia anomala*, *Pichia heedii*, *Pichia guilliermondii*, *Pichia kluyveri*, *Pichia membranifaciens*, *Pichia norvegensis*, *Pichia ohmeri*, *Pichia pastoris* (also known as *Komagataella phaffii*), *Yarrowia lipolytica*, *Pichia methanolica*, or *Pichia subpelliculosa*. A microorganism cell used in recombinant protein expression can be a filamentous fungal cell. A filamentous fungal cell can be, but is not limited to, a cell from *Trichoderma reesei* (also known as *Hypocrea jecorina*), *Aspergillus niger*, *Aspergillus*

oryzae, *Aspergillus tubingensis*, or *Ashbya gossypii* (also known as *Eremothecium gossypii*).

[0027] *P. pastoris* cells can be used in recombinant protein expression methods. *P. pastoris* cells have robust protein secretion; the ability to grow to high biomass density, allowing higher yields; have a high protein yield with up to 35 g/L yield for secreted proteins; have a low endogenous secretome, allowing for less downstream purification; grow in relatively inexpensive feedstock/media (e.g., glycerol or methanol); have high genetic tractability; have a proven ability to properly express and fold distantly-related eukaryotic (e.g., plant or animal) proteins; have a 90 minute doubling time; have an ability to grow in large-scale fermenters; and have an approximately one-week fermentation time from start to harvest.

[0028] A cell used in recombinant protein expression can be chosen because it displays a desired characteristic such as having a rapid doubling time, the ability to properly fold a target recombinant protein; the ability to make appropriate post-translational modifications to a target recombinant protein; the ability to efficiently secrete the target recombinant protein; the ability to be grown cheaply; the ability to be grown on easily accessible media; or genetic tractability. Alternatively, a cell used in recombinant protein expression may not secrete or may not efficiently secrete the target recombinant protein. In an example, cells that do not secrete or do not efficiently secrete the target recombinant protein may be lysed to release the protein. Cell lysis may include mechanical disruption, liquid homogenization, sonication, freeze/thaw cycling, manual grinding, or any combination thereof. In some examples, cells that secrete the target recombinant protein may be further lysed to increase protein yield.

[0029] During recombinant protein expression, a cell is contacted with a DNA sequence that can comprise an expression cassette. An expression cassette can be a component of vector DNA which can comprise at least one gene and/or at least one regulatory sequence.

[0030] An expression cassette can comprise at least one gene of interest. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more genes of interest. Alternatively, or in addition to, an expression cassette can comprise at least one additional gene other than a gene of interest. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more additional genes other than a gene of interest. An expression cassette can be a linearized plasmid. Alternatively, an expression cassette can be a linear PCR product.

[0031] A DNA sequence (e.g., a vector or an expression cassette) can be transfected into a cell using various chemical and physical methods such as electroporation, calcium phosphate exposure, spheroplasting, polyethylene glycol (PEG) exposure, alkali cation exposure, or liposome-based transfection.

[0032] A gene of interest can be an animal protein-coding gene. An animal protein-coding gene can encode a full length or a fragment or truncated version of a naturally occurring protein. Alternatively, or in addition to, an animal protein-coding gene can encode a modified version (e.g., a codon-optimized version) of a naturally occurring protein. Alternatively, an animal protein-coding gene can encode a synthetic or novel protein.

[0033] An animal protein-coding gene can encode an antibody, a contractile protein, an enzyme, a hormonal protein, a structural protein, a storage protein, or a transport

protein. An animal protein-coding gene can be specific to a particular type of tissue such as, but not limited to, connective tissue, epithelial tissue, muscle tissue, or nervous tissue.

[0034] An animal protein-coding gene can encode proteins represented in meat or other parts of animals that are eaten. Examples of proteins found in meat can include, but are not limited to: actin, myosin, myosin regulatory protein, albumin, carbonic anhydrase, myoglobin, collagen, elastin, titin, nebulin, sarcocalumenin, creatin kinase, phosphoglucosomutase, phosphoglycerate mutase, triosephosphate isomerase, alpha-actinin, glycogen phosphorylase, fructose-biphosphate adolase, glyceraldehyde-3-phosphate dehydrogenase, L-lactate dehydrogenase, tropomyosin, myosin-binding proteins, myomesin, beta-enolase, pyruvate kinase, phosphoglycerate kinase, plectin, ATP-dependent 6-phosphofructokinase, malate dehydrogenase, hemoglobin, filamin, aconitate hydratase, parvalbumin, heat shock cognate 71, troponin, calsequestrin, obscurin, glycogen debranching enzyme, elongation factor 1-alpha 2, aspartate aminotransferase, adenylate kinase isoenzyme, glucose-6-phosphate isomerase, desmin, AMP deaminase, and globulin.

[0035] An animal protein-coding gene can encode albumin. Albumin is a family of globular proteins, the most common of which are serum albumins. Albumin proteins are water-soluble and are commonly found in blood plasma. Albumin proteins differ from other blood proteins in that they are not glycosylated. Albumin proteins include, but are not limited to: serum albumin, alpha-fetoprotein, vitamin D-binding protein, afamin, ovalbumin and extracellular matrix protein 1.

[0036] An animal protein-coding gene can be a gene from a specific animal such as, but not limited to: a canine, a feline, a bird (e.g., a songbird, a pigeon, a chicken, a duck, a goose, a swan, a crow, a dove, a parrot, a turkey), a bovine (e.g., a cow, a bison), a camel, a goat, a pig, a deer, a sheep, a horse, a rodent (e.g., a mouse, a rat, a squirrel, a hamster, a guinea pig, a chipmunk, a gopher, a vole), a fish (e.g., a tuna, a salmon, a trout, a sardine, a tilefish, a halibut, a cod, a flounder), a rabbit, a lizard, a snake, or a frog.

[0037] An animal protein can comprise a mouse protein (Table 1). A mouse protein can comprise a mouse muscle protein. A mouse protein can comprise a mouse blood protein. A mouse protein can comprise a mouse albumin protein. A mouse protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99%, or about 100% sequence identity to any one of SEQ ID NOs: 1-53.

[0038] A food product may comprise a fragment or a truncation of a mouse protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 1-53. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 1-53. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 1-53. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of the length of any one of SEQ ID Nos: 1-53. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 1-53.

TABLE 1

Mouse Proteins		
SEQ ID NO	Description	Sequence
1	Mouse TITIN	MTTQAPMFTQPLQSVVVLEGSTATFEAHVSGSPVPEVSWFRDQGVISTSTLPGVQISFSDGRARLMI PAVTKANSGRYSLRATNGSGQATSTAE LLVTAETAPPNFSQRLQSM TVRQGSQVRLQVRVTGIPTPVVKFYRDGA EI QSSSLDFQISQBGDL YSLLIAEAYPEDSGTYSVNATNSVGRATSTAE LVVQGEEVVP AKKTKTIVSTAQISETRQTRIEKKIEAHFDA RSIATVEMVIDGATGQLPHKTPPRIPPKPKSRSPTPPSIAAKAQLARQQSPSPIRHSPS VRHVRAPTPSPVRSVPAGRISTSPIRSVKSPLLRKTQTTTMTATGPEVPPPPWKQEGYV ASSTEAMRETMTSSTQIRREERWEGRYGVQEQVTISGAAAAAASVSSSPTAGAI TTGTKEVKQETDKSAAVATVVAAVDMARVREPAISAVEQTAQR TTTTAVHIQPAQE QARKEAEKTAVTKVVVAADKAKEQELKSR TREVMVTTQEQTHISHEQIRKETEKAF VPKVVISATKAKEQETRI TGEITTKQBQKRITQETIRQETEEIAASMVVATAKSTKLE AAVGVQEETA IQDDQMH LTHEQMMKETRKT VVPKVIVATPKIKEQDLVSR SREAITT KRDQVQITQEKKRKEVET TALSTIAVATAKAKEQETV LRSREAMATRQEH IQVTHG QVGVGKKA EAVATVVAAVDQARVREP REPTHVEESH SQQT TLEYGYKEHISTTKVP EQPRRPASEPHVVPQAVKPAVIQAPSETHI KTTDQMGMHISSQVKKTTDISTERLVHV DKRPRTASPHFTVSKISVPKTEHG YEAS IAGSAIATLQKELSATSSTQK I TKSVKAPT V KPGETRVR AEPTSPQPFPADMP PD TYK SQAGIEVKKEVGVSISGSTVREEHFVLR GREAKVTETARVPAPAEVVP TPTLV SGLKNVTVI EGESV TLECHISGYPSPKVTWYR EDYQIESSIDFQITFQGGIARLMIREAFAEDSGRFTCSAVNEAGTVSTSCYLA VQVSEE FDKETTLTEKFATEEKR FVESRDVMTDTSITEEQAGPGEP AAPFFISKPVVQKLVEG GSVVFECQIGGNPKPHVYWKSGVPLTTGYRYKVSYNKQTGECRLVISMTFADDAG EYTVIRNKHGETSASASLLEEADYEALVK TQQEMLYQTQMS TFIQEPKVGEIAPGFA YSEYEKEYEKEQALIRKKMAKDTVMVRTFVEDQEFHISSEERLKEIEYRIKTTLEE LLEEDGEEKMAVDISESEAIESGFDIRIKNYRILEGMGVTFHCKMSGYPLPKIAWYKD GKRI RHGERYQMDFLQDGRASLRIPVVLPEDEGIY TAFASNIKGNAICSGKLYVEPAA PFSAPT YMPTEPAVSRIRSVSPRSLSRSP IRMS PAMSPARMSPARMSPARMSPARMSPARMS GRRLEETDESQLERLYKPVFVLPKASFKCLEGQTARFDLKVVGPRMPETFWFHNGQ QIVNDYTHKVVIKEDGTQSLIIVPASPSDSGEWTVVAQN RAGKSTISVLTVEAVEHQ IKPAFVEKLKNVINKEGSRLEMKV RATGNPNPDIVWLKNSDIIVPHKYPRIRIEGRGE AALKIDSIISQDSAWYTATAINKAGRDTTRCKVNVEVEFAEPEPERKLIIPRGTYRAKE IAAPELEPLHLRYGQEQWEEGDLYDKEKQKPPFKKLTSLRLKRFPAHFECRLTPI GDPTMVV EWLHDGKPLEAANRLRMINEFGYCSLDYGAAYSRDSGVI TCRATNKYG TDHTSATLIVKDEKSLVEESQLPDGKKGLQRIEELERMAHEGALTGVTTDQKEKQKP DIVLFPEPVRVLEGETARFR CRVTGYQPKNWYLNQLIRKSKRFRVRYDGIHYLDI VDCKSYDTGEVKVTAENPEGVTEHKVKLEIQOREDFRSVLRRAP EKPPEFHVHEPGK LQFEVQKVDRPVD TSETKEVVKLKRAERITHEKVS ESEELRSFKRRTEEGYEAIT AVELKSRKKDES YEELLKKTDELLHWTKELTEEEKKALAE EGKITIPTFKPERIELSP SMEAPKIFERIQSQT VGGSDAHFRVRVVGKPDPECEWYKNGVKIERSDRIYWYWP EDNVCELVIDRVT AEDSASIMVKAINIAGETSSHAFLLVQAKQLITFTQELQDVVAKE KDTMATFECETSEPFIKVWKYKDGIEVHAGDKYRMHSDRKVHFLSVLTIDTSDAED YSCVLVEDENIKTTAKLIVEGAVVEFVKELQDIEVPESYSGELECI SPENIBGKWYHN DVELKSN GKYSITSRRGRQNLTVKDVTKEDQGEYSFVVDGKKTCKLKMKPRPIAIL QGLSDQKVCEGDIQLEV KVSLENVEGVWMDGQEVQHS DRVHIVIDKQSHMLLIE DMTKEDAGNYSFTIPALGLSTSGNVSVYSVDVITPLKDVNVI EGTKAVLECKVSPVD VTSVKWYLNDEQIKPDDRQSVIKGTKQRLVINRTHASDEGPYKLMVGRVETSCNL SVEKIKIIRGLRDLTCTETQNVVFEVELSHSGIDVWNFKGKEIKPSSKYKIEAHGKIY KLTVLNMMKDD EGEYAFYAGENTTSGKLT VAGGAI STPLTDQTVAESQEA VFCEV ANPESEGEWLKDGKHLALSNNFRGESDGHKRRLVIAAAKLDDAGEYTYKVATSKTS AKLKVEAVKIKKTLRNLTVTETQDAVFSVELTHPDKGVQWIKNGVVLDSNDKYEI SVKGTLYSLKIKNCAMADESVYGFKLGRLGASARLHVETVKIIKKPKDVTALENATV TFEVSVSHDTPVPVKWPHKNVEIKPSDKHRLVSEKRVHKLMLQSI SPSDAGEYTAMV GQLECKAKLFVETLHITKTMKSIEVPETKAASFCEVSHFNVPMSMWLKNVGEIEMSE KFKIIVVQKHLQLIIMNTSTEDSAEYTFVCGNDQVSATLTVTPIMITSM LK DINAEK DTITFEVTVNYEGISYKWLKNGVEIKSTDRCQMR TTKLTHSLNIRNVHFGDAADYTF VAGKATSTATLYVEARHIEFRKHIKD IKVLEKKRAMFECEVSEPDITVQWMDKGQEL QIADRIKIQKEKYVHRLLI PSTRMSDAGKYTVVAGGNMSTANLFVEGRDVRIRSIKKE VQVIEKQRAVVEFEVNEDVD AHWYKDGIEIN FQVQERHQYVVERRIHRMFISETRH SDAGEYTFVAGRNRSSVTLYVNAPEPPQVLQELQPVTVQSGKPARFCAVISGRPQPKI SWYKEEQLLSTGPKCKFLHDGQEYTLLLIEAFPEDAAVYTCEAKNDYG VATTSASLS VEVPEVSPDQEMPVYPPAIVTLPQDTV TSEGRPARFQCQVSGTDLKVSWYCKDKKI KPSRFFRMTQFEDTYQLEIAEAYPEDEGT YAFVANNAVGVSSATLRLEAPESILHE RIGQQIEMEMKEIASLLSAEEDFQYSSDLRLPNANETLELSEPPARSTQFDSRQEGA APVFIRESIDVEISVEDVAKLSVTVTGCPKPKIQWFFNGMLLTPSADYKFVFDGDTHS LIILFTRFQDEGEYTC LASNEYKAVCSAHLRISPRGERSTEMESGEKKALEKPKGPCP PYFFKELKPVHCGPGIPAVFEYSVHGEPAPTVLWFKEDMPLYTSVCYTIHSPDGS GT FIVNDPQRGDSGLYLCKAQNLWGESTCAAELLVLPEDTDVPDASCKEESTLGVP GDF LETSARGPLVQGVDSRQEITAF AEGTISKAALIAEETLQLSYERSVDDSEVGTGVTIGA QKLPPVLS TPQGTGELPSIDGAVHTQPGRGPPPTLNLQAVQAQTTLPKEATLQFEPE EGVFPGASSAAQVSPVTIKPLITLTAE PKGNYPQSS TAAPDHALLS SVA AETLQLG EK KIP EVDKAQRALLLSQSLAEGCVESLEVDPVAVSNMRSEPPQVPPQHTCTEGKILMAS ADTLKSTGQDVALRTEEGKSLSFPLALBEEKQVLLKEEQSEVVAVPTSQTSKSEKEPEA IKGVKEVREQELLSKETLFPSPMPEEQRLHLKTQVRRALQA AVAREQANLFSEWL RNI

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		DKVEVTAVNFTQEPKRILCTYLITSVSSLTEELTVTIEDIDPQMANLETGLKDALCSIV CEERNILMAEDPRIHEEDKIDVQGGDRHLSAQKVETVIEAEADSKYLVSKEEVSW KVESQLKDGDTNEVPQAEATLKLAEEGTQKTSTEMSQQEEAEGTLADLCPAVLKKHLV DTISEEGDTHLTSSISNAKEVHWYFKGNLVPDGGFKCLKEQNAYTLVIEAVKTED EGEYVCEASNDSGAKTSAKLTVGRAAPVKKRIEPLVALGHLAKFTCEIQGAPN VRFQWFKAGREIYESDKCSIRSNYSVSSLEILRTQVVDCGEYTCASNEYGSVSCSTAT LTVTEAYPPTFLSRPKALTTFVGKAAKFLCTVSGTPVIEIIWQKDGAALSPSPDCRVTD ADNKHSLELNLTVQDRGIYSCKASNKFGADICQAEITIIDKPHFIKELEAVQSAINKK IHLEQCQVDEDRKVTITWSKDGQKLPAGKDYKIYFEDKIASLEIPLAKLKDSDGYTCTA SNEAGSSSSAAVAVREPPSFVKVDPSPYLMLPGESARLHCKLKGSPIQVTFWFKNN KELSSENTVRMSFVNSEAILDITDVKVDDSGTYSCEATNDVGSDESCSTEVIKEPPSFI KTLPEADIVRGANALLQCEIAGTGPFEVNWFKDKKQIRSSKKYRLFTQKTFFVYLEISS FNSADVGDYECVVANEVKGKGCVAATHLLKEPPTFVKKVDDFTALAGQTVTLQAAV RGSEPI SVMWMKGQEVIKEDGKIKMSFSNGVAVLTIPDVQISLGGKYTCLAENEAGS QTSVGLLIVKEPAKIERAELIQVTAGDPATLEYTVSGTPELKPKWYKDGRLVASKK YRISFKNNIAQLKFYSAEHLDSGQYTFEISNEVGSSSCETTFTVLDRIAPLFTKPLRN VDSVVGACRLDCKIAGSLPMRVSWFKDGKELTASDRYQIAFVEGTASLEISRVD NDAGNFTCRATNSVGSKDSSGALIVQEPSPFVTKPGSRDVLPGSAVCLKSAPQGSAPL TIKWFKGDKELVSGSCYITKETSESSLELYAVKTSDSGTCTCKVSNVAGSVECSAD LEVKEPATFIEKLEPSQLLKKDGTQLACKVTGTPPIKITWFANDRELRESSKHKMSF AESTAVLRLTDVAIEDSGEYMCEAQNEAGSDHCTGIVIVKESPYFTKEFKSIEVLKEY DVMLLAEVAGTPPFEITWFKDNTTLRSGRKYKTFLDQQLVSLQVLKFVAADAGEYQ CRVTNEVGSSSTCSARVTLREPPSFIKKIEATSSLRGGTAAQATLKGSLPITVTLWLDN DEITEDDNIRMTFENNVAASLYLSGIEVKHDKGYVCQAKNDAGIQRCSALLSVKEPATI MEEAVSIDVTQGD PATLQVKFSGTKEISAKWFKDGQELTLGPKYKISVTDVTSILKII TEKKDSGEYTFEVQNDVGRSSCKASINVLDLIIPPSFTKKLRKMSIKGSFIDLECIVA GSHPI SIQWPKDDQEISASDKHKFSFHDNTAFLEISQLEGTDSGTYTCSATNKAGHSQ CSGHLTVKEPPYFVEKPSQSDVNPGRVQLKALVGGTAPMTIKWFKDNKELHPGAA RSVWKDDTSTILELFSAKAADSGTYICQLSNDVGTSSKATIFVKEPPQFIKKPSVLV LRNGQSTTFECQVGTPEIRVSWYLDGNEITDLRKYGISFVDGLATFQISNARVENS TYVCEARNAGTASCSELKVKKEPPIFIRELEPVEVVKDSDELECEVMGTTPEFVTV LKNNKEIRSGKKTMEKMSVFYLHI TKCDPSDVGEYQCII ANEGGSCACSARVALK EPPSFIKKIENVTVTLKSSATFQSTVAGSPPI SITWLKDDQILEENDNVHISFEDSVATL QVRSDVNGHSGRYTCQAKNESGIERCYAFLLVQEPAQII EKAKSVDTVTEKDPVTLE VVAGTPELKVWLKDGKQIVPSRYFSMSFENNVA SFRIQSVMKQDSGQYTFKVEN FGSSSCDAYLRVLDDQDIPPSFTKKLTMDKVLGSSIHMECKVSGSLPISAQWPKDGKE ISTSAKYLVCHEMTVLEVSNLEEDTANYTCKVSNVAGDNACSGILTVKEPPSFLV KPERQQAIPDSTVEFKAVLKGT PPFKIKWLKDDVELVSGPKCFIGLGSTSFNLNYSV DSSKTGQYTCQVTDVGSDESCTTMLLVTEPPKFVKKLEASKIIKAGDSARLECKITGS PEIQVWYRNEHELTASDKYQMTFIDSVAVIQMNLSGTEDSGDFICEAQNPAGSTSC STKVIVKEPPVFSFPPIVETLKNTEVSLECELSGT PPFVWYKDKRQLRSSKKYK ASKNFHASIHILNVESTDIGEYHCKAONEVGSDACVCAVLEKPPKFIKLNLSLTVVA GEPAELQASIEGAQPI SVQWLKEKEEVI RESENIRISFVNNVATLQFAKVEPANAGKY CQVKNDGGVRENMATLTVLEPAVIEKAGSMVTVTGETCALECKVAGTPELSVEWY KDGLLTSSQKHKFSFYNKISSLKILSVEKEDAGTYTFQVQNNVKGSSCTAVVDVSD RMVPPSFTRRLKDTGGVLGTSCILECKVAGSSPISIAWFHEKTKIVSGAKYQTTFS VCTLQNLSDSSDMGSYTCVAANVAGSDECRALLTVQEPSPFVKEPELVLPGKNI TFTSVIRGTPPFKVGWFRGARELVKGNRCNIYFEDTVAELELFNIDISQSGEYTCVVS NNAGQASCTRLRVKEPATFVKKLSHSEVPGKSIILEGTYGTGLPISVTWKKDGVSI TPSERCNIVTTEKTCILEILSSTKGDAGHYSCEIENEAGRDACDALVSTLEPPYFVTE LEASVGSVSLQCQVAGTPEITVSWFKGDKLRS TPEYRTYFTNNVATLVFNKVSIN DSGEYTCMAENSIGTAASKTIFRIQERQLPPSPARQLKDI EQTVGLPVTLTCLRLNGSAP IQVCWYRDGVLLRDDENLQMSFVDNVATLKILQTDLSHSGQYSCSASNPLGTASSTA RLTAAREPKSPFFDIKPVSIDVIAGESADFECHVTGAQPMRVTSWKNKEIRPGGNYT ITCVGNTPHLRIKLVKGDSGQYTCQATNDVGKDMCSAQLSVKEPPKFIKKLDASKV AKQGESIQLECKISGSPEIKVVWFRNDSSELHESWKYNMSFVNSVALTINEASAEDTG DYICEAHNGVGDASCSTALKVKAPPVFTQKPPPVGALKGSDVILQCEISGT PPFVWV VKDRKQVRSSKKFKITSKNFDTSLHIFNLEAPDIGEYHCKATNEVGSDTCACTVFKKE PPRFVKKLSASTLIGDPVELQAVVEGFQPI SVVWLKDKGEVIRESENVIRISFVDNIAT LQLGSPASQSGKYVCQIKNDAGMRECSAVLTVLEPATIEKPEPMTVTGTGNPFTLE CVVAGTPELSAKWFKDRELSSGSRHHITFVRNLASLKIPSAEMNDKGLYTFEVENR VGKSSCTVSVHVS DRVPPSFVRRLKDTSATLGASVVLECRVSGSAPISVGWFLDGN EIISSPKQSSFADNVCTLTLSLEPSDTGAYTCVAANVAGQDESSAVLTVQEPSPFEQ TPDSEVLPGMSLFTTSVIRGTPPFKVKWFKGSRRELVSGEACTISLEDFVTELELLEVE PGQSGDYSLVTNDAGSASCTTHLTVKEPATFVKRLADTSVETGSPVILEATYSGTTP ISVSWMKNEYPLSQSPNCIGITTEKSSILEILESTIEDYAQYACLIENEAGQDICEALVS VLEPPYFIEPLEHVEAAIGEPITLQCKVDGTPEIRISWYKEHTKLRSAPAYKMQFKNN VASLVINKVDHSDVGEYTCKAENSVGAVASSAVLVIKERKLPPSFARKLKDVBHETLG FPAFAECRINGSEPLQVSWYKDGELLKDDANLQMSFVHHVATLQILQTDQSHVGQY NCSASNPLGTASSAKLILSEHEVPPFFDLKPVSDLALGESGSKCHVTGTAPIKITW AKDNREIRPGGNYKMTLVENTATLTVLKVAKGDAGQTCYASNAGKDCSQAQLG

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		VQEPFRFIKKLDQSRIVKQDEYTRYECKIGGSPEIKVLWYKDEVEIQESSKFRMSFEDS VAILEMHSLSVEDSGDYTCARNAAGSASSSTSLKVKEPPVFRKPPFVETLKGADV HLECELQGTPPFQVSWHKDKRELRSKKYKIMSENLLTSHILNVDTADIGEYQCKA TNDVGS DTCVGSVTMKAPPQFVKKLTDISTIIGKEVQLQTTIEGAEPISVAWFKDKGE IVRES DNIWISYSENIATLQFSRAEPANAGKYTCQIKNDAGMQECYATLSVLEPAIV EKPESIKVTTGDTCTLECTVSGTPELSTKWFKD GKELTSDNKYKISFFNKVSGLKIINV VPGDSGVYSFEVQNPVGKDSCKVSIQVSDRIIPPSFTRKLKETNGLSGSSVMECKVY GSPPISVLWFHDGNEISSGRKYQTTLTNTCALTVNMLEEADAGDYTCIATNVAGSD ECSAPLTVREPPSFVQKPDMDVLTGSNVTFSTIVKGSPPFTVSWFKGSTELVPGARC NVSLQDSVGELELFDVDTQSSEYTCIVSNEAGRASCCTRLFPVKAPAFVKRLNDYSI EKGKPLILEGTFSGTTPISVTVWKKNGINVIASQRCNITTEKSAILEILSSTVEDSGQYN CYIENASGKSDCSAQILILEPPYFVKQLEPVKVTVGDSASLQCLAGTPEIGVSWYKG DTKL RPTATCKMHFKNNVATLVFTQVDS SDSGEYICRAENSVGEVSSSTFLT VQEQK LPPSF SRQLRDVQETVGLPVVFECAVSGSEPI SVSWYKDGKPLKDS PNIQTSPLDNIAT LNIFKTRSLSGQYSCTATNPIGSASSGAKLILTEGKNPPFFDIPLAPMDAVVGESADL ECHVTGTQPIKVWAKDNREIRSGGNYQISYLENSAHLTIVKVDKDGSGGYTCYAV NEVGKDSCTAQLNIKERLIPPSFTKKLSETVEETEGNSFKLEGRVAGSQPITVAWYKN NVEIHPTSNCEIMFKNALLQVKRAMADAGLYTCKATNDAGSALCTSSIVIKEPK KPPVFDQHLAPVTASEGDSVQLSCHVQGSSEPIRIQWLKAGREVKPSDRCSFSFASGTA MLELKETAKADSGDYVCKASNVAGSDTSCKVTIKEKPAAPAAKAAVVGKLF VSEPQSIRVVEKTTATFIAKVGDPINPVKWTGKWRQLNQGGRI LHQKGD EAKLEI RDTTKTDSGLYRCVAFNKHGEIESNVNLQVDERKKQEKI EGDLRAMLKKT PALKKG SGEEEEIDIMELLKNVDPKEYEYARMYGITDPRGLLQAFELLKQS QEEETHRLEIEE LEKSERDEKEFEELVAFIQQLRTQTEPVTLIKDIENQTVLKDND AIFEIDIKINYPEIKLS WYKGETKLEPSNKYEISIDGRHTRLRVKNCQPKDQGNRLVCGPHIASAKLTVI EPA WERHLQDVTLKEGQCTCTMCQFSVPNVKSEWFRNGRVLKPKQGRVKTEVEHKVHKL TIADVRAEDQGQYCTKHEDLETS AELRIEAEPIQFTKRIQNIIVSEHQSATFCEVVSFD DAIVTWYKGPTELTESQKYNFRNDGRCHYMTIHNVTPDDEGVYSVIARLEPRGEARS TAEYLTLTKEIKLEMKPPDIPDSRVPITMPIRAVPPPEEIPPAVAPSIPLLLPLPEKKKPP AKRIEVTKKGVKKDTKKVVTKPKEEAPPPVAKKPPPTPMIPAKASEIIDVSSKAE VKITITTRKKEVHKEEAVYEREAEVYKKVHI EPWEEPYEELETEPYTEPYEPPY EPDEDEYEEIKVEAKKVHEEWEEDFEEGQYEREEGYDEGEEWE EIIYHEREIQV QKEVHEELHEKKIPAKVPEKKVPPKVVKKPVVEKVEKTRRMEEEKVQVIKVEPS KKIVPQKPSRTPVQEEIEIVKVPVHTKMMVISEEKMFFASHTEEVSVSVPEVQKIT VTEEKIHVAVSKKIEPPPKVPEPKKPVPEEVVPVPIPKKVEPPAAKVPEAPKKPVPEE KKPVPIPKKKEPAAPPQVPEAPKKPAPEEKIPVPVTKKKEAPPAKVPEVQKKVVT IAITQREESPPPAPEIPKKKVPPEEKRPVPRKEEVPPKVPVPPKKA VPAVPAIPK KAPPAEVS KKTVEEKRFAAEKLSVAVPQRVELMRHEEEWYTS EEEERVSVSV YREEERDEEEAEITEYEVLEPEEYVVEEKMHFISKKVEVEPAKVPEKKIIPKPVPAK IEEPPPTKVPEPPKKIVPEKKVPAPAPKVPVPAKAPESKRPVPEKRAPAEVGEIEPP TKVAERHMKITQEEKVLVAVTKKEAPPRARVPEEPKVAPEERFPKLKPRREEPPA KVTEVRKRAVKEEKVSI EVPKREPRPTKEVTVTEKKWSYTRREETVSEHREEEYED YEDYEEYKEFEYEPTEEDQYDEYAEREVEHYEEHEEYVTEPKKPVVPKPAQEPV AKPKAPPKVLKKA VPEEKAPLPIQKKLKLPPKAPPEPKKVVEEKIQISITKREKQV TEPVAKVPMKPRVVPPEAKIPAPTKEVAVPVVPVGPVKKRELEEVVVFKEEVEAHEE YIVEEEYVHVEEYVHKEEYVHEEYVHKEEYIHEEEHLHEEEETIAEEVVPVAP VKVPVVPKKVPPEKKPVVPKKEAPPAKVPEIPKKPEEKVPVPIPKKEKAPPAKV EVPKKPVPEEKPPVPVPPKVEPPPAKVPEVPKKPVPEKKVPAPTPKKVEAPPAKVPE PKKPIPEKKPTALLKKMEAPPKAPKKREVVPVVALPREEEEEVPFEEVPEEILP EEEVPS EEEAPPEVPEEEVLP EEEVLP EEEVLP EEEVLP EEEVLP EEEALPEIKPVKPK APVPKKTVPPEKKVPVPVPPKVEPPPPKVP EIKKKVPEKKVVPVPPKKEAPPTKVPEVS KKVEERRIIPKKEEVPPAEVYEEAEPTPEEIPPEPPSIEEEI VEEEEEEVLP PRAP VVKAVPEAPTVPKKA EAPPAKVPKKIPPEKVVPVQKKEAPPAKVPEKKKIPK VPVPKKEAVPPAKGKAVFEKKISVAYQEEELVQERIELELV EAKVEAEFEFEFHEVQ EYFEEEEFHEVEEPIRVEERRFQEEHKVEEVHVRVIEFLEAEVEVEYKPKIPPKKGEV SEKVI PPKKPTKVI PRKEPPAKVPEVT KKTVVEEKIRAPEEPKVPAPKAVEPKKITPE EKVREAVPKKPEVPPKVPVPPKKIQEEKL PVVLPEDTEIYMYEASEETVIEEHVTL PQKARLKVAKVAPPQT VVTEEKTYVTIRKTRETLALKESETTREAPELKS YKAVPE IPEPPSPEDLEII EDVLP EKRPPASKRRKTQLPTAPEAPREMPPEMNTFEEISVEPEMLPT QVLDTYQEA TVEKKT LRSRKKPELPSDEEVPEAPREVVAKKVLPQVPEVVPVKV PGAPKEVVSERKSL EEPKKPAVRPVTVPEEPKEV IPEKKVSLVPPKKPAAPPVTVPE APEEVFSEDEETLAPPQEP EAPPAKVPEAPKEVVPPEKKVSVVPPKPEAPPAKVPEAP KEAPEKKVPVAPKKKPEAPPVKVPEAPKKVPEKKLPVAAPKKPEAPAAEVPEVP KTAVPQKKIPEAIPPKPESPLEVPEVPKVEVTEPKKVPAAPTKPEIPPKVPEAPQAA VVEEKTPEALPKKAAAPVPVQVQETVPEKTRPVGPPKKPEATTVPVKVQKTIPEK TRPEAPPKRPEARVPEVPEKTRPMAPPKKPEATTLVPVEVQETVPEKTRPVGPPKK PEATTVPVPEVQETIPEKTRPPKKPEATVPVPEVQETVPEKTRPVGPPKKPEATTVS PEVQETIPEKTRPAAPPKKPEATVPEIPEKTRPEAPPKRPEATTVPVPEADQAVPE KKVPRVPKKVEAPPITVPEEPKEV IPEKKVSLVPPKKPAAPPVTVPEAPEEVFSEDEE TLAPPQEP EAPPAKVPEAPKEVVPPEKKVSVVPPKPEAPPAKVPEAPKEAPEKKVP VAPKKKPEAPPVKVPEAPKKVPEKKLPVAAPKKPEAPAAEVPEVPKAAVPQKKIPE

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		AIPPKESPPEVYEEPEEEIVPEEPPEAVEEPVPAAPPKVTPEPKKPVPEKKAPPAVV KKPEPPPAKVPEVPKEAPPEKKVPKKPEAPPAKVPEVPKEVVTEKKVAVPKKPEVPP AKVPEVPKKPVIEEKPAPVVEKVASPPAEVYEEPEEVTAESEEPAPAVEEEEEYEAAPP PAPVPEEPKKVPPEKKFPVVKPEAPPPKVPEVPKKAVPVKKVPVKKPEPEAEVPE VPKKLVVKKPEVPVTKTEVLPEKVPEAPKKITPEKKESVPVPEEPAAPPASVEETPE ETIYEEKATITIGRKETPPVEEREIEKFIQPEEPELEPEPEEIPVQPEPEKKVIEKPKLK RPPARPPSPKEDVKEKMFQKAVSKKKVPEKPEVVEKVEPAPLKVPATAEKKVRKLL PEPKPQKKEEVLKSVLRKKPEEEEPKVEPKKVEKAKKPEEPQPPKAVEVEAPPEPT PKERKVPPEAKVPEIKPAIPLPGPEPKPKPEPEVTKMKAPPIEPAPTPIAAPVTAPVVGK KAEAKPKDEAAKPGPIKGVAKKTPSPIEAERKKLRPGSGGEKPPDEAPFTYQLKAV PLKFVKEIKDIVLTEAESVSGSAIFECLVSPSTAITTWMKDGSNIRESPKHRFIADGKD RKLHIIDVQLSDAGEYTCVLRNLGNKEKTSTAKLIVEELPVRFVKTLEEEVTVVKGQPL YLSCELNKERDVVRKDKGKIIVVEKPGRIVPGVIGLMRALTINDADDTAGTYTVTV ENANNLECCSCVVKVEIIREWLVKPIRDQHVKPKGTAVFTCDIAKDTNPKWFKGYD EIPLEPNDKTEILKEGNHFLKVKKNAMPEDIDEYAVEIEGKRYPAKLTGLEREVELLK PIEDVTIYEKESASFDAEISEEDIPGEWKLKGELLRPSPTCEIKAEKGKRFLTLLHKVKL DQAGEVLYQACNAITTAITLVKEIELDFAVPLKDVTVPEKRQARFECVLTREANVIW SKGPDIIKASDKFDIADGKKHILVINDSQFDDEGVYTAEEVGKKTSAQLFVTGIRLKF ISPLEDQTVKEGQTATFVCELSHEKMHVVFKNDVKLHTRTVLMSSEGKTYKLEIR ETTLDDISQIKAQVKNLSSATANLVLEADPYFTVKLHDKTGVEKDEIILKCEVSKDVP VKWFKDGEEIVPSPKHSVKTDGLRRLIKIKKAEKDKGGEVYCDCGTDTTKANVTVE ARLIKVEKPLYGVFVGETARFEIELSEPDVHGQWKLKGEPLTASPDCEIIEDGKKH VLVLNQCQLDMTGESQAANAKSAANLVKELPLIFITPLSDVKVFEKDEAFECFV SREPKTFRWLKGTQEITGDDRFELIKDGRHSLVISAAPFEDEAKYMFEEADKRTSG KLIIIEGIRLKFLTPLKDVTAKERENAVFTVELSHDNI PVSWFKNDQRLHSTRKVSMDH EGKTHSITFKDLSIDDTSQIRVEAMGISSEAKLTVLEGDPYFTGKLQDVTGVEKDEVIL QCEISKADAPVKWFKDGKEIKPSKNVVIKADGKKRMLILKKALKSDIGQYTCDCGTD QTSGLKDIEDREIKLVRLYSVEVMEETETARFETIESEDDIHANWKLKGEALLQTEPC EIKKEEGKIHVLILHNCRLDQGGVDFQAANVKS SAHLRVKPRVIGLLRPLKDVTVTA GETATFDCELSYEDI PVWEYLKGGKLEPNDKVVTRSEGRVHTLTLDVLEDAAGEV QLTAKDFKTQANL FVKEPPVEFTKPLEDQTVEEEAATVLECEVSRENKVKWFKNG TEILKSKKYEIVADGRVRKLI IHGCTPEDIKTYTCDAKDFKTS CNLNVVPPHVEFLRPL TDLVQVEKETARFECEISKENEKQWFKDGAIEIKGKKYDIISKGAVRILVINKCLLN DEAYSCEVRTARTSGMLTVLEEAFTKNLANLEVSEGDITKLVCESKPGAEVIV YKGDEEIIETGRFEILTDRKRILIIQNAQLEDAGSYNCRPLPSRTDSKVVHELAAEFI SKPQNLEILEGEKAEFVCTISKESFEVQWKRDDQTLSEGDKYDIIADGKKRVLVVKD ATLQDMGTYYVMVGAARAAHLTVIEKLRIIVPLKDTKVKEQQEVVENCEVNTEGA KAKWFRNEEAFDSSKYIILQKDLVYTLRIRDARLDDQANFNVS LTNHREGENVKSAA NLIVEEEDLRIVEPLKDIETMEKKS VTFWCKVNRLNVLTKWTKNGEEVAFDNRISYRI DKYKHS LIIKDCGFPDEGEYVVTAGQDKSVAELLIIEAPTEFVEHLEDQTVTEFDDAV FSCQLSREKANVWKYRNGREIKGKKYKFEKDGSIHRLIIKDCRLEDECEYACGVED RKSRARLFVEEIPVEIIRPPQDILEAPGADVIFLAELNKDKVEVQWLRNMIIVQGDK HQMMSEGIHRLQICDIKPRDQGEYRFIAKDKEARAKLELAAAPKIKTADQDLVDDA GQPLTMVVPYDAYPKAEAEWFKENEPLSTKTVDTTAEQTSFRISEAKKDDKGRYKIV LQNKHKGAEGFINLQVIDVPGPVRLNLEVTEFTDGEVSLAWEEPLTDGGSKIIIGYVVER RDIKRKTWVLVTDRADESCEFTVTGLQGGVEYLFVRSARNRVGTGEPVETDSPEA RSKYDVPGPPLNVTITDVRNFGVSLTWEPPEYDGGAEITNYVIELRDKTSIRWDTAM TVRAEDLSATVTDVVEGQEYSFRVRAQNRIGVGKPSAATPFVKVADPIERPSPPVNL NASEQTQSSVQLTWEPPLKDGGSPIILGYIIRREEGKDNWIRCNMKVPPELTYYKVTGL QKGNKYLYRVS AENAAGVSDPSEILGPLTADDAFVEPTMDLSAFKDGLEIVPNPIKI LVPSTGYPRPKATWTFGDQVLEEGDRVKMKTISAYAELVISPSERTDKGIYTLTLENP VKSISGEINNVNIAPPSAPKELKFS DITKDSVHLTWEPDDDGGSPLTGYVVEKRDMS RKTWTKVMDFVTDLEFTVPDLVQGKEYLFKVCARNKCGGEPAYTDEPVNMSAPA TVPDPPENVKWRDRTANSIFLTDWPPKNDGGSRIKGYIVEKCPRGSDKWACGEPVP DTKMEVTGLEEGKWAYRVKALNRQGASKPSKPTTEEIQAQVDTQEAPEIFLDVKLLA GITVKAGTKIELPATVTGKPEPKITWTKADTLKPDQRIITENVPKKS TVTITDSKRSD TGTYIIIEAVNVCGRATAVVEVNLDPKGPAAFDITDVTNESCLLTWNPPRDDGGSK ITNYVVERKATDSVWHKLSSTVKDTNFKATKLTNKEYIFRVAENMYGVGPEVQ AAPIIAKYQFDPGPPPTLRLEPSDITKDAVTLTWCEPDDDGGSPIITGYWVERLDPDTDK WVRCKNMPVKDDTYRVKGLTNKKYFRVLAENLAGPGKPSRSTEPILIKDPIDPPW PPGKPTVKDIGKTLVLNWTKEHGGAKIESVVIEMLKGTDEWVRVAEGVPTTEH LLTGLMEGQEYSFRVRAVNKAGESEPSEPSDPVLCREKLYPPSPPRWLEVINITKNFA DLKWTVPKDGGSPIITNYIVEKRDVRRKGWQTVDTTVKDTKCTVTPLEGS LYVFR VAAENAIGQSDYTEIGDSVLAKDTFTTPGPPYALTVDVTKRHVDLKWEPKNDGG RPIQRYIIIEKKEKLGTRWVAGKTS GPD CNFRVTDVIEGTEVQFQVRAENAGVGHF SEPTEILSIEDTSPSPPLDLHVTAGRKHIAIAWKPEKNGGSPIIGYHVMCPVGTGE KWMVRNRPRIKDLKVEEGIVPDKEYVLRVRAVNAVGVSEPSEIENVVAKDPDC KPTIDLETHDIVVIEGEKLNIPVPFRAVPVPTVSWHKDGKEVKASDRLTMKNDHISAH LEVPKSVHADAGVYITITLENKLGSAATASINVKVI GLPGPKDIKASDITKSSCKLTWE PPEFDGGSPIILHYLLEERREAGRTYIIVMSGENKLSWTVKDLIPNGEYFFRVKAVNKI GGGEYIELKNPVI AQDPKQPPDPVDVEVHNPTAKAMTITWKPPLYDGGSKIMGYIIIE

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		KITKGEDRWKRCNEHLVPVLTYTAKGLEEGKEYQFRVRAENAAGIGEPSRATPPTKA VDPIDAPKVLIRTSLEVKGDEIALDATISGSPYPTITWIKDENVIPEEIKKRAAPPVR RKKGEAEEEEFPFSLPLTERLSINNSKQGESQLRIRDSLRPDHGQYMIKVENDHGVAKA PCSVSVLDTGPPPIINFVFEDIRKDSVLCKWEPLDDGGSEIINYTLEKKDKTKPDSDWI VITSTLRNCKYSVTKLIIEGKEYLFRVRAENRFGPGPPCVSKPLAKDPFEPDAPDKPI VEDVTSNSMLVKWNEPKDNGSPILGYWLEKREVNSTHWSRVNKTLLSSSLKTKVDGL LEGLTYVFRVCAENAAGPGKFSPPSDPKTARDPISPPGPPVPRVADTSSTTIELEWEP AFNGGGEIMGYFVDKQLVGTNEWSRCTEKMIVKQFTVKEIREGADYKLRVSAVNA AGEPPGETEPTVVAEPQEPPTVELDVSVKGGIIMAGKTLRIPAEVTRGPVPTKVWT IEEGELDKERVIIENVGTKSELIINKALRKDHGRYVITATNSCGSKFAAVRVEFVDPG PVLDLKPVVTRNKMCLLNWSDPADDDGGSDITGFIIERKDAKMHTWRQPIETERSKCD ITGLIEGQYKFRVIAKNKFGCGPPVEIGPILAVDPLGPPTSPELTYTERTKSTITLDW KEPRSDGGSPIQGYIEKRRHDKPDERVKNRCLPTTSFLVENLDEHQMYEPRVKAVN DVGESEPSLPLNVVIQDDEVPTIKLRLAVRGDTIKVKAGEPVNIPADVTGLPMPKIE WSKNEKVIDKPTDTLNIKKEEVRSEAKTELSIPKAAREDKGYTITASNRLGSVFRN VHVEVYDRPSPPRNLAVTDIKAESCYLTWDAPLDNGGSEITHYIIDKRASRKSEW BEVTNTAVERRYGIWKLIIPNGQYEFVRVAVNKYGISDECKSDKVVIQDPYRLPGPPG KPKVLERTKGSMLVSWTPPLDNGGSPITGYWLEKREEGGYWSRVSRAPITKVGLK GVEFNVPRLIEGVKYQFRAMAINAAGIGPPSEPSDPAVAGDPIYPPGPPSCPEVKDKT KSSISLAWKPPAKDGGSPKIGYIVEMQEEGTTDWKPVNEPDKLTACECVVNLKEL RKYRFRVKAVNEAGESEPSDTTGEIPATDIEQVEPVFIDIGAQDCLVCKAGSQVKIPA VIKGRPTPKSSWEFDGKAKKAMKDGVDIPEDAQLETAENSSVIIIECTRAHSGKYS ITAKNKAGQKTANCRVKVMDAGPPPKDLKVSDITRSGSRLSWKMPDDDDGGDRIGK YVIEKKTIDGKAWTKVNPNCGSTTFVVPDLISEQQYFFRVAENRFGIGPPAETIQRTT ARDPIYPDPPIKLIKGLITKNTVHLSWKPPKNDGGSPVTHYIVECLAWDPTGKKKEA WRQCNRRDVEELEFTVEDLIEGGYEFVRKAVNEAGVSKPSATVGPVIVKDQTCPPA IELKEPMEVEEGTDVNIIVAKIKGVPPPTLTWFKAPPKKPDSKEPVVYDTHVNKQVVD DTCVLVLPQSRSDTGLYSITAVNNLGTASKEMRLNVLGRPGPPVGPVKFESISADQM TLSWLPPKDDGGSKIINYVIEKREANRKTWVRVSSEPKECMYTIKPLLEGHEVYFRI MAQNKYIGIGELDSEPETARNLFSVPGAPDKPTVSSVTRNSMTVNWEEPEYDGGSPV TGYWLEMKDTSKRWRVNRDPIKAMTLGVSYKVTGLIEGSDYQFRVYAINAAGV GPASLPSDPVTARDPVAPPGPPFPKVTDWTKSSVDLEWSPPLKDDGGSKITGYIVEYKE EGKEWEKGDKEVRGKTGLVVTGLKEGAFYKFRVAVNVAGVGEPEVTDVIEMK DRIVSPDLQDASVRDRIIVHAGGVIRIAYVSGKPPPTVTWSMNERALPQEAIIETT AISSSMVIKNCQRSHQGVYSLAKNEGGERKKTIIVDVLDVPGVGPVIFLSDNLTNDS CKLTWFSPEDDGGSPITNYVIQKREARRAWTPVTYTVTRQNAVQGLIQGKSYFFR IAAENSIGMGPFVETPNALVIRDPITVPERPEDLEVKEVTNTVSLTWNPPKYDGGSEI INYVLESRLIGTEKPHKVTNDNLSRKYTVKGLKEGDTYEVYRSAVNIVGQKPSFC TKPITCKDELAPPTLDDLDFRDKLTVRVGESFALTGRYSKPKPKIDWFKDEADVLED DRTHIKTPTTLALEKTKAKRSDSGKYCVVENSTGSRKGFQVNVVDRPGPPVGPV IFDEVTKEYMVISWKPLDDGGSEITNYIEKKELGKDIWMPVTSASAKTCTCKVPKLL EGKDYIFRIHAENLYGISDPLVSDSMKAKDRFRVPDAPEQPVVTEVTKDSALVTWNK PNDGKPIITNYILEKRETMKRWVRVTKEPIHPYTKYRVDDLLEGQYEFVRSANENI GIGDPSPPSKPVFARDPIAKPSPPINPEAIDTTCNSELVTWQPPRHDDGGSKIIGYIVEYQ KVGDEEWRANHTPESCPETKYKVTGLRDSQSYKFRVLAVNEAGESDPAHVPEPVL VKDRLEPPELILANMAREQHIRVGDTLRLSAIKGVPPPKVTWKEDREAPTKAQID VTPVGSKLEIRNAAHEDGGIYSLTVENPAGTKTVSVKVLVLDKPGPPRDLLEVSEIRKD SCYLTWKEPLDDGGSVVTVYVVERKDVATAQWSPLSTTSKKKSHMAKHLTEGNQY LFRVAAENQYGRGPFVETPKPIKALDPLHPPGPPKDLHHVDVDKTEVSLVWNKPD DGGSPITGYLVEYQEEGDKDWIKFKTVKNLDCVVTGLQQGKTYRFRVKAENIIGLGL PDDTTIECQEKLVPPSVVELDVKLEGLVVKAGTTVRFPPIIRGVVPVTAKWTTDGTET KTDDHYTVETDSFSSVLTIKNLRKDTGEYQLTVSNAAGTKTAVHLTVLDVPGPPT GPINILDVTPPEYMTISWQPPKDDGGSPVINYIVEKQDTRKGTWGVVSAGSKLKLKV PHLQKGCYEVFRVKAENKMGVGPLDSIPTVAKHKFSPSPSPGKPVVTDITENAAAT SWTLPKSDGGSPITGYVVERREITGKWRVVKNTPIADLKFRVTGLYEGNTYEFVRFA ENLAGLSNPSPSSDPIKACRPIKPPGPPINPKLKDKSKEADLVWTKPLSDGGSPILGY VVEYQKPGTAQWDRINKDELIRQCAFRVPLGIEGNEYRFRIRAAINIVGEGEPRELAES VIAKDIHLHPEVELDVTCRDVITVRVGQTIIRILARVKGRPEPDIWSEKGVLVKDKR VDLIHLPRVELQIEAVRADHGKIIISAKNSSGHAQGSAINVLDRLPGPCQNLKVS VTKENCTISWENPLDNGGSEITNFIVEYRKPNQKGSIVASDVTKRLVKANLLANNE YFVRVCAENKVGVPPTIETKTPILAINPIDRGPENLHIAKDKTFVYLKWRPPDYD GGSNLSYHVERRLKGSADWERVHKGSIKETHYVMDKCVENQIYEFVRQTKNEGGE SDWVRETEVVVKEDLQKPVLDLKLSGVLTVKAGDITRLEAGVRGKPPFEVAVTKD KDATDLTRSPRVKIDTSAESSKPSLTAKRSDGGKYVITATNPAGSFVAYATVNVLD KPGVVRNLKITDVSSDRCTIRWDPPEDDGGCEIQNYILEKESKRMVWSTYSANVLT PSATVTRLIEGNEYIFRVRAENKIGTPPTESKPVIAKTKYDRPGRPDPEVTKVSKEE MTVVWNAPEYDGGSPITGYLEKKEKHAVRWVPVNSAIERRLKQVQNLPGHEY QFRVKAENEVIGIEPSLPSRPVVAKDPIEPPGPPNTFKVVDTTKSSITLAWGKPVYDG GAPIIGYVEMRPKIIADASPDEGWKRCNAAQLIRMEFTVTSLDENQYEFVRVCAQN QVGIRPAELKEAIKPEILEPPEIDLASMRKLVVRAGCPIRLFAIVRGRPAKPVTW RKVGIDNVVRKGQVDLVTMAFLVIPNSTRDDSGKYSLLVNPAGEKAVFVNVKVL

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		DTPGPVADLKVS DVT KTSCHVSWAPPENDGGSQVTHYIVEKREAERTKWSVTVTPEV KKT SFNV TNLVPGNEYFFRV TAVNEYGPGVPTDIPKPVLASDPLSEDP PPRKLEVTEM TKNSATLAWLPPLRDGGAKIDGYIISYREEDQPADRWTEYSVVKDL SLIVTGLKEGK KYKFRVAARNAVGVSM PREAEGVYEAKQLLPKILMPEQITIKAGKKLRVEAHVY GKPNPICWKKGDDDEVVTS SHLA IHKADGSSVLI IKD VTRKDSGYYSLTAE NSSGSDT QKIKVTVMDAPGPPQPPFDISEIDADACSLSWHIPLEDGGSNTNYIVEKCDVSRGDW VTALASVTKTS CRVGLIPGQEYIFRVRAENRFGISEPLTSPKMLAKFPFDVPSEPKNA RVTKVNKDCIFVAWDRPDSGGSPITGYLIERKERN SLLWVKANDTIVRSTEYPCAG LVEGLEYSFRIYALNKAGSSPPSKPTEYVVTARMPVDPGPKPEVVDVTKNASLIWARP KHDGGSRIIGYFVEACKLPDGK WVR CNTPHQIPQEEYTATGLEEN AQYQFRAIAKT AVNISQPSSESDPVTILAENVPRIELSVEMK SLLTVKAGTNVCLDATVFGKPMPTVS WKKD TTPIKQAEGIKMAMKRNCLTELEFSVNRKDSGDYTTITAE NSSGSKSATIKLKV LDKPGPPASVKINKMYADRAMLSWEPLEDGGSEITNYIIDKRETSRPNWAQVSATV PIT SCTVEKLI EGHEYQFRICAENKYGVGDPI LTEPAIAKNPYDPPGRCDPPVISNITKD HMTVSWKAPADDGGSPI TGYLVEKRETQAVNWT KVNRPVIERTLKATGLQEGTEY EFRVTAINKAGPGKPSDASKAVYAQDPLYPGPPAPPKVYDTRSSVSLSWGKPAYD GGSP IIGYLVEVKRADSDHWVR CNLPEKLQKTRFEVTGLMENTEYQFRVYAVNKIG YSDPSDVPDKHCPK DILIPPEGELDAELRKTLILRAGVTMRLYVPVKGRRPPPKITWSK PNVNLREIRGLDIKSTDFDTFLRCENVNKYDAGKYILTLENSCGKKEYTIVVKVLDTP GPPVNVTVKEVSKDSAYVTWDPPIIDGGSP IINVVVEKRDAERKSWSTVTTECSKTSF RVSNLEEGKSYFFRVFAENEYIGIDPGGETRDAVKASETPGPVVDL KALAITKSSCTIG WKKPRSDGGSRI TGYVVD FLTEENKQVRVMKSLSLQYSTKDLKEGKEYTFRVSAEN ENGE GTPSEIVVAKDDVVAPDLDLKDLPLCYLAKENS NFR LKIPKIGKPPAPSVSW KKGEDPLATDTRVSVESTAVNTTLVVYDCQKSDAGKYITLKNVAGTKEGTL SIKV VGKPGIP TGP IKFDEVTA EAMTLKWGPPKDDGGSEITNYVLEKRD S VNNKWVTCAS AVQKTTFRVTRLHGEI EYTFRVSAENKYGVGEGLKSEPIVAKHPFDVPDAPP PPNIVD VRHDSVSLTWDTPKKTGGSPITGYHIEFKERN SLLWKRANKTPIRMDFKVTGLTEG LEYEFVMAINLAGVGKPSLPSEPVVALDPI DPGKPEVISVTRNSVTLIWTEPKYDG GHKLTGYIVEKRDLP SKSWMKANHVNVDPCAFTVTDLVEGGKYEFRIRAKNTAGAI SAPSESTGTII CKDEYEAPTIVLDPTIKDGLTVKAGDSIVLSAISILGKPLPKS SWSRAG KDIRPSDIAQITSTPTSSMLTVKYATRKDAGEYTI TATNPFGTKEEHVKVSVLDVPGPP GPIEISNVSAEKATLTWTPPLEDGGSP I KAYVLEKRETSRLLTWTVVSEDIQACRHVVT KLIQGN EYLFRVSAVNHYGKGEPVQSEPVKMDRFGPPGPPGKPEISNVTKNATATVS WKRPTDDGGSEITGYHVERREKKGLRWVRATKTPVSDLRCKVTGLQEGNTYEFRVS AENRAGIGPPSDASNPVLMKDVA YPPGPPSNAHVTDTTKKSASLAWGKPHYDGGLE ITGYVVEHQVGGDAWIKDTTG TALRITQFVVVDLQTKEKYNFRISAINDAGVGPEA VINVEIVEKEVADPFDLDAELRRTL VVRAGLSIRIFVPIKGRPAPEVWT KDNINLKH RANIENTESFTLLIIP E CNRYDTGKFVMTIENPAGKKS GFVNVVRVLDTPGPVNLNRPTD ITKDSVTLHWDPLPIDGGSRI TNYIVEKREATRKSYSVT TTKCHKCTYKVTGLTEGCE YFFRVMAEN EYGVGEPTETTEPVRASEAPLPD SLNIMDITKNVSLAWPKPRHDGG SKITGYVIEAQRKSGSDQWTHISTVKGLECVVRNLTEGEEYTFQVMVNSAGRSAPRE SRPVI VKEQTM LPELDLRGIYQKLVIARAGDNIKVEIPVLGRPKPTVTWKKGDQILKQ TORVNVENTATSTILNINECV RSDSGPYPLTAKNTVGEVGDVITIQVHDIPGPPTGPIK FDEVSSDFVTF SWEPENDGGVPI SNVVEMRQTDSTTWVELATTVIRTTYKATRLT TGVEYQFRVRAQNR YGVGPGITSASV VANYPFKVP GPPGTPQVTA VKDSMTISWH EPLSDGGSP IALGYHIERKERNGILWQTVSKALVPGNIPKSTGLTDG IAYEFVRVIAENMA GKSKPSKPS EPMFALDPI DPGKPVPLNITRHTVALKWAKPEYTG GFKITSYVVEKRD LPNGRWLKANFSNILENEFTVSGLTEDAAYEF RVI AKNAAGAI SPPSEPSDAITCRDDL EAPRIMVDVRFKDTITL KAGEAFKLEADVSGRPPPTMEWTKDGKELEGTKLEIKIA DFSTHLINKDSRTSDGAYIL TATNPGGFAKHIFNVKVLDRPGPPEGPLAVSDVTSEK CVLSWLPPLDDGGAKIDHYIVQKRETSRLAWTNVATEVQVTKLKVTKLLKGNEYIF RVMAVNKYGVGEPLSEPVLA VDPYGPDP PPKNPEVTITKDSMVVCWGHPSDGG SEIINYIVERRDKAGQRWVKCNKKALTDLRFKVSGLTEGHEYEF RMAENAAGISAPS ATSPFYKACDITVFKPGPPGNPRVLDTSRSSISIAWNKPIYDGGSEITGYMVEIALPEED EWQVVT PPAGLKATSYTITSLIENQEYKIRIYAMNSEGLGEPALVPGTPKAEERMLPP EIELDADLRKVVTIRACCTLR L FVPIKGRPAPEVKWAREHGESLDKASIESTSSYTL LV VGNVNRFD SGKYILTVENSSGSKSAFVNVVRVLDTPGPPQNLKIKEVTKTSVTLTWEPP LLDGGSKIKNYIVEKRESTRKAYSTVATNCHKTSWKVDQLQEGCSYFVRVLAENEY GIGLPAETAESVKASERPLPGKI TLT DVTRNSVSLSWEKPEHDGGSRI LGYIVEMQSK GSDRWATCATVKVTEATITGLIQGEEYSFRVSAQNEKGISDPRQLSVPIAKDLVIPP AFKLLFNTFTVLAGEDLKIDVPIGRPPPAV TWKHDDIPLKQTRVNAESTENNSLLTI KEACREDVGHYTVKLTNSAGEATE TLNVIVLDKPGPPTGPVKMDEV TADSVTL SWE PPKYDGGSSINNYIVEKRDSTSTAWQIVSATVARTTIKACRLKTGCEYQFRIA AENRY GKSTYLNSEPVVAQY PFKVP GPPGTPFVTLASKDSMEVQWHEPVSDGSKVI GYHLE RKERN SILWVKLNKTPIQTKFKTTGLEEGIEYEF RVS AENIVGIGKPSKPS ECYAAHD PCDPPGRPEAIIVTRNSVTLQWKKPTYDGGSKI TGYIVEKKLEPDGRWMKASFTNIID TQFEVTLGLEDDRHYEF RVIARNAAGVFSEPS ESTGAI TARDEVEPPRISMDPKYRDT VVQAGESFKIDADIYKPIPTTQWVKGDQELSSSTARLEIKSTDFATSLSVKDAVRVDS GNYILKAKNVAGEKSVTINVKVLDRPGPPEGPV AISGVTA EKCTLAWKPLQDGGSD ITNYIVERRET SRLVWTLVDANVQTL SCKVLKLEGNEYIFRIMAVNKYGVGEPLSE SLIAKNPFVVPDAPKAP EVTAVTKDSMIVVWERPASDGGSEILGYVLEKRDKEGIRW

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		TRCHKRLIGELRLRVLTGLLENHNYEFRVSAENAAGLSESPSPSAYQKACDPIYKPGPP NNPKMVDVTRSSVFLSWTKPIYDGGCEIQGYIVEKCDVSVGEWMTCPPTGINKTNL EVEKLEKHEYNFRI CAINKAGVGEHADVPGPVMVEEKLAPDIDLDELRLKVINIRA GGSRLRFVPIKGRPTPEVKWQKVDGDIRDAAIIDVTSSTSLVLDNVRYSKGKTLT LENS SGTSAFVTVRVLDTPSPVNLKVTETIKDSVSIWEPPLLDGGS KIKNYIVEKR EATRKS YAAVVTNCHKNWIDQLQEGCSY YFRVTAENEYIGLGPARTADPIKVAE VPQPPGKITVDDVTRNSVSLSWTKPEHDGGSKI IQYIVEMQAKNTDKWSECARVKSL DAVITNL TQGE EYLFRVIAVNEKGRSDPRSLAVPI IAKDLVIEPDVRPAFSSYSVQVQG DLKIEVPISGRPKPSISWTKDGMPLKQTRINVTDSDLDTLSIKETHKDDGGQYGITV SNVVGQKTAIEIITLDKDPDPKGPVKFDEISAESITLSWNPPLYTGGCQITNYIVQKR DTTTTVWDVVSATVARTTLKVT KLKTGTEYQFRIFAENRYGQSFALESEPVVAQYYPY KEPGPPGTPFVTAISKESMVVQWHEPINNGGSPVIGYHLERKERN SILWTKVNKTIH DTQFKALNLEEGIEYEFVYAENIVGVGKASKNSECYVARDPCDPPGTPEAIIVKRNE ITLQWTKPVYDGGSMITGYIVEKRDLPEGRWMKASFNTV IETQFTVSGLTEDQRYEF RVI AKNAAGAI SKPSDSTGPITAKDEVELPRISMDPKFRDITVFNAGETFRLEADVHG KPLPTIEWL RGDKEIEESARCEIKNTDFKALLIVKDAIRIDGGQYILRASNVAGSKSFP VNVKVLDRPGPPEGPVQVTGVTA EKCTLAWSPLQDGGSDISHYVVEKRETSRLAW TVVASEVVTNSLKVTKLEGNKYIFRIMAVNKYGVGEPLASAPVLMKNPFVLPGPVK SLEVTNIAKDSMTVCWNRPSDGGSEIIGYIVEKRDRSGIRWIKCNKRRI TDLRLRV GLTDEH EYFRVSAENAAGVGEPSPATVYVKACDPVFKPGPPTNAHVVDTKNSITL AWSKPIYDGGSEILGYVVEICKADEEEWQIVTPQTGLRVTRFEIAKLIHQEYKIRVC ALNKVGLGEAASVPGTVKPEDKLEAPELDDSELKGI VVRAGGSARIHIPFKGRPTP EITWSKEEGEFTDKVQIEKGINFTQLSIDNCDRNDAGKYLKLENSSGSKSAFVTVKV LDTPGPPQNLA VKEVRKDSVLLVWEPPIIDGGA KVKNYVIDKRESTRKAYANVSSKC NKTSFRVENLT EGAIIYFRVMAENEFVGVPETSDAVKASEPPSPGKVTLDVDSQ TSASLMWEKPEHDGGSRLILGYVVMQPKGTEKWSVVAESKVCNAVVSGLSSGQEY QFRVKAYNEKGKSDPRVLGIPVIAKDLTIQPSFKLPNTYSVQAGEDLKI EIPVIGRPRP KISWVKDGEPLKQTRVNVEETATSTILHIKESKDDFGKYSVTATNSAGTATENLVS IVLEKPGPPVGPVKFDEVSADFVVISWEPPAYTGGCQISNYIVEKRDTTTTNWQMV ATVARTTIKISKKTGTEYQFRIFAENRYGKSTPLDSKPVVVQYPPKEPGPPGTPFVTS ISKDQMLVQWHEPVNDGSKVTGYHLEQKEKNSILWVKLNKIPIDTKFKTTLGLDE GLEYEFVSAENIVGIGKPSKVSECYVARDPCDPPGRPEAIVITRNSVTLKWKPVVD GGSKITGYIVEKDLDPGRWMKASFNTNVTEFTVTGLVEDQRYEFVRIARNAADN FSEPSSESGAITARDEIDAPNASLDPKYRDVIVHAGETFVLEADIRGKPIPIDIWSKDG NELEETAARMEIKSTLQKTTLIVKDCIRTDGGQYTLKLSNVGGTKTIPITVKVLD RRG PPEGPLKVTGVTA EKCYLAWNPPLQDGGASISHYIEKRETSRLSWTQVSNVQALN YKVTKL L PNEYIFRMAV NKYIGIGALESEPV IACNPYKRGPPSTPEASAITKDSM VLTWTRPVDDGGAIEGYILEKRDKEGIRWTKCNKKTLDLRFRTGLTEGHSYEF VAAENAAGVGEPSSESVFYRACDALYPPGPPSPNPKVTDTSRSSVSLAWN KPIYDGA PVRGYVIELKKAAD EWTCTPPSGLQGKQFTVTKLKENT EYNFRICAFNTEGVGE ATIPGSVVAQERMEAPEI EL DADLRKVVTLRASATLRLFVTIKGRPEPEVKWEKAEI LTERAQIEVTSSYTMLVIDNVTRFDSGRYNLTLENNSGSKTAFVNVVRVLDSPSAPVNL TIREVKDSVTL SWEPLIDGGAKITNYIVEKRETRKAYATITNNCTKNTPKIE NLQE GCSY YFRVLASNEYIGLPAETAEPVKVSEPLPPGRVTLVDVTRNTATIKWEKPESD GGSKITGYVVMQTKGSEKWSACTQVKTLETTISGLTAGEEYVFRVA AVNEKGRSD PRQLGVPIAKDIEIKPSVELPNTFNVKANDQLKIDIPFKGRPQATVAWKKDQVLR ETTRVNVASKT VTTLSIKEASREDVGTYELCVSNTAGSITVPITVIVLDRPGPPGPIRI DEVSCDNVSI SWNPPEYDGGCQISNYIVEKRETTSTTWQVVSQAVARTSIKIVRLTTG SEYQFRVCAENRYGKSSYESSA VAEYFPSPGPPGTPKVVHATKSTMVVSQVVP NDGGSQVIGYHLEYKERSSILWSKANKVLIADTQMKVSGLDEGLMYEYRVYAENIA GIGKCSKACEPVPARDPCDPPGQPEVTNITRKS VSLKWSKPRYDGGAKITGYIVERRE LPDGRWLKCNFTNVQETYFEVTELTEDQRYEFVRFARNAADSVSEPSSESTGPITVKD DVEAPRIMMDVKFRDVIIVKAGEVLKINADIAGRPLPVI SWAKDGVEIEERAKTEIVS TDYTTTLTVKDCVRRDTGQYVLT LKNVAGTRTMAVNCVLDKPGPPAGPLEINGLT AEKCSLSWGRPQEDGGADIDYIIVEKRETSRLAWTICEAELRTSCVKTLKLGNEY IFRVTGVNKYGVGEPLSMVAVKALDPFTTPSPPTSLEITSVTKDSMTLCWSRPETDGG SDISGYI IERREKNSLRWMRVNKKPVYDLRVKSTGLREGCEY EYRVFAENAAGLSLP SETSPLVRAEDPVLPSPPSKPIVDSGKTTITIGWVKPLFDGGAPITGYTV EYKKEET DWKVAIQSFRGTEY TMSGLTTGDEYVFRVRS LNKMGASDPSDSDPQVAKEREEEP VFDVDSEMRKTLIVKAGSSFTMTVPFRGRPIPNVSWSKPDTDLRTRAYIDSTDSRTLL TIENANRNDSGKYTLTIQNVLSAASMTFVVKVLDSPGPPANITVREVTKETAMLSWD VPENDGGAPVKNYHIEKREASKAWVSVTNNCNRLSYKVTNLQEGAIYYFRVSGEN EFGVGVP AETKEGVKITEKPSPEKLGVT SVSKDSVSLSWLKP EHDGGSRIHYVVEA LEKGQKTWVKCAVKTTHHVVSGLRESHEYFRVFAENQAGLSDPRELLLPVLIKD QLEPPEIDMKNFPSHTVYVRAGSNLKV DIPISGKPLPKVTL SRDGVPLKATMRENT EIT AENLTINL KESVTTAGRYEITAANS SGTTKTFINII VLD RPPGTPGPVAISDI TEESVTL KWEPKYDGGSHVNTYIVLKRETS AVWSEVSATVARTMIKVMKLTGEEYQPRIK AENREGISDHIDSVCVVVKLPYTTPGPSTP WVSNTRESITV GWHPEPVNNGGSAVIG YHLEMKDRNSILWQKANKMIIRTHFKVTTISAGLIYEFRVYAENAAGIGKPSHPSEP VLAI DACEPRNVRI TDISKNSVNLWQQPAFDGGSKITGYI VERRDLPGRWTKASF TNV IETQFTVSGLTQNSQYEFRVFARNVAGSVSNPSEVVGPI TCIDSYGGPVIDLPLEY

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		TEVVKYRAGTSVKLRAGISGKPEPTIEWYKDDKELQTNALVCVENSTDLASILIKDA NRLNSGSYELKLRNAMGSASATIRVQILDKPGPPGGPIEFKTVTAEKITLLWRPPADD GGAKITHYIVEKRETSRVVWSMVAENLEECIVTTTKIKGNEYVFRVRAVNKYGIGEP LESEPVVAKNAFVTPGPPSIPVTKITKNSMTVVWDRPTVDGGSEINGYFLERRDKKS LAWLKVLKETIRDTRQKVTGLTENS DYQYRVCANNAAGVGFSEPSDFYKAADPID PPGPPAKIRIADSTKSSITLGWSKPVYDGGSDVTGYVEMKQGDDEEWITIVSTRGEV RTTEYVVSNLKPGVNYFQVSAVNCAGQGEPITMTEPAQAKDVLLEEPEIDLDVALRT SVIAKAGEDVQLLIPFKGRPPPTVTWRKDEKNLGS DTRYSIQNTDSSSLLVIPQVTRN DTGKYILTIENGVGQPKSSTVSVKVLDTAACQKLQVKHVS LGTVTLWDPPLIDGG SPIINYVIEKRDATAKRTWSVSHKCSGTSFKVTDLSEKTPFFFRVLAENEIGIGEP CETT EPVKA AEVPAPIRDL SMKDS TKTSVVL SWTKP DFGGSIITDYLVERKKGGEQASH AGISKTCEIEIGQLKEQSVLEFRVSARNEKGQSDPVTIGPLTVKELVITPEVDLSEIPGA QISVRIGHNVHLELPYKGP KPSISWLKDGLPLKESEYVRF SKTENKITLSIKNSKKEH GGKYTVILDNAVCRNSFPITITLGPSPKPKGPIRFDEIKADSAIMSWDIPEDDGGGEIT CYSIEKREASQTNWKMVCSSVARTTFKVSNLVKDSEYQFRVRAENRYGVSEPLASNI IVAKHQFRI PGPPGKPV IYNVTSDGMSLTWDAPVYDGGSEVTGFHV EKKERN SILWQ RVNTSPISGREYRATGLIEGLDYQFRVYAENSAGLSSPSDP SKFTLAVSPVDPGPTDY IDVTRETITLKNWNPRLDGGSKIVAYSIEKRQGS DRWVR CNFTDVSE CQYTVTGLSPG DRYEFRIIARNAVGTISPPSQSSGLIMTRDENVPPTVEFGPEYFDGLVIKSGDSLRIKAL VQGRPVPRVTFKDGVEIERRMNMEITDVLGSTSLFVRDATRDHRGVYTV EAKNVS GSTKAEVTVKQDTPGKVVGPIRFTNITGEKMTLWWEAPLNDGCAPVTHYIEKRET SRLAWALIEDNCEALS YTAIKLITGNEYQFRISAVNKFVGVRPLESDPVVAQIYQTIPD APGVPEPSNVTGNSITLTWTRPESDGGSEIQHYILERREKKSTRWVKVISKRPISETRF KVTGLVEGNEYEFHVMAENAGIGPASGISRLIKCREPVNPPSAPS VVKVTDTSKTTV SLEWARPVFDGGMEIGYI IEMCKADLGDWHKVNT EPCVKTRYTVTDLQAGEEYKF RVSAVNGAGKGDSCEVTGTIKAVDRLSAPELDIDANFKQTHIVRAGVISIRLFIA YQGR PTPTAVWSKPD SNLSIRADIHTDSFSTLTVENCNNDAGKYTLTVENNSGKKSITFT VKVLDSPGPPGPI TFKDVT RGSATLMWDAPLLDGGARIHHYVIEKREASRRSWQVVS EKCTRQILKVS ELTEGV PYYFRVSAENEYGVGEPEYEMPEPIVATEQPAPRRRLDVVD TKSSAVLAWLKPDHDGGSRIITSY LLEMQRKGSDFWVEAGHTKQLTFTVERLVENTE YFPRVKAKN DAGYSEPREAFSSV IIEKPQIEPTADLTGINTQLITCKAGSTFTIDVPISG RPAPKV TWKLEEMRLKETDRMSIATTKDR TLT VKDSMRGDSGRYFLTLENTAGVK TFTITVVVIGRPGPV TGP IEVSSVSAESCVLSWTEPKDDGGTEITNYIEKRESGTTAW QLINSSVKRTQIKVTHLT KYKEYCFRVSS ENRFGVSKPLESVPIVAEHPFPVPSAPTRP EVYYV SANAMSIRWEEPYHDGGSKIVGYWVEKKERNITLWVKENKVPCL ECNKYV TGLVEGLE YQFRTYALNAAGVS KASEASRPIM AQNPVDP PGRPEVTDVTRSTVS LIW SAPVYDGGSKVGYI IERKPVS EVGDGRWLKCN YTVSDNFFT VTLSEGDYEFVRV LAKNAAGV I SKGSESTGPVTCRDEYAPPKAELDARLQGLDVTIRAGSDLVLDAAVG GKPEPKIIWTKGDKELDLCEKISLQYTGKRATAVIKYCDRSDSGKYTLTVKNASGTK SVSVMVKVLDSPGPGKLT VSRVTEEEKCTLAWSLPQEDGGAETHYIVERRETSRLN WVIVEGECLTASYV VTRLIKNNEYTFRVRAVNKYGLGV PVESEPIVARN SFTIP SQPGI PEEVGAGKEHIIQWTKPESDGGNEISNYLVDKREKKS LRWTRVNKDYVYDTRLK VTSLMEGCDYQFRVTA VNAAGNSEPS EASNFI SCREPSYTPGPPSAPRVDDTKRSISL AWTKPMYDGGTDIIGYVLEMQEKD TDQWCRVHTNATIRNNEFTVPDLKMGQKYSF RVAAVNAKMSDYSSETTAIEIPEVRL EIPDLELADDLKKTIVIRAGASLR LMSVSG RPSPVITWSKKGIDLANRAI DNTESYSLIIVDKVNR YDAGKYTIEAENQSGKKSATV LVKVYDTPGPCPSVSVKEVSRDSVTITWEIPTIDGGAPVNNYIEKREAA MRAFKTVT TKCSKTLYRISGLVEGTMYYFRVL PENIYGIGEP CETSDAVLVSEVLPVT KLEVV DV TKSTVTLAW EKPL YDGGSRLTGYVLEACKAGTERW MKVVT LKPTVLEHTV IISLNEG EQYLFRVRAQNEKGVSEPREIVTPVT VQDLRVLPTIDLSTMPQKTIHVPAGRPIELVPI TGRPPPTASWFFAGSKLRESERVTVETHTKVTKLTIRETTIRDTGEYTTLELKNVTGTS ETIKVIILDKPGPPTGPIKIDEIDATSVTISWEPPELDGGAPLSGYVVEQRDAHRPGWLP VSESVTRPTFKFTRLTEGNEYVFRVAATNRFIGSYLQSEVIECRSSISIPGPPETLQIFD VSRDGMTLTWYPEDDGGSQVTGYIVERKEVRADRWRVNKVPVMTMYRSTGLI EGLEYEHRVTAINARGTGKPSRSPKPTVAMDPIAPPKPKQNPRTVDTTRTSVSLAWS VPEDEGGSKVTGYLIE MQKVDQREWTKCNTPTPKIREYTLTHLPQGA EYFRVRLAC NAGGPGEPAEVPVTKVTEMLEYPDYELDERYQEGVFRVQGGVIRLTIPIKGKPPFV CKWTKEGQDISKRAMIATSETHTELVIKEADRNDSGTYDLVLENKCKKTVYIKVK VIGSPNTPEGPLEYDDI QARSVRVSWRPPADDGGADILGYILERREVPKAAWYTI DSR VRGTSLVVKG LKENVEYHFRVSAENQFGISKPLKSEEPVIPKTP LNPPEPPSPNPPEVLD VTKSSVLSWSRPKDDGGSRTGYI IERKETSDKWRHNKQTITMYTMTGLVP DAEQFRIIAQNDVGLSETSPASEPVVCKDPFDKPSQPGLEILSISKDSVTLQWEKPE CDGGEKILGYWVEYRQSGDSAWKKS NKERIKDRQFTIGGLLEATEYEFVRVFAENETG LSRPRRTAMSVKTLTSGEAPGVKEMADVTTLGEAAQLSCQIVGRPLPDIKWYRF GKELIQSRKYKMSDGRTHLTVMTDEQDEGEVYTCVATNEVEGEVSSSKLLQAA PQFHPGYPLKEKYGAVGSTLR LRVMIYIGRPVPAMTWFGQKLLQNSEKTI ENT EYH YTHLV MNVQRKTHAGKYKVL SNAFGTVDATLDVEIQDKPKDKPTGPVIEALLKN SVVISWKPPADDGGSWI TNVYVKECEAKEGAEWQLVSSAISVTTCRIVNLTENAGYY FRVSAQNTFGISEPLEVASIVI KSPFEKPGVPGKPTITAVTKDSCVVAWKPPASDGGGA KIRNYLYERREKKQNKWIAVTTEEIRETVFSVQNLIEGLEYEFRVKCENLGGSESEWSE ISEPVTPKSDVPIQAPHFKEELRNLNVR YQSNATLVCKVTGHPKPIVKWYRQGEKIIA

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		DGLKYRIQEFKGGYHQLIIASVTDDDATVYQVRATNQGGSVSGTASLEVEVPAKIHL PKTLEGMGAVHALRGEVVSIIKIPFSGKDPDVTWQKGQDLIDNNGHYQVIVTRSFSTSL VFSNGVERKDAGFYVVCANRFGIDQKTVELDVADVPDPPRGVKVSDVSRDSVNLT WTEPASDGGSKVTNYIVEKCATTAEERWLRVQARETRYTVINLFGKTSYQFRVIAEN KFGLSKPSEPSEPTVTTKEDKTRAMNYDDEVDRETTTAKASHSKTKELYEKYMAIE DLGRGEFGIVHRCVETSSKRTFMKFKVVKGTQDQVLVKKKEISILNIARHRNIIYLHES FESMEELVMIFEFISGLDIFERINTSAFELNEREIVSYVRQVCEALEFLHSQNI GHFDIRP ENIIYQTRKNSTIKIEFGQARQLKPGDNFRLLFTAPEYYAEVHQHDVVSTATDMWS LGTLVVYVLLSGINPFLAETNQMIENIMNAEYTFDEEAFKEISLEAMDFVDRLLVKER KSRMTASEALQHPWLKQRIDRVSTKVIKIRLKHRRYYHTLIKDLNMVVSAAISCGG AIRSQRGVSAKVVASIEIGPVSGQIMHAIGEAGGVVKYVCKIENYDQSTQVWTWYF GVRQLENSEKYEITYEDGVATMYVKDITKFDGDTYRCKVNDYGEDSSYAELEFVKG VREYDYCRRTKKVKRRTDAMRLLERPPEFTLPLYNKTAYVGENVRFVGTITVHP EPRVTWYKSGQKIKPGDDEKKTFFESDKGLYQLTINSVTTDDDAEYTVVARNKHGE DSCAKLTVTLHPPTTETTLRPMFKRLLANAECHGQSVCFEIRVSGIPAPTLKWEKD GQPLSLGPHIEIVHEGLDYALHIRTLPEDTGYRVATNTAGSTSCQAHLQVERLR YVKQEFQSKERERHVQKQIDKTLRMAEILSGTETVPLTPVAQEALEAAIYKPAVS TKTVKGEYRLQTEKKKEERKLMPYEVPEPRRFKQATVEEDQRIKQFVPMSDMKWY KKIRDQYEMPKLDRVQKRPKIRLSRWEQFVVMPLPRITDQYRPKWRIPKLTQD DLEMVRPARRTPSPDYDLYYYRRRRSLGDMSEDELLLPIDDLAMKRTTEERLRL EEEELEGFSAAPSRSPRPFLESSLRYSPPAHVKVEDRRDRFRYSTYHVPTKEETSTS YAELEERHAQASYRQPKLRQIRMAEKEEEELLRPVTTQRLSEYKSELDYMSKEEKS KKKSKRQVTEITEIEEYERIRRAQRESSSVSLRRRRSLSPITYIELMRPVSELIIRS HPRPAEEYEDDAERRSPPTERTPRSPSPVSSERSLSRPFERSARFDIFSRYESMKAALK TQKTSERKYEVLSQQPFTLDHAPRITLRMRSHRVPQGNTRFILNVQSKPTAEVKWY HNGVELQESSKIHYTNTSGVLTLEILDCQTEDGGTYRAVCTNYKGEASDYATLDVTG GAYTTYASQRRDEEVPKSVFPELTKEAYAVSSPKRTSELEAASSVREVKSQMTETR ESLSTYEHYASAEKMSATSEKSLSEKATVRKIKTTLAARILTKPRSTVHEGESARFS CDTDGEPVPTVTLWREGQVSTSAHQVTTTKYKSTFEISSVQASDEGNYSVVENS DGKQEAQFTLTQKARVIEKAVTSPPRVKSPEPRVKSPEVKSPEPRVKSPEPVTSHPK AVSPTEKPTKEGQHLVPSAPPKIQSLKAEASKDIAKLTCAVESSALCAKEVAWYK DGKKLKENGHPQPHYSDGTIELKIHNLSESDCGEYVCEVSGEGGTSKTSFQPTGQS FKSIEHQVSSISETTKSVQKTAESPEAKKQEPPIAPESISSKPVIVTGLRDTTVSSDSVAKF TIKVTGEPQPTITWTGDKAIAQGSKYKLSNKEEFLEILKTETSDGGLYACTVNTSA GSVSSCKLTIKAVKDTAEQKVSTQKTSEVTSQKKAQAEEISQKALTSEEEKMSSEVK SHETLAKEEASKVLIAEEVKRSAAASLEKSIHVHEEVTKTSQASEEVKTHAEIKTLSTQ MNIITKGQATLKANIAGATDVKWVNLGTLEPNSEYRYGVSGSDQTLTIKQASHRE EGILSCIGKTSQGVVKQDFDLTLEELSDAPSFITQPRSQINEGQNVLFSCSEVSGEPPS EIEWFKNNLPISISSNISVSRNRNVYTLERNAAVSDSGKYTIKAKNFHGQCSATASLT VLPVLEEPPREVVLKTSDDVSLHGSVSSQSVQMSASKQEAFFSFSSSASSMTEMFK ASMSAQMSMQESFVEMSSSSFMGKSMTQLESSTSRMLKAGGRGIPPKIEALPSDI SIDEGKVLTVACFTGEPTPEITWSCGGRKIQNQEQGRPHIENTDGLTTLIMDVQK QDGLYTLSLGNEFGSDSATVNIIRSM
2	Mouse MYH4 (Myosin-4)	MSSDAEMAVFGEAAPYLKSEKERIEAQNKPFDAKSSVFVVDAKESYVKATVQSRE GGKVTAKEGGATVTVKDDQVFSMNP PKYDKI EDMAMMTHLHEPAVLNKLKERY AAWMIYTYSGLFVTVNPKWLPVYNPEVVAAYRGKKRQEAAPHIFSIDNAYQFM LTDRENQSLITGESGAGKTVNTRKVIQYFATIAVTGDKKKKEATSGMKMGTLQDQII SANPLLEAFGNAKTVRNDNSRFGKFIIRIHFGATGLASADIEITYLLEKSRVTFQLKA ERSYHIFQYIMSNKKPELIEMLLITTNPYDFAYVSQGEITVPSIDQEEELMATDTAVDI LGFSADKVAIYKLTGAVMHYGNMKFKQKQREBQAEPDGTEVADKAAYLTSLSNA DLLKALCYPRVKVGNFYVTKGQTVQQVYNSVGALAKSMYKMLWMTVINQQQL DTKQPRQYFIGVLDIAGFEI FDNLTLEQLCINF TNEKLQQF FNHMFVLEQEEYKKEGI DWEFIDFGMDLAACIELIEKPMGIFSLIEECMFPKATDTSFKNKLYEQHLGKSNFQ KPKPAKGKAEHSLVHYAGTVDYNIGWLDKNDPLNETVVGLYQKSLGKTLAFL FSGGQAAEAEGGGGKGGKKGSSFTQVSALFRENLNKMLTNLSTHPHFVRCLIP NETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPRIILYADFKQRYKVLNLSAIEPG QFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGLEEMRDEKLAQLITRTQAV CRGYLMRVEFKMMERRESIFCIQYNVRAFNMVKNWPMWMLYFKIKPLLSAETE KEMANMKEDFEKAKEDLAKSEAKRKELEKMLVALMQEKNLQLQVQAEADGLAD AEERCQDLIKTKIQLEAKIKELTERAEDEEINAELETAKRKLDECESELKKDIDLEL TLAKVEKEKHATENKVKNLTEEMAGLDENIAKLTKKKALQEAHQQTLDLQAE DKVNTLTAKATKLEQQVDDLEGSLEQEKKLRLMDLERAKRKEGLDKLAQESTMDIE NDKQQLDEKLKKFEFMSNLQSKI EDEQALGMQLQKKIKELQARI EEL EEEIEAERA SRAKAEKQSRDLSRELEEISERLEEAGGATSAQIEMNKKRAEFQKMRRLLEEATLQ HEATAAALRKKHADSVAELGEQIDNLQRVKQKLEKEKSELKMEIDDLASNMTVSK AKGNLEKMCRTLEDQLESEVKTKEEQQLINELSTQKARLHTESGEFSRQLEKDA MVSQLSRGKQAFQQIEELKRQLEEEESAKNALAHALQSARHDCDLREQYEEEQE AKAEQRAMSKANSEVAQWRTKYETDAIQRTTEELEAKKLAQRLQDAEBHEAV NSKCALEKTKQRLQNEVEDLMIDVERSNAACAALDKQRNFDKVLAEWKQKYE TQAELEASQKESRLSTELFKVNAYEESLDQLETLKRENKNLQOEISDLTEQIAEGG

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		KHIHELEKI KKQIDQEKSELQASLEEEAASLEHEEGKILRI QLELNQVKSEIDRKIAEKD EEIDQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGLNEMEIQLNHANRQA AEAIRNLRNTQGMKDTQLHLDDALRGQDDLKEQLAMVERRANLMQAEIEELRAS LEQTERSRRVAEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIVQEAR NAEEKAKKAITDAAMMAEELKKEQDTSALHERMFKKNMEQTVKDLQHRLEDAEQL ALKGGKKQIQKLEARVRELENEVENEQKRNIEAVKGLRKHERRVKELTYQTEEDRK NVLRQLDLVDKLQTKVKAYKRQAEAEQSNVNLAKFRKIQHELEEEAERADIAES QVNKL RVKSREVHTKVI SEE
3	Mouse E9Q1W3 (Nebulin)	MADEEEYEEVIEYYTEETVYEEVPGETITEVYETTTTTRTSDYGQSETATPALAQPPA KPVERKKVIRKKVDSKFMTPYIQHSQKMQDLFSSNKYKENYEKAKGQPYAITSDTP ELRRIKKVQDQLSEVKYRMDGDVAKTICHVDEKAKDIEHAKKVSQQVSKVLYKQN WEDTKDKYLLPPDAPELVQAVKNTAMFSKKLYTEDWEADKALFYPYNDSPELRRV AQAQKALSDIAYKKGLTEQQTQFTSLPDPPEVEFAKKVTNQVSKQKYKEDYEKKVK GKWSETPCFEATARMNADNLSSRKYQEDFENMKDQIYFMQETETPEYKMNKQAGV AASKVKYQDYKKNKGADYNVLPASENPLLRQLKTAGNVLSKLYKENYEKTKA RSINYCETPKQLDVLQNFSSDTKYKDSYLNILGHYVGSFEDPYHTHCMKVTAQN SDKNYKAEYEEEDRGKGFPPQTI TQEYEAIKKLDQCKDHAYKVHPDKTKFTQVTDSP VLVQAQVNSKQLSDNLNYKAKHESEKFKCHIPDAPAFIQHRVNAVNLSDNVYKQDW EKS KAKKFDIKVDAILLAAKANSKNASDVMYKKDYKSKGKMIGALSINDDPKML HSLKTAKNQSDRLYRENYEKTAKSMNYCETPKYQLD TLLKNFSEAKYKDSYVKN MLGHYVGSFEDPYHTHCMKVS AQNSDKSYKAEYEEEDKGKCYFPQTI TQEYESIKKL DQCKDHTYKVHPDKTRFTAVTDPVLVLAQLNTKQLSDNLNYKAKHEGEKFKCSVP ADAEPIQHRVNAVNLSDNVYKHDWEKS KAKKFDIKVDAILLAAKANSKNASDV MYKKDYKSKGKMIGALSINDDPKMLHSLKTAKNQSDREYRKDYKESKTIYTAPLD MIQVTQAKKSQAIASDVYKHLVHNYSYPPDSINVDLAKKAYALQSDVEYKADYNS WMKCGGWVPPGSL EMEKAKRASEILNEKKYRQHPDTLKFTSIEDAPITVQSKINQAAQ RSDIAYKAKGEEILHKYNLPADLPQFIQAKVNAYNISENMYKADLKDLSKKGYDLRI DAIPIKAAKAAARQAASDIHYKKDYKAKGKMVGFSQLQDDPKLVHYMNVAKIQSD REYKKDYKTKYNTPHDMFDDVAAKKAQDVISNTNYKHS LHHYTYLPDAMDLE LSKNMMHIQSDNAYKEDYNTWMKGIWIPIGSLDQEKVKKAGDALS EKKYRQHPD TLKFTSIVDSPVMVQAKQNTQQASDILYKAKGEDVKHKYTLNPDLPQFLQAKCNAY NISDVICYKRDWHDLIAKGNVNLDAIPI TAAKSSRNIASDYKYKEAYEKAKGKHVG FRSLQDDPKLVHYMNVAKLQSDREYKKNYENTKTSYHTPGDMVSI TAARMAQDVA TNVNYKQPLHHYTYLPDALSL EHTNRNAQIQSDNVYKDEYNSFFKGIWIPIGSLEV EKAKKAGEALNERKYRQHPDTIKFTSVPDSMGVLAQQNTKQLSDNLNYKVEGEKM KHKYTMDPELPQFIQAKVNAINMSDAHYKADWKKTLAKGYDLRPAIPIVAAKSSR NIASDCYKYEAYEKARGKQIGFLSLQDDPKLVHYMNVAKIQSDREYKKGYEASKTR YHTPLDMFVS TAAKKSQEVATNSNYRQPHYNYTLLPDALNVEHSRNAMQIQSDNLY KSDFTHWMKGIWVPI DSLEVEKAKKAGEILSERKYRQHPDKLFTYSMDTMBQAL NKS NKL TMDKRLYTEKWNKDKTSIHVMPDTPDILLSRVNQITMSDKLYKAGWEEEEK TKGYDLKPDIAIKAAKASRDIASDYKYQAYEQSKGKHIGFRSLEDDPKLVHFMQ VAKIQSDREYKKGYEKSKTSFHTPVDMPFSVAAKKSQEVATNANYRNI IHTYNMLP DAMSFELAKNMMQIQSDNQYKADYADFMKGI GWLPLGSL EAEKNKKAMEI ISEKK YRQHPDTLKYSTLMDSMNMVLAQNNAKIMNKNYLKQAW EADKTIHIMPDIP EIVL AKANAINMSDKLYKLSLEESRKKGYDLRTDAIPIKAAKASRDIASDYKYKHSYEKEK GKMVGFRSLEDDPKLVHSMQVAKMQSDREYKKNYEKTKTSYHTPADMLSV TAAK DAQANI TNTNYKRLIHKYILLPDAMNIELTRNMNHIQSDNEYKQDYREWYRGLGWS PAGSLEVEKAKKATEYASDQKYRQHPSTFQFKKLTDSMDMVLAQN AHTMKNHLY TIDWNKDKTKIHVMPDTPDILQAKQNTLYSQKSYRLGWEEALKKGYDL PVDIAISV QLAKTSRDIASDFKYKQGYRKQLGHHIGFLSVQDDPKLVLSMNVAKMQSDREYK DFEKWTKYTSVPDMLGVVLAKKCQALVSDADYRNYLHQWTC LDPQNDVIAQAK VYELQSENMYSKDL EWL RGI GWSPLGSL EAEKNKRASEI ISEKKYRQPPDRNKFTSIP DAMDIVLAKTNAKNRSDRLYREAWDKDKTQIHIMPDTPDII LAKANLINTSDKLYKM GYEELKKGYDL PVDIAIPIKAAKASREIAS EYKYKEGFRKQLGHHIGARNIKDDPKM MWSMHVAKIQSDREYKDKFEKWKTKYSSPVDMLGVVLAKKCQTLVSDADYRNYL HQWTC LDPQNDVIHARQAYDLQSDNVYKADLQWLKIGWIPSGSLEDEKNKRATQI LSDHVYRQHPNPKFSSLMDSMPMVLAKNNAITMNHLYTEAWNKDKT TVHIMPD TPEVILAKQNKVNYSEKLYKLGLEEA KRKYDMRLDAIPIKTAKASRDIASDFKYKE GYRKQLGHHIGARGIHDDPKMMWSMHVAKIQSDREYKDKFEKWKTKYSSPVDML GVVLAKKCQTLVSDADYRNYLHQWTC LDPQNDVIHARQAYDLQSDNVYKADLQW MRGIGWVPIGSLDVEKCKRATEILSDKIYRQPPDKFKFTSVTDSLEQVLAKNNAITMN KRLYTEAWDKDKTQIHIMPDTP EIMLARMNKNVNYSELYKL ANEEAKKGYDLRSD AIPIVAASRDIASDYKYKDG YRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDRE YKDKFEKWKTKYSSPVDMLGVVLAKKCQILVSDIDYKHP LHEWTC LDPQNDVIHAR QAYDLQSDNVYKADLQWMRGIGWVPIGSLDVVKCKRAAEILSDNIYRQPPNKF SVTDSLEQVLAKSNALNMNKRLYTEAWDKDKI QIHVMPDTP EIMLARQNRINYS EYLKLANEEAKKGYDLRVDIAIPIVAAKASRDIASDYKYKGYRKQLGHHIGARNI EDDPKMMWSMHVAKIQSDREYKDKFEKWKTKYSSPVDMLGVVLAKKCQTLVSDV RNYLHQWTC LDPQNDVIHARQAYDLQSDNVYKADLQWMRGIGWVPIGSLDVVK RAAEILSDNIYRQPPNKFKFTSVTDSLEQVLAKSNALNMNKRLYTEAWDKDKTQIHI

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		MPDTPEIMLARQNKINYSESLYRQAMEEAKKEGYDLRSDAIPIVAASRDIASDYK YKEAYRQQLGHHIGARAIHDDPKMMWSLHIAKVQSDREYKKEFEKYKTRYSSPVD MLGIVLAKKCQTLVSDVDYKHPLHEWTCPLDQNDVIQARKAYDLQSDNLYKADLE WMKGIGWVPIGSLEVVKAKRATEILSDNIYRQRPDTLKFTSITDSLEQVLAKNNATM NKRLYTEAWDNDKKTIHVMPDTPPEIMLAKLNRINYSDKLYKVSLEEARRQGYDLRV DAIPIRSASASREIASDYKKEGYRQQLGHHIGARNIKDDPKMMWSMHVAKIQSDR EYKDFEKKWKTKYSSPVDMLGVVLAKKCQILVSDIDYKHPLHEWTCPLDQNDVIQA RKAYDLQSDAVYKADLEWLRGIGWVPIGSVEVEKVKRAGEILSDRKYRQPADQLKF TCITDTPPEIVLAKSNAITMSKHYLTEAWDADKASIHVMPDTPPEILLAKSNSANISHKL YTKGWDESKMKDYDLREDASIKSAKASRDIASDYKKEAYEKQKGHHIGAQSVED DPRILCAMHAGKIYSDREYKKEFKWKTKFSSPVDMLGVVLAKKCQTLVSDVDYR NYLHNWTCPLDQNDIIQAKKAYELQSDSVYKADLEWLRGIGWMPPEGSIEPTRVKG AQDLVNERLYRTRPEALKFTSIVDTPEVVLAKANSQMSERLYQEAWNKDKTNISLP LDTPAMLQAQINALQISNKLYQKDWEAKQGYDLRADAIEIKHAKASREIASEYK YKEGYRQQLGHHVGFRLQDDPKLVWSIHAAKIQSDREYKAYEKSKEGIYNTPLDM MSIVQAKKCQVLVSDVDYRNYLHQWTCPLDQNDVIQAKKAYELQSDAIYKADLEW LRGIGWMPPEGSPEVLRVKNAQEILCDSVYRTPVVKLYTSIVDTPEVVLAKSNAENIS IPKYREVVDKDKTSIHIMPDTPPEINLARANALNVSNKLYREGWDEVKMSCDVRLDAI PIQAAKASREIASDYKYLDEKQKGHYVGTLTARDNKRWALIAGKIQNEREYRL QWAKWKTKFQSPVDMLSILHSSKKCQTLVSDIDYRNYLHEWTCMPDQNDVIQAKKA YELQSDAVYKADLEWLRGIGWMPNDSSVSNHAKHAADIFSEKKYRTKIELTNFTPV DDRVDYVTAKQSSIEIRDDIKYRKDNNAIKSKYTLTETPLLHTAQEAARILDQYLYKE GWEKQKATGYILPPDAVPFVHAHSSDVQSELKYKAHVKQGRGHYVGVPTMRDDP KLWFEHAGQIQNDRLYKESYHKS KAKINIPADMVSVAAKEGQNLVSDIDYRQYL HQWTCPLDQNDVIQAKIAYDLQSDNVYKADLEWLRGIGWIPLDSVDHVRVTKNQE MVNQIKYKDALANYPNFTSVEDPPEIVLAKINSVQSDVKYKETFNKAQKGYIFSP DTPYISHSKDMKLYSTILYKGAWEGTKAYGYTLDERYIPIVGAHADLVNSSELKYK ETYEKQKGHYLAGKEISEFFGVVHCLDFQOMRSALNYRRDYEDTKANVHIPNDMM NHVLAKRCQYILSDLEYRHYFHQWTSLEPEPNVIRARNAQEILSDNVYKDDLNLWLK GIGCYVWDTPQILHAKKSYDLQSQLYTAASKENLQNYNLVTDTPVYVTLAQSGIN ASEVYKENVYHQIKDKYTTVLETADYDRTKHLKSLYSNLYKEAWDKVKATSYNLP TSTLSLTHAKNQKHLASHIKYREYEKFKALYTLPKSVDDDPNTARCLRVGKLNIDR LYRSVYEKNMKMINIVPDMVEMVTAKDSQKKVSEIDYRLHLHEWICHPLQVNSHV RKVTDQISDLYKDDLTLWLKIGCYVWDTPPEILHAKHAYDLRNDIKYKAHVQKTRN DYKLVTDTPVYVQAVKSGKQLSDAVYHYDVHVSVRGKVAPTTKTVDLDRALHAY KLQSENLYKKGLRFLPTGYRLPVDTPHFNHTKDTTRYMSSYFKYKEAYEHIKANGYT LGPKDVFPVHVRVNNVTSERLYRELHYHLKDKIHTTPTPEIRQVKKTQEAVALIY KSDFFKMQGHMISLPYTPQVLHCRYVGDITSDIKYKEDLQILKGLGCFLYDTPDMVR SRHLRKLWSNYLTDNARKMRDXYKVLDTPEYRKVQELKTHLSELVYRASGKKQ KSIFTSVPDTPDLLRAKRGQKLQSQYLYVELATKERPHHAGNQTTALQHAQKDVKD MVSEKKYKIQYKMKDKYTPVADTPILIRAKRAYWNASDLRYKETFOKTKGRVHT VKDALDIVYHRKVTDDISKVYKENYMSQLGIWSSIPDRPEHFHHRVATDAVSDVK YKEDLTWLKIGCYAYDTPDLTLAEQNKTLKYKYKEVFERTKSNFYVADCPIN RHFKYATQLMNEKKYRADYEQRKDKYHLVDEPRHLLAKTAGDQISQIKYRKKE KSKDKFTSVVDTPPEHLRTTKVNQISDILYKLEYNKARPRGYTTIHDTPMLLHVRKV KDEVSDLYKEVYQRNKSNTIKPDVAHIKAAKDAYKVNNTLDYKQYEAANKAY WKWTPDRPDIQAAKSSLQSDFEYKLDREFLKGCKLSVTDDKDMVLALRNSLIESD LKYKEKHVKERGSCHAVPDPQILLAKTVSNLVSENKYKDHVKKHLAQGSYTTLPE TRDTVHVKAIVKHVSDTNYKKKFVKEGKSNYSIMQEPPEVKHAMEVAKKQSDVA YKDKAKENLHYTTVADRDPDIKKATQAAKQASEVEYRAKHRKEGSHGLSMLGRPDIE MAKKAALSSQVYKRENFDEKEGKTPKYNPKDSQLYKTMKDANNLASEVYKAD LKKLHKPVTDMKESLIMHHVLNTSQLASSYQYKKNYKSKGHYHTIPDNLEQLHHR EATELQSIYKYKEKYEKERGKPMDFETPTYITAKESQQMQSGKEYRKDYBESI KGR NLGTGLEVTALLHVRHATKIASKEYRKDLEESIRGRGLTEMEDTPMDLRKAKNATQI LNEKEYKRDLELVKGRGLNANANETPDPMRARNATDIAAQIKYKQSAEMKANFT SVVDTPPEIIHAQQVKNLSSQKKYKEDAEKSMSSYETVLDTPEMQVRQENQKFNLSL QYQCDLKNSKGKITVVDTPPEILRVKENQKNFSSVLYKEDVSPGTAIGKTPPEMLRVK QTQDHISSVYKEAIGQGTPIPDLPVKRVKETQKHISVVMYKENPGTGITPTVTPPEIE RVKRNQENFSSVLYKESLGKGTPTAITPEMERVKRNQENFSSVLYKENMGKGTPLPV TPEMERVKNQENISSVLYKENVGKATPTVPTPEMQRVKRNQENISSVLYKENMGK ATPTPTPEMERVKRNQENFSSVLYKENMRKATPTVPTPMDERAKRNQENISSVLYS DSFRKQIQGKAAYVLDTPEMRRVRETQRHISTVKYHEDFEKHKGCFTPVVDTPITER VKKNTQDFSDINRYGIQRKVVEQKRNQDQDQITGLRVWRTNPGSVFDYDPAED NIQSRSLHMINVQARRSRDQSRASALSISGGEEKSEHSEAADQRLSTYSDGGAFFS ATSTAYKHVKTTLQPPQRRSSSVATQQTTLSSIPSHPS TAGKIFRAIYDIYAADADEVSF KDGAIVNAQAIDEGWMYGTVQRTGRTGMLPANYVEAI
4	Mouse ACTS (Actin)	MCDEDETALVCDNGSLVKAGFAGDDAPRAVFPISVGRPRHQGVMMGMQKDSY VGDEAQSKRGILTLKYP I EHGII TNWDDMEKIWHHTFYNELRVAPEEHPTLLTEAPLN PKANREKMTQIMFETPNVPAMYVAIQAVLSLYASGRTTGIVLDSGDGVTHNVP I YEG YALPHAIMRLDLAGRDLTDYLMKILTERGYSFVTTAEREIVRDIKEKLCYVALDFEN

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		EMATAASSSSLEKSYELPDGQVITIGNERFRCPETLFQPSFIGMESAGIHETTYNSIMK CDIDIRKDLYANNVMSGGTTMYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIG GSILASLSTFQQMWITKQEYDEAGPSIVHRKCF
5	Mouse KCRM (Creatine Kinase)	MPFGNTHNFKLNYKPKQEEYPDLSKHNNHMAKVLTPLYNKLRDKETPSGFTLDDV IQTGVDNPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGYKPTDKHKTDLNEH NLKGGDDLDPNYVLSRVRTGRSIGYTLPPHCSRGERRAVEKLSVEALNSLTGEFK GKYYPLKSMTEQEQQLIDHFLFDKPVSPLLASGMARDWPDARGIWHNDNKSFL VWVNEEDHLRVISMEKGGNMKEVFRFCVGLQKIEEIFKKAGHPFMWNEHLGYVL TCPNSNLGTGLRGVHVKLANLSKHPKFEEILTRLRLQKRGTGGVDTAAVGAVFDISN ADRLGSSEVEQVQLVVDGVKLMVEMKKLEKGQSIDDMI PAQK
6	Mouse ACTN3 (Alpha-Actinin- 3)	MMMVMQPEGLGAGEGPFSGGGGGEYMEQEEDWDRDLLDDPAWEKQQRKTFTAW CNSHLRKAGTQIENIEEDFRNGLKLMLLLEVISGERLPRPDKGKMRFHKIANVNKAL DFIASKGVKLVSIGAEEIVDGNLKMTLGMIWTIILRFAIQDISVEETSAKEGLLLWCQR KTAPYRNVNVQNFHTSWKDGALCALIHRHRPDLIDYAKLRKDDPIGNLNTAFEVA EKYLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFYHAFAGAEQAETAANRI CKVLAVN QENEKLMEEYEKLASELLEWIRRTVPWLENRVGEPSSAMQKLEDFRDYRRLHKP PRVQEKQCLEINFNTLQTKLRLSHRPAPMPSEGLVSDIANAWRGLEQVEKGYEDW LLSEIRRLQRLQHLAEKFQQKASLHEAWTRGKEEMLNQHDYESASLQEVREALLRRH EAFESDLAAHQDRVEHIAALAQELNELDYHEAASVNSRCQAICDQWNLGTLTQKR RDALERMEKLEETIDQLQLEFARRAAPFNWLDGAIEDLDQDVVLVHVSVEETQSLLT AHEQFKATLPEADRERGAILGIQGEIQKICQTYGLRPKSGNPYITLSSQDINNKWDTV RKLVPSTRDQTLQEELARQQVNERLRRQFAAQANAIGPWIQGVVEVGRLAAGLAGS LEEQMAGLRQQEQNI INYKSNIDRLEGDHQLLQESLVFDNKHTVYSMEHIRVWGEQ LLTSIARTINEVENQVLTTRDAKGLSQEQLNEFRASFNHFDKRKNGMMEPDDFRACLIS MGYDLGEVEFARIMTMDPNAAAGVVTQFQAFIDFMTRETAETDTAEQVVASFKILAG DKNYITPEELRRELPAEQAEYCIRRMAPYKGS GAPSGALDYVAFSSALYGESDL
7	Mouse MYH1 (Myosin-1)	MSSDAEMAVFGEAAPYLKSEKERIEAQNKPFDAKSSVFVVDAKESFVKATVQSRE GGKVTAKEGGTTVTVKDDQVYPMNPKYDKIEDMAMMTHLHEPAVLNLYNKERY AAWMIYTYSGLFCTVNPYKWL PVYNAEVVAAAYRGKKRQEAPPHIFSIDNAYQFM LTDRENQSLITGESGAGKTVNTRKVIQYFATIAVTGEKKKEEATSGKMQGTLEDQIIS ANPLLEAFGNAKTVNRDNSRFGKFI RIHFGTTGKLASADIETYLLEKSRVTFQLKAE RSYHIFYQIMSNKKPDLEIMLLITTNPYDYAFVSQGEITVPSIDQEEELMATDSADILG FTSDERSVIYKLTGAVMHYGNMKFKQKQREEQAEPDGTEVADKAAYLQNLNSADL LKALCYPRVKVNEYVTKGQTVQQVYNSVGALAKAVYEKMFWMVTRINQQLDT KQPRQYFIGVLDIAGFEIFDFNSLEQLCINFTEKLLQQFFNHHMFVLEQEYKKEGIE WEFIDFGMDLAACIELIEKPMGIFPSILEEECMFPKATDTSFKNKLYEQHLGKSNFPQK PKPAKGVEAHFSLVHYAGTVDYNIAGWLDKNKDPLNETVVGLYQKSSMKTLAYL FSGAAAAAEASGGGGGKGAKKKGSFQTVSALFRENLNKMLTNLRSTHPPFVRC IIPNETKTPGAMEHELVLHLQRCNGVLEGIRICRKGFPSPRI LYADFKQRYKVLNASAI P EGQFIDSKKASEKLLGSDIDHTQYKFGHTKVFFKAGLLGLLEEMRDDKLAQLITRQ AMCRGYLARVEYQKMVERRESIFCIQYNVRAFMMNVKHWPMKLYFKIKPPLKSAE TEKEMANMKEEFKAKENLAKAEAKRKELEEKVMALMQEKNLDLQVQSEADSL ADAERCDQLIKTKIQLEAKIKEVTERAEDEEINAELEAKRKELEDECSLEKDDIDD LELTAKVEKEKHATENKVNLTTEEMAGLDETI AKLTKEKKALQEAHQQTLDLQA EEDKVNTLTAKIKLEEQVDDLEGSLEQEKIRMDLERAKRKELEGLDKLAQESTMD VENDKQQLDEKLLKKEFEMSNLQSKI EDEQALGMQLQKKIKELQARIEELEEEIEAE RASRAKAEQSRDLSRELEIEERLEEAGGATSAQIEMNKKREAEFQKMRRDLEEAT LQHEATAATLRKKHADSVAELGEQIDNLQRVKQKLEKEKSEMKEIDDLASNMEVI SKSGNLEKMCRTLEDQVSELKTEEEQQRLINELTAQRGRLQTESGEYSRQLDEKD SLVSQLSRGKAFTQQIEELKRLQLEEEIKAKSALAHALQSSRHDCDLLREQYEEEQEA KAEQRAMSKANSEVAQWRTKYETDAIQRTTEELEAAKKLAQRLQDAEEHVEAVN AKCASLEKTKQRLQNEVEDLMI DVERTNAACAALDKQRNFDKILA EWKQKYEBT HAELEASQKESRSLSTELFKIKNAYEESLDHLETLKRENKNLQOEISDLTEQIAEGGK RIHELEKIKKQIEQEKSLEQAALAEAEASLEHEEGKILRIQLELNQVKSEIDRKIAEKDE EIDQLKRNHIRVVESMQSTLDAEIRSRNDAIRLKKMEGLDNEMEIQLNHSNRMAAE ALRNRYRNTQGI LKDTQLHLDDALRGQEDLKEQLAMVERRANLLQAEIEELRATLEQ TERSRIAEQELLDASERVQLLHTQNTSLINTKKKLETDISIQIGEMEDIVQEARNAAE KAKKAITDAAMAEELKKEQDTS AHLERMKNLEQTVKDLQHRLDEAEQALALKG GKKQIQKLEARVRELEGEVENEQKRNVEAI KGLRKHERRVKELTYQTTEEDRKNVLR LQDLVDKLSQSVKAYKRQAEAEQSNVNLAKFRKIQHELEEAERADIAESQVNK LRVKSREVHTKIISEE
8	Mouse PYGM (glycogen Phosphorylase)	MSRPLSDQDKRKQISVRGLAGVENVSELKKNFNRHLHFTLVKDRNVATPRDYFFAL AHTVRDHLVGRWIRTQQHYEKPRIYIYLSLEFFMGRITLQNTMTVNLALENACDE ATYQLGLDMEELEEEEDAGLGNGLGRLAACFLDSMATLGLAAYGYGIRYEFGIFN QKICGGWQMEEADDWLRVGNPWEKARPEFTLPVHFYGRVETSQGAKWVDTQVV LAMPYDTPVPGYRNNVNTMRLWSAKAPDNFNLKDFNVGGYIQAVLDRNLAENIS RVLYPNDNFFEGKELRLKQEYFVVAATLQDIIRRFKSSKPGSRDPVRTNFDAPDKVA

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		IQLNDTHPSLAIPELMRILVDLERLDWDKAWDVTVKTCAYTNHTVLPEALERWPVH LMETLLPRHLQIIYEINQRFNLNRVAAAPGDDVRLRRMSLVEEGAVKRINMAHLCTA GSHAVNGVARIHSEILKKTIFKDFYELEPHKFKQNTNGITPRRWLVLCNPGLAEVIAE RIGEDYISDLQDLRLKLLSYVDDEAFIRDVAKVKQENKLFKSAYLEREYKVHINPNSLF DVQVKRIHEYKRQLLNLCHIITLYNRIKREPFRFVPTIMIGGKAAPGYHMAKMIK LITAGDVVNHDPVAGDRLRVI FLENYRVSLAEKVIPAADLSEQISTAGTEASGTGMM KFMLNGALITGMDGANVEMAEAGEENFFIFGMRVEDVERLDQRGYNAQEYYDRI PELRQIEQLSSGFFSPKQPDLFKDIVNMLMHHDRFKVFADYEEYIKCQDKVSELYKN PREWTRMVRNIATSGKFFSSDRITIAQYAREIWGVEPSRQRLPAPDEKI
9	Mouse ALDOA (fructose- bisphosphate aldolase A)	MPHYPALTPEQKKELSIAHRIVAPGKGLAADESTGSIKRLQSIGTENTENRRFY RQLLLTADDRVNPICGGVILFHETLYQKADDGRPPQVIVKSKGGVVGIVKVDKGVVPL AGTNGETTTQGLDGLSERCAQYKKDGADPAKWRVCLKIGEHTPSALAIMENANVLA RYASICQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAVYKALSDDHVYLEGTLL KPNMVTGPHACTQKFSNEEIAMATVTALRRTPPAVTVGTFTLSGGQSEEEASINLNAI NKCPLLKPWALTFSYGRALQASALKAWGGKKENLKAQEYIKRALANSACQGK YTPSGQSGAAASESLFISNHAY
10	Mouse MYL 1 (myosin light chain)	MAPKKDVKKPAAAPAPAPAPAPAPAKPKEEKIDLSAIKIEFSKEQQEDFKEAFLLFDR TGECKITLSQVGDVLRALGNTPTNAEVKVLGNPSNEEMNAKKIEFEQFLPMMQAIIS MNKDQGGYEDFVEGLRVFDKEGNGTVMGAELRHVLATLGEKMKKEEVEALLAGQE DSGNCINYEAFVKHIMSV
11	Mouse G3P (glyceraldehyde- 3-phosphate dehydrogenase)	MVKVGUNGFGRIGRLVTRAAICSGKVEIVAINDPFIDLNYMVYMFQYDSTHGKFNG TVKAENGLVINGKPIITIFQERDPTNIKWGEAGAEYVVESTGVFTTMEKAGAHKGG AKRVIISAPSADAMPFVMGVNHEKYDNSLKIIVSNASCTTNCLAPLAKVIHDFNGIVE GLMTTVHAIATATQKTVDGPGSKLWRDGRGAQNIIPASTGAAKAVGKVIPELNGKL TGMARVPTPNVSVVDLTCLREKPAKYDDIKKVVQASEGPLKGIILGYTEDQVVS DFNNSHSSSTFDAGAGIALNDNFVKLISWYDNEYGYSNRVVDLMAYMASKE
12	Mouse TPM1 (tropomyosin alpha-1)	MDAIKKKMQLKLDKENALDRAEQAEADKKAEDRSKQLEDELVSQKLLKGT DELDKYSEALKDAQEKLELAEKATDAEADVASLNRRIQLVVEELDRAQERLATAL QKLEEAKEADESERGMKVI ESRAQKDEEKMEIQEIQLEAKHIAEDADRKYEEVAR KLVIIESDLERAEEAELSEGKCAELEELKTVTNLKSLEAQAEKYSQKEDKYEIEI KVLSDKLKEAETRAEFAERSVTLEKSIDDLEDELYAQKLYKAISELDHALNDMT SI
13	Mouse MYPC2 (myosin-binding protein C)	MPEAKPAKKASKGKDAPKEAPAKQTPPEPPKEAPPEDQSPSTAEPTGIFLKKPDSVS VETGKDAVILAKVNGKELPGKPTIKWFKGKWQELGSKSGARFIFKESHDSTSNVYTV ELHIGKVVLGDRGDRLEIKAKDVCDSFNVDEAPRQDSSGQSLESFKRSGDGKS EDAGELDFSGLLKKREVVEEKKKKDDDLGIPPEIWELLKGAKKSEYEKIAFQYG ITDLRGLMLKRLKAKVEVKKSAFTKKLDPAYQVDRGNKIKLVEISDPDLPLKWF KNGQEI KPSKYVFENVGKKRIILTINKCTLADDAAYEVAQDEKCTELFVKEPPVLI VTPLEDQOVFVGDVREMSVEVSEEGAQVMWMDGVEMTREDSYKARYRPFKDGK RHILIYSDVAQEDGGRYQVITNGGQCEALIVEEKQLEVLQDIADLTVAKAEQAVFK CEVSDKVTGKWKNGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDTYFVPDGYA LSLSAKLNFLKIEKVEYVPKQEPPIHLDCSGKTSDNSIVVAGNKLRLDVAITGEPPT ATWLRGDEVFTATEGRTHIEQRPDCCSFVIESAERSDEGRYTIKVTNPVGEDVASIFLR VVDVPDPPEAVRVTSVGEDWAILVWEPPKYDGGQPVTVGLMERKKKSGQRWMKIN FEVFTDTTYESTKMIEGVLYEMRVFAVNAIGVSGPSMNTKPFMPIAPTSAPQHLTVED VTDTTTTLKRWRPDRIGAGGIDGYLVEYCLEGSEEWVPANKEPVERCGFTVKDLPTG ARILFRVVGNIAGRSEPATLLQPVITREIVEQPKIRLPRHLRQTYIRKVGALNLVIPF QGKPRPQVVWTKGAPLDTSRVNVRTSDFDTVFVVRQAARSDSGEYELSVQIENMK DTATIRIRVVEKAGPAENVMVKEVWGTNALVWQPPKDDGNSEITGYFVQKADKK TMWFNVYEHNRHTSCTVSDLI VGNYYFRIFSENICGLSDSPGVSKNTARIKLTGITL KPLEYKEHDFRTAPKFLTPLMDRVVAGYTAALNCAVRGHPKPKVVMKMKMEIH EDPKFLITNYQGIILTLNIRRPSFPDAGTYSCTAFNELGEALAECKLDVVRVQ
14	Mouse MYPC3 (myosin-binding protein C)	MPEPGKKPVSAFNKKPRSAEVTAGSAVFEAETERSGVMVRWRQDGSITANDKY GLAEGKRHTLTVRDASPDQGSYAVIAGSSKVKFDLKVTEPAPPEKAESVAPGAP EEVPAPATELESVSSPEGSVSVTQDGSAAEHQAPDDPIGLFLMRPDGQEVTVGGS VFSARVAGASLLKPPVVKWFKGKWDLSKVGQHLQLHDSYDRASKVYLFELHITD AQTTSAGGYRCEVSTKDKFDCSNFNLTVHEAIGSGDLDLRSAPRRTSLAGAGRRTSD SHEDAGTPDFSSLLKKRDSFRDLSKLEAPAEEDVWEILRQAPPSEYERIAFQHGVEAC HRPLKRLKGMKQDEKSTAFQKKLEPAYQVNGHKHIRTVELADPDPAEVKWLKNG QEIQMSGSKYIFESVGAKTTLTISQCSLADDAAYQCVVGGEKCTELFVKEPPVLI TR SLEDQLVMVGQRVFEFECEVSEEGAQVWKLKDGVELTREETFYKRFKDKGRKHLII NEATLEDAGHYAVRTSGGQSLAELIVQEKKEVYQSIADLAVGAKDAQVFKCEVSD ENVRGVWLKNGKELVDPNRIKVSHIGRVHKLITIDVTPADEADYSFVPEGFACNL KLFHMEVKIDFVPRQEPKIHLDCEPGSTPDTIVVVTGNKLRLDVPISGDPAPTVMVQK TVTQGKKASAGPHDPADADAGADEEWVFDKLLCETEGRVVRVETTKDRSVFTVEGA

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		EKEDEGVYTVTVKNPVGEDQVNLTVKVIDVPDAPAAPKISNVGEDSCTVQWEPPAY DGGQPVVLGYILERKKKSYRWMLNFDLLRELSHEARMIEGVAYEMRVYAVNAV GMSRPSPASQPFMPPIGPPGEPTHLAVEDVSDTTVSLKWRPPERVAGGLDGYSV EYC QEGCSEWTPALQGLTERRSMLVKDLP TGARLLFRVRAHNVAGPGGPVITKEPVTVQ EILQRPRLQLPRHLRQTIQKKVGEPVNL LIPFQGKPRPQVWTWKEGQPLAGEVSI RNS PTDTILFIRAARRTHSGTYQVTVRIENMEDKATLILQIVDKPSPQDIRIVETWGFNVA LEWKPPQDDGNTIEWGYTVQKADKKTMEWFTVLEHYRRTHCVVSEL IIGNGYF RV FSHNMVGSSDKAAATKEPVFI PRPGIT YEP PKYKALDFSEAPSFTQPLANRS I IAGYNA ILCCA VRGSPKPKISWFKNGLDLGEDARFRMFCKQGVLTLEIRKPCPYDGGVYVCRA TNLQGEAQCECRLEVRVPQ
15	Mouse Q14BI5 (myomesin 2)	MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDQYALKKQATTQSSSQRSALTERSSSKRA SSQSSAGAMTCRLCAKRVSAEEEEVENENRYRSQAASYGEAKRQFLSELAQLEEN VHLARSQARDKLDKYFMEQTVEDNLAWERHSFEDRMSRAPEILVRLRSHTIWERM S VRLCFTVQGFPTPVVQWYKDGSLICQAGEPGKYRIESRYGVHTLEINRANFEDTATY SAVATNSHGQVSTNAAVVRRYRGEEEPFHSVGLPIGLPLSSVIPYTHFDVQFLEKFG VTFRREGETVTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTMFFGEGQASLS FSHLNKDDEGLYTLRIVSRGGVSDHSAFMFVRDADPLVTGAPGAPMDLQCHDANRD YVIVTWKPPNTTTESPVIGYFI DKCEVGTNNWVQCNDAPVKICKYPVTGLFEGRSYV FRVRAVNAGISRPSRISDAVALDPVDLRLRLQAIHLEGEKEIVIVQDDLEGDVQIPGP PTNVQASEVSRNYVVLSDWPPSPRGKDPLMYFIEKSAVSGSGSQVRNAQTA VRSPR YAVFDLAEGKSYVFRVLSANKHGLSDPSEITPPIQAQDMIVVPSAPGRVLASRNTKTS VVVQWDRPKHEEDLLGYVDCCVAGTNMWEPCNHKPIGYNRFVHGLTTGEQYIF RVKAVNAVGTSENSQSESEVIKVQAALTVPSHPYGI TLLNCDGHSMTLGWKVPKFSG GSAIIGYYL DKREVHHKNWHEVNSSPVKERILTVEGLTEGSLYEFKIAATNLAGIGQP SDPSEHKCEAWTDPEPGPAYDLTFCEVRDTSLVILWKAPVYSGSSPVSGYFVDFKEE DSGEWKTTEAATPNRYLKVCDLQQGKTYVFRIRAVNASGPGKPSDTSPEVLVEARP GTKEISAGVDEEGNI YLGFDCQEMTDASQFTWCKAYEEIAD EERFEVHTEGDHSKLY FKNPDKEDLGTYSVSVDTDGVSSSFVLDEEELERLMALSNEIKNPTIPLKSELAYEIEF DKGQVRFWLQAEHLSPDASFRFIINDREVSDSDTHRIKCDRSTGMIEMVMDRFTIENE GTYTQVQIDGKAKNQSSLVLIGDAFKAVLEEA EFQKEFLRKQGPHPFAEYLLHWDVT EECEVRLVCKVANTKRET VFKWLKDDVLYETETPPDLEKGVCELLI PKLSKKDHGE YKATLKDDRQGDVSVLEVGGKVVEDMILAMSRVCGASASPLKVLCTPEGIRLQCFM KYFTEEMKVSWYHKEAKISSSEHMRI GGSEMAWLQICEPTEKDKGYTFEIPDGKD SHQSRDLDSGQAFDEAYAEFQQLKAAFAEKNRGKVI GGLPDVVTIMEGKTLNLTCT TVFGNPDPEVWVFKNDKDI ELSHFVLVKMEQSKYVSLTIQGVTAEDSGKYSINVKNK YGGEKIDVTVSVYKHGEKIPDISPPQAKPKLI PASTSSD
16	Mouse ALBU (albumin)	MKWVTFLLLLFVSGSAFSGVFRREAHKSEIAHRYNDLGEQHFKGLVLI AFSQYLQK CSYDEHAKLVQEVTDFAKTCVADESAANCDKSLHTLFGDKLCAIPNLRENYGELAD CCTKQEPERNECFLQHKDDNPSLPPFERPEAEAMCTSFKENPTTFMGHYLHEVARRH PYFYAPELLYYAEQYNEILTQCCAEADKESCLTPKLDGVKEKALVSSVRQRMKCSS MQKFGERAFKAWAVARLSQTFPNADFABEITKLATDLTKVNKECHGDLLECADDR AELAKYMCENQATISSKLQTCCKPLLKAHCLSEVEHDTMPADLPAIAADFVEDQ EVCKNYAEAKDVFLGTFLYEYSRRHPDYSVSLLLRLAKKYEATLEKCCAEANPPAC YGTVLAEFQPLVEEPKNLVKTNCDLYEKLGEYGFQNAILVRYTQKAPQVSTPTPLVEA ARNLGRVGTCKCTLPEQRLPCVEDYLSAILNRVCLLHEKTPVSEHVTKCCSGSLVE RRPCFSALTVDETVVPKEFKAETFTFHSDICTLPEKEKQIKKQ TALAEVLKHKPKATA EQLKTVMDFAQFLDTCCKAADKDTCTFSTEGPNLVTRCKDALA
17	Mouse ENOB (beta-enolase)	MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALERLDGDKARY LGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKPMI ELDGTENKSKFGANAILG VSLAVCKAGAAEKGVPLYRHIADLAGNPDLVLPVPAFNVINGGSHAGNKLAMQEF MILPVGASSFKEAMRIGAEVYHHLKGVIKAKYKDATNVGDEGGFAPNILENNEAL ELLKTAIQAGYDPKVVIGMDVAASEFYRNGKYDLD FKSPDDPARHISGEKGLGLYK NFIQNYPVVSI ED PFDQDDWATWTSFLSGVDIQIVGDDLTVTNPKRIAQAVEKKACN CLLLKVNQIGSVTESIQACKLAQSNWGMVMVSHRSGETEDTFIADLVVGLCTGQIKT GAPCRSERLAKYNQLMRIEALGDKAVFAGRKFPNPKAK
18	Mouse MLRS (myosin regulatory light chain 2)	MAPKKAKRRAGAEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDLRDTFAAM GRLNVKNEELDAMMKEASGPINFTVFLTMFGEKLGADPEDVITGAFKVLDPDEGKG TIKKQFLEELLTTQCDRFSQEEIKNMWAAFPPDVGGNVYKNICYVITHGDAKDQE
19	Mouse KPYR (pyruvate kinase PKLR)	MSVQENELPQQLWPWIFKSQKDLAKSALS GAPGGPAGYLRRASVAQLTQELGTAFF QQQQLP AAMADTFLEHLCLLDIDSEPVAA RSTSI IATIGPASRSVDRLEKMIKAGMNI ARLNFSGSHHEYHAESI ANIREAAESFATSPLSYRPVAIALDTKGPEIRTVGLVGGGPES EVEIVKGSQVLVTVDPKFRTRGDAKT VVWDYHNITQVAVAGGRIYIDDG L I S LVVRK IGPEGLVTEVEHGGFLGNRKGVNLPNAEVDLPGLSEQD LLDLRFGEVHYVDIIFASFV RKASDVAVARDALGPEGRGIKISKI ENHEGVKFD EILEVSDGIMMARGDLGIEIPAE KVFLAQKMMIGRCNLAKGPVVCATQMLESMITKARPTRAETSDVANAVLDGADCI

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		MLSGETAKGSFPV EAVKMQHAIAREAEAAVYHRQLFEELRRAAPLSRDPTEVTAIGA VEASFKCCAAAIIVLTKTGRSAQLLSRYRPRAAVIAVTRSAQAARQVHLSRGVFPLLY REPPEAVWADDVDRRVQFGIESGKLRGFLRVGDLVIVVTGWPRGSGYTNIMRVLTIS
20	Mouse CAH3 (carbonic anhydrase 3)	MAKEWGYASHNGPDHWHELYPIAKGDNQSPIELHTKDIKHDPQLQPSWASYDPGSA KTI LNNGKTCRRVFD DTYDRSMLRGGLSGPYRLRQFHLHWGSSDDHGSEHTVDGV KYAAELHLVHWNPKYNTFGEALKQPDGIAVVGIFLKI GREKGEFQILLDALDKI KTK GKEAPFTHFDPSCLFPACRDYWTYHGSFTTPPCEECIVWLLLKPMPTVSSDQMAKLR SLFSSAENEPVPVLVGNWRPPQPVKGRVVRASFK
21	Mouse LDHA (L-lactate dehydrogenase A)	MATLKDQLIVNLLKKEEQAPQNKITVVGVGAVGMACAI S ILMKD LADELALVDVMD KLKGEMMDLQHGSLFLKTPKIVSSKDYCVTANSKLVI ITAGARQQEGESRLNLVQRN VNI FKF I I PNIVKYS PHCKLLIVSNPVD ILTYVAWKISGFPKNRVIGSGCNLDSARFRYL MGERLGVHALSCHGWVLGEHGDSSVPVWVGNNVAGVSLKSLNPELGTADAEQW KEVHKQVDSAYEVI KLKGYT SWAIGLSVADLAESIMKNLRRVHPISTMI KGLYGIN EDVFLSVPCLGQNGISDVVKVTLTPEEEARLKKSA DTLWGIQKELQF
22	Mouse G3UW82 (myosin, heavy polypeptide 2)	MSSDAEMAVFGEAAPYL RKSEKERIEAQNRPFD AKTSVFVAEPKESFVKGTIQSKDA GKVTVKTEAGATLTVKEDQIFPMNPPKYDKIEDMAMMTHLHEPAVL YNLKER YAA WMIYTYSGLCFVTVNPKWLPVYNPEVVAAYRGKKRQEAPPHI FSI SDNAYQFMLT DRENQS ILITGESGAGKTVNTRKVIQYFATIAVTGDKKKEEATSGKMQGTLEDQII SA NPLLEAFGNAKTVRNDNSSRFGKFIR IHFGTTGKLASAD IETYLLEKSRVTFQLKAER SYHIFYQITSNKKPELIEMLLIT TNPYDYPFVVSQGEISVAS ID DQEELMATDS AIDILGFT NDEKVS IYKLTGAVMHYGNMKFPKQKQREEQAE PDGTEVADKAAYLQGLNSADLLK ALCYPRVKVGNEYVT KGQTEQVTNAV GALAKAMY EKMP LWMVTRINQQLDTKQ PRQYF IGVLDIAGFEI P DFN SLEQLCINF TNEKLQQFFNHMHMFVLEQEEYKKEGIEWTF IDFGMDLAACIELIEKPMGIFS ILEEECMFPKATDTS FKNKLYEQHLGKSANFQPKPV VKGKAEAHFSLIHYAGTV DYNITGWLDKNKDPLNETVVGLYQKSSVKT LAYLFSGA QTAEAEASSGGA AKKGA KKKGS SFQTVSALFREN LNKLMTNL RSTHPHFVRCIIPNE TKTPGAMEHELVLHQ LRCNGVLEGIRICRKGFP SRILYADF KQRYKVLN ASAIPEGQY IDSKKASEKLLGS IDIDHTQYKFGHTKVFPKAGLLGLLEEMRDDKLAQLI TRTQAMC RGFLARVEYQKMVERRESIFCIQYNI RAFMNVKHWPMKLF FKI KPL LKSAETEKE MATMKEEFQTKDDLAKSEAKRKELEEKMSLLKEKNDLQ LQVQAEAGLADAE RCDQLIKTKIQLEAKI KEVTERAEDEE EINAEL TAKKRLEDECS ELKKD IDDELTLA KVEKEKHATENKVNLT EEMAGLDETI AKLTKEKKALQEAHQQTLD DLQAEEDKV NTLTAKIKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGDLKLAQESIMDIENEKQ QLDERLKKKEFEMS NLQSKI EDEQAI G IQLQKKIKELQARI EELEEEI EAERASRAKAE KQRSDLSRELEEEI SERLEEAGGATSAQI EMNKKREAEFQKMRDL E EATLQHEATAA TLRKKHADSVAE LGEQIDNLQRVKQKLEKEKSEMKMEIDDLASNVETVSKAGNLE KMCRTL EDQVSELKSKEEEQQR LINDLT TQRGR LQTESGEFSRQ LDEKEALVSQLSR GKQAF TQOI EELKRQLEEEV KAKNALAHALQSSRHDCDLLREQYEEBQESKAE LQR ALSKANSEVAQWR TKYETDAIQRTEEEBAKKLAQRLQAABEHVEAVNAK CASLE KTKQRLQNEVEDLMDV ERTNAACAALDKKQRNFDKI LAEWKQKYEE THAELEAS QKEARSLGT ELFKMKNAYEESLDQLETLKRENKNLQQEISDLTEQIAEGGKR IHELEK IKKQVEQEKC ELQAAL EEAASLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEIDQLK RNHIRRVESMQLD AEIRSRNDAIRIKKMEGDLNEMEQ LNLHNSNRMAEALRNRY NTQGILKDTQLHLDDALRGQEDLKEQLAMVERRANLLQAEIEELRATLEQTERS RKI AEQELLDASERVQLLHTQNTSLINTKKKLETDI SQIQGEMEDI VQEARNAE EKAKKAI TDAAMAEELKKEQD TSAHLERMKNMEQTVKDLQLRLDEAEQLALKGGKKQIQ KLEARVRELEGEVESEQKRNAEAVKGLRKHERRVKELTYQTEEDRKNI LRLQDLVD KLQAKVKS YKRQAEAE EQSNTNLSKFRKI QHELEEA EERADIAESQVNKL RVKSRE VHTKII SEE
23	Mouse Q6P6L5 (myosin-binding protein C)	MPEPTKKEENEVSAPAP PPEEPNKEKEAGTAPAKEWSLGES PAGGEEQDKQNANSQ L STLFVEKPQTG SVKVGANI TFVAKVRAEDLLRKPTVWFKGKWM DLASKAGKHLQ LKETFERQTRVYTFEMQII KAKENYAGNYRCEV TYKDKFDS CSFDEL VHESTGTT PNI DIRSAFKRS GEGQEDAGELDFSGLLKRREV KQQE EPEIDVWELLKNANPN EYEKI AF QYGITDLRGLKRLKRMRRVEKKSAAFAKILDPAYQVDKGKVR FVVELADPKLE VKWFKNGQEIRPSTKYIFEHKGNERIMFINNC SLTDDSEYVVTAGDEKCS TELFVREP PIMVTKQLEDMNAYCGERVEMEVEVSEDDANVKWFKNGEEIFPGPKSRYKIKVEGK KHTLII EGATKADSAEYSAMTTGGQS SAKLSVDLRPLKI TTP L TDQT VTKLGKEVCLKC EISENVPGKWTKNGLPVQEGERLKV VHKGR I HKLVIANAL I EDEGEYVFTPDAYNVP LSAKVHVIDPPKI ILDGLDADNTVTVIAGSKLRLEIPVTGEPPPKAIWSRADKAIMEGS GRIRAE SYPDS TLVIDVAERDDSGVYNINLKNEAGEAHASIKIKVVDIPDPVPAPNVT EVGDDWCIMNWEPPAYDGGSPILGYFIERKKKQSSRWMLNFDLCKETT FEPKKMIE GVAYEVRIFAVNAIGISKSPMSKPFVPLAVTSPPTLLAVDSVTDS SVTMKWRPPDI GAAGLDGYVLE YCFEGTEDWIVANTDLIDKT KPTINGLP TDAKIFVRVKAINAAGAS EPKYYSQPI LVKEIIEPPKIRIPRHLKQTYIRRVGEAVNLVIPFQ GKPRPELTWKKDGEE IDKNQINIRNSETDTIIFIRKAERSHSGKYDLQVKVDKYVENASIDIQIVDRPGPPQTVT IEDVWGENVALTWTPPKDDGNAAITGYTIQKADKKSMEWFTVIEHYHRTNATITEL VIGNEYFRVFAENMCGLS EDATMTKESAVIAKD GKIYKNPVYEDFNFT EAPMFTQP

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
24	Mouse PGM1 (phosphogluco-mutase-1)	LVNTYAIAGYNATLNCVSRGNPKPKITWMKNKVAIVDDPRYRMFSNQGVCTLEIRK PSPYDGGTYCCKAVNDLGTVEIECKLEVKVVAQ MVKIVTVKTQAYPDQKPGTSGRLKRKVKVQSNANYAENFIQSIVSTVEPALRQEATL VVGGDGRFYMTETAIQLIVRIAAANGIGRLVIGQNGILSTPAVSCIIRKIKAIIGGIILTASH NPGGPNGDFGIFKFNISNGGPAPPAITDKIPQISKTIIEYAI CPDLKVDLGVLGKQQFDLE NKFKPFTVEIVDSVEAYATMLRNI FDFNALKELLSGNRLKIRIDAMHGVVGPYVKKI LCEELGAPANSAVNCPLEDGFGHHPDNLTYAADLVETMKSGEHDFGAAFDGDG DRNMILKGHGFVNPSDSVAVIAANIFSIYPFQQTGVRGFARSMPTSGALDRVANAT KIALYETPTGWKFFGNLMDASKLSLCGEESFGTGSDHIREKDGWLAVLAWLSILATR KQSVEDILKDHQKFGRNFFTRYDYEVEAEGANKMMKDLEALMLDRSFVGKQFS ANDKVYTVKEADNFEYSDPVDGSI SKNQLRLIFADGSR IIFRLSGTGSAGATIRLYID SYEKDVAKINQDPQVMLAPLISIALKVSQQLQERTGRTAPTUIT
25	Mouse TPM2 (tropomyosin beta chain)	MDAIKKKMQLKLDKENAIDRAEQAEADKKQAEADRCQLEEEQALQKKLKGTE DEVEKYESVKDAQEKLEQAEEKATDAEADVASLNRRIQLVEEELDRAQERLATAL QKLEEAKAADSEERGMKVIENRAMKDEEKMEQEMQLKEAKHIAEDSDRYEEV ARKLVI LEGELERSEERA EVAESKCGDLEELKIVTNNLKSLEAQADKYSTKEDKYE EEIKLLEEKLEAETRAEFAERSVAKLEKTIDDDLEDEVYAQKMKYKAI SEELDNALN DITSL
26	Mouse TPIS (triosephosphate isomerase)	MAPTRKFFVGGNWKMNGRKKCLGELICTLNAANVPAGTEVVCAPPTAYIDFARQKL DPKIAVAAQNCYKVTNGAFTGEISPGMI KDLGATWVVLGHSERRHVFGESEDELIGQK VSHALAEGLGVIAICIGEKLDEREAGITEKVVFEQTKVIADNVKDWSKVVLAYEPVW AIGTGKTATPQQAQEVHEKLRGWLKSNVNDGVAQSTR I IYGGSVTGATCKELASQP DVDGFLVGGASLKPEFVDIINAKQ
27	Mouse PGAM2 (phosphogly-cerate mutase 2)	MTTHRLVMVRHGESLWNQENRFCGWFDAELSEKGAEEAKRGATAIKDAKIEPDI CY TSVLKRRAIRTLWTILDVTDQMWVPVVRTWRLNERHYGGLTGLNKAETAAKHGEEQ VKIWRRSFDPPTPPMDEKHNYYTISIKDRRYAGLKPEELPTCESLKD TIARALPFWNE EIA PKIKAGQVRVIAAHGNSLRGIVKHLEGMSDQAIMELNLP TGIPIVYELDQNLKPT KPMRFLGDEETVRKAMEAVAAQGGKAK
28	Mouse SRCA (sarcalumenin)	MKALLLLCCFLASLLLSGQAEVEDASEEAPLRDRSHIDKTLMLNEDKPADDYSAVLQ RLRLKIYHTSIKPLEQSYKYNELRQHEITDGEITSKPMVLFLGPWSVGKSTMINYLLGL EDTRYQLYTGAEPPTSEFTVLMHGPKLKTIEGIVMAADSARSFSPLEKFGQNPLEKLI GIEVPHKLLERVTFVDTPGI IENRKQQERGYPFNDVCQWFIDRADLIFVVFDP TKLDV GLELEMLFRQLKGRESQIRI ILNKADNLATQMLMRVYGALFWSLAPLINVTEPPRVY VSSFWPQDYKPDTHRELFLKEEISLLEDLNQVIENRLNKI AFIRQHAIRVRIHALVD RYLQTYKDKMTFFSDGELVFPKDIVEDPKFYIFKTI LAKTNVSKFDLPNREAYKDFPG INPISNFKLLSQQCSYMGGFLEKIERAITQELPSLLGSI GLGKNPGAPNCDKTGCGET PKNRYKKH
29	Mouse PGK 1 (phosphoglycerate kinase 1)	MSLSNKLTLDKLDVKGKRVVMRVD FNVPMKNNQITNNQRIKAAVPS IKFCLDNGAK SVVLMSHLGRPDGVPMPDKYSLEPVAELKSLLGKDVLF LKDCVGPVEVENACANPA AGTVILLENLRHFVEEBGKGKDGASGNKVKAEPAKIDAFRASLSKLGDVYVNDAFGT AHRAHSMVGYNLPQKAGGFLMKKELNYFAKALESPPRPLA ILGGAKVADKIQLIN NMLDKVNEMI IGGMAFTFLKVLNNMEIGTSLYDEEGAKIVKDLMSKAEKNGVKIT LPVDVFTADKFDENAKTGQATVASGIPAGWMLDCCGTESSKYAEAVGRAKQIVW NGPVGVFEWEAFARGTKSLMDEVVKATSRGCIT IIGGGDTATCCAKWNTEDKVSHV STGGGASLELLEGKVLPGVDALSNV
30	Mouse PLEC (plectin)	MVAGMLMPLDRLRAIYEVLPREGVMVAKDRRPRSLHPHPVGVNTLQVMRAMAS LKARGLVRETFAWCHFYYLTNIGIDHLRQYLHLPPEIVPASLQVRVRPVMV I PAR RRSPhVQTMQGPLGCPPKRGPLPAEDPAREERQVYRRKERE EGAPETPVVSATTVGT LARPGPEPATDERDRVQKKTFTKWNKHLIKHWRAEAQRHISDLYEDLRDGHNL ISLLEVLSGDSLPREKGRMRFHKLQNVQIALDYLHRHQV KLVNIRNDIADGNPKLT LGLIWTI I LHFQISDIQVSGQSEDMTAKELLLWSQRMVEGYQGLRCDNF TTSWRDG RLFNAI IHRHKPMLIDMNKVYRQTNLENLDQAFSVAERDLGVTRLLDPEDVDVPQD EKSII TYVSLYDAMPVRPGAQDGV RANELQLRWQEYRELVL LLLQWIRHHTAAFE RKFPSSFEIEI LWCQFLKFKETELPAKEADKNRSKVIYQSLEGAVQAGQLKIPPGYH LDVEKEWGKLVHAILEREKQLRSEFERLECLQRIVSKLQMEAGLCEEQLNQADALLQ SDIRLLASGKVAQRAGEVERDLKADGMIRLLFNDVQTLKDRHPQGEQMYRRVY RLHERLVAIRTEYNLRLKAGVGAPVTQVTLQSTQRRPELEDSTLRYLQDLLAWVEEN QRRIDS AEWGVDLPSVEAQLGSHRGMHQSI EEFRAKIERARND ESQSPATRGAYRD CLGRLLDQYAKLLNSKARLRSLES LHGFVAAATKELMWLNKEKEEEVGFWDSDRN TNMAAKKESYALMRELEMKEKKI KEIQNTGDRLLREDHPARPTVESFQAALQTQW SWMLQLCCCEIAHLKENTAYFQFFSDVREAEELQLKQETLRRKYS CDRITITVTRLE DLLQDAQDEKEQLNEYKGLHSLGAKRAKAI VQLKPRNPAHPVRGHVPL IAVCDYKQ VEVTVHKGDCQLVGPAQPSHWKVLSGSSSEAAVPSVCFVLP PPNQAQAEAVARLE AQHQALVTLWHQLHVDMSKSLLAWSLSRDIQLIRSWSLVTFRTLKPEEQRALRNL ELHYQAFRLDSQDAGGFGPEDRLVAEREYGS CSRHYQQLLQSLEQGEESRCQRCI SELKD IRLQLEACETRVTVHRLRLPLDKDPARECAQRIAEQQKAQAEVEGLGKGVARL

TABLE 1-continued

[illegible]

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		QDRVEQIAAIAQELNELDYHDAVNVNDRQCQKICDQWDRLGTLTQKRREALERTEKL LETIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEETIQLSITAHEQFKATLPEA DGERQSLAIQNEVEKVIQSYSIRISSSNPYSTVTMDLERNKWDKVKQLVPVRDQSLQ EELARQHANERLRQFAAQANAIGPWIQNKMEETARSSIQITGALEDQMNQLKQYEH NIINYKNNIDKLEGDHQLIQEALVFDNKHTNYTMEHIRVGWELLTTTIARTINEVETQI LTRDAKGITQEQMNEFRASFNHFDRRKNGLMDHEDFRACLI SMGYDLGEAEFARIM TLVDPNGQGTVTFQS FIDFMTR ETADTD AEQVIASFRILASDKPYILAEELRRELPPD QAQYCIKRMPPYSGPGSVPGALDYTAFSSSALYGESDL
32	Mouse PFKAM (ATP-dependent 6-phosphofructokinase)	MTHEEHHAAKTLGIGKAI AVLTSGGDAQGMNAAVRAVVRVGIFTGARVFFVHEGY QGLVDGGEHIREATWESVSMMLQLGGTVIGSARCKDFREREGRLRAAHNLVVRGIT NLCVIGGDGSLTGADTFRSEWSDLLNDLQDKGITAEATKSSYLNIVGLVGSIDNDF CGTDMTIGTDSALHRIVEIVDAITTTAQSHQRTFVLEVMGRHCGYLALVTSLSGAD WVFIPECPDDDWEEHLCRRLSETRTRGRSLNI IIVAEGAIDKNGKPI TSEDIKNLVVK RLGYDTRVTVLGHVQRGGTPSAFDRI LGSRMGEAVMALLEGTPDTPACVVSLSGN QAVRLPLMECVQVTKDVT KAMDEKRFDEAI KLRGRSFMNNWEVYKLLAHVRPPVS KGG LHTVAVMNVGAPAAGMNAAVRSTVRIGLIQGNRVLVVHDGFEGLAKGQIEEA GWSYVGGWTQGGGSKLGTKRTL PKKNLEQISANITKFNIQGLVIGGFEAYTGGLEL MEGRKQFDEL CIPFVVI PATVSNNVPGSDFSIGADTALNTI CTTCDRI KQSAAGTKRR VFIIETMGGYCGYLATMAGLAAGADAAYIPEEPFTIRD LQVNV EHLVQKMKTTVKR GLVLRNEKCNENYTTDFIFNL YSEEGKGFDSRKNVLGHMQGGGSPTPFDRNFATKM GAKAMNWMSGKIKESYRNGRIFANTPDSGCVLGMRKRALVFQPVTE LKDQTD FEH RIPKEQWWLKLRLILKILAKYEIDL DTS DHAHLEHISRKRSGEAAV
33	Mouse MDHM (malate dehydrogenase)	MLSALARPAGAALRRSFSTSAQNNAKVAVLGASGGIGQPLSLLLKNSPLVSR LTLYDI AHTPGVAADLSHIETRANVKGYLGPQLPCLKGCDVVVIPAGVPRKPGMTRDDL F NTNATIVATLTAACAQHCPEAMVCI IANPVNSTIPITAEVFKKHGVYNPNKIFGVTTL DIVRANTFV AELKGLD PARVNV PVIGGHAGKTI IPLISQCTPKVDFPDQLATLTGRIQ EAGTEVVKAKAGAGSATLSMAYAGARFVFSLV DAMNGKEGVVECSFVQSKETECT YFSTPLLLGKKGLEKNL GIGIKITPFEEKMIAEAIPELKASIKKGEDFVKNMK
34	Mouse MYG (myoglobin)	MGLSDGEQWLVLNVWGKVEADLAGHGQEVLI GLFKTHPETLDKFDKFNK LKSEED MKGSEDLKKHGCVTLTALGTILKKKGQHA AEIQPLAQSHATKHKIPVKYLEFISEIIIE VLKKRHS GDFGADAQGAMSKALELFRNDIAAKYKELGFQG
35	Mouse HBBB1 (hemoglobin subunit beta-1)	MVHLTDAEKA AVSCLWGKVNSDEVGGEALGRLLVVYPWTQR YFDSFGDLSSASAI MGNAKVKAHGKKVITAFNDGLNHLDS LKGT FASLS EHLCDKLHVDPENFRLLGNMI VIVLGHHLGKDFTPAQA AAFQKV VAGVATALAHKYH
36	Mouse FLNC (filamin-C)	MMNNSNYSDASGLGLVDEADEMPSTEKDLAEDAPWKKIQQNTFTRWCNEHLKCVG KRLTDLQRDLS DGLRLIALLEVL S QKRMRYKFHPRPNFRQMKLENV SVALEFLEREH IKLVSIDSKAIVDGNLKLILGLIWT LILHYSISMPMWEDEDEDARKQTPKQRL LGWI QNKVPQLPI TNFNRDWDQKALGALVDNCAPGLCPDWEAWDPNQPVQ NAREAMQ QADDWLGV PQVIAPEEIVDPNVDEHSVMTYLSQFPKAKLKP GAVPRSKQLNPKKAI A YGPGEI PQGNTVLQPAHFTVQTV DAVGVEGLVYIEDPEGHTEAAKVVPNMDKDRTY AVSYVPK VAGLHKVT VLFAGQNIERS PFEVNVGMALGDANKVSARGPGLPEGVN ANKPTYFDIYTAGAGTGDVAVVIVDPQGRRD TVEVALEDKGDNTFRCTYRPVMEGP HTVHVAFAGAPI TRSPFFVHVAECNPNACRASGRGLQPKGVRVKEVADFVKVFTKG AGSGELKVT VKGPKGTEEPVKVREAGDGVFECEYYPVVPGKYVVTITWGGYAI PRS PFEVQVSPEAGAQKVR AWPGL ETGQVGKSADFVVEAIGTEVGTLGFSIEG PSQAKI ECDDKGDGSCDVR YWPTPEGEYAVHVICDDEDIRDS PFIIAHIQPAPDCFPDKVKAF GPGLEPTGCIVDRPAEFTIDARAAGKDLKLYAQDADGCPIDI KVI PNGDGTFRCSYV PTKPIKHTIIVSWGGVNVPKSPFRVNVGEGSHPERVKVYGPGEKTLKAN EPTYFT VDCSEAGQGDV SIGIKCAPGVVGPVEADIDFDI IKNDNDTFTVKYTPPGAGHYTIMVL FANQEI PASPFHIVKVDPSHDASKVKAEGPGLSR TGVEVGKPTHFTVLT KGAGKALD VHFAGAAKGEAVRDFEIDNHDYSYTVKYTAVQQGNMAVTV TYGGDPVPSPFVV NVAPPLDLSKV KVGQLNSK VAVGQEQAFSVNTRGAGGQGLDVRMTSPSRRP IPCK LEPGGGAEAQAVYMPPEEGPYKVDI TYDGHVPVGPSPFAVEGLVLPDPSKVCA YGPG LKGLLVGT PAFPSIDTKGAGTGGLGLTVEGPCEAKIECQDNGDGSCAVSYLPTEPGE YTINILFAEAH IPGSPFKATIQPVFDPSKVRASGPGLERGKAGEAATFTVDCSEAGEAE LTI EILSDAGVKA EVLIQNNADGTYHITYSPAPPGTYITI TI KYGGHIPKFPTRVHVQPA VDTSGIKVSGPGV EPHGVLEVTTEFTVDARSLTATGGNHVTARVLNPSGAKTD TYV TDNGDGT YRVQYTAYEEGVHLV EVL YDEVAVPKSPFRVGVTEGCDPTRVRAFGPGL EGGLVNKANRFTVETRAGTGGLGLAIEGPSEAKMSCKDNKDGSC TVEYI PFTPGDY DVNI TFGGQPIPGSPFRVPVKD VDPGKVKCSGPGLTGVRARVPQFTFTVDCSQAGR APLQVAVLGPTGAEPVEVRDNGDGTHTVHYTPATDGPYTVAVKYADQEVPRSPFK IKVLPSHDASKVRASGPGLNASGIPASLPVEFTIDARDAGQGLLTQI LDPEGKPKKA NIRDNGDGT YTVSYLPDMSGRYTI TI KYGGDEI PYS PFR IHALPTGDASKCLVTV SIG HGLGACLGPRIQIG EETVITVDAKAAGKGVCTCTVSTPDGAELDVDVVENH DGTFDI YTTAPEPGKYVITIRFGGEHIPNSPFHVLACDPLPHVEEPAEMLQMRQPYAPLRPGTC PTHWATEEPVVPVEPLESMLRPFNLVIPFTVQKGELTGEVRMPSGKTARPNITDNKD

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		GTITVRYAPTEKGLHQMGIKYDGNHIPSPLQFYVDAINSRRHVSAYGPGLSHGMVVK PATFTIVTKDAGEGGLSLAVEGSPKAEITCKDNKDGTCTVSYLPTAPGDYSIIVRPDD KHIPGSPFTAKITGDDSMRTSQLNVGTS TDVSLKI TEGDLSQLTASIRAPSGNEEPCLL KRLPNRHIGISFTPKVEGEHVSVRKS GKHV TNSPPKILVGPSEIGDASKVRVWGKGL SEGQTFQVAEFIVDTRNAGYGGLGLSIEGPSKVDINCEDMEDGTCKVTCPTPEPGTYI INIKFADKHPVGPSPFTVVKVTGEGRMKESITRRRQAPS IATIGSTCDLNLKIPGNWFPQM VSAQERLTRTFTRSSHTYTRTERTEISKTRGGETKREVRVEESTQVGGDPPPAVFGDF LGRERLGSFGSITRQQEAGEASSQDMTAQVTSPSGKTEAAEIVEGEDSAYSVRVFPQE MGPHTVTVKYRGQHVPGSPFQFTVGPLGEGGAHKVRAGGTGLERGVAGVPAEFSIW TREAGAGGLSIAVEGSPKAEIAFEDRKDGS CGVSYVVQEPGDYEVSIKFNDHEIPDSP FVVPVASLSDARRLTVTSLQETGLKVNQPASFAVQLNGARGVIDARVHTPSGAVEE CYVSELDSDKHTIRFIPHENGVHSIDVKFNGAHI PGSPFKIRVGEQSQAGDPGLVSAY GPGLEGGTGVSSSEFIVNTQNAGSGALSVTIDGPSKVQLDCRECPGHVVITYTPMAP GNYLIAIKYGGPQHIVGSPFKAKVTGPRLSGGSHLHETSTVLVETVTKSSSSRGASYS IPKFSSDASKVVTGRGPLSQAFVGQKNSFTVDCSKAGTNMMMVGVHGP KTPCEEVY VKHMGNRVYNVTYTVKEKGYIILIVKWGDESVPSPFKVNVV
37	Mouse ACON (aconitase hydratase)	MAPYSLLVTRLQKALGVRQYHVASVLCQRAKVAMSHFEPSEYIRYDLLEKNINIVRK RLNRPLTSLSEKIVYGHLDPPANQEI ERGKTYLRLRPDRVAMQDATAQMAMLQFISSG LPKVAVPSTIHCDHLIEAQVGGKDLRRADINQEVYNFLATAGAKYGVGFWRPGS GIHQIILENYAYPGVLLIGTDSHTPNGGGLGGICIGVGGADAVIDVMAGIPWELKCPK VIGVKLTGSLSGWTS PKDVLKVAGILTVKGGTGAI VEYHGPVDSISCTGMATICNM GAEIGATTSVFPYNHRMKYLSKTGRTDIANLAEFKDHLVDPGCGYDQVIEINLN ELKPHINGPFTPDLAHPVADVGTVAEKEGWPLDIRVGLIGSCTNSSYEDMGRSAAVA KQALAHGLKCKSQFTITPGSEQIRATIERDGYAQILRDVGGIVLANACGPCIGQWDRK DIKKGEKNTIVTSYNRNFTGRNDANPETHAFVTSPEIVTALAIAGTLKFNPETDFLTG KDGKKFKLEAPDADELPRSDFDPGQDTYQHPPKDSGQRVDSPTSQRLQLLEPFDFK WDGDLEDLQLIKVKGKCTDHI SAAGPWLKFRGHLDNISNNLLIGAINIENGKANS VRNAVTFQFGVPDTPARYYKKGIRWVIGDENYEGSSREHAALPRHLGGRAIIT KSFARIHETNLKKQGLPLTFADPDSYINKIHPVDKLTIQGLKDFAPGKPLKCVIKHPN GTQETIILNHTFNETQIEWFRAGSALNRMKELQQ
38	Mouse PRVA (parvalbumin alpha)	MSMTDVLSAEDIKKAIGAFAAADSFDHKKFFQMVGLKKKNPDEVKKVFHILDKDKS GFIEDELGSILKGFSSDARDLSAKETKTLAAGDKDGDGKIGVEEFSTLVAES
39	Mouse HBA (hemoglobin subunit alpha)	MVLSGEDKSNIAAWGKIGGHGAEGYGAELERMFASFPTTKTYFPFHDVSHGSAQV KGHGKKVADALASAAGHLDDLPGALSALSDLHAHKLRVDPVNFKLLSHCLLVTLAS HHPADFTPAVHASLDKFLASVSTVLTSKYR
40	Mouse HBAZ (hemoglobin subunit zeta)	MSLMKNERAIMSMWEKMAAQAEPIGTETLERLFCSPQTKTYFPFHDVSHGSAQV AHGFKIMTAVGDAVKSIDNLSSALTKLSELHAYILRVDPVNFKLLSHCLLVTMAARF PADFTPEVHEAWDKFMSILSSILTEKYR
41	Mouse HSP7C (heat shock cognate 71)	MSKGPVAGIDLGTITYSCVGVFQHGKVEI IANDQGNRTTPSYVAFTDTERLIGDAKN QVAMNPNTVFDARLIGRRFDDAVVQSDMKHWPFPVNDAGRPKVQVEYKGET KSFYPEEVSSMVLTKMKEIAEAYLGKTVTNAVTVPAYFNDSQRQATKDAGTIAGL NVLRIINEPTAAAIAYGLDKKVGAEARNVLI FDLGGGTDFDVSILTIEDGIFEVKS TAGDT HLGGEDFDNRMVNHFIAEFKRKHKKDIS ENKRAVRRRLRTACERAKRTLSSSTQASIEI DSLYEGIDFYTSITRAREELNADLFRGTLPVEKALRDAKLDKSQIHDIVLVGGSTRI PKIQKLLQDFFNGKELNKSINPDEAVAYGAAVQAAILSGDKSENVDLDDVLTPLS LGIETAGGVMTVLIKRNTTIPKQTQFTTYSNQPGLVLIQVYEGERAMTKDNNLLG KFELTGIPAPRGVPQIEVTFDIDANGILNVSADV KSTGKENKITI TNDKGRLSKEDIER MVQEAKEYKADEKQDKVSSKNSLESYAFNMKATVEDEKLQGINDEDKQKILD KCNEIIISWLDKNQTAKEEFQKELEKVCNPIITKLYQSAGGMPGMPGGPFGGG APPSGGASSGPTIEEVD
42	Mouse TNNI2 (troponin I)	MGDEEKRNRATARRQHLKSVMLQIAATELEKEESRESEKENYLS EHCPLHIPGSM SEVQELCKQLHAKIDVAEEKYDMEVKVQKSSKELEDNMQKLFDLRGKFKRPPPLRR VRMSADAMLKALLGSKHKVCMDLRANLKQVKKEDTEKERDLRDVGWRKNIEEK SGMEGRKKMFESES
43	Mouse CASQ1 (calsequestrin-1)	MRATDRMGARAVSELRLALLFVLVLGTPLRGVQGEDGLDFPEYDGVDRVINVNAK NYKNVFKKYEVLLALLYHEPPEDDKASQRQFEMEELILELAAQVLEDKGVGFGLVDS EKDAAVAKKLGLTEEDSVYVFKGDEVIEYDGEFSADTLVEFLDLVLEDVPELIEGER ELQAFENIEDEIKLIGYFKSKDSEHYKAYEDAAEEFHPYIPFFATFDSKVAKKLTCLKN EIDFYEAEMEPMTIPDKPNSEEEIVSVEEHRSTLRKLKPESMYETWEDDLGDIHIV AFABEADPDGYEFLETCLKAVAQDNTENPDLSIIWIDPDDFPLLVVPWEKTFDIDLSAP QIGVVNVTDADSIWMEMDNEDLPSADELEDWLEDVLEGEINTEDDDDDDDDDDDD DDDDDD

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
44	Mouse OBSCN (obscurin)	MDHSFSGAPRFLTRPKAFVSVGKDATLSCQIVGNPTPHVSWEKDRQPVVEAGARFRL AQDGDVYRLTILDLALGDSGGYVCRARNAGEAFAAVGLRVDS EGTCAEQAPHFLL RPTSIRVREGADATFRCRVGGSPQPAVSWSKDGRRLGPPDAPHVRVEEHGESSALRI RSARPRDGGTYEVRAENPLGSASAAAALVVDSDAEVAGPPGTSTATLLAHLQQRRE AMRAEGIPSPPGAGTRTCTVTTEGKHARLSCFVTGEPKETVWKKDQQLVTEGRRH VVYEDEQENFVLKILFCKQSDRGLYTCTASNLVGQTYSSVLVVVREPTVPFKKRLQD LEVREKESATFQCEVAQPATEAAWFKETRLWASAKYDIEEGTERRLTVRNVSD DDAVYICETTEGSRVTAELSVQGNLTKRLPRKTAVRTGDTAIFWVELAVPEGPVRWL RNQEEMVAGGRVAITAEGTCHTLTIFQCTLEDMEGEVAFVSGGCRTTTQFCVSAAPRRP PLYPPADPVVKAKESSVTLSWAPPHGDRPVTIDGYVLEKRLKLGAYAWSRCHEAG WLATTEPTITGVAEEGDFQFRVSAINHFGQGPYLEFPPTMHLVPLAVKTPLKAVEA VEGGEVTFSDVLTVASSGEWELDGEALKASSIYVIRCDRTRHMLTIREVPARLHGAQ LKPFVANGIETSIQMVVRAALGLPSSKLPAAAAREVLAQLHHEAQLLAELSDQAAAVT WLKDGRELSLGPKYEMQVSAGRRALLVRDVAQDDAGLYECVSRGSR TAYQLLVQD ITDGYRDWGPAGPKHMKCKAGAKIARYLGSSCYRFLQYDKGVVHWEALDTR QGKGTSSCSLHEKPKLVFAKQQAHSEVKAEAGNSATLSCEVTQAQTEVTFWKDGK KLSSSSKVRMEASGCSRRLVVQAGKADAGEYSCAAGGQKLSFRLDVAEPKLVFAK GQQAHS EVKAEAGASATLSCEVAQAQTEVTFWKDGKLSSSSKVRMEASGCSRRL VVQAGKADAGEYSCAAGGQKLSFRLDVAEPKLVFAKEQQARSEVKAEAGASATL SCEVAQAQTEVTFWKDGKLSSSSKVRMEASGCSRRLVVQAGKADAGEYSCAAG GQKLSFRLDVTEPKLVFAKEQQARSEVKAEVGN SATLSCEVAQAQTEVTFWKDGK LSSSSKVRMEASGCSRRLVVQAGKADAGEYSCAAGGQKLSFRLDVAEPKLVFAKE QQARSEVKAEAGNSATLSCEVAQAQTEVTFWKDGKLSSSSKVRMEASGCSRRLVV QQAGKADAGEYSCAAGGQKLSFRLDVTEPKLVFAKEQQAHSEVKAEAGASATLSCE VAQAQTEVTFWKDGKLSSSSKVRMEASGCSRRLVVQAGKADAGEYSCAAGGQK LSFRLDVAEPKLVFAKEQQARSEVKAEAGASATLSCEVAQAQTEVTFWKDGKLS SSKVRVEASGCSRRLVVQAGKADAGEYSCAAGGQKLSFRLDVAEPKLVFAKEQQA NSEVKAEAGASATLSCEVAQAQTEVTFWKDGKLSSSSKVRVEASGCSRRLVVQAG KADAGEYSCAAGGQKLSFRLDVAEPKLAFAKEQQAHSEVKAEAGASATLSCEVAQ AQTEVTFWKDGKLSSSSKVRVEASGCSRRLVVQAGKADAGEYSCAAGGQKLSF RLDVAEPKLAFAKEQQAHSEVKAEAGASATLSCEVAQAQTEVTFWKDGKLSSSSK VRVEASGCSRRLVVQAGKADAGEYSCAAGGQKLSFRLDVAEPKLAFAKEQQAHSE VKAEAGASATLSCEVAQAQTEVTFWKDGKLSSSSKVRVEASGCSRRLVVQAG KADAGEYSCAAGGQKLSFRLDVAEPKLMPAKEQQAHSEVKAEAGASATLSCEVAQ AQTEVTFWKDGKLSSSSKVRVEASGCSRRLVVQAGKADAGEYSCAAGGQKLSF RLDVAEPKLVFAKEQQAHSEVKAEAGASATLSCEVAQAQTEVTFWKDGKLSSSSK VRVEASGCSRRLVVQAGKADAGEYSCAARGQKLSFRLDVADTRLMPAKEQQARL EVKAEAGNSATLSCEVAQAQTEVTFWKDGKLSSSSKVRVEASGCSRRLVVQAG KADAGEYSCAAGGQKLSFRLDVAEAESEQIPERPSRREPLVVEKETIILATIAAPSV AVTWLKDGVETIRRSKRHEATSLGDTHTLTVRGAQVLDIAIYSCRVGKEGQDFPVQV EEVAAKFSKPLEPVEGELGGTVMLVCELSPQEAEVWRCGNTQLRPGKRFQMTSEG PRRTLTVSGLREDDAEYVCESTRDRTSARLTVKVPVVKFTSGLSAMVAEEGQEA TFQCVVSPSDAGVTWYKDGMLQPSKEFVMVESGASRLTILGLTLEDAGQVTV EGASSAALRVREAPVLFKKKLEPQTVEERTSVTLEVELTRPWPEVKWTRNAAVLTP SENVEIRAEGARHCLVLRSGVGFADRGFFGCETPDDKTQAKLNVEMQVRLVRLQGE VEAKEQGTASMDVELSHAEVEGSWTRDGLRLQPGPKCHLAVQGPVHILTLALQPPQ DSGLVAFRAEGVHTSARLIVTELPSFTRVLQDVVATQKEKVTLECEL SRPVDVRWL KDGVELRAGKAIGVAGTCSRSLVYRCETGDQGVVCDALDAQTSASLRVQGRTY TLIFRRVLAEDAGEVKFVAENAESRAHLRVKELPVTLLRPLRDKIAMEKHRGVELEQ VSRASAQVRWFKGGVLEQSGPKYEVVSDGLYRKLVIN DVQPEDEDTYTC DAGNVK TSAQFFVEEQSITIVRGLKDMTVMEPAPAWFECETSI PSVRPPKWLGLKTVLQAGGN VGLEQDGTVHRLTLHKTCSTMTGPVHFTIGKSRSSAQLVVS DIPVLT RPLEPKAGRE LQSVVLSCDFRPAPKAVQWYKDDTPLSPSEKFKMALEGMALRILRLTPADAGVY RCQAGSAQSSAEVTVAREVTVIQPLQDAEAMEEGRVCFSCELSHKDEDEIWSLNGT PLYNDSFHEISHEGCLHTLVLSVRQADTGTVCATSPKVS SARLVVKGKPVVFLKA LDDVSAEERGTTLQCEVSDPEARVVWRKDGVELGPSDKYDFLHKAGARGLTVHD LSHEDAGLYTCQVGSKETQSKVSVHDLHVGITKRLKTVEVLEGESECSFECVLSHESPS DPAVTVGGKT VGGSGHFHAVRQGRKYTLTVKDAALSDAGEVFSVLGLTSKASLI IRERPVDITKPLEDQRTTLGEDVMLSCELSRAGTSVRWLKDGKAIRKSQKYDLLSEG TRAVLVVRKASLKDSGEYTCETEASKSTAKLCVEEKANRFT EELADLQVEEKGRAV FTCKTEHPASVTVWRKGLLELRASGKHVPSQEGTLTKLTINALERTDSDTYTCDIGQ ARTQARLLVHGQKVRVIEDLED TAVQEGSSAKFCRIAPADYGPVHWF LDKTPLHS NELNEITVQPGGYHVLTLRQLALKDSGTVYFEAGDQRTS AALRVTEKPSIFSRPLTDV TVTEGEDTLVCETTVDSSVRWTKDGKTLRPSARCQLSHEGCAQLLITATTPQDG GRYKCEIGGASSSIVRVHALPVRFRFESLKDVEVPGEKAATLRCVLSVAAPVEWRH GDDVLKSSNKYSRLQEGAVLELVIRDLQPDQSGQYSCSFGDQTTSATLTVKTSQAQF VGKLRNKEATEGTTVTLRCELKEAPVEWKKGTETLRNGDKYSLKQDGA VCELCIC SLLVADAGEYSCVCGQEKTSATLTVKALLVHFVRRLRSKEATEGDTTTLQCELSKAA PVEWRKGTETLRDGRYSLKQDGA VCELCIRSLTIADAGEYLCTCGQEKTSATLTVR ALPAKFKDSLKNEATEGTTATLSCELSKAAPVTWKKGP KTLQSGDKYVLRQDGA V CGLQIHGLTMADAGEYSCVCGQEKTSATLTVRGLPAKFIEDLRSQEATEGATAILRC

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		ELSKAAPVEWRKGETLKDGDYTLRQDGAVCELQIRGLAVVDGTYSLSLPTKFTF GLRNEEATEGTMATLRCQMSKAAPVEWRKGETLRDGDYSLRQDGAMCELQIRG LTIEDSGEYTCVCGQEKTSATLSVKALPSRFIEDLRSQEATEGTMATLRCQMSKTAPV EWKKGSETLRDGGYSLRQDGPVCELQICDLAVEDAGEYSCVCGQEKTSATLSIKAL PPRFIEDLRSQEATEGTMATLRCQMSKAAPVEWRKGETLGDGGYSLRQNGAVCE LQIHDLAVEDTGEYSCVCGQEKTSATLNVKALPPRFIEDLRSQEATEGTMATLRCQM SKAAPVEWRKGETLRDGGYSLRQDGAVCELQIHDLVDVEDAGQYSCVCGQEKTS AVLTVDALPPKFTBGLKKEATEGTMVTLRCQMSKEATVEWRKGAKTSLSDGGYSL LRQDGAMCELQICGLAVEDAGEYSCVCGQEKTSATLSVKALPPRFIEDLRSQEATEG TMATLRCQMSKAAPVEWRKGETLRDGDYSLRQDGAVCELQIRDLAVEDAGEYL CVCGQEKTSATLSVKALPPRFIEDLRSQEATEGTMATLRCQMSKAAPVEWRKGSKT LRDGDYSLRQDGAMCELQICDLAVEDTGDYSCVCGQEKTSATLSVKALPPRFIEDL RSQEAREGTVALRCRMSKAAPVEWRKGETLKDGDYSLRQEGNLCELQIRDLAV EDTEEYSCVCGQEKTSATLSVKALPAKFIEDLRSQEAPESSTVTLRCKLSKASVVM KKGSETLRNGARYSLRQDGAVCELEIRDLTVEDTGEYSCVCGQERTSATLSIMAPQV VFQQPLQNLQAEEGSMASLRCELSVPNAAMVWSKGLLEQGDTRREARQQGCVAE LLLRDLRREDAGEYSCVCGQTSATLMVTAAPVRFLELQAQDVEDGATARLRCE LSREAVSVEWRKGSLLQFPKAKYQMVQEGTTAELLVHGVBEQEDAGEYTCDAHTQ SIARLSVRAPKPKFKTDLQSTEQEAGGTARLCCQLSEAEPTPVQWLKEGVELHVGS KYEMRRQGAVALLEHGLEAKDTGEYACLVGGQKTLASLRVKEPEVTIVRGLVDM VQADEDVEFTCKVSQAGATDVQWHLQGLPLQSNVETAVLADGCTHVLQKLGVT LEDAGTVSFHVGGLSSAQLTVRVPEVTVLEPLKDVQLSEGDHAFRCRLSRASGQE ARWALGVPLQCNEMNDITVEQGTLYLLTLHKVTLEDAGTITLQVGSSEAEQLKV TAKNTVLRGLENVDALEGGEALFECQLSQPEVAHAHTWLLDDEPVHTSEKVEVYFE NGLRHLLELKNLKPQDSRVTFLAGDVVTSAPLTVRGWRLEVLPEPHDASVKAGMQ VRFTCILSEAVPVGEATWYINGAAIQPDDTDWTVTTDGSHHALTLNAQPOHAGEVT FAARDAVASARLSVLALPDDPEDAEVVGSRDHSVTLWSVAPMSDGGGLCGYRVE MKEASTGQWQLCHDLVPGPECVVDDLVPKTYRFRVAAGPAGAGEPVHLPQMV KIAPAPAPAPAPAPAPETRQAVVGEDICLELEVAADGGEVVMHKGTERIQPGGHF EVLSRGQRQMLVIKGFRTEDQGEYRCGPIQGLPSSGAATFNVMTSGSGDEVPAQPS LPPEAAQEGDLHLLEALARKRRMSREPTLDSISELPEEDSRVQHLRQEAEEETAPDLS EGYSTADELARTGEADLSHTSSDDESAGTPSLVTYLKKAGGPGISPLASKHEAQVT TSVKPKQKQEPVVPCTPPPGDLSAADLMDPSLDKAAVKIQAAPKGYKVRKEMKQOE GPVFSRTFGDTEAQVGDVLRLECVLATKTDMRACWLKDGIELTDGRHYHIDQLKDG TCSLLVTGLAPTDSGRYTCQVSTKSGRVSHSACVVVSGTSEAESSSGGELDDAFRR AARLHRLFRITKSPAELEEELFLSADEGPGEPEPADWQTYREDENFVCIRFESLAE ARQAVTCFRNMFATMGIGVEISLGEQGRGVEMRIKQVAPTVPVPLAKTPGLQTS DAAPVFLTELQNDVQDGYPMSFDCVVTGQPVPSVRWFKDKLLEEDDHVMINED QQGGHQLIITAVVPADMGVYRCLAENSMGVSSSTKAELRVELTSTDYDTAADATETS SYPSAQGYLSSREQGETSEDEGLPQVLEELKDLQVAPGTRLAKFQLKVGYPAPKL YWFKDGQPLTSDHIRMKTDKTLHTLEIVSVTREDSGQYAAIYISNAVGAAYSARLL VRGPSEPEEKPSADVHERLVPPRILEKFTPKKVKRGSSTIFSVKVEGHPAPSVHVLKE EAEKGLWIGPDTGYTMASSSKQHSVLVLDVGRQHGTYTCTIATNPAGQALCSAS LHIISGLAKEERQERVKEALISSFLQGSTQAVSAQMSSESAGFADLVGQSGESLVAEAA HSHLSLAEVGTETFLQKLTQITEMVSAKISQAKLQVPGGDSDEETKTPSASPRHGRS RPSSSVQESSSESDEGDSRGEIPIYVVTADYLPGLGAEQDAIILREGQYVEVLDLAPL RWLVRTKPTKSSPSRQGWSPAYLDKRLKLSPEWGPTAEPEFPEAVSEDEYRTRLS SVIQELLSSSEQAVFVGLQFLESHHMKHLERSPRVPAAVASQKTVIFRNVQDISHFHSSF LKELQCGCTDDDVAMCFIKNQEAFKYLEFLVGRVQAESVVVSTPVQEFYKKYAEE TLSAKDPTQPPPPPLQHYLEQPVVERVQKYQALLKELIRNKARNRQNCALLEQAYAVV SALPQRAENKLHVSLENYPGTLEALGEPVIRQGHFIVWEGAPGARMPPKGNHRHVF LFRNYLVIKPRRDSRTDTSYVFRNMMLSSIDLNDQVEGDDRAFEVWHEREDSV RKYLLQARTVVIKNSWVKEICGIQORLAQPVWRPPEFEELADCTAELGETVKLACR VTGTPKPIVSWYKDGKPVVEDPHHILIEDPDGSCITLILNDLTGIDSGQYMCFAASAAG NASTLGKILVQVPPRFVNVKVRATPFVEGEDAQITCTVEGAPYQPIRWYKDGTLAPG NRYRMLNEPRSGVLVLVIAASKEDLGHYECLELVNRLGSTRGGGELYMQSPALRAR DQHHREQIQAAVEDTSVEGSAHSAQDGADQQAASVLWRLLGSEALGPSGDLNTR QSEPPAFEEAASQIPGAASGTPESVQPGTHKLEQETTSQSGSQGWTPVIRVEGTAWPG AGTQQLLLDVHSQVIMETTQRTYVQCAPDTGVTRAPSMQVTIEDVQVQVGDMAQF DAVIEGHPPIVITWYKGSTQLTSSARLSQRQDGTYSVLVTDVAPHDAGVYTCVANN AGGQVLCKAELLVHGGDKLDAENQVYRRKLHSFYDVQEEIGRGVFGVVRVQKHG NKMFCAAKFIPLRSKTRAQAYQERDILATLGHPLVTGLLDQFETRKTLLILILELCSSEE LLDRLFKKGVVTEAEVKVYIQQLVEGLHYLHSHGILHLDIKPPNIMLVHPAREDIKIC DFGPAQKITPSEPQYSKYGSPFVSPETIEQNVS EGSDIWMGVISYLSLTCSSPFAE SDRATLLNVLEGRVSWSSPTAAHLSSEDAKDFIKATLQRTPRARPSTSQCLAHWPFLK SMPAEAAHINTKQLKFLRLARSRWQRSLMSYKSLVMRSIPPELLQGPPDSPSLGVARH LRGEASGASSSSSSDNELAPFARAKSLPPSPVTHSPLLHPRGFLRPSASLPETEASMP TADAAVPAS PQSAGPPASPGCVPRHSVSSLFYQQAEGEAGERNKTSAGAKRHPARRR HLLKGGYIARALPLGLREPLMEYSLLEEEAAREEQASLMTKTPSFETALRLPSSSVREV PGRSHSLDNPPVTGSPSEACKEQLLFPSTGLTHETAKDRGHKEGFLQESVFPFPM SGDSRPKQEGSSQDSCRGPASSCHSELGSGSQEGCGPPSSQSLGSLPPQSLKKELST

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		SCGPLFSEQPQAAPPFTQVSPLLGSEKEBPQDGLSEGPVVPVSSSPGSASQVDASLDTE GLSEAGDTCDFTPPPQRPQEATTRKFSLESRGYAGVAGYGTFAFGGDAGGMLGQ GPLWARMAWAVSQSSEEQDEAATESPQPLESLGPIAEASGVPLRTSPSLTPWEEVEQ VSLVQIRDLSGDAEAADTISLDISEVDPAYLNLSLDYDIKYLPEFEMI FRRVPKPIEQPE SPGSETEAGQGLADFLEEAWPWPGLGLRAGLEITEEPEEPGDLEALLGEAAVGRK RKWSPSRGLFQFPGRCLSGEEPVELGLRQVRKASMAHISRILKGRPEGPEREPPRKK AGLASFRLSGLKGRDQELSDEAVVLGQSVTLACQVLAQPTAQATWSKDGVLLESSG HLLISSTLKNFQLLTILVVKEDDLGTYTCCVSNPLGTAVTTGVLRAERPSSSPRPEVG ELYKDAVLLVWKPVESCGPVTYIVQCCIEGGSWTTLASDISDCCYLTGKLSRGMMYI FRTACVSKAGMPYSSPSEQVLLGGPNHLASEEESRGRPAQLLPSTKTFAFQMQRIR GRFSVVRQCREKASGRALAAKIVPYQPEDKTAVLREYEALKRLHHPHLAQLHAAYL SPRHLVLILELCSGPELLPSLAERESYESDVKDYLWQMLSATQYLHAQHILHLDLRS ENMMVTYNNLLKVIDLGNQSLDQEKVPAPENFKDYLETMAPELLEGGGAVPQTDI WAIGVTAIFIMLSGEYPESSEGTDLQKGLRKGLIRLSRCYAGLSGGAVAFLLQSSSLCA QPWGRPCASTCLQCGWLTEEGTGSRTPTVTFPTVRLRAFVREKRALLKKNL AQVR
45	Mouse F8VPN4 (glycogen debranching enzyme)	MEHSKQIRILLNEMEKLEKTLFRLEQGFELQFRLGPTLQGKAVTVYTNPLPGEAFN REKFRSLDWNPTEREDDSKCYKHLHQSGSFQYYFLQGNEKSGGGYIVVDPILRI GVDNHVLPDLCVTLQTFIAKCLGPFDEWESRLRVAKESGYNMIHFTPLQTLGLSRSC YSLADQLELNPDFSRSKRYTWSVQQLVEKLRKWNILCI TDVVYNHTATNSKWIL EHPESAYNLVNSPHLKPAWVLDRLWHFSCDVADGKYREKGVPALIENDQHMNCI RKIIWEDIFPRIQLWEFFQVDVHKAVEQFRLLSQENRRVTKSEPKHEHLKIIQDPEYRR RGCAVDMDTALATFI PHDNGPAIEECNWFRRKLEELNSEKHHLTSCHEQAVNC LLGNVFFERLAGHGPKLGPVTRKYPLVTRYFTFPFGEMALSAEALIHLPDKACFLM AHNGWVMGDDPLRNFAEPGSDVYLRRLEICWGDVSVKLRYNKPEDCPYLWAHMK KYTEITATHFQGVRLDNCHSTPLHVAEYMLDAARKLQPNLYVVAELFTGSEELDNI F VTRLGISSLIREAMSAYNSHEEGRLVYRYGGEPVGSFVQPCPLPLMPAIAHALFMDIT HDNECPIVHRSAYDALPSTTVVSMACCASGSTRGYDELVPHQISVVAERFYTQWNP GASPADTGDVNVHSGIIAARCAINRLHQELGAKGFIQVYVDQVDEDIVAVTRHSPSIH QSVVAVSRTAFRNPKTSFYSEKVPQMCIPGKIEEVVLEARTIERNTKPYKKDENING MPNMTVELREHIQLHESKIVRQAGVATKGPNEYIQEIEFENLSPGSVI IPRVSLDPHAQ VAVGILRNHLTQFSSHFKSGSLAVDNADPILKIPFASIASKLTALBNQVLYRCESEEQ EDGGGCYDIPNWSLKYAGLQGLMSVLAIRPKNDLGHPPCENLRSGLWMDIVVSG RLISRSGSIAEVGKWLQAMFFYLKQIPRYLIPCYFDAILIGAYTTLLDVAWKQMSFV QNGSTFVKHLSLGSVQMGVGKCPCLPLSPSLLDVPCRLNEITKEKEQCCASLAAG LPHFSSGLFRCWGRDTFIALRGMLLVTRYLEARNIILAFASTLRHGLIPNLLGEGTY ARYNCRDAVWWWLQCIQDYCRTVPNGLDILKCPVSRMYPTDDSAPLPAGTLDQPLF EVIQEAQRMHQGIQFRERNAGPQIDRNMKDEGNITAGIDEETGFVYGGNRPNCGT WMDKMGESDRARNRGI PATPRDGSAVEIVGLCKSAVRWLLLELSKKNIFPYHEVRVK RHGKVAVSYDEWNRKI QNNFEKL FHVSEDPSPNEKHPNLVHKRGYIKDSYGASS PWCDYQLRPNFTIAMVVAPELFTA EKAWKALEIAEKKLLGPLGMKTLDPDDMVYC GVYDNALDNDNYNLARGFNHYHQGPWLWPIGYFLRAKLYFSKMMGPETAAKTVFL VKNVLSRHYVHLERSPWKGLPELTNENGQYCPFSCEQAWSMAVVLETLYDL
46	Mouse EF1A2 (elongation factor 1-alpha 2)	MGKEKTHINIVVIGHVDSGKSTTGHILIYKCGGIDKRTIEKFEKEAAEMGKGSFKYA WVLDKKAERERGITIDISLWKPETTKYYITIIDAPGHRDFIKNMITGTSQADCAVLIV AAGVGEFEAGISKNGQTREHALLAYTLGVKQLIVGVNKMDSSTEPAYSEKRYDEIVK EVSAYIKKIGYNPATVPFVPI SGWHGDNMLEPSNMPWFKGWKVERKEGNASGVSL LEALDTILPPTRPDKPLRLPLQDVYKIGGIGTVPVGRVETGILRPGMVVTFAPVNIIT EVKSVMHEALS EALPGDNVGFNVKNVSVKDIRRGNVCGDSKADPPQEEAQPTSQ VIILNHPQGISAGYSPIVICHTAHIAKFAELKEKIDRRSGKKLEDNPKSLKSGDAAI V EMVPGKPMCVESFSQYPPPLGRFAVRDMRQTVAVGVIKNVEKKS GGAGKVTKSAQK AQKAGK
47	Mouse AATC (aspartate aminotransferase)	MAPPSVFAQVPQAPPVLVFKLTADFRDDPDRKVNLVGAYRTDESQPWVLPVVRK VEQKIANDNSLNHEYLPILGLAEFRSCASRLVLGDNLSAIRENRVGVQSLGGTGAL RIGADFLGRWYNGTDNKNTPIYVSPTWENHNNAVFSAGFKDIRPYCYWDAEKRL DLQGFNLNDLENAPEFSIFVLHACAHNPTGTDPTPEQWKQIAAVMQRRFLPPFFDSAY QGFSAGDLEKDAWAI RYFVSEGFELFCAQSFKNFGLYNERVGNLTVVGKESDSVLR VLSQMEKIVRI TWSNPPAQGARIVAATLSDPELFKEWKGNVKTMAIRILTMRSERLA RLEALKTPGTWSHITEQIGMFSFTGLNPKQVEYLVNEKHIIYLLPSGRINMCGLTTKNL DYVATSIHEAVTKIQ
48	Mouse KAD1 (adeny late kinase isoenzyme 1)	MEEKLKKAKIIFVVGPGSGKGTQCEKIVQKYGYTHLSTGDLRAEVSSGSEKGLL SAIMEKGELVPLDVLDMRLDAMLAKVDSSNGFLIDGYPREVKQGEFEQKIGQPTL LLYVDAGAETMTQRLLRGETSGRVDNEETIKRLETTYNATEPVISFYDKRGIVR KVNAEGTVDTVPSEVCTYLDLSLK

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
49	Mouse G6PI (glucose-6-phosphate isomerase)	MAALTRNPQFQKLEWHRANSANLKLRELFEADPERFNNFSLNLTNHHGILVDYS KNLVNKEVMQMLVELAKSRGVEAARDNMFSGSKINYTENRAVLHVALRNRSNTP KVDGKDVMEVNRVLDKMKSFQQRVRSQDWKGYTGKSIDTIINIGIGSDLGPLMV TEALKPYSKGGPRVWFVSNIDGTHIAKTASLSLPETSLFIIASKTFTTQETITNAETAKE WFLEAAKDPSAVAKHFVALSTNTAKVKEFGIDPQNMFEFWDWVGGRYSLWSAIGL SIALHVGDFHFEQLLSGAHWMDQHFLKTPLEKNAPVLLALLGIWYINCYGCETHALL PYDQYMHRAAFYQQGDMESNGKYITKSGARVDHQTGPIVWGEFPTNGQHAFYQL IHQGTMKMIPCDFLIPVQQTQHPPIRKGLHHKILLANFLAQTEALMKGLPPEARKELQAA GKSPEDLEKLLPHKVFEGRNPTNSIVFTKLTPFILGALIAMYEHKIFVQGIMWDINSFD QWGVLELKGQLAKKIEPELEGSSAVTSHDSSNGLISFIKQQQRTKLE
50	Mouse DESM (desmin)	MSQAYSSSQRVSSYRRFTFGGAPGFSGLSPLSSPVFPFRAGFGTKGSSSSMTSRVYQVSR TSGGAGGLGLSRRLGTTTRAPSYGAGELLDFSLADAVNQEFLATRNEKVELQELN DRFANYIEKVRFLEQQNAALAAEVNRLKGREPTRVAELYEEEMRELRRQVEVLTNQ RARVDVERDNLIDDLQRLKAKLQEEIQLREEAENNLAAFRADVDAATLARIDLERRI ESLNEEIAFLKKVHEEEIRELQAQLQEQVQVEMDMSKPDLTAALRDIRAQYETIAA KNISEAEWYKSKVSDLTQAANKNNDALRQAKQEMMEYRHQIQSYTCIDALKGT NDSLMRQMRLEDRFASEANGYQDNIAARLEEEIRHLKDEMARHLREYQDLLNVKM ALDVEIATYRKLLEGEESRINLPIQTFSALNFRETSPEQRGSEVHTKKTVMIKTITETRD GEVVSEATQQQHEVL
51	Mouse KCRS (creatine kinase S-type)	MASAFSKLLTGRNASLLFTTLGTSALTGYLLNRQKVSADAREQHKLFPPSADYPDL RKHNNCMAECLTPTIYAKLRNKMTPSGYTLDDQCIQTGVNDNPGHPFIKTVMGVAGDE ESYEVFADLFDVPVILKLRHNGYDPRVMKHPTDLDAKITHGQFDERYVLSRVRTRGRSI RGLSLPACSRARERREVENVAITALEGLKGDLAGRYYKLSMTEQDQORLIDDHFLF DKPVSPLLTACAGMARDWPDARGIWHNYDKTFLIWIINEEDHTRVISMEKGGNMKRVF ERFCRGLKEVERLIQERGWEFMWRNLGYILTCPSNLGTGLRAGVHVIRPKLSKDPFR FSKILENLRLQKRGTTGGVDTAAVADVVDISNIDRIGRSEVELVQIVIDGVNYLVDCEK KLERGQDIKVPPLPQFGRK
52	Mouse E9PWQ3 (collagen, type VI, alpha 3)	MRKHRHLPLVAVFSLLSGIATTHAQHQGDVKNAAAADIVFLVDSSWSAGKDRFLL VQEFSLSDVVESLAVGDNDFHFAVRLNGNPHEFLNLTYSKQEVLSHIVNMSYIGG SNQTGKGLVYIHSHLTEASGSRAADGVQVVIIVLTDGQSEDGFALPSAELKSADVN VFAVGVEGADERALGEVASEPLSMHVFNLENVTSLHGLVGNLVSCIHSVNPERRAG DKESLKDITAQDSADIIIFLIDGSQNTGNANFVDVIRDFLVNVLRLSVGNQQVQVGVV QYSEEPITMPSLNSYPSKAAVLDAVKGLSLVGGESANIGQALDFVVENHFRAGGSR VEEGVPQVLVLISAGPSDEIRDSVVALKQASVFSFGLGAQAASRAELQHIATDDSLV FTVPEFRSFGDLQEQILPPLVGVQQRHIVLQPPAIVTQVMEVNKRDIVFLVDGSSSLG PSNFNAIRDVTRVVIQRLEIGQDLVQVSVAQYADTVKPEFYLNSTYTNKRDAITAVRK MRALNGSALYTGSSLDVFRNNLFTSSAGHRAAEGVPKLLVLITGGKSLDEVSPAQE LKRGSIMALAVGSKAADEDELKEIAFDSLSLVFIPAEFRPAPLQNMPLSLMAPLRLTLTG TTEVHVNRKDIIFLLDGSNDVGNKNNFPYVRDFVTNLVNSLDVGSNDIRVGLVQFSDT PVTEFSLDTYQTKSELLAHLRRLQLKGGSGNLNAGSALSIIHANHFTAGGSRTRHEV PQLLLLLMAGPSEDAYLQAANALVRSGLVTFVCGVTNRADKAELEHIAFNPSLVYLM DDFRSLPSLQQLIQLPTTYVSGGVQEVPLSQPESKRDILFLPDGSVNVLGQPPAVRDF LYRIIEELDVKPDGTRVAIAQFSDVRLSRSFSEHQTKAEILNLVKKMKLKTGKALNL GYALDYALRNIIFRSAGSRIEDNVQQFLVLLVAGRSSDAVAGPASSLKQRGVVPFIF QAKNANPSELEQIVLSPAFILAAESLPKIGDLQSQIVSLLKAEQSGSPVSGEKDVVFLI DGSEGVRSFGFPLLDKFVQRVVESLDVGPDRVRVALVQYSRTRPEFYLNSHMDQQG VISAIRRLTLGGPTPNTGAALFEVLRNLTSSTGSRIAEGVPQLLIVLTAEPSSGDDVRG PSVVLKQGGAVPIGIGIGNADISEMQTISPIPDFAVAIPTFRELGTQQVVISERVQLNRE ELSSLPILTPSTGAGVSGKKDVVFLIDGSRNAGPEFQYIRTLIERIVEYLDIGFDTRV AVIQFSEDSKMEFPLNAHFSKDEVQNAVRLRPKGGSQVYIGNALEYVLKNIQRPPL GSRIEBGPQFLVLISSGKSDDDEVDDSAVELKQFQVAPLTIARHTDQEEVLKISLSPFY VYSVSTFRELPRLEQKLLTPITTLTSQQIHQILASTRYPPSVVESDAADIVFLIDSSDAV KPDGIAHIRDFVSRIVRRLNIGPSKVRIGVVQFSNDVFPEFYLKTHKSQSSVLEAIRRLR FKGGSPLNTGRALFEFARNLFVKSAGSRIEDGVPQHLVFLGKGSQDDVARHAQVIS SSGIVSLGIGDRNIDRTDLQITNDPRLVFTVREFRELPIIEERVMLSFGPSGATPQPPG VDLPSPSRPEKKKADIVFLLDGSINFRRDSFQEVLRFASEIVDVTYEDGDSIRVGLVQY NSDPTDEFFLRDFSTKRQIIDAINKVYKGRHANTRVGIEHLLRNHFVPEAGSRLE RVPQIAFVITGGKSVEDAQDVSALTQKGVKVFVAVGVNRNIDSEEVGKIASNSATAFR VGSVQELSELSETVLETLHDAMHETLCPGVTDVSKACNLEVILGFDGSRDQNVFVSQ KGLSKVDIILNRSIQIRISCSGNQLPTVRVSVMANTPSGPVEAFDFAEYQPELFEKF RNMRSQRPPVLTADTLKLYQNKFRQSSPDTVKVVIFHTDGDGDMADLYRASIELR QAGAQUALVLGLERVANLERLMLHEFGRGFMYDRPLRLNLLDLDELAEQLDNIAE KACCGVPCKCSGERGDRGPIGSGPKGISGEDYRGYPGDEGGPGERGPPGVNGTQG FQGCPGQRGVKGSRGFPGEKGELGEIGLDGLDGEEDKGLPGSSGEGKSGPGRRGDK GPKGDKGERGDVGIRGDPGDSGRDSQQRGPKGETGDI GPMGLPGRDGI PGSPGDPGK DGGSGRRGPAGAKNRRGGPGQPGFEGEQGTRGSGQPPPGIGPPGLIGEQQIGPGRGG GGTAGAPGERGRTGPLGRKGEPGEPGPKGSGINRGRPRGETGDDGRDGVGSEGRRGK

TABLE 1-continued

Mouse Proteins		
SEQ ID NO	Description	Sequence
		KGERGFGPYGPGKGTGPEPGADGPPGPKGIRGRGRNSGPPGATGQKGDQGYPGPSGH KGNRGDSVDQCALIQSIRDKPCPCYGLECPVFPTLAFALDTSSEGVTDPTFSRMRV LLGIVGDLTIAESNCPRGARVAVVTYNNEVTTEIRFADSKKKSALLDSIQNLQVALTS KQQSLETAMSFVARNTFKRVRSGLMRKVAVFSSNKPTRASPOLREAVLKLSDAGIT PLPLTSQEDRQLINALQIINNTAVGHALVLPARRDLTDFLKNVLTCHVCLD ICNIDPSC GFGSWRPSFRDRRAAGSDVDIDLAFILDSSEATTLFQFNEMKKYIGYVIRQLDLSDPD KASQHFARVAVVQOSTYESVDNASVPPVKVEFSLTDYGAKEKLLDLFSRRMTQLQG TMGLGNAIEYTIENI FESAPNPRDLKIMVLMLTGDMQRQQLLEAQRAI LQACKGYF FVVLGIGRKNVNIKEVYSFASEPNDFVFFKFDVDS TELNNEEPLMRFGRLPSFVSSENAF YLPDDLKQCDWFGQDQPVKNGVKFGHKQINPPHTANSSLT SKVVTMTMKPVTTTKP TAIVNLPPAKPAPAPAPAPQVLAQPDPAKPAQARPAKPAKPAKPAKPAKPAKPAKPAKPA PAQTASVRPAPAKPAPPQAAAAPVPAKPAVPAQAPAPPQAAAAPVPAKPAVPAQPA PPQPAKPAKPAKPAVPAQPAQAAAQPMAPQVLT KSAAVKPAKPAKPAKPAKPAKPAKPA ATATARPALAAKPAKPAATRPLAAAIRPVATKPEAPRQQAAPKPAKPAKPAKPAKPAK VSREVQVSEVTENSARLHWERPEPSSFFFYDLTVTSAHQSLVLRQNLTVTDRVIGG LLAGQLYHVVVVSYLQSQVRAIYQGSFNTKTKTQPPPLQAARASSSTINLMVNT EPL FLT KTDICKLSRDAGTCVDPKLLWHYDLESKSKCRFWYGGCGGNFRHFSQEECEK MCSFDLLV
53	Mouse AMD1 (AMP deaminase 1)	MPLFKLTGQKQKIDAMRSFAEKVFASEVKDEGGRHEISPFVDVEICPISLHEMQAHI FHMENLSMDGRKRRFRQGRKTVNLSIQSETSSTKLSHIEEFISSSPYTESVPDFQRVQI TGDYASGVTVEDFEVVCGLYRALCIREKYMQKSFQRFPKTPSKYLNRINDGEALVG NESFYVPVTPPPKKGEDPFRTEDL PANLGYHLKMKAGVIYIYPDEAAANDKEDPKPY YPNLDDFLDDMNFLLLALIAQGVPKTYAHRRLKFLSSKQVQHMLNEMDELKELKNN PHRDFYNYCRKVDLTHIAAACMNQKHLRLRIKSYHIDADRVVSTYKLSKLTDELKFLA KLMNMPHLDLTVDSLVDHAGRQTQRFQDKFNDKYNPVGASELRDLYLKLTDTNNGE YFATIIKEVGADLVEAKYQHAEPRLSYIGRSPDEWNKLSWVFCNRIYCPNMTWMIQ VPRIYDVFRSKNLPFHFGKMLENI FLFPVEATINPQAHPDL SVFLKHI TGFDSVDDSEK HSGHMFSSKSPKPEEWTMENNPSTYYAYMYANI TVLNSLRKERGMNTFLFRFPK GEAGALTHLMTAFMIADNISHGLNLKKSVPVLQYLFFLAQIP IAMSPLSNNSLFLEYAK NPFLDPLQKGLMISLSTDDPMQPHFTKEPLMEEYIAAQVFKLSTCDMCEVARN SVL QCGI SHEEKAKFLGNLYLEEGPVGNDIRRTNVAQIRMAVRYETWCYELNLIAEGLKA TE

[0039] An animal protein can comprise a rabbit protein (Table 2). A rabbit protein can comprise a rabbit muscle protein. A rabbit protein can comprise a rabbit blood protein. A rabbit protein can comprise a rabbit albumin protein. A rabbit protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99%, or about 100% sequence identity to any one of SEQ ID NOs: 54-78. A rabbit albumin protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97%

at least or about 98% at least or about 99%, or about 100% sequence identity to SEQ ID NO: 54-78.

[0040] A food product may comprise a fragment or a truncation of a rabbit protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 54-78. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 54-78. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 54-78. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 54-78. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 54-78.

TABLE 2

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
54	Rabbit KCRM (Creatine Kinase M-type)	FGNTHNKYKLNKYKSEEEYPDLSKHNNHMAKVLTDPDYKKLRDKETPSGFTLDDVIQTGVDNPNPGHPFMTITVGCVAGDEESYTVFKDLDPDPIIQDRHGGFKEPTDKHKTDLNNHENLKGDDDLDPHYVLSRVRTGRSIKGYTLPPCHSRGERRAVEKLSVEALNSLTGEFGKGYPLKSMTEEQEQQQLIDDHFLDPKPVSPLLLASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFRFCVGLQKIEEIFKKAGHPFPMWNEHLGYVLTCPNSNLGTGLRGGVHVKLAKLSKHPKFEILT

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		RLRLQKRGTGGVDTAAGVSFVDISNADRLGSSEVEQVQLVVDGVKLMVEME KKLEKGQSIDDMI PAQK
55	Rabbit GIU4H8 (Actinin Alpha 3)	MMVMVQPEGLGAREGPPAGGGGGGEYMEQEEDWDRDLLLDPAWEKQQRK TFTAWCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPRPDKGKMRFH KI ANVNKALDFIASKGVKLVSIGAEIVDGNLKM TLMGIWTII LRFAIQDISVE ETSAKEGLLLWCQRKTAPYRNVVNQNFHTSWKDG LALCALIHRHRPDLIDY AKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAIMTYVSCFY HAFAGAEQAETAANRICKVLAVNQENKLMEEYEKLASELLEWIRRTVPWL ENRVGEPSSMSAMQRKLEDFRDYRRLHKKPRVQEKQCLEINFNTLQTKRLLSH RPAPMPSEGLVSDIANAWRGLEQVEKGYEDWLLSEIRRLQRLQHLAEKFPQ KASLHETWTRGKEDMLTQRDYESASLQEVRRALLRRHEAFESDLAAHQDRVE HIAALAEQELNELDYHEAASVNSRCQAICDQWDNLGTLTQKRRDALERMEKL LETIDQLQLEFARRAAPFNWLDGAI EDLQDVVLLVHSVEETQSLLTAHQDFK ATLPEDRERGA ILGIQGEIQKICQTYGLRPSSTNPYINLSPQDINTKWDTVRK LVPSRDQTLQEELARQQMNERLRQFAAQANAVGPWIQAKVEEVGRLAAGL AGSLEEQ MAGLRQQEQNI INYKSNIDRLEGDHQLLQESLVFDNKHTVYSMEH IRVGWEQLLTSIARTINEVENQVLTDRDAKLSQEQLNEFRASFNHFDKRKNG MMEPDDFRACLISMYDLGEVEFARIMTMDPNAAGVVTQAFIDFMTRETAE TDTAEQVVASFKILAGDKNYITPEELRRELPAEQAEYCIHRMAPYKGAGAPA GALDYVAFSSALYGESDL
56	Rabbit PYGM (Glycogen Phosphorylase, muscle form)	SRPLSDQEKRKQISVRGLAGVENVTELKKNFNRHLHFTLVKDRNVATPRDYY FALAHTVRDHLVGRWIRTQQHYEYKDPKRIYYLSLEFYMGRTLQNTMVNLA LENACDEATYQLGLDMEEL EEEEDAGLGNGLGRLAACFLDSMATLGLAA YGYGIRYEFGIFNQKICGGWQMEEDDWLR YGNPWEKARPEFTLPVHFYGR VEHTSQGAKWVDTPVVLAMPYDTPVPGYRNVNVTMLWSAKAPNDFNL KDFNVGGYIQAVLDRNLAEINISRLVYPNDFFEKGLRLKQEYFVVAATLQD IIRREFKSKFGCRDPVRTNFDAFPDKVAIQLN DTHPSLAIPELMRVLVDLERLD WDKAEVTVKTCAYTNHTVLPALERWPVHLETL LPRHLQIIYEINQRFLN RVAAAFPGDVLRLRRMSLVEEGAVKRINMAHL CIAGSHAVNGVARIHSEILK KTIKDFYELEPHKFNKNTNGITPRRWLVLCNPGLAEIIAERIGEEYISDLDQLR KL LSYVDDEAFIRDVAKVKQENKLF AAYLEREYKVHINPSLFDVQVKRIH EYKRLQLNCLHVI TLYNRIKKEPNKFVVPRTVMI GGKAAAPGYHMAKMI I KLIT AIGDVVNHDPPVGDRLRVI FLENYRVSLAEKVIPAADLSEQISTAGTEASGTG NMKFMNLGALTIGTMDGANVEMAEAEAGEENFFIFGMRVEDVDRLDQRGYN AQEYYDRIPELRQII EQLSSGFFSPKQPDLPKDIVNMLMHHDRFKVFADYEEY VKCQERVSA LYKNPREWTRMVIRNIATSGKFSSDRTIAQYARETWGVPEPSRQR LPAPDEKIP
57	Rabbit ALDOA (Fructose- Bisphosphate Aldolase A)	HSHPALTPEQKELSDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTENR RFYRQLLLTADDRVNPCIGGVILFHETLYQKADGRPPFPQV IKS KGGVVGIKV DKGVVPLAGTNGETTQGLDGLSERCAQYKKGADGAFKWR CVLKI GEHTPS ALAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAV YKALSDHHIYLEGTL LKPNMVTGPHACTQKYSHEEIAMATV TALARRTVPPAV TGVTFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQASALKAWGGK KENLKAQEYVVKRALANS LACQGYTPSGQAGAAASESLFISNHAY
58	Rabbit MYL1 (Myosin Light Chain, Skeletal Muscle Isoform)	APKKDVKKPAAAAAPAPAPAPAPAPAKPKEEKIDLSAIKIEFSKEQQDEFK EAFLLYDRGTGDSKITLSQVGDVLRALGTNP TNAEVKKVLGNPSNEEMNAKKI EFEQFLPMLQAI SNNKQGTYEDFVEGLRVFDKEGNGTVMGAE LRHVLATL GEKMKEEVEALMAGQEDSNGCINYEAFVKHIMS I
59	Rabbit G3P (Glyceraldehyde- 3-Phosphate Dehydrogenase)	MVKGVNGFGRIGRLVTRAAFNSGKVDVVAINDPFI DLHYMVYMFQYDSTH GKPHGTVKAENGLVINGKAITI PQERDPANI KWGDAGAEYVVESTGVFTTM EKAGAH LKGGAKRVI ISAPSADAPMFVMGVNHEKYDNSLKIVSNASCTTNCL APLAKVIHDHFGIVEGLMTTVHAI TATQKTVDGPSGLWRDGRGAAQNI IPAS TGAAKAVGKVIPELNGLKTGMAFRVPTPNVSVVDLTCRLEKAAKYDDIKKV VKQASEGLPKGILGYTEDQVVSDFNSATHSS TFDAGAGIALNDHFVKLISWY DNEFGYSNRVVDLMVHMASKE
60	Rabbit TPM1 (Tropomyosin Alpha-1 Chain)	MDAIKKMQMLKLDKENALDRAEQAEADKKAEDRSKQLEDELVSLQKKL KGTEDEL DKYSEALKDAQEKL ELAEKKATDAEADVASLNRR IQLVEEELDRA QERLATALQKLEEAKEADESERGMKVI ESRAQKDDEKMETQEIQ LKEAKHI AEDADRKYEEVARKLVI IESDLERAEEAEELS EGKCAELEELKTVINNLKSL EAQAEKYSQEDKYEEI KVLSDKLKEAETRAEFAERSVTKLEKS IDDLDEL YAQKLKYKAISEELDHALNDMTSI

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
61	Rabbit Myosin- Binding protein C (fast-type)	EAKPAARKAPKGDAPKEEPPKEAAEPPKEAPPEDQSPTAEPTGIFLKCPD SVSVETGKDAVIMAKLNGKELPAKPTIKWFKGKWLGLSGSKSGARFSFKETHD AASNVSYLELHIGKVLGDRGDYRLEVAKADVCDSCSFNIDVEAPRQDSAGQ SLESFKRAGDAKPDGAGELDFSGLLKKREVVEEEKKKKDDDLGIPPEIWE LLKGAKKSEYERIAFYQGITDLRGMLKRLKAKVEVKKSAFTKKLDPAYQ VDRGNKVKLVVEISDPDLPLKWFKNQGEIKPSLKYYFENVGKKRILTIKCTL ADDAAYEVVNDEKCFTELFKPEPPVLIVTPLEDQQVFVGDRVEMSVSEVSEE GAQVMWMDGVELTREGSFKSNYRKKDKGRHILYSEVNLEDRGHYQVM TNGGQCEAELIVEEKQLEVLQDIADLTVKASEQAMFKCEVSDEKVTGKWKY NGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDYTFVPDGYALSLSAKLNFL EIKVEYVPKQEPPIHLDCSGKTSDNAIVVAGNKLRLDVSITGEPPPVAWTM KGDEVFTVTEGRVRIEQRPDSSSFVIESAERGDEGRYTIKVTNPAGEDVATVM LRVVDVDPPEAVRVTSGEDWAILVWDPPKYDGGQPVTYLLERKKKGSQ RWMKLNFEVFTETTYESTKMI EGILYEMRVFAVNAIGVSQPSMNTKPFMPIAP TSAPQHLTVEDVTDTTTLKWRPPDRIGAGIDGYLVEYCEGSDEWIPANK EPVERCGFTVKGLPTGAKILFRVIGVNIAGRSEPA TLVQPV TIREIVEQPKIRLP RHLRQTYVRKVGEHINLVI PFQGGKPRPQVWVTGGAPVDTSRVHVRTSDFT VFFVREAARSDSGEYELSVQIENMKDTATVRI RVVEKAGPAENVMVKEVWG TNALVEWQPPKDNNGSEITGYLVQKADKKTMEWFTVYEHNRHTSCTVSDLI VGNEYFRVYTENVCGLSDSPGVSKNHTAHILKTGITFKPFYEKHEDFRSPPKF LTPLIDRVVAGYAAALNCAVRGHPKPKVVMKNKVEIHEDPKFLMTNHQ GVLTLNIRRSPFDAGTYS CRAFTNELGEALAECLDVRVPQ
62	Rabbit G1T4T6 (Myomesin 2)	MSLVAVPFYQKRHKHFDQSYRNIQT KYLMDEYTAKKRASTQMS THKASTQ MATHKSALQKASSQTSSGGTTCRLCAKRLSAAEEEEQESKNRYSRLVASIGE AKRQRFLELAQLEESVHLARSQARDVVDKYALQQMVEDKMAWERHVFEE RLSRAPEILLVQLRSHTVNEKMSVKLCFTVQGFPTPVVQWYKDGSLICQAGEP GKYRIDSKYGVHTLEISRASFDDTATYS AVATNAHGQVSTNAAVVVRIRG DEEPFYSVGLPIGYVLVKESKWTMFFGEGQASLSFSLNKDDGLYTLRIVS RGGVSDHS AFLFVRDADPLVTGAPGAPMDLQCHDANRDYIVTWKPPNTTT ESPVIGYFVDRCEVGTNNWVQCNDAPVICKYPTVGLYEGRSYIFRVRVANS AGISRP SRVSDAVAALDPADLRRLQAIHLEGEKEVVIYQDDLEGDVQVGPPT NVHASEISRNYVVL SWPEPPVPRGKEPLMYFIEKSVVSGSGWQRVNAQTAVRS PRYAVFDLAEGKPYVFRVL SANKHGLSDPSEITSPIQAQDTIAVPSAPGRVLAS RNTKTSVVVQWDRPKHEEDLLGYVDDCCVAGTNAWEP CNHKPIGYNRFVV HGLTTGEQYIFRVKAVNALGTSENSQSEVIVKQAALTVPSPHPYGVTLNCD GHSMTLGWKVPKFSGGSPILGYVVDKREVHHKNWHEVNSSPIQERILTVEGL TEGSFYEFKIAATNLAGIGQPSDPSHFKECAWTAPEPGPAYDLTFCEVRDTSL VMLWKAPVYSGSPVSGYFVDFKEEDSGEWSTVNEAATPNRYLKVSDLHQ RTYVFRVRAVNASGVGKPSDVSEPVLEARPGTKEISAGVDEEGNIYLGFD QEMTDASHFTWCKSYEEIADPERFHVETKGDHSLYFKNPKADLGTYSVSV SDTDGVSSSFVLDEEELERLLALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQT EPLSPDASFRFVINDREVSDSETHRMKCDKASGIEMVMDFRTIENEGTYTVQI HDGKAKNQSSVLIGDAFRKVL EEA EFQRKEFLRKQGP HFAEYLHWDVTEEC EVRLICKVANTKKETVFKWLKDDVLYETETPPDLEKGVCELLIPKLSKKDHD EYKATLKD ERGQDASILEIAGKVYEDMILAMSRVCGMSASPLKILCTPEGIRL QC FMKYFTEEMRVNWHKDAKISSSEHMRI GGSSEDMAWLQICEPTEKD K GK YTFEIFDGKDNHQRSLDLSGQAFDEAFAEFQRLKAAAF AEKNRGKVI GGLPD VVTIMEGKTLNLTCTVFGNPDPEVVWFKNDQDIELSEHFSVKVEQAKYVSLTI KGVTSSEDSGKYSINVKNKYGGEKIDVTVSVYKHGEAIP EVAPPQQA KPKLIPA SASATAP
63	Rabbit ENOB (Beta-enolase)	AMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALERLDGDK SRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVKFMIELDGTENKSKF GANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNHDLVLPVAFNVINGGSH AGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVIKAKYKGDATNVGD EGGFAPNILENNEALELLKTAIQAGYPDKVVI GMDVAASEFHRNGYDLDF KSPDDPARHITGQKLGEYKSF IKNYPVVSIEDPFDQDDWATWTSFLSGVDIQI VGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQSNWGV MVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMRIEALGD KAVFAGRKFRNPKAK
64	Rabbit MLRS (Myosin Regulatory Light Chain 2, Skeletal Muscle Isoform Type 2)	APKKAKRRAAEEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDLRDT FAAMGRLNVKNEELDAMMKEASGPINFTVFLTMFGKELKGADPEDVITGAF KVLDPGKGTIKKQFLEELLTTQCDRFSQE EIKNMWAAFPDPVGGNVYDKNI CYVITHGDAKDQE

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
65	Truncation of Rabbit G1SJN7 (Myosin 4)	IFSISDNAYQFMLTDRENQSILITGESGAGKTVNTRVIQYFATIAITGDKKKEE PTPGKMQGTLEDQII SANPLLEAFGNKATVRNDNSSRFGKFI RHFGATGKLAS ADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITNPNYDFAFVS QGEITVPSIDDOEELMATDSAVDILGFTSDEKVAIYKLTGAVMHYGNLKFQK KQREEQAEPDGTVEADKAAYLTSLSNADLLKALCYPRVKVGNEYVTKGQTV QQVYNAV GALAKAIYEKMFLLWVTRINQQLDTKQPRQYFIGVLDIAGFEIFD FNSLEQLCINFNTNEKLQOFFNHHMFVLEQEYKKEGIEWEFIDFGMDLAACIE LIEKPMGIFSILEEEMFPKATDTSFKNKLYEQHLGKSNNFQKPKPAKGKAEA HFSLVHYAGTVDYNI TGWLDKNKDPLNETVVGLYQKSGMKTALFLFSGAQT GEEGGGGGKGGKKGSSFTQVSALFRENLNKMLTNLRSTHPPHVRCLIPNE TKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPSRILYADFKQRYKVLNASAI PEGQFIDSKKASEKLKLSIDVDHTQYKFGHTKVFVKAGLLGLLEEMRDEKLA QLITRTQAI CRGYLTRVEFKMMERRESIFCIQYNIRAFMNVKHWPMKLYF KIKPLLKSAETEKEMANMKKEEFKTKELAKSEAKRKELEEKMAVMQEK DLQLQVQAEADGLADAEERCDQLIKTKIQLEAKI KEVTERAEDEEINAEFTA KKRLEDECSSELKKDIDDLLETLAKVEKEKHATENKVNLTTEEMAGLDENIA KLTKKKALQEAHQOTLDDLQAEEDKVNTLTAKTKLEQQVDDLEGSLEQE KKLRMDLERAKRKEGLDLKLAQESTMDIENDKQQLDEKLKKKEFEMTNLQS KIEDEQALAMQLQKKIKELQARIEELEEIEAERASRAKAEKQSDLSRELEEI SERLEEAGGATSAQIEMNKKREAEFQKMRDLLEATLQHEATAAALRKKHA DSVAELGEQIDNLQRVKQKLEKEKSELKMEIDDLAGNMTVSKAGNLEK CRTLEDQLSEVKTKEEHQRLINELSAQKARLHTESGEFSRQLEKDMVQS LSRGKQAFQTQIEELKRQLEEBETKAKSALAHALQSSRHDCDLLREQYEEEQE AKAELQRAMSKANSEVSQWRTKYETDAIQRTTEEEAKKKLAQRLQDAEEH VEAVNSKASLEKTKQRLQNE
66	Truncation of Rabbit G1SJN7 (Myosin 4)	IFYQIMSNKKPELIEMLLITNPNYDFAFVSQGEITVPSIDDOEELMATDSAVDIL GFTSDEKVAIYKLTGAVMHYGNLKFQKQKQREEQAEPDGTVEADKAAYLTS NSADLLKALCYPRVKVGNEYVTKGQTVQQVYNAV GALAKAIYEKMFLLW VTRINQQLDTKQPRQYFIGVLDIAGFEIFD FNSLEQLCINFNTNEKLQOFFNHHM FVLEQEYKKEGIEWEFIDFGMDLAACIE LIEKPMGIFSILEEEMFPKATDTSF KNKLYEQHLGKSNNFQKPKPAKGKAEAHFSLVHYAGTVDYNI TGWLDKNK DPLNETVVGLYQKSGMKTALFLFSGAQTGEEGGGGGKGGKKGSSFTQVS ALFRENLNKMLTNLRSTHPPHVRCLIPNETKTPGAMEHELVLHQLRCNGVLE GIRICRKGFPSRILYADFKQRYKVLNASAIPEGQFIDSKKASEKLKLSIDVDHT QYKFGHTKVFVKAGLLGLLEEMRDEKLAQLITRTQAI CRGYLTRVEFKMM RRESIFCIQYNIRAFMNVKHWPMKLYFKIKPLLKSAETEKEMANMKKEEFK TKEELAKSEAKRKELEEKMAVMQEKNDLQLQVQAEADGLADAEERCDQLI KTKIQLEAKI KEVTERAEDEEINAEFTAKKRLEDECSSELKKDIDDLLETLAK VEKEKHATENKVNLTTEEMAGLDENIAKLTKKKALQEAHQOTLDDLQAE DKVNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKEGLDLKLAQES TMDIENDKQQLDEKLKKKEFEMTNLQSKIEDEQALAMQLQKKIKELQARIEE LEEEIEAERASRAKAEKQSDLSRELEEISERLEEAGGATSAQIEMNKKREAEF QKMRDLLEATLQHEATAAALRKKHADSVAEELGEQIDNLQRVKQKLEKEK ELKMEIDDLAGNMTVSKAGNLEKMCRTLEDQLSEVKTKEEHQRLINEL SAQKARLHTESGEFSRQLEKDMVQSLSRGKQAFQTQIEELKRQLEEBETKAK SALAHALQSSRHDCDLLREQYEEEQEAKAELQRAMSKANSEVSQWRTKYET DAIQRTTEEEAKKKLAQRLQDAEEHVEAVNSKASLEKTKQRLQNEVEDL MIDVERSNAACAALDK
67	Truncation of Rabbit G1SJN7 (Myosin 4)	TLDDLQAEEDKVNTLTAKTKLEQQVDDLEGSLEQEKKLRMDLERAKRKE GDLLKLAQESTMDIENDKQQLDEKLKKKEFEMTNLQSKIEDEQALAMQLQKKI KELQARIEELEEIEAERASRAKAEKQSDLSRELEEISERLEEAGGATSAQIE MNKKREAEFQKMRDLLEATLQHEATAAALRKKHADSVAEELGEQIDNLQRV KQKLEKEKSELKMEIDDLAGNMTVSKAGNLEKMCRTLEDQLSEVKTKEE EHQRLINELSAQKARLHTESGEFSRQLEKDMVQSLSRGKQAFQTQIEELKR QLEEBETKAKSALAHALQSSRHDCDLLREQYEEEQEAKAELQRAMSKANSEVS QWRTKYETDAIQRTTEEEAKKKLAQRLQDAEEHVEAVNSKASLEKTKQ RLQNEVEDLMIDVERSNAACAALDKQRFKVLAEWKHKYEETQAELEASQ KESRSLSTEVFKVNAYEESLDHLETLKRENKNLQOEISDLTEQIAESGKHIHE LEKVKQIDQEKSELQALAEAGSLEHEEGKILRIQLELNQVKSEIDRKIAEK DEEIDQLKRNLHVRVVEQMSTLDAEIRSRNDALRIKKKMEGDLNEMEIQLNH ANRQAAEAIKNLRNTQGILKDTQLHLDDAVRGQDDLKEQLA
68	Truncation of Rabbit G1SJN7 (Myosin 4)	LAQESTMDIENDKQQLDEKLKKKEFEMTNLQSKIEDEQALAMQLQKKIKELQ ARIEELEEIEAERASRAKAEKQSDLSRELEEISERLEEAGGATSAQIEMNKK REAEFQKMRDLLEATLQHEATAAALRKKHADSVAEELGEQIDNLQRVKQK EKEKSELKMEIDDLAGNMTVSKAGNLEKMCRTLEDQLSEVKTKEEHQ RLINELSAQKARLHTESGEFSRQLEKDMVQSLSRGKQAFQTQIEELKRQLEE ETKAKSALAHALQSSRHDCDLLREQYEEEQEAKAELQRAMSKANSEVSQW RTKYETDAIQRTTEEEAKKKLAQRLQDAEEHVEAVNSKASLEKTKQRLQNE

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		VEDLMIDVERSNAACALDDKKQRNFDKVLAEWKHKYEETQAELEASQKESR SLSTEVFKVKNAYEESLDHLETLKRENKLNQQEISDLTEQIAESGKHIHELEKV KKQIDQEKSELQAALAEAEAGSLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEI DQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKKMEGDLNEMEIQLNHANR QAABAIKNLRNTQGILKDTQLHLDDAVRGQDDLKEQLAMVERRANLMQAEI EELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTKKKLETDISQIQG EMEDIVQEARNAEKAKKAITDAAMMAEELKKEQDTSAHLERMKKNMEQT VKDLQHRLEAEQLALGGKKQIQKLEARVRELENEVESQKRNVEAVKGL RKHERRVKELTYQTEDRKNVLRQLQDLVDKLQSKVKAYKRQAEAEAEQSN
69	Truncation of Rabbit MYH4 (Myosin 4)	VFVADPKESFVKATVQSREGGKVTAKTEAGATVTVKEDQVFPMPNPYDKI EDMAMMTHLHEPAVLYNLKERYAAWMIYTYSGLFCVTVNPKWLPVYNAE VVTAYRGKKRQEAPPHIFISIDNAYQFMLTDRENQSILITGESGAGKTVNTKR VIQYFATIAVTGDKKKEEATSGKMGTLEDQIISANPLLEAFGNAKTVRNDNS SRFGKFIRIHFGTGTGLASADIETYLEKSRVTFQLKAERSYHIFYQIMSNKKP DLIEMLLITTPYDYAFVSQGEITVPSIDDEELMATDSASIDILGFTSDERSIY KLTGAVMHYGNMKFKQKQREEQAEPDGTVEADKAAYLQSLNSADLLKALC YPRVKVGNIEYTKGQTVQOVYNAV GALAKAVYEKMFWMVTRINQQLDT KQPRQYFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQFPNHHMFVLEQEEYKK EGIEWEFIDFGMDLAACIELIEKPMGIFSI LEEECMPFKATDTSFKNKLYEQHL GKSNNFQKPKPAKGKVEAHFSLVHYAGTVDYNITGWLDKNKDPLNETVVGL YQKSAMKTLAFLFTGTAAAEAEAGGGKGGKGGSSPQTVSALPRENLNKL TNLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIIRICRKGFP SRILYADFQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVFFKA GLGLGLEEMRDDKLAQLITRTQAMCRGFLARVEYKKMVERRESIFCIQYNIR AFMNVKHWPMKLYFKIKPLLSAETEKEMANMKEEFEKTESLAKAEAK RKELEEKMVALMQEKNDLQLQVQAEADSLADAEERCQDLIKTKIQLEAKIKE VTERAEDEEEINAELTAKKRKLEDECESELKKDIDDELTLAKVEKEKHATENK VKNLTEEMAGLDETI AKLTKEKKALQEAHQTLDDDLQAEEDKVNTLTAKT KLEQQVDDLEGSLEQEKIRMDLERAKRKLEGLDKLAQESTMDIENDKQQL DEKLKKKEFEMS NLQSKIEDEQALAMQLQKKIKELQARIEELEEEIEAERASR AKAEKQRSDLRELEEI SERLEEAGGATSAQIEMNKKREAEFQKMRDLDEA TLQHEATAATLRKKHADSVAELGEQIDNLQRVKQKLEKEKSELKMEIDDLAS NMETVSKAKGNLEKMCRTL
70	Truncation of Rabbit Nebulin	TLVSDVDYRNYLHQWTCMPDQNDVIQAKKAYELQSDAVYKADLEWLRGIG WMPSDSVSVNHAKHAADIFSEKKYRTKIEITLNFPTVDDRVDYVTAKQS GEILDDIKYRKDWNSIKSKYTLTETPLLHTVQEAARILDQYLYKESWEKQKATGYI LPPDAVPFVHAHSSSDVQSELKYKAEHVQQRGHYVGVPTMRDDPKLVWFE HAGQIQNDRLYKEDYHKTAKINIPADMVSVVAAKEGQNLVSDIDYRNYLH QWTCHPDQNDVIQARKAYDLQSDNVYKADLEWLRGIGWIPLDSVDHVRVT KNQEMVNQIKYKDALDNYPNFTSVVDPPIVLAKINAVNQSDVKYKETFN KAKGKYTFSPDTPHISHSKDMGKLYSTILYKGAWEGTKAYGYTLDERYIP IVGAKHADFNSELKYKETYEKQKGHYLAGKVISEFPVGVHCLDFQKMRSVLNY RRDYENTKANVHIPNDMMNHVLAKKCQYILSDLEYRHYFHQWTSLPDEPKV LLARNAQEILSDNVYKDDLNLWLGKIGCYVVDTPQILHAKKSYDLQSQ LQYTAAKGLENQNYNLVTDTPLYVTALQSGINASEVKYKENYHQIKDYTT VLETVDYDRTRNLKDLYSNLYKEAWNKVATSYILPTSTLSLTHAKNQKHLASHIK YREEYEKFKALYTLPKSVEDDPNTARCLRVGKLNIDRLYRSVYEKNMKI HIHVPDMVEMVTAKDTQKKVSEIDYRLHLHEWICHPDQVNSHVRKVTDQIS DIVYKDDLNLWLGIGCYVVDTPAILHAKHAYDLRNDIKYKAHMMKTRNDYKLV TDPVYVQAVKSGKQLSDAVYHHDYVHVSVRGKVAPTTKTVDLERALHAYK LQSENLYKTSRLTLP TG YRLPGDTPHFKHNDIRYMSYFKYKEVYEH IKANGYTLPNDVPFVNVRVKNKVTSERLYRELYHKLKDKIHTPDTP EIRQVKKTQEA VSELIYKSDFFPKMGHMSLPYTPHVLHCRYVGDITS DIKYKEDLQILKGLGCFLYDTPDMVRSRHLRKLWSNLYLTDNARKMRD KYKVLDTPYRKVQELKTHLSELVYRASGRKQKSIFTSVPDTPDLLRAKRGQKLQSQYL VVELATKERPHHHAGNQTTALKHARDVKDMVSEKKYKVQYEKMKDKYTP IPDTPILIRAKNAYWNASDLRYKETFPQTKGKYHTVKDALDIVYHRKVTD DISKVYKENYMRHLGIWRSIPDRPEHFHHRAVTDAVSDIKYKEDLTWLGIGCYAYDTPDF TLAEKNKTLYSKYKYKEVFERTKSDPKYVADCPINRHFKYATQLMNEKKYR ADYEQRDKYQLVDEPRHLLAKTAGDQISQIKYRKQYKSKDKFTSIVDTP EHLRTTKVKNQISDILYKLEYNKAKPRGYTTIHDTPMLLHVRKVKEVSDLK YKEVYQRNKSNTIKPDAVHIKAAKDAYKVTNLDYKKQYEANKAHRWT PDRPDFIQAAKSSSQSDFEYKLDREPLKGCKLSVTDDKDMVLALKNSLIESD LKYKEKHVKERGSCHAVPDTPQILAKTVSNLV
71	Truncation of Rabbit Nebulin	EHTRNVNQIQSDNVYKDEYNNFFKGIWIPIGSLEVEKVKKAGDALNEKKYR QHPDTIKFTSVPSMGMVLAQHNTKQLSDNLKYVEGEKVHKYTMDEVPVQ FIQARVNAFNLSDAHAKDKKTLAKGYDLRPDAIPIVAAKSSRNIA SDCKYKEAYEKAGKQIGFLSLQDDPKLVHYMNVAKIQSDREYRKG YEASKTRYHTPLDMLSVTAAKKSQEVATNANYKQPFHNYTLLPDALP VEHSRNAMQIQSDN

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		LYKSDFTGWMKGIGWIPQISLEVEKAKKAGEILSEKRYRQHPEKLKFTYAMD TMEQALNKANKLNMDKRLYTKKWDKDTTIHVMPDTPDILLSRVNQITMSD KLYRAGWEEEEKKGYDLRPDAISIKAAKASRDIASDYKYKQAYEQAKGKHI GFRSLEDDPKLVHFMQVAKMQSDREYKAYEKSSTSFHTPVDMLSVVAAK KSQEVATNANYRNVIHTYNMLPDAMGFELAKNMMQIQSDNQYKADYADFM KGIGWLPLGSLAEKKNKAMEIIEKKYRQHPDTLKYSTLMDSMNMVLAQN NAKIMNEHLYKQAWHEADKTKVHIMPDIPOIVLAKANAINMSDKLYKLSMEE AKKKGYDLRTDAIPIKAAKASRDIASDYKYKYNIEKQKGMVGFRLSLEDDP KLVHSMQVAKMQSDREYKKNYEKTKTSYHTPADMLSVTAAKDAQANITNT NYKHLIHKYIILLPDAMSLELSRNMNRQSDNEYKQDYNEWYKGLGWSFAGS LDVEKAKKATEYASDQYRQHPNSNFQPKKLTDSMDMVLAKQNAHTMKNHL YTIWNKDKTKIHVMPDTPDILQAKQNQTLFSQKFKYKLGWEEALKKGVDLP VDAISVQLAKASRDIASDFKYKQGYRQGLGHHVGFRLQDDPKLVLSMNVA KMQSERYKKDFEKWTKFSSPVDMLGVVLAKKQCALVSDVDYKKNYLHQ WTCLPDQNDVIHAKKAYELQSENMYKSDLLEWLKGIWSPGLSLEAEKNKRA SEILSEKKYRQPPDNTNKTSPIDAMDIVLAKTNAKNRSDRLYREAWDKDKTQ VHIMPDTPDIILAKANLINTSDKLYRMGYEELKKKGVDLPVDAIPIKAAKASR EIASDYKYKEGFRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFE KWKTKFSSPVDMLGVVLAKKQCLVSDIDYKKNYLHQWTCLPDQNDVTHAR HAYDLQSDNLYKSDLQWLKGIWIPGTSIEDEKNKRATQILSDHVYRQHPDK PKFSLMDSIPMVLAKNNATNMNHLRYTEAWDKDKTTVHIMPDTPEVLLAK QNQVNCSEKLYKLGLEETKRKGDMRVDAIPIKAAKASRDIAEFKYKEGYR KQLGHHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWTKFSSPVDML LGVVLAKKQCALVSDVDYKKNYLHQWTCLPDQNDVMHARQAYDLQSDNLY KSDLQWLRGIGWVPIGSLDVEKC
72	Truncation of Rabbit Nebulin	DVTHARHAYDLQSDNLYKSDLQWLKGIWIPGTSIEDEKNKRATQILSDHVY RQHPDKFKFSLMDSIPMVLAKNNATNMNHLRYTEAWDKDKTTVHIMPDTPE EVLLAKQNVNCSKLYKLGLEETKRKGDMRVDAIPIKAAKASRDIAEFKY YKEGYRQGLGHHIGARAIHDDPKMMWSMHVAKIQSDREYKKDFEKWTKF SSPVDMLGVVLAKKQCALVSDVDYKKNYLHQWTCLPDQNDVMHARQAYDL QSDNLYKSDLQWLRGIGWVPIGSLDVEKCKKATEILSDKIYRQPPDRFKFTSV TDSLQVLAKNNAINMNKRLYTEAWDKDKTQIHIMPDTPEIMLARMNKINYS ESLYKLANEAAKKGYDLRSDAIPIVAASRDIAEDYKYKGYRQGLGHHI GARDIKDDPKMMWSMHVAKIQSDREYKKDFEKWTKFSSPVDMLGVVLAK KCQTLVSDIDYKKNYLHQWTCLPDQNDVMHARQAYDLQSDNLYKSDLQWLR GIGWVPIGSLVAKCKRAAEILSDNIYRQPPDKLFTSVSDSLQVLAKNNAIN MNKRLYTEAWDKDKTQIHIMPDTPEIMLARQNKINYSKLYKLANEAAKKG YDLRSDAIPIVAASRDIAEDYKYKGYRQGLGHHIGARDIKDDPKMMWS MHVAKIQSDREYKKDFEKWTKFSSPVDMLGVVLAKKQCLVSDIDYKKNYL HQWTCLPDQNDVMHARQAYDLQSDNLYKSDLQWLRGIGWVPIGSLDLVVC KRAAEILSDNIYRQSPDKLFTSVTDSLQVLAKNNAINMNKRLYTEAWDKD KTQIHMPDTPPEIMLARQNMNYSNFYRQAMEAAKKEGYDLRSDAIPIVA AKASRDIAEDYKYKEAYRQGLGHHIGARAIHDDPKMMWSLHIAKQSDREY KKEFEKYKTRYSSPVDMLGIVLAKKQCLVSDVDYKKNLLHEWTCLPDQNDV IQARKAYDLQSDNLYKSDLQWLRGIGWVPIGSLVVKAKRATEILSDNLYRT RPDTLKFTSVTDSLQVLAKNNAINMNKRLYTEAWENDKKTIVHIMPDTPEIM LAKLNKINYSKLYKLALAEESKKEGYDLRVDAIPIQAASRDIAEDYKYKE GYRQGLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKKDFEKWTKFSSP VMDLGVVLAKKQCALVSDIDYKKNLHEWTCLPDQNDVIQARKAYDLQSDAV YKADLEWLRGIGWVPIGSVEVEKVRAGEILSDRKYRQPADKLFTSITDTP IVLAKSNALTMSKHLYTEAWADKTSIHVMPDTPPEILLAKSNSANISQKLYTK GWDESKMKDYDLKADAIPISAKASRDIAEDYKYKEAYEKQKGGHIGAQSIE DDPKIMCAIHA
73	Truncation of Rabbit TTN	DKRETSRPNWAQVSANVPITSCSVEKLEIGHEYQFRI CAENKYGVGDPVFTPEP AIAKNPYDPPGRCDPPVISNVTKDHTVSWKPPADDGGSPI TGYLEKRETHA VNWTKVNRKPIVIERIKATGLQEGTEYEFVTAINKAGPGKPSDASKAVYAQ DPLYPGPPAPFPKYDTRSSVSLSWGKPAYDGGSPITGYLVEVKRADTNDW VRCNLPLQKLQKTRFEVTLGMENTEYQFRVYAVNKGYSDDPSDVPDKHCPKD ILIPPEGELDADLRKTLILRAGVTMLRYVVPVKGRPPPKITWSKPNVNLRRIGL DIKSTDFDTFLRCENVNKYDAGKYILTLENSCGKKEYTIVVKVLDTPGPPNV TVKEISRDSAYITWDPPIVDGGSPITINVVVEKRDAERKSWSVTTECSKTSFRV SNLEEGKSYFRVFAENYEGIDPGETRDVAKASETPGPVVDLKVLTVTKSSC NIGWKPRSDGGSRTGYVVDFTLEENKQVRVMKSLSLQYSTKDLNEGKQY TFRVSAENENGEGTPSEITVVAKDDVAPDLDLKDLPLCYLAKENSFRLKI PIKGPAPSVTWKKGEDPLATDTRVSVSSAVNTTLVVYDCQKSDAGKYTIT LKNVAGTKEGTLISKVVGKPGIPTGPIKFDEVTAETLKWGPPKDDGGSEITN YILEKRDVNNKWWTCASAVQKTTFRVTRLHEGMEYTFRVSAENKYGVGEG LKSEPIVAKHPFDVPDAPPPNIVDVRHDSVSLTWTDPRTKGGSPITGYHIEFK ERNSLLWKRANKTPIRMKDFKVTGLTEGLEYEFRVMAINLAGVGKPSLSEPE VVALDPIDPPGKPEVINVTRNSVTLIWTEPKYDGGHKLTYIVEKRDLPSTW

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		MKANHINVPDCAFTVTDLVEGGKYEFRIRAKNTAGAISAPSESTGTIICKDEYE APTIVLDPITKDGTLTIKAGDTIVLNAISILGKPLPKSSWSKAGDIRPSDITQITS TPTSSMLTVKYASRKDAGEYITATNPFGTKEEHVRVTVLDVPGPPGPIEISNV SAEKATLTWTPPLEDGGSPIKSYVLEKRETSRLLTWTVVAEDIQSCRHVVTCLI QGNEYLFRVSAVNHYGKGEVQSEPVKMVDRFGPPGPPGKPEVSNVTKNTA TVSWKRPTDDGGSEITGYVVERREKKGLRWVRATKTPVSDLRCKVTGLQEG NTYEFRVSAENRAGIGPPSDASNYVLMKDVAYPGPPSNARVTDTTKKSASL AWGKPHYDGGLEITGYVVEHQKVGDETVVKDTTGPALRI TEFVVPDLHTKE KYNFRISAINDAGVGEPAVDPDVEIVEREMAPDFELDAELRRTLVRAGLSIRI FVPIKGRPAPEVTTWKDDINLKTRANIENTESFTLLI IPECNRYDTGKFVMTIEN PAGKKS GFVNVRVLDTPGPVLNLRPTDITKDSVTLHWDPLIDGGSRTITNYIV EKREATRKSYSYTVTTKCHKCT
74	Truncation of Rabbit KCRM (Creatine Kinase M- Type)	DKETPSGFTLDDVIQTGVNDPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDR HGGFKPTDKHKTDLNHENLKGDDLDPHYVLSRVRTGRS IKGYTLPPhCSR GERRAVEKLSVEALNSLTGEFGKGYPLKSMTEQEQQQLIDDHFLDPKPVSP LLASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVI SMEKGGNMKEVFR RFCVGLQKIEEIFKKAGHPFMWNEHLGYVLTCPSNLTGLRGGVHVLAHLS KHPKFEEILTRLRLQKRGTTGGVDTAAVGSVFDISNADRLGSSEVEQVQLVVD GVKLMVEMEKLEKGQS IDDMIPAQK
75	Truncation of Rabbit GIU4H8 (Actinin Alpha 3)	GGGGGGEYMEQEEDWDRDLLDPAWEKQQRKTFTAWCNSHLRKAGTQIEN IEEDFRNLKLMLELVISGERLPRPDGKMRPHKIANVNKALDFIASKGVKL VSI GAEEIVDGNLKM TLMGIWTIILRFAIQDISVEETS AKEGLLLWCQRKTAPY RNVNVQNFHTSWKDGLALCALIHRHRPDLIDYAKLRKDDPIGNLNTAFEVAE KYLDIPKMLDAEDIVNTPKPDKAIMTYVSCFYHAFAGAEQAETAANRIKCV LAVNQENKLMEEYEKLASELLEWIRRTVPWLENRVGEPSSAMQRKLEDF RDYRRLHKPPRVQEKQLEINFNTLQTKLRLSHRPAPMPSEGLVSDIANAW RGLEQVEKGYEDWLLSEIRRLQRLQHLAEKFQOKASLHETWTRGKEDMLTQ RDYESASLQEVRRALLRRHEAFESDLAAHQDRVEHIAALAQELNELDYEHAAS VNSRCQAI CDQWDLNLGTLTQKRRLDALERMEKLETTIDQLQLEFARRAAPENN WLDGAI EDLQDVVLVHVSVEETQSLLTAHDQFKATLPEADRERGAILGIQGEIQ KICQTYGLRPSSTNPYINLSPQDINTKWDTVRKLVP SRDQTLQEELARQQMNE RLRRQFAAQANAVGPWIAKVEEVGRLAAGLAGSLEEQMAGLRQEQNIIN YKSNIDRLEGDHQLLQESLVFDNKHTVYSMEHIRVGWEQLLTSIARTINEVEN QVLTRDAKGLSQEQLNEFRASFNFDRKRNGMMEPDDFRACLISMYDLGEV EFARIMTMVDPAAGVVTFAFIDFMTR
76	Truncation of Rabbit PYGM (Glycogen Phosphorylase, Muscle Form)	IYYLSLEFYMGRTLQNTMVNLALENACDEATYQLGLDMEELEEIEEDAGLGN GGLGRLAACFLDSMATLGLAAYGYGIRYEFGINQKICGGWQMEEADDWLR YGNPWEKARPEFTLPVHFYGRVEHTSQGAKWVDTPVVLAMPYDTPVPGYR NNVNTMRLWSAKAPDNFNLKDFNVGGYIQAVLDRLNLAENISRLVLPNDNF FEGKELRLKQEYFVVAATLQDIIRRFKSSKFGCRDPVRTNFDAPFDKVAIQLN DTHPSLAIPELMRVLVDLERLDWDKAWETVKTCAYNHTVLPALERWPV HLLETLLPRHLQI IYEINQRFNLNRVAAFPDVRDLRRMSLVEEGAVKRINMA HLCIAGSHAVNGVARIHSEILKKTIFKDFYELEPHKFNKNTNGITPRRWLVLCN PGLAEI IAERIGEEYISDLDQLRKLLSYDDDEAFIRDVAKVKQENKLFPAAYLE REYKVHINPNSLFDVQVKRIHEYKRQLNCLHVI TLYNR IKKEPNKFVVPRTV MIGGKAAPGYHMAKMI IKLITAGDVVNHD PVVGDRLRVIFLENYRVSLAEK VI PAADLSEQI STAGTEASGTGNMKFMLNGALTIGTMDGANVEMAE EAGEEN FFIFGMVRVEDVDRLDQRGYNAQEYDRIPELRQIIEQLSSGFFSPKQPDLFKDI VNMLMHHDRFKVFADYEEYVKCQERVSAKYKNPREWTRMIRNIATSGKFS SDRTIAQYAREIIGVPEPSRQRLPAPDEKIP
77	Truncation of Rabbit ALDOA (Fructose- Bisphosphate Aldolase A)	SHPALTPQKKELSDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTENRRF YRQLLLTADDRVNPCIGGVILFHETLYQKADDRPFPQV IKS KGGVVGIKVDK GVVPLAGTNGETTTQGLDGLSERCAQYKKGADFAKWRVCVLKI GEHTPSAL AIMENANVLARYASICQNGIVPIVEPEILPDGDHDLKRCQYVTEKVLAAVYK ALSDHHIYLEGTLTKPNMVT PGHACTQKYSHEEIAMATVTALRRTPVPAVVG VTFLSGGQSEEEASINLNAINKCPLLKFWALTFSYGRALQASALKAWGGKKE NLKAAQEYVVKRALANS LACQGYKTPSGQAGAAAS
78	Rabbit ALBU (Albumin)	EAHKSEIAHRENDVGEHF IGLVLITFSQYLQKCPYEEHAKLVKEVTDLAKAC VADESAANCDSLSLHDIFGDKICALPSLRDTYGDVADCEKKEPERNECFLHH KDDKPDLPFFARPEADVLCKAFHDDKAFFGHYLYEVARRHPYFYAPELLEY AQKYKAILTECCEAADKGACLT PKLDAL EGKSLISAAQERLRCASIQKFGDRA YKAWALVRLSQRFPKADFTDISKI VTDLT KVHKECCCHGDLLECADDRADLAK YMEHQETISSHLKECCDKPILEKAHCTYGLHNDET PAVAEFEFVEDKDV CKNYEEAKDLFLGKFLYEYSRRHPDYSVVL LRLGKAYEATLKCCCATDDP HACYAKVLDEFQPLVDEPNKLVKQNCELYEQLDYFNQNALLVRYTKKVPQ VSTPTLVEISRSLGKVGSKCKHPEAERLPCVEDYLSVVLNRLCVLHEKTPVS EKVTKCCSESLVDRRPCFSALGPDETVVPKEFNAETFTFHADICTLPETERKIK

TABLE 2-continued

Rabbit Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		KQTALVELVKHKPHATNDQLKTVVGFEPTALLDKCCSAEDKEACFAVEGPKL VESSKATLG

[0041] An animal protein can comprise a cow protein (Table 3). A cow protein can comprise a cow muscle protein. A cow protein can comprise a cow blood protein. A cow protein can comprise a cow albumin protein. A cow protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 79-93.

[0042] A food product may comprise a fragment or a truncation of a cow protein. The food product may include

a fragment or truncation of any one of SEQ ID Nos: 79-93. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 79-93. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 79-93. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 79-93. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 79-93.

TABLE 3

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
79	Cow KCRM (Creatine kinase M-type)	MPFGNTHNKHKLNFKAEEYPDLSKHNNHMAKALTLEIYKKLRDKETPSG FTLDDVIQTGVDNPGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGFKP TDKHKTDLNHENLKGDDLDPNYVLSRVRTGRSICKGYALPPHCSRGERR AVEKLSVEALNSLTGEFGKYYPLKSMTEQEQQLIDHFLFDKPVSPLLL ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVI SMEKGGNMKEVFR RFCVGLQKIEEIFFKAGHPFMNEHLGYVLTCPSNLGTGLRGGVHVKLAH LSKHPKFEEILTRLRLQKRGTTGGVDTAAGSVFVDVSNADRLGSSEVEQVQL VVDGVKLMVEMEKKLEKQGSIDDMI PAQK
80	Cow ACTN3 (Alpha-actinin-3)	MMMVLPQEGLTGEGFPAGGRGGGEYMEQEEDWDRDLLDPAWEKQQR KFTTAWCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPRPDKGM RFHKIANVNKALDFIASKGVKLVSIGAEIIVDGNLKMTLGMIWTIILRFQAIQ DISVEETSAKEGLLLWCQRKTPAPYRNVNVQNFHTSWKDGALCALIHRHR PDLIDYAKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDKAIM TYVSCFYHAFAGAEQAETAANRICKVLAVNQENKLMEEYEKLASELLAW IRRTVPWLENRVGEPSSMSAMQRKLEDFRDYRRLHKPPRVQEKQCLEINFNT LQTKLRLSHRPAPFMPSEGLVSDIANAWRGLEQAEKGYEDWLLSEIRRLQR LQHLAEKFQKASLHEAWTRGKEDMLSQRDYETASLQEVRLALRRHEAFE SDLAHQDRVEHIAALAQELNELDYHEAASVNSRCQAI CDQWDNLGTITQ KRRDALERMEKLETTIDQLQLEFARRAAPFNWLDGAVEDLQDVWLVS VEETQSLVTAHDQFKATLPEADRERGA ILGIQGEIQKI CQTYGLRPSSTNPYI TLTPQDINTKWDTVRKLVPSRDQMLQEELTRQQVNERLRQFAAQANAIG PWIQGVVEVGRLAAGMAGSL EEQ MAGLRQQEQNI INYKSNIDRLLEGDHQ LLQESLVFDNKHTVYSMEHIRVGEQQLTSTIARTINEVENQVLTDRDAKGLS QEQLEFRASFNFHDKRNGMMEPD DFRACLISMGYDLGEVEFARIMTMV DPNAAAGVVTFQAFIDFMTRETAETDTAEQVVASFKILAGDKNYITAEELRR ELPAEQAEYCIRRMAYKYGAGAPAGALDYVAFSSALYGESDL
81	Cow PYGM (Glycogen phosphorylase, muscle form)	MSRPLTDQEKRKQISVRGLAGVENVTELKKNFNRLHFTLVKDRNVATPR DYFALAYTVRDHLVGRWIRTQQHYEYKDPKRIYYLSLEFYIGRTLQNTM VNLALENACDEATYQLGLDMEELIEEDAGLNGGLGRLAACFLDSMAT LGAAAYGYGIRYEFQINQKISGGWQMEEDDWLRVGNPWKARPEFTLP VHFYGRVEHTSQGAKWVDTVQVVLAMPYDTPVPGYRNNVNTMRLWSAK APNDPNLKD FNVGGYI QAVLDRNLAEISRVLYPNNDNFEGKELRLKQEYF VVAATLQDIIRRFKSSKFGCLDPVRTNFDAFPDKVAIQNLNTHPSLAIPELM RILVDQERLEWEKAWEVTVKTCAYTNHTVLP EALERWPVHLIETLLPRHLQ IIYEINQRFNLNRAAFAFGDVRRLRMSLVEEGAVKRIINMAHLCTAGSHAV NGVARIHSEILKKTIFKDFYELPHKFKQNTNGITPRRWLMCNPLGAETIAE RIGEEYIADLDQLRKLKLSYVDDSEFIRDVAKVKQENKLFSAYLEKEYKVHI NPNSLFDIQVKRIHEYKRLNLCLHVI TLYNRIKKEPNKFFVPRVTVMIGGA APGYHMAKMI IKLITAGDVVNHDPVVGDRLRVIFLENYRVSLAEKVI PAA DLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENFFIF GMRVEDVERLDQKGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPDFKDIV

TABLE 3-continued

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		NMLMHDRFKVFADYEEYIKCQERVSALYKNPREWTRMVRINIATSGKFS SDRTIAQYAREIWGVEPTRQRMPAPDEKI
82	Cow A6QLL8 (Fructose- biphosphate aldolase)	MPHQYPALTPEQKKELCDIAHRIVAPGKGILAADESTGSIKRLQSIGTENTE ENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDGRPFQVIKAKGGV VGIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKGADFAKWRCVULKI GEHTPSSLAIMENANVLARYASICQONGIVPIVEPEILPDGDHDLKRCQYVT EKVLAAYKALSDHHIYLEGTLKPNMVTGPHACTQKYSHEEIAMATVTA LRRTPPAVPGITIFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS ALKWGGKKENLKAQEEYVKRALANSLACQGGKYTPSGKAGAAASESLFI SNHAY
83	Cow MYL1 (Myosin light chain 1/3, skeletal muscle isoform)	MAPKKDVKKPAAAAAPAPAPAPAPAPAPAPPPKEEKIDLSAIKIEFSKQQQDE FKEAFLFLDRTGECKITLSQVGDVLRALGNTPTNAEVKKVLGNPSNEEMNA KKIEFEQFLPMLQAI SNKDGQTYEDFVEGLRVFDKENGTVMGAE LRHV LATLGEKMKEEVEALMAGQEDSNGCINYEAFVKHIMSN
84	Cow G3P (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKVGVNGFGRIGRLVTRAAFNNSGKVDIVAINDPFIDLHVMVMFYDST HGKFNQTVKAENKLVINGKAITIFQERDPANIKWGDAGAEYVVESTGVFT TMEKAGAHKGGAKRVIISAPSADAPMFVMGVNHEKYNNTLKI VSNASCT TNCLAPLAKVIHDHFGIVEGLMTTVHAI TATQKTVDGPSGKLWRDGRGAA QNIIPASTGAAKAVGVKVIPELNGKLTGMAPRVPTPNVSVVDLTCLREKPAK YDEIKKVVQKASEGPLKGI LGYTEDQVVSDFNSDTHSSSTFDAGAGIALND HFVKLISWYDNEFGYSNRVVDLMVHMASKE
85	Cow TPM1 (Tropomyosin alpha-1 chain)	MDAIKKMQMLKLDKENALDRAEQAEADKKAEDRSKQLEDELVS LQKK LKATEDELDKYSEALKDAQEKLELAEKATDAEADVASLNRRIQ LVEEEL DRAQERLATALQKLEEAKEADESERGMKVIESRAQKDEEKMEIQEIQLKE AKHIAEDADRKYEEVARKLVIIESDLERAERAE LSEGKCAELEELKTVTN NLKSLAQAEKYSQKEDKYEIEIKVLSDKLEAETRAEFAERSVTKLEKSID DLEDELYAQKLKYKAISEELDHALNDMTSI
86	Cow E1BNV1 (Myosin binding protein C2)	MPEAKPAKKAPAGKDAAAKAPKATPQEAAPAPPTPEAPEDQSPTAEFP TGVFLKKPDSVSVETGKDTIVAKLNGKELPAKPAVKWFKGKWLGLGSKS GARFSFKESHDAASNVTIELHITKVVLGDRGDYRIEVKAKDFCDSCAFNID VEAPRQDSAGQSLESFKRSGEAKSDTAGELDFSGLLKKRQVVEEEKKKK DDDDLGIPEIWEELKGAKKSEYERIAFYQGITDLRGMKLRLKAKAVEVKK SAAPTCKLDPAYQVDRGNKIKLVVEISDPDLPLKWFKNQGEIKPSSKYVFE NVGKKRILTIKCTLADDAAYEVVVKDEKCFTEL FVKEPPVLI VTPLEDQQ VFVGDREMAVEVSEGAQVMWLKDGVELTREDTSFKARYRLKDKGKRHI LIYSEVTMEDKGHYQVMTNGGQCEAELIVEEKQLEVLQDIADLT VKASEQ AVFKCEVSDEKVTGKWKNGVEVRPSKRITISHTGRFHKLVIDDVRPEDEG DYTFVPDGYALSLSAKLKFLKIEIKVEYVPKQEPPIHLDCSGQTSENAIVVVA GNKLRMDVSI TGEPRPVATWMKEDVFTGTEGRVRIEQRGDISSFVIESAER GDEGRYTIKVTPVGEDVASILLKVVDVDPPEAVRVTSVGEDWAILVWEP PKYDGGRPVTGYLLERKKKSGQRWMLNFEVFTTETTYESTNMI EGILYEM RVFAVNVIGVSQPSMNTKPFMPIAPTSEPQHLTVEDVTDTTTTLKWRPDPKI GAGGIDGYLIEYCVESDEWIPANTEPVERCGFTVKNLPTGAKITFRVGVV NIAGRSQPATLAQPVITIREIVEQPKIRLPRHLRQTYVRKVGHEHLNLVIPFQ GK PRPQVVTWKGAPIDTSHVHVRTSDFDTVFVRQAARSDSGEYELTVQIEN MKDTATVCIRVVEKAGPPQNVMVKEVGTNALVEWQPPKDDGNSEITGY FVQKADKKTMEWFTVYERNRHTSCTVSDLIMGNEYFRVYSENVCGLSDL PGVSKNTARIVKTGMTLKLPEYKEHDFRTPPKFLTPLPDRVVVAGYAAALN CAVRGHPKPKVVMKNKMEIREDPKFLMTNQGVLTNLNIRRSPSPFDSGT SCRAVNELGEALAECKLEVRVQ
87	Cow E1BF23 (Myomesin 2)	MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDEYSAKKKRASAQSSLRQSVT RKSSSQRESSRAALGESTLGETTLGETSLGDTTLGRTTCRLCAKRMSALEE EAQERKRRYQSMVAAYGEAKRERYLS ELAQLEEDVHVARTHARDQLDLRL ALQHAGRHLHHAVDRLAWERHSFEERISRAPEILVRLRSHTVWERMSV KLCFTVQGFPTPVVQWYKDGSLICQAGEPGKYRIDSKYGVHTLEINRADFD DTATYSAVATNVHGQVSTNAAVVVRFRGDEEPPHVSGLPIGLPLSSVIPYT HFDVQFIEKFGVTFRREGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKE SKWTKMFFGEGQASLSFSLNKKDDEGLYTLRIVSRGGITDHS AFLFVRDAD PLVTGAPGAPMDVHCHDANRDYIVITWKPPNATKESPIGYFIDRCEVGTD NWIQCNDAPVKICKYPVTGLFEGRSYIFRVRAVNSAGISRPSRVSEAVAALD PVDLRLQAVHLEGEKDIVVYQEELEGEVQIPGPPTNVHASETSRTYVVLS WDPEPRGKEPLMYFIEKSMVSGSGWQRVNAQTAVRSPRYAVFDLAEGKP YVFRVLSANKHGLSDPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVVVQW DRPKHEEDLLGYVDCSVAGSNVWEP CNHKPIGYNRFVHGLTTGEQYIF RVKAVNAVGTSENSQESDVIKVQAALTVPSHPYGITLLNCDGHSMILGWK VPKFSGGSPILGYVVDKREAHKKNWHEVNSSPLKERILTVEGLTEGSLYEF

TABLE 3-continued

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		KIAAANMAGIGQPSDPSSELFKCEAWTMPEPGPAYDLTFCEVRDTSVLVLLWK PPVYSGSSPVSGYFVDYKEEDSGEWLTVHETATPHRYLKVCDDLHQGKTYV FRVRVAVNASGVGRPSDTSEPVLEARPGTKEVSAGVDEEGNVYLSFDCPEV TDASHFTWCKSYEDIAADDDRKFVETAGDHSKLYFKNLDKEDLGTVSVSVS DTDGVSSSFVLDEEELERLKALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQ AEHLSPDSSYRFIINDREVADSETHRIKCDKSTGIIEMVMDRFTIDNEGTYTV QIQDGKAKSQSSSLVLIGDAFKAFLKEAEFQRKEFLRKQGPFAEYHLHWDVT EECEVRLVCKVANTKKETVFKWLKDDVLYETEKMPDLEKGVCELLIPKLS KKDHGEYKATLKDDRGQDVSTLEIAGKVYEDMILAMSRVCGASASPLKIL CTPEGIRLQCFMKYFTEEMKVNWYHKDAKITSSEHMRIGGNEQMAWLQIC EPTEKDGKTYTFEILDGKKNHQRSLDLSGQAFDEAFAEFQQLKAAAFAEKN RGKVIIGLDPVVTIMEGKTLNLTCTVFGNPDPEVVWFKNDKDIELSDHFSV KVEQKQYVSMITKGVTSSEDSGKYSIHVKNKYGEKIDVTVSVYKHGEIIPD VVLQQAKPKLLPAAAPSPA
88	Cow ENOB (Beta-enolase)	MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALERD GDKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGT NKSKEFGANAILGVSLAVCKAGAAEKGVPLYRHIAADLAGNPELILPVPAFNVI NGGSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVIKAKYKGD ATNVGDEGGFAPNILENNEALELLKTAIQAAGYDPKVVI GMDVAASEFYRN GKYDLDFKSPDDPARHISGEKLGELYKNFIKNYPVVSIEDPFDQDDWATWT SFLSGVNIQIVGDDLTVTNPKRIQAQVEKKACNCLLLKVNQIGSVTESIQAC KLAQSGNGWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKY NQLMRIEALGDKAVFAGRKFRNP KAK
89	Cow MLRS (Myosin regulatory light chain 2)	MAPKAKRRAAAEGSSSVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL RDTFAAMGRLNVKNEELDAMMKEASGPINFVFLNMFGEKLGADPEDVI TGAFKVLDPGKGTIKKKFLEELLTTQCDRFSQEEIKNMWAAFPDVGGNV DYKNICYVITHGDAKDQE
90	Cow MYH1 (Myosin-1)	MSSDQEMAVFGEAAPYLKSEKERIEAQNKPFDAKTSVFVADPKESFVKAT VQSREGGKVTAKEAGATVTVKEDQVFPMPNPPFKDI EDMAMMTHLHEP AVLYNLKERYAAWMIYTYSLFCVTVNPKWLVPYNAEVVTAIRGKKRQ EAPPHIFSISDNAYQFMLTDRENQSI LITGESGAGKTVNTKRVIQYFATIAVT GEKKKEEPTSGMKQGTLEDQIISANPLEAFGNAKTVRNDNSSRFGKFI R I H FGTTGKLASADIETYLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLIT TNPYDYAYVSQGEITVPSIDQEEELMATDS AIEILGFTSDERVS IYKLTGAV MHYGNLKFQKQREEQAEPDGTVEADKAAYLQGLNSADLLKALCYPRVK VGNEFVTKGQTVQVYNVAVGALAKAVYDKMFLWMVARINQQLDTKQPR QYFI GVLDIAGFEIFDFNSLEQLCINFTEKLGQFFNHHMFVLQE EYKKEGI EWEFIDFGMDLAACIELIEKPMGIFSILEEECMFPKATDMSFKNKLYEQHLG KSNNFQKPKPAKGKAEAHFSLIHYAGTVDYNI TGWLDKNKDPLNETVVGL YQKSSVKTLLALLFGSPASGEAEGGPKKGGKKGSSPQTVSALFRENLNKLM TNLRSHTPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRGFPFSR ILYADFKQRYKVLNASAIPEQGQFIDSKKASEKLLASIDVDHTQYKFGHTKVF FKAGLLGLEEMRDEKLAQLITRTQARCRGFLARVEYQKMYERRESIFCIQ YNVRAFNMNVKHPWMLYFKIKPLLKSAETEKEMANMKEEFKTEELA KSEAKRKELEEKMTLTQEKNDLQLQVQSEADALADAEEERCDQLIKTKIQL EAKIKEVTERAEDEEINAELTAKKRKLEDECSSELKDIIDLELTLAKVEKE KHATENKVNLTTEEMAGLDETIAKLTKEKKALQEAHQQTLDLQAEEDK VNTLTAKTKLEQQVDDLEGSLEQEKLRMDLERAKRKEGDLKLAQEST MDIENDQQLDEKLKKKEFEMSNLQSKI EDEQALAMQLQKKIKELQARIEE LEEEIEAERASRAKAEKQRSDLRSRELEEISERLEEAGGATSAQIEMNKKREA EFQKMRDLLEEATLQHEATAAALRKKHADSVAELEGQIDNLQRVKQKLEK EKSEMKEIDDLASNMTVSKAKGNLEKMCRALEDQSELKTKEDQEQRL INDLTTQARLQTESGEFSRQLDEKDALVSQLSRGKQAFQQTQIIEELKRQLEE EIKAKSALAHALQSARHCDLLREQYEEEBGKAEALQRAMSKANSEVAQ WRTKYETDAIQRT EELEEAKKKLQRLQDAEEHVEAVNAKASLEKTKQR LQNEVEDLMIDVERTNAACAALDKQRNFDKILSEWKQKYEETHAELEAS QKESRSLSTELFKIKNAYEESLDQLET LKRENKNLQQETSDLTEQIAEGGKRI HELEKVKQVBEQKSEIQAAL EEAASLEHEEGKILRIQLELNQVKSEIDRKI AEKDEEIDQLKRNHIRIVESMQSTLDAEIRSRNDAIRLKKMEGDLNEMETIQ LNHANMAAEALKNYRSTQAILKDTQIHLDDALRGQEDLKEQLAMVERR ANLLQAEIEELRATLEQTERSRIAEQELLDASERVQLLHTQNTSLINTKKK LETDITQIQGEMEDIQ EARNABEAKKAITDAAMMAEELKKEQDTS AHLE RMKKNLEQTVKDLQHRLEAEQLALGKGGKQIQKLEARVRELEGEVESEQ KRNVEAVGLRKHERRVKELTYQTEEDRKNILRLQDLVDKLQAKVKS YKR QAEAEAEQSNVNL SKFRKLQHELEEAERADIAESQVNKL RVKSRREVHTKII SEE

TABLE 3-continued

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
91	Cow A0A3Q1M707 (Nebulin)	MPNFQTRTRFYETTTTTSDBEQSETSRPALAQPVPPEKPVVERKRVIRKKVDP SKFMTPIYIAHSQKMQNLFSTNKYKENYEKAGKPYAITTDTPELRRIKKVQ DQLESEVKYRVDGDAKTIHVDEKAKDIEHAKKVSQQVSKVLYKQNWED TKDKYLLPPDAPELVQAIKNTAMFSKKLYTEDWEADKTMFYYPYNDSPELR RVAQAQKALSDIAYKGLAEQQTQFTSLPDPDIEFAKKVTNQVSKQKYKE DYENKVKGWSETPCFEIIATARMNSNNISTKKYQEDFEHMKDQIYFMQTE TPEYKVNKQAGVAASKVKYKQDYENKKGKADYNVLPASENPLLRQLKAA GDALSDKLYKENYEKTKAKSINYCETPKFKLDTVLHNFSSDTKYKDSYLLKD ILGHYVGSYEDPYHTHCMRVSQAQNSDKNYKAEYBEDRGKGFPPQTITQEY EAIKKLDQCKDHTYKVHPDKTKFTQVTDSPVLMQAQVNSKQLSDLNKYA KHENEKFKCHIPPDTPAFIQHKLNAYNLSNIIYKHDWEKTKAKKFDIKVDA IPLLAANKANTKIASDVMYKKDYEKSKGKMIGALSINDDPKMLHSLKTAKN QSDRLYKENYEKTKAKSMNYCETPKYQLDTLLKNFSEAKYKDSYVQNVL GHYIGSFEDPYQVHCLKISAQNSDKNYKAEYEDKGKCYFPQTITQEYEAIK KLDQCKDHTYKVHPDKTKFTAVTDTPVLLQQLNTKQLSDLNKYKAKHEG EKFKCHIPADAPQFIQHRVNAYNLSNIIYKHDWEKTKAKKFDIKVDAIPLL AAKANTKIASDVMYKKDYEKSKGKMIGALSINDDPKMLHSLKTAKNQSD HEYRKDYEKSKTIYTAPLDMLPLTHAKKSQAIASDVDYKHLHNSYPPDS VNVDLAKAYALQSDVEYKADYNSWMKGCWMPFGSLEMEKAKRASDI LNEKKYRQHPDTLKFTSIEDAPIIVQSKINQAQRSDVAYKAKGEEVIHKYSL PADLPQFIQAKVNAYNISENLYKADLKDLSKKGYDLRIDAPIKAAKAAARQ AASDVQYKKDYEKAGKMGVGFQSLQDDPKLVHYMNVAKIQSDREYKKA YEKTKTRHNTPHDMVNI VAAKKAQDVASNVNYKHSLLHYYTLPDAMDLE LSKNMMHIQSDNVYKEDYNNWMKGIGWIPIGSLEVEKVKKAGDALNEKK YRQHPDTLKFTSIVDSPVMVQAKQNTQQVSDILYKAGEDVKHKYTMSPD LPQLQAKCNAYNLSVICYKRDWHDLIAKGTNVLGDAIPITAAKASRNIA DYKYKEAYEKAGKQVGFPSLQDDPKLVHYMNVAKLQSDREYKKNYEN TKTSYHTPGDMVSI TAAKMAQDVATNVNYKQPIHHYTYLPDALSLHIRN VNQIQSDNVYKDEYNHFFKGMGWIPIGSLEVEKVKKAGDALNEKKYRQHP DTIKFTSVPDSMGVLAQHNTKQLSDLNKYVEGEKVKHKYTMDDPVPQFI QAKVNAYNMSDSHYKADWKKTLAKGYDLRPDAIPIVAASSRNIASDFKY KEAYEKTGKQIGFRSLQDDPKLVHFMNVAKMQSDREYKKGYEASCTKY HTPLDMLSVTAAAKSQEVATNTNYKQPFHHYTLPLDALNVEHSRNAMQIQ SDNLYKSDFTNWMKGIGWLPLESLEVEKAKKAGEILSEKKYRQHPKELKFT YAMDTMEQALNKSINKLIMDKRLYTEKWNKDKTTIHVMPDTPDILLSRVNO ITMSDKLYKAGWEEKKKGIDLRPDAISIKAAASRDIASDYKYKQAYEQ AKGKQIGFRSLEDDEPKLVHFMQVAKMQSDREYKKAKEKSTSPQTPVDML SVVAAKKSQEVATNANYRNVIHTYNMLPDAMSLELAKNMMQIQSDNQYK ADYADFMKGIGWLPGLSLEAEKNKKAMEIISEKKYRQHPDTLKYSTLMDS MNMVLAKNNAKIMNEHLYKQAWEDKTKVHIMPDIPIILAKANAINISD KLYKLSLEAEKKGIDLRDAIPIKAAASRDIASDYKYKHSYKERGKM VGFRLSLEDDEPKLVHSMQVAKMQSDREYKKNYEKTKTSYHTPADMLSVA AKDAQANI TNTNYKHLIHKYILLPDAMNIIQLSKNMRIQSDNEYKQDYNE WYKGLGWSAGSLEVEKAKKATEYASDQKYRQHPSPNFQFTKLNDMSDMV LAKQNAHTMNKYLTYVDWNKDKTKIHVMPDTPDILQAKQNTMYSQKL YKLGWEEALKKGYDLPVDAISVQAASRDIVSDFKYKQYRKQIGHHIG FRSLQDDPKLVLSMNVAKMQSERYKKDFEKWTKFSSPVDMLGVVLAK KCQALVSDVDYKYNLHGWTCLPDQNDVIHAKKAYDLQSENLYKSDLEWL KGIGWSPLGSLEAEKNKRASEIISEKKYRQPPDRNKFTSIPDAMDIVLAKTN AKNRSDILYREAWDKDKTQIHIMPDTPDII LAKANLINTSKFYRMGYEELR KKGIDLPVDAIPIKAAASREIASSEYKKEGFRMQLGHHIGARNIKDDPKM MWSMHVAKIQSDREYKKDFEKWTKFSSPVDMLGVVLAKKQTLVSDID YKNYLHQWTCLPDQSDVIHARRAYDLQSDNLYKSDLQWLRGIGWLPSSGL EDEKNKRASQILSDHVYRQHPDQKFSSLMDSIPMVLAKNNAITMNHRLYT EAWDKDKTTHIMPDTPEVLLAKQNIINSEKLYKLGLDEAKRKGDMRI DAIPIRAASRDIASSEFKYKEGYRRQLGHHIGARAIHDDPKMMWSMHVA KIQSDREYKKDFEKWTKFSSPVDMLGVVLAKKQTLVSDIDYKNYLHQ WTCLPDQSDVIHARQAYDLQSDNLYKSDLQWLRGIGWVPIGSLDVEKSKR ASEILSDKLYRQPPDKFKFTSVTDSLEQVLAKNNAINMKNRLYTEAWDKD KTQVHIMPDTPEITLARTNKVNYSENLYKLAHEEAKKGYDLRSDAIPIVA AKASRDII SDYKYKGYRQLGHHIGARDIKDDPKMMWSMHVAKIQSDRE YKKDFEKWTKFSSPVDMLGVVLAKKQTLVSDIDYKNYLHQWTCLPDQ SDVIHARQAYDLQSDNLYKSDLQWLRGIGWVPIGSDVVKCKRAAEILSD NLRYQPPDTLKFTSVPSLEQVLAKNNAINMKNRLYTEAWDKDKTQIHIMP DTPEIMLARQNKLYSETLYKLANEAEKKGYDLRSDAIPIVAASRDII S DYKYKGYRQLGHHIGARDIKDDPKMMWSMHVAKIQSDREYKKDFEK WTKFSSPVDMLGVVLAKKQTLVSDIDYKNYLHQWTCLPDQSDVIHARR AYDLQSDNIIYKSDLQWLRGIGWVPIGSDVVKCKRAAEILSDNLYRQRPDT LKFTSVPSLEQVLAKNNAINMKNRLYTEAWDKDKTQIHIMPDTPEITLA RQNKINYSNLYRQAMEAEKKEGYDLRSDAIPIVAASREIASDYKKEA YRQQLGHHIGARAIHDDPKMMWSVHVAKMQSDREYKKDFEKYKTRFSSP

TABLE 3-continued

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		VDMLGIVLAKKCQTLVSDVDYKHLHEWTCCLPDQNDIIHARKAYDLQSDN LYKSDLEWLKGIWVPIGSVEVLKAKRAGEILSDNIYRQRPDTLKFTSVTDS PEQVLAKNNAINMNKRLYTEAWDNDKKTIHVMPDTPETMLAKLNRINYSD KLYKLALLESERKEGYDLRLDAIPIQAAKASREIASDYKKEGYRKQLGHHI GARNIKDDPKMMWSIHAGKLQSDLEYKDFEKWTKFSSPVDMMGLVQA KKCQILVSDIDYKNLLHEWTCCLPDQNDIIQARKAYDLQSDAIYKADLEWLR GIGWVPIDSVGVEHAKRAGEILSERKYRQPAVQLKFTSITDTPETVLAKNNA LNVSKHLYTEAWDADKTSIHVMPDTPETLLAKSNSANISHKLYTKGWDESK MKDYDLRADAI SIKSAKASRDIASDYKKEAYEKHKGGHHIGARSVEDDPRI MCAMNAGRIQSEREYKKEFKWKTKFSSPVDMLGILLAKKCQTLVSDIDY RNYLHHWTCCLPDQNDIIQARKAYDLQSDNVYKSDLEWLRGIGWMEGSVE MNRVKNAQDLINERRYRTRPEALTFTSIVDTPEVVLAKANSLOMSEKLYQE AWNKDKTNISIPSDTPLILQAQINAMQISNKLQKDWDEAKQKGYDLRAD AIEIKHAKASREIASDYKKEGYRKQLGHHVGFRTLQDDPKLVWSIHAAKI QSDREYRKAYEKSIGIHNTPLDMMSIVQAKKCQVLVSDIDYRNYLHQWIC LPDQNDVIQAKKAYELQSDAIYKADLEWLRGIGWMEGSPEVLRVKNAQH IFRDSVYRTPVVVKLYTSIVDTPEVVLAKSNAENISIPKYREVWDKDKTSIHI MPDTPETINLARTNALNVSNKLYREGWDEMKMSCDVRDLDAIPIQAAKASREI ASDYKYLDEKQKGGHYVGTRTARDNKKIRWALITAGKIQNEREYRLHWA KWKSKFQSPDMLSI EHSKNSQTLVSDIDYRHYLHQWTCMPDQNDVIQAR KAYDLQSDAIYKADLEWLRGIGWMPADSVSVNHAKHMADIFNEKKYRTKI ETLSFTPVDDRVDVYTAHKSGEILNDIKYRKDWNTKSKYTLTETPOLHTA QEAARI LDQYLYKESWEKQKATGYILPPDAVPFVHAHHSQVQSELKYKA EHVKQKGGHYVGTPMRDDPKLVWFEHAGQIQNDRLYKENYHKTAKAKIHI PPDMVSVLAAKEGQALASDIDYRNYLHQWICHPDQNDVIQARKAYDLQSD NIYKADLEWLRGIGWIPDLSVDHVRVTRNQEMMNQIKYKDALANYPNFT SVVDPPEIVLAKINSVNSQDVKYKETFNKLIKGYIFSPDTPYITHSKDMEKL YSTILYKRAWDGTYAGYTLDERYIPIVGAHADFVNSELKYKETYEKLLK GHYLAGKEISEFPNVVHCLDFQKMRSLLNRYRRHYEDTKANVHI PQDMMNH VLAKRCQYILSDLEYRHYFHQWTSLEPEPNVVRVRAHQEILSDNVYKDDL NWLKIGICYVWDTPOILHAKKSYDLQSQILYTAAGKENLKNYNLVTDTPL YVTALQSGINASEVKYKENYHQTKDKYTTVLETVDYDRIKNLKDLPSSNLY KEAWDKVKATSYILPPNTVSLTHAKNQKYMASHIKYREEYEKFKALYTLP RSVEDDPNTARCLRVGKFNIDRLYRSVYEKNKMKIHI VPMVMEMVTAKDS QKVSSEIDYRLHLHEWICHPDLQVNSHVRKVTQDISDIVYKDDLTLWLKIG CYVWDTPEILHAKHAYDLRNDIKYKAHVLKTRNNYKLVTDTPVYVQAVK SGKQLSDAVYHYDVVHSIRGRVAPTTKTVDLDRALHAYKLQSENLYRKAG LHALPTGYRLPVDTPHFKHTKDRYMSYFYKEVYEHMKAYGYTLGPN DVPFVNVRRVNNITSERLYRQLYHKLKDKIHTTPDTPETIRQVKKTQEAASEL IYKSDFFKMQGHMISLPYTPQVLHCRYVDITSDIKYKEDLQVLRGMGCF YDTPDMVRSRHLRKLWSHYLYTDKARKMRDKYKVVLDTPEYRKVQELKT HLSSELVYRAAGRKQKSI FTSPDTPDLTRAKRGQKLQSQYLYVELATKERP HHHAGNQTTALKHARHVKDMVSENKYKIQYKMKDKYTPVPDTPILIRAK RAYWNASDLRYKETFPQTKGKYHTVKDALDIVYHRTVTDHISKIKYKENY MSQLGIWRSIPDRPEHFHRAVTDVAVSDVKYQDLTLWLKIGCYAYDTPD FTLAEKNKTLISKYKYKEVFERTKSNFKYVADCPINRHFKFATQLMNEKK YRADYEQKDKYHLVDEPRHLLAKIAGDQISQIKYRKNYEETDKKFTSIV DTPPEHLRTTKVNKQISDILYKLEYNAKPRGYTTIHDTPMLLHVRYKDEV SDLKYKEVYQRTKSNCTIEPDAVHIKAAKDAYKVNTNLDYKKKYEATKAY WKWTPDRPDFIQAAKSTLQQSDFEYKLDREYLGKCKLSVTDDKDMVLAL KNSIESDLKYKEKHVKERGSCHAVPDPQILLAKTVSSLVSENKYKSYVKK HLAQGSYTTLPETRDTIHVKEVTKNVSDTNKKKFVKEKGKSNYSIMLEPP DVKHAMDVAKKQSNVAYKDAKENLHYTTVADRPDIKKAQAAKQASE VEYRAKHKRKEGSHGLSMLGRPDIEMAKKAAKLSSQVQYRENFNKEKGKTP KYNPKDSQLYKVMKDANTLASEVKYKADLKKLHKPVTDMKESLIMNHVL NTSHLASSYQYKKNYEKSNGHYHTIPDNLEQLHLKEATELQSI VKYKEKYE KERGKPMDFETPTYITAKESQQMQSGKEYRKDYEESIKGRNLTGLEVT PA LLHVKYATKIASKEYRKDLEESIRGKGLSEMEDTPMLRAKNATQILNEK EYKRDLELEVKGRLNAMANETPDLRARNATDIASQIKYKQSAEMEKAN FTSVVDTPEIIHAQQVKNLSSQKKYKEDAEKCMSYYETVLDTPEMQVRREN QKNFSLQYQYDLKNSKGKITVVQDTPETILRVKENQKNFSSVLYKEDVSPG TAIGKTPEMMRVKQTQDHISSVKYKEVIGQGTPIPDLPVKRVKQTQKHIS VMYKENLGTGIPVTPETIERVKRNQENFSSNLSLKHEVTSKSIILTKYNSLK LPPHTPLPPVLYKENLKGKGTPTPI TPEMERVKRNQENFSSVLYKENMGKGT LPVTPEMERIKHNQENISSVLYKENMRKATPTPTPEMERAKRNQENISSV YSDSFRKQIQGKAAYVLDTPEMRVRVRETQRHISTVKYHEDEFKHKGCFTPV VTDPI TERVKKNTQDPSDICYRGIQRKVVEMEQKRNDDQETITGLRVWRT NPGSVFDYDPAEDNIQSRSLHMINAQARRSREQSRASGLSISGGEEKSEH SEAAHLSTYSDGGVFFSAASTGYKHARTTELPQQRSSSVATQQTTVSSIPSH PSTAGKIFRAMYDYMADADEVSKGDIAIVNVQAIDEGWMYGTVQRTG RTGMPLPANYVEAI

TABLE 3-continued

Cow Proteins		
SEQ ID NO	Description	Amino Acid Sequence
92	Cow E1BP87 (Myosin heavy chain 4)	MSSDQEMAIFGEAAPYLKSEKERIEAQNKPDAKTSVFVVDPKESFVKAT VQSRGGKVTAKTEAGATVTVKEDQVFPMPNPKYDKIEDMAMMTHLHEP AVLYNLKERYAAWMIYTYSGLFCTVNPYKWLVPYNAEVVTAYRGKKRQ EAPPHIFSISDNAYQFMLTGELDVSIGESGAGKTVNTKRVIQYFATIAVTGEK KKEEPTSGKMQGTLEDQIIISANPLLEAFGNAKTVRNDNSRFGKFIRIHFGTT GKLASADIETYLLEKSRVTFQLKAERSYHI FYQIMSNKKPELIEMLLITTNPY DYAYVSQGEITVPSIDDEELIATDS AIDILGFTSDERVSIYKLTGAVMHYGN LKFKQKQREEQAEPDGT EVADKAAYLQSLNSADLLKALCYPRVKVGNFV TKGQTVQQVYNVAVGALAKAVYEKMFLWMVTRINQQLDTKQPRQYFIGVL DIAGFEIIFDFNSLEQLCINF TNEKLQQFFNHMPVLEQEEYKKEGIEWEFIDF GMDLAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNNFQK PKPAKGKAEAHFSLVHYAGTVDYNIAGWLDKNKDPLNETVVGLYQKSAM KTLAFLFAGGPSAEEGGGKKGGKKGS SFQTVSALFRENLNKMLTNLRS THPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRGFPSPRILYAD FKQRYKVLNASAIPEGQFIDSKKASEKLGSIDIDHTQYKFGHTKVFFKAGL LGILEEMRDEKLAQLITRTQAICRGFLMRVEFRKMMERRESIFCIQYNVRAF MNVKHWPMKLYFKIKPLLKSAETEKEMANMKEEFKTEELAKSEAKR KELEEKMTLMQEKNDLQLQVQSEADGLADAEERCDQLIKTKIQLEAKIKE LTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLLETLAKVEKEKHATE NKVKNLTEEMAGLDENIAKLTKEKKALQEAHQQLDDDLQAEEDKVNTLT KSKTKLEQQVEDLEGSLEQEKLRMDLERAKRKLEGLDKLAQESTMDIEN DKQQLDEKLKKKEFEMS NLQSKI EDEQALAMQLQKKI KELQARIEELEEEIE AERASRAKAEKLRSDLSRELEEISERLEEAGGATSAQIEMNKKREAEFQKM RRDLEEATLQHEATAAALRKKHADSVAELGEQIDNLQRVKQKLEKEKSEM KMEIDDLASNMTVSKAGNLEKMCRTLEDQLSEVKSKEEHHQRLINELSV QKAHLHTESGEFSRQLEEKDALVSQLSRGKQAFQIIEELKRQLEEBTKAK SALAHALQSARHDCDLLREQYEEBQEGKAE LQRALSKANSEVAQWRTKYE TDAIQRTTEELEEAKKLAQRLQDAEEHVEV VNAKCASLEKTKQRLQNEVE DLMLDVERSNAACAALDKQRNFDKVLSEWKQKYEETQADLEASQKESR SLSTELFKVKNAYEESLDQLETLKRENKLNQOEISDLTEQIAEGGKHIHELE KVKKQIEQEKGELQAAL EEAASLEHEEGKILRIQLELNQVKS EIDRKIAEK DEEIDQLKRNLHRVVE SMQSTLDAEIRSRNDALRIKKMEGDLNEMEIQLN HANRQATEAIKNLRNTQGV LKDTQLHLDDAIRGQDDLKEQLAMVERRAN LMQAEIEBLRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTKKKLE TDITQIQGEMEDIVQEARNAEKAKKAITDAAMMAEELKKEQD TSAHLER MKKNMEQTVKDLQHR LDEAEQLALKGKKQIQKLEARVRELENEVENEQ KRNAEAVKGLRKHERRVKELTYQTEEDRK NVRLQDLVDKQLQSKVKAYK RQAEAEAEQSNVNLAKFRKLQHELEEAERADIAESQVNKL RVKSRREVHTK VISEE
93	Cow ALBU (Albumin)	MKWVTFISLLLLFSSAYSRGVFRDTHKSEIAHRFKDLGEEHFKGLVLIASF QYLQQCPFDEHVKL VNELTEFAKTCVADESHAGCEKSLHTLFGDELCKVA SLRETYGDMADCEKQE PERNECF LSHKDDSPDLPKLPDPNTLCDEFKAD EKKFWGKLYE IARRHPYFYAPELLYYANKYNGVFQECQAEDEKGAACLLP KIETMREKVLASSARQLRCASIQKFGERALKAWSVARLSQKFPKAEFVEV TKLVTDLT KVHKECCHGDLLECADDRADLAKYICDNQDTISSKLKECCDKP LLEKSHCIAEVEKDAIPENLPPLTADFAEDKDVCKNYQEA KDAFLGSFLYE YSRRHP EYAVSVLLRLAKEYEATLEECCAKDDPHACYSTVPDKLKLHVDE PQNLIKQNC DQFEKLGEYGFQNALIVRYTRKVPQVSTPTLVEVSRSLGKVG TRCCTKPESERMPCTEDYLSLILNRLCVLHEKTPVSEKVT KCCTESLVNRRP CFSALTPDETYPKAFDEKLFTFHADICTLPDTEKQIKKQTALVELLKHKPK ATEEQLKTVMENFVAFVDKCAADDKEACFAVEGPKLVVSTQTALA

[0043] An animal protein can comprise a sheep protein (Table 4). A sheep protein can comprise a sheep muscle protein. A sheep protein can comprise a sheep blood protein. A sheep protein can comprise a sheep albumin protein. A sheep protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 94-108.

[0044] A food product may comprise a fragment or a truncation of a sheep protein. The food product may include

a fragment or truncation of any one of SEQ ID Nos: 94-108. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 94-108. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 94-108. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 94-108. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 94-108.

TABLE 4

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
94	Sheep W5PJ69 (Creatine kinase)	MPFGNTHNKHKLNFKAEEYPDLSKHNHMAKALTLEIYKKLRDKETPSG FTLDDVIQTGVDPNGHPFIMTVGCVAGDEESYTVFKDLFDPIIQDRHGGFKP TDKHKTDLNHENLKGDDLDPHYVLSRVRTGRSISKYALPPHCSRGERR AVEKLSVEALNSLTGEFGKGYPLKSMTEQEQQLIDHFLFDKPVSPLLL ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMKEKGNMKEVFR RFCVGLQKIEEI FKKAGHPFMWNEHLGYVLTCPNLTGLRGGVHVKLAH LSKHPKFEEI LTRLRLQKRGTTGGVDTAAGSVFVDSNADRLGSSEVEQVQL VVDGVKLMVEMKKLEKQSIDDMI PAQK
95	Sheep W5P9P1 (Actinin alpha 3)	MMVLQPEGLGTGEGPFAGGGRGGGEYMEQEEDWDRDLLLDPANEKQQ RKTFTAWCNHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPRDPKKG MRPHKIANVNKALDFIASKGVKLVSIGAEIIVDGNLKMILMIWTIILRFAI QDISVEETSAKEGLLLWCQRKTAPYRNVNQNFHTSWKDGLALCALIHRH RPDLIDYAKLRKDDPIGNLNTAFEAKEYLDIPKMLDAEDIVNTPKPDKAI MTYVSCFYHAFAGAEQAETAANRICKVLAVNQENKLMEEYEKLASLELLE WIRRTVPWLENRVGEPSSAMQKLEDPRDYRRLHKPPRVQEKQCLEINF NTLQTKLRLSHRPAPMPSEGLVSDIANAWRGLEQAKEYEDWLLSEIRRL QRLQHLAEKFQKASLHEAWTRGKEEMLSQRDYETASLQEVRAALLRRHEA FESDLAAHQDRVEHIAALAEQLNELDYHEAASVNSRCQAI CDQWNLGTGL TQKRRDALERMEKLELLETIDQLQLEFARRAAPFNWLDGAVEDLQDVWL HSVVEETQSLVTAHDQFKATLPEADRERGAIIQGEIQKI CQTYGLRPSSTNP YIALTPQDINTKWDTVRKLVP SRDQMLQBELTRQQMNERLRQFAAQANA IGPWIQGVVEGGRLAGMAGSLEEQMAGLRQQEQNI INYKSNIDRLEGDH QLLQESLVPDNKHTVYSMEHIRVGWEQLLTSIARTINEVENQVLTDRACKGL SQEQLNEFRASFNFDRKRNGMMEPDDFRACLISMGYDLGEVEFARIMTM VDPNAGVVTTFQAFIDFMTRETAETDTAEQVVASFKILAGDKNYITAEELR RELPAEQARRAPDRGAGAPAGALDYVAFSSALYGESDL
96	Sheep PYGM (Glycogen phosphorylase)	MSRPLTDQEKRKQISVRGLAGVENVTLEKKNFNRHLHFTLVKDRNVATPR DYYPALAYTVRDHLVGRWIRTQQHYEYKDPKRIYYLSLEFYIGRTLQNTM VNLALENACDEATYQLGLDMEELIEEEDAGLGNGLGRLAACFLDSMAT LGLAAYGYGIRIYEFQIFNQKISGGWQMEEDDWLRYGNPWEKARPEFTLP VHFYGRVEHTSQGAKWVDTPVVLAMPYDTPVPGYRNNVNMTMLWSAK APNDFNLKDFNVGGYIQAVALDRNLAEISRVLYPNDNFFEGKELRLKQEYF VVAATLQDIIIRRFKSSKFGCLDPVRTNFDAPFDKVAIQLNDTHPSLAIPELM RILVDQERLEWEKAWEVTVKTCAYTNHTVLPALERWPVHLIETLLPRHLQ IIYEINQERFLNRVAAFPDGDVDRLRMSLVEEGAVKRINMAHLCIAGSHAV NGVARIHSEILKKTIFKDFYELEPHKFNQKTNGITPRRWLMCNPGDLAEVIA ERIGEEYIADLDQLRKLLSYVDDSFIRDVAKVKQENKLFSAYLEKEYKV HINPNSLFDIQVKRIHEYKRQLLNLCHVITLYNRIKKEPNKFFVPRVTMIGGK AAPGYHMAKMIIIRLITAGDVVNHDVVGDRLRVIFLENYRVSLAEKVIPAA DLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENFFIF GMRVEDVERLDQKGYNAQEYDRIPELRHII DQLSSGFFSPKQPDLPKDIVN MLMHDRFKVFADYEEYVKQERVSALYKNPREWTRMVIIRNIATSGKFSS DRTIAQYAREIWGEPTQRMPAPDEKI
97	Sheep A0A6P7DGM3 (Fructose- biphosphate aldolase)	MPHQYPALTPEQKKELCDIAHRIVAPGKGIILADESTGSI AKRLQSIGTENTE ENRRFYRQLLLTADDRVNPCIGGVILFHETLYQKADDRPFPQVIKAGGV VGIKVDKGVVPLAGTNGETTTQGLDGLSERCAQYKKGADFAKWRCLVKI GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT EKVLAAYKALSDHHIYLEGTLKPNMVTGPHACTQKYSHHEIAMATVTA LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS ALKAWGGKKENLKAQEYVKRALANSLACQGGKYTPSGKAGAAASESLFI SNHAY

TABLE 4-continued

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
98	Sheep W5QD16 (Myosin light chain 1)	ILLNMAPKKDVKKPAAAAAPAPAPAPAPAPAPPKEEKIDLSAIKIEFSK QQQDEFKEAFLLYDRTGDGKI TLSQVGDVLRALGTPNPTNAEVKKVMGNPS NEEMNAKKIEFEQFLPMLQAI SNNKDQGTIEDFVEGLRVFDKEGNGTVMG AELRHVLATLGEKMKKEEVEALMAGQEDSNGCINYEAFVKHIMSN
99	Fragment of Sheep G3P (Glyceraldehyde-3-phosphate dehydrogenase)	IGRLVTRAAFNTEGKVDIVAINDPFIDLHYMVVMFYDSTHGKFGHTVKAEN GKLIVINGKAITIFQERDPANIKWGDAGAEYVVESTGVFTTMEKAGAHKLG GAKKVI ISAPSADAPMFVMGVNHEKYNNLTKI VSNASCTTNCLAPLAKVIH DHFGIPEGLMTTVHAI TATQKTVDGPSGLWRDGRGAQNII PASTGAAGA VGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAKYDEIKKVVQKASE GPLKGLGYTEDQVVS CDFNSDTHSSTFDAGAGIALNDHFVKLISWYDNEF GYSNRVVDLMVHMASKE
100	Sheep W5QI03 (Tropomyosin 1)	MDAIIKKMQMLKLDKENALDRAEQAEADKKAEDRSKQLEDELVSLQKK LKATEDELKYS EALKDAQEKLELAEKKATDAEADVASLNRRIQLVVEEL DRAQERLATALQKLEEAKEADESERGMKVIESRAQKDEEKMEIQEIQLEKE AKHIAEDADRKYEEVARKLVI IESDLERAEERAELS EGQVRQLEEQLRIMDQ TLKALMAEDKYSQKEDKYEIEIKVLSDKLKEAETRAEFAERSVTKLEKSI DDELELYAQKLYKKAISEELDHALNDMTSIESGSCQRRKP
101	Sheep W5PX04 (Myosin binding protein C2)	MPEAKPAKKAPAGKDAAKAPAPKEAPPQEPAPAPSTEAPPEDQSPTEEEP TGVFLKKPDSVSVENGKDTVIVAKLNGKELPAKPAVKWFKGKWLGLGSKS GARFSFKESHDAASNVTVELHITKVVLGDRGDYRIEVKAKDFCDSCSTFNI DVESPRQDSAGQSLSEFKRSSEAKSDTAGELDFSGLLKKRQVVEEEKKKKK DDDDLGPPEIWEELKGAKKSEYER IAPQYGITDLRGLMKRLKAKVEVKK SAAFTKKLDPAYQVDKGNKIKLVVEISDPDLPLKWFKNQGEIKPSSKYVFE NVGKKRILITINKCTLADDAAYEVVVKDEKCFTEL FVKVEPPVLI VTPLEDQ QVFPVGNRVEMAVEVSEEGAQVMWLKDGVELTREDSPKSRYLKKGDKRH ILIYAEVALEDKGYKVMTNGDQCEALIVEEKQLEVLQDIADLTVKASEQ AVFKCEVSDKVTGKWKYKNGVEVRPSKRITISHTGRIHKLVIDVRPEDEG DYTFVPDGYALSLSAKLNLFLGAGLSPLPPEPPKIHLCDSGQTSSENSVVVAG NKLRLDVSITGEPRPVATWMKEDEVFTGTGGRVRIEQRGDISSFVIESAERG DEGRYTIKVTPAGEDVASILKVVVDVDPPEAVRVTSVGEDWAVLVWEP PKYDGRPVTGYLLERKKKGSQRWMLNFEVFTETTYESTNMI EGILYEM RVFAVNAIGVSQPSMNTKPFMPIAPTSEPQHLTVEDVTDTTTLKWRPPEKI GAGGIDGYLIEYCVEGSDEWIPANTEPVERCGFTVKNLPTGAKITFRVVG NIAGRSQPATLAQPVTIREIVEQPKIRLPHRLRQTYVRKVGEHLNLVIPFQKG PRPQVVTGKGAPIDTNHVVHRTSDFDTVFFVRQAARSDSGEYELTVQIEN MKDTATIRIRVVEKAGPPENVMVKEVWGTNALVVEQPPRDDGNS EITGYL VQKADKKTMEWFTVYERNRHTSCTVSDLIMGNEYFRVYS ENICGLSDLP GVSKNTARIVKTGMTLKLPEYKEHDFRTPPKFLTPLFDRVVVAGYAAALNC AVRGHPKPKVVMKKNKMEIREDPKFLMTNHHQGVLTNLNRRPS PFDSGTYS CRAVNELGEALAECKLEVRGEP
102	Sheep W5P6W4 (Myomesin 2)	MSLVAVPFYQKRKHFDQSYRNIQTRYLLDEYAAKKKRASQSSLRQSVT RKSSSQRESSRAALGETTLGETTLGRTTCPLCAKRMSSAL EEEAQERKRRY QSMVAAYGEAKRERYLSLAQLEEDVHVARTHAREELDRALQAHAGRHG LHHAVDDRLAWERHWFEDRISRAP EILVRLRSHTVWERMSVKLCFTVQGF PTPVVQWYKDGSLIQAGEPGKYRIDSKYGVHTLEINRADFDDTATYSAVA TNVHGQVSTNAAVVRRFRGDEEPFHSVGLPIGLPLSSVI PYTHFDVQFIEK FGVTFRRREGETLTLKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTMFF GEGQASLSFSLNKDDEGLYTLRIVSRGGVTDHSAFLFVRDADPLVTGAPG APMDVHCHDANRDYIVITWKPPNTTKESPVIGYFIDRCEVGTDNWVQCND APVKICKYPVTGLFEGRSYIFRVRVNSAGISRPSRVSEAVAALDPVDLRLRL QAVHLEGEKDVIYQEDLEGEVQIPGPPTNVHAS ETSRTYVVLSWDPPEPRG KEPLMYFIEKSMVSGSGWQRVNAQTAVRSPRYAVFDLAEGKPYVFRVL SA NKHGLSEPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVVQWDRPKHEED LLGYYVDCSVAGSNVWEP CNHKPIGYNRFVHGLTTEGEQYIFRVKAVNAV PTSENSQESDVIKVAALTVP SHPYGITLLNCDGHSMILGWKVPKFSGGSPI LGYYIDKREAHHKNWHEVNSSPLKERILKVEGLTEGSLYEFKIAANMAG MGSQQTLELFCKEAWTMPEPGPAYDLTFCEVRDTSVLVLLWKPPVYSGSSPVS GYFVDYKEEDSGEWLTVNEVTPHRYLKVCDLHQKQTYVFRVRVNASG VGRPSDTS E PVLVEARPGTKEVSAGVDEEGNIYLSFDCPEVTDASHFTWCK SYEDIADDRFKVETAGDHSKLYFKNLDKEDLGTYSVSVSDTDGVSSSFVL DEEELERLKALSNEIKNPTIPLKSELAYEIDFKGQVRFWLQAEHLSPDSSYRF IINDREVADSETHRIKCDKSTGVIMVMDRFTVDNEGTYTVQIQDGKAKSQ SSLVLIGDAFAKAVLEAEFQRKEFLRKQGPFAEYLNHWDVTEECEVRLVCK VANTKKETVFKWLKDDVLYETEKMPDLEKGVCELLIPKLSKKDHGEYKAT LKDDRQGDVSTLEIAGKVYEDMILAMSRVCGASASPLKILCTPEGIRLQCF MKYFTEEMKVNWYHKDAKIASSEHMRIGGSEEMAWLQICEPTEKDKGKY TFEILDGKKNHQRSLDLSGAQDEAFAEFQQLKAAFAEKNRGKVI GGLPD VVTIMGKTLNLNCTVFGNPDPEVWFKNDKIDELSDHFSVKVEQAKYVS

TABLE 4-continued

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		MTIKGVTSSEDSGKYSMHVKNKYGGEEKIDVTVSVYKHGEEIPDVVLPQQAK PKLLPAAAPSPAH
103	Sheep W5P663 (Phosphopyruvate hydratase)	MAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYEALERLD GDKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTE NKSKEGANAILGVSLAVCKAGAAEKGVPLRYHIADLAGNPELILPVPFNV I NGGSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVIKAKYKGD ATNVGDEGGFAPNI LENNEALELLKTAIQAGYPDKVVI GMDVAASEFYRN GKYDLDFKSPDDPARHISGEKLGELYKSF IKNYPVVSIEDPFDQDDWATWT SFLSGVNIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESTIQAC KLAQSNNGWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKY NQLMRIEALGDKAVFAGRKFRNPKAK
104	Sheep W5P7V0 (Myosin light chain)	QAPKKAARRAAAEAGGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL RDTFAAMGRNLNVKNEELDAMMKASGPINFTVFLTMFGEKLGADPEDVI TGAFKVLDPGKGTIKKQFLEELLTTQCDRFSQEEIRNMWAAFPDPVGGNV DYKNICYVITHGDAKDQE
105	Sheep W5PPG6 (Myosin heavy chain 1)	MSSDQEMAI FGEAAPYLKSEKERIEAQNKPDAKTSVFVADPKESFVKAT VQSREGGKVTAKTEAGATVTVKEDQVFPMPNPKFDKIEDMAMMTHLHEP AVLYNLKERYAANMIYTSGLFCVTVNPNYKWLVPVNAEVVAAYRGKKR QEAPPHIFISIDNAYQFMLTDRENQSI LITGESGAGKTVNTKRVIQYFATIAV TGEKKKEEPTSGKMQGTLEDQII SANPLLEAFGNAKTVRNDNSSRFGKFIRI HFGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLI TTNPYDYAFVSQGEITVPSIDDQEEELMATDSAIEILGFTSDERSVIYKLTGAV MHYGNLKFQKQREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVK VGNFEVTKGQTVBQVYNAV GALAKAIYDKMFLWMVTRINQQLDTKQPRQ YFIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFPNHHMFVLEQEEYKKEGIE WEFIDFGMDLAACIELIEKPMGIFSL EEEECMPKATDTSFKNKLYEQHLGKS NNFQKPKPAKGKAAEHFSLIHYAGTV DYNITGWLDKNKDPLNETVVGLYQ KSSVKTALFLFSGAASAEAEAGGAKGKGGKSSFPQTVSALPRENLNKL TNLRSTHPHFVRCIIPNETKTPGAMEHELVLHLQRCNGVLEGIRICRKGFP ILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKV FKAGLLGLEEMRDEKLAQLITRTQARCRGFLARVEYQRMVERRESIFCIQ YNVRAFNMVKNHWPMKLYFKI KPLLSAETEKEMANMKEEFEKTKLELA KSEAKRKELEEKMTLTQEKNDLQLQVQSEADALADAEERCDQLIKTKIQL EAKIKEVTERAEDEEEINAELTAKKRKLEDECSSELKDDIDDLLELTAKVEKE KHATENKVNLTTEEMAGLDETIAKLTKKKALQEAHQQTLDLQAEEDK VNTLTAKTKLEQQVDDLEGSLEQEKKL RMDLERAKRKL EGDLLKLAQEST MDIENDKQQLDEKLKKKEFEMS NLQSKIEDEQALGMQLQKKIKELQARIEE LEEEIEAERASRAKAEQQRSDLSRELEESERLEEAGGATSQAIEMNKKREA EFQKMRRDL EEA TLQHEATAALRKQGGVGEQLDNLQVRKQKLEKEKS EMKMEIDDLASNMTVSKAKGNLEKMCRTLEDQSELKTKEDEQQRLIND LTTQARLQTESGEFSRQLEDKDALVSQLSRGKQAF TQQIEELKRQLEEEIK AKSALAHALQ SARHDCDLLREQYEEBQEAELQ RALS KANSEVAQWRT KYETDAIQRT EELEAKKLAQRLQDAEEHVEAVNAK CASLEKTKQRLQN EVEDLMIDVERTNAACAALDKQRFNFKI LSEWKQYEEETHAELEASQKE SRSLSTELFKIKNAYESLDQLET LKRENKNLOPTEEISDLTEQIAEGGKRIH ELEKVKKQVEQEKSEIQAALEAEASLEHEEGKILRIQLELNQVKSEIDRKIA EKDEEIDQLKRNHIRVVESMQSTLDAEIRSRNDAILKKKMEGDLNEMEIQ LNHANRMAAEALKNYRNTQAILKDTQLHLDDALRGQEDLKEQLAMVERR ANLLQAEIEELRATLEQTERSRIAEQELLDASERVQLLHTQNTSLINTKKK LETDITQIQGEMEDIIQEARNABEEKAKKAITDAAMMAEELKKEQDTS AHLE RMKKNLEQTVKDLQHRLEAEQLALKGKKQIQKLEARVRELEAEVESEQ KRNVTVKGLRKHERRVKELTYQTEEDRKNIILRLQDLVDKLQAKVSKYKR QAEAEAEQSNVNL SKFRKLQHELEEAERADIAESQVNLRVKSRVHTKII SEE
106	Sheep W5PFV9 (Nebulin)	MADDEEYEEVVEYYTEETVYEEVPGETITEFYETTTTRTS DYEQSDTSRPA AKPVPEKPKVERKVRIRKKVDPKFMTPYIAHSQKMQDLFSTNKYKENYEK AKGQPYAITTDTPELRRIKKVQDQLSEVKYRTDGDVAKTICHVDEKAKDIE HAKKVSQQVSKVLYKQNWEDTKDYLLPPDAPELVQAIKNTAMFSKKLY TEDWEADKTLFYPYNDSPELRRVAQAQKALSDITYKKGLAEQQTQFTSLPD PPDIEFAKKVTNQVSKQYKEDYENKVKGKWS ETPCFEIA TARMNSQNI ST KKYQEDFENTKDQIYFMQTETPEYKVNQAGVAASKVKYQDYENKKGK ADYNVLPASENPLLRQLKAAGDALSDKLYKENYEKTKAKSINYCETPKFKL DTVLHNFSSDTKYKDSYLNKILGHYVGSYEDPYHTHCMRVSAQNSDKNYK AEYEEDRGKGFPPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTQVTDSPV LVQAQVNSKQSLDNLNKAHENEKFKCHIPPDTPAFIQHKVNAYNLSNIIY KHDWEKTKAKKFDIKVDAIPLLA AKANTKIASDVMYKKDY EKS KGMIG ALSINDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQDLTL LKNFSEAKYKDSYVQNVLGHYIGSFEDPYREHCMKVAAQNSDKNYKAEY

TABLE 4-continued

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		EEDKGKCYFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTAVTDTPTVLLQ AQLNTKQLSDLNKAKHEGEKFKCHIPADAPQFIQHRVNAYNLSdni YKH DWEKTKAKKFDIKVDAIPLLAAKANTKIASDVMYKKDYEKSKGMIGALS INDDPKMLHSLKTAKNQSDREYRKDEYKSKTIYTAPLDMLPLTHAKKSQEI ASDVDYKHLHNYSPDSDVNVDLAKKAYALQSDVEYKADYNSWMKGC GWVPFGSLEMEKAKRASDILNEKKYRQHPDTLKFTSIEDAPIIVQSKINQAO RSDVAYKAKGEEVIHKYSLPADLPQFIQAKVNAYNISENMYKADLKDLK KGYDLRIDAIPIKAAKAAQAASDVQYKKDYKAKGKMVGFSQSLQDDPKL VHYMNAKIQSDREYKDYKTKTRHNTPHDMVNIIVAAKKAQDVASNV NYKHSLHHYTYLPDAMDLELSKNMMHIQSDNVYKEDYNNWMKGIWPI GSLEVEKVKAGDALNEKKYRQHPDTLKFTSIVDSPVMVQAKQNTQQVSD ILYKAKGEDVKHYKTMSPDLQFLQAKCNAYNLSVCYKRDWHDLIAKG TNVLGDAIPIATAKASRNIASDYKYKEAYEKAGKQVGFRLQDDPKLVH YMNVALQSDREYKKNYENTKTSYHTPGDMVSIATAKMAQDVATNVNY KQPIHHYTYLPDALSLHIRNVNIQSDKVYKDEYNHFFKIGIWIPIGSLEV EKVKKAGDALNEKKYRQHPDTIKFTSVPSMGMLAQHNTKQLSDLNK VEGEKVHKYTYLPDVPQFIQAKVNAFNMSDSHYKADWKKTLAKGYDLR PDAIPIVAAKSSRNIASDFKYKEGYEKAGKQIGFRSLQDDPKLVHFMNVA KMQSDREYKKGYEASKTKYHTPLDMLSVTAAKKSQEVATNTNYKQPFHH YTLPLDALNVESRNAMQIQSDNLYKSDFTSWMKGIWMPLESLEVEKAK KAGEILSEKKYRQHPDKFTYAMDTEQALNKSNIIMDKRRTYKKN KDKTTIHVMPDTPDILLSRVNQITMSDKLYKAGWEEKKKGYDLRDAIPI KAARASRDIASDYKYQAYEQAGKQIGFRSLEDDEPKLVHFMQVAKMQS DREYKAYEKSSTSFHTPVDMLSVVAAKKSQEVATNANYNVNIHNVMNL PDAMSLAKNMMQIQSDNQYKADYADFMKGIWLPGLSLEAEKNKA MEIISEKKYRQHPDTLKYSTLMDSMNVLAKNNAKIMNEHLYQAWHAD KTKVHIMPDIPIILAKANAINMSDKLYKLSLEAAKKKGYDLRTDAIPIKAA KASRDIASDYKYKHNYEKERGMVGFRLSLEDDEPKLVHSMQVAKMQSDRE YKKNYEKTKTSYHTPADMLSVTAAKDAQANITNTNYKHLIHKYIILPDAM NIQLSKNMNRLQSDNEYKQDYNEWYKGLGWSAGSLEVEKAKRATEYAS DQKYRQHPSPNFQPSKLNDMSDMVLAKQNAHTMNKLYTYVDWNKDKTKI HVPDTPDILQAKQNTIYSQKLYKLGWEEALKKGYDLVDAISVQAAKA SRDIVSDFKYKQGYRKQIGHHIGFRSLQDDPKLVLSMNVAKMQSREYKK DFEKWTKFSSPVDMLGVVLAKKQALVSDVDYKNYLHGWTCLPDQND VIHAKAYDLQSENLYKSDLEWLKGIWSPGLSLEAEKNKRASEIISEKKY RQPPDRNKFTSIPDAMDIVLAKTNAKNRSDILYREAWDKDKTQVHIMPDP DII LAKANLINASDKFYRMGYEELRKKGYDLVDAIPIKAAKASREIASSEYK YKEGFRKQLGHHIGARNIKDDPKMMWSMHVAKIQSDREYKDFEKWTK FSSPVDMLGVVLAKKQTLVSDIDYKNYLHQWTCCLPDQSDVIHARRAYDL QSDNLYKSDLQWLKGIWLPGLSLEDEKNKRAQILSDHVRQHPDKFKFS SLMDSIPMVLAKNNAITMNHRLYTEAWDKDKTTVHIMPDPGVLLAKQNG INYSEKLYKGLDEAKRKGDMRIDAIPIRAAKASRDIASEFKYKEGYRKQ LGHHIGARAIHDDPKMMWSMHVAKIQSDREYKDFEKWTKFSSPVDML GVVLAKKQTLVSDIDYKNYLHQWTCCLPDQSDVIHARQAYDLQSDNVYK SDLQWLKGIWVPIGSLDVECKRASEILSDKLYRQPPDKFKFTSVTDSLEQ VLAKNNAINMNKRLYTEAWDKDKTQVHIMPDPTEITLARMKNVYSESLEY KLAHEBAKKKGYDLRSDAIPIVAAKASRDII SDYKYKGYRKQLGHHIGAR DIQDDPKMMWSMHVAKIQSDREYKDFEKWTKFSSPVDMLGVVLAKK QTLVSDIDYKNYLHQWTCCLPDQSDVIHARRAYDLQSDNVYKSDLQWLRI GWVPIGSDVDVCKRAAEILSDNLYRQPPDTLKFTSVSPDSLEQVLAKNNAI NMNKRLYTEAWDKDKTQVHIMPDPTEIMLARQNKLNYSETLYLANEEA KKKGYDLRSDAIPIVAAKASRDII SDYKYKGYRKQLGHHIGARDIKDDPK MMWSMHVAKIQSDREYKDFEKWTKFSSPVDMLGVVLAKKQTLVSDI DYKNYLHQWTCCLPDQSDVIHARQAYDLQSDNIYKSDLQWLRIGWVPIGS VDVVKCKRAAEILSDNLYRQPPDTLKFTSVSPDSLEQVLAKNNAINMNKRL YTEAWDKDKTQIHMPDPTEITLARQNKINYSNLYRQAMEBAKKEGYDL RSDAIPIVAAKASREIASDYKYKEAYRKQLGHHIGARAIHDDPKMMWSVH VAKMQSDREYKDFEKYKTRFSSPVDMLGIVLAKKQTLVSDVDYKHLPH EWTCLPDQNDIIHARKAYDLQSDNLYKSDLEWLKGIWVPIGSDVDVKK RAGEILSDNIYRQPPDTLKFTSVTDSPEQVLAKNNAINMSKRLYTEAWDND KKTIVHMPDPTEIMLAKLNRINYSKLYKLALESKEGYDLRLDAIPIQAA KASREIASDYKYKEGYRKQLGHHIGARNIKDDPKMMWSIATKLQSDREY KKDFEKWTKFSSPVDMMGLVQAKKQILVSDIDYKNLLEHWICLPDQND VIQARKAYDLQSDAIYKADLEWLRIGWVPIGSDVGVENVKRAEILSERKY RQPAVQLKFTSITDTPTEIVLAKNNAINMSKHLYTEAWDADKTSIHVMPDP EILLAKSNSANISHKLYTKGWDESKMKDYDLRADAI SIKASASRDIASDY KYKEAYEKQKGGHHIGARSVEDDPKIMCAMNAGKIQSREYKKEFQKWK KFSSPVDMLGILQAKKQILVSDIDYRNYLHHWTCCLPDQNDIIQARKAYDL QSDAIYKADLEWLRIGWMEGSPVLRVKNQAQIFRDSVYRTPVVKLY TSIVDTPPEVVLAKSNAENISIPKYREVWDKDKTSIHIMPDPTEINLARTNALN VSNKLYREGWDEMMSCDVRLDAIPIQAAKASREIASDYKYKLDHEKQKQ HYVGTRTARDDNKIRWALJAGKIQNEREYRMHWAKWSKFPSPDMLSI E

TABLE 4-continued

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		HSKNSQTLVSDIDYRHYLHQWTCMPDQNDVIQARKAYDLQSDAIYKADLE WLRGIGWMPADSVSVNHAKHMGDI FNEKKYRTKIE TLNFTPVDDRVDYVT AKHSGEILNDIKYRKDWN DTKSKYTLTETPQLHAAQEAARILDQYLYKES WEKQKATGYILPPDAVPFVHAHHSQSDVQSELKYKAEHVQKKGHYVGVPT MRDDPKLVWF EHAGQIQNDRLYKESYHKTKAKIHIPPDMVSVLAAKEGQA LASDIDYRTY LHQWICH PDQNDVIQARKAYDLQSDNIYKADLEWLRGIGW MPLDSVDHVRVTRNQEMVNQIKYKKDALANYPNFTSVVDPPEIVLAKINA VNQSDVKYKETFNKLIKGYIFSPDTPYI THSKDMEKLYSTILYKRAWEGT KAYGYTLDERYIPIVGAHAD FVNS ELKYKETYEKLKGHYLAGKEISEFPN VVHCLDFQKMRSLNRYRRHYEDTKANVHIPQDMMNHVLAKRCQYILSDL EYRHYFHQWTS LPEEPNVVRVHAQEI LSDNVYKDDLNLWKGIGCYVVD TPQILHAKKSYDLQSQILYTAGKENLKNYNLVTDTPLVYTALQSGINASE VKYKENYHQIKDKYTTVLETVDYDRTRNLKDLFSSNLYKEAWDKVKATS YILPPNTVSLTHAKNQKHMASVYKYREEYDKFKALYTLPRSVEDDPNTAR CLRVGKFNIDRLYRSVYEKNMNIHIVPDMVMEMVTAKDSQKKVSEIDYRL HLHEWICH PDLQVNSHVRKVTDQISDIVYKDDL TWLKGIGCYVVDTP EILH AKHAYDLRNDIKYKAHV LKTRNNYKLVTDTPVYVQAVKSGKQLSDAVYH YDYVHSIRGRVAPTTKTVDL RALHAYKLQSENLYRKAGLHALPTGYRLP VDTPHFKHTKDRYMS SYFYKEVYEHMKAYGTYLGPNDVPFVNVRVN NITSERLYRQLYHKLDKDIHTPDTPEIRQVKKTQEA VSELIYKSDFFKMQG LRKLWSHLYTDKARKMRDKYQVVLDTPEYRTVQELKTHLSELVYRASG RKQKSIFTSVPDTPDLIRAKRGQKLQSQYLYVELATKERPHHHAGNQTTAL KHARHV KDMVSENKYKIQYEKMKDKYTPVPDTPILIRAKRAYWNASDLR YKETFOKTKGRYHTVKDALDIVYHRTVTDHISKIKYKENYMSQLGIWRSIP DRPEHFFHRAVTDASDVKYKADLTWLKGIGCYAYDTPDFTLAEKNKTLY SKYKYKEVFERTKSNFYVADCPINRHPKFATQLMNEKKYRADYEQRKDK YHLVVD EPRHLLAKIAGDQISQIRYRKNYEKTDKFSTSIVDTP EHLRRTTKVN KQISDILYKLEYNKAKPRGYTTIHDTPMLLHVVRKVKDEVS DLKYKEVYQRT KSNCTIEPDAVHIKAAKDAYKVNTNLDYKKQYEATKAYWKWTPDRPDFIQ AAKSTLQQSDFEYKLDREY LKGC KLSVTD DDMVLALKNSIIESDLKYKEK HVKERGSCHAVPDT PQILLAKTVSSLVSENKYQYVKKHLAQS YTTLPET RDTIHVKEVTKNVSDTNYKKKFVKEKGKSNYSIMLEPPDV KHAMDVAKK QSNMKA VYKKDAKENLHYTTVADRPDIKKATQAAKQASEVEYRAKHRKE GSHGLSMLGRPDIEMAKKAAKLS SQV KYRENFNKEKGKTPKYNPKDSQLY KVMKDANTLASEV KYKADLKLHKPVTDMKESLIMNHV LNTSHLASSYQ YKKNYEKSGHYHTIPDNLEQLHLKEATELQSI V KYKEKYEKERGKPM LD FETPTYITAKESQQMQSGKEYRRDY EESI KGRNLTGLEVT PALLHVKYATKI ASEKEYRKDL EESIRGKGLSEMEDTPDMLRAKNATQILNEKEYKRDLELEV KGRGLNAMANETPDFLAR NATDIASQIKYKQSAEMEKANFTSVVDTP EII HAQQVKNLSSQKKYKEDAEKCMSY YETVLDTP EMQVRVRENQKNFSSLQY QYDLKNSKGKITVQDTP EILRVKENQKNFSSVL YKEDVSPGTAIGKTP EM MRVKQTQDHIS SVKYKEVIGQGTPIPDLP EVKRVKQTQKHIS SVMYKENLG TGIPTPVTP EIERVKRNQENFSSVL YKENLGKGTPTPI TPEMERVKRNQENFS SILYKENLSKGTPLPVTPEMERVKRNQENFSSVL YKENVGKGTPTPVTPEM QRVKRNQENIS SVLYKENLGKATPTPTPEMERVKRNQENFSSVL YKENMR KATPTPVTPEMERAKRNQENIS SVLYSDSFRKQIQGKAAYVLDTP EMRRVR ETQRHISTVKYHEDFEKHGCFPTPVTDPI TERVKKNMQD FSDISYRGQRK VVEMEQRNDQDQETVTGLRVWRTNPGSVFDYDPAEDNIQSRSLHMINAQ AQRRSREQRSASGLSVSGGEEKSEHSEAAHLS TYSDGGVFFSATSTGYKH ARTTELPQQRSSSVATQQT TVSSIPSHPSTAGKIFRMYDYMAADAEV SF KGDIAIVNVQAIDEGW MYGTVQRTGRTGMLPANYVEAI
107	Sheep W5PIU1 (Myosin-4)	MSSDQEMAI FGEAAPYLKSEKERIEAQNKPPDAKTSV FVADPKESFVKAT VQSREGGKVTA KTEAGATVTVKEDQVFP MNPPKFDKIEDMAMMTHLHEP GVLN LKERYAAWMIY TYSGLFCVTVPNYKWL PVPVYNAEVV TAYRGKKRQ EAPPHIFSISDNAYQFMLTDRENQS ILITGESGAGKTVNTKRVIQYFATIAVT GEKKKEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIRIH FGTTGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLIT TNPYDYAFVSQGEITVPSID DQEELMATDSAI EILGFTSDERVS IYKLTGAVM HYGNLKFQKQREEQAE PDGTEVADKAAYLQSLNSADLLKALCYPRVKV GNEFVTKGQTVQVYNAV GALAKAVYKMF LWMVTRINQQLDTKQPRQ YFIGVLDIAGFEIFDFNSLEQLCINF TNEKLQQFFNHHMFVLEQEYKKEGIE WEPIDFGMDLAAELIEIEKPMGIFSILEEEECMF PKATDTSFKNKLYEQHLGKS NNFQKPKPAKGKAEAHFSLVHYAGTVDYN IAGWLDKNKDPLNETVVG LY QKSAMKTLAFLFAGGPSAEVHVSEGGGKGKGGKKGSS FQTVSALFR ENL NKLMTNLRLRSTHPHFVRCIIPNETKTPGAMEHELVLHQ LRCNGVLEGI RCRK GFPSRI LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSDIDHTQYKFGH TKVFFKAGLLGILEEMRDEKL AQLITRTQAI CRGPLTRVEFRKMMSRESI FCIQYNVRAF MNV KHPW MKLYFKIKP LLSAET EKEMANMKEEF EKT EELAKSEAKRKELEEKMTLMOEQKNDLQLQVQSEADGLADAEERC DQLIK TKIQLEAKIKELTERAEDEE EINAELTAKRKLEDECS ELKKDIDDELTLAK VEKEKHATENKVNKLTEEMAGLDENIAKLTK EKKALQEAHQQTLDL LQA

TABLE 4-continued

Sheep Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		EEDKVNLTLSKSTKLEQQVDDLEGSLEQEKKLRMDLERAKRKLEGLDLKLA QESTMDIENDKQQLDEKLKKKEFEMSNLQSKI EDEQALGMQLQKKIKELQ ARIEELEEETEAERASRAKAEQSRDL SRELEETSERLEEAGGATSAQIEMNK KREAEFQKMRDLLEATLQHEATAALRKKHADSVAELESEQIDNLQRVKQ KLEEEGSLNLTQALDI ISNFKI INRYYYIEGNLEKMCRTLEDQLSEVKSKEEE HQRLINELSAQKARLHTESGEFSRQLDEKDALVSQLSRGKQAFQTQIIEELKR QLEEETKAKSALAHALQSARHDCDLLREQYEEEQEAKAELQRSMSKANSE VAQWRTRYETDAIQRTTEELEEAKKLAQRLQEAEEHVEAVNAKCASEKT KQRLQNEVEDLMLDVERSNAACAALDKQRNFDKVLSEWKQKYEETQAE LEASQKESRSLSTELFKVKNAYESLQDQLETLKRENKNLQQEISDLTEQIAE GGKHIHELEKVKKQIEQEKGELOAALEEAASLEHEEGKILRIQLELNQVKS EIDRKIAEKDEEIDQLKRNHLRVVESMQSTLDAEIRSRNDALRIKKMEGDL NEMEIQLNHANRQATEATKNLRNTQGVLDKDTQLHLDDAIRGQDDLKEQLA MVERRANLQAEIEELRASLEQTERSRRAEQELLDASERVQLLHTQNTSL INTKKKLETDITRVHGSVVGSGIISKHLTFVYISVGCAPQAAMMAEELKKEQ DTSALHERMKKNLEQTVKDLQHRLEAEQLALKGGKQIQKLEARVRELE NEVENEQKRNAEAVKGLRKHERRVKELTYQTEEDRKNVRLQDLVDKLQ SKVKAYKRQAEAEQANVNLSKFRKLQHELEEAERADIAESQVNKL RV KSEVHTKVI SEE
108	Sheep ALBU (Albumin)	MKWVTFISLLLLFSSAYSRCVFRRDTHKSEIAHRENDLGEENFQGLVLIAFS QYLQQCPFDEHVKLKELTEFAKTCVADESHAGCDKSLHTLFGDELCKVA TLRETYGDMADCCKEQEPERNECFLNHKDDSPDLPKLKPEPDTLCAEFKAD EKKFWGKYLEVARRHPYFYAPPELLYYANKYNGVFQECQAECDKGAACL PKIDAMREKVLASSARQLRCASIQKFGERALKAWSVARLSQKFPKADFTD VTKIVTDLTKVHKECCHGDLLECCADDRADLAKYICDHQDALSSKLKECCD KPVLEKSHCIAEVDKDAVPENLPLTADFAEDKEVCNKYQEAQDVFLGSFL YEYSRRHPYAVSVLLRLAKEYEATLEDCCAKEDPHACYATVFDKLKHLV DEPQNLIKKNCELFEKHGEYGFQNALIVRYTRKAPQVSTPTLVEISRLGKV GTKCCAKPESERMPCTEDYLSLILNRLCVLHEKTPVSEKVTCCCTESLVNRR PCFSDTLTDETYVPKPFDEKFFTFHADICTLPDTEKQIKKQ TALVELLKHKP KATDEQLKTVMENFVAFVDKCCAADKQECFVLEGPCLVASTQAALA

[0045] An animal protein can comprise a pig protein (Table 5). A pig protein can comprise a pig muscle protein. A pig protein can comprise a pig blood protein. A pig protein can comprise a pig albumin protein. A pig protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 109-123.

[0046] A food product may comprise a fragment or a truncation of a pig protein. The food product may include a

fragment or truncation of any one of SEQ ID Nos: 109-123. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 109-123. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 109-123. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 109-123. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 109-123.

TABLE 5

Pig Proteins		
SEQ ID NO	Description	Amino Acid Sequence
109	Pig KCRM (Creatine kinase M-type)	MPFGNTHNKYKLNFKAE EYPDL SKHNNHMAKALTLEIYKKLRDKETPSGF TLDDVIQTGVDNPGHPFIMTVGCVAGDEESYVVF KDLPDPIIQDRHGGYKPT DKHKTDLNHNELKGGDDLDPNYVLSSRVTRGRSIKGYTLPPHCSRGERRAV EKLSVEALNSLTGEFKGKYPLKSMTEQEQQLIDDHFLFDKVPSPLLLASG MARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMEKGGNMKEVFRFCV GLQKIEEIFKKAGHPFMWNEHLGYVLTCPNSLGTGLRGGVHVKLHLKSKHP KFEEILTRLRLQKRGTTGGVDTAAVGSVFDVSNADRLGSSEVEQVQLVVDGV KLMVEMKKLEKGQSIDDMI PAQK

TABLE5-continued

Pig Proteins		
SEQ ID NO	Description	Amino Acid Sequence
110	Pig A0A480REY2 (Actinin alpha 3)	MMVMQPEGLGTREGPFAGGSGGGGEYMEQEEDWDRDLLDPAWEKQQ RKTFTAWCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPRPDGKGM RPHKIANVNKALDFIASKGVKLVSIGAEIIVDGNLKMTLGMWITIIIRFAIQDI SVEETSAKEGLLLWCQRKTAPYRNVNQNFHTSWKDGLALCALIHRHRPDL IDYAKLRKDDPIGNLNTAFEVAEKYLDIPKMLDAEDIVNTPKPDEKAIMTYV SCFYHAFAGAEQAETAANRI CKVLAVNQENEKLMEEYEKLASELLEWIRRT VPWLENRVGEPMSAMQRKLEDFRDYRRLHKPPRVQEKQLEINFNTLQTK LRLSHRPAPFMPSEGLVSDIANAWRGLEQVEKGYEDWLLSEIRRLQRLQHL AEKFQQKASLHEAWTRGKEDMLSQRDYESASLQEVALLRRHEAFESDLA AHQDRVEHIAALAQELNELDYHEAASVNSRCQAI CDQWDNLGTLTQKR RD ALERMEKLL ETIDQLQLEFARRAAPPNNWLDGAVEDLQDVWL VHSVEETQS LVTAHEQFKATLPEADRERGA ILGIQGEIQKICQTYGLRPSSTNPYIALSPQDI NTKWDTVRKLVP SRDQMLQEELTRQQVNRLRQFAAQANAIGPWIQGV EEVGRLAAGMAGSLEEQMAGLRQQEQNI INYKSNIDRLEGDHQLLQESLVF DNKHTVYSMEHIRVGEQLLTSIARTINEVENQVLRDQAKGLSQEQNLNFR ASFNHFDKRKNGMMEPDDFRACLI SMGYDLGEVEFARIMTMVDPNAGV TFQAFIDFMTRETAETDTAEQVVASFKI LAGDKNYITAEELRRELPAEQAEYC IRRMAPYKGAGAPAGALDYVAFSSALYGESDL
111	Pig A0A5G2QU59 (Alpha-1, 4 glucan phosphorylase)	MAMSRPLTDQEKRKQISVRGLAGVENVAELKKNFNRLHFTLVKDRNVAN ARDYYFALAYTVRDHLVGRWIRTQQHYEKPRIYYLSLEFYMGRTLQN TMVNLALENACDEATYQLGLDMEELEEMEDAGLNGGLGRLAACFLDSM ATLGLAAYGYGIRYEPGIFNQKITGGWQMEEDDWLRYGNAWEKARPEFT LPVHFYGRVEHTNQGAKVWDTVVVLAMPYDTPVPGYRNNVNMTMLWSA KAPNDFNLKDFNVGGYIQA VLDNRNLAENISRVLYPNDNFPEGKELRLKQEFY VVAATLQDI IRRFKSKFGCRDPVRTSDFADPKVAIQLNDTHPSLAIPELMRI LVDEERLEWEKANDVTVRTCAYTNHTVLP EALERWPVHLMETLLPRHLQII YEINQRFNLNRVAAAYPGD VDLRRMSLVEEGAVKRINMAHLCIAGSHAVNG VARIHSEILKKTIFKDFYELEPHKFNKNTNGITPRRWLVMCNPGLAEVIAERI GEEYIADLDQLRKLLSYVDDEAFIRDVAKVKQENKLFSAYLERYKVHINP NSLFDIQVKRIHEYKRLNCLHVIITLYNRIKKEPNKFVVPRTVMIGGKAAP GYHMAKMI I KLITAGDVVNHD PVVGDRLRVIFLENYRVSLAEKVI PAADLS EQISTAGTEASGTGNMKFMLNGAL TIGTMDGANVEMAEAGEENFFIFGMR VEDVERLDQKGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPD LFKDIVNMLM HHRDFKVFADYEDYIKCQERVSA LYKNPREWTRMVRNIATSGKFS SDRITIA QYAREI WGEPSRQLPAPDEKI
112	Pig A0A286ZYX8 (Fructose- bisphosphate aldolase)	MPHQYPALTPQKKELSDIAHRIVAPGKGILAADESTGSI AKRLQSIGTENTE ENRRFYRQLLLTADDRVNPCIGGVILFPHETLYQKADGRFPQVIKAKGGVV GIKVDKGVVPLAGTNGETTQGLDGLSERCAQYKKGADFAKWR CVLKIG EHTPSSLAIMENANVLARYASICQQVGLQNGIVPIVEPEILPDGDHDLKRCQY VTEKVLAAVYKALSDHHIYLEGTL LKPNMVT PGHACTQKYSHEEIAMATVT ALRRTVPPAVPGITFLSGGGQSEEEASINLNAINKCPLLKPWALTFSGRALQA SALKAWGGKENLKAQEYVKRALANS LACQGYTPSGKAGAAASESLF ISNHAY
113	Pig A0A287BJF1 (Myosin light chain 1)	MAPKKDVKKPAAAAAPAPAPAPAPAPAPAPPPKEEKIDLSAIKSFSADQIAEFK EAFLLFDRTGECKITLSQVGDVLRALGTNPTNAEVKKVLGNPSNEEMNAKKI EFEQFLPMLQAISNNKDQGSYEDFVEGLRVFDKEGNGTVMGAELRHVLATL GEKMKEEEVEALMAGQEDSNGCINYEAFVKHIMS I
114	Pig G3P (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKGVNGFGRIGRLVTRAAFN SGKVDIVAINDPFIDLHYMVYMFQYDSTH GKPHGTVKAENGLVINGKAITIFQERDPANIKWDAGATYVVVESTGVFTT MEKAGAHLKGGAKRVIISAPSADAPMFVGMVNHEKYDNLKIVSNASCTTN CLAPLAKVIHDHFGIVEGLMTTVHAI TATQKTVDGPSGLWRDGRGAAQNI I PASTGAAKAVGKVIPELNGKLTGMAPRVPTPNVSVVDLTCRLEKPAKYDDI KKVVQKASEGPKGILGYTEDQVVSDFNSDTHSSTFDAGAGIALNDHFVK LISWYDNEFGYSNRVVDLMVHMASKE
115	Pig TPM1 (Tropomyosin alpha-1 chain)	MDAIKKMQMLKLDKENALDRAEQAEADKKAEDRSKRLEDELVSLQKK LKATEDELDKYSEAPKDAQEKLEAEKKATDAEADVASLNRRIQLV EEEEL RAQERLATALQKLEEAKEADESERGMKVI ESRAQDDEKMEIQEIQLKEA KHIAEDADRYEEVARKLVIIESDLERAEBRAELSEKCAELEEEELKTVINNL KSLAQAEKYSQEDKYEEIKVLSDKLKEAETRAEFAERSVTKLEKSIDDL EDELAYQKLKYKAISEELDHALNDMTSI

TABLE5-continued

Pig Proteins		
SEQ ID NO	Description	Amino Acid Sequence
116	Pig A0A5G2QB13 (Myosin binding protein C2)	MPEAKPATKKAPAGKDAAPKAPKEAPPKGAPEEPPKEAPPEDQSPATAEEPTG IFLKKPDSVSVENGKDALIVAKVNGKELPGKPEIKWFKGWLELGSKSGARF SFKESHDSASNVTLELRIGKVVLGDRGDYRLEVAKAKDFCDSCAFNIDVEAP RQDSAGQSLESFKRSGEGKPEGAGELDFSGLLKKRQVVEEEKKKKDDDDL GIPPEIWELLKGAKKSEYEKIAFQYGITDLRGLMLKRLKAKVEVKKSAFTK KLDPAYQVDQGNKIKLVVEISDPDLPLKWYKNGQEIKPSSKYVFENVGKKRI LCINKCTLADDAAYEVAVKDEKCFTELFVREPPVLIVTPLEDQQVYVGDRE MTVEVSEGAQVMWLKDGVELTREDSFKSRYRFPKDKGRHILYSDVSL RGRYQVMTNGGQCEAELSVEEKQLEVLQDIADLTVKASEQAVFKCEVSDEK VTGKWYKNGVEVRPSKRITISHVGRFHKLVIDDVRPEDEGDYTFVPDGYALS LSAKLNFLEIKVEYVPKQEPPIHLDCSGKTSENAVVVVAGNKLRLDVSITG EPPPVATWMKEDEVFSGTEGRVRLERQADISSFVIESAERGDEGRYTIKVNTN AGEDVASIFLRVVDVDPDPEAVRVTTVGEDWAILVWEPPKYDGGQPVTYL LERKKKGSQRWMLNFEVFTETTYESTNMIEGILYEMRVFAVNAIGVSQPSV NTKPFMPIAPTSEPQHLTVEDVDTTTTLKWRPPDRIGAGGIDGYLVEYCV GSEWIPANTEPVERCGFTVKNLPTGARITFRVVGNIAGRSQPATLAQPVTI REIVGEPPRQSSVLIRASPDSLVATLSVPLRRVPQPYAEPAPGIVSPEQPKIRL PRHLRQTYIRKVGHEHLNLVIPPQGKPRPVVWTKGGAPIDTSRVHVRTSNFD TVFVVRQAARSDSGEYELSVQIENMKDTATVHIRVVEKAGPPQNVVMVKEV WGTNALVWQPPKDDGNTEITGYFVQKADKKTMEWPNVVEHNRHTSCTV SDLIMGNEYFRVYSENI CGLSDSPGVSKNTARILKTGITFKLPEFKEHDFRTP PKFLTPLPDRVVVAGYTAALNCAVRGHPKPKVVMKNQMEIGEDPKFLMT NHQGVLTNLIRRPSPDSTYSCRASTSWGRWPWPSASWKSECRSEAPASTQG SWLLSPDPLPEPLSLSRVITRFLSRTO
117	Pig A0A287BSB6 (Uncharacterized protein)	TMQHVTKMSLVAVPFYQKRHRHFDQSYRNIQTRYLLDEYAAKKRASTQS SSQKSVTQESSQRESSSERASSQTLLGGTTCRLCAKRVSTALEEEQERKRR YHSLAAYGEAKRERYLSLAQLEEDVHLARTHARDKLDRLFALQHVDTA LRHAVGDRLLAWERHSFEDRISRAPETLVRLRSHTVWERMVSVLKCFYVQGFPT PVVQWYKDGSLICQAGEPGKYRIESKYGVHTLEINRADFDDTATYSAVATN VHGQVSTNAAVVRRFHDDEEPFHSVGLPIRLPLSSVIPYTHFDVQLEKFGV TFRREGETLTCLKTLVTPDLKRVQPRAEWYRDDVLLKESKWTMFFGEGQ ASLSFSHLNKDDGLYTLRIVSRGGISDHSALFVRDADPLVMGAPGAPMDV HCHDANRDYVIVTWKPNNTTKESPVLGYFIDRCEVGTDNWVQCNDAPVKIC KYPVSGLVEGRSYTFRVRAVNSAGVSRPRASEAVAALDPLDLKRLQGIHW EGDKDIVIYQDDLEGEVQIPRPPTNVHASEISRTYVVLSDWPTPRGKEPLMY FIEKSVVSGTQWQRENAQTAVRSPRYAVFDLAEKSYVFRVLANKHGLSD PSEITSPVIAQDITVPSAPGRVLASRNTKTSVVVQWDQPKHGEDLLGYVVD CCVAGSNVWEPCHNHKPIGYNRFFVHGLTTGEQYIFRVKAVNAVGTSENSQE SDVIKVAALTVPSHPYGITLLNCDGHSMTLGWKVPKFSGGSPVIGYVYDKR EGHHKNWHEVNSSPLKERILTVEGLTEGSTYEFKIAAPTWWGSGCRRIPVSS SSVRPGQRLSQVPTSGWVGTGFFLVLLSLYRRYLLGSGEGHGD
118	Pig ENOB (Beta-enolase)	MAMQKIFAREILDSRGNTVEVDLHTAKGRFRAAVPSGASTGIYEALERD DKSRYLGKGVLKAVEHINKTLGPALLEKKLSVVDQEKVDKFMIELDGTENK SKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPDVLVPVPAFNING GSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVIKGKYGKD NVGDEGGFAPNILENNEALELLKTAIQAGYDPKVVIGMDVAASEFYRNGK YLDLFKSPDDPSRHI TGEKLGELYKSFINKYPVVSIEDPFDQDDWKTWTSF GVNIQIVGDDLTVTNLKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQ SNGWGMVMSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMR IEEALGDKAVFAGRKFFNP
119	Pig Q5XLD2 (Myosin light chain)	MAPKKAKRRAAEGSSNVFSMFQDTQIQEFKEAFTVIDQNRDGIIDKEDLRD TFAMGRNLNVKNEELDAMMKEASGPINFTVFLTMFGEKLGADPEDVITGA FKVLDPEGKGTIKKHFLLEELLTQCDRFSQEEIKNMWAAFPDPVGGNVYK NICYVITHGDAKDQE
120	Pig MYH1 (Myosin-1)	MSSDQEMAI FGEAAPYLKSEKRIEAOQNPFDAKTSVFVAEPKESFVKGT QSREGKVTVKTEAGATLTVKEDQVFPMPNPPKFDKIEDMAMMTHLHEPAV LYNLKERYAAMMIYYSGLFCVTNPNYKWLVPVNAEVVTAIRGKKRQEAP PHIFISIDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTGEKK KEEPTSGKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFGKFIIRIHFGTTG KLASADIETYLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITNPYDY AFVSQGEITVPSIDQEEELMATDSAEILGFTSDERSVSIYKLTGAVMHYGNLK FKQKQREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVKVGNFVTK GQTVQVYNVAVGALAKAVYDKMFLWMVTRINQQLDTKQPRQYFIGVLDIA GFETFDNSLEQLCINFTNEKLQFFNHHMFVLEQEEYKKEGIEWEFIDFGMD LAACIELIEKPMGIFISILEEEMFPAKATDTSFKNKLYEQHLGKSNFQKPKPA KGKVEAHFSLIHYAGTVYNIITGWLDKNKDPLNETVVGLYQKSSVKTALFL FTGAGADAEGGGKGGKKKGSSFTVSALFRENLNKMLTNLRSTHHPH

TABLE5-continued

Pig Proteins		
SEQ ID NO Description	Amino Acid Sequence	
	VRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPSPRILYADFKORY KVLNASAIPEGQFIDSKKASEKLLGSIDIDHTQYKFGHTKVFFKAGLLGLLEE MRDEKLAQLITRTQARCRGFLARVEYQKMVERRESIFCIQYNIRAFMNVKH WPWMKLYFKIKPLLKSAETEKEMANMKEEFKTKESLAKAEAKRKELEEK MVALMQEKNLQLQVQAEADSLADAEERCQDLIKTKIQLEAKIKEVTERAE DEEINAELTAKKRKLEDECSSELKDDIDDLLETLAKVEKEKHATENKVNLT EEMAGLDEITAKLTKEKKALQEAHQQTLDLQAEEDKVNTLTAKTKLEQ QVDDLEGSLEQEKKLMDLERAKRKEGDLKLAQESTMDIENDKQQLDEK LKKKEFEMSNLQSKI EDEQALAMQLKKIKELQARIEEEEEIEAERASRAK AEKQRSDLSRELEESERLEEAGGATSAQIEMNKKREAEFQKMRDLLEATL QHEATAATLRKKHADSVAEELGEQIDNLQVRVKQKLEKEKSEMKEIDDLAS NMETVSKAKGNLEKMCRTLQDLSELKTEEEQQLINDLTAQARLQTES GEYSRQLDEKDTLVSQLSRGKQAFQTQIEELKRQLEEEIKAKSALAHAVQSS RHCDLLREQYEEQEAELQRAMSKANSEVAQWRTKYETDAIQRTTELE EAKKKLAQRLQDAEEHVEAVNAKASLEKTKQRLQNEVEDLMIDVERSNA ACAALDKKQRNFDKILAEWKQKYEETHAELEASQKESRSLSTELFKVNAY EESLDQLETLKRENKLNQOEISDLTEQIAEGGKRIHELEKIKKQVEQEKSEIQ AALEEAASLEHEEGKILIRIQLELNQVKSEVDRKIAEKDEEIDQLKRNHVRV VESMQSMLDAEIRSRNDAILKKKMEGDLNEMEIQLNHANRMAAEALRNY RNTQGIKDTQIHLDDALRSQEDLKEQLAMVERRANLLQAEIEELRATLEQT ERSRKVAEQELLDASERVQLLHTQNTSLINTKKKLETDISIQIGEMEDI IQEA RNAEEKAKKAITDAAMMAELKKEQDTSALHERMKNLEQTVKDLQHRL DEAEQLALKGKKQIQKLEARVRELEGEVESEQKRNVETVKGRLKHERRVK ELTYQTEEDRKNI LRLQDLVDKLQAKVKS YKRQAEAEQSNVNLSKFRKL QHELEEAERADIAESQVNKLVRKSEVHTKIISEE	
121 Pig F1SHX0 (Nebulin)	MADDEEYEEVVEYYTEETVYEEVPGETITEVYETTTTSTSGVEQSDTSQPAV AQPMPSPKPVERRKVIKKVDPSKFMPYIAHSQKMQDLFSNKKYKKNYEKA GKPYAITTDTPELRRIKKVQDQLSEVKYRIDGDVAKTICHVDEKAKDIEHA KKVSQQVSKVLQKQNWEDTKDKYLLPPDAPELVQAIKNTAMFSKKLYTED WEADKSLFYPYNDSPELRRVAQAQKALSDIAYKKGHAEQQTQFTSLPDPPI EFACKVTNQVSQKQYKEDYENKIKGKWS ETPCFEIAATARMNAKNI STRKYQ EDFENMKDQIYFMQETETPEYKVNKQAGTAASKVKYKEDYEKNKGKADYN VLPASENPLLRHLKAAGDALSDKRYKENYEKTKAKSINYCETPKFKLDTVL HNFSSDTKYKDSYLNILGHYVGSFEDPYHTCMKVAEENS DNKYAEYE EDRGKGFFPQITIQEYEAIKKLDQCKDHTYKVHPDKTKFTQVTDSPVLVQA QVNSKQLSDLNKAKHENEKFKCHIPPDPAPFIQHKVNAYNLSDNVYKHD WEKTKAKKFDIKVDAIPLLAAKANTKNTSDVMYKKDYKSKGMIGALSIN DDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQLDTLLKNFS EAKYKDSYVQNVGLGHYIGSFEDPYQLHCMKVSQNSDNKYAEYEEDKGK CYFPQITI QEYEAIKKLDQCKDHTYKVHPDKTKFTAVTDPVLLQAQLNTK QLSDLNKAKHEGEKFKCNI PADAPQFIQHRVNAYNLSDNVYKHDWEKTK AKKFDIKVDAIPLLAAKANTKNTSDVMYKKDYKSKGMIGALSINDDPKM LHSLKTAKNQSDREYRKDYKSKTIYTAPLDMLQVTQAKSKQAIASDVYK HLLHNYSYPPDSINVDLAKKAYALQSDVEYKADYNSWMKGC GWPFGSLE MEKAKRASDILNEKKYRQHPETLKFTSIEDAPIVQS KINQAQRSDIAYKAKG EEI IHKYSLPADLPFIQAKVNAYNISSEMYKADLKDLSKGYDLRTDAIP IKA AKAARQAASDVQYKDYKAKGKMVGFSQLQDDPKLVHYMNVAKIQSDR EYKDYKTKTYNTPHDMFNVVAAKQADVASNVNRYHSLHHYTYLPD AMDLELSKNMMHIQSDNAYKEDYNTWMKGI GWPFGSLEVEKVKKAGDA LNEKKYRQHPDTLKFTSIVDSPVMVQAKQNTQVSDILYKAKGEDVKHKYT MGPDLPQFLQAKCNAYNISDV CYKRDWHDLI AKGNVNLGDAIPIA AAKASR NIASDYKYKEAYEKAGKHVGFRSLQDDPKLVHYMNVAKLQSDREYKKN YENTKTSYHTPGDMVSI TAAKMAQDVATNVNYPQIHHYTYLPDALSL EHT RNVNQIQSDNVYKDEYNHFFKGIGWIPFGSLEVEKVKKAGDALNERKYRQH PDTIKFTSVPSMGVLAQHNTKQLSDLNKVEGEKLLKHXYTMDPDVPQFI QAKVNAINMSDAHYKADWKKTLAKGYDLPDAIP I VAAKSSRNIASDFK YK EAYEKAGKQVGFRLQDDPKLVHYMNVAKIQSDREYKKGYEASKTRYHT PLDMLSVTAAKKSQEVATNTNYKQPFHHYTLPLDALNVEHSRNMQIQSDN LYKSDFTNWMKGI GWPFGSLEVEKAKKAGEILSEKKYRQHPDKLFTYAM DTMEQALNKS NKLIMDKRLYTEKWNKDKTTHVMPDTPDILLSRVNIITLS DKLYKAGWEEKKGYDLRPDAIS IKAARASRDIASDYKYQAYEQAKGK HIGFRSLEDDPKLVHFMQVAKMQSDREYKAYEKS KTSFHTPVDMLSVVA AKKSQEVATNANYRNVIHTYNMPLDAMS FELAKNMMQLQSDNQYKADYA DFMKIGIWLPLGSLAEKKNKAMEI ISEKKYRQHPDTLKYSTLMDSMNMYL AKNNAKIMNEHLYKQAWADKTKVHIMPDI PQI I LAKANAVNMSDKLYKLS LEESKKKGYDLRTDAISIQA AAKASRDIASDYKYKHSYKERGKMVGFRSLED DPKLHSMQVAKMQSDREYKKS YEKTKTSYHTPADMLSVTAARDAQANIT NTNYKHLIHKYI LLPDAMNIQLSKNMMHIQSDNEYKQDYNECYKGLGWSPA GSLEVEKAKKATEYASDQKYRQHP SNFQFQKLNDSDMVLAKQNAHTMN KLYTADWNKD KTKI HVMPTDPI LQAKQNTLFSQKLYKLGWEEALKKG YDLPVDAISVQLAKVSRDNASDPKYKG YRKQLGHGIFGRNLQDDPKLVLS	

TABLE5-continued

Pig Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		MNVAKMQSEREYKKDFEKWTKFTSPVDMLGVVLAKKCCQALVSDVDYKN YHLQWTCPLPDQNDVIAKRSYELQSENLYKSDLEWLRGIGWSPLGSLEAEK NKRASEIISEKKYRQPPDRNKFTSIDAMDIVLAKTNAKNRSDRLYREAWDK DKTQVHIMPDPDIIILAKANLINSCLKYRMGYEELKKKGVDLPVDAIPIKA AKASREIASYKYKEGFRKQLGHHVGARNIEDDPKMMWSMHVAKIQSDRE YKKDFEKWTKFSSPVDMLGVVLAKKCCQTLVSDIDYKNYLQWTCPLDQS DVIHARRAYDLQSDNLYKSDLQWLKGIGWLPSSGSLEDEKNKRATQILSDHV YRQHPDKFKFSSLMDSIPMVLAKNNAITMNRRLYTEAWDKDKTTVHIMPDT PEVLLAKQNKINYSEKLYKLGLEAEKRKGYDMRLDAIPRAAKASRDIASEF KYKEGYRQQLGHHIGARAIHDDPKMMWSMHVAKVQSDREYKKDFEKWKT KFSSPVDMLGVVLAKKCCQTLVSDIDYKNYLQWTCPLDQSDVIHARRAYDL QSDNMYKSDLQWLRGIGWVPIGSLDVEKCKRATEILSDKIYRQPPDTFKFTS VPDSLEQVLAKNNAINMNRRLYTEAWDKDKTQVHIMPDPTEIMLARQNRL NYSETLYKLANEBAKKGYDLRSDAIPIVAASRDIVSDYKYKDGYSRQQL GHHIGARNIEDDPKMMWSMHVAKIQSDREYKKDFEKWTKFSSPVDMLGL VQAKKCCQILVSDVDYKNYLQWTCPLPDQNDVIHARQAYDLQSDNIYKSDL QWLKGIGWVPIGSDVVKCKRAAEILSDHIYRQPPDKLFTSVPDSLEQVLA KNNAINMNRRLYTEAWDKDKTQIHIMPDPTEIMLARQNKINYSESRYQAM EEAKKEGYDLRSDAIPIVAASRDIASDYKYKEAYRQQLGHHIGARAVHD DPKMMWSLHVAKMQSDREYKKDFEKYKTRFSSPVDMLGIVLAKKCCQTLVS DVDYKHPLEHWTCPLDQNDIIHARKAYDLQSDNLYKSDLEWMLGIGWIPLG SVEVVKAKRAGEILSDNIYRQRPDTLFTSVTDSPELVLAKNNALNMNRRL YTEAWDNKKTIHVPDPTEIMLAKLNRINYSDKLYRLALEESKKEGCDLR VDAIPIQAAKASREIASDYKYKEAYRQQLGHHIGARNIEDDPKMMWSIHAA KIQSDREYKKDFEKWTKRFPSPVDMLGLVQAKKCCQILVSDIDYKHPLEHWIC LPDQNDIIQARKAYDLQSDAIYKADLEWLRGIGWVPLGSVEVEKVKRAGEIL SERKYRQPADQLKFTCIITDPEIVLAKNNALTMSKHYLTAWDTDKTSIHV MPDTPTEILLAKSNSANISQKLYTKGWDESKMDYDLKADAI SIKSAKASRDI ASDYKYKEAYEKQKGHHIGAQSVEDDPKIMCAIHAGKIQSEREYKKDFQKW KTKFTSPVDMLGILLAKKCCQTLVSDIDYRNYLHHWTCPLDQNDIIQARKAYE LQSDALYKADLEWLRGIGWMPQGSPEVLRVKNAQEILRDSVYRTPVVKLK YTSIVDTPEVVLAKSNAENISIPKYREVWDKDKTSVHIMPDPTEINLARANAL NVSNKIYREGWDEMMSCDVRLDAIPIQAAKASREIASDYKYKLDHEKQKG HYVGTLTARDNKRIRWAL IAGKIQNEREYRLHWAKWSKFPQSPADMLSITH SKNCQALVSDIDYRNYLQWTCPLDQHDVIAQARKAYDLQSDAIYKADLEW LRGIGWMPSDSVSVNHAKHAADIFSEKKYRTKIETLNFTPVDDRVDYVTAK QSGETLNDIKYRKDNDTKSKYTLTETPQLHAAQEAARILDQYLYKESWEK QKATGYILPPDAVPFVHAHSSDVQSELKYKADHVKKQGHYVGVPTMRDD PKLVVFEHAGQIQNDRLYKEHYHKTAKAINIPDMVSVVAAKEGQILVSDID YRNYLQWTCPLPDQNDVIAQARKAYDLQSDNLYKADLEWLRGIGWIPLDSV DHRVTRNQEMVNIQIKYKDALANYPNFTSVVDPPAVVLAKINSVNSDV KYKETFNKLLKGKYIFSPDTPYITHSKDMGKLYSTILYKGAWEGTKAYGYTL DERYIPIVGAHADFVNSSELKYKETYEKQKGHYLAGKEISEFPNVVHCLDFQ KMRSVLNYRRHYEDTKANVHIPNDMMNHVLAKRCQYILSDLEYRHYFHQ WTSLEPEPNVIRARNAQEILSDNVYKDDLNLWLGIGCYVWDTPQILHAKKS YDLQSQILYIADGKENLKNYNMVTDTPLYVTALQSGINASEVKYKENYHQI KDKYTTVLETVDYDRTRNLKDLYSNVYKEAWDKVKATSYILPSNTVSLTH AKNQKHLASHIKYREEYEKFKALYTLPKSVDDDPNTARCLRVGKFNIDRLY RSVYEKNKMKIHMVPDMVEMVTAKDSQKKVSEIDYRLHLHEWICHPLDQV NSHVRKVTDQISDIVYKDDLNLWLGIGCYVWDTPTEILHAKHAYDLRNDIKY QAHVKKTRNNYKLVTDTPLYVQAVKSGKQLSDAVYHYDVVHVSVRGKVAP TTKTVDLDRALHAYKLQSENLYRKDGLNSLPTGYRLPGDTPHFKHNKDIRY MSYFYKYKEVYEHFKANGYTLGPKDVPFVHVRVNNVTSERLYRQLYHKL KDKIHTTDPDTPETIRQVKKTQEAVALIYKSDFFKMQGHMISMPYTPHVLHCR YVGDITSIDIKYKEDLQILRGLGCFLYDTPDMVRSRHLRKLWSNYLYTDNAR KMRDKYKVVLDTPYRKVQELKTHLSELVYRAAGKKQKSIFTSPVDPDPL RAKRGQKLQSQYLYVELATKERPHHAGNQTTALKHARDVKDMVSEKKY KVQYKMKDKYTPVPDTPILIRAKKAYWNASDLRYKETFOKTKGKYHTVK DALDIYVHRKVTDHI SKI KYKENYMSQLGIWRSIPDRPEHFFHRAVTDVSD VKYKEDLTWLKGIGCYAYDTPDFTLAEKNKTLYSKYKYKEVFERTKSDPKY EADCPINRHPKFATQLMNEKKYRADYEQRDKYQLVDEPRHLLAKIAGD QISQIKYRKNYEKTKDKFTSVVDTPPEHLRTTKVNKQISDILYKLEYNAKPR GYTTIHDTPMLLHVRVKDEVSDLKYKEVYQRTKSNCTIEPDVAVHIKAAKD AYKVNTNLDYKQYEAANKAHWRWTPDRPDPFIAAKSSLQSQSDFEYKLDRE FLGCKLSVTDDKDMVLALRNSVIESDLKYKEKHVKERGTCHAVPDTPQIL LAKTVSNLVSENKYKRHVKKHLAQGSYTTLPETRTDVHVKEVTKHVSNTN YKKKFVKERKSNYSIMLEPPEVKHAMEVAKQSDVAYKKDAKENLHYTT VADRPDIKKATQAAKQASEVEYRAKHRKEGSHGLSMLGRPDIEMAKKAAT LSSQVKYRENFNKEGKTPKYNPKESQLYKIMKDANNLASEVKYKADLKK LHKPVTDMKESLIMNHVLNTHLASSYQYKKNYEKSKGHYHTIPDNLEQLH LKEATELQSIYKYKEKYEKERGKPLDFETPTYITAKESQQMQSGKEYRKD YEESI KGRNLTGLEVTALLHVKYATKIASKEYRKDLSEIRGKGLSEMEDT

TABLE5-continued

Pig Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		PDMLRAKNATQILNEKEYKRDLELEVKGRGLNAMANETPDFMRARNATDI ASQIKYKQLAEMEKANFTSVVDTPETIHAQQVKNLSSQKKYKEDAECMSY YETVLDTPEMQRVRENQKNFSLQYQCDLKNKSGKITVVQDTPETILRVKEN QKNFSSVLYKEDVSPGTAIGKTPEMMRVKQTQDHISVVKYKETIGQGTPIPD LPEVVRKVKETQKHISVVMYKENLGTGIPVTPETIERVKRNQENFSSVLYKEN LGKGTPTPIPEMERVKRNQENFSSVLYKENMGKGTPLPVTPEMERVKHNQ ENISSILYKENVGKGTPTPTPEMQRVKRNQENISSVIYKENLGKATPTPTPTPE MERVKRNQENFSSVLYKENMRKATPTPTPEMERAKRNQENISSVLYSDSF RKQIQGKAAVLDTPEMRRVRETQRHISTVKYHEDFEKHKGCTPVVTDPTIT ERVKNTQDFSDINRYGIQRKVVEMEQKRNQDQETITGLRVWRTNPGSVF DYDPAEDNIQSRSLHMLNAQAQRRSREQRSASALSISGGEEKSEHSEARL STYSDGGVFFSATSAAKXKAKTTELPPQQRSSSVATQQTTVSSIPSHPSTAGKI FRAMYDYMAADADEVSFRDGAIVNVQAIDEGWMYGTVQRTGRTGMLPA NYVEAI
122	Pig MYH4 (Myosin-4)	MSSDQEMAI FGEAAPYLKSEKIERIAQNKPFAKTSVFVAEPKESFVKGTV QSREGGKVTVKTEAGATLTVKEDQVFPMPNPPKFDKIEDMAMMTHLHEPAV LYNLKERYAAWMIYYSGLFCVTVNPKWLPVYNAEVVTAAYRGKKRQEAEP PHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTGEKK KEEPTPGMQGTLEDQII SANPLLEAFGNAKTVRNDNSSRFGKFI RIHFGTTG KLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITNPNYDY AFVSQGEITVPSIDQEEELMATDSAI EILGFTSDERSVSIYKLTGAVMHYGNLK FKQKQREEQAEPDGT EVADKAAYLQGLNSADLLKALCYPRVKVGNFVTK GQTVQQVYNAV GALAKAVYDKMFLWMVTRINQQLDTKQPRQYFIGVLDIA GFETIDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIEWEFIDFGMD LAACIELIEKPMGIFSILEEECMFPKATDTSFKNKLYEQHLGKSNFQKPKPA KGKAAEHFSLIHYAGTVDYNITGWLDKNKDLNETVVGLYQKSSVKTLAFL FAERQSSEEGGTTKGGKKKGSSFTVSALFRENLNKLMNTNLRSTHPHFVRCI IPNETKTPGAMEHELVLHLQRCNGVLEGIRICRKGFPFRIYADFKQRYKVLN ASAIPEGQFIDSKKASEKLLGSDIDHTQYKFGHTKVFFKAGLLGTLEEMRDE KLAQLITRTQAMCRGFLMRVEFRKMMERRESIFCIQYNIRAFMNVKHPW MKLYFKIKP LLLKSAETEKEMANMKEEFEKTKEDLAKSEAKRKELEEKVAL MQEKNDLQLQVQAEADGLADAEERCDQLIKTKIQLEAKI KEVTERAEDEEEI NAELTAKRKLDEECSELKKDIDDLTLAKVEKEKHATENKVKNLTEEMA GLDENIAKLTKKALQEAHQQLDLDQAEEDEKVNLTAKAKTLEQQVDD LEGSLEQEKLRMDLERAKRKLGLDLKLAQESTMDIENDKQQLDEKLKKKE FEMSNLQSKIEDEQALAMQLQKKIKELQARTEELEEEIEAERASRAKAEQR SDLSRELEESERLEEAGGATSQIEMNKKRAEFQKMRDL EEATLQHEAT AAALRKKHADSVAELEGQIDNLQRVKQKLEKEKSELKMEIDDLASNMETVS KAKGNLEKMCRTL EDQLSEVTKKEEHHQRLINELSAQKARLQTESGEFSRQL DEKEALVSQLSRGKAFTQQIEELKRQLEETKAKSALAHAVQSSRHDCDLL REQVEEQEAKAELQRAMSKANSEVAQWRTKYETDAIQRTTEELEAAKKL AQRLQDAEEHVEAVNAKASLEKTKQRLQNEVEDLMLDVERSNAACAALD KKQRNFDKILAEWKHYEETQAELEASQKESRSLSTELFKVKNAYEESLDQ LETLKRENKNLQQEISDLTEQIAEGGKHIEHELEKVKKQIEQKSELQAAL EEA EASLEHEEGKILRIQLELNQVKSEIDRKIAEKDEEIDQMKRNHIRVVEISMOST LDAEIRSRNDALRIKKMEGDLNEMETQLNHANRQATEAIRNLNRTQGVLK DTQLHLDDAIRGQDDLKEQLAMVERRANLQAEIEELRASLEQTERSRVA EQELLDASERVQLLHTQNTSLINTKKKLETDISQIQGEMEDIVQEARNAEKA KKAITDAAMMAEELKKEQDTSALHERMKKNMEQTVKDLQHRLEDEAEQLA LKGGKKQIQKLEARVRELENEVENEQKRNVEAVKGLRKHERRVKELTYQT EEDRKNVLRQLDVLQSKVKAYKQAEAEAEQSNVNL SKFRKLQHELEEE ABERADIAESQVKNLVRKSVREVHTKVIS EE
123	Pig ALBU (Albumin)	MKWVTFISLLFLFSSAYSRGVFRDRTYKSEIAHRFKDLGEQYFKGLVLIAFSQ HLQQCYPYEEHVKLVRVETFAKTCVADES AENCDSIHTLFGDKLCAIPSLR EHYGDLADCCEKEEPERNECFLQHKNDNPDIPKLKPDVVALCADFQDEQK FWGKYLIEIARRHPYFYAPELLYYAI IYKDVFSECCQAADKAACLLPKIEHL REKVLTSAAKQRLKASIQKFGERAFKAWSLARLSQRFPKADFTESKIVTDL AKVHKECCGHDLLECADDRADLAKYICENQDTISTKLKECCDKPLLEKSHCI AEAKRDEL PADLNPLEHDFVEDKEVCNRYKEAKHVFLGTFLYEYSRRHPDY SVSLLRIAKIYEATLEDCAKEDPPACYATVPDKFQPLVDEPKNLIKQNCEL FEKLGEYGFQNALIVRYTKKVPQVSTPTLVEVARKLGLVGSRCCKRPEEERL SCAEDYLSVLNRLCVLHEKTPVSEKVTCCTESLVNRRPCFSALTPDETYK PKEFVEGTFTFHADLCTLPEDEKQIKKQTALVELLKHKPHATEEQRLRTVLGN FAAFVQKCCAAPDHEACFAVEGKFKVIEIRGILA

[0047] An animal protein can comprise a bison protein (Table 6). A bison protein can comprise a bison muscle protein. A bison protein can comprise a bison blood protein. A bison protein can comprise a bison albumin protein. A bison protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 124-138.

[0048] A food product may comprise a fragment or a truncation of a bison protein. The food product may include

a fragment or truncation of any one of SEQ ID Nos: 124-138. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 124-138. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 124-138. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 124-138. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 124-138.

TABLE 6

Bison Proteins		
SEQ ID NO	Description	Amino Acid Sequence
124	Bison A0A6P3IIV2 (Creatine kinase)	MPFGNTHNKHKNFKAEEEEYPDLSKHNNHMAKALTLEIYKKLRDKETPSG FTLDDVIQTGVNDNPGHPFIMTVGCVAGDEESYTVFKDLFDPPIIQDRHGFGFKP TDKHKTDLNHENLKGDDLDPNYVLSRVRTGRSISKYALPPHCSRGERR AVEKLSVEALNSLTGEFGKYYPLKSMTEQEQQQLIDDHFLFDKPVSPLLL ASGMARDWPDARGIWHNDNKSFLVWVNEEDHLRVISMCKGNMKEVFR RFCVGLQKIEEIFFKAGHPFMWNEHLGYVLTCPSNLGTGLRGVHVHKLAAH LSKHPKFEEILTRLRLQKRGTTGGVDTAAVGSVDFVSNADRLGSSEVEQVQL VVDGVKLMVEMEKLEKQSIDDMIPAQK
125	Bison A0A6P3J4W6 (Alpha-actinin-2)	MNQIEPGVQYNYVYEDDEYMIQEEEDRDLDDPAWEKQQRKTFTAWCN SHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKPDGKMRPHKIANVN KALDYIASKGVKLVSIGAEIIVDGNVMTLGMWTII LRFAIQDISVEETSAK EGLLLWCQRKTAPYRNVNIQNFTSWKDGGLCALIHRHRPDLIDYSKLNK DDPIGNINLAMEIAEKHLDI PKMLDAEDIVNTPKPDRAIMTYVSCFYHAPA GABQAETAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPWLENRT PEKTMQAMQKKLEDFRDYRRKHKPKPVQEKCCLEINFNTLQTKLRISNRPA FMPSEGMVSDIAGAWQRLEQAEGYEEWLLNEIRRLERVEHLAEKFRQK ASTHETWAYGKEQILLQKDYESSLTLEVRALLRKHEAFESDLAAHQDRVE QIAAIAQELNELDYHDAVNVDRCQKICDQWDRGLTTLTQKRREALERTEK LLETIDQLHLEFAKRAAPFNWMEGAMEDLQDMFIVHSIEEIQSLITAHEQF KATLPEADGERQSLAIQNEVEKVIQSYISIRISSNPYSTVTVDIEIRSKWDKV KQLVPIRDQSLQEELARQHANERLRRQFAAQANAIGPWIQNKMEETARSSIQ ITGALEDQMNQLKQYEHNIINYKNIDKLEGDHQLIQEALVFDNKHTNYTM EHIRVGWELLTTIARTINEVETQILTRDAKGI TQEQMNEFRASFNHFDRRK NGLMDHEDFRACLISMGYDLGEAEFARIMTLVDPNQGGTQVTFQSFIDFMTIR ETADTDTAEQVIASFRIASDKPYILAEEELRRELPPDQAQYCIKRMPAYSGP GSVPGALDYTAFSSALYGESDL
126	Bison A0A6P3HAI4 (Alpha-1,4 glucan phosphorylase)	MSRPLTDQEKRKQISVRGLAGVENVTTELKKNFNRLHFTLVKDRNVATPR DYYFALAYTVRDHLVGRWIRTQQHYEKPRLYYLSLEFYIGRTLQNTM VNLALENACDEATYQLGLDMEELEEIEEDAGLNGGLGRLAACFLDSMAT LGLAAYGYGIRYEFQIFNQKISGGWQMEEDDWLRVGNPWEKARPEFTLP VHFYGRVEHTSQGAKWVDVTQVVLAMPYDTPVPGYRNNVNMTMLWSAK APNDFNLKDFNVGGYIQAVALDRNLAENI SRVLYPNDNFFEGKELRLKQEYF VVAATLQDIIIRRFKSKFGCLDPVRTNFDAPDKVAIQNLNDTHPSLAIPELM RI LVDQERLEWEKAWVTVKTCAYTNHTVLPEALERWPVHLIETLLPRHLQ IIYEINQRFNLNRVAAAFPGDVDRLLRMSLVEEGAVKRINMAHLCIAGSHAV NGVARIHSEILKKTIFKDFYELEPHKFQNKNTNGITPRRWLMVNCNPLAETIAE RIGEEYIADLDQLRKLLSYVDDSFIRDVAKVKQENKLFSAYLEKEYKVHI NPNSLFDIQVKRIHEYKQRLNCLHVIITLYNRICKPEPNKFFVPRVTVMIGGKA APGYHMAKMIIKLITAGDVVNHDVVGDRLLRVIFLENYRVSLAEKVI PAA DLSEQISTAGTEASGTGNMFKMLNGALTIGTMDGANVEMAEAEAGEENFFIF GMRVEDVERLDQKGYNAQEYDRIPELRHI IDQLSSGFFSPKQPDLFKDIVN MLMHHDRFKVFADYEEYVKQERVSYLYKNPREWTRMVI RNATSGKFPSS DRTIAQYAREIWGVEPTRQRPAPDEKI
127	Bison A0A6P3H8G9 (Fructose- bisphosphate aldolase)	MPHQYPALTPEQKELCDIAHRIVAPGKGLAADESTGSI AKRLQSIGTENTE ENRRFYRQLLLLTADRVNPCI GGVILFHETLYQKADGRFPQVIKAKGGV VGIKVDKGVVPLAGTNGETTQGLDGLSERCAQYKKGADFAKWCVLKI GEHTPSSLAIMENANVLARYASICQQNGIPIVPEBELPDGDHDLKRCQVVT EKVLAAYKALSDHHIYLEGTLKPNMVT PGHACTQKYSHEEIAMATVTA LRRTPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS ALKAWGGKKENLKAAQEYVKRALANSLACQGYTPSGKAGAAASESLFI SNHAY

TABLE6-continued

Bison Proteins		
SEQ ID NO	Description	Amino Acid Sequence
128	Bison A0A6P3J7N1 (Myosin light chain 1/3)	MAPKKDVKNPAAAAAPAPAPAPAPAPPKEEKIDLSAIKIEFSKQQQDE FKEAFLLFDRTGECKITLSQVGDVLRALGTNPNTAEVKKVLGNPSNEEMNA KKIEFEQFLPMLQAI SNNKDQGT YEDFVEGLRVFDKEGNGTVMGAELRHV LATLGEKMKEEVEALMAGQEDSNGCINYEAFVKHIMSN
129	Bison A0A6P3J4R7 (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKVGVNGFGHIGRLVTRAAFN SGKVDVIAINDPFIDLHYIVVMFYDTH GKFHGT VKAENGKLIINGKAVTIFQERDPASIKWGDAGAEYVVESTGVFTT MEKAGAHKGGAKSVIISAPSADAPMFVLGMNHEKYNNNLKIVSNASCTT NCLAPXAKVIHDHFGIVEGLMTTFHAITVTQKTV DGP SGKLWHDGGGAAQ NIIPASTGA AKAVGKVIPELNRKLTGMACVPTPNVSVMDLTYHLEKPAKY DEIKVVKQTSSEGPL EGI LGYTQDQVVS CDFNSDSYXSTFDAGAGIALNDH FVKLSWYDNEFGYSNRVVDLMVHMASKE
130	Bison A0A6P3GY04 (Tropomyosin alpha-1 chain isoform X2)	MDAIKKMQMLKLDKENALDRAEQAEADKXVPVPHGHSQLEDELVS LQKK LKATEDELDKYSEALKDAQEKL ELAEKKATDAEADVASLNRRIQLVVEEL DRAQERLATALQKLEEA EKADES ERGMKVI ESRAQKDEEKMEIQEIQLKE AKHIAEDADRKYEEVARKLVIIESDLERAEEAEELSEGKCAELEELKTVTN NLKSLEAQAEKYSQKEDKYEEI KVLSDKLKEAETRAEFAERSVTKLEKSID DLEDELYAQKLKYKAISEELDHALNDMTSM
131	Bison A0A6P3IB01 (Myosin-binding protein C, fast type)	MPEAKPAAKKAPAGKDAAAKPAPKEATPQEAAPAPTGEIDGAPAGRAEP GGSSAASRVVPPRPLLRSVFPTAPGLTALPYSQPCRAPQRFQGGGLTVPLPP HSPLLPGKDTVIVAKLNGKELPAKPAVKWFKGWLELGSKSGARFSFKESH DAASNVTYIELHITKVVLGDRGDYRIEVKAKDFCDSCAFNIDVEAPRQDSA GQSLESFPRSGEAKSDTAGELDFSGLLKKRQVVEEEKKKKDDDLGIPPEI WELLGAKKSEYERIAFYGI TDLRGMLKRLKKAKVEVKKSAFTKKLDP AYQVDRGNKIKLVEIISDPDLPLKWFKNQGEIKPSSKYVFENVGKKRI LTIN KCTLADDAAYEVVVKDEKCFTEL FVKEPPVLIVTPLEDQQV FVGDRVEMA VEVSEGAQVMWLKDGVELTREDSPKARYRLKKDGKRHIL IYSEVTMEDK GHYQVMTNGGQCEAELIVEEKQLEVLQDIADLT VKASEQAVFKCEVSDEK VTGKWKYKNGVEVRPSKRITISHTGRFHKLVIDDVRPEDEGDTFVVPDGYAL SLSAKLFL EIKVEYVPKQEPPIHLDCSGQTS ENAIVV VAGNKLMDVSI T GEPRPVATWMKEDEVFTGT EGRVHIEQRGDISSFVIESAERGDEGRYTIKVT NPVGEDVASILLKVVDPDPPEAVRVTSVGEDWAVLVWEPPKYDGGRPVT GYLLERKKKGSQRWMLNFEVFTTETTYESTNMI EGILYEMRVFAVNVIGVS QPSMNTKPFMPIAPTSEPHRLTVEDVTD TTTTLKWRPPDKIGAGGIDGYLIE YCVESGSEWIPANTEPVERCGFTVKNLPTGAKITFRVVG VNIAGRSQPATL AQPVTIREIVEQPKIRLPRHLRQTYVRKVGEHLNLVIPQGGKPRPQVVTGK GAPIDTSHVHVRTSDFTVFFVRQAARSDSGEYELTVQIENMKD TATVCIR VVEKAGPPQNVVMVKEVWG TNALVEWQPPKDDGNSEITGYFVQKADKKT MEWFTVYERNRHTSCTVSDLMGNEYFRVYSENVCGLSDLPGVSKNTARI VKTGMTLKLPEYKEHDFRTPPKFLTPLPDRVVVAGYAAALNC AVRGHKPK KVVMKNKMEIREDPKFLMTNQGGVLTNLNRRSPFDSGTYS CRAVNELG EALAECKLEVRVPQ
132	Bison A0A6P3GV86 (Myomesin-2)	MSLVAVPFYQKRHKHFDQSYRNIQTRYLLDEYSAKKKRASAQSS LQSVT RKSSSQRESSRAALGESTLG ETTLG ETS LGD T T LGR T T CRLCAKRMSSALEE EAQERKRRYQSMVAAYGEILVRLRSHVTWERM SVKLCFTVQGFP TPVVO WYKDGLSLICQAGEPGKYRIDSKYGVHTLEINRADPDDTATYS AVATNVHG QVSTNAAVVRRFRGDEEPFHSVGLPIGLPLSSVIPTHFVQFIEKFGVTFR REGETLT LKCTLLVTPDLKRVQPRAEWYRDDVLLKESKWTMFFGEGQAS LSFSLHMKDDEGLYTLRIVSRGGITDHSAPLFVRDADPLVTGAPGAPMDVH CHDANRDYVIVTWKPPNTTKESPVIGYFIDRCEVGTDNW IQCNDAPVKICK YPVTGLFEGRSYIFRMRAVNSAGISRPSRVSEAVAALDPVDLRR LQAVHLE GEKDVIYQEELEGEVQIPGPPTNVHAS ETSRTYVVLSDWPPPEPRGKEPLMY FIEKSMVSGSWGQRVNAQTAVRSPRYAVFDLAE GKPYVFRVLSANKHGLS DPSEITAPIQAQDTIVVPSAPGRVLASRNTRTSVVQWDRPKHEEDLLGY VDCSVAGSNVGEPCNHKPIGYNRFVVHGLTTGEQYIFRVKAVNAVGTSENS QESDVIKVQAALTVP SHPYGITLLNCDGHSMILGWKVPKFSGGSPI LGYV DKREAHKKNWHEVNSSPLKERILTVEGLTEGSLYEFKIAAANMAGIQGPSD PSELFKCEAWTMPEPGPAYDLTFCEVRDTS LVLWKPPVYSGSSPVSGYFV DYKEEDSGEWLTVHETATPHRYLKVCDLHQGKTYVFRVRAVNASGVGRP SDTSEPVLEARP GTKEVSAGVDEEGNVYLSFDCPEVTDASHFTWCKSYED IADDDRPFKVETAGDHSKLYFKNLDKEDLGTYSVSVSDTDGVSSSFVLDEE LERLKALSNEIKNPTIPLKSELAYEIFDKGQVRFWLQAEHLSPDSSYRPIIND REVADSETHRIKCDKSTGIIEMVMDRFTIDNEGTYTVQIQDGKAKSQSSLVL IGDAFKAVLKEAEFQRKEFLRKQGPFAEYLLHWDVTECEVRLVCKVANT KKETVPKWLKDDVLYETEKMPDLEKGVCELLIPKLSKKDHGEYKATLKDD RGQDVSTLEIAGKVYEDMILAMSRVCGASASPLKILCTPEGIRLQCFMKYFT EEMKVNWYHKDAKITSSEHMRIGGSEQMAWLQICEPTEKDKGKYTFEILD DKKNHQRS LDSLGAQFDEAFAEFQQLKAAFAEKNRGKVIIGGLPDVVTIM EGKTLNLTCTVFGNPDPEVWVFNKDKDIELSDHFSVKVEQAKVVSMTIKG

TABLE6-continued

Bison Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		VTSEDSGKYSIHVKNKYGGEKIDVTVSVYKHGEEIPDVVLPQQAKPKLLPA AAPSPA
133	Bison A0A6P3HSX8 (Phosphopyruvate hydratase)	MRTLAMAMQKIFAREILDSRGNPTVEVDLHTAKGRFRAAVPSGASTGIYE ALELRDGDKSRYLKGVKAVEHINKTLGPALLEKKLSVVDQEKVDKPMI ELDGTENKSKFGANAILGVSLAVCKAGAAEKGVPLVRHIADLAGNPELILP VPAPFNVINGGSHAGNKLAMQEFMILPVGASSFREAMRIGAEVYHHLKGVK AKYGDATNVGDEGGFAPNILENNEALELLKTAIQAAAGYDPKVVIGMDVA ASEFYRNGKYDLDPKSPDDPARHISGEKLGEYKNFIKNYPVVSIEDPFDQD DWTWTSFSLSGVNIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSV TESIQACKLAQSNVGWVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRS ERLAKYNQLMRIEALGDKAVFAGRKFRNPKAK
134	Bison A0A6P3H0Z6 (Myosin regulatory light chain 2)	MAPKKAARRAAEGSSSVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKEDL RDTFAAMGRNLNVKNEELDAMMKEASGPINFVTLNMFGEKLKGADPEDVI TGAFKVLDPGEGKGTIKQFLEELLTTQCDRFSHEEIKNMWAAPPDPVGGNV DYKNICYVITHGDAKDQE
135	Bison A0A6P3HQ42 (Myosin-1)	MSSDQEMAI FGEAAPYLKSEKERIEAQNKPDAKTSVFVADPKESFVKAT VQSREGGKVTAKTEAGATVTVKEDQVFPMPNPKFDKIEDMAMMTHLHEP AVLYNLKERYAANMITYTSGLFCVTVNPKYKWLVPVYNAEVVTAIRGKKRQ EAPPHIFSI SDNAYQFMLTDRENQSILITGESGAGKTVNTRKVIQYFATIAVT GEKKKEEPTSGKMQGTLEDQIIISANPLLEAFGNAKTVRNDNSSRFGKFIIRIH FGTTGKLASADIEITYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLIT TNPYDAYVVSQGEITVPSIDQEEELMATDSAEILGFTSDERVS IYKLTGAV MHYGNLKFQKQREEQAEPDGTEVADKAAYLQGLNSADLLKALCYPRVK VGNEFVTKGQTVQVYNAV GALAKAVYDKMFLWMVARINQQLDTKQPR QYFIGVLDIAGFEIFDFNSLEQLCINFTEKLLQQFFNHMHMFVLEQBEYKKEGI EWEFIDFGMDLAAICELIEKPMGIFSI LEEECMFPKATDMSFKNLYEQHLG KSNMFQKPKPAKGAEAHFSLIHYAGTVVDYNTGWLDKNKDPNLNETVVGL YQKSSVKTLLALLFSGPASGEAEGGPKKGGKKKGSSFTVSALFRENLNKLM TNLRSTHPPHVRCCIIPNETKTPGAMEHELVLHQLR CNGVLEGIRICRKGFP SR ILYADFKQRYKVLNASAIPEGQFIDSKKASEKLLASIDVDHTQYKFGHTKVF FKAGLLGLLEMRDEKLAQLITRTQARCRGFLARVEYQKMERRESIFCIQ YNVRAFMMNVKHPWMMKLYFKIKPLLKSAETE KEMANMKEEFEKTEELA KSEAKRKELEEKMVTLTQEKNDLQLQVQSEADALADAEEERCQDLIKTKIQL EAKIKVTERAEDEEINAEILTAKKRKLEDECESELKKDIDDLLETLAKVEKE KHATENKVKNL TEEMAGLDETI AKLTKEKKALQEAHQQTLDLDAQEEDK VNTLTAKTKLEQQVDDLEGSLEQEKLRMDLERAKRKL EGDLLKLAQEST MDIENDKQQLDEKLKKKEFEMS NLQSKI EDEQALAMQLQKKIKELQARIEE LEEEIEAERASRAKAEQSRDLSRELEEISERLEEAGGATSAQIEMNKKREA EFQKMRDLLEBATLQHEATAAALRKKHADSV AELGEQIDNLQRVQKQLEK EKSEMKMEIDDLASNMETVSKAGNLEKMCRALEDQLSELKTKEDQQRLL INDLTTQARLQTESGEFSRQLDEKDALVSQLSRGKQAFQTQIIEELKRQLEE EIKAKSALAHALQ SARHDCDLLREQYEEEQEGKAE LQRAMSKANSEVAQ WRTKYETDAIQRTTELEEAKKLAQRLQDAEEHVEAVNAKCASELEKTKQR LQNEVEDLMIDVERTNAACAALDKQRNFDKILSEWKQKYETHAELEAS QKESRSLSTELFKIKNAYEESLDQLETLKRENKNLQQEI SDLTEQIAEGGKRI HELEKVKQVEQEKSEIQAALEEAASLEHEEGKILRIQLELNQVKSEIDRKI AEKDEEIDQLKRNHIRVVEQMSTLDAEIRSRNDAIRLKKKMEGDLNEME QLNHANRMAAEALKNYRNTQAILKDTQIHLDDALRGQEDLKEQLAMVER RANLLQAEIEELRATLEQTERRKIAEQELLDSERVQLLHTQNTSLINTKK KLETDITQIQGEMEDI IQEARNABEAKKAITDAAMMAEELKKEQD TSAHL ERMKNLEQTVKDLQHRLDEAEQALQKGGKKQIKQKLEARVRELEGEVESE QKRNV EAVKGLRKHERRVKELTYQTEEDRKNILRLQDLVDKLQAKVKS Y KRQAEAEAEQSNVNLSKFRKLQHELEAEERADIAESQVNKL RVK SREVHT KIISEE
136	Bison A0A6P3H861 (Nebulin isoform X11)	MADDEEYEEVVEYYTEETVYKEVPGETITGFYETTTTRTSDYEQSETSRPAL AQPVPEKPVERRKRVIRKKVDPSKFMPPIAHSQKMQLNLFSTNKYKENYEK AKGKPYAITTDTPELRRIKKVQDQLSEVKYRVDGDAKTICHVDEKADIE HAKKVSQQVSKVLYKQNWEDTKDKYLLPPDAPELVQAIKNTAMFSKKLY TEDWEADKTMFYPYNDSPELRRVAQAQKALSDIAYKKGLAEQQQTQFTSLP DPPDI EFAKVTNQQVSKQYKEDYENKVKGWSETPCFEATARMNSNNIS TKKYQEDFEHMKDQIYFMQTETPEYKVNKQAGVAASKVKYQDYEKNK GKADYNVLPASENPLRLQLKAAGDALSDKLYKENYEKTKAKSINYCETPK FKLDTVLHNFSSDTKYDSYLKDLGHYVGSYEDPYHTHCMRVSAQNSDK NYKAEYEEDRGKGFFPQTITQEYEAIKKLDQCKDHTYKVHPDKTKFTQVT DSPVLMAQAVNSQLSDLNKAKHENEKFKCHIPDTPAFIQHKL NAYNLS DNIYKHDWEKTAKKFDIKVDAIPLLAAKANTKIASDMYKDKYEKSGKG MIGALSINDDPKMLHSLKTAKNQSDRLYKENYEKTKAKSMNYCETPKYQL

TABLE6-continued

Bison Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		DTLLKNFSEAKYKDSYVQNVLGHYIGSFEDPYQVHCLKISAQNSDKNYKA EYEEDKGKCYFPQITITQEYEAIKKLDQCKDHTYKVHPDKTKFTAVTDTPLV LQAQLNTKQLSDLNKAKHEGEKFKCHIPADAPQFIQHRVNAYNLSDNIYK HDWEKTKAKKFDIKVDAIPLLAAKANTKIASDVMYKKDYKSKGMIGAL SINDDPKMLHSLKTAKNQSDHEYRKDYKSKTIYTAPLDMPLTHAKKSQ AIASDVDYKHLHNYSPDPSVNVDLAKKAYALQSDVEYKADYNSWMKG CGWMPFGSLEMEKAKRASDI LNEKKYRQHPDTLKFTSIEDAPIIVQSKINQA QRSDVAYKAGGEEVIHKYSLPADLPQFIQAKVNAYNISENLYKADLKDLK KGYDLRIDAIPIKAAKAAQAASDVQYKKDYEKAKGMVGFQSLQDDPKL VHYMNVAKIQSDREYKAYEKTTRHNTPHDMVNI VAAKAQDVASNV NYKHS LHHYTYLPDAMDLELSKNMMHIRSDNVYKEDYNNWMKGIWPII GSLEVEKVKKAGDALNEKKYRQHPDTLKFTSIVDSPVMVQAKQNTQQVSD ILYKAGGEDVKHKYTMSPDLQFLQAKCNAYNLSDVCYKRDWHDLIAK TNVLGDAIPITAAKASRNIAADYKYKEAYEKAGKQVGFRLQDDPKLVH YMNVAQLQSDREYKKNYENTKTSYHTPGDMVSI TAAKMAQDVATNVNY KQPIHHYTYLPDALSLHIRNVNIQSDNVYKDEYNHFFKGMGWIPIGSLEV EKVKAGDALNEKKYRQHPDTIKFTSVPSMGMLAQHNTKQLSDLNK VEGEKVKHKYTMDDVPQFIQAKVNAYNMSDSHYKADWKKTLAKGYDL RPDAIPIVAAKSSRNIAADF KYKEAYEKTGKQIGFRSLQDDPKLVHFMNV AKMQSDREYKKGYEARKTKYHTPLDMLSVTAAKKSQEVATNTNYKQPPH HYTLLPDALNVEHSRNAMQIQSDNLYKSDFTNWMKGIWLPLESLVEKA KKAGEILSEKKYRQHPDKLFTYAMD TMEQALNKNKLIMDKRLYTEKW NKDKTTIHVMPDTPDILLSRVNI TMSDKLYKAGWEEKKGYDLRPDAIS IKAARASRDIAADYKYQAYEQAKGRQIGFRSLEDDEPKLVHFMQVAKMQS DREYKAYEKS KTSFHTPVDMLSVVAAKKSQEVATNANYRNVIHTYNNML PDAMSLELAKNMMQIQSDNLYKADYADFMKGIWLPGLGLEAEKNKKA MEIISEKKYRQHPDTLKYSTLMDSMNMVLAKNNAKIMNEHLYKQAEAD KTKVHIMPDIPIILAKANAINISDKLYKLSLEEAKKGYDLRTDAIPKAAK ASRDIAADYKYKHSYKGERKGMVGFRLSEDDPKLVHSMQVAKMQSDREY KKNYEKTKTSYHTPADMLSVTAAKDAQANITNTNYKHLIHKYILLPDAMNI QLSKNNMRIQSDNEYKQDYNEWYKGLGWSPAGSLEVEKAKKATEYASDQ KYRQHPSPNFQFTKLNDSDMVLAKQNAHTMNKLYTIDWNKDKTKIHV MPDTPDILQAKQNTMYSQKLYKLGWEEALKGYDLPVDAISVQAAKAS RDIVSDFKYKQGYRKQIGHHIGFRSLQDDPKLVLSMNVAKMQSREYKDD FEKWKTFSSPVDMLGVVLAKKQALVSDVDYKNYLHGWTCLPDQNDVI HAKKAYDLQSENLYKSDLEWLKGIWSPGLGLEAEKNKRASEI ISEKKYRQ PPDRNKFTSIPDAMDIVLAKTNAKNRSDILYREAWDKDKTQIHIMPDTPDII AKANLINTSDKFYRMGYEELRKGYDLPVDAIPKAAKASREIASEYKYKE GFRMQLGHHIGARNIKDDPKMWSMHVAKIQSDREYKDDFEKWKTFSS PVDMLGVVLAKKQCTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQS DNLYKSDLQWLRGIGWLPSPGSLDEKNKRASQILSDHVYRQHPDQFKFSSL MDSIPMVLAKNNAITMNHRLYTEAWDKDKTTVHIMPDTPEVLLAKQNI YSEKLYKLGLDEAKRKGYDMRIDAIPIRAAKASRDIASEFKYKEGYRRQLG HHIGARAIHDDPKMWSMHVAKIQSDREYKDDFEKWKTFSSPVDMLGV VLAKKQCTLVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQSDNVYKSDL QWLRGIGWVPIGSLDVEKSKRASEILSDKLYRQPPDKFKFTSVTDSLEQVLA KNNAINMNKRLYTEAWDKDKTQVHIMPDTPEITLARTNKVNYSENLYKLA HEEAKKGYDLRSDAIPIVAAKASRDII SDYKYKGYRKQLGHHIGARDIK DDPKMWSMHVAKIQSDREYKDDFEKWKTFSSPVDMLGVVLAKKQCT LVSDIDYKNYLHQWTCLPDQSDVIHARRAYDLQSDNVYKSDLQWLRGIG WVPIESVDVVKCKRAAEILSDNLYRQPPDTLKFTSVPSLEQVLAKNNAIN MNKRLYTEAWDKDKTQIHIMPDTPEIMLARQNKLNYS ETLYKLANEAAK KGYDLRSDAIPIVAAKASRDII SDYKYKGYRKQLGHHIGARDIKDDPKMM WSMHVAKIQSDREYKDDFEKWKTFSSPVDMLGVVLAKKQCTLVSDIDY RNYLHQWTCLPDQSDVIHARQAYDLQSDNIYKSDLQWLRGIGWVPIGSVD VVKCKRAAEILSDNLYRQRPDTLKFTSVPSLEQVLAKNNAINMNKRLYTE AWDKDKTQIHIMPDTPEITLARQNKINYS ENLYRQAMEEAKKEGYDLRSD AIPIVAAKASREIASDYKYKEAYRKQLGHHIGARAIHDDPKMWSVHVAK MQSDREYKDDFEKYKTRFSSPVDMLGIVLAKKQCTLVSDVDYKHLHEWT CLPDQNDIIHARKAYDLQSDNLYKSDLEWLKGIWVPIGSVEVLKAKRAGE ILSDNIYRQRPDTLKFTSVTDSPEQVLAKNNAINMNKRLYTEAWDNDKTTI HVMPDTPEIMLAKLNRINYSDKLYKLAL EESRKEGYDLRLDAIPKAAKAS REIASDYKYKEGYRKQLGHHIGARNIKDDPKMWSIHAGKIQSDLEYKDD FEKWKTFSSPVDMMGLVQAKKQCTLVSDIDYKNNLHEWTCLPDQNDIIQ ARKAYDLQSDAIYKADLEWLRGIGWVPI DSVGVEHAKRAGEILSERKYRQ AVQLKFTSITDTP EIVLAKNNALNVSKHLYTEAWDADKTSIHVMPDTPEILL AKSNSANISHKLYTKGWDESKMKDYDLRADAISIKSAKASRDIAADYKYKE AYEKHKGHHIGARSVEDDPRIMCAMNAGRIQSEREYKKEFKWKTFSSP VDMLGILLAKKQCTLVSDIDYRNYLHHWTCLPDQNDIIQARKAYDLQSDAI YKADLEWLRGIGWMPGSPVLRVKNQAQHIFRDSVYRTPVVKLYKTSIVD TPEVVLAKSNAENISIPKYREVWDKDKTSIHIMPDTPEINLARTNALNVSNK LYREGWDEMKMSCDVRDLDAIPKAAKASREIASDYKYL DHEKQKGHYV

TABLE6-continued

Bison Proteins		
SEQ ID NO Description	Amino Acid Sequence	
	GTRTARDNKKIRWALIAGKIQNEREYRLHWAKWKS KFQSPDMLSEHSK NSQTLVSDIDYRHYLHQWTCMPDQNDVIQARKAYDLQSDAIYKADLEWL RGIGWMPADSVSVNHAKHMADI FNEKKYRTK IETLSFTPVDDRVDYVTAK HSGEILNDIKYRKDNDTKSKYTLTETPQLHTAQEAARILDQYLYKESWEK QKATGYILPPDAVPFVHAHSGDVQSELKYAEHVQKKGHYVGVPTMRD DPKLVWFEHAGQIQNDRLYKENYHKTAKIHI PPDMVSVLAAKEGQALAS DIDYRNYLHQWICHDPQNDVIQARKAYDLQSDNIYKADLEWL RGIGWIPL DSVDHVRVTRNQEMMNQIKYKDALANYPNFTSVVDPPEIVLAKINSVNQ SDVKYKETFNKLIKGYIFSPDTPYI THSKDMEKLYSTILYKRAWDGTKAY GYTLDERYIPIVGAKHADPVNSELKYKETEYELKGHYLAGKEISEFPNVVH CLDFQKMRSLNRYRRHYEDTKANVHIPQDMNHVLAKRCQYILSDLEYR HYFHQWTSLEPEPNVVRHAQEILSDNVYKDDLNLWLGIGCVVWDTPOI LHAKKSYDLQSQILYTAAGKENLKNYNLVTDTPLYVTALQSGINASEVKY KENYHQIKDKYTTVLETVDYDRINKLKLFSNLYKEAWDKVKATS YILFP NTVSLTHAKNQKYMASHIKYREEYEKFKALYTLPRSVEDDPNTARCLRVG KFNIDRLYRSVYEKNMKIHI VPDMMVEMVTAKDSQKKVSEIDYRLHLHEWI CHPDQVNSHVRKVTDOI SDIVYKDDLTLWLGIGCVVWDTPEILHAKHAY DLRNDIKYKAHLKTRNNYKLVTDTPVYVQAVKSGKQLSDAVYHYDYVH SIRGRVAPT TKTVDLDRALHAYKLQSENLYRKAGLHALPTGYRLPVDTPHF KHTKDTRYMSSYFKYKEVYEHMKAYGYTLGPNVDPFVNVRVNNITSERL YRQLYHKLKDKIHTTDPDTP EIRQVKKTQEAVSELIYKSDFFKMQGHI SLPY TPQVLHCRYVGDITSDIKYKEDLQVLRGMGCFLYDTPDMVRSRHLRKLWS HYLTDKARKMRDKYKVLDTPEYRKVQELKTHLSELVYRAAGRKQKSI F TSVPTDPLTRAKRGQKLQSQYLYVELATKERPHHAGNQTTALKHARHV KDMVSENKYKI QYKMKDKYTPVPDTPILIRAKRAYWNASDLRYKETFOK TKGKYHTVKDALDIVYHRTVTDHVS KIKYKENYMSQLGIWRSIPDRPEHFH HRAVTDVAVSDVKYKQDLTLWLGIGCYAYDTPDFTLAEKNKTLYSKYKYK EVFERTKSNFKYVADCPINRHFKFATQLMNEKKYRADYEQRKDKYHLVVD EPRHLLAKIAGDQISQIKYRKNYEETKDKFTSIVDTP EHLRTTKVKNQISDIL YKLEYNAKAPRGYTTIHDTPMLLHVVRKVKDEVSDLYKVEYQRTKSNCTI EPDAVHIKAAKDAYKVNNTLDYKKKYEATKAYWKWTPDRPDFIQAAKST LQQSDFEYKLDREYLGKGLSVTDDKDMVLALKNSIIESDLKYKEKHVKE RGSCHAVPDTPQILLAKTVSSLVSENKYKSYVKKHLAQGSYTTLPETRTDIIH VKEVTKNVSDTNYKKKPVKEKGKSNYSIMLEPPDVKHAMDAVAKQSNVA YKKDAKENLHYTTVADRPDIKATQAQKQASEVEYRAKHRKEGSHGLSM LGRPDIEMAKKAAKLSSQVQYRENFNKEKGKTPKYNPKDSQLYKVMKDA NTLASEVKYKADLKKLHKPV TDMKESLIMNHVLNTSHLASSYQYKKNYEK SKGHYHTIPDNLEQLHLKEATELQSI VKYKEKYEKERKPMPLDFETPTYI TA KESQMQSGKEYRKDYEESI KGRNLTGLEVT PALLHVKYATKIASKEYRK DLEESIRGKGLSEMEDTPDMLRAKNATQILNEKEYKRDLEEVKGRGLNA MANETPDFLRARNATDIASQIKYKQSAEMEKANFTSVVDTP EIIHAQQVKIN LSSQKKYKEDAEKCMSYYETVLDTP EMQVRVRENQKNFSLQYQYDLKNS KGKITVVQDTP EILRVKENQKNFSSVLYKEDVSPGTAIGKTPEMMRVKQTQ DHISSVKYKEVIGQGTPI PDLPEVKRVKQTQKHISVVMYKENLGITGPTPVT EIERVKRNQENFSSVLYKENLGKGTPTPIPEMERVKRNQENFSSVLYKEN MGKGTPLPVPTEMERIKHNQENISSVLYKENVKGKGTPTPTPEMERVKRNQ ENISSVLYKENLGKATPTPTPEMERVKRNQENFSSVLYKENMRKATPTPV TPEMERAKRNQENISSVLYSDSFRKQIQGKAAYVLDTPEMRRVRETQRHIS TVKYHEDFEKHKGCFPTPVVTDPI TERVKKNTQDQSDICYRGIQRKVVMEO KRNDQDQETITGLRVWRTNPGSVFDYDPAEDNIQSRSLHMINAQARRSRE QSRASAGLSISGGEEKSEHSEAAHLSTYSDGGVFFSAASTGYKHARTTELPO QRSSSVATQQTTVSSIPSHPSTAGKIFRMYDYMAADAEVSFKDGDIAIVN VQAIDEGWYGTVQRTGRTGMLPANYVEAI	
137	Bison A0A6P3HPK5 (Myosin-4)	MSSDQEMAI FGEAAPYLRSKSEKRIEAQNKPFAKTSVFVVDPKESFVKAT VQSRREGKVTAKTEAGATVTVKEDQIFPMNPPKYDKI EDMAMMTHLHEPA VLYNLKERYAAWMIYTYSGLFCVTVNPKWL PVYNAEVVYAYRGKKRQE APPHIFSIDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIAVTG EKKKEEPTSGKMQGTLEDQIISANPLLEAFNGAKTVRNDNSRFGKFIIRHF GTTGKLASADIETYLEKSRVTFQLKAERSYHIFYQIMSNKKPELIEMLLITT NPYDYAFVSQGEITVPSIDQEEELIATDS AIDILGFTSDERSVIYKLTGAVMH YGNLKFKQKQREEQAEPDGTVEADKAAYLQSLNSADLLKALCYPRVKVG NEFVTKGQTVQQVYNAGALAKAVYEKMFLLMVTRINQQLDTKQPRQY FIGVLDIAGFEIFDNFSLQQLCINFTNEKLQQFFNHHMFVLEQEYKKEGIEW EFIDFGMDLAACIELIEKPMGIFSI LEEECMPFKATDTSFKNKL YEQHLGKSN NFQKPKPAKGAEAHFSLVHYAGTVDYN IAGWLDKNKDLPLNETVVGLYQ KSAMKTLAFLFAGGSAEEGGGKGGKGGKSSFTVSALFRENLNKLMT NLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGPPSRI LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSDIDHTQYKFGHTKVFF KAGLLGILEMRDEKLAQLITRTQAI CRGFLMRVEFRKMMERRESIFCIQYN VRAFPMNVKHPWMKLYFKIKPLLSAET EKEMANMKEEFEKTEELAKS EAKRKELEEKMVTLMQEKNDLQLQVQSEADGLADAERCDQLIKTKIQLE

TABLE6-continued

Bison Proteins		
SEQ ID NO Description	Amino Acid Sequence	
	AKIKELTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLLELTAKVEKEK HATENKVKNLTEEMAGLDENIAKLTKKKALQEAHQQTLDLQAEEDKV NTLTKSKTKLEQQVDDLEGSLEQEKLRMDLERAKRKLEGLDLAQESTM DIENDKQQLDEKLKKKEFEMSNLQSKIEDEQALAMQLQKKIKELQARIEEL EEEIEAERASRAKAEKQSRDLSRELEISERLEEAGGATSAQIEMNKKREAE FQKMRDLLEATLQHEATAAALRKKHADSVAEELGEQIDNLQVRVKQKLEKE KSEMKMEIDDLASNMTVSKAKGNLEKMCRTLEDQLSEVSKSEEEHQRLI NELSAQKARLHAESGEFSRQLEEKDALVSQLSRGKQAFQIIEELKRQLEE ETKAKSALAHALQSARHDCDLLREQYEEEGKAEQALQALSKANSEVAQ WRTKYETDAIQRTTEELEEAKKLAQRLQDAEEHVEAVNAKASLEKTKQR LQNEVEDLMLDVERSNAACAALDKQRNFDKVLSEWKQKYEETQAELEA SQKESRSLSTELPKVKNACEESLDQLETLKRENKNLQOEISDLTEQIAEGGK HIHELEKVKQIEQEKGEQALAEAEASLEHEEGKILRIQLELNQVKSEIDR KIAEKDEEIDQLKRNHLRVMESTQSTLDAEIRSRNDALRIKKMEGDLNEM EIQLNHANRQATEAIKNLRNTQGVLDKDTQLHLDDAIRGQDDLKEQLAMVE RRANLMQAEIEELRASLEQTERSRRVAEQELLDASERVQLLHTQNTSLINTK KLETDITQIQGEMEDILQEARNAEEKAKKAITDAAMMAEELKKEQDTS HLERMKKNMBQTVKDLQHRLEDAEQALALGKGGKQIQKLEARVRELENEV ENEQKRNAEAVKGLRKHRRVKELTYQTEEYRKNVLRQLDVLVKLQSKV KAYKRQAEAEQSNVNLAKFRKLQHELEAEERADIAESQVNLKRVKSR EVHTKVISSE	
138 Bison A0A6P3HBK1 (Serum albumin)	MKWVTFISLLLFSSAYSRGVFRDRTHKSEIAHRFKDLGEEHFKGLVLIAFS QYLQCCPFDEHVKLVNELTEFAKTCVADESHAGCEKSLHTLFGDELCKVA SLRETYGNMADCEKQEPERNECFLSHKDDSPDLKPKPDPNTLCDEFKAD EKKFWGKYLIEIARRHPYFYAPELLYANKYNGVQECQAEKKGACLLP KIETMREKVLTSARQLRCASIQKFGERALKAWSVARLSQKFPKAEFVEV TKLVTDLTQVHKECCGDLLECADDRADLAKYICDNQDTISSKLKECCDKP LLEKSHCIAEVEKDAVPENLPLTADFAEDKDVCKNYQAKDAFLGSPFYE YSRRHPEYAVSVLLRLAKEYEATLEECCAKDDPHACYSTVFDKLKHLVDE PQNLIKQNCQDFEKLGEYGFQNALIVRYTRKVPQVSTPTLVEVSRLGKVG TRCCTKPESERPCTEDYLSLI LNRLCVLHEKTPVSEKVTCCTESLVNRRP CFSALTPDETYVPKAFDEKLFTHADICTLPDTEKQIKKQALVELLKHKPK ATEEQLKTMENFVAFVDKCCAADDKEACFAVEGPKLVVSTQTALA	

[0049] An animal protein can comprise a deer protein (Table 7). A deer protein can comprise a deer muscle protein. A deer protein can comprise a deer blood protein. A deer protein can comprise a deer albumin protein. A deer protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 139-147.

[0050] A food product may comprise a fragment or a truncation of a deer protein. The food product may include

a fragment or truncation of any one of SEQ ID Nos: 139-147. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 139-147. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 139-147. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 139-147. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 139-147.

TABLE7

Deer Proteins		
SEQ ID NO Description	Amino Acid Sequence	
139 Deer A0A212DH10 (ACTN3)	MLDAEDIVNTPKPDEKAIMTYVSCFYHAFAGAEQAETAANRICKVLAVNQ ENEKLMEEYEKLASELLEWIRRTVPWLENRVGEPSSAMQKLEDFRDYR RLHKPPRVQEKQCLEINFNTLQTKLRLSHRPAPFMPSEGLVSDIAKAWRGL EQVEKGYEDWLLSEIRRLQLQLHLAEKFPQKASLHEAWTRGKEEMLSQRD YETASLQEVFALLRRHEAFESDLAAHQDRVEHIAALAQELNELDYHEAAS VNSRCQAICDQWDNLGTLTQKRRDALERMEKLELTDQLQLEFARRAAPF NNWLDGAVEDLQDVVLVHVSVEETQSLVTAHDQFKATLPEADRERGAILGI QGEIQIKCQTYGLRPSSTNPYIALTPQDINTKWDTVRKLVPSRDQMLQEBELT RQQVNERLRQFAAQANAIGPWIQGKVEEVGRLAAGMAGSLEEQMAGLR QQEQNIINYKSNIDRLEGDHQLLQESLVFDNKHTVYSMEHIRVGWEQLLTISI ARTINEVENQVLRDAKGLS-QEQLNEFRASFNHFDKRKNGMMEPDDFRAC	

TABLE7-continued

Deer Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		LISMGYDLGEVEFARIMTMVDPNAAGVVTFQAFIDFMTRETAETDTAEQV VASFKILAGDKNYITAEELRRELPAEQAEYCIRRMAPPYKGAGAPAGALDYV AFSSALYGESDL
140	Deer A0A212DGZ3 (Alpha-1,4 glucan phosphorylase)	MSRPLTDQEKRKQISVRGLAGVENVTELKKNFNRLHFTLVKDRNVATPR DYYFALAYTVRDHLVGRWIRTQQHYEKKDPKRIYYLSLEFYMGRTLQNTM VNLALENACDEATYQLGLDMEELEEIEEDAGLNGGLGRLAACFLDSMAT LGLAAYGYGIRYEFQIFNQKISGGWQMEEDDLRYGNPWKARPEFTLP VHFYGRVEHTSQGAKWVDTQVVLAMPYDTPVPGYRNNVNTMRLWSAK APNDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF VVAATLQDIIRRFKSSKFGCLDPVAIQNDTHPSLAIPELMRILVDQERLEWE KAWEVTVKTCAYTNHTVLPALERWPVHLIETLLPRHLQIIYEINQRLNRV AAAPFGDVDRLRMRSLVEEGAVKRINMAHLICAGSHAVNGVARIHSEILKK TMTSKWSLINSRIRPMGSPGLGAGWLCRIGEEYIADLDQLRKLLSYVDDSEF IRDVAKVKQENKLKFSAYLEKEYKVHINPNSLFDIQVKRIHEYKRLNCL HAAPGYHMAKMIIKLITAIGDVVNHDPPVGDRLRVIFLENYRVSLAEKVIP AADLSEQISTAGTEASGTGNMKFMLNGALTI GTMDGANVEMAEAGEENF FIFGMRVEDVERLDQRGYNAQEYYDRIPELRHVIDQLSSGFFSPKQPDLFKD IVNMLMHHDREFKVFADYEEYVKCQERVSALYKNPREWTRMVRNIATSGK FSSDRITIAQYAREIHWGVEPNRQRMPPADEKI
141	Deer A0A212CZ49 (Fructose- bisphosphate aldolase)	MAQRKPEGSGFHMTCLSMALAYSFRPDANIQPHPQLGNTQKQTELKELT GTMPHQYPALTPQKKELCDIAHRIVAPGKGILAADESTGSIAKRLQSIGTE NTEENRRFYRQLLLTADDRVNPCIIGGVLIFHETLYQKADGRFPQVVKAK GGVVDKGVPVPLAGTNGETTTQGLDGLSERCAQYKKGADFAKWRCLVKI GEHTPSSLAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQYVT EKVLAAYKALSDHHIYLEGTLKPNMVTGPHACTQKYSHEEIAMATVTA LRRTVPPAVPGITFLSGGQSEEEASINLNAINKCPLLKPWALTFSYGRALQAS ALKAWGGKENLKAQEEYVKRALANSLACQGYTPSGKAGAAASESLFI SNHAY
142	Fragment of Deer A0A212D3R8 (MYL1)	PAPAPPKEEKIDLSAIIKIEFSKQQQDEFKEAFLFDRTGCEKITLNQVGDVLR ALGTNPTNAEVKKVLGNPSNEEMNAKKIEFEQFLPMLQAISSNKDQGTYE DFVEGLRVFDKKEGNGTVMGAELRHVLATLGEKMKKEEVEALMAGQEDSN GCINYEAFVKHIMSN
143	Deer A0A212CCI7 (GAPDH)	MFVMGVNHEKYNNTLKVSNASCTTNCLAPLAKVIHDHFGIVEGLMTTVH AITATQKTVDGSPGKLWRDGRGAAQNIIPASTGAAKAVGKVIPELNGKLTG MAFRVPTPNVSVVDLTCRLEKPKAYDEIKKVVKQASEGPLKGILGYTEDQV VSCDFNSDTHSSTFDAGAGIALNDHFVKLISWYDNEFGYSNRVVDLMVHM ASKE
144	Deer A0A212D927 (Phosphopyruvate hydratase)	MRTLAMAMQKIFAREILDSRGNPTEVDLHTAKGRFRAAVPSGASTGIYE ALELRDGDKSRYLKGKVLKAVEHINKTLGPALLEKASLGRLYLLPPAKFGA NAILGVSLAVCKAGAAEKGVPLYRHIALDAGNPELILPVPAFNVINGGSHA GNKLAMQEFMILPVGASSFKEAMRIGAEVYHHLKGVIKAKYKGDATNVG DEGGFAPNILENNEALELKTATQAGGYDPKVIIGMDVAASEFYRNGKYDL DFKSPDDPARHISGEKLGELYKNFIKNYPVVSIEDPFDQDDWATWTSFLSGV NIQIVGDDLTVTNPKRIAQAVEKKACNCLLLKVNQIGSVTESIQACKLAQSN GWGVMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMRIE EALGDKAVFAGRKFRNPKAK
145	Deer A0A212CY74 (MYLPF)	MFDQTQIQEFKEAPKAKRRAAEGGSSSVFSMFDQTQIQEFKEAFTVIDQ NRDGIIDKEDLRDTFAAMGRNLNVKNEELDAMMKEASGPINFVFLNMFGE KLKGADPEDVITGAFKVLDPGKGTIKKQFLEELLTTQCDRFSQEEMRNM WAAPPDPVGGNVYKNIYVITHGDAKDQE
146	Deer A0A212CNJ3 (Actin)	MCEEDSTALVCDNGSLCKAGFAGDDAPRAVFPISVGRPRHQGVVMVGM GQKDSYVGDEAQSKRGILTLKYPIDHEGII TNWDDMEKIMFETFNVPAMYVA IQAVLSLYASGRTTGIVLDSGDGVTHNVPIYEGYALPHAIMRLDLAQRDLT DYLMLKILTERGYSFVTTAEREIVRDIKEKLCYVALDFENEMATAASSSSLEK SYELPDGQVITIGNERFRCPETLFPQPSFIGMESAGIHETTYSIMKCDIDIRKD LYANNVLSGGTTMYPGIADRMQKEITALAPSTMKIKI IAPPERKYSVWIGGS ILASLSTFQQMWISKQEYDEAGPSIVHRKCF
147	Deer A0A212D5P0 (ALB)	MKWVTFISLLLLFSSAYSRGVFRDRDTHKSEIAHRFKDLGEDNFQGLVLIAFS QYLQCCPFDEHVKLVELTDFAKECVADESHAGCEKSLHTLFGDELCKVA SLRETYGDMADCEKQEPERNECFLKHKDDNPDLKLPKPDPTLCAEFKA DEKKFWGKYLIEVARRHPYFYAPELLYYANKYNGVFQECQADDKAACL LPKIDTMRKVLTSARQLKCAISIQKFGERALKAWTVALLSQKFPKADFA EVTKI VTDLT KVHKECCHGDLLECADDRADLAKYICDNQDTLSSKLKECC DKPVLEKSHCIAETIEKDAVPENLPPLTADFAEDKEVCNKYQEAQDVFLGTF

TABLE7-continued

Deer Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		LYEYARRHPEYAVSVLLRVAKEYEATLEDCCAKDDPHACYATVFDKLQHL VDEPQNLIKQNCDLFEKVGEYGFQNALIVRYTKKAPQVSTPTLVEVSRNLG RVGTKCCTKPESERMSCTEDYLSVLNRLCVLHEKTPVSEKVKCTESTLV NRRPCFSALTLDETYVPKAFDEKLFTFHADICTVPAEKQIKQTALVELLK HKPKATDEQLKTVMENFVAFVDKCCAADVKEACFAVEPTLRVRGKKEK

[0051] An animal protein can comprise a chicken protein (Table 8). A chicken protein can comprise a chicken muscle protein. A chicken protein can comprise a chicken blood protein. A chicken protein can comprise a chicken albumin protein. A chicken protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 148-161.

[0052] A food product may comprise a fragment or a truncation of a chicken protein. The food product may

include a fragment or truncation of any one of SEQ ID Nos: 148-161. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 148-161. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 148-161. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 148-161. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 148-161.

TABLE8

Chicken Proteins		
SEQ ID NO	Description	Amino Acid Sequence
148	Chicken KCRM (Creatine kinase M-type)	MPFSSTHNKHKLFKFSAEFEFPDLSKHNHMAKVLTPELYKRLRDKETPSGF TLDDVIQTGVDPNGHPFIMTVGCVAGDEESYEVFKDLFDPVQDRHGGYKP TDKHRDTDLNHNENLKGDDLDPKYVLSRVRTGRSIRKGYSLPPHCSRGERRA VEKLSVEALNSLEGEFKGRYYPKAMTEQEQQLIDHFLFDKPVSPLLLA SGMARDWPDARGIWHNDNKTFLVWVNEEDHLRVISMCKGNMKEVFRF CVGLKKIETIFKKAGHPFMTEHLGYILTCPSNLGTGLRGVHVKLPLKSQ HPKFEEILHRLRLQKRGTGGVDAAVGAVFDISNADRLGFSEVEQVMVV DGVKLMVEMEKKLEQNQPIDDMIPAQK
149	Chicken (ACTN2 Alpha-actinin-2)	MNSMNQIETNMQYTYNYEEDYMTQEEWDRDLLDPAWEKQQRKTFTA WCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKPDGRGKMRPHKI ANVNKALDYIASKGVKLVSTIGAEIIVDGNVKMTLGMWITIIIRFAIQDISVE ETSAKEGLLLWCQRKTAPYRNVNIQNPHLSWKDGLGLCALIHRHPDLIDY SKLNKDDPIGNINLAMEIAEKHLIDPKMLDAEDIVNTPKPDERAIMTYVSCF YHAFAGAEQAEATAANRIKVLAVNQENERLMEEYERLASELLEWIRRTIPW LENRTPEKTMQAMQKKLEDPRDYRRKHKPPKVQEKQLEINFNTLQTKLRI SNRPAPMPSEGMVSDIAGAWQRLQAEKGYEWWLLNEIRRLERLEHLAE KFRQKASTHEQWAYGKEQILLQKDYESASLTVRAMLRKHEAFESDLAAH QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGTLTQKREAL ERTEKLETDIDQLHLEFAKRAAPFNWMEGAMEDLQDMFIVHSIEEIQSLIS AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNYPYSTVTVEEIRT KWEKVQLVPQDQSLQEELARQHANERLRQFAAQANVIGPWITQKME ETARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVFDN KHTNYTMEHIRVGVWELLTTIARTINEVETQILTRDAKITQEQMNDFRASF NHFDRRKNGLMDDHDFRACLISMGYDLGAEAFARIMSLVDPNGQGTVTPO SFIDFMTRETADTDTAEQVIASFRILASDKPYILADELRRELPEPQAYQYCIKR MPQYTGPGSVPGALDYTSFSSALYGESDL
150	Chicken E1BSN7 (Alpha-1,4 glucan phosphorylase)	MAAPLSDGERRKQISVRGIAGLDVAEVRKSFNRHLHFTLVKDRNVATPR DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM VNLGLQNACDEAIYQLGLDLEELIEEEDAGLNGGLGLRLAACFLDSMATL GLAAYGYGIRYEFGI FNQKIIDGWQVEEADDWLYRGNPWEKARPEYMLPV HFGYRVEHSPGEGVKWVDTVVLAMPYDTPVPGYKNNTVNTMRLWSAKA PNDFNQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQYFV VAATLQDIIIRRFKSSKFGCDPVRTSFETFPDKVAIQNLNDTHPALSIPELMRIL VDVEKVDWDKAWIEITKQTCAYTNHTVLPEALERWPVSMFPEKLLPRHLEII YALNQMLDRVAALYPGDVDRLRMSVIEEGDCKRINMAHLCVIGSHAV NGVARIHSDIVKNTVFKDFYELEPEKFQNKNTNGITPRRLLLLCNPLADLVIA EKIGEDFVTDLSQLKKLLDFINNETHIRDAVKQENKLFSAYLEERYKVK INPSSMFDIQVKRIHEYKRQLLNCLHAITLYNIRSDPSKSFVPRTIMIGGKA

TABLE8 - continued

Chicken Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		APGYHMAKMIIKLITSIGEVVNNDPYVGDKLVIFLENYRVSLAEKVIPAAD LSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIFG MRVEEVEALDRRGYNAREYYDRIPELRQAIDQISSGFFSPRDPGCFRDVVN MLMHDRFVKVFADYEAYIKCQGGVDQLFMDPREWTKKVIIRNIACSGKFFS DRTITEYAREIIGVEPSATKIPPPNLPD
151	Chicken R4GM10 (Fructose- bisphosphate aldolase)	MTHQYPALTAEQKKELSDIALRIVAPGKGILAADESVGSMAKRLNQIGVEN TEENRLRYRQILFSADSRVKKCIGGVIFFHETMYQKADDGTPFVQMIKDKGI VVGIKVDKGVVPLAGTDGETTQGLDGLSERCAQYKKDGADFAKWRCVL KISDNTPSALAIMENANVLARYASICQQNGIVPIVEPEILPDGDHDLKRCQY VTEKVLAAVYKALSDHHVYLEGTLKPNMVTPGHSCPTKYSPEEIAMATV TALRRTVPPAVPGVTFLSGGQSEEEASINLNAINTCPLVRPWALTFYGRAL QASALSAWRGQQRDNANAATEEFVKRAEVNGLAALGKYEGSGDDSGAAGQ SLYVANHAY
152	Chicken MLE1 (Myosin light chain 1)	MAPKKDVKKPAAAAAPAPAPAPAPAPAPAKPKPAIDLKSIKIEFSKEQQD DFKEAFLFDRTGDAKITLSQVGDIVRALGQNPTNAEINKILGNPSKEEMNA KKITFEFLPMLQAAANNKDQGTFFEDFVEGLRVFDKEGNGTVMGAELRHV LATLGEKMTBEEVEELMKGQEDSNGCINYEAFVKHIMSV
153	Chicken G3P (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKGVNGFGFRIGRLVTRAAVLSGKVQVVAINDPFIDLNYVMVMFKYDST HGHPKGTVKAENGKLVINGHAITFQERDPSNWKWADAGAEYVVESTGVFT TMEKAGAHKGGAKRVIIISAPSADAPMFVMGVNHEKYDKSLKIVSNASCT TNCLAPLAKVIHDNFGIVEGLMTTVHAITATQKTVDGPSGLWRDGRGAA QNIIPASTGAAKAVGKVIPELNGKLTGMAFRVPTPNVSVVDLTCRLKPAK YDDIKRVVKAADGPLKGLGYTEDQVVSDFNGDSHSSFTFDAGAGIALND HFVKLVSWYDNEFGYSNRVVDLMVHMASKE
154	Chicken TPM1 (Tropomyosin alpha-1 chain)	MDAIKKMQMLKLDKENALDRAEQAEADKAAEERSKQLEDELVALQK KLKGTEDLDKYSESLKDAQEKLELADKKATDAESEVASLNRRIQLVVEEL DRAQERLATALQKLEEAKEAASESERGMKVIENRAQKDEEKMEIQEIQKE AKHIAEADRKYEEVARKLVIIEGDLERAEEAELSESKCAELEELKTVTN NLKSLEAQAEKYSQKEDKYEIEIKVLTDLKLEAETRAEFAERSVTKLEKSID DLEDELYAQLKYKAISEELDHALNDMTSI;
155	Chicken MYPC2 (Myosin-binding protein C)	MPEPSKAAPKKEAKKKKEKKKEKKEAPPQEHKDEAPDDVHPPETPDPEGL FLSKPQNMVVESEGRDVTVSARVAGAAPCAPAVKFKGWAEELGDKSAR CRLRHSVDDDKVHTFELTITKVAMGDRGDYRCEVTAKEQKDSFSDIDVEA PRSEGNVLQAFKRTGEGKDDTAGELDFSGLLKKREVQVEEKKKKKDEDD QPPPEIWELLKGVTKKSEYERIAFYGYITDLRGLMKRLKKVHVEPKKSEAFI KRLDPAYQVDKGNKI KLVVLSDPDLPLKWKYKNGQLKPKSTKYVFENVGL KRILT I HKCSLADDAAYECRVNDEKCFTEVFVKEPPVTTVRGLEDQVVVG DRVVLEAEVSEEGAQVMWLKDGVDVTRDDAFKYRFKKDGKKHFLINEA ELSDSAHYKIMTNGGESEAELESVEEKQLEVLQDMADLTVKASEQAVFKCE VSDEKVTGRWFRNGVEVPSKRIHISHNGRFHKLVIDDVRPEDEGDTTFIPD GYALSLSAKLNFLEIKVEYVPKQEPPIHLDCSGKAAENTIIVVAGNKVRL DVPISEGPAPTVTWKRGDQLFTATEGRVHIDSQADLSSFVI ESAERSDEGRY CITVTNPVGEDSATLHVRVVDVDPDPPQSVRVTSVGEDWAVLSWEAPPFDG GMPITGYLMERKKKSGMRWMKLNFEVFPDITYESTKMIEGVFYEMRVFA VNAIGVSPSLNTQPFMPIAPTSEPTHVLEDVDTTATIKWRPPERIGAGG VDGYLVEWCREGSNEWVAANTELVERCGLTARGLPTGERLLFRVISVNMA GKSPPTMAQPVTIREIVERPKIRLPRHLRQTYIRRVGEQVNLVIPFQGKPRP QVTSREGGALPAEVQTRTSDVDSVFFIRSAARPLSGNYEMRVIDNMEDC ATLRLRVVERPGPPQAVRVMEVWGSNALLQWEPKDDGNAEISGYTVQK ADTRTMEWFTVLEHSRPTCTVSELVMGNEYRFRVYSENVCGTSQEPATS HNTARIAKEGLTLKMVPYKERDLRAAPQFLTPLVDRSVVAGYTVTLNCAY RGHPKPKVTWLNKSVEIGADPKFLSRHGLGVLSLLIRRPFPDGGTYGCRA VNEMGEATTECRLDVRVPQ
156	Chicken MPSF (M-protein)	MSSVAVPFYQRRHKHFDQSYRNIQTRYVLEEYAAKAAASRQAAYESTGL GKTTCLRCARRARSLAHEAMQESRKRTHQKSHASDEKRIKFASELSSLER EIHMARHHAREQLDRLAIQRMVEENMALERHVVVEEKI SRAPEILVRLRSH VWEKMSVRLCFTVQGFPSPVVQWYKNEELITPASDPAKYSVENKYGHVHL HINRADFDSDATYSAVATNIHGQASTNCAVVRRFRSEEPHPAGIMPFHLP LSYDVCFTHFDVQFLEKFGVTFATEGETLTLKCSVLVTPELKRLRPRAEWY RDDVLIKDSKWTLYFGEGQAALSFTHLNKDDGLYTLRMVTKGGVNECS AFLFVRDADALIAGAPGAPMDVKCHDANRDYIVTWKPPNTTSQNPVIGY FVDKCEVGLENWVQCNDAPVKICKYPVTGLYEGRSYIFRRAVNSAGISRP SRVSEPVAAALDPVDLERTQTVHVDEGRKIVISKDDLEGDIQIPGPPTNVHAS EISKTYVVLVSWDPPVPRGREPLTYFIEKSMVSGSGWQRVNAQVAVKSPRYA VFDLAEKQYVFRVLSANKHGISDPSEITEPIQPQDIVVPSAPGRVATRNT KTSVVVQWDPKHEENLYGYIIDYSVVGSNQWEPANHKPINYNRFVVHG

TABLE8 -continued

Chicken Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		LETGEQYIFRVKAVNAVGFSENSQSESAIKVQAALTCPSYPHGITLLNCDGH SMTLGWKAPKYSGGSPILGYIIDKREANHNWHEVNSSVISRTIYTVEDLT EDIFYEFKIAAANVVGIGHPSDPSEHFCKAWTMPEGPAYDLTVCEVRNT SLVLLWKAPVYEGKSPITGYLVVDYKEVDTEDWITANEKPTSHRYFKVTDLH QGHTYVFKVRAVNDAGVGKSSEISEPVFVEASPGTKEIFSGVDEEGNIYLG ECKEATDASHFLWGKSYEEIEDSDKFKIETKGDHSLKLYFKHPDKSDLGTYCI SVSDTDGVSSSVFLDEEELERLMTLSNEIKNPTIPLKSELAYEVLDKGEVRF WIAESLSPNSTYRFVINDKEVENGRHKISCDHSNGIEMVMDKFTIDNEG TYTVQIQDGKAKNQSSLVLIGDAFKAILAESLQKKEFLRKQGPHEFLY EVTEECVLLACKIANTKKETVFKWYRNGSGIDVDEAPDLQKGECHLTVPK LSRKDEGVYKATLSDDRHDVSTLELSGKVYNDIILALSRVSGKTASPLKIL CTEGIRLQCFKYYNEEMKVTVSHRESKISSGKMKIGGEDVAVLQITE PTEKDKGNYTFEIFSDESFKRITLDLSGQAFDDALTEFQRLKAAAFKAKNR GKVIIGLPDVTIMDGKTLNLTCTVFGNPDPEVWFKNDKALELNEHYLV SLEQGYASLTIKGVTSEDSGKYSIVVKNKYGGETVDVTVSVYRHGEKIPE VNQQQLAKPRLIPPSSST
157	Chicken ENOB (Beta-enolase)	MSIQKIHAREILDSRGEPTEVDLHTAKGHFRAAVPSGASTGIHEALEPRDG DKRFLGKGVLKAVEHINKTIGPALIEKKISVVEQEKIDKVMIEDGTENKS KFGANILGVSLAVCKAGAAEKGVPLRYHIADLAGNTEILPVPAPNVING GSHAGNKLAMQEFMVLVPGAAAFHDMRVGAEVYHSLKGVIKAKYKGD ATNVGEGGEGFAPNILDNHEALELLKAAIAQAGYTDKVVIGMDVAASEFCR DGRYHLDKSPPHTKRYITGEQLGEIYRGFIKDYPPVVSIEDPFDQDDWEAW KRFPVHVDIQVVGDDLTVTNPKRIAHGAEQHACNCLLLKVNQIGSVTESIQ ACKLAQSHGWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLA KYNQLMRIEALGDKAKFAGRKFRNPKAK
158	Chicken MYL11 (Myosin regulatory light chain 11)	MAPKAKRRAAEGSSNVFSMFDQTQIQEFKEAFTVIDQNRDGIIDKDDLRE TFAAMGRNLNVKNEELDAMIKEASGPINFVTFLTMFGEKLGADPEDVIMG AFKVLDPDGKSGIKKSFLLELLTTQCDRFTPEEIKNMWAAFPDPVAGNVDY KNICYVITHGEDKEGE
159	Chicken MYH1B (Myosin-1B)	MATDADMAIFGEAAPYLRLKSEKERIEAQNKPFDKASSVVFVHAKESYVKST IQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMTHLHEPAV LYNLKERYAAWMIYYSGLFCVTNPKWLPVYNPEVVLAYRGKKRQEA PPHIFSISDNAYQFMLTDRENQSILITGESGAGKTVNTKRVIQYFATIASGD KKKEEQPAGKMQGTLEDQIISANLLEAFGNAKTVNRDNRNRRGKFIIRIHFG ATGLASADIBTYLLEKSRVTPQLKAERSYHIFYQIMSNKKPELIELMLITTN PYDYQVVSQGEITVPSINDQEELMATDSIDILGFTPDEKTAIYKLTGAVMH YGNLKFQKQREEQAEPPGTEVADKAAAYLMGLNSADLLKALCYPRVKVG NEYVTKGQTVQQVYNSVGALAKSVFEKMFLLMVVRINQQLDTKQPRQYF IGVLDIAGFEIFDFNSLEQLCINFTEKLLQFFNHHMFVLEQEEYKKEGIEWE FIDFGMDLAACIELIEKPMGIFSILEECCMFPKATDTSFNKLYDQHLGKSN FQKPKPGKGAEAHFLSVHYAGTVDYNI TGWLEKNKDPNETVVGLYQKS SLKTLALLFASVGGAAESGAGGKGGKGGSSFTQSALFRENLNKLMS NLRSTPHPHVRCIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPRI LYADFPQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKV FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLMRVEFKKMMERRESIFCIQ YNVRAFPMNVKHWPWMKLFKKIKPLLSAASEKEMANMKKEFEKTEELA KSEAKRKELEEKMVSLQEKNDLQLQVQAEADGLADAEERCDQLIKTKIQ LEAKIKELTERAEDEEEMNAELTAKKRKLEDECSSELKIDDLLELTAKVE KEKHATENKVNLTTEMAALDETI AKLTKEKKALQEAHQQTLDLQAEED KVNTLTAKTKLEQQVDDLEGSLEQEKLRMDLERAKRKLEGLDKMTQE STMDLENDKQQLDEKLKKDFEISQISQKIEDEQALGMQLQKKIKELQARIE EEEEEIEAERTSRAKAEKHRADLSRELEIEISLERLEAGGATAAQIDMNKKRE AEFQKMRDLLEATLQHEATAAALRKHADSTADVGEQIDNLQVRVKQL EKEKSELKMEIDDLASNMESVSKAKANLEKMCRSLEDQLSEIKTEEEQOR TINDISAQKARLQTESGEYSRQVEEKDALISQLSRGKQAPTQQIEELKRHLEE EIKAKCPAHALQSARHCDLLREQYEEQEAQKELQALSKANSEVAQW RTKYETDAIQRTTELEAKKKLAQRLODAEEHVEAVNSKASLEKTKQRL QNEVEDLMIDVERSNAAALDKKQKNFDKILSEWKQKYEETQAELEASQ KESRSLSTELFKMKNAYEESLDHLETLKRENKLQOEISDLTEQIAEGGKAI HELEKVKQIEBQKSELQTALEBAEASLEHEEGKILRVQLELNQVKSIDRK IAEKDEEIDQLKRNLFRVDSMQSTLDAEIRSRNEALRLKKKMEGLDNEIEI QLSHANRQAAEAQKNLRNTQGVLDKDTQIHLLDALRSQEDLKEQVAMVER RANLLQAEIEELRAALEQTERSARKVAEQELLDASERVQLLHTQNTSLINTKK KLESDISIQISEMEDTIQEARNAEEKAKKAITDAAMMAEELKKEQDTSALH ERMKNLDQTVKDLQHRLDEAEQLALGGKKQIQKLEARVRELEGEVDA EQKRSABAVKGVRRYERRVKELTYQSEEDRKNVLRQLVDLKLQMKVKS YKRQAEAEELSNNVLSKFRKIQHELEEAERADIAESQVKNLRAKSREIGK KAESSE

TABLE8-continued

Chicken Proteins		
SEQ ID NO	Description	Amino Acid Sequence
160	Chicken R4GIG1 (Uncharacterized protein)	MATDADMAIFGEAAPYLRLKSEKERIEAQNKPFDAKSSVFVVHAKESYVKST IQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPAV LYNLKERYAAWMIYTYSGLFCVTNPKWLPVYNPEVVLAYRGKKRQEA PPHIFSI SDNAYQFMLTDRENQSI LITGESGAGKT VNTKRV IQYFATIAASGD KKKEEQPAGKMQGTLEDQII SANPLLEAFGNAKTVRNDNSSRFGK FIRIHFG ATGKLASAD IETYLLEKSRVTFQLKAERSYHIFQIMSNKKPEL IEMLLITTN PYDYQYVSQGEITVPSINDQEELMATDS AIDILGFTPDEKTAIYKLTGAVMH YGNLKFKQKQREEQAEPDGT EVADKAAYLMGLNSADLLKALCYPRVKVG NEYVTKGQTVQQVYNSVGALAKSVFEKMFLWMVVRINQQLDTKQPRQYF IGVLDIAGFEI PFDNSLEQLCINF TNEKLQQFFNHMFVLEQEYKKEGIEWE FIDFGMDLAACIELIEKPMGIFSI LEEECMPFKATDTSFKNKLYDQHLGKSNN FQKPKPGKGAEAHFSLVHYAGTVDYNI TGWLEKNKDPLNETVVGLYQKS SLKTLALLFASVGGAEAESGAGGKGGKGGKSSFTQVSALFRENLNKLMS NLRSTHPHFVRC LIPNETKTPGAMEHELVLHQLRCNGVLEGI RIRCRKGFP SRI LYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVF FKAGLLGLLEEMRDEKLAQLITRTQARCRGFLMRVEFKMMERRESIFCTQ YNVRAFNMNVKHPWMLFFKIKPLLSAESEKEMANMKKEFEKTKKEELA KSEAKRKELEEKMVSLLEQEKNDLQLVQAEADGLADAERCDQLIKTKIQ LEAKI KELTSRAEDEEMNAELTAKKRKLEDECSELKDDIDDL ETLAKVE KEKHATENKVKNLTEEMAALDETI AKLTKEKKALQEAHQQTLDLQAEED KVNTLTAKTKLEQQVDDLEGSLEQEKKL RMDLERAKRKLEGLDKMTQE STMDLENDKQQLD EKLKKDFEISQIQSKIEDEQALGMQLQKKI KELQARIE ELEEEIEAERTSRAKAEKHRADLSRELEISERLEEAGGATAAQIDMNKKRE AEFQKMRRDLEEATLQHEATAAALRKKHADSTAELGEQIDNLQRVKQKLE KEKSELKMEIDDLASNMESVSKAKANLEKMCRSLEDQLSEIKTKEEEQQRT INDISAQKARLQTESGEYSRQVEEKDALISQLSRGKAFTQQIEELKRHLEEE IKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSEVAQWR TKYETDAIQRT EELEBAKKLAQRLQDAEEHVEAVNSKCASLEKTKQRLQ NEVEDLMIDVERSNAACAALDKKQKNFDKILSEWKQKY EETQAELEASQK ESRSLSTELFKMKNAYEESLDHLETLKRENKNLQQEISDLTEQIAEGGKAH ELEKVKQIEQEKSELQAAL EEAASLEHEEGKILRVQLELNQVKSIDIRKI AEKDEEIDQLKRNHLRVVDSMQSTLD AEIRSRNEALRLKKKMEGDLNEIEI QLSHANRQAEEAQKNLRNTQGV LKDTQIHLDDALRSQEDLKEQVAMVER RANLLQAEIEELRAALEQT ERSRKVAEQELLDASERVQLLHTQNTSLINTKK KLESDISQIQSEMEDTIQEARNAEKAKKAITDAAMMAEELKKEQD TSAHL ERMKKNLDQTVKDLQHRLDEAEQLALKGKKQIQKLEARVRELEGEVDA EQKRSABAVKGVRYERRVKELTYQSEEDRKNVLRQLQDLVDKLMKVKS YKRQAEEAEELS NVNLSKFRKI QHELEEAERADIAESQV NKLRAKSREIGK KAESEE
161	Chicken ALBU (Albumin)	MKWVTLISFIFLFSSATSRNLQRFARDAEHKSEIAHRYNDLKEETFKAVAMI TFAQYLQRC SYEGLSKLVKD VVDLAQKCVANEDAPECSKPLPSIILDEICQV EKL RDSYGAMADCCSKADPERNECFLSFKVSQPDFVQPYQRPASDVICQY QDNRVSF LGHF IYSVARRHPFLYAPAILSFAVDFEHALQSCCKESDVGACLD TKEIVMREKAKGVS VKQQYFCGILKQFGDRVFQARQLIYLSQKYPKAPFSE VSKFVHDSIGVHKECC EGDMECEMDMARMMSNLCSQQDVFSGKIKDCC EKPIVERSQCIMEAEFDEKPADLPSLVEKYIEDKEVCKSFEAGHDAFMAEFV YEYSRRHPEFSIQLIMRIAKGYESLLEKCKTDNPAECYANAQEQLNQHIKE TQDVVKTNC DLLHDHGEADFLKSI LIRYTKMPQVPTDLLLETGKKMTTIG TKCCQLGEDRRMACSEGYLSIVIHDTCRKQETTPINDNVSQCCSQLYANRR PCFTAMGVDTKYVPPFPNPMFSFDEKLCSAPAEEREVGQMKLLINLIK RK PQMTEEQIKTIADGFTAMVDKCKQSDINTCFGEEGANLIVQSRATLGIGA

[0053] An animal protein can comprise a turkey protein (Table 9). A turkey protein can comprise a turkey muscle protein. A turkey protein can comprise a turkey blood protein. A turkey protein can comprise a turkey albumin protein. A turkey protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 162-174.

[0054] A food product may comprise a fragment or a truncation of a turkey protein. The food product may include

a fragment or truncation of any one of SEQ ID Nos: 162-174. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 162-174. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 162-174. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 162-174. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 162-174.

TABLE9

Turkey Proteins		
SEQ ID NO	Description	Amino Acid Sequence
162	Turkey G1NL16 (Creatine kinase)	MPFSNSHNLLKMKHSADDEFPDLSAHNNHMAKVLTLDLKYKLRDRQTSSG FTLDDVIQTGVDPNPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYK PTDEHKTDNLADNLQGGDDLDPNYVLSSRVRTGRSIRGFCFPHCSRGERRA IEKLSVEALGSLGGDLKGKYALRNMTDAEQQLIDDHFLFDKPVSPLLLA SGMARDWPDARGIWHNDNKTFLVWINEEDHLRVISMQKGGNMKEVFTRF CTGLTQIETLFFKSKNYEFMWNPHLGYILTCPSNLGTGLRAGVHIKLPNLGK HEKFGEVLKRLRLQKRGTTGGVDTAAGGVDFVSNADRLGFSEVELVQMV VDGVKLLIEMEKRLEKQGSIDDLMPAQK
163	Turkey A0A803Y9V3 (Actinin alpha 2)	MNSMNQIETNMQYTYNYEEDYMTQEEEDWRDLLLLDPAWEKQQRKTFTA WCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKDRGKMRPHKI ANVNKALDYIASKGVKLVSIGAEEIVDGNVKMTLGMITWIIILRFAIQDISVE ETSAKEGLLLWCQRKTAPYRNVINQNFHLSWKDGLGLCALIHRHRPDLIDY SKLNKDDPIGNINLAMEIAEKHLDPKMLDAEDIVNTPKPDRAIMTYVSCF YHAFAGAEQAETAANRICVLAVNQENERLMEEYERLASELLEWIRRTIPW LENRTPEKTMQAMQKLEDPRDYRRKHKPPKVQEKQCQLEINFNTLQTKLRI SNRPAPFMPSEGMVSDIAGAWQRLEQAEGYEEWLLNEIRLERLEHLAE KFRQKASTHEQWAYGKEQILLQKDYESASLTVRAMLRKHEAFESDLAAH QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGTLTQKRRREAL ERTEKLETTIDQLHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLIS AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEEIRT KWEKVQQLVPQRDQSLQBELARQHANERLRQFAAQANVIGPWIQTKME EIARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVEDN KHTNYTMEHIRVGVWELLTTIARTINEVETQILTRDAKITQEQMNDFRASF NHFDRRKNGLMDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTTFQ SFIDFMTRETADTDAEQVIASFRILASDKPYILADELRRELPEPEQAQYCIKR MPQYTGPGSVPGALDYTSFSSALYGESDL
164	Turkey A0A803Y8V3 (Alpha-1,4 glucan phosphorylase)	MAAPLSDGERRKQISVRGIAGLDVAEVRKSFNRHLHFTLVKDRNVATPR DYFFALAHTVRDHLVGRWIRTQQHYERDPKRIYYLSLEFYMGRTLQNTM VNLGLQACDEAIYQLGLDLEELLEEIEEDAGLGNGLGLRLAACFLDSMATL GLAAYGYGIRYEPGIFNQKIVDGWQVEEADDWLRYGNPWKARPEYMLP VHFYGRVHSPEGVKWDVTQVVLAMPYDTPVPGYKNNTVNTMRLWSAK APNDFNLQEFNVGDYIEAVLDRNLAENISRVLYPNDNFFEGKELRLKQEYF VVAATLQDIIRRFKSSKFGCQDPVRTSFETFPDKVAIQNLNDTHPALSIPELMR ILVDVEKVDWDKAWIEITKQTCAYTNHTVLPALERWPVSMFEKLLPRHLEI IYALNQMHLDRAALYPGDIIDRLRRMSVIEEGDCKRINMAHLGCVIGSHAVN GVARIHSDIVKNTVFKDFYELDPEKFNKNTNGITPRRWLLLCNPGGLADVIAE KIGEDFVTDLSQLKKLLDFIDDETFFIRDVAKVKQENKLFKSAYLEEYKVKI NPSSMFDIQVKRIHEYKRLNCLHAIITLYNRKSDPSKSFVPRTIMIGKAA PGYHMAKMIKLIITSGEVVNNDPYVGDKLKVI FLENYRVS LAEKVIPAADL SQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFIFG MRVEEVEALDRQGYNAREYYDIRPELRQAIDQISSGFFSPRDPGCFRDRVVN MLMHHDRFKVFADYEAYIKCQGVQDQLFMDPREWTKKVIIRNIACSGKFSS DRTITEYAREIWGVEPSGDVFPAAWEA
165	Turkey GIN2A4 (Fructose-bisphosphate aldolase)	MTHQYPALTAEQKKELSDIALRIVAPGKGI LAADESVGSMKRLNQIGVEN TEENRRLYRQILFSADSRVKKCIGGVIFFHETMYQKADGTPPVQMIDKGI VVGIKVDKGVVPLAGTDGTTTTQGLDGLSERCAQYKKGDGADPAKWRCVL KISDNTPSALAIMENANVLARYASICQNGIVIVIVEPEILPDGDHDLKRCQY VTEKVLAAVYKALSDHHVYLEGTLKPNMVTGHSCTPKYSPEEIAMATV TALRRTPVPAVPGVTFLSGGQSEEEASINLNAINTCPLVRPWALTFSYGRAL QASALSAWRGQRDNANAATEEFVKRAEVNGLAALGKYEGSGDDSGAAGQ SLYVANHAY

TABLE9-continued

Turkey Proteins		
SEQ ID NO	Description	Amino Acid Sequence
166	Turkey Q6W5H1 (Myosin essential light chain isoform 1)	MAPKKDVKKPAAAAAPAPAPAPAPAPAKPKPAIDLKSIIKIEFSKEQQD DFKEAFLFPDKTGDAKITLSQVGDIVRALGQNPTNAEMNKILGNPSKEEMN AKKITFEELPMLQAAANNKDQGTIEDFVGLRVFDKEGNGTVMGAELRH VLATLGEKMTREEVEELMKGQEDSNGCINYEAFVKHIMSV
167	Turkey G1NMR6 Glyceraldehyde- (3-phosphate dehydrogenase)	MVKGVNGFGRIGRLVTRAAVLSGKVQVVAINDPFIDLNYVMVMFKYDST HGHEFGTVKAENGLVINGHAI TIFQERDPSNIKWADAGAEYVVESTGVFT TMEKAGAHKGGAKRVII SAPSADAPMFVMGVNHEKYDKSLKIVSNASCT TNCLAPLAKVIHDFGIVEGLMTTVHAI TATQKTVDGPGSKLWRDGRGAA QNIIPASTGAAKAVGKVI PELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK YDDIKRVVKAADGPLKGILGYTEDQVVSDFNGDSHSSTFDAGAGIALND HFVKLVSWYDNEFGYSNRVVDLMVHMASKE
168	Turkey A0A803XNQ2 (Tropomyosin 3)	MEAIKKMQMLKLDKENALDRAEQAEAEQQAEEERSKQLEDELAAMQK KLKGTEDLDKYSEALKDAQEKLELAEKKAADAEVASLNRRIQLVEEE LDRAQERLATALQKLEAEKAADSESRGMKVIENRALKDEEKMELOEIQ KEAKHIAEEADRYEEVARKLVIIEGDLERTEERAELAESKCSLEEEELKNV TNNLKSLEAQAEKYSQKEDKYEEI KILTDLKKEATRAEFAERSVAKLEK TIDDLDELVAQKLKYKAISEELDHALNDMTSI
169	Turkey A0A803YPL8 (Myosin-binding protein C)	MCLQLFVFSLEPPPPSFLKNKELTFVYCLHKALMVECWCHSICSSPTLADQI REVVGLESLSVSGKCILERIKHWLIIDIFLFLCISVMSIGESPPEETEKHDDSQ RSTLFIEKPQSGSVSGGNI TFIKVEAKDLLRKPNVWFKGKWMDLASKA GKHLQLKESFERHTKIHTFEMHII QAKENYAGNYRCEVSYKDKFDSCSFDL EVTESQAAPSIDIRSAFKRSGDGQDDAGELDFSGLLKRREIKQQEEEEPEVD VWELLKNANPSEYEKIAFYGITDLRGLMLKRLKMRREVKKSAAFSGKGLD PAYQVDKGGKVRFMVELADPTVELKWKYKNGQEIRPSAKYIFEHKGNGRIL FINNCTMTDDARYYV TAGDEKCS TELFVRDPPILVTKGLEDTNTYVGERVE LSCEVSEDDANVWKFNGVELTNEPKSRYRI KVEGKHTLIIEEAAKNDNA TYSVMTTGGQSEAKLSVDLRPLKISLALQDTVRLGQEIHLKCEISENVEGK WYKNGQLIEASDRVKLYHKGRIHRFI IASAAVDDEGEYMFVPDAYNINIPC KVHVVDPPKLHLDGLGENNTVTVAGTKLRLEIPITGEPTPKVMWSKGDK WITDSGRIRAEYSDSSCLVIDTAEREDSGPFRITLKNAGEEDSALINIKVVD VPDPPQAPNVTEVGEDWCMTWEPPANDGGSPILGYFIERKKKQSSRWMR LNFELCKETTFEPKKMIEGVAYEVRFVAVNAIGTSKPSMPKSFVPLAVTSP PTLLAVDSVTDTSVTMKWRPPDHI GAAGLDGYVVEYCFEGTDEWIVANPE LTDKTKFTINGLPTGSKI LVRVKAVNAAGESEPRYHPQPI LVKEVIEPPKIRL PRHLKQTYTRRVGETVNLVIPFQVRTKNTDRKTLNTHFSYATASYLLFLYV NLGYWTA FVNI LHPEGKSHDKASGIIHSYLGTKIGHYHMSQSN
170	Turkey G3URS1 (Myomesin 1)	MNVKGEISAYASLVVKRYKGEIDASHLHAGTFSMPLSLGVTPHGFASRFEI KFIDKFEVMPGREGETMSLGCTVVISPDIKRFKPIEWYRNGVLITPSKWVQ MHW TGERASLT LTHANKEDGLYTLRVVMGEYEEYSGYV FVRDADAEI PGAPGAPLDVQCLDANKDYIVSWKQPAVDGNGNPI LGYFIDRCEVGTTHW TQCNETPVKPARFPVTGLIEGRSYIFRVRVNRAGVSLPSRVSEAVAALDPA DRARLRSHPSAPWTGQIIVTEEEPAEGVIPGPPTDLQVIEATKNYVVLWKP PGQRGHEGIMYFVEKCIAGTEDWQRVNTEIPVKS PRFAVFDLSEKSYCFR VRCNSAGIGEPSEATEATVVDKLDIPKPPGRIMPTRNTDTSVVVTWTEPP DAKELVGYYIESSVVGSGHWEP CNNNPVKGTRFICHGLVNGEKYIFRVRV NAAGLSEFSQDSEPIEVQAAIAAPSPPYDITVLESVRDSMVLGWKQPKATGG TEITGYVNYREVIGVPGKWEANI KAISERAYRIENLKENMVYQFQVAA ANLAGVGTPSLPSKPFKEEWTIAVPGPPHDVITYTEVRKDSLVLWKEPVY TRGSP IAGYIIDMKETEAKEEHRSVNEKPLQKKFLKIAGLTEGVSYVFRV RATNQAGVGRPSDITDPVVAETRPGTKEVVNVDDGVI SLNFECQMS NSKFPVSKNYEPIEDDSRLDIDSKGGKS KAI FKNLGEDDLGIYSCVVTDTDG VSSGYTIDEEEMKRLLALSHERKFPTIPLASELAVEILEKGEVRFWLQAEKLS GNAKANFVNDKEIFNGEKYKMKVDQRTGLVEMIMDKLEDKDEGTYTFQ LQDGKATNQSSSLVLIGDVFKKL RDEAEFQRKEWHRKQGP HFDQLGWEV TDECNVMLKCKVANI KKETHIVWYKDDREIMVDEEHD FDKGVCTLLISEFS KKDTGIYEVILKDDRDKVSELKLTDTAFADLMNEVCRRIALSATDLKIQST AEGIRLYSFVTYYVDDL RVGVVHNDTQIRYTRVKTGVTGEQIWLQINEPT PQDKGKYTELMELFDGKTGHKKTVDLSGQAFDEAFAPQLKQAAIAEKNRA RVLGGLPDVVTIQEGKALNLSCTVWGDPTEPSWLKNEKSFVSDANCILKF EAGKNVSFTISAVSTHDSGKYSLVVKNKYGTETSDVTIVITVSGQTD TAK HTVSI SQSQNNITVSKKKKAT TMSKPQKEKKDKIPHVDEIIATEETS DSKEIPRVEKEKNTENNQTAKKTHGKAAAARTVP

TABLE9-continued

Turkey Proteins		
SEQ ID NO	Description	Amino Acid Sequence
171	Turkey A0A803YQ52 (Phosphopyruvate hydratase)	MSILKIQAREIFDSRGNPTVEVDLYTNKGLFRAAVPSGASTGIYEALERLDN DKTRYLGKGVSKAVEHVNKT IAPALISKNNVVEQEKIDKLMLEMDGTEN KSKFGANAILGVSLAVCKAGAAEKGVPLRYHIADLAGNPEVILPVPAPNVI NGGSHAGNKLAMQEFMILPVGADNFKEAMRIGAEVYHNKKNVIKEKYGK DATNVGDEGGFAPNILENKEALELLKTAIGKAGYSKVVIGMDVAASEFYR DGKYDLDFKSPDDPSRYISPDQLADLYKGFVNKNYPLVSIEDPFDQDDWAA WKKFTASVGIQVVGDDLTVTNPKRIAKAVEEKS CNCLLLKVNQIGSVTESL QACKLAQSNWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERL AKYNQLLRIEELGSKARFAGRNFRNPRIN
172	Turkey G1N0W1 (Myosin light chain 2)	MAPKKAKKRIEGANSNVFSMFEEQAQIQEFKEAFTIMDQNRDGFIDKADLRD TFAALGRLNVKNEEIDEMIKIAPGPINFTVFLTMFGEKLGKADPEETILNAP KVFDPEGKGLKSAYIKEMLMTQGERFSQEEIDQMFAAPPDVSNGNLDYKNL VHVITHGEEKD
173	Turkey GIMSQ0 (Myosin heavy chain 2)	MSSDAEMAI FGEAAPYLKSEKERIEAQNRPFDAKAACFVVDQKQMYVKG TIQSREGGKVTVKTYDDTTVTVKDDVFPNMPKPKYDKIEDMAMMTHLHEP AVLYNLKERYAAWMIYTYSGLCVTVNPKWLPVYNPEVVS GYRGKKRQ EAPPHIFSISDNAYQFMLTDRDNQSILITGESGAGKTVNTRKVIQYFATIAVT GEKKKQOPSKMQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFKGFIRIHF GTTGKLASADIETYLLEKSRVTFQLASERSYHIFYQIMSNKKPELIGEYLLLI TTNPYDFTYVSQGEITVASIDDELLATDNAVDILGFSPEKVGMYKLTGA VMHYGNMKFKQKQREEQAEPDGTVEADKAAYLMLNSAELLKALCYPR VKVGNEYVTKGQNVQQVYNSVGLAKSVYEKMFWMVTRINQQLDTKQ PRQHFIVGLDIAGFEIFDFNSLEQLCINFNTEKLQQFNNHMFVLEQEEYKKE GIDWEFIDFGMDLAACIELIEKPMGIFSILEECCMPKATDTSFKNKLVDQHL GKSNNFQKPKPAKGKAEAHFSLIHYAGTVDYNI TGWLEKNKDPLNETVVG LYQKSLKLLSPLFSNYAGAETSDSGSGKKGGKKGGSFQTVSAVPRENLN KLMTNLRSTHPPHVRCLIPNETKTPGEMDHVLMHQLRCNGVLEGIIRK GFPSRILYADFKQRYKILNASAIPEGQFIDSKKASEKLSSIDVDHNQYRFGH TKVFFKAGLLGLEEMRDEKLVSLITHMQAMCRGYLMRTEFKKMNARRES IYIIQYNVRAFMNVKHPWMKLYFKMKPMLKSVESEKEMANMKEEFECT KEELAKSEAKRKELEEMVSLLEKENDLQLQVQSESENLADEERCEGLIK SKIQLEARIKELNERMEDEEMNAELTAKKRKLEDECESELKKDIDDELTLA KVEKEKHATENKVNLTTEMAALDENISKLTKEKKALQEAHQQTLEYLQA EEDKVSILTKIKTKLEQQVDDLEGSLEQERKQRMDLERAKRKLGDGLKMS QDSIMDLKNDKQQTDERLKKKDFEISHLQSKIEDEQAQSSQLQKKIKELQA RTEELEEEIEAERANCAKTEKQRADLSRELEESERLEEAGGATAAQIEMNK KREAEFQKMRRDL EEATLQHEATAAALRKKHADSTAELEGBQIDNLQRVKQ KLEKEKSELKMEIDDLASNMESVSKSNLEKMCRALEDQYSEVRTKDDE HMRILNDMNTQKTRLTENGELTRQLEEKESLISQLTRGQAITQQTEELK RQLEENKAKNALAHGLQSAHDCDLLREQYEELEAKGELQRALEKANS EVAQWRTKYETDAIQCMEELED AKKLAQRLQDSEEQIEAVNSKCSLEK TKQRLQGEVDDLMIAVERSNAACAAPDKKQKQNFDMVAEAWKQKYQESQ AELEAAQKESRTLSTEIFRMKNAYEEVLDQVETVRRENKLNQOEISDLTEQ LAEAGKANHGLEKAKKQTEQEKYDLQAAL EAEVSL EHEEGKILRVQLEL NQVKSGVDRRS AEKDEEIQQLKKNHQRVLESMTMLDAEIRSRNDALRLK KKMEGDLNDMEIQLSRANCQIAEIQKHLKGMQGQLKDSQLHLDDALREND DLKEQLAVVERRNNLMMTELEQMRAALEQTERARKVSEQLTDTSERVQI LHSQNTSLNLT KKKLEVDITHLQNEVEDSIQEASNAEEKAKKAITDAAMMA EELKKEQDTSALERMKNRLEQMVKDLQHRLDEAEQLTLKGGKKQLQKL EARIHELENELDIEQKRGTESLKGARKYERRLKELTYQSEEDKKNILRLQNL VDKLQLKV KAYKQAEAEAEQASTNLSRCRKTQHELEAEERADIAECQV NKLRAKSRDTGRQKSLSELGDNTPLPVIWHNMCNIDRPTTPQVILLHQ
174	Turkey GINCR2 (Albumin)	MKWVTLISFIFLFSSATSRNLQRVARDAEHKSEIAHRYNDLKEETFKAVAMI TFAQYQLQRCSDGLSKLVKD VDLAQKCVANEDAPCSKSLPSIILDEICQV EKL RDSYGAMADCCSKADPERNECFLSFKVSPDFIQPYQRPADVICKEY QDNRVSLLGHF IYSVARRNP MYAPTILSFAADYEHALQSCCQESDVGACL DAKETAMREKAKKISVKQYSCGILKKGFGDRVFAEKLVRLSQKYPKAPFS EVAKLVRSEKDVHKECCEGDMVECDMDMAEIIINMCSKQDVFSSKIKGCC EKPIMERSQICIMEAEFDEKPADLP SLVEKYIQDKEVCKSFEAGHD AFMSEFV YEYSRRHPEFSTQLILRIAKGYESLLEKCCCKTDNPAECYANAQDQLNQHIKE TQDVVKTNCDDL SAHGEDPDLKSLIRYTKMPQVSTETLLEIGKKMTAIGT KCCQLSEDRMACSEGYSIVIHDMCRRQETTPINDNVSHCCSDSYAHRRP CFTAMGVDTKYVPPFPNPMFSD EKLCSAPAEEREVGQMKLLINLIKRP QMTTEEQIKTIADGFTAMVDKCKQSDINTCFGEEGANLIVQSRATLIGIV

[0055] An animal protein can comprise a duck protein (Table 10 and Table 11). A duck protein can comprise a duck muscle protein. A duck protein can comprise a duck blood protein. A duck protein can comprise a duck albumin protein. A duck protein can comprise a protein from a mallard duck (*Anas platyrhynchos*). A duck protein can comprise a protein from a Northern mallard duck (*Anas platyrhynchos platyrhynchos*). A duck protein can comprise a protein from a Muscovy duck (*Cairina moschata*). A duck protein can comprise a protein from a domestic Muscovy duck (*Cairina moschata domestica*). A duck protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 175-200.

[0056] A food product may comprise a fragment or a truncation of a duck protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 175-200. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 175-200. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 175-200. A length of the fragment or truncation that is at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 175-200. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 175-200.

TABLE 10

Northern Mallard Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
175	Northern Mallard Duck A0A493TFK1 (Creatine kinase)	MPFSNSHNLLKMKHSATDEFPDLSAHNNHMAKVLTLDLYKKLRDKQTPSG FTLDDVIQTGVDPNPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYK PTDQHKTDNLNAGNLQGGDDLDPNYVLSSRVRTGRSIRGFCLPPHCSRGERR AIEKLSVEALGSLGGDLKGKYVALRNMNTDAEQQLIDDFLFDKPVSPLLL ASGMARDWPDARGIWHNDNKTFLVWINEEDHLRVISMQKGGNMKEVFTFR FCTGLTQIENLFKSKNHEFMWNPFLGYILTCPSNLGTGLRAGVHIKPLHLGK HEKFGVVLKRLRLQKRGTTGGVDTAAGGVGVDVSNADRLGFSEVELVQMV VDGVKLLIEMEKRLKGGQSIDDLIPAQK
176	Northern Mallard Duck A0A493TCE5 (Actinin alpha 2)	MNSMNQIETNMQYTYNYEEDEYMTQEEWDRDLLLDPAAWKQQRKTFTA WCNSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLKPDRGKMRPHKI ANVNKALDYIASKGVKLVSIGAEEIVDGNVKMTLGMWITIILRFQIDISVE ETSAKEGLLLWCQRKTAPYRNVIQNFHLSWKDGLGLCALIHRHRPDLIDY SKLNKDDPIGNINLAMEIAEKHLDIPKMLDAEDIVNTPKPDERAINTVVSFC YHAFAGAEQAETAANRICKVLAVNQENRMLMEEYERLASELLEWIRRTIPW LENRTPEKTMQAMQKLEDPRDYRRKHKKPKVQEKCCLEINFNTLQTKLRI SNRPAPMPSEGMVSDIAGAWQRLQAEKGYEEWLLNEIRRLERLEHLAE KFRQKASTHEQWAYGKEQILLQKDYASALTEVRAMLKHEAFESDLAAH QDRVEQIAAIAQELNELDYHDAASVNDRCQKICDQWDSLGTLTQKRREAL ETEKLLLETIDQLHLEFAKRAAPFNWMEGAMEDLQDMFIVHSIEEIQSLIS AHDQFKATLPEADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEIIRT KWEKVKQLVPQRDQSLQEEELARQHANERLRRQFAAQANVIGPWIQTKME ETARSSIEMTGPLEDQMNQLKQYEQNIINYKHNDKLEGDHQLIQEALVEDN KHTNYTMEHIRVGEWELLTTIARTINEVETQILTRDAKGTQEQMNDFRASF NHFDRRKNGLMDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTFPQ SFIDFMTRETADTDTAEQVIASFRILASDKPYILADELRLRELPPQQAQYCIKR MPPYTGPGSVPGALDYTSFSSALYGESDL
177	Northern Mallard Duck U3ISG6 (Alpha-1,4 glucan phosphorylase)	MAAPLSDGEEKRKQISVRGIAGLDVAEVRKSPNRHLHFTLVKDRNVATPR DYFFALAHTRVDRHLVGRWIRTQQHYIERDPKRIYYLSLEFYMGRTLQNTM VNLGLQACDEAIYQLGLDLEEEIEEDAGLNGGLGRLAACFLDSMATL GLAAYGYGIRYEFGIFNQKIVDGVQVEEADDWLRYGNPWEKARPEYMLP VHFYGRVEHTPEGVKWVDQVVLAMPYDTPVPGYKNTVTNMLWSAK APNDFNLQEFNVGDYIEAVLDRNLAEISRVLYPNDFEGEKRLRKQYEF VVAATLQDIIRRFKSSKFGCRDPVTSFETFPDKVAIQLNDRTHPALSIPELMR ILVDVEKVDWKAWETKRTCAYTNTHTVLPALERWPVSMFEKLLPRHLEI IYALNQMHLDRAALYPGDVDRLRMSVIEEGDCKRINMAHLCVIGSHAV NGVARIHSDIVKHSVFKDFYELEPEKFQNKNTNGITPRRWLLCNPLGLADVIA EKIGEGFVTDLSQLKLLDFIDNEMFIRDVAKVKQENKLFSAYLEEQYK KINPSSMFDVQVKRIHEYKRQLLNLCHAITLYNRIRSDPSKSFVPRTIMIGK AAPGYHMAKMIKILITSIGEVINNDPYVGDKLVIFLENYRVSLAEKVIAPAA DLSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFI GMRVEDVEALDRKGYNALEYERLPELRQAIDQISSGFFSPRDPGCFRDVV NMLMHDRFKVPADYEAYIKCQGVQDQLFMDPREWTKKVIIRNIACSGKFS SDRTITEYAQEIWGVPSSTKIPPPNIPRD

TABLE 10-continued

Northern Mallard Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
178	Northern Mallard Duck U3IHG8 (Fructose-bisphosphate aldolase)	MTHQFPALSP EQKKALS DIAQRIVASGKGILAADES VGTMG NRLQRINVEN TEENRRAFREILFSSDTSINQSIGGVILFHETLYQKDSGKPPAIKKEKGI VVG IKLKDGTSPLAGTNGETTIEGLDGLSERCAQYKKGVDGFKWRAVLKITST TPSQLAIQENANTLARYASICQQNGLVPIVEPEVLPGDHDLDQRCQYVTEK VLAAYKALNDHHVYLEGTLLKPNMVTAGHACTKKYTPQDVAAATVTTL LRTVPAAVPGICFLSGGQSEEEASLNLNAMNQSSLPKPWKLTFSYGRALQA SALAALWGKSENKKAQAEFRKRAQINSACRGQYVVS GKTDAATQSLF TASYTY
179	Northern Mallard Duck U3IA79 (Myosin light chain 1)	MAPKKDVKKPAAAAAPAPAPAPAPAPAKPKPAIDLKS IKIEFSKEQQD DFKEAFLLFDRTGDAKITLSQVGDIVRALQONPTNAEINKILGNPSKEEMNA KKITTFEEFLPMLQAAANNDQGTGYEDFVEGLRVFDKEGNGTVMGAE LRHV LATLGEKMT EEEVDLMKGQEDSNGCINYEAFVKHILSV
180	Northern Mallard Duck A0A493TQC9 (Glyceraldehyde-3-phosphate dehydrogenase)	MVKGVNGFGRIGRLVTRAAILSGKVQVVAINDPFIDLNYMVYMFKYDST HGHRGT VKAENGKLVINGNAITIFQERDPSNIKWADAGAEYVVESTGVFT TMEKAGAH LKGGAKRVII SAPSADAPMFVMGVNHEKYDKSLKIVSNASCT TNCLAPLAKVIHDNFGIVEGLMTTVHAI TATQKTVDGPSGKLWRDGRGAA QNIIPASTGA AKAVGKVI PELNGKLTGMAFRVPTPNVSVDDLTCRLEKPAK YDDIKRVVKAADGPLKGI LAYTEDQVVS CDFNGDSHSSTFDAGAGIALND HFVKLVSWYDNEYGYSNRVLALSTLLHAANI PPSSR
181	Northern Mallard Duck A0A493TRM1 (Tropomyosin 3)	MEAIKKKMQMLKLDKENALDRAEQAEAEQQA EERSKQLEDELAAMQK KLKGTED ELDKYSEALKDAQEKLELA EKKAADEAEVASLNRRIQLVEEE LDRAQERLATALQKLEAEAKADESERGMKVIENRALKDEEKMELOEIQL KEAKHIAEEADRKYE EVARKLVII EGD LERTEERAELAESKCSLEEEELKNV TNNLSLEAQAEKYSQKEDKYEEIEKILTDKLEAE TRAEFAERSVAKLEK TIDDELELYAQKLKYKAI SEELDHALNDMTSI
182	Northern Mallard Duck U3J9W2 (Myosin binding protein C3)	MPEPVKKT VSAFTKKPKTTEVTAGSTAVFEA TEKASVVKWQ RAGTEIT DSEKYV IKAENKHS LTINNVVKEDDVTYAVIAGSSKVKFELKVKEPEKSE TVTPAEAAPAPAA SELPAPPAESSQNT E VPSAGTQPEEPADPIGLFVTRPQDG EVTVGGNITFTAKVAGESLLKKPSVKWFKGKWM DLASKVGHQLQLHDSY DRNNKVYTFE LIEI EANMTFAGGYRCEVSTKDKFDSN FNLT VNEAPLSGD LDIRAAPRR TT EGHDEAGELNFSALLKKRGDGKTESQPDVDVWEILRKAPP SEYEKIAFQYGITDLRGM LKRLKRIKKEEKSTAF LKKLDPAYQVDKGQKI KLMVEVANPDADV KWLKNGQEIQVSGSRYIFEAVGNKRILTINHCSLADD AAYECVVGEESFTELFVKEPPI LITRPLEDQMV MVGERVEFECEVSEEGAT VKWEKDGVELTREETF KYRFKDKGKKQYLI INESTKEDSGHYTVKTNGGV SVAELIVQEKKLEVYQSIADLT VKARDQAVFKCEVSDENVKGIWLKNGKE VVPDERIKI SHIGRIHKLTI EDVTPGDEADYSFIPQGFAYNLSAKLQFLEV KID FVPREPPKIHLDCLGQSPDTI VVVGANKLR LLDVPI SGDPPTPTVIWQVNNK TDLLRND DSMTPSSENSDLNTDSKLLFSESEGRVRVEKHEDHCVFIEGAEK EDEGVYRVIVKNPVGEDKADITVKVIDVDPDPEAPKISNIGEDYCTVQWQPP KYDGGQPV LGYILERKKKKS YRWMLNFDDLKELTYEAKRMIEGVVYEM RIYAVNSIGMSRSPASQPFMPIAPPSEPTHTFTVEDVSDSTVALKWRPPERIG AGGLDGYIVEYCKEGSTEWIPALPGLTERTSALIKDLVTGDKLHFR IKA INL AGESGAIIKEPI TVQEILQRPKIWLPRHLRQTLVKKVGETINIVIPFQKPPRP KITWTKDQSLDAKDVGIRNSNTDTILFIRKAE LHHS GAYEITLQIENMTDK VAITIQI IDKPGPPQNIK LTVWGFNAALEWTPPQDDGNAQILGYTVQKAD KKTMEWYTVFDHYRTNCV VSELI MGNEYFRVRFSENL CGLSETAATTKN PAYIQKTGTTYKPPCYKEHDFSEPPKFTHPLVNRSVIAGYNTT LSCAVRGIP KPKIYWKVKNV DLSGDAKYRMFSKQGVLTLEIRKPTPF DGGGYTCKAVNE CGEAIECKLDVRVPQ
183	Northern Mallard Duck A0A493TFW8 (Myomesin 2)	MSSLAVPFYQRRHKHFDQSYRNIQTRYVLEEYAARKAASRQAAYYESSGL GKTCRLCARRARTLAHETMQESRKRTHEQKSHASEEKRIRFAS ELSALEK EIHTARHRAREQLDRLAIQRMVEENMAMERH AVEERI SRAP EILVRLRSHT VWEKMSVSLCFTVQGFP TPVVQWYKNEELITPASDPAKYRVESKYGVHVL HINRADFDDSATYSAVATNVHGQASTNCAVVRFRFREGEPHAGIMPFHL PLSYDVCFTHFDVQFL EKFGVT FATEGETLTLKCSVLVTP ELKRLRPRAEW YRDDVLIKDSKWT KLYFGEGQAALSFTHLNKDDEGLYTLRMITKGGVNEC SAYLFVRDADALIAGAPGAPMDVKCHDANRDYIVTWKPNPTTSQSPVIG YFVDKCEVGLENWVQCNDAPVKICKYPVTGLYEGRSYIFRVRAVN NAGIS RPSRVSEPVAAALDPVDLERTQTIHVDEGRKIVISKDDLEGDIQIPGPPTNVHA SEISKTYVVLSDPPVPRGREPLTYPIE KSMVGSGSQRVNAQVAVKSPRY AVFDLAE GKPYPVFRVVSANKHGISDPSEITEPIQPQDTVVPVSAPGRVVATR NTKTSVVVQWDKPKHEEDLYGYIIDYSVVGSNHWEPSNHKPIKYNRFVVH GLETGEQYIFRVKAVNAVGFSENSQSEAIKVQAALT CPSYPHGITLLNCDG HSMTLGWKAPKYSGGSPILGY YMDKREANHQNWHEVNSSLITRTIYTVED

TABLE 10-continued

Northern Mallard Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		LTEDAFYEFKIAAANIVGIGQSPDSEHFCKAWTMEPGPAYDLTVCEVR NTSLVLLWKAPVYEGKSPITGYLVVYKEVDKEDWITANEKPTAQRYPKVT DLHEGHTYVFKVRAVNDAGVGKSEVSEPVLEAKPGTKEIFSGVDDDEGNI YLAFECKEATDASHFAWGKSYEEIEDSDKYKIETEGDHSKLYFKNPDKSDL GTYSISVSDTDGVSSSFVMDDEELERLMAISNEIKNPTIPLKSELAYEVLEKG EVRFWIIQAESLSPDSTFRFVINDKEVENS DRHKINCDRPNGIEMVMDFRTID NEGTYTVQIQDGKAKNQSSLVLIGDAFKAVLAESLQRKEFLRKQGPFPSE FLYWEVTEECEVLLACKVANTKKETVFKWYKDGSGIDVDEMPDLQKGECE HLSIPLRSKDEGVYKATLSDDRGHVSTLELSGKVYDDIILALSRVSGKTA SPLKILCTEEGIRLQCFLKYNEEMKISWSHRESKISSGEKMKIGGGENIAWL QITEPTKDKGNYTFEIFGDKESFKRTLDSLGGQAFDDAYAEFQRLKAAAFAPAE KNRGKVIIGLPDVVTIMDGKTLNLTCTVFGNPDPEVVWFKNDALELNEH YVVSLEQGGYASLTIKGVTSEDSGKFSIHVKNKYGGETVDVTVSVYRHGEK VPEVKPGQLAKPRPIPPSEDSS
184	Northern Mallard Duck A0A493T6U6 (Phosphopyruvate hydratase)	MPERSWIPVGTPLLRWTCTHTKACFEQRSPAVPLVSTKHWSSETTSPVSL EKGLSVVDQEKIDNLMLEMDGTENKSKFGANAILGVSLAVCKAGAAEKDV PLYRHIADLAGNSDLILPVPAPNVINGGSHAGNKLAMQEFMILPVGAESFRD AMRIGAEVYHNLKSVIKEKYGKDATNVGDEGGFAPNILENSEALELLKEAI DKAGYTDKIVIGMDVAASEFYRDGKYDLDFKSPDDPSRYISADELGDLYQS FVRDYPVVSIEDPFDQDDWEAWSKFTANVGIQIVGDDLTVTNPKRIERAVE EKACNCLLLKVNQIGSVTEAIQACKLAQENGWGMVSHRSGETEDTFIADL VVGLCTGQIKTGAPCRSERLAKYNQLMRIEELGDEARFAGHNFRNPSVL
185	Northern Mallard Duck U3I9B1 (Myosin light chain 2)	MQGENHRAPKKAKKRIEGANSNVFSMFEQAQIQEFKEAFTIMDQNRDGFID KADLRDTFAALGRLNVKNEEIDEMIKEAPGINFTVFLTMFGKELKGADPE ETILNAPKVPDPEGKGLKSAYIKEMLMTQGERFSQEEVDQMFAAPPDVS NLDYKNLVHVI THGEEKD
186	Northern Mallard Duck U3J7T0 (Myosin motor domain-containing protein)	FSIISDNAYQFMLTDRENQSIILITGESGAGKTVNTRKVIQYPATIAASGEKKK EEQSGMKQGTLEDQIISANPLLEAFGNAKTVRNDNSSRFKGFIRIHFAGTGK LASADIETYLLEKSRVTFQLKAERSYHIFQIMSNKKPELIDMLLITNPDYD HFVSQGEITVASINDQEELMATDSAIIDLGTADKVAIKYLTGAVMHYGN LKFKQKQREEQAEPDGEVADKAAAYLMGLNSADLLKALCYPRVKVGN VTKGTQVQQVNNISVGLAKAVYEKMFLLMWVRINQQLDTKQPRQYFIGV LDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIEWEFID FGMDLAACIELIEKPMGIFSIIEEEMFPKATDTSFKNKLYDQHLGKSSNFQ KPKPAKGKVEAHFSLVHYAGTVDYNTGWLEKNKDPNLNETVIGLYQKSSV KTLALLFANYGGAEINPQKIIKIDTFKSSSFQTVSALFRENLMKMLTNLRSTH PHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGIRICRKGFPSSVLYADF KQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFGHTKVFVKAGL LGLLEEMRDEKLAQLITRTQARCRGFLMRVEYQRMVERRESIFCIQYNVRA FMNVKHWPMKLFKIKPLLSAESEKEMANMKEFEKTEELAKSEAK KKELEEKMKVLVQEKNDLQLQVQAEADALADAEEERCQDLIKTKIQLEAKI KEVTERAEDEEEINAELTAKKRKLEDECSSELKKDIDDLTLAKVEKEKHA TENKVNLTTEMAALDETIKLTKEKKALQEAHQQTLDLQAEEDKVNTL TKAKTKLEQQVDDLEGSLEQEKRLMDLERAKRKLQEGDLKLSQDSIMDL NDKQQDLDEKLKKDFEISQISQIKIEDEQALGMQLKKIKELQARIEELEEIE AERTSRAKAEKHRADLSRELEEISERLEEAGGATAAQIEMNKKRAEFQKM RRDLEATLQHEATAAALRKKHADSTAELEGEQIDNLQRVKQKLEKEKSEL KMEIDDLASNMESVSKANLEKMCRTLEDQLSEIKTKEEHQRMINDLTA QRARLQTESGEYSRQVEEKDSQISQLSRKQAFQTQIIEELKRHLEEEIKAKN ALAHALQSARHDCDLLREQYEEQEAKEGLQRLSKANSEVAQWRTKYE TDAIQRTEELEEAKKLAQRLQDAEEHVAVNAKASLEKTKQRLQNEVE DLMIDVERSNAACAALDKQKNEDKILAEWKQKYETQAELEASQKESRS LSTELFKMKNAYEESLDHLETLKRENKQLQOEISDLTEQIAEGGKAIHELEK VKKQIQEKSEIQAALAEAEASLEHEEGKILRLQLELNQVKSIEDRKIAEKDE EIDQLKRNLHRLIVESMQSTLDAEIRSRNEALRLKKMEGLDNEMEIQLSHA NRQAAEAQKNLRNTQGVLDKDTQIHLDDALRTQEDLKEQVAMVRRANLL QAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSLINTKKKLET DITQIQSEMEDAIQEARNAEEKAKKAITDASMMABELKKEQDTSAH

TABLE 10-continued

Northern Mallard Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
187	Northern Mallard Duck U3IU92 (Albumin)	MITFAQYLQRCSDGLSKLVKDVVDLAQKCVANEDAPECSKSLPSIFLDEIC QVEKLRSYGAMADCCAASDPERNECFLSFKIPQPDFVQPYQRPASDVICK EYEDNRVSFLGHFIYSVARRNPFLYAPTILGVAADYEHALKSCCKESDVGA CLDGKETGIREKVKKISVKKQYSCGILKKFGDRIFQADKLALLSQYPKASF AEISKLIHDVKDYKECEGDMVECMDRAELVNYMCSKHEAFSSKIKDC CEKPIVERSQCIMEAEFDEKPADLPSLVEKYIQDKEVCKSFEGHDAFMSEF VVEYSRRHPEFSTQLILRIAKGYETLLEKCKTDNPAECYANAQDQLNEHIK ETQDVVKTNCDLLTTHGEADFLKALLIRYTKMPQVSTETLLEIGKMTGV GTKCKCLPEDRRLLPCSEGYLSMVIQDMCRRQETSPINDNVSHCCSDSYAYR RPCFSAMGVDTKYVPPFPNPDVFNFDEKLCSAPAEEAEGQMKLLINLIK KPMTEEQIKTIADGFTAMVDKCKQSDIETCFGEEGANLIVQSRATLGIGA

TABLE 11

Domestic Muscovy Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
188	Domestic Muscovy Duck A0A8C3BBL6 (Creatine kinase)	MCCLKCQQGFWDPCAPAVLARGPCFGCGALVTAVGSAMPFSNSHNLKLM KHSATDEFPDLSAHNNHMAKVLTLDLYKKLRDKQTPSGFTLDDVIQTGVD NPGHPFIMTVGCVAGDEESYEVFKELFDPVIEDRHGGYKPTDQHKTDLNAG NLQGGDDLDPNYVLSSRVRTGRSIRGFCLPPHCSRGERRAIKLSVEALGSL GGDLKGKYYALRNMTDAEQQLIDHFLFDKPVSPLLLASGMARDWPD RGIWHNDNKTFVLWINEEDHLRVI SMQKGGNMKEVFTRFCTGLTQIENLFK SKNHEFMWNPHLGYILTCPSNLGTGLRAGVHIKPLHLGKHEKFGVLLKLR LQKRGTGGVDTAAGGVFVSNADRLGFSEVELVQMVVDGVKLLIEMEK RLEKGQSIDDLIPAQK
189	Domestic Muscovy Duck A0A8C3BEA7 (Actinin alpha 2)	RAGDKPQDEYMTQEEWDRDLLLDPANEKQQRKTFTAWCNSHLRKAGT QIENIEEDFRNGLKMLLLEVISGERLPKPDGRGKMRFKIANVKNALDYIAS KGVKLVSIGAEIIVDGNVKMTLGMIIITILRFAIQDISVEETSAKEGLLLWC QRKTAPYRNVNIQNFHLSWKDGLGLCALIHRHRPDLIDYSKLNKDDPIGNI NLAMEIAEKHLIDIPKMLDAEDIVNTPKPDERAIMTYVSCFYHAFAGAEQAE TAANRICKVLAVNQENERLMEEYERLASELLEWIRRTIPWLENRTPEKTMQ AMQKLEDFRDYRRKHKPKVKQEKQLEINFNTLQTKLRISNRPAPMPSEG KMVSDIAGAWQRLQAEKGYEEWLLNEIRLERLEHLAEKFRQKASTHEQ WAYGKEQILLQKDYASASLTEVRAMLRKHEAFESDLAAHQDRVEQIAAIA QELNELDYHDAASVNDRCQKICDQWDSLGLTQKRREALERTEKLLLETIDQ LHLEFAKRAAPFNNWMEGAMEDLQDMFIVHSIEEIQSLI TAHDQFKATLPE ADGERQAILSIQNEVEKVIQSYSMRISASNPYSTVTVEEIRTKWEKVKQLVP QRDQSLQEELARQHANERLRRQFAAQANVIGPWIQTKMEEIARSSIEMTGP LEDQMNLKQYEQNI INYKHNDKLEGDHQLIQEALVFDNKHTNYTMEHIR VGWELLTTIARTINEVETQILTRDAKGITQEQMNDFRASFNFHFRKNGL MDHDDFRACLISMGYDLGEAEFARIMSLVDPNGQGTVTQSFIDFMTRETA DSTDTEQVIASFRILASDKPYILADELRRELPPQEAQYCIKRMPPYTGPGSV GALDYTSFSALYGESDL
190	Domestic Muscovy Duck A0A8C3BW27 (Alpha-1,4 glucan phosphorylase)	MAAPLSDGEKRRQISVRGIAGLGDVAEVRKSFNRHLHFTLVKDRNVATPR DYFFALAHTRVDHLVGWRITQQHYHERDPKRIYYLSLEFYMGRTLQNTM VNLGLQNACDEAIYQLGLDLEEEIEEDAGLGNGGLGRLAACFLDSMATL GLAAYGYGIRYEFGI FNQKIVDGWQVEEADDWLRVGNPWEKARPEYMLP VHFYGRVEHNPEGVKWDVTQVVLAMPYDTPVPGYKNNTVNTMRLWSAK APNDFNLQEFNVGDYIEAVLDRNLAENI SRVLYPNDNFFEGKELRLKQYEF VVAATLQDI IRRFKSKFGCRDPVRTSFETFPDKVAIQNDTHPALSIPELMR ILVDVEKVDWDKAWIITKRTCAYTNHTVLPALERWPVSMFEKLLPRHLEI IYALNQMHLDRAALYPGDVDRLRMSVIEEGDCKRINMAHLVIGSHAV NGVARIHSDIVKHSVFKDFYELEPEKFNQNTNGITPRRWLLLCNPGLADVIA EKIGEGFVTDLSQLKLLDFIDNKMFI RDVAKVKQENKLFKSAYLEEYKVV KINPSSMFDVQVKRIHEYKRQLNLCHAITLYNRI RSDPSKSFVPRTIMIGGK AAPGYHMAKMI IKLITSIGEVINNDPYVGDKLVIFLENYRVSLAEKVIPAA DLSQQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAEENLFI GMRVEDVEALDRKGYNALEYERLPELRQAIDQISSGFFSPGDPGCFRVDV NMLMHDRFKVFADYEAYIKCQGGVDQLFMDPWETKVKVIRNIACSGKF SSDRITITEYAQEIWGVESPSTKIPPNIPRD

TABLE 11-continued

Domestic Muscovy Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
191	Domestic Muscovy Duck A0A8C3CD90 (Fructose- bisphosphate aldolase)	MTHQFPALSPBQKKALS DIAQRIVASGKGILAADES VGTMGNRLQRINVEN TEENRRRAFREILFSSDTSINQSIGGVILFHETLYQKDSGKPPPAI I KEKGI VVG IKLDKGTSPLAGTNGETTIEGLDGLSERCAQYKKDGVDFGKWRAVLKITST TPSQLAIQENANTLARYASICQONGLVPIVEPEVLDPGDHDLQRCQYVTEK VLAAYKALNDHHVYLEGTLLKPNMVTAGHACTKKYTPQDVAAATVTTL LRTVPAAPVPGICFLSGGQSEEEASLNLNAMNQSSLPKPWKLTFSYGRALQA SALAALWLGKPKENKKAQAEAFRKRAQINS LACRGQYVVS GKTDAATQSLF TASYTY
192	Domestic Muscovy Duck A0A8C3CJ8 (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKGVNGFGRIGRLVTRAAILSGKVQVVAINDPFIDLNYVMVFMKYDST HGHRGTVKAENGLVINGNAITIFQERDPSNIKWADAGAEYVVESTGVFT TMEKAGAHKGGAKRVII SAPSADAPMFVMGVNHEKYDKSLKIVSNASCT TNCLAPLAKVIHDNFGIVEGLMTTVHAI TATQKTVDPGSGKLWRDGRGAA QNIIPASTGAAGVGVKVI PELNGKLTGMAFRVPTPNVSVVDLTCRLEKPAK YDDIKRVVKAADGPKLGILAYTEDQVVS CD FNGD SHSSTFDAGAGIALND HFVKLVSWYDNEYGYSNRVVDLMVHMASKE
193	Domestic Muscovy Duck A0A8C3GR18 (Tropomyosin 1)	MDAIKKMQMLKLDKENALDRAEQAEADKKAEEERSKQLEDELVALQK KLKGTEDLDKYESL KDAQEKL ELADKKATDAESEVASLNRRIQLVEEEL DRAQERLATALQKLEEA KAADES ERGMKVI ENRAQKDEEKMEIQETQLKE AKHIAEEADRKYEEVARKLVII ESDLERAEEAELSESKCAELEELKTVTN NLKSLAQAEKYSQKEDKYEEEIKVLTDKLKEAETRAEFAERSVTKLEKSID DLEDELYAQKLKYKAISEELDHALNDMTSM
194	Domestic Muscovy Duck A0A8C3CSW2 (Myosin binding protein C3)	YPTTKPKTTEVTAGSTAVFEATEKAGVKVKWQAGTEITDSEKYV IKAEG NKHSLTINN VVKEDDVTYAVIAGSSKVKFELVKVEPEKSETVTPAEAAPAP AASELPPAPAESQNTVEVPSAGTQPEEPVDP IGLFVTRPQDGEVTVGGNITFT AKVAGESLLKKPSVKWFKGKWMDLASKVQGHQLQHDSDYRNNKYVTFE LEII EANMTFAGGYRCEVSTKDKFDSN FNLT VNEIYHLCHDIVSPFP SKTN YIFLLLRSTEGHDEAGELNFSALLKKRGDGKTESQPDVDVWEILRKAPPSEY EKIAFYQGI TDLRGLKRLKRI KKEEKKS TAPLKKLDPAYQVDKGQKI KLM VEVANPDADV KWLKNGQEIQVSGRYI FEAVGNKRI LT INHCSLADDAAYEC VVGEEKSFTEL FVKEPPI LITHPLEDQMVMVGERVEFECEVSEEGATV KWE KDGVELTREETF KYRFPKDGKKQYLI INESTKEDSGHYTVKTNGGVSAEL IVQEKKLEVYQSIADLTVKARDQAVFKCEVSDENVKGIWLKNGKEVVPDE RIKISHIGRIHKLTI EDVTPGDEADYSFIPQGFAYNLSAKLQFLEV KIDFVPRE EPPKIHLDCLGQSPDTIVV VAGNKLRLDVPISGDPTPTV IWQKVNKKTDLRL SNDDSMTPSENSDLNTDSKLLFESEGRVRVEKHEDHCVFIEGAEKEDEGV YRVIVKNPVGEDKADITV KVIDVPDPPEAPKISNIGEDYCTVQWQPPKYDG GQPVLG YILERKKKSYRWMRLNFDLLKELTYEAKRMIEGVVYEMRIYAV NSIGMSRPSPASQPFMPIAPPSEPTHTFTVEDVSDSTVALKWRPPERIGAGGLD GYIVEYCKEGSTEWI PALPGLTERTSALIKDLVTGDKLHFR IKA INLAGESGA AI I KEPI TVQEILQRPKI WLPRLRQTLVKKVGETINIVIPQGKPRPKITWTQK DGQSLDAKDVGIRNSNTD TILFIRKAELHHS GAYEITLQIENMTDKVAITIQII DKPGPPQNI KLTDVWGFNAALEWTPPQDDGNAQILGYTVQKADKKTMEW YTVDFHYRR TNCVVS ELIMGNEYFRFVSENLCGLSETAATTKNPAYIQKT GTTYKPPCYKEHDFSEPPKFTHPLVNRSVIAGYNTT LSCAVRGI PKPKIYWY KNKMDLSGD AKYRMFSKQGVLTLEIRKPTPFDDGGFYTC KAVNERGEAEIEC KLDVRGNHMSIMHNTNTAVK
195	Domestic Muscovy Duck A0A8C3CXN6 (Myomesin 2)	MSSLAVPFYQRRHKHFDQSYRNIQTRYVLEEYAARKAASRQAAYYESSGL GKTTCLRCARRARTLAHETMQESRKR THEQKSHASEEKRIRFASELSALEK EIHTARHRAREQLDRLAI QRMVEENMAMERHAVEERI SRAPEILVRLRSHT VWEKMSVSLCFTVQGFPPTPVVQWYKNEELITPASDPKYRVESKYGVHVL HINRADFDDSATYSAVATNVHGQASTNCAVVRFRFREGEEPHPAGIMPFHL PLSYDVCFTHFDVQFLEKFGVT FATEGETLT LKCSVLVTPELKRLRPRAEW YRDDVLIKDSKWT KLYFGEGQAALSFTHL NKDDEGLYTLRMITKGGVNEC SAYLFVRDADALIAGAPGAPMDVKCHDANRDYIVTWKPNPTTSQSPVIG YFVDKCEVGLNWNVQCNDAPVKICKYPVTGLYEGRSYIFRVRAVNAGIS RPSRVSEPVAAALDPVDLERTQT IHVDEGRKVISKDDLEGDIQIPGPPTNVHA SEISKTYVVLSDPPVPRGREPLTYIEKSMVSGSGWQRVNAQVAVKSPRY AVFDLAEKGKPYVFRVVSANKHGISDPSEITEPIQPQD TVVVP SAGRRVATR NTKTSVVVQWDPKPKHEEDLYGYIIDYSVVGSNHWEPSNHNKPIKYNRFVVH GLETGEQYIFRVKAVNAVGFSENSQSEAIKVQAALT CPSYPHGITLLNCDG HSMTLGWKAPKYSGGSPILGYYMDKREANHQNWHEVNSSLITRTIYTVED LTEDAFYEFKIAAANIVGIGQSPDSEHFKCAWMTPEPGPAYDLTVCEVR NTSLVLLWKAPVYEGKSPITGYLV DYKEVDKEDWITANEKPTAQRYFKVT DLHEGHTYVFKVRAVNDAGVGKSSEVSEPV LVEAKPGTKEIFSGVDDEGNI YLAFECKEATDASHFAWGKSYEEIEDSKYK IETEGDRSKLYFKNPDKSDL GTYSISVSDTDGVSSSFVMDDEELERLMAISNEIKNPTIPLKSELAYEVLKGG EVRFWIQAESLSPDSTFRFVINDKEVENS DRHKINC DRPNGI IEMVMDRFTID

TABLE 11-continued

Domestic Muscovy Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		NEGTYTVQIQDGKAKNQSSSLVLIGDAFKAVLAESSELQRKEFLRKQGPFPSE FLYWEVTEEECVLLACKVANTKKETVFKWYKDGSGIDVDEMPLQKGEC HLSIPRLSRKDEGVYKATLSDDRGHVSTLELSGKVYDDIILALSRVSGKTA SPLKILCTEEGIRLQCFLKYNEEMKISWSHRESKISSGEKMKIGGENIAWL QITEPTKDKGNYTFEIFGDKESFKRTLDSLGGQAFDDAYAEFQRLKAAAFAE KNRGKVIIGLPDVVTIMDGKTLNLTCTVFGNPDPEVVWFKNDKALELNEH YVVSLEQGKYASLTIKGVTSEDSGKFSIHVKNKYGGETVDVTVSVYRHGEK VPEVKPGQLAKPRPIPPSEGSS
196	Domestic Muscovy Duck A0A8C3CT67 (Phosphopyruvate hydratase)	MVFKMSILKIHAREIFDSRGNPTVEVDLYTNKGLFRAAVPSGASTGIYEALE LRDNDKTRYMGKGVSKAVEHVNKTIPALISKNNVVEQEIKDKLMLDM DGSENKSKFGANAILGVSLAVCKAGAAEKGVPLYRHIADLAGNPEVILPVP AFNVINGGSHAGNKLAMQEFMILPVGADSFKEAMRIGAEVYHNLKNVIKE KYGKDATNVGDEGGFAPNILENKEALELLKTAIGKAGYSDKVVI GMDVAA SEFYRDGKYDLDFKSPDDPSRYISPDQLADLYKGFVKNYPVVSIEDPFDQD DWAAWKKFTGSVGIQVVGDDLTVTNPKRIAKAVEEKS CNCLLLKVNQIGS VTESLQACKLAQSNGWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPC RSERLAKYNQLLRIEELGSKARFAGRNFRNPRIN
197	Domestic Muscovy Duck A0A8C3BXQ0 (Myosin light chain 2)	MAPKKAKKRIEGANSNVFSMFEQAQIQEFKEAFTIMDQNRDGFIDKADLRD TFAALGRNLNVKNEEIDEMIKEAPGPINFVFLTMPGEKLGADPEETILNAF KVFDPEGKGLKSAYIKEMLMTQGERFSQEEVDQMFAAPPDVS GNLDYKN LVHVITHGEEKD
198	Domestic Muscovy Duck A0A8C3BJT4 (Uncharacterized protein)	MTSPDAEMAAFGEAAPYLKSEKERIEAQNKPFDAKSSVFVAHPKESFVKG TIQSKESGKVTVKTEGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLHEPA VLYNLKERYAAWMIYTYSGLFCVTVPYKWL PVYNPEVVLAYRGKKRQE APPHIFISIDNAYQFMLTDRENQSILITGESGAGKT VNTKRVIQYFATIAASG EKKKEEQSGKMQGTLEDQII SANPLLEAFGNAKTVNRDNSSRFGKFIRIHFG ATGKLASADIETYLLEKSRVTFQLKAERSYHIFYQIMSNKKPELIDMLLITTN PYDYHFVSQGEITVASINDQEELMATDS AIDILGFTADEKVAIYKLTGAVMH YGNLKFKQKQREEQAEPDGT EVADKAAYLMGLNSADLLKALCYPRVKVG NEYVTKGQTVQQVNNSVGALAKAVYEKMLWMVVRINQQLDTKQPRQY FIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHMHFVLEQEEYKKEGIEW EFIDFGMDLAACIELIEKPMGIFSI LEEECMFPAKATDTSFKNKLYDQHLGKSS NFQKPKPAKGKAEAHFSLVHYAGTV DYNITGWLEKNKDPLNETVIGLYQK SSVKTLALLFANYGGAEAEASGGGASGGKKGKKKGS SFQTVSALFRENL NKLMTNLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGRICRK GFPSRVLYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFG HTKVFFKAGLLGLLEEMRDEKLAQLITRTQARCRGF LMRVEYQRMVERRE SIFCIQYNVRAFMMNVKHWPMKLF FFKIKPLLKSAESEKEMANMKEEFEKT KEELAKSEAKRKELEEKMKLVQEKNDLQLQVQAEADALADAERCDQLI KTKIQLEAKIKEVTERAEDEEEINAELTAKKRKLEDECSELKKDIDDLLELTL AKVEKEKHATENKVKNLTEEMASLDETI AKLTKEKKALQEAHQOTLDDLQ AEEDKVNTLTAKATKLEQQVDDLEGSLEQEKKLRMDLERAKRLEGLDKL SQDTIMDLENDKQQLDEKLEKKKDFEISQIQSKI EDEQALGMQLQKKIKELQ ARIEELEEIEAERTSRAKAEKHRADLSRELEEISERLEEAGGATAAQVEMN KKREAEFQKMRRDLEEATLQHEATAATLRKKHADSTAE LGEQIDNLQRVK QKLEKESSELKMEIDDLASNMESVSKAKANLEKMCRTLEDQLSEIKTKEEE HQRMINDLTAQRARLQTESGEYSRQVEEKDSQISQLSRSKQAF TQQIEELKR HLEEEIKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRALSKANSE VAQWRTKYETDAIQRTEELEEAKKLAQRLQDAEEHVEAVNAK CASLEKT KQRLQNEVEDLMIDVERTNAACAALDKKQKNFDKILA EWKQKYEETQAE LEASQKESRSLSTELFKMKNAYEESLDHLETLKRENKNLQQEISDLTEQIAE GGKAIHELEKVKKQIEQEKSEIQAALEEAEEASLEHEEGKILRLQLELNVKS EIDRKIAEKDEEIDQLKRNLHRIVESMQSTLDAEIRSRNEALRLKKKMEGDL NEMEIQLSHANRQAEEAQKNLRNTQAVLKD TQIHLDDALRTQEDLKEQVA

TABLE 11-continued

Domestic Muscovy Duck Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		MVERRANLLQAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSL INTKKKLETDIVQIQGEMEETIQEARNAEKAKKAITDASMMAEELKKEQD TSAHLERMKNLDQTVKDLQHRLEAEQLALGGKKQIQKLEARVRELEG DVDAAEQKRSABAVKGVKRYERRVKELTYQSEEDRKNI LRLQDLVDKLQM KVKS YKRQAEAEELSNVNL SKFRKI QHELEEAERADIAESQVNKLRAKS REFHGKKIEEEE
199	Domestic Muscovy Duck A0A8C3BJT4 (Uncharacterized protein)	MTSPDAEMAAFGAAPPYLRKSEKERIEAQNKPFDAKSSVFAHPKESFVKG TIQSKESGKVTVKTEGGETLTVKEDQIFSMNPPKYDKIEDMAMMTHLEPA VLNKLKERYAAWMIYTYSGLCVTVNPKWLPVYNPEVVLAYRGKKRQE APPHFISIDNAYQFMLTDRENQSILITGESGAGKTVNTRKVIQYFATIAASG EKKKEEQSGKMGGTLEDQIIISANPLLEAFGNAKTVRNDNSSRFGKFI R IHFG ATGKLASADIETYLLKSRVTFQKLAERSYHIFYQIMSNKKPELIDMLLI TTN PYDYHFVSQGEITVASINDQEEELMATDS AID ILGFTADEKVAIYKLTGAVMH YGNLKFQKQREEQAE PDGTEVADKAA YLMGLNSADLLKALCYPRVKVG NEYVTKGGTVQQVNSVSGALAKAVYEKMLWVVRINQQLDTKQPRQY FIGVLDIAGFEIFDFNSLEQLCINFTNEKLQQFFNHHMFVLEQEEYKKEGIEW EFIDFGMDLAACIELIEKPMGIFSI LEEECMFPAKTDTSFKNLYDQHLGKSS NFOQKPKPAKGKAEAHFSLVHYAGTV DYNITGWLEKNKDPLNETVIGLYQK SSVKT LALLFANYGGAEAEASGGGASGGKGGKGGSSFTV SALPRENL NKLMTNLRSTHPHFVRCIIPNETKTPGAMEHELVLHQLRCNGVLEGI R ICRK GFPSRVLYADFKQRYKVLNASAIPEGQFIDSKKASEKLLGSIDVDHTQYKFG HTKVFFKAGLLGLEEMRDEKLAQLI TRTQARCRGFLMRVEYQRMVERRE SIFCIQYNVRAFMMNVKHWPMKLF FFIKPLLKSAESEKEMANMKEEF EKT KEELAKSEAKRKELEEKMKLVQEKNDLQLQVQAEADALADAEERCDQLI KTKIQLEAKI KEVTERAEDEE INAE LTA KRKLEDECESELKKDIDDL E LTL AKVEKEKHATENKVKNL TEEMASLDETI AKLTKEKKALQEAHQOTLDD LQ AEEDKVNTLTAKTKLEQQVDDLEGSLEQEKLRMDLERAKRLEGLDKL SQDTIMDLENDKQQLDEKLKKDFEISQIQSKI EDEQALGMQLQKKIKELQ ARIEELEEEIEAERTSRAKAEKHRADLSRELEIEISERLEAGGATAAQVEMN KKREAEFPQKMRDL E EATLQHEATAATLRKKHADSTAE LGEQIDNLQRVK QKLEKESELKMEIDDLASNMEVS KAKANLEKMCRTLEDQ LSEIKTKEEE HQRMINDLTAQRARLQTESGEYSRQVEEKDSQISQLSRSKQAF TQQIEELKR HLEEEIKAKNALAHALQSARHDCDLLREQYEEEQEAKGELQRLASKANSE VAQWRKYETDAIQRTEELEEAKKLAQRLQDAEEHVEAVNAKASLEKT KQRLQNEVEDLMIDVERTNAACAALDKKQKNFDKI LAEWKQKYEETQAE LEASQKESRSLSTELFKMKNAYEESLDHLET LRKRNKLNQQEISDLTEQIAE GGKAIHELEKVKKQIEQEKSEIQAAL EEA EASLEHEEGKILRLQLELNQVKS EIDRKIAEKDEEIDQLKRNHLRIVESMQSTLDAEIRSRNEALRLKKMMEGDL NEMEIQLSHANRQAAEQKLNRLNTQAVLKD TQIHLDDALRTQEDLKEQVA MVERRANLLQAEIEELRAALEQTERSARKVAEQELMDASERVQLLHTQNTSL INTKKKLETDIVQIQGEMEETIQEARNAEKAKKAITDASMMAEELKKEQD TSAHLERMKNLDQTVKDLQHRLEAEQLALGGKKQIQKLEARVRELEG DVDAAEQKRSABAVKGVKRYERRVKELTYQSEEDRKNI LRLQDLVDKLQM KVKS YKRQAEAEELSNVNL SKFRKI QHELEEAERADIAESQVNKLRAKS REFHGKKIEEEE
200	Domestic Muscovy Duck A0A8C3CNB8 (Albumin)	MKWVTLISFIFLSSATSRNLQRVARDADHKSEIAHRYNDLKEETFKAVAMI TFAQYLQRCSDGLSKLVKDVVDLAQKCVANEDAPECSKSLPSIFLDEICQ VEKLDSYGAMADCCAKSDPERNECFLSFKI PQPDFVQPYQRPASDVICKE YEDNRVSFLGHFIYSVARRNPFLYAPTILGVAADYEHALKSCCKESDVGAC LDGKETAI REKVKKISVKQQYSCGILQKFGDRIFQADKLAFLSQKYPKASFT EISKL IHDVKDVYKECCGDMVBCMDRAELVNYI CSKHESFSSKIKDCC E KPIVERSQCIMEAEFDEKPADLPSLVEKYIQDKVCKSF EAGHDAFMS E FVY EYSRRHPEFSTQLILRIAKGYETLLEKCKCTDNPAECYANAQDQLNEHIKET QDVVKTNCDDL TTHGEADFLKALLIRYTKMPQVSTETLLEIGKKMTGVG TKCCKLPEDRRLPCEGYLSMVIQDMCRRQETSPINDNVSHCCSDSYAYRR PCFSAMGVDTKYVPPFPNPEMFNFDEKLCSAPAEERAAGMKLLVNLIKRK PQMTTEEIKTIADGFTAMVKCKCKQSDIETCFGEEGANLIVQSRATLGIGA

[0057] An animal protein can comprise a salmon protein (Table 12). A salmon protein can comprise a salmon muscle protein. A salmon protein can comprise a salmon blood protein. A salmon protein can comprise a salmon albumin protein. A salmon protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about

97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 201-216.

[0058] A food product may comprise a fragment or a truncation of a salmon protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 201-216. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 201-216. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%,

or more identical to a portion of a sequence of any one of SEQ ID Nos: 201-216. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of

SEQ ID Nos: 201-216. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 201-216.

TABLE 12

Salmon Proteins		
SEQ ID NO	Description	Amino Acid Sequence
201	Salmon B5DGP0 (Creatine kinase)	MPFGNTHNNFKLNFKVEEEYPDLTKHNNHMAKVLTKDMYAKLRDKQTPSG FTLDDVVIQTGVNDNPGHPFIMTVGCVAGDEESYEIFKDLLDPIISDRHSYGKPT DKHKTDLNFENLKGDDLDPNYVLSRVRTGRS IKGYTLPPHNSRGERRAV EKLSVEALNTLDGEFGKGYPLNKMTDAEQEQLIADHFLFDKPVSPDLLGA GMARDWPDARGIWNDAKSFVLVWNEEDHLRVISMEKGGNMKEVFRRFC VGLKR IEETFKKHNHGFWMNEHLGYVLTCPSNLGTGLRGGVHVKLPKLSH PKFEEILTRLRLQKRGTTGGVDTASVGGVFDISNADRLGSSEVDQVQMVVDG VKLMVEMKKLEKGEAIDGMIPAQK
202	Salmon A0A1S3RGX6 (Alpha-actinin-3)	MTAIEQTQVQYGSYMTSEQVYMTQEDDWRDRLDPAWEKQQRKTFTAWC NSHLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKPKDGKMRPHKIANV NKALDFICSGKVKLVSIGABEIVDGNVKMTLGMWITIILRFQIDISVEETSAK EGLLLWCQRKTAPYRNVNVQNFHISWKDGLALCALIHRHRPDLIDYSKLRK DDPMGNLNTAF EVAEKYLDIPKMLDAEDIVNTPKPEKAIMTYVSCFYHAF AGAEQAETAANRICKVLAVNQENKLMEEYEKLASELLEWIRRTIPWLENR VAEQTMRAQQKLEDFRDYRRVHKPKPVQEKQLEINFNTLQTKLRLSNRP AFMPSEGMVSDIANAWKGLEQVEKGYEEWLLTEIRRLERLDHLAEKFKQK SSLHESWTSKGVELLMSMKDYESASLMEIRALMRKHEAFESDLAAHQDRVEQ IAAIAQELNELDYHDAVTINARCQGICDQWDNLGTLTQKRRLSLERVEKLW ETIDQLYLEFAKRAAPFNWMDGAMEDLQDMFIVHSIEEIQSLITAHQDFKA TLPEADKERVATMGIQNEIVKIAQTYGIKLSGVNPTNLSPQDITDKWDAVK HLVPLRDQMLQEEVARQQSNERLRRQFAAQANIIGPWIQTKMEEISHVSVDI AGSLEEQMSNLKQYEQNIINYKCNIDKLEGDHQLSQESLIFDNKHTNYTMEH VRVGWEQLLTTIARTINEVENQILTRDAKGISQEQLENFRAFNHFDRKRNG MMDPDDFRACLISMGYDLGEVEFARIMTLVDSNNTGVTQAFIDFMTRET AETDTADQVMASFILASDKTYITVEELRRELPPQADYICISRMTSYIGSGAP PGALDIYSFSSALYGESDL
203	Salmon C0H8W5 (Alpha-1,4 glucan phosphorylase)	MPKPLTDQDKRKQISVRGLAGVENVSDLKTNFNRHLHFTLVKDRNVATKR DYYFALANTVRDHLVGRWIRTQQHYEYKDKPRVYYLSLEFYMGRTLQNTM VNLALENACDEATYQLGLDMEELQDIEEDAGLNGGLGRLAACFLDSMAS LGLAAYGYGIRYEFGIFNQKIVNGWQVEADDWLRYGNPWEKARPEYMRP VHFYGRVEHTPEGVKWVDTQVVLALPYDTPVPGYRNVVNTMRLWSAKA PCDFNLKDFNVGGYIQAVLDRNLAENISRVLYPNDNFFEGKELRLKQYFVV AATLQDIVRRFKSSKFGSTEVRVDLSTLPDKVAIQLNDRTHPAMAIPELMRIL LDTEHQTWKAWDICTRTCAYTNHTVLPEALERWPTDLLQNLPRHLEIIEYE INRRHLERIAKLYPGDHLRRLMSLVEEGDQKKINMAHLCIVGSHAVNGVA RIHSDILKATLFKDFYEVDPHKFQNKTNGITPRRWLMCNPLGAELVIERIGE DYIRDLQDKKLLTFVDDSLIHDIAKVKQENKLFSAYLEEHYKVKINPNS MFDVQVKRIHEYKRQLNLCLHIITLYNRIKKEPNKNWTPRTIMIGGAAPGY HTAKLIIRLITAIGEIVNHDTVVGDRLKVIPLFENYKVTLAEKIIPASDLSEQIST AGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEENLFI FGMRVDDV DAMDKSGYDAMDYYNRIPELQAMDQIAGGFSPDQHDLFKDIVNMLLHH DRFKVFADYEAYIKCQEKVSALYKNPKEWTKMVIHNIAGCGKFSSDRTISQY AREIWGMESLERIPAPDDPRQ
204	Salmon C0H911 (Fructose- bisphosphate aldolase)	MPHSFPFLSTEQKKELSDIAQKIVAPGKGI LADESTGSVAKRFQSINAENTE ENRRLYRQLLTADRVEPCIGGVIFFHETLYQKTDGKLFPPQLIKERDMVVG KVDKGVVPLAGTNGETTQGLDGLYERCAQYKKDGADFAKWRCLVKTIDT TPSELAIKENANVLARYASICQMHGIVPIVEPEILPDGDHDLKRCQYVTEKVL AAVYKALSDHHVYLEGTLKPNMVTGPHSCSQKYSNQEIAMATVTALRRT VPPAVPGITFLSGGQSEEEASINLNAMNQCP LHRPWAITFSYGRALQASALK AWGGKPKENGKACQEEFIKRALNNSKACQKGYVSSGDKGAAGGESLYVANH AY
205	Salmon B5DH12 (Myosin light chain 1-1)	MAPKKDAKAPAKAEPAPAPAPPEPDVVAAPPLDLSTVKVEFTPDQQ EDYKEAFGLFDRVGDADKAYNQVADIMRALGQNPTNKEVRKVLGNPDEED MAAKRLEFEAFPLMLQHI VNDPNKGTEDDYVEGLRVFDKEGNGTVMGAEL RIVLGLTGKMSAEIDALMQGQEDENGSI NFEAFVKHIMSI

TABLE 12-continued

Salmon Proteins		
SEQ ID NO	Description	Amino Acid Sequence
206	Salmon B5DGR3 (Glyceraldehyde- 3-phosphate dehydrogenase)	MVKVGVNGFGRIGRLVTRAAFHSSKKGVEIVAINDPFIDLDYVMVMFYDST HGRFHEVKAEGGKLVIDGHKI TVPHERDPANI KWGDAGATYVVESTGVFT TIEKASTHLKGGAKRVVISAPSADAPFMVGMVNHKEYDNSLKVVSNASCTT NCLAPLAKVIHDNYHIIIEGLMSTVHAVTATQKTVDGPGSKLWRDGRGASQN IIPASTGAAKAVGKVIPELNGKITGMAFRVPTPNVSVVDLTVRLEKPASYDAI KKVVKAAADGPMKGI LGYTEQQVVSDFNGDTHSSIFDAGAGIALNDHFVK LVTWYDNEFGYSNRVIDLMAHMATKE
207	Salmon Q91472 (Fast myotomal muscle tropomyosin)	MDAIKKMQMLKLDKENALDRAEGAEGDKKAAEDKSKQLEDDLVALQKK LKGTEDLDKYSSESLKDAQEKLEVAEKTATDAEADVASLNRRIQLVEEELD RAQERLATALTKLEEAKEADESERGMKVIENTRASKEEKMLQDIQLKEA KHIAEEADRKYEEVARKLVIIESDLERTERAELS ESKCSELEELKTVTNL KSLAQAEKYSQKEDKYEIEIKVLTDKLEAEATRAEFAERSVAKLEKTIDDL EDELQAQLKYKAISEELDNALNDMTSI
208	Salmon A0A1S3RC27 (Myosin-binding protein C)	MPEAKKPAAKTDKAAAAPADKEARSAIEEGSECRDELPPDETVPGGQSELT GLFVERPESTTVIKGNVTFVAKVDSDDLRLKPNMKWLKGLWLDLGSKAG KHVQFKEAYDRNSKVITYEMSIIKVVEGDAGGYRCEVTSKDKCDSCTFEVT VEAVQEEQPANILDAFKRSGDAGEDAGDLDFSALLKKREKKARDEPKEDVD VWEILKDAKPCDYEKIAFDYGITDLRGLLRLKKMKKVEPKSDAPLKKLE QCYSVDKGGKRTQMVELHDPNVQVKLNGVVEIKPSAKYVFEVGNKRTL TINKCNLSDDAAYECVVGEESFTEVFVKEPPVTITKLLDDVHVVVGEKVEF ECEVSEEGANVWMDGVELTKDGKIRIKKDGKHTLVINEATIEDIGMY VYTNGGESKGELEVEAKELEVLQSIADLSVKACEQAMFKCEVSDEKVVVK WFKDGVVFKGNRIKMSHIGRIHKLTIDDVKPDQEGNYTFVPEGYALSLSAK LNFIEIKIEYVPRQDPPKIHLDTSSTGSKNTIVVVGANKLRDLVEITGEPVPTV CWMKGDTVISEAERVRVETRTTSSFVIEGAERPDGQYSIIVTNPAGEDRA ELTVKIVDVPNPPENVRCMGVGEDTATIWDPPKPDGGAPIKGYLMERKKQ GSSRWTKLNFVEFESTTYEAKKMI EGVFVEMRVFAVNGIGISQPSGNSKPFM PIAPTSEPTRLTVEDVTDSTCALKWRPPERVAGGVDGYII EWCKEGEDNWV AANKPEVDKNTYRVKGLPTGEKLLFRVAMNIIAGRSPCTMKQSVTIREIM EYPKIRLPRQLRTKFKRVGKINLVI PFQGGKPRPVVNWLDGEPLENKSVGI RTSDFDITLFI RSAERDHS GTYTL SVQIDNMQDKADIHIQIVDKPGPPINVMVT DVWGFNAALEWKPKDGTGNTDITGYTIQKADKKTQGWFTVYEHNRPSCT ASDLVMGNEYSFRVFSENICGLSDEVAFSKNTAIIGKTDLEYNKPAFKERDM SSSPKFTAPLVDRVVVAGYSTAISCAVRAYPKAKI VWMKNKMIIGEDPKFLM QNNQGVLTNLIRKPGQFDGGKYSCAINDLGEDEVCELRLEVRVLKEKKEGD EEKK
209	Salmon A0A1S3L6A0 (Myomesin-2 isoform X1)	MATKVMHFNQKHVSHVSHSSHTTKHVVKESSRKSSSKSVNQVQHTHA TEAMAEPAITVPAPRQRSADIEEYQRVSSHVKGGLASIQKELHRMRVVTK TQVENIAIKREVQEMMHKKMTLSTETDKAPDFMVALRPHTVWEKTPVKLF CTVDGHPRPVIVKWKGGKVPDPLSAPGKYKIESKYGVHSLIISRCMVSDTAE YSAVATNSHGSTTSKASVIVKRPAGGAYGSCQLFGQVPHLTVIHHSKLEITM LDRFGVSFGTEGGSISLVCTMVVVVDLPNVPPLAQWYRDDKLLKAGKLAEI KVGGAATLTLPHLAKDDEGLYTLRMWTKDGTTEHSAYLFVKDAAPSVAG APGAPISVKAFDINSYVLVAVKPPNTTNEAAITGYFVDKRESGSATWSQCN DAPVQICKYPVHGLNVGHAYHFRVRAVNSAGISRPSRESKVTALDPAERE RLQVIKLDGKHEVVIKDDDLEGYVTLPGIPTDMHASEIFKSYIVLSWKPPSPR GRAPLWYII EKSVSGTGAWRINTAVKLQSPRYPVFDLEAGKKYQFRVYSE NLYGASEASKPTDPIEKVDEHGVP SAGQV VATRNTKSSVFVQWDPKHPN HLMGYIIDASLVGSKTWAPCNHKKPYKNTRFVVHGLTPGETYVFRVQAVNV YGLSDESQESTPIAVEPALTPPSAPHGITMLSCDGASMI IAWNSPKHRGGSKI NAYYVDKRDADSLAWKEVNSAPTNTRTYTV DGLTEGTFYEFKVQAGNLAG VGVPSPASTPLKCEAWTS AVPGPAYDLAFREVGRGSLVILWKAPVYTGASA VTGYIVEMAKKGHHYHPLNEEAIGHCYLQVTGLEAGAETFKVKAVNAEG VGKASQTSEPVFAKALPGTQEIQCQVDEDTGDI FLSPFEACEISETSNFAWSKN YKEISDCTRVSIETKGKHSKLT FVNPDVSDLGAFSVHVTDDGV SASHTVTE DALNTMLELSYNIRHPVPLKHQLNVEVLEKGHVRFWLQCLKMSPAVNRYF IVNDKEVTS SSSHKISHDVNTGVIQMTVDHFSKASEATYTVQIHDGRAKNQS SLVLVGDFKAALKEAEFQRSEHIRKQGPFAEYLSYHVDDDC TVLLVCKV ANVKKETVFWHFKDEEIVPETPPNVMSGVALPISLFSRKDQGHYKAKLSD DRGKDTSVFDISGQVDDIINAIAHAGSSASELVLCQCTPEGIRLQCYMKYYT EEIRTAWLHKDSKISSSEKMRIAGTAEMAMQIREPSEKEKGHSYI QIMDAT KSHTRTFDL SGQAYTDAYEEYLR LKAAAFKAKNRGRVVGGLPDVVTIMEQ KTLSLTCTVWGDPSPVETWFKNENEVSTEHAKITLEANKFASLTI TAVTSE DSGKYSINVRNKYGGFEVETVSVYKQGESPEPKMKGGRGATPAPLASKTPA KTPTAQTPTPSLKSPTPSLKSPTPSLRSPAKSPTTPKSPTPPRRVKSPSPARSL KSPTPPRK

TABLE 12-continued

Salmon Proteins		
SEQ ID NO	Description	Amino Acid Sequence
210	Salmon ENOB (Beta-enolase)	MSITKIHAREILD SRGNPTVEVDLYTAKGRFRAAVPSGASTGVHEALELRDGD DKSRYLKGKTVKAVDHVNKDIAAKLIEKKFSVVDQEKIDHFMLELDGTENK SKFGANAILGVSLAVCKAGAAEKGVPLRYHIADLAGHKDVI LPCPAFNVING GSHAGNKLAMQEFMILPIGASNPFHEAMRIGAEVYHNLKNVI KAKYKDATN VGDEGGFAPNILENNEALELLKTAIEKAGYDPKIIIGMDVAASEFYKAGKYD LDFKSPDDPARYITDGLDGLDYKSFIKGYVPVQSIEDPFDQDDAAWTKFTAA VDIQVVGDDLTVTNPKRIQQAVEKKACNCLLLKVNQIGSVTESIKACKLAQS NGWGMVSHRSGETEDTFIADLVVGLCTGQIKTGAPCRSERLAKYNQLMRI EELGAKAKFAGKDYRHPKIN
211	Salmon Q7ZZN0 (Myosin regulatory light chain 2)	MAPKKAARRGAAAEAGSSNVFSMFEQSQIQEYKEAFTIIDQNRDGIISKDDL RDVLASMGQLNVKNEELEAMVKEASGPINFVFLTMFGEKLGADPEDVIV SAFKVLDPEATGFIKKEFLQELLTTQCDRFSAEEMKNLWAAFPDPVAGNVD YKQICVITHGEEKEE
212	Salmon A0A1S3LA68 (Myosin heavy chain)	MTTLGLTEGHPAKYQHLTNLVP AISYELGQKKERVIMSTDAEMAAFGPAAI YLRKPERERIASQAAPFDAKTAFFVVEPKEMYLKGVLSQKEGKATVKTLCL NKVLTVKEDDIHPMNPPKYDKIEDMAMMTHLNEATVLYNLKERYAAWMI YTYSGLPFCVTVNPKWLPVYDSVVVNAYRGKKRIEAPPHIFSISDNAYQFML TDHENQSILITGESGAGKTVNTKRVIQYFATIAVAGGKKEQTAAATSGKIKG SLEDQIIAANPLLEAYGNAKTVRNDNSSRFGKFI RIRHFGTTGKLASADIETYLL EKSRVTFQLSAERSYHIFYQLMTGHQPQLLEALLITNPDYDPI ISHGDI AVKS IDDT EEF IATDTAIDILGFTAEEKIGIYKLTGSVMHHGAMKFKQKQREEQAE DGT E EADKISYLMGLNSADLLKALCYPRVKVGNEMVTKGQTPVQVNNSTM ALCKSVYEKMF LWMVVRINEMLDTKQSRQFFIGVLDIAGFEIFDYNLSLEQLC INFTEKLQQFNFHMFVLEQEEYKKEGIEWAFIDFGMDLAACIELIEKPMGI FSILEEEMCPPKASDITTFKSKLYDQHGKTKAFKPKPGKGAEAHFALVHY AGTVDYNVTGWLDKNKDPLNDSVVQLYQKSSVLLAMLVVGAAASQADD AARGGGKKKKKGSFQTVSALFRENLGKLMTNLRSTHPHFVRCLIPNESKTP GLMENFLVIHQLR CNVLEGI RICKRGFP SRIQYGD FKR YRVNLNASVIPDQG FIDNKA SEKLLGSI DV DKSQYKF GHTKVFFKAGLLGTLEEMRDEKLATLV MTQALCRGYVMRKEFVKMMARREAIYSIQYNIRSF MNV KHWPMKLYFKI KPLLKSAETEKEMAAMKENFEKMKEDLAKALAKIKGLEEKIVSLLOENNDL QLQIAAEENTLCDAEERCEGLIKSKI QMEAKLKETFERLEDEEELNAELTAKK RKL EDECS ELK KDI DDL E L T L A K V E K E K H A T E N K V K N L T E E M S S Q D E S L I K L TKEKKALQEVHQVLDLQAEEDKVNNTLAKTKLEQQVDDLEGSLEQEK KVRMDLERAKRKL EGD L K L A L E S L M D L E N D K Q Q S D E K I K K D F E T S Q L S K I E D E Q A L G A Q L Q K K I K E L Q A R I E E L E E E I E A E R A A R A K V E K Q R S D L S R L E E I S ERLEEAGGATSVQIEMSKKREAEFQKLRRDLEESTLQHEATAAALRRKQAD TV A E L G E Q I D N L Q R V K Q K L E K E S Y K M E I D D L S S N M E A I A K A G N L E K M C R S L E D Q L S E L K T K N D E H V R Q L N D I N V Q R A R L L T E N G E L G R Q M E E K S L V S Q L T R A K Q T F T Q Q I E D L K R Q I E E E V K A K N A L A H G L Q S A R H D C D L L R E Q Y E E E Q E A K A E L Q R A L S K A N S E V A Q W R S R Y E T D A I Q R T E E L D A K K L T Q R L Q D A Q E A V E A T N A K C A S L E K T Q R L L G E V E D L M I D V E R A N A S A A Q L D K K Q R N F D K V L S E W K Q K Y E E G Q A E L E G S L K E A R S L S T E L F P K M K N S Y E E A L D H L E T L K R E N K N L Q H E I S D L T E L V G E S G K S I H E L D K A K K T V E N E K A E I Q A A L E E A E G T L E H E E S K I L R I Q L E L N Q I K G E V D R K L A E K D E E T E Q I K R N S Q R M M D S M Q S T L D S E I R S R N D A M R V K K M E G D L N E M E I Q L S H A N R Q A A E S Q K Q L R N V Q G A L K D A Q L Q L D D A I R V A D M K E Q V A M V E R R N N L M M A E I E E L R A A L E Q T E R G R K V A E Q E L V D A S E R V G L L H S Q N T S L I N T K K K L E V D L T Q V Q G E M E D T V Q E A R N A E E K A K K A I T D A M M A E E L K K E Q D T S S H L E R M K K N L E V T V K D L Q H R L D E A E N L A M K G G K K Q L Q K L E A R V R E L E S E V D A E Q K R G V E A I K G V R K Y E R R V K E L T Y Q T E E D N K N V S R L Q D L V D R L Q L K M K A Y K R A E D A E E Q S N I H L S R Y R K V Q H E L E A Q E R A D I S E S Q V N K L R A K S R E I G S K G N V A E E
213	Salmon A0A1S3NZV0 (Nebulin isoform X4)	MDENEEFEFEFEFEFEIIEEYFEETVTSITSHQTTSRAAAVTQQPGVPVRRVV RKVKVDTSKFMTPPYLAHSQKMLDLFSYNKYRRAYDKAKGKPYAITADTP EMMRIKKAQEQLESEVKYRMEGNKVRCTSLYDGEAREIAHVKHVSELISKVL YRAKWDETKDRYLLPPDAPELVLA VKNAANYSKKLYTEQWDEDKTCYP YSDSPELRVRVAKAQEVLSDILYKKGAERTAKYTS LAVPNDMELAKKNFEH RSDLKYHEDYNKNVKGHWCEPTYIEVAVARVAMDNLSNMYTAAIEDIK DQINFMTETPEYKQHKKAGHAASSIKYKKDYENSKSVADYNVLPATDNPL LRQLRYAGTILSDKVYKASYEKS RSGSSINCYDTPKFKMDSVVQQFSDAHYK DKYENEVKGHYIGSYEDIYMQHCKMEEIKSEQNYKADYEDIKTRCFYPQT MSPEYEAHKKVAQCKDKVYRQHPD TVKFTQVTDSPVQLQAAINAKQLSDL NYHAAYELD KFKCALPPDYPPFIQSRANAYNLSDS CYRYDWEKV KAKQFEV KGDAISILAARAHTNIASDVYKKEFEKNGKMGV GALS IHDDPKILHSMHV AKIQSDREYKKNYEKTKKYITTLDMMSNTLAKKSQGIASMTGYRTIHNRY LLPHDAMSLDLAKKANIIQSDNEYHSDYNNSTKGTGWVAFGSMEEVAKK AGDILNERKYRQHPDTPPFTSIADHPVMMQAQVNQIMRSDLHYKKGLEETH QKYSPLPHNDPQFLQAKCNAYNISKNYKQSWEEETIAKGYDMRPDAISVIHA

TABLE 12-continued

Salmon Proteins		
SEQ ID NO Description	Amino Acid Sequence	
	KHGRHIASDIQYKKDYVKGKGKQIGCRNLQDDPLLVHYMEVARLQSDKNY KKDYHKSCLKYHSPVDMMSVVHAKHASTVQTFAGYRKIQHHYSLPLGAM NLLEARTMNTNASDVNYKADWENSVKGTGWIPIGSIPIVELAKQAGAVLDEH KYRQHPENFKFSSKMDGMNLMELCKHNQKIMNVQEQYKASGEKFLHTYNLPA DAPEFMQAKCNAVNI SKNYTYAHRQDILKGHVKADAIPI LAARACTNIAS NYKYKLAYEKGRGKHVGFRNVKDDPLLVHYMEVAKMQSEQNYKKDYHK SKLKYHSPVDMMSLVHAKHASAAQTFAGYRKS LHNYYTLPEAMPVQLART MMEIQSNNAYSKSDYNNMCRGSGWVPIGSMQVELAKTAMAARDEQGYRVH PSTLKFTSLTGQMMALAMTNTKLLDDASYRASGEKFKHTYNLPADSPQFL QAKCNAQTMSESHYKHKWVEDIAKGYDIRPDAISVIHAKHGRHIASDIQYK KDYVKGKGKQIGCRNLQDDPLLVHYMEVAKMQSDKNYKKDYHKSCLKYH SPVDMMSLEHAKHASTVQTFAGYRKIQHHYTLTADSMNLALARTMNASSS DYEYKSDWNNSVRGTGWIPIGSLPVELAKVGGDIQSEQKYRTHPSKFQFKKL MDSMDIVLATDNNKIMNKQEYTKQWHNDKLTVHVMPDAPEI VLAKANAIN MSQKTYKAGLAKMYKNGYDLKVDAIAIQAAKQGTKIASDYKYKLSHRMN RGKQIGCRNLQDDPLLVHYMEVARLQSDKNYKKDYHKSCLKYHSPVDMM SVVHAKHASTVQTFAGYRKIQHHYTLADAMNVELAKMQMTIASDNLKYS DYETSVKGTGWIPIGSLDVEKAKTAGLIGSEKLYRQHPSSFKFTKDMHSM DV VLANANNAIMNKQTYVDANKDKTSVHIMPDSEVVLARANKINC SATLY KLANEQDKKGYDLRNDIAIALAAKQGTSIASDFKYRAGYRKQVGHHIGAR CVEDPLMLLAINSAKIASDALYKDFHQSKTKFHLPLDMLAFELAKKNQIQ VNHAKYVTRLHNWTC LDPDSTDVVHARHVYDIASDAVYKADLKCLQGIGW VPIGSLDVEKAKKAGEVLSEKLYRQHPSNFKFTSSVQDIGMVLAKQNNDNA NKQAYTEAWETDKTKNHI MPDAMDIVLANANKSNYSWNQYRKAMVEDK KKGYDMRNDIAISIAAKHGREGISDHKYKSSYRKQVGHHIGARCVEDDPLLV LALNSAKIASDALYKDFHQSKTKFHLPLDMLAFELCKKNQIQVNDFNRYRTH LHNWTC LDPDSTDVVHARHVYDIASEAVYRADMKWLRGIGWVPTGSLDVEK AKKAGEILSEKKYRQHPSTLDFKSLPDGMDIVLAKHNGQIMNKRQYVEQW DADKIKIHVGNELPELV LAKANAITMSNKLKQGFDETIKKGYLLPVDIAISV KAAKHGREIISDYKYKAGYSKQVGHHIGARCVEDDPLMLLAINSAKIASDAL YKDFHQSKTKFHLPLDMLAFELCKKNQIQVNDFNRYRTHLHNWTC LDPDSTD VVHARHVYDLHSDAVYRADMEFIRGTGWVPIGSLDVLKAKAAEILSDRLY KQKPD SLKYTTDMHSMMPVLAKANAEIINKGNVYQW EADKIKIHVGNEL PELV LAKANAINNSYKLYKAELEELYKKGYNLKSDCIAIKAAKHGREIISDY KYKAGYSEQVGHHIGARCVEDDPLIMLALNSAKIASDALYKDFHQSKTKF HLPVDMLTIELAKKCQMQVNDFNRYRTRLHNWTC LDPDSSDVVQARKVYALH SDAVYRADMEWLKGTGWVPIGSLDVEKAKKAGEILSERKYREHPSNFKFTA TTDDMPYALATANAQIMDKRQYVQW EADKIQIHVGNELPELV LAKANAI TMSNKL YREGFNEMIGQGYHLDKDAIAVQHAKKSREIISDYKYKAGFRNQ GHHIGAHCVEDDPLLVLAANAARISSDINYYKDFHLTKTKFSIPLDMMSFDL AKKSQFIQEFYRTKLHNWTCMPDNTDVVRARKVYSLHSDAVYRADLEW LRGCGWLPQGSVDVIAKNAQTI LNERLYRQPPSSIKFTSVVDSPEVVL SKQ NADILSDRYREVWDEKTN IHLNNDLPSLVLSKANAIASIN KLYTKDWA DKAGYHIKEDAI AVLKARKSRDIISDYKYHEGYRKQVGHHIGARCVEDDPL IMLALNSAKIASDALYKDFNKSCTKFHL PVDMLTIELAKKCQVQVNDFNRY RTHLHNWTC LDPDSTDVVQARKTYD LQSDAVYKADLEWLRGCGWNP HESV ELVKXHAQAVLADRGYRVKLDQQKFTAPVDRIDMVC AKNAEILNEANY RALWHTEKTKYTTDTPGLATAREVAKNLHPKLYTKDWDKVKATGFFMPG DAVPIKQCIHNNKVQSDHKKYKADYRKQAGHHIGARCVEDDPLIMLAINSAKI ASDALYKDFNKSCTKFHL PVDMLTIELAKKCQVQVNDHRYRTRLHNWTC LPDSTDVTQARKAYDLQSDVYKADLEWLRGCGWVAADCVDHV KVRNA QKIMNERLYKKDAKNFAKYTHIVDRPEIILATINAFNLSDLKYKETFN A EK GKYIGSEDTPQLAHSREVAKVISEKYKLAWDESKATGYQLDAHYIPVVG A KKTREIVSDVKYHDTHEKAKGHYLAGTLVDFPEVMRCGEQEKQKGLRDYH KEYNTNKAKNH I PADII PNMAKRCQDMLS DVLRYTRLHNWTC HPDQDII RARKNEILSDVYKDDLNIWIKIGCYAWDTPELVRAKNAHL LQSDLK YK EEGKKEFNYSIVTDPVYV TALLGHTWASDLNRYEAYHKDKHMYTTILDT YNYAHNATLKD IYSTKTYTSAWDKIKAKDYTLPSDSHALKHAQKQKVI LSN VKYREDYEFKSLYTLPKSLED DPQTLRCLKAGKMGLDRLYKADYEKTA KNHIPDMLDVL SARQTQNYVSEIGYRKHLHQWICMPDMQVYVQARKVNE QLSDIFYKDDLNLRLGIGCYAWDTP EILRVKAADKLQSDNKYRAKGIEAFK DYSVVIDTPYETAKLSSQNLSDLHYRNDYATNVKGTNSSPAVTV DTERAR LANFNQSDNLYKEANKNFMP TGYSLPHDNPLMKQAKQNSIVASSVKYKND YEKSKATNYTMDPDGVSFVNIRNANKVTNPRLYRELYEKGKGIHSTYDTP EIRQVKMTQEA VSDLCYKEKYFANRGTMIPMGVTPQMIHCNHVNEITSDLR YKEDLLWLRLGIGCFLYDTPEMSTVRNITKFRSNYEVEAKKNHANFTVVLDT PEYQRVTELKTHMSDLMYRAQGNVEKTKCTTTLDSLDVKRVKWAQQLSN KMYMTLAS KERAHFTPEKDTPI MSHARAMKVVS EIKYKDQYKMKHKY TSIHDTPI LVRAKAYLQSSDLRYKESFERSKGHYHSVKDALDIVCHRRVTD DISEVKYREKYINSLGTWRSMPDRPEFFHSRIIGENVSDIKYKEDLDWLKGIG CYVWDRPEMVQAEKNKVLFSERHYKETFEKNRHRFKYTC DTPFFQAAKKA SVI INDRHYRAAYEKTDKYSSVLDDPRNLLAKENAKNSQWKYREQYERA	

TABLE 12-continued

Salmon Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		KNKFTSILETPEHESHKRSKAI GDNVYKMDYNKNKAGYTLGHDTPHSTHM KKVKETSVLKYEKLYEKSQAQINMDPESFEIRAAKEAYKINISNLDYRKKYQ ATKNKWIWTVDRPDFIQAANKFQQSDIEYTDKEMMKGCVISIVDDKLAV LAKMNADLSNVKYREKFEKAGQYKTVDTPTNLHAKSVRILASESKYKLA GKMEQQTGGQFTTLPTRDTIHAQDMGKLVSEKQYKQKFEKEKSKYNSM RVPPDVQHAMDVAKNQSNFAYRQEAQAKLNYTSVAMRPD IMRATQASKL ASNIGYREKARQEAHGGSLVHRPDIALATAVSKLNSQVYKAKYDKEKGL RPKYDMKESVYKTKDAHQLASEVYKGDLLKHLKPVTDMAESLSMQH NLSTS KLSDDYQYKKKHEDSKGHYLM IADSPEQLHHKEASELQSHVYKKEK YEKAKGKAMLD FETPTVYVTAKEAQM QSADYHKEYEESIKGKNLSGLEV TPAMLHVRHATKIASERDYRDDLEEGVKGKGLTVFEETPELLRARNATAILS QKEYKKTLETEIKKGMLALATDTPDFMRARNATDILSQKYKQTAEADRA SYTSVIDTPDILHAQNIRNMVSQKKYREEAEKCMSHYVPLDTPEMMRVRE NQKNFSTIKYKDLLCRCTAISDLPEVKRVREMQKNISSVLYKDSAKGTPQV FTPEMERAKRNQE QISSVLYSDSFRKQVQGAAFVLDTPEMRRVKETQRNIS GVKYHQDFEKS KGSFTPTTSDPVM DRVKNKTQEFSDINRYGIQRRVVEMEK RRVVEHEQETVTDLRVWRNTPGSVFDYDPAEDNIQSRSLHMSVQAQRRS KEHSRSTSAMS GVGDEKSEVSEDPAHHVSLYSNGFATSSMGYTHTKTVELQ QRSSSVATQQTTVSSVPSHPSTTGKTVRAMYDYSSADMDEVSPKGDVIVN VQSIDEGWMYGTQVRSGKTGMLPANYVEAI
214	Salmon Q78BU2 (Actin alpha 1-1)	MCDDDETTALVCDNGSGLVKAGFAGDDAPRAVFPPIVGRPRHQGVMMGM GQKDSYVGDEAQS KRGILTLYKPIEHGIITNWDMEKIWHHTFYNELRVAPE EHP TLLTEAPLNPKANREKMTQIMFETFNVPAMYVAIQAVLSLYASGRTTGI VLDSDGDVTHNVP IYEGYALPHAIMRLDLAGRDLDYLMKILTERGYSFVTT AEREIVRDIKEKLCYVALDFENEMATAASSSSLEKSYELPDGQVITIGNERFR CPETLFQPSFIGMESAGIHETAYNSIMKCDIDIRKDYANNVLSGGTTMYPGI ADRMQKEITALAPSTMKI KI IAPPERKYSVWIGGSILASLSTFQAMWITKQEY DEAGPSIVHRKCF
215	Salmon ALBU2 (Albumin 2)	MQWLSVCSLLVLLSVLSRSQAQNICTIFTEAKEDGFKSLILVGLAQNLDPST LGDLVPLIAEALAMGVKCCSDTPPEDCERDVADLFQSAVCSSETLVEKNDLK MCCEKTA AERTHCFVDHAKIIPRDL SLKAE LPAADQCEDFKKDHKAFVGRF IFKFSKSNPMLPPHVLAIAKGYGEVLTTCGEAEAQTCFDTKKATFQHAIA KRVAELKSLCIVHKYGD RVVKA KLVQYSQKMPQASFQEMAGMVDKI V ATVAPCCSGDMVTCMKERKTLVDEVCADES VLSRAAGLSACCKEDAVHRG SCVEAMKPD PKPDGLSEHYDVHADIAAVCQTFTKTPDVAMGKL VYEISVRH PESSQVILRFAKEAEQALLQCCDMEDHAEVCVTALAGSDIDKKITDETYY KKMCAAEAAVSDDNFEKSMVYYTRIMPQASFDQLHVMSETVHVDVLHAC CKDEPHGFVLP CAEEKLTD AIDATCDDYDPSSINPHIAHCCNQSYSMRRHCIL AIQPDTEFTPELDASSFHMGP ELCTKDSKDLLSGKKLLYGVRHKTITIED HLKTI STKYHTMKDKCAEDQAACFTEEAPKLVS EAEVLKV
216	Salmon ALBU1 (Albumin 1)	MQWLSVCSLLVLLSVLSRSQAQNICTIFTEAKEDGFKSLILVGLAQNLDPST LGDLVPLIAEALAMGVKCCSDTPPEDCERDVADLFQSAVCSSETLVEKNDLK MCCEKTA AERTHCFVDHAKIIPRDL SLKAE LPAADQCEDFKKDHKAFVGRF IFKFSKSNPMLPPHVLAIAKGYGEVLTTCGEAEAQTCFDTKKATFQHAV MKRVAELRSLCIVHKYGD RVVKA KLVQYSQKMPQASFQEMAGMVDKI V VATVAPCCSGDMVTCMKERKTLVDEVCADES VLSRAAGLSACCKEDAVHR GSCEAMKPD PKPDGLSEHYDIHADIAAVCQTFTKTPDVAMGKL VYEISVR HPSSQVILRFAKEAEQALLQCCDMEDHAEVCVTALAGSDIDKKITDETYY YKKMCAAEAAVSDDNFEKSMVYYTRIMPQASFDQLHVMSETVHVDVLHA CKDEQGHFVLP CAEEKLTD AIDATCDDYDPSSINPHIAHCCNQSYSMRRHC ILAIQPDTEFTPELDASSFHMGP ELCTKDSKDLLSGKKLLYGVRHKTITIED EDHLKTI STKYHTMKDKCAEDQAACFTEEAPKLVS EAEVLKV

[0059] An animal protein can comprise a cod protein (Table 13). A cod protein can comprise a cod muscle protein. A cod protein can comprise a cod blood protein. A cod protein can comprise a cod albumin protein. A cod protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to any one of SEQ ID NO: 217-231.

[0060] A food product may comprise a fragment or a truncation of a cod protein. The food product may include a

fragment or truncation of any one of SEQ ID Nos: 217-231. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 217-231. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 217-231. A length of the fragment or truncation may be at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 217-231. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 217-231.

TABLE 13

Cod Proteins		
SEQ ID NO	Description	Amino Acid Sequence
217	Cod A0A8C4Z9R4 (Creatine kinase)	MAKNCHNDYKMKFSGDEEFPDLSLHNNHMAKVLTKDIYNKLRGKSTPSGF TVDDIIQTGVNDNPGHPFIMTVGCVAGDEESYEVFKDLLDPIISDRHSGYKPT DKHRTDLNFENLKGDDLDPAYVLSRVRTGRSIRKGTLPHPNSRGERRTIE KLSTIEALATLSGEFKGKYPLNGMTDKQDQLINDHFLFDKPVSPLLTCAG MARDWPDARGIWHNDAKTFLVWVNEDHLRVISMQQGGNMKEVFRFC TGLLKIETFFKKHNGFMWNEHLGYVLTCPSNLGTGLRGVHVKLKPLST HAKFEEILTRLRLQKRGTTGGVDTASVGGVFDISNADRLGSSEVDQVQMVV DGVKLMVAMEKKLEKSEIDDMIPAQK
218	Cod A0A8C5D617 (Actinin alpha 3b)	PHKLLQTNGKGMIMLAYMTQEEGWDRDLLDPAWEKQQRKTFTAWCNS HLRKAGTQIENIEEDFRNGLKMLLLEVISGERLPKPKDKGKMRPHKIANVN KALDYICSGVKLVSIGAEEIVDGNAKMTLGMINTIILRFATQDISVEETSAK EGLLLWCQRKTAPYRNVNQNFHISWKGDLALCALIHRHRPDLIDYSKLR KDDPIGNLNTAFEVAEKFLDIPKMLDAEDIVNTPKPKDEKAIMTYVSCFYHAF AGAEQAEATAANRIKVLAVNQENKLMEEYEKLASELLEWIRRTIPWLENR AAEQTRAMQQKLEDPRDYRRVHKPPRVQEKQCLEINFNTLQTKLRLSNR PAPMPSEGMVSDIANAWKGLEQVEKGYEEWLLTEIRRLERLDHLAEKFK QKCSLHETWTSGKEELLSQKDFESASLMEIRALMRKHEAFESDLAAHQDR VEQIAAIAQELNELDYHDAVTNARCQICDQWDNLGTLTQKRRDSLERY EKLWETIDQLYLEFAKRAAPPNNWMDGAMEDLQDMFIVHSIEEQSLITAH DQFKATLPEADKERMATMGIQSEILKIAQTYGIKLSGINPYTNLSPQDISNK WDAVKHLVPLRDQMLQEEVARQQANERLRRQFAAQANII GPWIQTKMEEI GHVSADITGSLEDQMNLLKQYEQNIINYKSNIDKLEGDHQLSQESLIFDNKH TNYTMEHIRVGEWQLLTIARTINEVENQILTRDAKGISQEQLENEFRASFMH FDRKRNGMMDPDDFRACLSMGYDLGEVEFARIMTLVDANNVTGVVTFQAF IDFMTRETAETDTAEQVMASFKILASDKSYITVDELRRRELPLEQAECISRM TRYVGADGTTGALDYISFSSALYGESDL
219	Cod A0A8C5D074 (Alpha-1,4 glucan phosphorylase)	MAKPLTDQEKRKQISVRGLAGVENVTDIKSNFNRLHYTLVKDRNVATKR DYYFALANTVRDHLVGRWIRTQQHYEYKDPKRVYLLSLEYFMGRTLQNT MVNLALENACDEATYQLGLDMEELQDIEEDAGLNGGLGRLAACFLDSM ASLGLAAYGYGIRYEFQIFNQKIVDGWQVEADDWLRYPNWEKARPEY MRPVHYYGRVEHTPGGVKWVDTQVVLALPYDTPVPGYRNNVNMTMLW SAKAPCDFNLKDFNVGGYIQAFLDRNLAENISRVLYPNDFEKGELRLKQ EYFVVAATLQDIIIRRFKSSKFGSTEFVRMDLSLTPDKVAIQLNDTHPALAIPE LMRILMDLEELPWEKAWDIVVRTCAYTNHTVLPEALERWPIDLQFNLLPRH LDIVYEINRRHLERVAQQFPGDEGLRRLMSLIEEGDHKKINMAHLCIVGSH AVNGVARIHSEIIKDITVFKDFYEMDPEKFQNKNTNGITPRRWLVMSNPGLAE VIAERIGDDYIRLDQLKNLLEFVDDDSLIRDVAKVKQENKMKFSAYLEE YKVKINPESMFDVQVKRIHEYKRQLNLCHIIITFYNRIKKQPNKAWTPRTIM IGGKAAPGYHTAKMIIKLIITAIGDIVNHDAVVGDRLKVI FLENYKVTLAEKI I PASDLSEQISTAGTEASGTGNMKFMLNGALTIGTMDGANVEMAEAGEEN LFIFGMRVDDVKAMDQTYDAKSYDRIPELQKAMDQISGGFFSPGQPDLF KDIVNMLMNHDRFKVPADYEEYIKCQDKVSALYKDPKEWTKMVRNIGSS GKFSSDRTIAQYAREIWMGMEPSLEKIPAPDDPRD
220	Cod A0A8C5F6L1 (Fructose-bisphosphate aldolase)	MPHAYPFLSTEQKKELSDIAKKIVAPGKGILAADESTGSVAKRFQGINAENT EENRRLYRQLLFTADDRVAPCIGGVIFFHETLYQKTDDGKLFQILIRERGQV VGIKVDKGVMLAGTDGETTTQGLDGLYERCAQYKKDGADFAKWRCVLK ITENTPSELAIMENANVLARYASICQMVRRHGIVPIVEPEILPDGDHDLQRTQ YVTEKVLAAVYKALSDHHVYLEGTLTKPNMVTAGHSCTRKYSAQEVAMA TVTALRRTPPAVPGVTFSLGGQSEEQASINLNAINQCPLGRPWALTFSYGR ALQASALKAWGGKKENGKACQEEYIKRALNNNLASQKGYVSSGDKGAAG GESLYVANHAY
221	Cod A0A8C4YT70 (Myosin, light chain 1, alkali; skeletal, fast)	MAPKKDVKAPAAAAKAEPKAKAEPAPFVAVPAPKTVDSLAVKVEFTPD QMEDYREAFGLFDRVGNKVCYNQIADIMRALGQNPTNKEVKAILGNPSD EDMNSKRVDFFEGFLPMMQTI VNSPNKGTLDYVEGLRVFDKEGNGTVMG AELRIVLSTLGEKMTEAEIDALMQGQEDENGCI NYESFVKHIMSI
222	Cod Q8AWX8 (Glyceraldehyde-3-phosphate dehydrogenase)	MVKVINGFGRIGRLVTRAGFQSKKVEIVAINDPFIDLEYMVYMFKYDSTH GRFHGEVKIEGDKLVIDGHKISVFHERDPANIKWGDAGAQQVVESTGVFTT IEKASAHKGGAKRVII SAPSADAPMFVMGVNHEKYENSLQVVSNASCTTN CLAPLAKVIHDNFGIVEGLMSTVHAI TATQKTVDGPSGKLWRDGRGASQNI IPASTGAAKAVGKVIPELNGKITGMAFRVTPPNVSVVDLTVRLEKPAKYDDI KKVIAKASEGPMKILGYTEDQVSTDFNGDIRSSIFDAGAGIALNDHFVKL VTWYDNEFGYSNRVIDLMAHMSCKE

TABLE 13-continued

Cod Proteins		
SEQ ID NO	Description	Amino Acid Sequence
223	Cod A0A8C5CX2 (Tropomyosin 1, alpha)	MDAIKKKMQLKLDKENALDRAEQAEQDKKAAEDRSKQLEDDLVALQK KLKGTEDLDKYSEALKDAQEKLELAEKKAADAEASLNRRIQLVEEEL DRAQERLATALTKLEEAKEADESERGMKVIENRAMKDEEKMELQEIQLK EAKHIAEEADRKYEEVARKLVIIESDLERTEERAELSEGKCELEELKTVT NNLKSLEAQAEKYSQKEDKYEEI KVLTDKLEAEETRAEFAERSVAKLEKT IDDLLEDELYAQKLKYKAISEELDHALNDMTSI
224	Cod A0A8C5B6F3 (Myosin binding protein Cb)	MPEAADAGELPTAEEDSGDEEGQGTEL TGLFVERPQENNVIIAGADVITA KVDSTTLTRKPTVKWLKGKWLDSLASKAGKHLQFKETYDRETKIYTYEMK VIKAKVGAAGAAGAYRCEVTSKDKFDSASFETVEAAQSQSDVDILSSFKR ADAGEDDGNLDFSALLKAAKKKRPEDDKPVIDVWELLKSAHPSEYEKIA FEYGITDLRGLMKRLKKMAEKPSEAFKRMESCYSVEKGKKIVLTCEVF DPNVQVKWLKNGQEI KGS AKYIMETIGNIRKLTIMKTTLADDAAYECVVG EDKCFTEVFVKEPPVTITKLMDDYHVVGGERVEFEVEVSEGANAMWFFE DIEIPKDTNPAYKRIKKDGRKHLIIIEAGLEDIGMYHCWTNGGHTKGELE VEEKELVLQDIADLTVKATDQAIKCEVSDEKVTGKWFKDGVVLPDER HKMTHIGRFHKLIDNVKPDAGDYTFIPNGYALSLSAKLNFLIEIKIDYVPR QDPPKIHLDTTGNMVSQNTIIVVAGNKLRLDVEITGEPATVVWSKGEVIT ATEGRVRVDNRKTLSSFVI EGAERED EANYTICVTNPAGEDKAKLFVKIVD VPDPPQNVKCLSVGEDSSSMVWEPPAFDGGAPLKGYLMEKKGSSRWIT KLNFDVYEATTYEA KRMIEGVYEMRVFAINSIGMSQPSITSKPFMPLAPTS EPTRLLVHDVTDGTCTLKWMAPERIGAGGLDGYII EYCKEGETEWWVANK ELCERQGVVVRGLPVGEKINFRVFAKNLAGVSPATLAQPVTVREIVDLPKI RLPRDLRTKYIRKVGDKINLVIPFQGGKPRPVVWKYKDGAPVDEKIVGVRS TVDSILFIRSAERDHSKYELVVQIENMEDRATIDLRVVERPGPPTKVKVTE VWGFNAALEWEAPKDDGNCEITGYTIQKADKKTMEWYTI EHNRRPSCTS SDLIMGNEYMFRVYSENLVGPSEEPGQS KNTAVIGKTGLAYNPPEFKEKDV SCIPTFTQPLLD RSVVAGYSTAMSCSVKGFPKPKIIVWMKNKMI LGGDPKYI MLNNQGVLTNLIRKPSTFDGGRYSCMAVNDLGQALVECKLEIRIPTDPTAE VK
225	Cod A0A8C5F718 (Myomesin 2b)	MAMRSGNYNHEYYHKQSKYVVKEYSSSESVDYQHKAAVRIQGVVRK KMVEKYVREEPMIRGPKFVVRRLRSHTVFENSPIKLPCTVEGFPVPIV TWYKD GVILEHGSAGRYI VDDNVGMRSLEIPRCGTDDTAQYSAVASNIHGQASSQA SII VKRFRQEDESCLYAWLPYTVPLPSIQYTKINITFTEKFPVFASEGGSSAS LTATMHLEPNLANLQPDSSQWYRDDTRLVDSKWKVINTGRGTSTLTIPNLPK DDEGLYTLRLVTKGGTAEHSAFISVSDGPPPVVGAPGAPMDIRIHDANRDY VIVSWKPPNNTTTEGPILGYFVDKCEVGTENWAQCNDHPIKICKYPVPLFEG HSYYFRVRAVNARGISKPSRMSDPIAALDPTFERLHALKLGGLDVVTYD DDMEAEKGKPPGVSSGIYASETDRTYVVLWSKAPAYSSKAPMWYIIEKQV GTAAWQVRVNTQVPIRSPRYAVFDLAEGKEYQFRVLSANMYGTSEPESESKL IQTQALKGVPSPAGQVISTRETDTSLIIQWAPPKEPNNLIGYYIDQCVKGSK DWTSAHKKPHSSTKFVVSGLTTGETYVFRVQA INELGLSDSESQSSPLSVKA ALTSPTAPYDVALLGCDGQSMVLNWKPLHTGGTA INAYFVDKRRSGAT MWREVHIPAVTERVYQVEGLTQGA IYQFRVYAANLAGLGKSSKASADFSC EAWTMPPEPGPAYDLTASEVRDTSVLVLEWKKPVYYGSGPI TGYHVVEYAKKG SSDWTRANETA VSHRYLKVGLEEGAKYVFRVRGVNEAGVGMA SLPSDS VVAKALAGAEVSCSVDEKSGDIVFAFESCQMNVGSEFLWKKNYKEITEFS SSIIVHTEGNLCKLMFKNVAVKEDFGTYSVSVSHTDGVSSSCSISAEELQKM LELSYSIRHPIIPLKSELAYKILERGRMRFWLQAEIISKVYTKFFANNKELS GADPNKMAHDVSTGVIEFILDHFTEDSEGTYTCQITDGGGAQSSSLVLIGDA FKAALAEADYQRREYIRVKEGPHFSEFLTVKVGEDCSVTLVCKVANLKES EFHWYKEDTEIIPDVKPDLSGSGVCKLPIHLFSRKTGTGVYKATISDDR GKDMS QVDISGKVFDDVINKINQLAGASAEELVINCTATGIQLQCHMKYTTSEMNI HWMHGESKVGHSEKCVIGTPAMATMEVIEPNDDKGVYSIVI TDLENTN KRTLDLSGVDYDKAYA QFQKLKAEAYAE MNRGKVMGGLPDVVTIMEKK TSLTCTVCGDPKPLVSWLKNNDVEPDNQVVS LDQGKFACTLINGIGM DDSGKYTMIVQNKYGGECVDIVVS VYRHGDTIPVAKPNLTAKTII PPKMPIE LPQPKTAAPSAPPASPAPPAKAVGGRGMSPTPTRK
226	Cod A0A8C5C5F8 (Phosphopyruvate hydratase)	MKSNPLTSNTGEHAS YLLHVNGVMSITKIHAREILDSRGNPTVEVDLWTAK GLFRAAVPSGASTGIHEALELRDGDKTRYLGKGTQKAVDHVNKDISPKLIE KNFSVVDQEKIDKFMLELDGTENKSKPGANAILGVSLAVCKAGAAEKGV LYRHIA DLAGNKD VILPVP AFNVINGGSHAGNKLAMQEFMILPVGAANFHE AMRIGAEVYHNLKNVIKAKYKDATNVGDEGGFAPNILENNEALELLKSAI EKAGYDPKIIIGMDVAASEFYRSGKYDLDFKSPDDPARHISGKELGDLYKTF IANYPVQSIEDPFDQDDWANWADFTA AVDIQVVGDDLTVTNPKRIQQAVE KKACNCLLLKVNQIGSVTESIQACKLAQSNGWGMVSHRSGETEDTFISDL VVG LCTGQIKTGAPCRSERLSKYNLMRIEELGDKAKFAGKDYRHPKIN

TABLE 13-continued

Cod Proteins		
SEQ ID NO	Description	Amino Acid Sequence
227	Cod A0A8C4ZUF5 (Myosin light chain)	MAPKKAKRRAGGSGSSNVFSMFESQSIQIYEKFAFTIIDQNRDGIISKDDL RDVLSMGGQINVKNEELEAMVKEASGPINFVFLTMPGEKLGADPEDVIV AAFKILDPEATGSIKKEFLQELLTTQCDRFSPEEIKNMWSAFPDPVAGNV KNICYVITHGEEKEE
228	Cod A0A8C5FJN9 (Myosin, heavy chain b)	IMGDGEMQCFGEAAVYLRKPERERLEAQTSPFDAKTAFFVSEPKEMYLG VLQSRDGGKATVKTLCDKTLTVKEEEIYPMNPKYDKIEDMAMMTHLNEP TVLYNLKERYAAWMIYTYSGLCFVTVNPKWLPVYDSVVNAYRGKKRIE APPHIFSIDNAYQAMLTDRNQSLITGESGAGKTVNTKRVIQYFATIAVSG PKKTEQVAGKMGGLEDQIIAANPLLEAYGNAKTVRNDNSSRFKGFIRIHF ATSGKLASADIETYLLEKSRVTFQLSASERSYHIFYQLMTGHKPELIDALLITK NPFYHMSIQGEITVKSIDDVVEEFATDDAIDILGFTADEKIGIYKLTGAVLH HGNMKFKQKQREEQAEPDGNEEADKIGYLMGLNSADVLKALCYPRVKVG NEYVTGQCTCPQVHNSVMALCKSVYEKMFLWMVVRINEMLDTKQPRQFF IGVLDIAGFEIFNYSNLEQLCINFTEKLLQOFFFNHMFVLEQEEYKKEGIEW EFIDFGMDLAACIELIEKPMGIFSIIEECMFPKASDTTFKNKLHDQHLGKTK AFKPKPAKGAAEHFSLVHYAGTVDNITGWLDKNKDPLNDSVIQLYQK SSVKLLALLYASHALPDADAASKKGAKKKGGSPQTVSALPRENLGKLMTN LRSTHPHFVRCLIPNESKTPGLMENFLVIHQLRNGVLEGIRICRKGFPSPRVL YGDQKQYKVLNASVIEPGQFIDNKKASEKLLGSIDVDSQYKFGHTKVFF KAGLLGALCEMRDEKLAALVTLTQALCRGFNMNRKEFKMMERREAVPTI QYNIRSFMNVKTWPWMKVYFKIKPLLKSAETEKEMAQMKETFGTKEDL AKALSKKKELEEMVSLLEKNDLLQVQSEAEATLSDEERCEGLIKSKIQL EARLKETSERLEDEEEINSELTAKKRKLEDECSSELKKDIDLELTLAKVEKE KHATENKVKNLTEEMSSQDEIIAKLTKEKKALQEAHQQTLDLQAEEDKV NTLSKAKCKLEQQVDDLEGSLEQEKRLRMDLERAKRKLQGLDLKLAQESIM DLENDKQSSDEKIKKRDFFETSQLLNKIEDEQTLQAQFQKKIKELQARIEELE EEIEAERAARAKVEKQRADLSRELEEISERLEEAGGATSLQIEMNKKREAEF QKLRRDLEESTLQHEATSAALRKKQADSVAELGEQIDNLQRVKQKLEKEK SECKMEIDDLSSNMEAVAKSKANLEKLARTLEDQLSELKSKNDEHVRQLN DLSVORARLQTENGEFGQQEKEALISQLTRGQAYTQQIEELKRHIEEV KAKNALAHSVQSAHRDCDLREQFEEQEAELQRGMSKANSEVAQWR SKYETDAIQRTTEEEAKKLAQRLQDAEEAVEATNSKASLEKTQORLLG EVEDLMIDVERANALANLDDKQRFNDKVLAEWKQKYEEQAELEGAQK EARSLSTEMFKLNSYEESSLDQLETLKRENKNLQOEISDLSQMGETGKTIH ELEKGKKTVEVEKSELQTSLEEAETLEHEESKILRVQLELTQIKGEVDRKM AEKDEEMEQIKRNSLRVIDSMQSNLSEIRSRNDALRIKKMEGDLNEMEI QLSHANRQASEAQQLRNVQGLKDAQLHLDDALRGQEDMKEQVAMVE RRNNMLAEIEELRAALEQTERGRKVAEQELVDASERLGLLHNSQNTSLINT KKLEETDLIQIQGEVEDTVQEARNAEEKAKKAITDAAMMAEELKKEQDTS SHLERMKKNLELTVKDLQHRLEDAESLAMKGGKKQLQKLEARVRELETEC EAEQRRGSEAIKGVRYKERVKELSYQCDEDKKNICRLQDLVDKLQLKMK AYKRQAESEEQANTHLSRYRKVQHELEEAQERADVSESQVNLKRTKSRN GITKIIN
229	Cod A0A8C5F7Y0 (Nebulin)	MATAGMPTRKVRKRVAVDTSKFMTPYLAHSMKMLEQYSHNMYRKAYDL SRGTPPVHIADTPMVIRKAQEQLSEVKYRMENKARAGSLYDGEAKEIA HVHMSSELLSKVLRYQKWDETCDRYLLPPDAPELVLAVKNAANYSKKLY TEDWDLEKASFLPYSDSPELRRVAKAEALSDIQYKKGYDQRLKFTSLAD RPDVELAKKVFTQRSDLKYKEDYNKNVKGQWCETPYFDVATARVAMENL SSRRYTQDWEAAKDQINFMTETPVYDTNKKAGTSASSVKYRKYDEKTK DKSDYNTLPATENPLLRQLRYAGTILSDKVYKKNYEKSRGFSINYCDTPKF KMDSVLKKFTDTHYQDKYKNEVQGHYIGSYEDIYMLHCQKVEGFKSEQN YKAEYEDIKTRCFPPQTITPEYEVDDKLAQCKDKVYRQHPDKVFTQVTD PVQQAALNAIQLSIDINYKAKYEQTKTNSLPADYPPFIQSRVNAFNLSENC YKYNWEKEKAKRFEMKGDASISILAAHAHTNIASDVYKKEYEKNRGRMV GALSIHDDPKILHSVHVAKIVSNREYTKDYEKVKTKYHTSLDMSVTLAK KSQAIASMTGYKKISNRYLMPYDAILDVLAKKANSIQSDNEYKSDYNQYTK GSPWITFGSMDVELAKKAGEILSEKKYRKHPDTPVFTAIDHHPVMLQARVN QLQRSDFYKKGLEEIHQKYSLPIDMPQLLARANSYINISKNYKAAWEET ISKGYDLKLDAISIKAAKAAARHAASDIQYKKAIEKARGHHVGFNRIKDDPL LVHYMQVAKMQSEKNYKDYHKSCLKYHSPVDMMSLVHAKQASKAQT YAGYRKIQHHYSLMPDAMNVLARKMNNTDASDCEYKSEWNNYVRGTGW VPIGSIGVETAKIGGQIQSENKYRTHPSTPFKSKLMDSMDELATQAYTSAW EKDKRTIHVMPDPSPEIVLAKANAITMSDVREYYYKLHQNGYHLKGDAIPIV AAKLGTKIASNYKYKLDYEKARGHHVGFNRIKDDPLLVHYMQVAKMQSE KNYKDYKAKLYHSPVDMMSLVHAKASKAQTYAGYRKIPHTYFLPL DNMQLQLCRKMQEIQSDNEYKSEYNNLYKGTGWVPIGSIDVELAKTANSA LREHGYYRQHPSSFKFTSKTSDMDMVLALANSKIIDQAAYKASGEKVKHTY HLPADAPEFIQAKFNAQTMKSHYKQKWEHIAKGYDMRPDAISILHAKQ GRHIASDIQYKKAIEKARGHHVGFNRIKDDPLLVHYMQVAKMQSEKNYK

TABLE 13-continued

Cod Proteins		
SEQ ID NO Description	Amino Acid Sequence	
	KDYEKAKLYHSPVDMMSLVHAKESKAQTYAGYRKIQHHYNLMPDAM NLVLARKMNTDASDCEYKSEWNNYVRGTGWVPIGSIGVETAKIGGEIQSE NKYRTHPSTFKFSKLMDSMDLELATANNKIMNKDKRTIHVMPDSPEIVLAK ANAITMSDKLYKAGVMEMAKKGYHLKGDAIPIVAAKLGTKIASNYKYKL DYEKARGHHVGFRNIKDDPLLVHYMQVAKMQSERNYTKDYQSKLKYHS PVDMSLVHAKESKAQTYAGYRKIEHHYNLMPDAMNLVLARKMNTDA SDCEYKSEWNNYVRGTGWVPIGSIGVETAKIGGEIQSENKYRTHPSTFKFSK LMDSMDLELATANNKIMNKQAYTSAWEKDKRTIHVMPDSPEIVLAKANA TMSDKLYKAGVMEMAKKGYHLKGDAIPIVAAKLGTKIASNYKYKLDYEK ARGHHVGFRCIQDDPLLVHYMQLAKQQSEKNYKKDYEKAKLYHSPVDM MSLVHAKDATKAQTYAGYRKIQHHYSLPDAMNLVLARNMQVNIQSDNQY KSDYESYCKGIWPIGSLDVEKAKIAGKIQSEKLYRQHPSNLKTCKDMFSM DLTLATANNQIMNKVLLALYCVDTLMNILSSGNNLTNNFVVSFSLKYKL DNELSKKKGHNVRKDAIAI LAAKNCTTASDYKYKAGYRKQVGHGIGARC VEDDPLLVLALNSARIASDALYKKDFNKSSTKFNLVPMMAFELAKKNQI QVNHANYITRLHNWTCPLDSTDVVNARRAYEIQSDNYKEDQKLMQGGIG WVPIGSLDVEKAKKAGEILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQI TNKKAYHDAWDKEKSNIHIMPDAMDVLAKANRANYSIKRYKQANEDAK KKGYDLRNDAGIIAARAQRNIISDIKYKAGYRKQVGHGIGARCVEDDPLL MLALNSARIASDLYKKDFNKSSTKFNLVPMMAFELAKKNQIQVNHAN YITRLHNWTCPLDSTDVVNARRAYEIQSDNYKEDQKLMQGGIGWVPIGSL DVEKAKKAGEILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQITNKKAY HDAWDKEKSNIHIMPDAMDVLAKANRANYSIKGYKQANEDAKKKGYD LRNDAGIIAARAQRNIISDIKYKAGYRKQVGHGIGARCVEDDPLLVLALNS ARIASDLYKKDFNKSSTKFNLVPMMAFELAKKNQIQVNHANYITRLHN WTCPLDSTDVVNARRAYEIQSDNYKEDQKLMQGGIGWVPIGSLDVEKAKK AGQILSEHLYRKHPSNYKFTSKCDAMPLVLAKANAQITNKKAYHEAWDKE KSNIHIMPDAMDVLAKANRANYSIKKYKEANESKKKGYDLRNDAGIIA ARAQRNIVSDYKYKAGYRKQVGHGIGARCVEDDPLLVLALNSARIASDAL YKKDFNKSSTKFNLVPMMAFELAKKNQIQVNHANYITRLHNWTCPLDST DVVHARHVYEIQSDAVYKADLKLWLGMGWVPIGSLDVEKTKKAGEILSE RKYRQPASSHKFTSPLDAMEIVLAKNNALTMNQRLYTEAWEKDKTKLHIT SETPEIILSQQNAINMSNKKYKQGYEDTIKKGYLPGDAISVKSAXHSRNIIS DYKYKAGYRTQVGHGIGARCVEDDPLLMLALNSARIASDLYKKDFNKS TKFNLVPMMAFELAKKNQIQVNHANYITRLHNWTCPLDSTDVVNARKA YDLQSDAVYKADMEFMKGIWVPIGSLDVVKAKHAGKILSDHLKYQKTD KLKYTSDTKAMPMLLAKANAEMVNKSSYVAWEKDKIKTHLIPDLPEILL SRSNAYNLSNKMVKEDVVKMFRKGYHLKGDAIAI VAAKKGREIISDIKYK AGYRKQVGHGIGALSVRDDPLIMLALNSARIASDALYKKDFNQSKTKFHL VDMLTIELAKKCCQVNDNFYRTRLHNWTCPLDSTDVVQARKAYDLQSD AVYKADMEWIKGQGWVPIGSLDVEKAKKAAEILSERKYREHPSKFKFTAK TDDMPYALAKANAIEIMNKQSYVAWESDKTKLHIMPDTPEIILGQQNKL NSRKYYREAYEETLKKGYLLPKDAISVLAARASRNII SDYKYKAGFRKQCG HHIGALSQDDPLLMAANAAKIFSANLYKQDPFKTKTKYKLPVDMVAF DLAKTAQSHMKEVTYRTRLHNWTCMPDNTDVVHARKVYDLHSDAVYRA DMEWIRGCGWSPQSDVDVVKARNAQKILNENLYRQAPSTVKFTSVVDLPA IVLSKQNAENISEMKYTAAWNKDKLTFHLKPDIPSVQLAKANAINISNLYT KAWNQKAKAYHIKEDTIAVLKARASRDII SDYKYKTGYRKQVGHGIGAL SVRDDPLIMLALNSARIASDALYKKDFNQSKTKFHLVDMTLTIELAKKCCQI QVNDNFYRTRLHNWTCPLDSTDVVQARKAYDLQSDAVYKADMEWIRGCG WVPHECVDVVKVKAQKILADRGRVRLDTQKFTAPVDRMDFVCAKNA AEVLNDAKYREAWHTDKSKFSLVDTPLGATSREAVAKIMHPKLYTKDWDK VKATGYFMPGDAVPIKQCTQTTKVQSSHKYLEGYRKQVGHGIGALSVD PLIMLAMSAKIASDALYKKDFNKSSTKFHLVDMTLTIELAKKCCQIQVND FYRTRLHNWTCPLDSTDVVNARKAYDLQSDVYKADLEWIRGCGWVPAE SVHHVKVRKAQEI LNERLYKKGAIENFSKYTHVDRPEILARVAAANLSD LKYKETFNNEKGHYIGSEDTPQLAHSRECAKNISEAKYHESHEKAKGHY ANTLVDFPEVVRGGEQEKAKALRVYQKEYHESKTNNHIPSDIIANQVAKRC QDMLSDVIYRTHLHQWTCPPDQLDTRARKGNEILSDVYKEDLTWMKG MGCYVWDTPEIVRAKSYELQSELKYKEAGKKEFNYSIVTDTPVYVTAI L GSTWASELNRYEAYHKEKHLTYTILDTHDYARCHDFKHFFSNKIYTA AWDKIKAGKYKIPHDSHAMHAKQKQVILSGVYKEDYEKFKSLYSLPKCLQ D DPATARCIIKAGKLNLDRLYKEDYEKTKAKNHI PADMLDIVSGREVQRN VSEINRKRHLHQVCLPDMQVVFQARKVNDQLSDIFYKDDLNLRLGIGCYA WDTPELRVKTAGDQQSENKYKAKAVEAYKDYTVVLDTPYETAKQNAV NLSDLHYRLDYANVKGNTTAPAVTMETERARLANYNQSDNLYKEANKN SMATGYLMTNDTPVVKQVKNNSLVASSVYKAAAYEMMKATGYTLHPEG VTFTNTRKVNQVINPRLYREQYKEDKDKIHSTLDTPDIRQVKMNQEHLS ELCYREKYNNTKGQLINMPMPQLMHCQHVNQITSDRLYKEDLLWLRLG L GCFMYDTPEMVNRNITKFKNNYQVEAKKNHKSFPVLDTPPEYQRVTE L KSHMSMVSPSTNTSLNTLDSVDIQRSKWAQQLSNKFLYTDLASKERA HFSQVEVETPAMGLARTMKVTNSNYKKEQYKMKHRYTSIADTPILIRAKKAYLQSS	

TABLE 13-continued

Cod Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		DLRYKETFERSKGYHTVKDALDILCHRRVTDDISEVKYREKYINSLGTWK SIPDRPEFFFNRI VGENVSSVKYKEDLEWMKGIGCFVWDTPEMVQADINKT LYSGRLYKASFENRYNFKYTCDPFFHAAKNASSLINERSYRSSYEKNKD KFTITTTDDPRYQLARENKMSQAKYKSRAKELLKSGCNELHRPDILTALYQ SVLGSKWYREQYERVKSKFTSILETPEYESHKRSKKIGDIVRMEYNKNK AKGYSLPYDTPHQNMMKKVQALTSNLKYREVYEKAKSQINMDPDFAFEIRA AKEAYKNTTNLDYKKKYEATKNKIWTVDPRDFISHSRNTLQQSDVEYKY DKEMMKGCVIPVDDKI TLAKKGCEMASDIKYKEKYEKAKGHYNPVADS PQILHAKAVRTLASESKYEASKKEMQSGSFTTLPETRDTVHSKEVNKLISG KLYKSKFEKEKGQSKYNRMIVPPDVQHAMDVAKSQSSYAYKRDADNLH YTTTADRPDIRKATQAAKLISDVGYREKARAEASRGGS LTYPDIALATEVS KLTSLVDAKSALHGAASVDTLQMRHSKKMSALTSNVKYHEKFDKEMKGG KPKYDLKDSKIYQTLKDATNLASEVKYKGDLLKHLKHPVTDMAESLSMQH NLSTSKLSSDYRYKKKFKDSKGHYLV IADTPEQLHHKEATELQSSVSGPSLC GVHVCKDYKDYEEGMRGKNLCGLEVTPAMIHVRHATKIASEREYRKDL EEGVKGGLSAPETPEHLRAKNATHILSREYKKSLELEIKGKGLALAT DTPDFMRARNATDILSQTKYKHTAELDRASFTSVIDTPDIIHAQQMRNIISQN KYKAEAGKTM SHYVPVFDTPEMQVRVRENQKNFSTLQYQADHKIKGKVLT VKDTPEMLRVKENTKNFSSVLYKDSKAKGTPVVFTPEMERVKRNQEHISSV LYSDSFRKQVQGKAAPVLDTPEMRRVKESQLNISGVRYHQDHEKSKGSFT AMASYPEMERVKKNTQEFSDINRYGIQRRVVEMERRRVVEHEQETTTDLR VWRTNPGSVFVDYPAEDNIQSRSLHMMSVHAQRRSKEHSRSTSAMSGPAD EKSEMSHEGDHMMSTYSNGFGASSVGQTTKTVELQQRSSSVATQOTTVS SIPSHPTTGKTVRAMYDYGADNDEVTFKGDGDIVNVQSIDEGWMYGTV QRTGKTGMLPANYEAV
230	Cod A0A8C5ARQ5 (Myosin heavy chain, fast skeletal muscle-like)	MSTDAEMAVYGQAAIYLRKPEKERLEAQSKPFDAKSAAYVTDVKELYLKC TILKKEGGKVTVKVHDEEEKTVKDDDVTPMNPVKYDKIEDMAMMTHLN EASVLYNLKERYAAWMIYYSGLFCATVNPYKWLVPVDQHVVCAYRGKK RMEAPPHIFSVSDNAYQFMLTDRENQSVLITGESGAGKT VNTKRIQYFATI SVGGDKKKDIGAGKIQGSLDEQIIAANPLLEAYGNAKTVRNDNSSRFGKFI IHFGTSGKLSSAD IETYLLEKSRVTFQLPAERGYHI FYQMNTNHHKPEIEMTL ITTNPYDFPTCSMGQIAVASIDDKELVATDTAIDILGFNNDEKMGYKLTG AVLHHGNMKFKQKQREEQAEPEGTEDADKVAYLLGLNSADMLKGLCYPR VKVGN EYVTKGQTPQVLNSVSALAKSIYERMFLWMVIRINQMLDTKQPR SSFI GVLDIAGFEIFDYN S MEQLCINF TNEKLQQFFNHHMFVLQE EYKKEGI IWEFIDFGMDLAACIELIEKPMGIFSI LEEECMPPKASDVTFKAKLYDQHLGK TKAFEKPKPAKGAEAHFSLVHYAGTV DYNITGWL DKNK DPLNDSVVQL YQKSGVKLLPVLYPPVASEDSTKKGGKKKGSMQTVSSQFRENGLKMTN LRSTHPHFVRCLIPNESKTPGLMENFLVIHQLR CNGLVLEGIRICRGFPSPRIY ADFKORYKVLNASVIP EGQFIDNKKASEKL LGSIDVNHEEYKFGHTKVFFK AGLLGTLEEMRDEKLATLVGM IQALCRGFQMRKEFKMMERRHAVYTIQ YNIRSF MNVKTWPWMKVYKIKPLLSAETEKELSAMKENYEKMKDLA TALAKKLELEEKMVSLLEQKNDLSLQVASEGENLND AEERCEGLIKSKIQL EARLKETTERLEDEEEMNADLTAKKRKLEDECESELKKDIDDL E LTLAKVEK EKHATENKVKNLT EEMASQDESVAKLSKEKKALQEAHQQTLDL DQAEED KVNTLSKAKTKLEQQVDDLEGSLEQEKLRMDLERSKRKLEGLDKLQAES IMDLENDKQQSDEKIKKDFETSQLLSKIEDEQSLAIQLQKKIKELQARIEEL EEEIEAERAAKVEKQRADLSRELEISERLEEAGGATSQAIEMNKKREAE FQKLRRDL EESTLQHEATAAALRKKQADSV AELGEQIDNLQRVKQKLEKE KSEYKMEIDDLSSNMETVAKAKGNLEKICRTLEDQLSEIKAKSDENS RQIND IGAQRARLLTENGEFGRQLEKEALVSQLTRGKQAYTQQIEELKRQIEEEVK AKNALAHGVQSARHDCDLLREQFEEEQEAKAELQRGMSKANGEVAQWRS KYETDAIQRT EEELESKKLAQRLQEAEEQIEAVNSK CASLEKTKQRLQGE VEDLMVDVERANGLAANLDKKQNFQVLA EWKQKYEEGQAELEGTOK EARSLS TELFKMKNSYEALDQLETLKRENKNLQQEISDLSEQIGETGKSIH ELEKSKQVETEKTEIQTAL EEAEGTLEHEESKILRVQLELNQIKGEVDRKL AEKDEEME QIKRNSLRITDSMQSTLDSEVRSRNDALRIKKKMEGDLNEMI QLSHANRQA AEAQQLRNVQQLKDAQLHLDDAVRGGEDLKEQAAMVD RRNGLMVAEIEELRAALEQTERGRKVAEQELIDASERVGLLHSONTSLNLT KKKLETDLVQVQGEVDDTVQEARNAEEKAKKAITDAAMMAEELKKEQDT SSHLERMKNLEVTVKDLQHR LDEAENLAMKGGKKQLQKLESRVRELETE VEGEQRRGVD AVKGVRYERRVKELTYQTEEDKKNGARLQDLVDKLQLK VKAYKRQAEAEAEQANSYLSKCRKVQHELEEAERADIAETQVNKLRSKS RDGGKGDPAE
231	Cod Q78AY8 (Actin alpha 1)	MCDDEETTALVCDNGSGLVKAGFAGDDAPRAVFP SIVGRPRHQGVMMVMG GQKDSYVGDEAQS KRGILTLYPIEHGIITNWDMEKIWHHTFYNELRVAP EEHPTLLTEAPLNPKANREKMTQIMFETFNVPAMYVAIQAVLSLYASGRIT GIVLSDSGDGVTHNVPVYEGYALPHAIMRLDLAGRD L TDYLMKILTERGYSF VTTAEREIVRDIKEKLCYVALDFENEMATAASSSSLEKSYELPDGQVITIGN ERFRCPETLFPSPFIGMESAGIHETAYNSIMKCDIDIRKDL YANNVLSGGTT

TABLE 13-continued

Cod Proteins		
SEQ ID NO	Description	Amino Acid Sequence
		MYPGIADRMQKEITALAPSTMKIKIIAPPERKYSVWIGGSILASLSTFQQMWI SKQEYDEAGPSIVHRKCF

[0061] An animal protein can comprise a tuna protein (Table 14). A tuna protein can comprise a tuna muscle protein. A tuna protein can comprise a tuna blood protein. A tuna protein can comprise a tuna albumin protein. A tuna protein can comprise a sequence that has at least or about 70%, at least or about 75%, at least or about 80%, at least or about 90%, at least or about 91%, at least or about 92%, at least or about 93%, at least or about 94%, at least or about 95%, at least or about 96%, at least or about 97% at least or about 98% at least or about 99% or about 100% sequence identity to SEQ ID NO: 232.

[0062] A food product may comprise a fragment or a truncation of a tuna protein. The food product may include a fragment or truncation of any one of SEQ ID Nos: 232. A sequence of the fragment or truncation may be identical or substantially identical to a portion of any one of SEQ ID Nos: 232. A sequence of the fragment or truncation may be at least about 70%, 80%, 90%, 95%, 98%, or more identical to a portion of a sequence of any one of SEQ ID Nos: 232. A length of the fragment or truncation that is at least about 10%, 15%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or more of a length of any one of SEQ ID Nos: 232. A length of the fragment or truncation may be less than or equal to about 90%, 80%, 70%, 60%, 50%, 40%, 30%, 20%, 15%, 10%, or less than a length of any one of SEQ ID Nos: 232.

TABLE 14

Tuna Proteins		
SEQ ID NO	Description	Amino Acid Sequence
232	Tuna XP_042264497.1 (actin alpha 1, skeletal muscle b)	MCDDDETALVCDNGSGLVKAGFAGDDA PRAVFPSIVGRPRHQGVVMGMGQKDSYV GDEAQSKRGILTLKYPIEHGIITNWDDM EKIWHHTFYNELRVAPEEHPTLLTEAPL NPKANREKMTQIMFETFNVPAMYVAIQ VLSLYASGRTTGIVLDAGDGVTHNVPVY EGYALPHAIMRLDLAGRDLTDYLMKILT ERGSYFVTTAEREIVRDIKEKLCYVALD FENEMATAASSSLEKSYELPDGQVITI GNERFRCPETLFPQPSFIGMESAGIHETA YNSIMKCDIDIRKDLANNVLSGGTMY PGIADRMQKEITALAPSTMKIKIIAPPE RKYSVWIGGSILASLSTFQQMWISKQEY DEAGPSIVHRKCF

[0063] An additional gene that is not a gene of interest can be a gene that modulates protein expression. A gene that modulates protein expression can enhance protein expression, decrease protein expression, stop protein expression, or maintain protein expression.

[0064] A gene that modulates protein expression can increase protein expression by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least

about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A gene that modulates protein expression can increase protein expression by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0065] A gene that modulates protein expression can increase protein expression by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A gene that modulates protein expression can increase protein expression by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0066] A gene that modulates protein expression can decrease protein expression by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A gene that

or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0073] An expression cassette can comprise at least one regulatory sequence. A regulatory sequence can comprise a promoter, a repressor, an enhancer, a terminator, a 5' UTR, a 3' UTR or a combination thereof. A regulatory sequence can be part of a coding sequence. Alternatively, a regulatory sequence can be a non-coding sequence. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more regulatory sequences.

[0074] A promoter can be an endogenous *P. pastoris* promoter or variant thereof. Alternatively, a promoter can be a yeast promoter from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a promoter can be a synthetic or novel promoter. A promoter can be an AOX1 promoter, an ADH3 promoter, a TEF1 promoter, a PGK1 promoter, a TPI1 promoter, a PHO89 promoter, a THI11 promoter, an AOD promoter, or a GAP promoter.

[0075] An enhancer can be an endogenous *P. pastoris* enhancer or variant thereof. Alternatively, an enhancer can be a yeast enhancer from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, an enhancer can be a synthetic or novel enhancer. An enhancer can be an AOX1 enhancer or a GAP enhancer.

[0076] A repressor can be an endogenous *P. pastoris* repressor or variant thereof. Alternatively, a repressor can be a yeast repressor from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a repressor can be a synthetic or novel repressor. A repressor can be an AOX1 repressor or a GAP repressor.

[0077] A terminator can be an endogenous *P. pastoris* terminator or variant thereof. Alternatively, a terminator can be a yeast terminator from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a terminator can be a synthetic or novel terminator. A terminator can be an AOX1 terminator, a RPL15A terminator, a RPL41B terminator, a DAS1 terminator, a VMA2 terminator, a SRP54 terminator, a FUR4 terminator, a PLB3 terminator, a GIC1 terminator, a FBP1 terminator, a CYC1 terminator, a IDP1 terminator, a PRC1 terminator, a RPL3 terminator, a PGK1 terminator, a YPT31 terminator, an ASP3-1 terminator, a TPO4 terminator, a RPS28A terminator, a VMA16 terminator, a LSC2 terminator, a DIT1 terminator, a MIG2 terminator, an ADH2 terminator, an ADH3 terminator, an ICY2 terminator, a PHO89 terminator, a THI11 terminator, an AOD terminator, or a GAP terminator.

[0078] A UTR can be a 5' UTR or a 3' UTR. A UTR can be an endogenous *P. pastoris* UTR or variant thereof. Alternatively, a UTR can be a yeast UTR from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a UTR can be a synthetic or novel UTR. A UTR can be an

AOX1 UTR, a RPL15A UTR, a RPL41B UTR, a DAS1 UTR, a VMA2 UTR, a SRP54 UTR, a FUR4 UTR, a PLB3 UTR, a GIC1 UTR, a FBP1 UTR, a CYC1 UTR, an IDP1 UTR, a PRC1 UTR, a RPL3 UTR, a PGK1 UTR, a YPT31 UTR, an ASP3-1 UTR, a TPO4 UTR, a RPS28A UTR, a VMA16 UTR, a LSC2 UTR, a DIT1 UTR, a MIG2 UTR, an ADH2 UTR, an ADH3 UTR, a TEF1 UTR, a PGK1 UTR, a TPI1 UTR, a PHO89 UTR, a THI11 UTR, an AOD UTR, or a GAP UTR.

[0079] An expression cassette can comprise a signal sequence. A signal sequence (e.g., a secretion sequence or a nuclear localization sequence) can aid with and/or induce protein secretion. An expression cassette can comprise at least 1, at least 2, at least 3, at least 4, at least 5, at least 6, at least 7, at least 8, at least 9, at least 10 or more signal sequences.

[0080] A signal sequence can be an endogenous *P. pastoris* signal sequence or variant thereof. Alternatively, a signal sequence can be a yeast signal sequence from a different species (e.g., *S. cerevisiae*) or variant thereof. Alternatively, a signal sequence can be a synthetic or novel promoter. A signal sequence can be an amylase signal sequence, an inulinase signal sequence, an invertase signal sequence, a lysozyme signal sequence, an albumin signal sequence, a glucoamylase signal sequence, or a killer protein signal sequence, a *S. cerevisiae* Ost1 signal sequence, a *S. cerevisiae* alpha mating factor, a partial *S. cerevisiae* alpha mating factor, or a *Pichia* Exg1p signal sequence.

[0081] A regulatory sequence can modulate protein yield. A regulatory sequence can enhance protein yield, decrease protein yield, or maintain protein yield.

[0082] A regulatory sequence that modulates protein yield can increase protein yield by at least about 5%, at least about 10%, at least about 20%, at least about 30%, at least about 40%, at least about 50%, at least about 60%, at least about 70%, at least about 80%, at least about 90%, at least about 100%, at least about 150%, at least about 200%, at least about 300%, at least about 400%, at least about 500%, at least about 600%, at least about 700%, at least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A regulatory sequence that modulates protein yield can increase protein yield by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0083] A regulatory sequence that modulates protein yield can increase protein yield by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at

least about 800%, at least about 900% at least about 1,000%, at least about 5,000%, at least about 10,000% or more. A regulatory sequence that modulates protein secretion can decrease protein secretion by at most about 10,000%, at most about 5,000%, at most about 1,000%, at most about 900%, at most about 800%, at most about 700%, at most about 600%, at most about 500%, at most about 400%, at most about 300%, at most about 200%, at most about 150%, at most about 100%, at most about 90%, at most about 80%, at most about 70%, at most about 60%, at most about 50%, at most about 40%, at most about 30%, at most about 20%, at most about 10%, at most about 5%, or less.

[0090] A regulatory sequence that modulates protein secretion can decrease protein secretion by at least or up to about 1-fold, at least or up to about 2-fold, at least or up to about 3-fold, at least or up to about 4-fold, at least or up to about 5-fold, at least or up to about 6-fold, at least or up to about 7-fold, at least or up to about 8-fold, at least or up to about 9-fold, at least or up to about 10-fold, at least or up to about 20-fold, at least or up to about 30-fold, at least or up to about 40-fold, at least or up to about 50-fold, at least or up to about 60-fold, at least or up to about 70-fold, at least or up to about 80-fold, at least or up to about 90-fold, at least or up to about 100-fold, at least or up to about 500-fold, at least or up to about 1,000-fold, at least or up to about 5,000-fold, or at least or up to about 10,000-fold. A regulatory sequence that modulates protein secretion can decrease protein secretion by at most or less than about 10,000-fold, at most or less than about 5,000-fold, at most or less than about 1,000-fold, at most or less than about 500-fold, at most or less than about 100-fold, at most or less than about 90-fold, at most or less than about 80-fold, at most or less than about 70-fold, at most or less than about 60-fold, at most or less than about 50-fold, at most or less than about 40-fold, at most or less than about 30-fold, at most or less than about 20-fold, at most or less than about 10-fold, at most or less than about 9-fold, at most or less than about 8-fold, at most or less than about 7-fold, at most or less than about 6-fold, at most or less than about 5-fold, at most or less than about 4-fold, at most or less than about 3-fold, at most or less than about 2-fold, or at most or less than about 1-fold.

[0091] A food product or substance can be comprised of the recombinant proteins described herein. The food product or substance may comprise recombinant proteins described herein provided in an isolated or purified form. Alternatively, or in addition to, the recombinant proteins described herein may be provided to the food product or substance as whole cells comprising the recombinant proteins. A food product or substance can be a food for animals. A food product or substance can be food for a pet (e.g., companion animal), a farm or agricultural animal, a wild animal, or a human. A food product or substance can be food for an obligate carnivore (e.g., a feline), which requires animal protein as a part of the diet.

[0092] A food product can be a wet puree. Alternatively, a food product can be a dry pellet. A food product can replace an animal or human diet. Alternatively, a food product can supplement an animal or human diet.

[0093] A food product can contain additional ingredients such as, but not limited to, animal-based meat, amino acid supplements (e.g., taurine), vitamins (e.g., vitamin E), fats, nutritional yeast, *spirulina*, agar, xanthan gum, kelp powder, squash powder, sweet potato powder, *papaya* powder, flax-

seed oil, coconut oil, probiotics, cranberry powder, wheat grass powder, turmeric, coconut flour, chitosan, pumpkin puree, inulin powder, guar gum, green tea extract, koji, stabilizers, vegetables, fruits, microbes, fillers, or flavoring compounds.

EXAMPLES

Example 1: Recombinant Protein Expression

[0094] In this experiment, a *Pichia pastoris* cell was transfected with a DNA sequence, resulting in recombinant protein expression.

[0095] Host cells were chosen based on predetermined characteristics such as, but not limited to, doubling time, ability to properly fold and make appropriate post-translational modifications on recombinant proteins, protein secretion, ability to grow on cost-effective and/or easily available media/feedstock, or genetic tractability. Host cells that do not efficiently secrete proteins may be lysed to release the target recombinant proteins.

[0096] *Pichia* cells were grown overnight in YPD media in baffled flasks at 30° C. to an optical density at 600 nm wavelength (OD 600) of about 1 to 2.6. Cells were collected by centrifugation and resuspended at room temperature for 30 min in 100 mM lithium acetate (LiOAc), 10 mM dithiothreitol (DTT), 0.6M sorbitol, and 10 mM Tris-HCl pH 7.5, at 1e8 cells/ml. Cells were pelleted and washed four times in 1 mL 1M sorbitol. Cells were resuspended in 1M sorbitol at a final concentration of 1e9 cells/ml. 40 µL of cell suspension was mixed with 1 µL (100-1 ug) of DNA, transferred to a 0.2 cm gap cuvette, and incubated on ice for 5 minutes.

[0097] The expression cassette used comprised a promoter, a 5' untranslated region, a signal sequence for protein secretion, an open reading frame (ORF) of an animal muscle protein, a 3' UTR, and a transcriptional termination and 3' end processing sequence. The construct used a *P. pastoris* AOX1 promoter, a *P. pastoris* 5' UTR, a *P. pastoris* 3' UTR, a *S. cerevisiae* alpha mating factor, a rabbit serum albumin ORF, and a *P. pastoris* terminator.

[0098] The expression cassette was transfected into the *P. pastoris* host cell using electroporation at 1.5 kV, 200 ohms, and 25 uF, where the expression cassette was integrated into the genome of *P. pastoris*. Electroporated cells were diluted in 1 mL 1M sorbitol. Cells were transferred to 15 mL round-bottom tubes with 1 mL YPD media and incubated while shaking at 30° C. for 1 hour.

[0099] Cells were plated on YPDS plates and incubated at 30° C. Once the cells' endogenous protein expression machinery expressed the protein of interest, the cells secreted the protein of interest into the media, which was harvested and purified for further use.

Example 2: Analysis of Secreted Proteins

[0100] In this experiment, genes of interest were cloned into plasmids containing the GAP1 promoter, *Saccharomyces cerevisiae* alpha-Mating factor signal sequence, and a zeocin resistance gene. Plasmids were linearized and transformed into *Pichia pastoris* cells using electroporation. Transformed cells were spread onto YPDS plates containing 100 micrograms per milliliter (ug/mL) zeocin and incubated at 30° C. for 3-4 days. Colonies were restreaked to single colony twice. To produce secreted protein, cells were resus-

pended in 4 milliliters (mL) of buffered complex glycerol medium (BMGY) (rich glycerol) and cultured shaking at 250 revolutions per minute (rpm) and 30° C. for 4 days. Every 24 hours, 200 microliters (uL) of culture was removed and centrifuged to separate cells from supernatant. A 10 uL aliquot of supernatant was used to analyze secreted proteins on a sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) gel.

[0101] FIG. 1 shows a Coomassie stained SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit albumin or wild-type cells. Wildtype cells lack the construct for secreted recombinant rabbit albumin. Lane 1 **101** shows a band at 66 kilo Daltons (kDa), which corresponds to the recombinant rabbit albumin. Lane 2 **102**, which correspond to the wild-type media, lacks a band at 66 kDa. Lane 3 **103** shows a bovine serum albumin marker with a band at 66 kDa.

[0102] FIG. 2 shows a Coomassie stained sodium SDS-PAGE gel of an example media from *Pichia pastoris* cells including a construct for secreted recombinant rabbit Tropomyosin alpha-1 chain or wildtype cells. The wildtype cells lack the construct. Lane 2 **201** shows the wildtype expressed protein from wildtype cells. Lanes 3-5 **202** show secreted protein expression for Tropomyosin alpha-1 chain. Lanes 3-5 show a band **203** at 33 kDa corresponding to the Tropomyosin alpha-1 chain.

[0103] For proteins produced in a bioreactor, cells were resuspended in YPG medium and grown in shake flasks at 30° C. for 16 hours. The cells were transferred to seed fermenters and grown at 30° C. for another 16 hours. The cultures were transferred to a 100 liter (L) fermenter and batch fed a glycerol-based media for 92 hours at 30° C. The culture was then centrifuged to separate solid and liquid phases, filtered, and spray dried to produce the final protein powder.

[0104] FIG. 3 shows a Coomassie stained SDS-PAGE gel of an example recombinant rabbit albumin. Lane 1 **301** shows an example recombinant rabbit albumin (rALBU-RABIT) grown from a bioreactor. The rabbit albumin peptides were purified and spray dried. The protein powder was resuspended and run on an SDS-PAGE gel. Lane 2 **302** shows a bovine serum albumin marker.

Example 3: Cat Food with Rabbit Proteins

[0105] In this example, cat food is made using rabbit proteins. Rabbits are naturally hunted by felines in the wild, and rabbit proteins are hypoallergenic for cats. A recombinant rabbit protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0106] Twenty-five rabbit proteins are produced using the recombinant protein expression method of Example 1. The twenty-five rabbit proteins tested are found in Table 2.

[0107] The rabbit proteins are harvested from the recombinant protein expression method and produced into a wet pureed cat treat. Additionally, the rabbit proteins are harvested from the recombinant protein expression method and produced into a dry extruded cat food kibble.

Example 4: Pet Food with Cow Proteins

[0108] In this example, pet food is made using cow proteins. A recombinant cow protein is produced and used as

an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0109] The cow proteins are produced using the recombinant protein expression method of Example 1. The cow proteins tested are found in Table 3.

Example 5: Pet Food with Sheep Proteins

[0110] In this example, pet food is made using sheep proteins. A recombinant sheep protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0111] The sheep proteins are produced using the recombinant protein expression method of Example 1. The sheep proteins tested are found in Table 4.

Example 6: Pet Food with Pig Proteins

[0112] In this example, pet food is made using pig proteins. A recombinant pig protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0113] The pig proteins are produced using the recombinant protein expression method of Example 1. The pig proteins tested are found in Table 5.

Example 7: Pet Food with Bison Proteins

[0114] In this example, pet food is made using bison proteins. A recombinant bison protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0115] The bison proteins are produced using the recombinant protein expression method of Example 1. The bison proteins tested are found in Table 6.

Example 8: Pet Food with Deer Proteins

[0116] In this example, pet food is made using deer proteins. A recombinant deer protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0117] The deer proteins are produced using the recombinant protein expression method of Example 1. The deer proteins tested are found in Table 7.

Example 9: Pet Food with Chicken Proteins

[0118] In this example, pet food is made using chicken proteins. A recombinant chicken protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0119] The chicken proteins are produced using the recombinant protein expression method of Example 1. The chicken proteins tested is found in Table 8.

Example 10: Pet Food with Turkey Proteins

[0120] In this example, cat food is made using turkey proteins. A recombinant turkey protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0121] The turkey proteins are produced using the recombinant protein expression method of Example 1. The turkey proteins tested are found in Table 9.

Example 11: Cat Food with Duck Proteins

[0122] In this example, cat food is made using duck proteins. Ducks are naturally hunted by felines in the wild, and duck proteins are hypoallergenic for cats. A recombinant duck protein is produced and used as an ingredient in both wet and dry cat food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0123] The duck proteins are produced using the recombinant protein expression method of Example 1. The duck proteins tested are found in Table 10 and Table 11.

Example 12: Pet Food with Salmon Proteins

[0124] In this example, pet food is made using salmon proteins. A recombinant salmon protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0125] The salmon proteins are produced using the recombinant protein expression method of Example 1. The salmon proteins tested are found in Table 12.

Example 13: Pet Food with Cod Proteins

[0126] In this example, pet food is made using cod proteins. A recombinant cod protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0127] The cod proteins are produced using the recombinant protein expression method of Example 1. The cod proteins tested are found in Table 13.

Example 14: Pet Food with Tuna Proteins

[0128] In this example, pet food is made using tuna proteins. A recombinant tuna protein is produced and used as an ingredient in both wet and dry pet food, snacks, and/or treats. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution. For dry food, ingredients are mixed and extruded.

[0129] The tuna proteins are produced using the recombinant protein expression method of Example 1. The tuna protein tested is found in Table 14.

Example 15: Food Product for Human Consumption

[0130] In this example, a food product for human consumption is made using any of the proteins described in Tables 2-14. A recombinant protein from Tables 2-14 is produced and provided in a food, beverage, or supplement. In some formulations, the protein is in dried powder form. In some formulations, the protein is in a solution.

[0131] While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. It is not intended that the invention be limited by the specific examples provided within the specification. While the invention has been described with reference to the aforementioned specification, the descriptions and illustrations of the embodiments herein are not meant to be construed in a limiting sense. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. Furthermore, it shall be understood that all aspects of the invention are not limited to the specific depictions, configurations or relative proportions set forth herein which depend upon a variety of conditions and variables. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention. It is therefore contemplated that the invention shall also cover any such alternatives, modifications, variations or equivalents. It is intended that the following claims define the scope of the invention and that methods and structures within the scope of these claims and their equivalents be covered thereby.

SEQUENCE LISTING

Sequence total quantity: 232

SEQ ID NO: 1 moltype = AA length = 35213
FEATURE Location/Qualifiers
source 1..35213
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 1

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VRLQVRVTGI PTPVVKFYRD GAEIQSSLDF QISQEGDLYS LLIAEAYPED SGTYSVNATN 180
SVGRATSTAE LVVQGEVVP AKKTKTIVST AQISETRQTR IEKKIEAHFD ARSIATVEMV 240
IDGATGQLPH KTPPRIPPKP KSRSPTPPSI AAKAQLARQQ SPSPIRHSFS PVRHVRAPTP 300
SPVRSVSPAG RISTSPIRSV KSPLLIRKTQ TTTMATGPEV PPPWKQEGYV ASSTEAMRE 360
TTMTSSTQIR REERWEGRYG VQEQVTISGA AAAAASASVS SSFTAGAITT GTKEVKQETD 420
KSAAVATVVA AVDMARVREP AISAVEQTAQ RTTTTAVHIQ PAQEQRKEA EKTAVTKVVV 480
AADKAKEQEL KSRTRVMTV TQEQTHISHE QIRKETEKAF VPKVVISATK AKEQETRITG 540
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-continued

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PTHVEESHQ	QTTLEYGYKE	HISTTKVPEQ	PRRPASEPHV	VPQAVKPAVI	QAPSETHIKT	780
TDQMGMHIS	QVKKTTDIST	ERLVHVDKRP	RTASPHFTVS	KISVPKTEHG	YEASIAGSAI	840
ATLQKELSAT	SSTQKITKSV	KAPTVKPGET	RVRAEPTPSP	QFPFADMPPP	DTYKSQAGIE	900
VKKEVGVSI	GSTVREEHFE	VLRGREAKVT	ETARVPAPAE	VPVTPPTLVS	GLKNVTVIEG	960
ESVTLLECHIS	GYPSPKWTWY	REDYQIESSI	DFQITFQGGI	ARLMIREAFA	EDSGRPTCSA	1020
VNEAGTVSTS	CYLAVQVSEE	FDKETTLETEK	FATEEKRFEV	SRDVMTDTS	ITEEQAGPGE	1080
PAAPFFISKP	VVQKLVEGGS	VVFECQIGGN	PKPHVYWKKS	GVPLTTGYRY	KVSYNKQTGE	1140
CRLVISMTFA	DDAGEYTTIVI	RNKHGETSAS	ASLLEADYE	ALVKTQQEML	YQTQMSTFIQ	1200
EPKVGEIAPG	FAYSEYEKEY	EKEQALIRKK	MAKDTVMVRT	FVEDQEFHIS	SFEERLIKEI	1260
EYRIKTTLE	ELLEEDGEEK	MAVDISESEA	IESGFDIRIK	NYRILEGMGV	TFHCKMSGYP	1320
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LYVEPAAPFS	APTYMPTPEA	VSRIRSVSPR	SLSRSPIRMS	PAMSPARMSP	ARMSPARMSP	1440
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DIEVPESYSG	ELECIISPEN	IEGKWYHNDV	ELKSNKGYSI	TSRRGRQNL	VKDVTKEQDQ	2340
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ISVKGTLVSL	KIKNCAMADE	SVYGLKGLRL	GASARLHVET	VKI IKKPKDV	TALENATVTF	2820
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AFVANNAVQ	VSSATATRL	APESILHERI	QQOIEMEMKE	IASLLSAEED	FQTYSSDLRL	3480
PNANETLELL	SEPPARSTQF	DSRQEGAAPV	FIREISDVEI	SVEDVAKLSV	TVTGC PKPKI	3540
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VREPPSFVVK	VDPSYLVLP	ESARLHCKLK	GSPVIQVTFW	KNNKELSESN	TVRMSFVNSE	4680
AILDITDVKV	DDSGTYSCEA	TNDVGS DCS	TEVVIKEPPS	PIKTLEPADI	VRGANALLQC	4740
EIAGTGPFVE	NWFKDKKQIR	SSKKYRLFTQ	KTFVYLEISS	FNSADVGDYE	CVVANEVGKC	4800
GCVATHLLKE	PPTFVKVDD	FTALAGQTVT	LQAAVRGSEP	ISMVMWKQGE	VIKEDGKIKM	4860
SFSNGVAVLT	IPDVQISLGG	KYTCLAENEA	GSQTSVGELI	VKEPAKIIER	AELIQVTAGD	4920
PATLEYTVSG	TPELKPWKYK	DGRPLVASKK	YRISFKNNIA	QLKFYSLEHL	DSGQYTFEIS	4980
NEVGSSSCET	TFTVLDRDIA	PLFTKPLRNV	DSVVGACRL	CKIAGSLPM	RVSWFKDGKE	5040
LTAASDRYQIA	FVEGTASLEI	SRVDMNDAGN	FTCRATNSVG	SKDSSGALIV	QEPSPFVTKP	5100
GSRDVLPGSA	VCLKSAFQGS	APLTIKWKFG	DKELVSGGSC	YITKETSESS	LELYAVKTS	5160

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SGTYTCKVSN	VAGSVECSAD	LFVKEPATFI	EKLEPSQLLK	KDGTQLACK	VTGTPPIKIT	5220
WFANDRELRE	SSKHKMSFAE	STAVLRLTDV	AIEDSGEYMC	EAQNEAGSDH	CTGIVIVKES	5280
PYFTKEFKSI	EVLKEYDVML	LAEVAGTPPF	EITWFKDNTT	LRSGRKYKTF	LQDQLVSLQV	5340
LKFVAADAGE	YQCRVTNEVG	SSTCSARVTL	REPPSFIKKI	EATSSLRGGT	AAFQATLKGS	5400
LPITVTWLKD	NDEITEDDNI	RMTFENNVA	LYLSGIEVKH	DGKYVCQAKN	DAGIQRCAL	5460
LSVKEPATIM	BEAVSIDVTQ	GDPATLQVKF	SGTKEISAKW	FKDQBELTLG	PKYKISVTD	5520
VSILKIISTE	KDSEGYTFE	VQNDVGRSSC	KASINVLDLI	IPPSFTKKLR	KMDSIKGSFI	5580
DLECIVAGSH	PISIQWFKDD	QEISASDKHK	FSFHDNTAFL	EISQLEGTDS	GTYTCSATNK	5640
AGHSQCSGHL	TVKEPPYFVE	KPOSQDVNPG	TRVQLKALVG	GTAPMTIKWF	KDNKELHPGA	5700
ARSVWKDDTS	TILELFSAKA	ADSGTYICQL	SNDVGTTSK	ATIFVKEPPQ	FIKKPSVPLV	5760
LRNGQSTTPE	CQVTGTPEIR	VSUWLDGNEI	TDLRKYGISF	VDGLATFQIS	NARVENSGETY	5820
VCEARNDAQT	ASCSIEKLKVK	EPPIFIRELE	PVEVVKDSV	ELECEVMGTT	PFEVTLKNN	5880
KEIRSGKKYT	MSEKMSVYFL	HITKCDPSDV	GEYQCIANE	GGSCACARV	ALKEPPSFIK	5940
KIENVTTVLK	SSATFQSTVA	GSPPISTIWL	KDDQILEEND	NVHISFEDSV	ATLQVRSVDN	6000
GHSGRYTQCA	KNESGIERCY	APLLVQEPAQ	IEKAKSVDV	TEKDPVTLEC	VVAGTPELKV	6060
KWLKDGKQIV	PSRYFSMSFE	NNVASFRIQS	VMKQDSGQYT	FKVENDFGSS	SCDAYLRVLD	6120
QDIPPSFTTK	LTKMDKVLGS	SIHMECKVSG	SLPISAQWFK	DGKEISTSATK	YRLVCHENTV	6180
SLEVSNLELE	DTANYTCKVS	NVAGDNACSG	ILTVKEPPSF	LVKPERQQA	PDSTVEFKAV	6240
LKGTPPFKIK	WLKDDVSLVS	GPKCFIGLEG	STSFLNLVSV	DSSKTQGYTC	QVTNDVSGDS	6300
CTTMLLVTEP	PKFVKKLEAS	KIIKAGDSAR	LECKITGSPE	IQVWVYRNEH	ELTASDKYQM	6360
TFIDSVAVIQ	MNSLGTEDSG	DFICEAQNPA	GSTSCSTKVI	VKEPPVSSSF	PPIVETLKNT	6420
EVSLECELGG	TPPEVLEVYK	DKRQLRSSKK	YKVASKNFHA	SIHILNVEST	DIGEYHCKAQ	6480
NEVGSDACVC	AVKLKEPPKF	ISKLNSLTVV	AGEPAELQAS	IEGAQPISVQ	WLKEKEEVIR	6540
ESENIRISFV	NNVATLQFAK	VEPANAGKYI	CQVKNDDGVR	ENMATLTVLE	PAVIEEKAGS	6600
MTVTVGETCA	LECKVAGTPE	LSVEWYKDGK	LLTSSQKHKF	SFYNKISSLK	ILSVEKEDAG	6660
TYTFQVQNNV	GKSSCTAVVD	VSDRMVPPSF	TRRLKDTGGV	LGTSCILECK	VAGSSPISIA	6720
WFHEKTKIVS	GAKYQTTFS	NVCTQLQNSL	DSSDMGSYTC	VAANVAGSDE	CRALLTVQEP	6780
PSFVKEPEPL	EVLPGKNITF	TSVIRGTPPF	KVGWFRGARE	LVKGNRCNIY	FEDTVAELEL	6840
FNIDISQSGE	YTCVVSNNAG	QASCTTRLFV	KEPATFVKKL	SDHSVEPGKS	IILEGTYTGT	6900
LPISVTWKKD	GVSTPISERC	NIVTTEKTCI	LEILSSTKGD	AGHYSCIEEN	EAGRDACDAL	6960
VSTLEPPYFV	GDSVSLQCVQ	AGTPEITVSW	FKGDTKLRS	PEYRTYFTNN		7020
VATLVFNKVS	INDSGEYTCM	AENSIGTAAS	KTIIFRIQERQ	LPPSFARQLK	DIEQTVGLPV	7080
TLTCLRLNGSA	PIQVCWYRDG	VLLRDDENLQ	MSFVDNVATL	KILQTDLSHS	GQYSCASNP	7140
LGTASSTARL	TAREPKKSPF	FDIKPVSIDV	IAGESADFE	HVTGAQPMRV	TWSKDNKEIR	7200
PGGNYTITCV	GNTPHLRILK	VGKGDGSGQYT	CQATNDVGKD	MCSAQLSVKE	PPKFIKKLDA	7260
SKVAKQGESI	QLECKISGSP	EIKVWVFRND	SELHESWKYN	MSFVNSVALL	TINEASAEDT	7320
GDYICEAHNG	VGDASCSTAL	KPPVPVGA	SDVILQCEIS	GTPPFVFWV		7380
KDRKQVRSSK	KFKITSKNFD	TSLHIFNLEA	PDIGEYHCKA	TNEVGSDTCA	CTVKFKEPPR	7440
FVKKLSAST	LIGDPVLEQA	VVEGFPQISV	VWLKDKGEVI	RESENVRISE	VDNIATLQLG	7500
SPEASQSGKY	VCCIKNDAGM	RECSAVLTVL	EPATIVEKEPE	PMTVTGNPF	TLECVVAGTP	7560
ELSAKWFKDG	RELSSGSRHH	ITFVRNLASL	KIPSAEMNDK	GLYTFEVENR	VGKSSCTVSV	7620
HVSDRVVPPS	FVRRLKDTSA	TLGASVLEEC	RVSGSAPISV	GWFLDGNIEI	SSPKCQSSFA	7680
DNVCTLTLS	LEPQDGTAYT	CVAAVAGQD	ESSAVLTVQE	PPSFEQTPDS	VEVLPGMSLT	7740
FTSVIRGTPP	PKVKWFKGSR	ELVSGEACTI	SLEDVFTLE	LLEVEPGQSG	DYSCLVNTDA	7800
GSASCTTHLF	VKEPATFVKR	LADTSVETGS	PIVLEATYSG	TPPISVSWMK	NEYPLSQSPN	7860
CGITTEKES	ILELESTIE	DYAQYACLIE	NEAGQDICEA	LVSLEPPYF	IEPLEHVEAA	7920
IGEPITLQCK	VDGTPEIRIS	WYKEHTKLRS	APAYKMQFKN	NVASLVINKV	DHSDVGEYTC	7980
KAENSVGAVA	SASVVLVIER	KLPPSPARKL	KDVHETLGFP	VAFECRINGS	EPLQVSWYKD	8040
GELLKDDANL	QMSFVHHVAF	LQILQTDQSH	VQYNCASAN	PLGTASSAK	LILSEHEVPP	8100
FFDLKPVSV	LALGESGSFK	CHVTGTAPIK	ITWAKDNREI	RPGGNYKMTL	VENTATLTVL	8160
KVAKGDAGQY	TCYASNAGK	DSCSAQLGVQ	EPFRFIKKLD	QSRIVKQDEY	TRYECKIGGS	8220
PEIKVLWYKD	VEVEQESSKF	RMSFEDSVAI	LEMHSLSVED	SGDYTCERN	AAGSASSSTS	8280
LKVKEPPVFR	KKPPFVETLK	GADVHLECEL	QGTPPFQVSW	HKDKRELRS	KKYKIMSEN	8340
LTSIHILNVD	TADIGEYQCK	ATNDVGSDDC	VGSVTMKAPP	QFVKKLTDIS	TIIGKEVQLQ	8400
TTIEGAEPIS	VAWFKDKGEI	VRESNIIWIS	YSENIATLQF	SRAEPANAGK	YTCQIKNDAG	8460
MQECYATLSV	LEPAAIIVEKP	ESIKVTTGDT	CTLECTVSGT	PELSTKWFKD	GKELTSDNKY	8520
KISFFNKVSG	LKIINVVPID	SGVYSFEVQN	PVGKDSCKVS	IQVSDRIIPP	SFTRKLKETN	8580
GLSGSSVME	CKVYGSPPIS	VLWFHDGNEI	SSGRKYQTTL	TDNTCALTVN	MLEEADAGDY	8640
TCIATNVAGS	DECSAPLTVR	EPPSFVQKPD	PMDVLTGSNV	TFTSIVKGS	PFTVSWFKGS	8700
TELVPGARCN	VSLQDSVGL	ELPDVDTQS	GEYTCIVSNE	AGRASCTTRL	FVKAPAFVK	8760
RLNDYSIEKG	KPLILEGTF	GTPPISVTWK	KNGINVIASQ	RCNITTEKS	AILEILSSTV	8820
EDSGQYNCYI	ENASGKSDCS	AQILILEPPY	FVKQLEPVKV	TVGDSASLQC	QLAGTPEIGV	8880
SWYKGDTKLR	PTATCKMHFK	NNVATLVFTQ	VDSSDSGEYI	CRAENSVGEV	SSSTPLTVQE	8940
QKLPPSFSRQ	LRDQVETVLG	PVVFECAVSG	SEPIVSWSYK	DGKPLKDSPN	IQTSFLDNIA	9000
TLNIFKTDRS	LSGQYSSTAT	NPIGSASSGA	KLILTEGKNP	PFFDIPLAPM	DAVVGESADL	9060
ECHVTGTQPI	KVTWAKDNRE	IRSGGNYQIS	YLENSAHLTI	VKVDKGDSGQ	YTCYAVNEVG	9120
KDSCTAQLNI	KERLIPPSFT	KKLSETVEET	EGNSPKLEGR	VAGSQPIVTA	WYKNNVEIHP	9180
TSNCEIMFKN	NALLLQVKRA	SMADAGLYTC	KATNDAGSAL	CTSSIVIKEP	KKPPVFDQHL	9240
APVTASEGDS	VQLSCHVQGS	EPIRIQWLKA	GREVKPSDR	SFSFASGTAM	LELKETAKAD	9300
SGDYVCKASN	VAGSDTSKCK	VTIKEKPA	PAAKAAVDG	KLFPVSEFQS	IRVVEKTAT	9360
FIAKVGDDPI	PNVKWTKGKW	RQLNQGGRL	IHQKGDEAKL	EIRDTTKTDS	GLYRVAFNK	9420
HGEIESNVNL	QVDERKKQEK	IEGDLRAMLK	KTPALKKGS	EEEEIDIMEL	LKNVDPKEYE	9480
KYARMYGTID	FRGLLQAFEL	LKQSQEEETH	RLIEIELEKS	ERDEKEFEEL	VAFIQRLTQ	9540
TEPVTLIKDI	ENQTVLKDN	AIFEIDIKIN	YPEIKLSWKY	GTEKLEPSNK	YEISIDGRH	9600
TLRVKNCQPK	DQGNRYLVCG	PHIASAKLTV	IEPAWERHLQ	DVTLKEGQTC	TMTCQFSVPN	9660
VKSEWFRNGR	VLKPQGRVKT	EVEHKVHKLT	IADVRAEDQG	QYTKCHEDLE	TSABELRIAE	9720

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PIQFTKRIQN	IVVSEHQSAT	FECEVSFDDA	IVTWYKGPT	LTESQKYNFR	NDGRCHYMTI	9780
HNVTPDDEGV	YSVIARLEPR	GEARSTAEALY	LTTKEIKLEM	KPPDIPDSRV	PIPTMPIRAV	9840
PPEEIPPAVA	PSIPLLLPLP	EEKKPPAKRI	EVTKKGVKID	TKKVVTKEPE	EAPPPPVAKK	9900
PPPTPMIPA	KASEIIDVSS	KAEVKITTI	TRKKEVHKEK	EAVYEREEAV	YEKKVHIEPW	9960
EEPYEELTE	PYTEPYEOPY	YEEPDYEE	IKVEAKQVH	EEWEEDFEEG	QEYEREERY	10020
DEGEWEWEI	YHEREIIQVQ	KEVHEELHEK	KIPAKVPEKK	VPPPKVVKP	VVEKVEKTTR	10080
RMEEKQVQI	KVPEVSKIV	PQKPSRTPVQ	EEIEVKVPA	VHTKKMVIS	EKMFFASHT	10140
EEVSVSPEV	QKKTVTTEKI	HVAVSKKIEP	PKKVPEPPK	PVPEEVVPV	IPKKVEPPAA	10200
KVPEAPKKPV	PEKKPVPI	KKKEPAAPPQ	VPEAPKKPAP	EEKIPVPVTK	KKEAPPAKVP	10260
EVQKKVVTVE	KIAIITQERE	SPPPAVPEIP	KKKVPEEKRP	VPRKEEVPPP	KVPVPKKAV	10320
PEAVVPAPIP	KKAPPRAEVS	KKTVVEEKRF	AABEKL SAV	PQRVELMRHE	EEEWYSEEE	10380
ERVSVSVYRE	EERDEEEAEI	TEYEVLEEPE	EYVVEEKMHF	ISKKEVEEPA	KVPEKKIIPK	10440
PKVPAKIEEP	PPTKVPEPPK	KIVPEKKVPA	PAPKKVPPAK	APESKRPVP	EKRAPAEVVG	10500
IEEPPPTKVA	ERHMKITQEE	KVLVAVTKKE	APPRARVPEE	PKKVAPEERF	PKLKPRREEE	10560
PPAKVTEVRK	RAVKEEKVSI	EVPKREPRPT	KEVTVTEEEK	WSYTREEETV	SEHREEEYED	10620
YEDYEEYKEF	EYEPTEIEPE	QYDEYAEERV	EHYEEHEEYV	TEPKKVPVVK	PAQEPVPAKP	10680
KAPPPKVLKK	AVPEEKAPLP	IQKKLKPLPP	KAPPEPKKV	EEKIQISITK	REKQOVTEPV	10740
AKVPMKPKPV	VPEAKIPAPT	KEVAVPVRVP	GVPPKKELEE	VVVFKEEVEA	HEEYIVEEEE	10800
EVVHEEEYVH	KEEVVEHEEY	VHKEEYIHIE	EEHLHEEEET	IAEEVVPVA	PVKVPVVPKK	10860
PVPEEKPV	VPKKKEAPPA	KVPEIPKKPE	EKVVPVPIPK	EKAPPAKVPE	VPKKPVPEEK	10920
PRVVPVKVE	PPPAKVEVP	KKPVPEKKVP	APTPKKVEAP	PAKVPEVPKK	PIPEKKKPTA	10980
LLKMEAPPP	KAPKKREVPV	VPVALPREEE	EEEVPFEEVP	EEELPEEEV	PSEEEAPPEE	11040
VPPEEEVL	EEEEVLPEEE	EVLPEEEVQ	PEEEALPEIK	PKVPKPAPVP	KKTVPPEKKVP	11100
VVPVKVVEP	PPPKVPEIKK	KVPEKKVVP	KKEEAPPTKV	PEVSKKVEER	RIIPPKEEEV	11160
PPAEVVEEAE	EPTPEEIEPE	PPSIEEEIIV	EEEEEEVL	PPRAPEVVKK	AVPEAPTVP	11220
KKAEAPPAKV	PKKIPEEKVP	VPVQKKEAPP	AKVPEKKKIP	EKKVPVPKKE	AVPPAKGKAV	11280
PEEKISVAYQ	QEEVLQERIE	LELVEAKVEE	AFEEEEFHEV	QEYFEEEFH	EVEEFIRVEE	11340
RRFQEEHKVE	EVHVEIEFLE	AEEVEVYEKP	KIPPKKGPEV	SEKVIPPKKP	PTKVIPIRKEP	11400
PAKVPEVTCK	TVVEEKIRAP	EEPKVPAKPA	PEVPPKTIPE	EKVREAVPKK	PEVPPPKVPE	11460
VPKKIIQEEK	LPVVLPEDE	IYMYEASEET	VIEEEHVTLP	QKARLKVAKV	PAPPQTVVTE	11520
EKTYVTIRKT	TTREALKESE	TTREAFPELK	SYKAVPEIPE	PPSPEDLEII	EDVLPEKRRP	11580
ASKRRKTQLP	TAPEAPREMP	PEMNTFEEIS	VEPEMLPTQV	LDTYQEATVE	KKTLRISRKK	11640
PELPSDEEVP	EAPREVAVAKK	KVLPPQVPEV	VPVKVPGAPK	EVVSEKRSLE	EPKPKPAVRP	11700
VTVPEEPKEE	EPTPEEIEPE	PKKPAAPPVT	VPEAPEEVFS	EDEETLAPPQ	EPEAPPAKVP	11760
EAPKEVVPKE	KVSVVPKKP	EAPPAKVPEA	PKEAAPEKKV	PVAPKKKPEA	PPVKVPEAPK	11820
KVVPEKKLV	AAPKKPAPA	AEPVEVPKTA	VQKKIPEAI	PPKPESPPLE	VPEVPPEKEV	11880
PEKKVPAAP	PEAKVPEVP	PEAPQAAVVE	EKTPEALPKK	AEAAPVPVPQ	VQETVPEKTR	11940
PVGPPKKPEA	TTVPVPKQK	TIPEKTRPEA	PPKRPEARTV	PETVPEKTRP	MAPPKKPEAT	12000
TLVPPEVQET	VPEKTRPVGP	PKKPEATTVP	VPEVQETIPE	KTRPPKKPEA	TPVPVPEVQE	12060
TVPEKTRPVG	PKKPEATTVP	SVPEVQETIP	EKTRPAAPPK	KPEATAVPET	IPEKTRPEAP	12120
PKRPEATTVP	VPEADQAVVP	EKKVPRVPPK	KVEAPPITVP	EEPKEVIEPK	KVSLVPPKKP	12180
AAPPVTVPEA	PEEVFSDEEE	TLAPPQPEEA	PPAKVPEAPK	EVVPEKKVSV	VPPKKPEAPP	12240
AKVPEAPKEA	APKKEVPPV	KKKPEAPPVK	VPEAPKKVVP	EKKLPVAPAK	KPEAPAAEV	12300
EVPKAAVPQK	KIPEAIPPK	ESPPPEVYEE	PEEEIVPEEP	PEEAEEVPV	APPPKVTEPP	12360
KKPVPEKKAP	PAVKKPEPP	PAKVPEVPKE	APPEKKVPPK	KPEAPPKVP	EVPEVVPTEK	12420
KVAVPKKPEV	PEPAKVEVPK	KPVIEEKPAI	PVVEKVASPP	AEVYEEPEEV	TAEEEEPAPA	12480
VEEEYEAPP	PPAPVPEEPK	KVVPKEKFPV	IKKPEAPPPK	VPEVPKAVP	VKKVPVVKKP	12540
EPPEAEVPEV	PKKLVVPKKE	PVPVTKKTEV	LPEKVPEAPK	KITPEKKESV	VPPEEPPEAP	12600
ASVEETPEET	IYEEKATITI	GRKETPPVEE	REIEKFIQPE	EPELEPEPEE	IPVQEPPEEK	12660
KVIEKPKLKP	RPPARPPSP	KEDVKEKMFQ	LKAVSKKKVP	EKPEVVEKVE	PAPLKVPAT	12720
KKVRKLLEP	KPQKPEEVL	KSVLRKKPEE	EEKVPEPKV	EKAKKPEEPQ	PPPKAVEVEA	12780
PPPTPKERK	PEPAKVEVP	KPAIPLGPE	PKPKPEPEVK	TMKAPPIEPA	PTPIAAPVTA	12840
PVVGKAEAK	PKDEAAKPKG	PIKGVAKKTP	SPIEAERKKL	RPGSGGKPP	DEAPPTYQLK	12900
AVPLKFVKEI	KDIVLTAES	VGSSAIFEC	VSPSTAITTW	MKDGSNIRE	PKHRFIADGK	12960
DRKLHIIDVQ	LSDAGEYTCV	LRLGNKEKTS	TAKLIVEELP	VRFKVTLLEE	VTVVKGQPLY	13020
LSCENLKERD	VWRKDGKIV	VEKPGRIVPG	VIGLMRALTI	NDADDTAGT	YTVTVENANN	13080
LECSCVVKV	EIIREWLVPK	IRDQHVKPKG	TAFTVCTDIK	DTPNIKWFKG	YDEIPLPEND	13140
KTEILKEGNH	LFLKVNAMP	EDIDEYAVEI	EGKRYPAKLT	LGEREVELLK	PIEDVTIYEK	13200
ESASFDAEIS	EEDIPGEWKL	KGELLRPSPT	CEIKAEGGKR	FLTLHKVKLD	QAGEVLYQAC	13260
NAITTAITLV	KEIELDFAVP	LKDVTVPKEK	QARFECVLTR	EANVIWSKGP	DIKASDKFDP	13320
IADGKKHIL	VINDSQFDE	GVYTAEEVKG	KTSAQLFVGT	IRLKFISPLE	DQTVKEGQTA	13380
TFVCELSHEK	MHVWFKNVD	KLHTTRTVLM	SSEGKTYKLE	IRETTLDDIS	QIKAQVKNL	13440
STANLKVLEA	DPYPTVKLHD	KTGVEKDEII	LKCEVSKDVP	VKWFKDGEEI	VSPKHSVKT	13500
DGLRLIKIK	KAEKLDKGEY	VDCGTDTTK	ANVTVEARLI	KVEKPLYGVE	VFVGETARFE	13560
IELSEPDVHG	QWKLKGEPLT	ASPDCEIIE	GKKHVLVLYN	CQLDMTGEIS	FQAANAKSAA	13620
NLKVKEPLLI	FITPLSDVKV	FEKDEAKFEC	EVSREPKTFR	WLKGTQIEITG	DDRFELIKDG	13680
TRHSLVIKSA	AFDEAKYMF	EADDKRTSGK	LIIEGIRLKP	LTPDKDVTA	ERENAVFTVE	13740
LSHDNIPVSW	FKNDQRHLTS	KRVSMDDEGK	THSITFKDLS	IDDTSQIRVE	AMGISSEAKL	13800
TVLEGDPYFT	GKLQDYTGVE	KDEVILQCEI	SKADAPVKWF	KDGKEIKPSK	NVVIKADGKK	13860
RMLILKKALK	SDIGQYTCDC	GTQTSGLKD	IEDREIKLVR	PLYSVEVMT	ETARFETEIS	13920
EDDIHANWKL	KGALLQHTPE	CEIKEEGKIH	VLILHNCRLD	QTGGVDFOAA	NVKSSAHLRV	13980
KPRVIGLLRP	LKDVTVTAGE	TATFDCELSY	EDIPVEWYLK	GKKLEPNDKV	VTRSEGRVHT	14040
LTLRDVKLED	AGEVQLTAKD	FKTQANLQVK	EPPVEFTKPL	EDQTVVEEAT	AVLECEVSRE	14100
NAKVWKFKNG	TEILKSKKYE	IVADGRVRKL	IIHGCTPEDI	KTYTCDAKDF	KTSCNLNVVP	14160
PHVEFLRPLT	DLQVKEKETA	RFCESISKEN	EKVQWFKDGA	EIKKGKKYDI	ISKGAVRILV	14220
INKCLLNDEA	BYSCEVRTAR	TSGMLTVLEE	EAVFTKNLAN	LEVSEGDITK	LVCEVSKPGA	14280

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EVIWYKGDEE	IETGRFEIL	TDGRKRILII	QNAQLEDAGS	YNCRLPSSRT	DSKVKVHELA	14340
AEFISKPQNL	EILEGEKAEP	VCTISKESE	VQWKRDDQTL	ESGDKYDIIA	DGKKRVLVVK	14400
DATLQDMGT	VVMVGAAARA	AHLTVIEKLR	IIVPLKDTKV	KEQQEVVENC	EVNTEGAKAK	14460
WFRNEEAFD	SSKYIILQKD	LVYTLRIRDA	RLDDQANFNV	SLTNHRGENV	KSAANLIVEE	14520
EDLRIVEPLK	DIETMEKKS	TFWCKVNRNL	VTWKTKNGE	EVAFDNRISY	RIDKYKHSLL	14580
IKDCGFPDGE	EYVVTAGQDK	SVAELLIIIEA	PTEFVEHLED	QTVTEFDDAV	FSCQLSREKA	14640
NVKWYRNGRE	IKCKGKYKFE	KDGSIHRLII	KDCRLEDECE	YACGVEDRKS	RARLFVEEIP	14700
VEIIRPPQDI	LEAPGADVIF	LABELNKDKVE	VQWLRNNMIV	VQGDKHQMMS	EGKIHRLQIC	14760
DIKPRDQGEY	RFIAKDKEAR	AKLELAAAPK	IKTADQDLVV	DAGQPLTMVV	PYDAYPKAEA	14820
EWFKENEPLS	TKTVDTTAEQ	TSFRISEAKK	DDKGRYKIVL	QNKHGKAEFG	INLQVIDVPG	14880
PVRNLEVTTT	PDGEVSLAWE	EPLTDGGSKI	IGYVVERRDI	KRKTWVLVTD	RADSCEFTVT	14940
GLQKGGVEYL	FRVSARNRVG	TGEPVETDSP	VEARSKYDVP	GPPLNVTITD	VNRFGVSLTW	15000
EPPEYDGGAE	ITNYVIELRD	KTSIRWDTAM	TVRAEDLSAT	VTDVVEGQEQ	SFRVRAQNRI	15060
GVGKPSAATP	FVKVADPIER	PSPPVNLNAS	EQTSQSSVQLT	WEPPLKDGGS	PILGYIIEER	15120
EEGKDNWIRC	NMKPVPELTY	KVTGLQKGNK	YLRYVSAENA	AGVSDPSEIL	GPLTADDAFV	15180
EPTMDLSAFK	DGLEVIVFNP	IKLIVPSTGY	PRPKATWTFG	DQVLEEGDRV	KMKTISAYAE	15240
LVISPSERTD	KGIYTLTLEN	PVKISISGEIN	VNVIAPPSAP	KELKFSBITK	DSVHLTWEP	15300
DDGGGSPLTG	YVVEKRDMSR	KTWTKVMDVF	TDLEFTVDDL	VQGEYELFKV	CARNKCGPGE	15360
PAYTDEPVNM	SAPATVPDDP	ENVKWRDRTA	NSIFLTWDDP	KNDGGSRIKQ	YIVEKCPGRS	15420
DKWVACGEPV	PDTKMEVTGL	EEGKWYAYRV	KALNRQGASK	PSKPTTEEIQ	VDTQEAPEIF	15480
LDVKLLAGIT	VKAGTKIELP	ATVTGKPEPK	ITWTKADTLL	KPDQRITIE	VPKSTVTIT	15540
DSKRSDTGT	IEAVNVENVL	ATAVVEVNL	DKPGPPAADF	ITDVTNESCL	LTWNPPRDDG	15600
GSKITNYVVE	RKATSDSVWH	KLSSTVKDTN	FKATKLTPNK	EYIFRVAEEN	MYGVGEPVQA	15660
APIIAKYQFD	PRGPPTRLPE	SDITKDAVTL	TWCEPDDGG	SPITGYWVER	LDPDTDKWVR	15720
CNKMVPKDDT	YRVKGLTNKK	KYRFRVLAEN	LAPGPKPSRS	TEPILIKDPI	DPWPWPCKPT	15780
VKDIGKTSLV	LNWTKPEHDG	GAKIESYVIE	MLKTGTDEWV	RVAEGVPTTE	HLLTGLMEGQ	15840
EYSFRVAVN	KAGESESEPE	SDPVLCREKL	YPPSPRRWLE	VINITKNTAD	LKWTVPEKDG	15900
GSPITNYIIE	KRDVRRKGWQ	TVDDTVKDTK	CTVTLPLTEG	LYVFRVAEEN	AIGQSDYTEI	15960
GDSVLAKDTF	TTPGPPYALT	VVDVTKRHVD	LKWEPPKNDG	GRPIQRYIIE	KKEKLGTRWV	16020
KAGKTSKDPD	NFRVTDVIEG	TEVQFQVRAE	NEAGVGHSEP	PTEILSIEDP	TSPSPPLDL	16080
HVTDAGRKH	AIAMKPKYKE	GGSPIIGYHV	EMCPVGTEKW	MRVNSRPIKD	LKFKVEEGIV	16140
PDKEYVLRVR	AVNAVGVSEP	SEISENVVAK	DPDCKPTIDL	ETHDIVVIEG	EKLNIPIVPPR	16200
AVPVPTVSWH	KDGKEVKASD	RLTMKNDHIS	AHLEVPKSVH	ADAGVYTITL	ENKLGASAT	16260
INVKVIPLPG	TKSDIKASDI	TKSSCKLTWE	PPEFDGGSPI	LHYVLERREA	GRRTYIPVMS	16320
GENKLSWTVK	DLIPNGEYFF	RVKAVNKIGG	GEYIELKNPV	IAQDPKQPPD	PPVDVEVHNP	16380
TAKAMITWVK	PPLVDGGSKI	MGYIIEIKIT	GEDRWKRCNE	HLVPVLTYTA	KGLEEGKEYQ	16440
FRVRAENAAG	KLEPVSRAPIV	TKAVDPIDAP	KVILRTSLEV	KRGDEIALDA	TISGSPYPTI	16500
TIWKDENVIV	PEEIKKRAAP	PVRRKKGEAE	EEEPFSLPLT	ERLSINNSKQ	GESQLRIRDS	16560
LRPDHGQYMI	KVENDHGVAK	APCSVSVLDT	PGPPINFVFE	DIRKDSVLCK	WEPPLDDGGS	16620
EIINTYLEKK	KDTPKPSDWI	VITSTLRNCK	YSVTKLIEGK	EYLFVRVAEN	RFGPGPPCVS	16680
KPLLAKPPE	PPDAPDKPIV	EDVTSNSMLV	KWNEPKDNGS	PILGYWLEKR	EVNSTHWSRV	16740
NKTLSSSLKT	KVDGLLEGLT	YVPRVCAENA	AGPGKFSPPS	DPKTARDPIS	PPGPPVPRVA	16800
DTSTTIELE	WEPAPFNGGG	EIMGYFVDKQ	LVGTNEWSRC	TEKMIKVRQF	TVKEIREGAD	16860
YKLRVSAVNA	AGEGPPGETE	PVTVAEPQEP	PTVELDVSVK	GGIQUIMAGKT	LRIPAEVVTGR	16920
PVPTKVTWIE	EGELDKERVI	IENVGTSEKL	IKKNALRKDH	GRYVITATNS	CGSKFAAVRV	16980
EVFDVPGPVL	DLKPVVTNRK	MCLLNWSDPA	DDGSDITGFI	IERKDKAMH	TWRQPIETER	17040
SKCDITGLIE	GQEQYKFRVIA	KNKFGCGPPV	EIGPILAVDP	LGPPTSPEPL	TYTERTKSTI	17100
TLDWKEPRSD	GGSPIQGYII	EKRHRDKPDF	ERVNKRLCPT	TSPLVENLDE	HQMYEPRVKA	17160
VNDVGESEPS	PLPNVVIQDD	EVPTTIKRLR	AVRGDTIKVK	AGEPVNIPAD	VTGLPMPKIE	17220
WSKNEKVIDK	PTDTLNTIKE	EVSRSSEAKTE	LSIPKAARED	KGTYTITASN	RLGSVFRNVH	17280
VEVYDRPSGP	RNLAVTDIKA	ESCYLTWDAP	LDNGGSEITH	YIIDKRDAER	KKSEWEVNT	17340
TAVERRYGIW	KLIPNGQYEF	VRVAVNKYGI	SDECKSDKVV	IQDPYRLPGP	PGKPKVLERT	17400
KGSMLVSWTP	PLDNGGSPIT	GYWLEKREEG	GTYSWSRVSA	PITKVGLKGV	EFNVPRILIEG	17460
VKYQFRAMAI	NAAGIGPPSE	PSDPAVAGDP	IYPPGPPSPC	EVKDKTKSSI	SLAWKPPAKD	17520
GGSPIKGYIV	EMQEEGTTDW	KPVNEPKDKL	TACECVVNL	KELRKYRFRV	KAVNEAGESE	17580
PSDTTGEIPA	TDIQEVPEVF	IDIGAQDCLV	CKAGSQVKIP	AVIKGRPTPK	SSWEFDGKAK	17640
KAMKDGVDHI	PEDAQLETAE	NSSVIIIEPC	TRAHSGKYSI	TAKNKAGQKT	ANCRVKVMDA	17700
PGPPKDLKVS	DITRGSRLS	WKMPDDDDGD	RIKGYVIEKK	TIDGKAWTKV	NPNCGSTTFV	17760
VPDLISEQQY	FFRVRAENRF	GIGPPAETIQ	RTTARDPIYP	PDPPIKLKIG	LITKNTVHLS	17820
WKPPKNDGGS	PVTHYIVECL	AWDPTGKKKE	AWRQCNRRDV	EELEFTVEDL	IEGGEYEFVRV	17880
KAVNEAGVSK	PSATVGVIV	KDQTCPPAIE	LKEFMEVEEG	TDVNIIVAKIK	GVFPPTLTWF	17940
KAPPKKPKDSK	EPVVYDTHVN	KQVDDTCTCL	VIPQSRSDT	GLYSITAVNN	LGTASKEMRL	18000
NVLGRPGPPV	GPIKFESISA	DQMTLSWLPP	KDDGGSKITN	YVIEKREANR	KTWVRVSSEP	18060
KECMYTIPLK	LEGHEYVFR	PLDSEPETAR	NLFSVPGAPD	KPTVSSVTRN		18120
SMTVNWEPEE	YDGGSPVTGY	WLEMKDTTSK	RWKRVRNDPI	KAMTLGVSYK	VTGLIEGSDY	18180
QPRVYAINAA	GVGPASLPSD	PVTARDPVAP	PGPPFPKVD	WTKSSVDLEW	SPPLKDGGSK	18240
ITGYIVEYKE	EGKEEWEKKG	DKEVRGTLKV	VTGLKEGAFY	KFRVRAVNV	GVGEPGEVTD	18300
VIEMKDRIVS	PDLQLDASVR	DRIVVHAGGV	IRIAYVSGK	PPPTVTWSMN	ERALPQEAAR	18360
ETTAISSSMV	IKNCQRSHQG	VYSLLAQNEG	GERKKTIIIV	VLDVPGPVGI	PFLSDNLNTD	18420
SCKLTFWSP	DDGGSPTITNY	VIQKREARR	AWPTVITYVT	RQNTATVQGLI	QGKSYFFRIA	18480
AENSIGMGPF	DPITVPERPE	DLEVKEVTKN	TVSLTWNPPK	YDGGSEIINY		18540
VLESRLIGTE	KFKHVTNDNL	LSRKYTVKGL	KEGDTYEYRV	SAVNIVGQKG	PSFCTKPITC	18600
KDELAPPTLD	LDFRDKLTVR	VGESFALTGR	YSGKPKPKID	WFKDEADVLE	DDRTHIKTTP	18660
TTLALEKTKA	KRSDSGKYCV	VVENSTGSRK	GFCQVNVVDR	PGPPVGPVIF	DEVTKEYMVI	18720
SWKPLDDGG	SEIKNYIIIEK	KELGKDIWMP	VTSASAKTTC	KVPKLLLEGK	YIFRIHAENL	18780
YGISDPLVSD	SMKAKDRFRV	PDAPQPVVVT	EVTKDSALVT	WNKPNDDGGK	ITNYILEKRE	18840

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TMSKRWVRVT	KEPIHPYTKY	RVPDLLEGCO	YEFVSAENE	IGIGDPSPPS	KPVFARDPIA	18900
KSPPPINPEA	IDTTCNSVEL	TWQPPRHDGG	SKILGYIVEY	QKVGDEEWRR	ANHTPESCE	18960
TKYKVTGLRD	GQSYKFRVLA	VNEAGESDPA	HVPEPVLVKD	RLEPPELILD	ANMAREQHIR	19020
VGDTLRLSAI	IKGVFPFKVT	WKKEDREAPT	KAQIDVTPVG	SKLEIRNAAH	EDGGIYSLTV	19080
ENPAGTKTVS	VKVLVLDKPG	PPRDLEVSEI	RKDSCYLTWK	EPLDDGGSVV	TNYVVERKDV	19140
ATAQWSPLST	TSKKKSMAK	HLTEGNQYLF	RVAAENQYGR	GPFFVETPKPI	KALDPLHPPG	19200
PPKDLHHVDV	DKTEVSLVWN	KPDRDGGSPI	TGYLVEYQEE	GDKDWIKFKT	VKNLDCVVTG	19260
LQQGKTYRFR	VKAENIIGLG	LPDTTIPIEC	QEKLVPPSVE	LVDVKLIEGLV	VKAGTTVRFP	19320
AIIRGVVPVT	AKWTTDGTET	KTDHHTVET	DSFSSVLTIK	NCLRKDTGEY	QLTVSNAAGT	19380
KTVAVHLTVL	DVPGPPTGPI	NILDVTPEYM	TISWQPPKDD	GGSPVINYIV	EKQDTRKGTW	19440
GVVSAGSSKL	KLKVPHLQKG	CEYVFRVKA	NKMGVGPPLD	SIPTVAKHKF	SPPSPGKPV	19500
VTDITENAT	VSWTLPKSDG	GSPITGYVE	RREITGWVR	VNKTPIADLK	FRVTGLYEGN	19560
TYEFRVFAEN	LAGLSNPSPS	SDPIKACRPI	KPPGPPINPK	LKDKSKESAD	LVWTKPLSDG	19620
GSPILGYVVE	YQKPGTAQWD	RINKDELIRQ	CAFRVPGLE	GNEYRFRIRA	ANIVGEGERP	19680
ELAESVIAKD	ILHPPVEVEL	VTCDRVITVR	VQGTIRILAR	VKGRPEPDIT	WSKEGKVLVK	19740
DKRVDLIHDL	PRVELQYKEA	VRADHGKYII	SAKNSSGHAQ	GSAIVNVLD	PGCQNLKVS	19800
NVTKENCTIS	WENPLDNGGS	EITNFIVEYR	KPNQKGWSIV	ASDVTKRLVK	ANLLANNEY	19860
FRVCAENKVG	VGPTIETKTP	ILAINPIDRP	GEPENLHIAD	KGKTFVYLKW	RRPDYDGGSP	19920
NLSYHVHERR	KGSADWERVH	KGSIKETHYM	VDKCVENQIY	EFVRQTKNEG	GESDWVREE	19980
VVVKEDLQKP	VLDLKLSGVL	TVKAGDTIRL	EAGVRGKPPF	EVAWTKDKDA	TDLTRSPRVK	20040
IDTSAESSKF	SLTKAKRSDG	GKYVITATNP	AGSFVAYATV	NVLDPKPGFVR	NLKITDVSSD	20100
RCTIRWDPEE	DGGECEIKNY	ILEKCESKRM	VWSTYSANVL	TPSATVTRLI	EGNEYIFRVR	20160
AENKIGTGPP	TESKPVIAKT	KYDRPGRPDP	PEVTKVSKEE	MTVVWNAPEY	DGGKSITGYY	20220
LEKKEKHAVR	WVPVKNKAIP	ERRLKVQNL	PGHEYQFRVK	AENEVGIGEP	SLPSRPVAK	20280
DPIEPPGPPT	NFKVVDTTKS	SITLAWGKPV	YDGGAPIIGY	VVEMRPKIAD	ASPDEGWKRC	20340
NAAAQLIRME	FTVTSLDENQ	EYEFVRCAQN	QVGIGRPAEL	KEAIKPKEIL	EPPEIDLDS	20400
MRKLVVVRAG	CPIRLFAIVR	GRPAPKVTVR	KVGIDNVVRK	QGVLDVDTMA	FLVIPNSTRD	20460
DSGKYSLTIV	NPAGEKAVFV	NVKVLDTPGP	VADLKVS DVT	KTSCHVSWAP	PENDGGSVT	20520
HYIVEKREAE	RKTWSTVTP	VKKTSPNVN	LVPNGEYFFR	VTAVNEYGPG	VPTDIPKPV	20580
ASDPLSEDP	PRKLEVTEMT	KNSATLAWLP	PLRDGGAKID	GYIISYREED	QPADRWTEYS	20640
VVKDLSLIVT	GLKGLKKEA	RVAARNAVGV	SMPREAGVY	EAKQLLPPK	ILMPEQITIK	20700
AGKKLRVEAH	VYGKPNPICK	WKKGDDEVVT	SSHAIHKAD	GSSVLIIDKV	TRKDSGYYS	20760
TAENSSGSDT	QKIKVTVMDA	PGPPQPPFDI	SEIDADACSL	SWHIPLEDDG	SNITNYIVEK	20820
CDVSRGDWVT	ALQSVYTKTSC	RVGKLI PGQE	YIFRVRAENR	FGISEPLTSP	KMLAKFPFDV	20880
PSEPKNARVT	KVNKDCIFVA	WDRPDS DGG	PITGYLIERK	ERNSLWVKA	NDTIVRSTY	20940
PCAGLVEGLE	YSFRIYALNK	AGSSPPSKPT	EYVTARMPVD	PPGKPEVVDV	TKNSASLIWA	21000
RPKHDDGSGR	IGYFVEACKL	PGDKWVRWNT	TPHQIPQEEY	TATGLEENAQ	YQFRAIKTA	21060
VNISQPSEPS	DPVTILAENV	PPRIELSVEM	KSLLTVKAGT	NVCLDATVFG	KPMPTVSWKK	21120
DTTPIKQAE	IKMAMKRNLC	TLLELFSVNRK	DSGDTYTAE	SSSSGKSATI	KLKVLDPKGP	21180
PASVKINKMY	ADRAMLWSEP	PLEDGGSEIT	NYIIDKRETS	RPNWAQVSAT	VPITSTCTVEK	21240
LIEGHEYQFR	ICAENKYGVG	DPILTEPAIA	KNPYDPPGRC	DPPVISNITK	DHMTVSWKAP	21300
ADDGGSPI TG	YLVEKRETOA	VNMTKVNRPK	VIERTLKATG	LQEGTEYEFR	VTAINKAGPG	21360
KPSDASKAVY	AQDPLYPPGP	PAPPKVYDVT	RSSVLSWKG	PAYDGGSPII	GYLVEVKRAD	21420
SDHWVRCLNP	EKLQKTRFEV	TGLMENTEYQ	FRVYAVNKG	YSDPSDVPDK	HCPKDILIPP	21480
EGELDAELRK	TLILRAGVTM	RLVVPVKGPR	PKKITWSKPN	VNLRRERIGL	IKSTDFTDPL	21540
RCENVNKYDA	GKYLTLLENS	CGKKEYTIVV	KVLDTPGPPV	NVTVKEVSKD	SAYVTWDPPI	21600
IDGGSPIINY	VVEKRDAERK	SWSTVTTECS	KTSFRVSNLE	EGKSYFFRVF	AENEYIGIGDP	21660
GETRDAVKAS	ETPGPVVDLK	ALAITKSSCT	IGWKKPRSDG	GSRTIGYVVD	FLTEENKQWR	21720
VMKSLSLQYS	TKDLKEGKEY	TFRVSAENEN	GEGTPSEIVV	VAKDDVVAPD	LDLKDLPDL	21780
YLAKENSFR	LKIPIKGKPA	PSVSWKKGED	PLATDTRVSV	ESTAVNTTLV	VYDCQKSDAG	21840
KYITILKNVA	GTKEGTSLIK	VVGKPGIPTG	PIKPFDEVTAE	AMTLKWGPCK	DDGGSITNY	21900
VLEKRDVNN	KVWLTCAVAV	KTTFRVTRLH	EGIEYTFRVS	AENKYGVGEG	LKSEPIVAKH	21960
PFDPVDPAPP	PNIVDVRHDS	VSLTWTDPKK	TGGSPITGYH	IEFKERNSL	WKRANKTPIR	22020
MKDFKVTGLT	EGLEYEFRVM	AINLAGVGKP	SLPSEPVAL	DPIDPPGKPE	VISVTRNSVT	22080
LIWTEPKYDG	GKHLTGIVIE	KRDLPSKSWM	KANHVNVPDC	AFTVTDLVEG	GKYEFRIRAK	22140
NTAGAISAPS	ESTGTIICKD	EYEAPTIVLD	PTIKDGLTVK	AGDSIVLSAI	SILGKPLPKS	22200
SWSRAGKDIR	PSDIAQITST	PTSSMLTVKY	ATRKDAGEYT	ITATNPPGTK	EEHVKVSVDL	22260
VPGPPIE	SNVSAEKATL	TWTPPLEDDG	SPIKAYVLEK	RETSRLWTV	VSEDIQACRH	22320
VVTKLIOGNE	YLFVRSVAVN	YGKGEVQSE	PVKMVD RFGP	PGPPGKPEIS	NVTKNATATVS	22380
WKRPTDDGGS	EITGYHVERR	EKKGLRWVRA	TKTPVSDLR	KVTGLQEGNT	YEFVSAENR	22440
AGIGPPSDAS	NPVLMKDVAY	PPGPPSNAHV	TDTTKKSASL	AWGKPHYDGG	LEITGVVVEH	22500
QKVGDDAWIK	DTTGALTALRIT	QFVVPDLQTK	EKYNFRISAI	NDAGVGEPAP	IPNVEIVEKE	22560
VAPDFELDAE	LRRTLVVRAG	LSRIFVPIK	GRPAPVETWT	KDNINLKHRA	NIENTESFTL	22620
LIIEPCNRYD	TGKFVMTIPE	PAGKKS GFVN	VRVLDTPGPV	LNLRPDTITK	DSVTLHWDLP	22680
LIDGGSRI TN	YIVEKREATR	KSYSTVTTKC	HKCTYKVTGL	TEGCEYFFRV	MAENEYGVGE	22740
PTETTEPVRA	SEAPLPDPSL	NIMDITKNTV	SLAWPKPRHD	GGSKI TGYVI	EAQRKGS DQW	22800
THISTVKGLE	CVVRNLTEGE	EYTFQVMAVN	SAGRSAPRES	RPVIVKEQTM	LPELDRLGIY	22860
QKLVIARAGD	NKIVEIPLVG	RPKPTVTWKK	GDQILKQTOR	VNVENTATST	ILNINECVRS	22920
DSGPYPLTAK	NTVGEVGDVI	TIQVHDIPGP	PTGPIKFDEV	SSDFVTFWSWE	PPENDGGVPI	22980
SNYVEMRQT	DSTTWRELAT	TVIRTTYKAT	RLTTGVEYQF	RVRAQNR YGV	GPGITSASV	23040
ANYPFKVPGP	PQTPOVTAVT	KDSMTISWHE	PLSDGGSPI	GYHIERKERN	GILWQTVSKA	23100
LVPGNIFKST	GLTDGIAEYF	RVIAENMAGK	SKPSKPSEPM	FALDPIDPPG	KPVPLNITRH	23160
TVALKWAKPE	YTGGFKITSY	VVEKRDLPNG	RWLKANFSNI	LENEFTVSG	TEDAAEYFRV	23220
IAKNAAGAIS	PPSEPSDAIT	CRDDLEAPRI	MVDVRFKDTI	TLKAGEAFKL	EADVSGRPPP	23280
TMEWTKDGKE	LEGTKGLEIK	IADFSTHLIN	KDSSRTDSGA	YILTATNPGG	FAKHIFNVKV	23340
LDRPGPPEGP	LAVSDVTSEK	CVLSWLPLPD	DGGAKIDHYI	VQKRETSRLA	WTNVATEVQV	23400

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TKLKVTKLK	GNEYIFRVMA	VNKYGVGEPL	ESEPVLAUDP	YGPDPKPNP	EVTITKDSM	23460
VVCWGHPPDS	GGSEIINYIV	ERRDKAGQRW	VKCNKKALTD	LRFKVSGLTE	GHEYEFIRMA	23520
ENAAGISAPS	ATSPFYKACD	TVFKPGPPGN	PRVLDTSRSS	ISIAWNKPIY	DGGSEITGYM	23580
VEIALPEEDE	WQVVTTPAGL	KATSYTITSL	IEHQEYKIRI	YAMNSEGLGE	PALVPGTPKA	23640
EERMLPPEIE	LDADLRKVVT	IRACCTLRLF	VPIKGRPAPE	VKWAREHGES	LDKASIESTS	23700
SYTLVLVGNV	NRFDGSKYIL	TVENSSGSKS	AFVNVRLDT	PGPPQNLKIK	EVTKTSVTLT	23760
WEPLLDGGS	KIKNYIVEKR	ESTRKAYSTV	ATNCHKTSWK	VDQLQEGCSY	YFRVLAENEY	23820
GIGLPAETA	SVKASERPLP	PGKITLTDTV	RNSVLSWEK	PEHDGGSRL	GYIVEMQSKG	23880
SDRWATCATV	KVTEATITGL	IQGEEYSFRV	SAQNEKGISD	PRQLSVPVIA	KDLVIPPAPK	23940
LLFNTFTVLA	GEDLKIDVPF	IGRPPPAVTV	HKDDIPLKQT	TRVNAESTEN	NSLLTIKEAC	24000
REDVGHYTVK	LTNSAGEATE	TLNVIVLDKP	GPPTGPVKMD	EVTADSVTSL	WEPPKYDGGG	24060
SINNYIVEKR	DTSTTAQIV	SATVARTTIK	ACRLKTGCEY	QFRIAAENRY	GKSTYLNSEP	24120
VVAQYPFKVP	GPPGTPFVTL	AKSDSMEVQW	HEPVSDDGSK	VIGYHLERKE	RNSILWVKLN	24180
KTPIPQTKFK	TTGLEEGIEY	EFRVSAENIV	GIGKPSKPSE	CYAAHDPCDP	PGRPEAIIVT	24240
RNSVTLQWK	PTYDGGSKIT	GYIVEKKELP	DGRWMKASFT	NIIDTQFEVT	GLLEDHRYEF	24300
RVIARNAAGV	PSEPSGSGTA	ITARDEVEPP	RISMDPKYRD	TVVQAGESF	KIDAIYKGP	24360
IPTTQWVKGD	QELSSTARLE	IKSTDFATSL	SVKDAVRVDS	GNYLKAKNV	AGEKSVTINV	24420
KVLDRPGPPE	GPVAISGVTA	EKCTLAWKPP	LQDGGSDITN	YIVERRRST	LVWTLVDANV	24480
QTLSCVKVLK	GEGNKIDVFI	MAYNKYGVGE	PLESESLIAK	NPFVVPDAPK	APEVTAVTKD	24540
SMIVVWERPA	SDGGSEILGY	VLEKRDKEGI	RWTRCHKRLI	GELRLRVITGL	LENHNYEFRV	24600
SAENAAGLSE	PSPPSAVQKA	CDPIYKPGPP	NNPKVMDVTR	SSVFLSWTKP	IYDGGCEIQG	24660
YIVEKCDVSV	GWMCTVPTPT	GINKTNLEVE	KLLEKHEYNF	RICAINKAGV	GEHADVPGPV	24720
MVEEKLEAPD	IDLDLELRKV	INIRAGGSLR	LFVPIKGRPT	PEVKWGVKVDG	DIRDAAIIDV	24780
TSSFTSLVLD	NVNRYDSGKY	LTLENSSSGT	KSAPVTVRVL	DTPSPVNLK	VTEITKDSVS	24840
ITWEPPLLDG	GSKIKNYIVE	KREATRKSVA	AVVTNCHKNS	WKIDQLQEGC	SYFRVTAEN	24900
EYGIGLPART	ADPIKVAEVP	QPPGKITVDD	VTRNSVLSW	TKPEHDGGSK	I IQYIVEMQA	24960
KNTDKWSECA	RVKSLDAVIT	NLTQGEELYF	RVIAYNEKGR	SDPRS LAVPI	IAKDLVIEPD	25020
VRPAFSSYSV	QVGQDLKIEV	PISGRPKPSI	SWTKDGMPLK	QTTINVTDS	LDLTTLSIKE	25080
THKDDGGQYG	ITVSNVVGQK	TAAIEIITLD	KDPPKGPVK	PDEISAESIT	LSWNPPLYTG	25140
GCQITNYIVQ	KRDPTTTVTD	VVSATVARTT	LKVTKLKTGT	EYQFRIFAEN	RYQGSFALES	25200
EPVVAQYPYK	EPGPPGTFPV	TATSKESMVV	QWHEPINNGG	SPVIGYHLER	KERNISILWTK	25260
VNKTIHDTQ	FKALNLEEGI	EYEFVRVYAE	IVGVGKASKN	SECYVARDPC	DPPGTPEAII	25320
VKRNEITLQW	TKPVYDGGSM	ITGYIVEKRD	LPEGRWMKAS	FTNVIETQFT	VSGLTEDQRY	25380
EPRVIAKNAA	GAIKSPSDST	GPITAKDEVE	LPRIISMDPKF	RDTIVVNAGE	TFRLEADVHG	25440
KPLPTIEWLR	GDKEIEESAR	CEIKNTDFKA	LLIVKDAIRI	DGGQYILRAS	NVAGSKSPFV	25500
NKVLDLRPGP	PEGPVQVTGV	TAEKCTLAWS	PPLQDGGSDI	SHYVVEKRET	SRLAVTVVAS	25560
EVVTSNLKVT	KLLEGNDLKI	RIMAVNKYGV	GEPLASAPVL	MKNPFVLPGP	PKSLEVNTIA	25620
KDSMTVCWNR	PDSDDGGSEII	GYIVEKDRS	GIRWIKCNKR	RITDLRLRVT	GLTEDHEYEF	25680
RVSAENAAAG	GEPSPATVYV	KACDPVFKPG	PPTNAHVVDT	TKNSITLAWS	KPIYDGGSEI	25740
LGYYVEICKA	DEEWEQIVTP	QTGLRVTRFE	IAKLIHQEY	KIRVCALNKK	GLGEAASVPG	25800
TVKPEDKLEA	PELDLDSSEL	KGIVVRAGGS	ARIHIPFKGR	PTPEITWSKE	EGEFTDKVQI	25860
EKGINFOTLS	IDNCDRNDAG	KYTLKLENSS	GSKSAFVTVK	VLDTPGPPQN	LAVKEVRKDS	25920
VLLVWEPPII	DGGAKVKNYV	IDKRESTRKA	YANVSSCKNK	TSFRVENLTE	GAIIYFRVMA	25980
ENEFVGVPT	ETSDAVKASE	PPSPPGKVTL	TDVSQTSASL	MWEKPEHDGG	SRLGYVVMEM	26040
QPKGTEKWSV	VAESKVCNAV	VSGLSSGQEY	QFRVKAYNEK	GKSDPRVLGI	PVIAKDLTIQ	26100
PSFKLFNTY	SVQMGEDLKI	EIPVIGRPRP	KISWVKDGEP	LKQTRVNVTE	ETATSTILHI	26160
KESSKDDFGK	YSVTATNSAG	TATENLSVIV	LEKPGPPVGP	VKFDEVSADF	VVISWEPPAY	26220
TGGCQISNYI	VEKRDTTTTN	WQMVSAVTR	TTIKISKLKT	GTEYQFRIFA	ENRYGKSTPL	26280
DSKPVVVQYP	FKEPGPGPTP	FVTSISKDQM	LVQWHEPVND	GGSKVTGYHL	EQKEKNSILW	26340
VKLNKIPIQD	TKFKTTGLDE	GLEVEFRVSA	ENIVGIGKPS	KVSECYVARD	PCDPPGRPEA	26400
IVITRNSVTL	KWKPKVYDGG	SKITGYIVEK	KDLPDGRWMK	ASFNTNVETE	FTVTGLVEDQ	26460
RYEFRVIANR	AADNFSEPE	SSGAITARDE	IDAPNASLDP	KYRDVIVVHA	GETFVLEADI	26520
RGKPIPIDII	SKDGNELEET	AARMEIKSTL	QKTTLIVKDC	IRTDGGQYTL	KLSNVGGTKT	26580
IPITVKVLDL	PGPPEGPLKV	TGVTAEKCYL	AWNPLQDGG	ASISHYIEEK	RETSRLSWTQ	26640
VSNEVQALNY	KVTKLLPGNE	YIFRVMAVNV	YGIGEALESE	PVIACNPYKR	PGPSTPEAS	26700
AITKDSMVL	WTRPVDDGGA	EIEGYILEKR	DKEGIRWTKC	NKKTLTDLRF	RVTGLTEGHS	26760
YEFRVAENAA	AGVGEPSEPS	VFYRACDALY	PPGPPSNPKV	TDTSRSSVSL	AWNPKIYDGG	26820
APVRGVVIEL	KKAADEWTTT	CTPPSGLQKQ	QFTVTKLKEN	TEYNFRICAF	NTEGVGEPAT	26880
IPGSVVAQER	MEAPEIELDA	DLRKVVTLRA	SATLRLFVTI	KGRPEPEVKW	EKAEGILTER	26940
AQIEVTSSTY	MLVIDNVTRF	DSGRYNLTLE	NNSGSKTAFV	NVRVLDSPSA	PVNLTIREVK	27000
KDSVTLSEWP	PLIDGGAKIT	NYIVEKRETT	RKAYATITNN	CTKNTPFKIEN	LQEGCSYTFR	27060
VLASNEYGIG	LPAETAEPVK	VSEPPPLPPGR	VTLDVTRNT	ATIKWEKPES	DGGSKITGYV	27120
VEMQTKGSEK	WSACTQVKTL	ETTISGLTAG	EEYVFRVA	NEKGRSDPRQ	LGVPIAKDI	27180
EIKPSVELPF	NTFNVKANDQ	LKIDIPFKGR	QOATVAVKKD	GQVLRRETRV	NVASSKTVTT	27240
LSIKEASRED	VGTYELCVSN	TAGSITVPI	VIVLDRPGPP	GPIRIDEVSC	DNVSIWNPP	27300
EYDGGCQISN	YIVEKRETT	TTQVVSQAV	ARTSIKIVRL	TTGSEYQFRV	CAENRYGKSS	27360
YSESSAVVAE	YDPSPPGPPG	TPKVWHATKS	TMVVSQVQPV	NDGGSQVIGY	HLEYKERSI	27420
LWSKANKVLI	ADTQMVKYSG	DEGLMYEYRV	YAENIAGIGK	CSKACEPVPA	RDCDPPGQP	27480
EVTNITRKSV	SLKWSKPRYD	GGAKITGYIV	ERRELDPGRW	LKCNFTNVQE	TYFEVTELTE	27540
DQRYEFRVFA	RNAADSVSEP	SESTGPITVK	DDVEAPRIMM	DVKFRDVIV	KAGEVLKINA	27600
DIAGRPLPVI	SWAKDGVIE	ERAKTEIVST	DYTTTLTVKD	CVRRDTGQYV	LTLKNVAGTR	27660
TMAVNCKVLD	KPGPPAGPLE	INGLTAEKCS	LSWGRPQEDG	GADIDYIIVE	KRETSRLAWT	27720
ICEAELRTTS	CKVTKLLKGN	EYIFRVTVGN	KYGVGEPLS	MAVKALDPFT	TPSPPTSLEI	27780
TSVTKDSMTL	CWSRPETDGG	SDISGYIER	REKNSLRWMR	VNKKPVYDLR	VKSTGLREGG	27840
EYEVYFAEN	AGLSLPSSET	SPLVRAEDPV	FLPSPPSKPK	IVDSGKTTIT	IGWVKPLFDG	27900
GAPITGYTVE	YKKSSEEDWK	VAIQSFRGTE	YTMISGLTTGD	EYVFRVRSLN	KMGASDPDS	27960

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SDPQVAKERE	EEPVDVDSE	MRKTLIVKAG	SSFTMTVPFR	GRPIPNVSW	KPDTDLRTRA	28020
YIDSTDSRTL	LTENANRND	SGKYTLTIQN	VLSAASMTFV	VKVLDSPGPP	ANITVREVTK	28080
ETAMLSWDVP	ENDGGAPVKN	YHIEKREASK	KAWVSVTNNC	NRLSYKVTNL	QEGAIYYFRV	28140
SGENEFVGUV	PAETKEGVKI	TEKPSPEKL	GVTSVSKDSV	SLSWLKPEHD	GGSRIIHYVV	28200
EALEKGQKTW	VKCAVVKTH	HVVSGLRESH	EYFFRVFAEN	QAGLSDPREL	LLPVLIKDQL	28260
EPPEIDMKNF	PSHTVYVRAG	SNLKVDIPIS	GKPLPKVTL	RDGVPLKATM	RFNTEITAEN	28320
LTINLKESVT	TDAGRYEITA	ANSSGTTKTF	INIIVLDRPG	PPTGPVAISD	ITEESVTLKW	28380
EPPKYDGGSH	VTNYIVLKRE	TSTAVWSEVS	ATVARTMIKV	MKLTGTGEYQ	FRIKAENRFG	28440
ISDHIDSVCV	VVKLPYTPG	PPSTPWVSNV	TRESITVGNH	EPVSNNGSAV	TGYHLEMKDR	28500
NSILWQKANK	MIIRTHFKV	TTISAGLIYE	FRVYAENAAG	IGKPSHPSEP	VLAIDACEPP	28560
RNVRTIDISK	NSVNLWQQP	AFDGGSKITG	YIVERRDLPD	GRWTKASFTN	VIETQFTVSG	28620
LTQNSQYEFR	VFARNAVGSV	SNPSEVVGPI	TCIDSYGGPV	IDLPLEYTEV	VKYRAGTSVK	28680
LRAGISGKPE	PTIEWYKDDK	ELQTNALVCV	ENSTDLASIL	IKDANRLNSG	SYELKLRNAM	28740
GSASATIRVQ	ILDKPGPPGG	PIEFKTVTAE	KITLLWRPPA	DGGGAKITHY	IVEKRETSRV	28800
VMSMVAENLE	BCIVTTTKII	KGNEYVFRVR	AVNKYGIGEP	LESEPVVAKN	AFVTPGPPSI	28860
PEVTKITKMS	MTVVWDRPTV	DGGSEINGYF	LERRDKKSLA	WLKVLKETIR	DTRQKVTLGT	28920
ENSQYQYRVC	AVNAAGVGPF	SEPSDFYKAA	DPIDPPGPPA	KIRIADSTKS	SITLGWSKPV	28980
YDGGSDVTVQ	VVEMKQGDDE	EWITIVSTRGE	VRTTEYVVS	LKPGVNYFYQ	VSAVNCAGQG	29040
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LQVKHVS LGT	VTLWLDPLI	DGGSPIINYV	IEKRDKATRT	WSVSVHKCSG	TSFKVTDLSE	29220
KTPFFFRVLA	ENETIGIEPC	ETTEPVKAAE	VPAPIRDLSM	KDSTKTSVVL	SWTKPDDFGG	29280
SIITDYLVER	KGKGEQAWSH	AGISKTCIEI	IGQLKEQSVL	EFRVSARNEK	GQSDPVTIGP	29340
LTVKELVITP	EVDLSEIPGA	QISVRIGHNV	HLELPYKGGP	KPSISWLKDG	LPLKESEYVR	29400
FSKTENKITL	SIKNSKKRHH	GKTYTVILDNA	VCRNSFPITI	ITLGPSPKPK	GPIRFDEIKA	29460
DSAIMSWDIP	EDDGGGEITC	YSIEKREASQ	TNWMVCSSV	ARTTFKVSNL	VKDSEYQFRV	29520
RAENRYGVSE	PLASNIIVAK	HQPRIPGPPG	KPIYINVTSD	GMSLTWDAPV	YDGGSEVTGF	29580
HVEKKERNIS	LWQRVNTSPI	SGREYRATGL	IEGLDYQFRV	YAENSAGLSS	PSDPSKFTLA	29640
VSPVDPPGTP	DYIDVTRITI	TLKWNPLLRD	GGSKIVAYSI	EKRQGS DRWV	RCNFTDVSEC	29700
QYTVTGLSPG	DRYEFRIIAR	NAVGTISPPS	QSSGLIMTRD	ENVPTTVEFG	PEYFDGLVIK	29760
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EAKNVSGSTK	AEVTVKVQDT	PGKVVGPIRF	TNITGKMTL	WWEAPLNDGC	APVTHYIIEK	29880
RETSRLAWAL	IEDNCEALSY	TATKLITGNE	YQFRISAVNK	FGVGRPLESD	PVVAQIQYTI	29940
PDAPGVPEPS	NTGTIKAVTL	WTRPESDGG	EIQHYILERR	EKKSTRWVKV	ISKRPISETR	30000
FKVTGLVEGN	EYEPHVMAEN	AAGIGPASGI	SRLIKREPV	NPPSAPSVVK	VTDTSKTTVS	30060
LEWARPVDPG	GMEIIGYIIE	MCKADLGWDH	KVNTEPCVKT	RYTVDLQAG	EEYKFRVSAV	30120
NGAGKGDSC	VTGTIKAVTL	LSAPELDIDA	NFKQTHIVRA	GVSIIRLFIAI	QGRPTPTAVW	30180
SKPDSNLSIR	ADIHTTDSFS	TLTVENCNRN	DAGKYTLTVE	NNSGKKSITF	TVKVLDSGPG	30240
PGPITFKDVT	RGSATLWMDA	PLLDGGARIH	HYVIEKREAS	RRSQVQVSEK	CTRQILKVSE	30300
LTEGVPPYYFR	VSAENYGVST	EPYEMPEPIV	ATEQPAPRRR	LDVVDTSKSS	AVLAWLKPDPH	30360
DGGSRTISYL	LEMROKGSDF	WVEAGHTKQL	TFTVERLVEN	TEYEFVRKAK	NDAGYSEPRE	30420
AFSSVIEKEP	QIEPTADLTG	ITNQLITCKA	GSTFTIDVPI	SGRPAPKVTV	KLEEMRLKET	30480
DRMSIATTDK	RTTLTVKDSM	RGDSGRYFLT	LENTAGVKTF	TITVVVIGRP	GPVTPGPIEV	30540
SVSAESCVL	WTEPKDDGGT	EITNYIVEKR	ESGTTAWQLI	NSSVKRTQIK	VTHLTKEYKEY	30600
CFRVSSENRF	GVSKPLESVP	IVAEPHFVPP	SAPTRPEVYY	VSANAMSIRW	EEPYHDGGSK	30660
IVGYWVEKKE	RNTLLWKEN	KVPCLECNKY	VTGLVEGLE	QFRTYALNAA	GVSKASEASR	30720
PIMANQNPVD	PGRPEVTDVT	RSTVSLIWSA	PVYDGGSKV	GYIIERKPV	EVGDGRWLKC	30780
NYTIVSDNPF	VTVALSBDGT	YEPRVLAKNA	AGVISKGS	TGPVTCRDEY	APPKAELDAR	30840
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RSDSGKYTLT	VKNASGTSKV	SVMVKVLDSP	GPCGKLTVSR	VTEEKCTLAW	SLPQEDGGAE	30960
ITHYIVERRE	TSRLNWIVVE	GECLTASYVV	TRLIKNNEYT	FRVRAVNVKY	LGVPEVSEPI	31020
VARNSTPTPS	QGTIPEEVGA	GKEHIIQWT	KPESDGGNEI	SNYLVDKREK	KSLRWTRVKN	31080
DYVVYDTRLK	VTSLMEGCDY	QFRVTAVNAA	GNSEPSSEAS	FISCREPSYT	PGPPSAPRVV	31140
DTTKRSISLA	WTKPMYDGGT	DIIGYVLEMQ	EKDTDQWCRV	HTNATIRNNE	FTVPDLKMGQ	31200
KYSFRVAAVN	AKGMSDYSET	TAEIEPVERL	EIPDLELADD	LKKTIVIRVAG	ASLRLMVSVS	31260
GRPSPVITWS	KKGIDLANRA	IIDNTESYSL	LIVDKVNRD	AGKYTIEAEN	QSGKKSATVL	31320
VKVYDTPGPG	PSVSVEKVS	DSVTITWEIP	TIDGGAPVNN	YIEKREAAAM	RAFKTVTTKC	31380
SKTLYRISGL	VEGTMYFRV	LPENIYGIGE	PCETSDAVLV	SEVPLVPTKL	EVVDVTKSTV	31440
TLAWEKPLYD	GGSLRTGYVL	EACKAGTERW	MKVVTLKPTV	LEHTVISLNE	GEQYLFRVRA	31500
QNEKGVSEPR	EIVTPVTQD	LRVLPTIDLS	TMPQKTIHVP	AGRPILVLP	ITGRPPPTAS	31560
WFFAGSKLRE	SERVTVETHT	KVTKLTIRET	TIRDTEGYTL	ELKNVTGTT	ETIKVIIIDK	31620
PGPPTGPIKI	DEIDATSVTI	SWEPPELDGG	APLSGYVVEQ	RDHRPGWLP	VSESVTRPTF	31680
KFTRLTEGNE	VYFRVAATNR	FGGSGYLQSE	VIECRSSISI	PGPPETLQIF	DVSRDGMTLT	31740
WYPPEDDGG	QVTGYIVERK	EVRAWRVRV	NKVPVMTTRY	RSTGLIEGLE	YEHRTAINA	31800
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QKVDQREWTK	CNTTPTKIRE	YTLTHLPQGA	EYRFRVLACN	AGGPGEPAEV	PGTVKVTEML	31920
EYDPYELDER	YQGVFVRQG	GVIRLTIPIK	GKPPPVCKWT	KEGQDISKRA	MIATSETHTE	31980
LVIKADNRND	SGTYDLVLN	KCGKKTIVIK	VKVIQSPNTP	EGPLEYDDIQ	ARSVRVSWRP	32040
PADDGGADIL	GYILERREVP	KAAWYTIDSR	VRGTSLVVKG	LKENVEYHFR	VSAENQFGIS	32100
KPLKSEEPVI	PKTPLNPPEP	PSNPPEVLDV	TKSSVSLWS	RPKDDGGSRV	TGYIIEKRET	32160
STDKWVRHKN	QIITMTTVTV	TGLVPDAEYQ	FRIIAQNDVG	LSETSPASEP	VVCKDPFDKP	32220
SQPGELEILS	ISKDSVTLQW	EKPECDDGKE	ILGYWVEYRQ	SGDSAWKKSN	KERIKDRQFT	32280
IGLLEATEY	EFVRFAENET	GLSRPRRTAM	SVTKTLTSGE	APGVRKEMAD	VTTKLGEAAQ	32340
LSCQIVGRPL	PDIKWYRFGK	ELIQSRKYKM	SSDGRTHTLT	VMTDEQDEEG	VYTCVATNEV	32400
GEVSSSKLL	LQAAPQPHQN	YPLIKEYYGA	VGSTLRLHVM	YIGRPVPAMT	WPHGQKLLQN	32460
SEKITIENTE	HYTHLVMKNV	QRKTHAGKYK	VQLSNAGFTV	DATLDVEIQD	KPKTKPTGPIV	32520

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AGYYFRVSAQ	NTFGISEPLE	VASIVIIKSP	FEKPGVPGKP	TITAVTKDSC	VVAWKPPASD	32640
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GLKYRIQEFK	GGYHQLIIAS	VTDDDATVYQ	VRATNQGGSV	SGTASLEVEV	PAKIHLPKTL	32820
EGMGAVHALR	GEVVSIIKPF	SGKPDVPITW	QKGQDLIDNN	GHYQVIVTRS	FTSLVFSNGV	32880
ERKDAGFVVV	CAKNRFGIDQ	KTYELDVADV	PDPGRGVKVS	DVSRDSVNL	WTEPASDGG	32940
KVTNYIVEKC	ATTAERWLRV	GQARETRYTV	INLFGKTSYQ	FRVIAENKFG	LSKPSEPSEP	33000
TVTKEDEKTRA	MNYDDEVDET	REVTTTKASH	SKTKELYEKY	MIAEDLGRGE	FGIVHRCVET	33060
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IFERINTSAF	ELNEREIVSY	VRQVCEALEF	LHSQNIHFDP	IRPENIIYQT	RKNSTIKIIE	33180
FQQAQRLKPG	DNFRLLFTAP	EYIYAEVHQH	DVVSSTATDMW	SLGTLVYVLL	SGINPFLAET	33240
NQQMIIENIMN	AEYTFDEEAF	KEISLEAMDF	VDRLLVKERR	SRMTASEALQ	HPWLKQRIDR	33300
VSTKVIRTLK	HRRYYHTLIK	KDLNMVVSAA	RISCGGAIRS	QRGVSVAKVK	VASIEIGPVS	33360
GGIMHAIGEE	GGYVKYVCKI	ENYDQSTQVT	WYFGVRQLEN	SEKYEITYED	GVATMVVKDI	33420
TKFDDGTYRC	KVNDYQYEM	SYAELFVKGV	REVYDYYCRR	TKVKRRRTDA	MRLLRPPPEF	33480
TLPLYNKTAY	VGENVRFGVT	ITVHPEPRVT	WYKSGQKIKP	GDDEKKYTFE	SDKGLYQLTI	33540
NSVTTDDDAE	YTVVARNKHG	EDSCAKLTV	TLHPPPTETT	LRPMFKRLLA	NAECHGQSV	33600
CPEIRVSGIP	APTLKWEKDG	QPLSLGPHIE	IVHEGLDYIA	LHIRDTLPED	TGYRVATATN	33660
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EALREAAAILY	KPAVSTKTVK	GEYRLQTEEK	KEERKLRLMPY	EVPEPRRFKQ	ATVEEDQRIK	33780
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IPKLTQDDLE	MVRPARRRTP	SPDYDLYYR	RRRSLGDMS	DEELLPLIDD	YLAMKRTEEE	33900
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STSYAELRER	HAQASYRQPK	LQRIMAEKE	EEELLRPVTT	TQRLSEYKSE	LDYMSKEEKS	34020
KKKSKRQROV	TEITEIEEY	EISRAQRES	SSSVSRLLR	RRSLSPYIE	LMRPVSELIR	34080
SHPRPAEYEE	DDAERRSPTP	ERTRPRSPSP	VSSERSLSRF	ERSARFDIFS	RYESMKAALK	34140
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KARVIEKAVT	SPPRVKSP	RKVSPEVTKS	PKRVKSPPEV	TSHPKAVSPT	ETKPTKEGQH	34500
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AESPEAKKQE	PIAPESISSK	PVIVTGLRDT	TVSSDSVAKF	TIKVTGEPQP	TITWTQDGA	34680
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VSTQKTSEVT	SQKKASAQEE	ISQKALTSEE	IKMSEVKSH	TLAIKEEASK	VLIAEEVKRS	34800
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LNGTELNPSE	EYRYGVSGSD	ITVPSIDQ	REEGILSCIG	KTSQGVKQCV	FDLTLSEELS	34920
DAPSFTQPR	SQININEQNV	LFSCEVSGEP	SPEIEWFKNN	LPISISSNIS	VSRSRNVYTL	34980
ETRNVAAVSD	GKYTIKAKNF	HGQCSATASL	TVLPLVEEPP	REVVLKTSDD	VSLHGSVSSQ	35040
SVQMSASKQE	ASFSFSFSSS	ASSMETMKFA	SMSAQSMSSM	QESFVEMSSS	SFMGKSSMTQ	35100
LESSTSRMLK	AGGRGIPPKI	EALPSDISID	EGKVLTVACA	FTGEPTPEIT	WSCGGRKIQN	35160
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 organism = Mus musculus

SEQUENCE: 2

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SGAGKTVNTK	RVIQYFATIA	VTGDKKKEEA	TSGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSRFGFKF	IRIHFGATGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTNP	YDFAYVSQGE	ITVPSIDQ	ELMATDTAVD	ILGFSADEKV	AIYKLTGAVM	360
HYGNMKFKQK	QREQAEPD	TEVADKAAYL	TSLSADLLK	ALCYPRVKVG	NEYVTKGQTV	420
QQVYNSVGAL	AKSMYEMFPL	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDFTLEQLC	480
INFNTNEKLQ	FFNHHMFVLE	QEEYKKEGID	WEFIDFGMDL	AACIELIEKP	MGIFSIILEE	540
CMFPKATDTS	FKNKLYEQHL	GKSNFQKPK	PAKGKAEAHF	SLVHYAGTVD	YNIIGWLDKN	600
KDPLNETVVG	LYQSGGLKTL	APLFSGGQAA	EAEGGGGKKG	GKKKGSSFTQ	VSALEPRLNL	660
KLMTNLKSTH	PHFVRCLTPN	ETKTPGAMEH	ELVLHQRLCN	GVLEGIRICR	KGFPSTRILYA	720
DFKQRYKVLN	ASAIPEQOFI	DSKKASEKLL	GSIDIDHTQY	KFGHTKVFFK	AGLLGTLEEM	780
RDEKLAQLIT	RTQAVCRGYL	MRVEFKKMM	RRESIFCIQY	NVRAFMNVKH	WPWMKLYFKI	840
KPLLKSAETE	KEMANMKEDF	EKAKEDLAKS	EAKRKELEEK	MVALMQEKND	LQLQVQAEAD	900
GLADAERCD	QLIKTKKIQLE	AKIKELTERA	EDEEINAEEL	TAKKRKLEDE	CSELKKDIDD	960
LELTAKVEK	EKHATENKVK	NLTEEMAGLD	ENIAKLTKKE	KALQEAHQQT	LDLQAEEDK	1020
VNTLTAKATK	LEQQVDLLEG	SLBQEKKLRL	DLERAKRKL	GDLLKLAQEST	MDIENDKQQL	1080
DEKLKKKEFE	MSNLQKIED	EQALGMQLQK	KIKELQARIE	ELEEEIEAER	ASRAKAEKQR	1140
SDLSRELEEI	SERLEEAGGA	TSAQIEMNKK	REAEFQKMRR	DLEEATLQHE	ATAAALRKKH	1200
ADSVAELEGQ	IDNLQVRVQK	LEKEKSELKM	EIDDLASNME	TVSKAKGNLE	KMCRTLEDQL	1260
SEVKTKEEBQ	QRLINELSTQ	KARLHTESGE	FSRQLDEKDA	MVSQLSRGKQ	AFTQQIEELK	1320
RQLEESKAK	NALAHALQSA	RHDCDLLEQ	YEEEQEAKAE	LQRAMSKANS	EVAQWRKYKE	1380
TDAIQRTTEL	BEAKKLAQR	LQDAEEHVEA	VNSKCSLEK	TKQLQLQNEVE	DLMDIVERSN	1440

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KILRIQLELN	QVKSEIDRKI	AEKDEEIDQL	KRNHLRVVES	MQSTLDAEIR	SRNDALRIKK	1620
KMEGDLNEME	IQLNHANRQA	AEAIRNLRNT	QGMLKDTQLH	DDALRGQDD	LKEQLAMVER	1680
RANLMQAEIE	ELRASLEQTE	RSRRVAEQEL	LDASERVQLL	HTQNTSLINT	KKKLETDISQ	1740
IQGEMEDIVQ	EARNAAEKAK	KAITDAAMMA	EELKKEQDTS	AHLERMKNM	EQTVKDLQHR	1800
LDEAEQLALK	GKKQIQKLE	ARVRELENEV	ENEQKRNIEA	VKGLRKHERR	VKELTYQTEE	1860
DRKNVLRLOD	LVDKLQTKVK	AYKRQAEAE	EQSNVNLAKF	RKIQHELEEA	EERADIAESQ	1920
VNKLVRVKSRE	VHTKVISSEE					1939

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 organism = Mus musculus

SEQUENCE: 3

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VERKKVIRKK	VDSSKFMTPT	IQHSQKMQDL	FSSNKYKENY	EKAKGQPYAI	TSDTPELRRI	120
KKVQDQLSEV	KYRMDGDVAK	TICHVDEKAK	DIEHAKKVSQ	QVSKVLYKQN	WEDTKDKYLL	180
PPDAPELVQA	VKN TAMFSKK	LYTEDWEADK	ALFYPPYNDSP	ELRRVAQAQK	ALSDIAYKKG	240
LTEQQQTQFTS	LPDPPEVEFA	KKVTNQVSKQ	KYKEDYEKKV	KGKWSETPCF	EIATARMNAD	300
NLSSRKYQED	FENMKDQIYF	MQTETPEYKM	NKQAGVAASK	VKYKQDYEKN	KGKADYNVLP	360
ASENPLLRQL	KTAGNVLSDK	LYKENYEKTK	ARSINYCETP	KFQLD TVLQN	FSSDTKYKDS	420
YLKNILGHYV	GSPEDPYHTH	CMKVTAQNSD	KNYKAEEYED	RGKGFPPQTI	TQYEYAIKKL	480
DQCKDHAYKV	HPDKTKFTQV	TDSPVLVQAQ	VNSKQLSDLN	YKAKHESEKF	KCHIPPDAPA	540
FIQHRVNAYN	LSDNVYKQDW	EKS KAKKFDI	KVDAIPLLAA	KANSKNASDV	MYKKDYEKSK	600
GKMI GALSIN	DDPKMLHSLK	TAKNQSDRLY	RENYEKT KAK	SMNYCETPKY	QLDTLLKNFS	660
EAKYKDSYVK	NMLGHYVGSF	EDPYHTHCMK	VSAQNSDKSY	KAEEYEDKKG	CYFPQTITQE	720
YESIKKLDQC	KDHTYKVHPD	KTRFTAVTDT	PVLLQAQLNT	KQLSDLNYKA	KHEGEKFKCS	780
VPADAPEFTQ	HRVNAVNLSD	NVYKHDWEKS	KAKKFDIKVD	AIPLLAAKAN	SKNASDVMYK	840
KDYEKSKGKM	IGALSINDDP	KMLHSLKTAK	NQSDREYRKD	YEKSKTIYTA	PLDMIQVTQA	900
KKSQAIASDV	DYKHLVHNYS	YPPDSINVDL	AKKAYALQSD	VEYKADYNWS	MKGCGWVPFG	960
SLEMEKAKRA	SEILNEKKYR	QHPDTLKFTS	IEDAPITVQS	KINQAQRSDI	AYKAKGEEIL	1020
HKYNLPADLP	QFIQAKVNAY	NISENMYKAD	LKDL SKKGYD	LRIDAIPIKA	AKAARQAASD	1080
IHYKKDYEKA	KGKMGVGFQSL	QDDPKLVHYM	NVAKIQSDRE	YKKDYEKTKT	KYNTPHDMFD	1140
VVAAKKAQDV	ISNTNYKHSI	HHYTYLPDAM	DLELSKNMMH	IQSDNAYKED	YNTWMKGIGW	1200
IPIGSLDQEK	QFIDPLGSLA	KKYRQHPD TL	KFTSIVDSPV	MVQAKQNTQQ	ASDILYKAKG	1260
EDVKHKYTLN	PDL PQFLQAK	CNAYNISDVC	YKRDWHDLIA	KGNNVLADAI	PITAAKSSRN	1320
IASDYKYKEA	YEKAKGKHVG	FRSLQDDPKL	VHYMNVAKLQ	SDREYKKNYE	NTKTSYHTPG	1380
DMVSITAAARM	AQDVATNVNY	KQPLHHYTYL	PDALSLEHTR	NANQIQSDNV	YKDEYNSFFK	1440
GIGWIPIGSL	EVEKAKKAGE	ALNERKYRQH	PDTIKFTSVP	DSMGMVLAQQ	NTKQLSDLNY	1500
KVEGEKMKHK	YTMDEPLPQF	IQAKVNAINM	SDAHYKADWK	KTLAGGYDLR	PDAIPIVAAK	1560
SSRN IASDCK	YKEAYEKARG	KQIGFLSLQD	DPKLVHYMNV	AKIQSDREYK	KGYEASKTRY	1620
HTPLDMFSVT	AAKKSQEVAT	NSNYRQPYHN	YTLPLDALNV	EHSRNAMQIQ	SDNLYKSDFT	1680
HWMKGIGWVP	IDSLEVEKAK	KAGEILSERK	YRQHPPELKF	TYSMDTMEQA	LNKSNKLTMD	1740
KRLYTEKWNK	KDTSIHVMPD	TPDILLSRVN	QITMSDKLYK	AGWEEETKKG	YDLKPDAIAI	1800
KAASASRDIA	SDYKYQAYE	QSKGKHIGFR	SLEDDPKLVH	FMQVAKIQSD	REYKKGYEKS	1860
KTSFHTPVDM	FSVVAAKSKQ	EVATNANYRN	I IHTYNMLPD	AMSFELAKNM	MQIQSDNQYK	1920
ADYADFMKGI	GWLPLGSLEA	EKNKKAMEII	SEKKYRQHPD	TLKYSTLMDS	MNMVLAQNNA	1980
KIMNKNLYKQ	AWEADKTKIH	IMPDIPEIVL	AKANAINMSD	KLYKLSLEES	RKKGYDLRTD	2040
ATPIKAAKAS	RDIASDYKYK	HSYEKEKGKM	VGRSLEDDEP	KLVSMSQVAK	MQSDREYKKN	2100
YEKTKTSYHT	PMMLSLVTA	KDAQANITNT	NYKRLIHKYI	LLPDAMNIEL	TRNMNHIQSD	2160
NEYKQDYREW	YRGLGWSPAG	SLEVEKAKKA	TEYASDQKYR	QHPSTFQFKK	LTDSDMDVLA	2220
KQNAHTMKNK	LYTTIDWNKDK	TKIHVMPDTP	DILQAKQNTQ	LYSQKSYRLG	WEALKKGYD	2280
LPVDAISVQL	AKTSRDIASD	FKYKQGYRQK	LGHHIGFLSV	QDDPKLVLSM	NVAKMQSDRE	2340
YKKDFEKWKT	KYTSPVDMLG	VVLAKKQCAL	VSDADYRNYL	HQWTCPLDQN	DVIQAKKVYE	2400
LQSENMYKSD	LEWLRLGIGWS	PLGSLEAEKN	KRASEIISEK	KYRQPPDRNK	FTSIPDAMD I	2460
VLAKTNAKNR	SDRLYREAWD	KDKTQIHIMP	DTPDILAKA	NLINTSDKLY	KMGYEELKKK	2520
GYDLPVDAIP	IKAAKASREI	ASEYKYKEGF	RKQLGHHIGA	RNIKDDPKMM	WSMHVAKIQS	2580
DREYKDFEKF	WTKTYSSPVD	MLGVVLAKKC	QTLVSDADYR	NYLHQWTCPL	DQNDVIHARQ	2640
AYDLQSDNVY	KADLQWLKGI	GWIPSGSLED	EKNKRATQIL	SDHVYRQHPN	KFKFSSLMDS	2700
MPMVLAKNNA	ITMNNHLYTE	AWNKDKTTHV	IMPDTPEVIL	AKQNKVNYSE	KLYKLGLEEA	2760
KRKGVDMLRD	AIPKTKAKAS	RDIASDFKYK	EGYRKQLGHH	IGARGIHDDP	KMMWSMHVAK	2820
IQSDREYKSD	FEKWTKYSS	PVGMLGVVLA	KKCQTLVSDA	DYRNYLHQWT	CLPDQNDVIH	2880
ARQAYDLQSD	NMYKSDILQWM	RGIGWVPIGS	LDVEKCKRAT	EILSDKIYRQ	PPDKFKFTSV	2940
TDSLQVQLAK	NNATIMNKRL	YTEAWDKDKT	QIHIMPDTPE	IMLARMNKVN	YSESLYKLAN	3000
EEAKKKGIDL	RSDAIPIVAA	KASRDIASDY	KYKDGGRKQL	GHHIGARNIK	DDPKMMWSMH	3060
VAKIQSDREY	KQDFEKKWTK	YSSPVDMLGV	VLAKKCQILV	SDIDYKHPLH	EWTCPLDQND	3120
VIHARQAYDL	QSDNVYKSDL	QWMRGIGWVP	IGSLDVVKCK	RAAEILSDNI	YRQPPNKKFKF	3180
TSVTDSLQV	LAKSNALNMN	KRLYTEAWDK	DKIQIHVMPD	TPEIMLARQN	RINYSESLYK	3240
LANEAEKKKG	YDLRVDAIMPI	VAAKASRDIA	SDYKYKGYR	KQLGHHIGAR	NIEDDPKMMW	3300
SMHVAKIQSD	REYKDFEKF	KTKYSSPVD	LGVLAKKQ	TLVSDVDYRN	YLHQWTCPLD	3360
QNDVIHARQA	YDLQSDNVYK	SDLQWMRGIG	WVPIGSLDVV	KCKRAAEILS	DNIYRQPPNK	3420
FKFTSVTDSL	EQVLAKSNAL	NMNKRLYTEA	WDKDKTQIHI	MPDTPEIMLA	RQNKINYSES	3480
LYRQAMEEAK	KEGYDLRSDA	IPIVAAKASR	DIASDYKYKE	AYRKQLGHHI	GARAIHDDPK	3540
MMWSLHIAKV	QSDREYKKEF	EYKTRYSSP	VDMLGIVLAK	KCQTLVSDVD	YKHPLEHWTC	3600

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LPDQNDVIA	RKAYDLQSDN	LYKADLEWMK	GIGWVPIGSL	EVVKAKRATE	ILSDNIYRQR	3660
PDTLKFTSIT	DSLEQVLAKN	NAITMNRKLY	TEAWDNDKKT	IHVMPDTPET	MLAKLNRINY	3720
SDKLYKVSLE	EARRQGYDLR	VDAIPIRSAK	ASREIASDYK	YKEGYRKQLG	HHIGARNIKD	3780
DPKMMWSMHV	AKIQSDREYK	KDFEKWKTKY	SSPVDMLGVV	LAKKCQILVS	DIDYKHLPE	3840
WTCLPDQNDV	IQARKAYDLQ	SDAVYKADLE	WLRGIGWVPI	GSVEVEKVKR	AGEILSDRKY	3900
RQPAQQLKFT	CITDTPETVL	AKSNAITMSK	HLYTEAWDAD	KASIHVMPDT	PEILLAKSNS	3960
ANISHKLYTK	GWDESKMKDY	DLREDAISIK	SAKASRDIA	DYKYKEAYEK	QKGHHIGAQS	4020
VEDDPRILCA	MHAGKIYSR	EYKKEPQKWK	TKFSSPVDML	GVVLAKKCQT	LVSDVDYRNY	4080
LHNWTCCLPDQ	NDIIQAKKAY	ELQSDSVYKA	DLEWLRGIGW	MPEGSIMETR	VKGAQDLVNE	4140
RLYRTRPEAL	KFTSIVDTPE	VVLAKANSLO	MSERLYQEAW	NKDKTNISLP	LDTPAMLQAQ	4200
INALQISNKL	YQKDWDEAKQ	KGYDLRADAI	EIKHAKASRE	IASEYKYKEG	YRKQLGHHVG	4260
FRSLQDDPKL	VWSIHAAKIQ	SDREYKKAYE	KSKGIYNTPL	DMMSIVQAKK	CQVLVSDVDY	4320
RNYLHQWTCCL	PDQNDVIAQ	KAYELQSDAI	YKADLEWLRG	IGWMPGSGPE	VLRVKNAQEI	4380
LCDSVYRTPV	VKLKYTSIVD	TPEVVLAKSN	AENISIPKYR	EVWDKDKTSI	HIMPDTPEIN	4440
LARANALNVS	NKLYREGWDE	VKMSCDVRLD	AIPIQAAKAS	REIASDYKYK	LDHEKQKQGHY	4500
VGTLTARDND	KIRWALTAGK	IQNEREYRLQ	WAKWKTKFQS	PVDMLSLHS	KCKTLVSDI	4560
DYRNYLHEWT	CMPDQNDVIAQ	AKKAYELQSD	AVYKADLEWL	RGIGWMPNDS	VSVNHAKHAA	4620
DIFSEKKYRT	KIETLNFTPV	DDRVDYVTAK	QSSEIRDDIK	YRKDWNIAKS	KYTLTETPLL	4680
HTAQEAARIL	DQYLYKSLP	QKATGYILP	PDAVPFVHAH	HSSDVQSELK	YKAEHVQKRG	4740
HYVGVPMTMR	DPKLVWFEHA	GQIQNDRLYK	ESYHKS KAKI	NIPADMVSVV	AAKEGQNLVS	4800
DDYRQYLLHQ	WTCHPDQNDV	IQAKIAYDLQ	SDNVYKADLE	WLRGIGWIPL	DSVDHVRVTK	4860
NQEMVNQIKY	KKDALANYPN	FTSVEDPPEI	VLAKINSVNQ	SDVKYKETFN	KAKGKYIFSP	4920
DTPYISHSKD	MGKLYSTILY	KGAWEGTKAY	GYTLDERYIP	IVGAKHADLV	NSELKYKETY	4980
EKQKGHYLAG	KEISEFPQGV	HCLDFQOMRS	ALNYRRDYED	TKANVHIPND	MMNHVLAKRC	5040
QYILSDLEVR	HYFQWTLSE	EENPVIRARN	AQEILSDNVY	KDDLNLWLGK	GCYVWDTPQI	5100
LHAKKSYDLQ	SQLQYTAASK	ENLQNYNLVT	DTPVYVVTALQ	SGINASEVKY	KENYHQIKDK	5160
YTTVLETADY	DRTKHLKSLY	SSNIYKEAWD	KVKATSYNLP	TSTLSLTHAK	NQKHLASHIK	5220
YREEYEKPKA	LYTLPKSVDD	DPNTARCLRV	GKLNIDRLYR	SVYEKKNMKI	NIVPDMVEMV	5280
TAKDSQKKVS	EIDYRLHLHE	WICHPDQVN	SHVRKVTDQI	SDIVYKDDLT	WLKGIGCYVW	5340
DTPEILHAKH	AYDLRNDIKY	KAHVQKTRND	YKLVTDTPVY	VQAVKSGKQL	SDAVYHYDYV	5400
HSVRGKVAPT	TKTYDRLKLE	HAYKLQSEN	YKGLRFLPT	GYRLPVDTPH	FNHTKDRYTM	5460
SSYFYKKEAY	EHKANGYTL	GPKDVFPVHV	RRVNNVTSER	LYRELYHKLK	DKIHTTPDTP	5520
ETRQVKKTQE	AVSBLIYKSD	PFKMQGHMIS	LPYTPQVLHC	RYVGIDTSDI	KYKEDQLILK	5580
GLGCFLYDTP	DMVRSRHLRK	LWSNLYTDN	ARKMRDKYKV	VLDTPYRKV	QELKTHLSEL	5640
VYRASGKKQK	SIFTSVPDTP	DLRLAKRGQK	LQSQYLYVEL	ATKERPHHHA	GNQTTALQHA	5700
KDVKDMVSEK	KYKIQYKEMK	DKYTPVADTP	ILIRAKRAYW	NASDLRYKET	FQKTKGRYHT	5760
VKDALDIVHV	RKVDDISKV	KYKENYMSQL	GIWSSIPDRP	EHFHHRAVTD	AVSDVKYKED	5820
LTWLKIGICY	AYDTPDLTLA	EQNKTLYSKY	KYKEVFERTK	SNFKYVADCP	INRHFKYATQ	5880
LMNEKRYRAD	YEQRKDKYHL	VVDEPRHLLA	KTAGDQISQI	KYRKKEKESK	DKFTSVVDTP	5940
EHLRTTKVNK	QISDILYKLE	YNKARPRGYT	TIHDTPMLLH	VRKVKDEVSD	LKYKEYVQRN	6000
KSNCTIKPDA	VHIAAKADAY	KVNTNLDYKK	QYEANKAYWK	WTPDRPDFIQ	AAKSSLQQSD	6060
FEYKLDREPL	KGCKLSVTDD	KDMVLALRNS	LIESDLKYKE	KHVKERGSCH	AVPDTPQILL	6120
AKTVSNLVSE	NKYKDHLVKKH	LAQGSYTTLP	ETRDTHVHKA	VTKHVSDTNY	KKKFVKEKKG	6180
SNYSIMQEP	EVKHAMEVAK	KQSDVAYKKD	AKENLHYTTV	ADRPDIKKAT	QAAKQASEVE	6240
YRAKHREKGS	HGLSMLGRPD	IEMAKKAAKL	SSQVKYRENF	DKEGKGTPKY	NPKDSQLYKT	6300
MKDANNLASE	VKYKADLKKL	HKPVTDMMES	LIMHHVLNLS	QLASSYQYKK	NYEKS KGHYH	6360
TIPDNLEQLH	HREATELQSI	VKYKEKYEKE	RGKPMDFET	PTYITAKESQ	QMOSGKEYRK	6420
YDEESIKGRN	LTGLEVPAL	LHVRHATKIA	SEKEYRKDLE	ESIRGRGLTE	MEDTPMLRA	6480
KNATQILNKK	YKRDLELEV	KGRGLNAMAN	ETPDFMRARN	ATDIASQIKY	KQSAEMEKAN	6540
FTSVVDTPET	IHAQQVKNL	SQKKYKEDAE	KSMSSYETVL	DTPEMQVRRE	NQKNFSLLOQ	6600
QCDLKNSKQK	ITVQDTPET	LRVKNQKNF	SSVLYKEDVS	PGTAIGKTPE	MLRVKQTDH	6660
ISSVKYKEAI	GQGTPIPDLP	EVKRVKETQK	HISSVMYKEN	PGTGIPPTVT	PEIERVKRNQ	6720
ENFSSVLYKE	SLGKGTPTAI	TPEMERVKRN	QENFSSVLYK	ENMGKGTPLP	VTPEMERVKH	6780
NQENISSVLY	KENVGKATPT	PVTPEMQVRK	RNQNENISSV	YKENMGKATP	TPFTPEMERV	6840
KRNQENFSSV	LYKENMRKAT	PTPVTDMER	AKRNQENISS	VLYSDSFRKQ	IQGKAAYVLD	6900
TPEMRRVRET	QRHISTVKYH	EDFEKHKGCF	TPVVTDPITE	RVKKNQDQFS	DINYRGIQRK	6960
VVEMEQRKND	QDQDITGLR	VWRTNPGSVF	DYDPAEDNIQ	SRLHMINVQ	AQRSSRDQSR	7020
SASALSISGG	EEKSEHSEAA	DQRLSTYSYG	GAFFSATSTA	YKHVKTTLQ	QQRSSSVATQ	7080
QTTLSIPSH	PSTAGKIFRA	IYDYIAADAD	EVSPKDGDAI	VNAQAIDEGW	MYGTVQRTGR	7140
TGMLPANYVE	AI					7152

SEQ ID NO: 4 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 4

MCDEDETTAL	VCDNGSGLVK	AGFAGDDAPR	AVFPSIVGRP	RHQGVVMVGMG	QKDSYVGVDEA	60
QSKRGILTLLK	YPIEHGIITN	WDDMEKIWHH	TFYNELRVAP	EEHPTLLTEA	PLNPKANREK	120
MTQIMFETFN	VPAMYVAIQA	VLSLYASGRT	TGIVLSDSGD	VTHNVPIYEG	YALPHAIMRL	180
DLAGRDLDYD	LMKLLTERGY	SFVTTAEREI	VRDIKEKLCY	VALDFENEMA	TAASSSSLEK	240
SYELPDGQVI	TIGNERFRCP	ETLFQPSFIG	MESAGIHETT	YNSIMKCDID	IRKDLYANNV	300
MSGGTTMYPG	IADRMQKEIT	ALAPSTMKIK	IIAPPERKYS	VWIGGSILAS	LSTFQQMWT	360
KQEYDEAGPS	IVHRKCF					377

SEQ ID NO: 5 moltype = AA length = 381

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FEATURE
source Location/Qualifiers
1..381
mol_type = protein
organism = Mus musculus

SEQUENCE: 5

MPFGNTHNKF	KLNYKPQEY	PDLSKHNNHM	AKVLTPDLYN	KLRDKETPSG	FTLDDVIQTG	60
VDPNGHPFIM	TVGCVAGDEE	SYTVFKDLFD	PIIQDRHGGY	KPTDKHKTDL	NHENLKGDD	120
LDPNYVLSSR	VRTGRSIKGY	TLPPHCSRGE	RRAVEKLSVE	ALNSLTGEFK	GKYYPLKSMT	180
EQEQQLIDD	HFLFDKPVSP	LLLASGMARD	WPDARGIWHN	DNKSFLVWVN	EEDHLRVISM	240
EKGGMKEVF	RRFCVLQKI	EEIFKKAGHP	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLA	300
NLSKHPKFEE	ILTRLRLQKR	GTGGVDTA AV	GAVFDISNAD	RLGSSEVEQV	QLVVDGVKLM	360
VEMEKKLEKG	QSIDDMIPAQ	K				381

SEQ ID NO: 6 moltype = AA length = 900

FEATURE
source Location/Qualifiers
1..900
mol_type = protein
organism = Mus musculus

SEQUENCE: 6

MMVMQPEGL	GAGEGPFSGG	GGGEYMEQEE	DWDRDLLLDP	AWKQQRKTF	TAWCNShLRK	60
AGTQIENIEE	DFRNLKLM	LLEVISGERL	PRPDGKMR	HKIANVNKAL	DFIASKGVKL	120
VSIGAEIIVD	GNLKMTLGM	WTIILRFAIQ	DISVEETS	AK EGLLLWCQ	RK TAPYRNVVQ	180
NFHTSWKDG	LALCALIHR	RH PDLIDYAK	LR KDDPIGNL	NT AFEVAEKY	LD IPKMLDAE	240
VNTPKPD	KA IMTVVSCF	YH AFAGAEQ	AE ANRICKV	LA VNQENEL	ME EYEKLASE	300
EWIRRTVP	WL ENRVGEPS	MS AMQRKLE	DFR DYRRLH	KPPR VQEK	CQLEIN FNTLQ	360
SHRPAFMP	SE GKLVS	DIANA WRGLEQ	VEKG YEDWLL	SEIR RLQRLQ	HAE KFOQKAS	420
AWTRGKEE	ML NQHDY	ESASL QEVRA	LLRRH EAFESD	LAH QDRVEH	IAAL AQELN	480
EAASVNSR	QC AICDQ	WNLG TLTQK	RRDAL ERMEK	LLETI DQLQ	LEFARR AAPF	540
AIEDLQDV	WL VHSVEE	TQSL LTAHEQ	PKAT LPEADR	ERGA ILGIQ	GEIQK ICQTY	600
SGNPYITL	SS QDINN	KWDTV RKLVP	SRDQT LQEE	LARQQV NERL	RRQFAA QANA	660
GKVEEVGR	LA AGLAG	SLEE Q MAGLR	QQEQN IIN	YKSNIDR LEGD	HQLLQE SLV	720
YSMEHIRV	GW EQLLT	SIART INEVEN	QVLT RDAKGL	SQEQ LNEFR	ASFNH FDRK	780
PDDFRACL	IS MGYDL	GVEF ARIMT	MVDPN AAGV	VTQAF IDFM	TRETA ETD	840
FKILAGDK	NY ITPEE	LRREL PAQAE	YICIR RMAPY	KSSGA PSGA	LDYVAF SS	900

SEQ ID NO: 7 moltype = AA length = 1942

FEATURE
source Location/Qualifiers
1..1942
mol_type = protein
organism = Mus musculus

SEQUENCE: 7

MSSDAEMAV	FGEAAPYL	RKSEKERIE	AQNKFDAK	SSVFV VDAKES	PFKA TVQSRE	GGKV 60
TAKTEGGTT	VTVKDDQ	VYPM NPPKYD	KIED MAMMTH	LHEP AVL	VNLKERY AAWMI	YTYSG 120
LPCVTVN	PYK WLPVNA	EVV AAYRGK	KRQE APPHIF	SISD NAYQF	MLTDR ENQSIL	ITGE 180
SGAGKT	VNTK RVIQY	FATIA VTGEK	KKEEA TSGKM	QGTLE DQII	ISANPLL EAFGN	AKTVR 240
NDNSSR	FGFGR IRIH	FGTTGK LASAD	IETVL LEKSR	VTFQL KAERS	YHIFY QIMSN	KKPD 300
IEMLLIT	TNP YDYAF	VSQGE ITVPS	IDDQE ELMAT	DSID ILGFT	SDERV SIYKLT	GAVM 360
HYGNMK	FKQK QREEQ	AEFDG TEVAD	KAAYL QNLNS	ADLLK ALCY	PRVKVG NEYVT	KGQTV 420
QQVNSV	QAL LYQSS	MKTL MVMTR	INQQL DTKQ	PRQYFI GVL	DIAGFEI FDFNS	LEQLC 480
INFTEK	LQQ FFNH	HMFVLE QBEEY	KKEGIE WEFID	FGMDL AACIE	LIEKP MGIFS	ILEEE 540
CMFPKAT	DS FKNK	LYBQHL GKSNN	FQPKP PAKGK	VEAHF SLVHY	AGTVD YNIAG	WLDKN 600
KDPLNET	TVG LYQSS	MKTL AYLFS	GAAAA AEAES	GSGGG KKGAK	KGSS FQTVS	ALFRE 660
NLNKLM	TNLR STPHF	VRCI IPNET	KTPGA MEHEL	VVLHQL RCNGV	LEGIR ICRK	GFPRI 720
LYADFK	QRYK VLNAS	AIPG QFIDS	KKASE KLLGS	IDIDH TQYK	FGHTKV FFKAG	LLGL 780
EEMRDD	KLAQ LITR	TQAMCR GYLAR	VEYQK MVERRE	SIFC IQYNV	RAFMN VKHWP	WMKLY 840
FKIKPL	LKSA ETEK	EMANMK EEF	EKAKENL AKAE	AKRKEL EEK	MVALMQE KNDL	QLQVQS 900
EADSLA	DAEE RCDQ	LITKI QLEAK	IK EVT ERAE	DEEIN AELT	AKRKEL EDEC	SELK 960
IDDL	ELTLAK VEKE	KHATEN KVKNL	TEEMA GLDET	IAKLT KEKKA	LQEAH QQTLD	DLQAE 1020
EDKVNT	LTKA KIKLE	QQVDD LEGS	LEQEK IRMD	LRAKR KLEGD	LKLAQ ESTMD	VENDK 1080
QQLEK	LKKK EFEMS	NLQSK IEDEQ	ALGMQ LQKKI	KELQA RIEE	LEEEIE AERAS	RAKAE 1140
KQRSD	LSREL LEIS	ERLEEA GGAT	SAQIEM NKKRE	AEFQK MRRD	LEEATL QHEAT	AATLR 1200
KKHAD	SVAEL GEQID	NLQVR KQKLE	KEKSE MKMEI	DDLAS NMEVI	SKSG NLEKM	CRCTL 1260
DQVSEL	KTYE EEQRL	INEL TAQR	GRQLTE SGEYS	RQLE KDSL	VSQLSR GKQ	AFTQ 1320
ELKRQ	LEEEI KAKS	ALAHAL QSSRH	DCDLL REQY	EEQEAE KAE	LQRAMSK ANSE	VAQWRT 1380
KYETDA	IQRT EELE	EAKKKL AQRL	QDAEEH VEA	VNAKCA LEKTK	QRLQN EVEL	MIDVE 1440
RTNAA	CAALD KKQ	RNFQKIL AEWK	QKYEET HAELE	ASQKE SRSL	STELFK IKNAY	EESLD 1500
HLET	LKRENK NLQ	QEISDLE EQIA	EAGGKRI HELE	KIKKI EQEKS	ELQAA LEEAE	ASLEH 1560
EEGKIL	RIQL ELN	QVKEID RKIA	EKDDEI DQKRN	HIRV VESMQ	STLDA EIRSR	NDAIR 1620
LKKKME	GDLN EMEI	QLNHSN RMAAE	ALRNY RNTQ	GILKDT QLHL	DDALRG QEDL	KEQLAM 1680
VERRAN	LQAE IEEL	RATLE QTERS	RKIAE QELLD	ASERV QLLHT	QNTSL INTK	KKLETD 1740
ISQIQ	GEMED IVQ	EARNAE KAKKA	ITDAA MMAE	ELKKEQ DTS	AHLERMK KNLE	QTVKDL 1800
QHRLDE	AEQL ALKGG	KKQIQ KLEAR	VRELE GEVENE	QQRN VEA	IKGLRKH ERV	KELTYQ 1860
TEEDRK	NVLR LQD	LVDLQS KVKAY	KRQAE EABEQ	SNVNL AKFR	KIQHEL EEAE	ERADIA 1920
ESQV	NKLRVK SRE	VHTKIIS EE				1942

SEQ ID NO: 8 moltype = AA length = 842

-continued

FEATURE
source

Location/Qualifiers
1..842
mol_type = protein
organism = Mus musculus

SEQUENCE: 8

MSRPLSDQDK	RKQISVRGLA	GVENVSELKK	NFNRHLHFTL	VKDRNVATPR	DYYPALAHTV	60
RDHLVGRWIR	TQQHYEYKDP	KRIYYLSLEF	YMGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELEEIEEDA	GLNGNGGLRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	ICGGWQMEEA	180
DDWLRYGNPW	EKARPEFTLP	VHFGYGRVEHT	SQGAKWVDTP	VVLAMPYDTP	VPGYRNNVVN	240
TMRLWSAKAP	NDFNLKDFNV	GGYIQAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQEFV	300
VAATLQDIIR	RFKSSKFGSR	DPVRTNFDAP	PDKVAIQLND	THPSLAIPEL	MRILVDLERL	360
DWDKAWDVT	KTCAYTNHTV	LPEALERWPV	HLMETLLPRH	LQIIYEINQR	FLNRVAAAFP	420
GDVDRLRMS	LVEEGAVKRI	NMAHLCIAGS	HAVNGVARIH	SEILKKTIFK	DFYELEPHKF	480
QNKTNGITPR	RWLVLCPNGL	AEVIAERIGE	DYISDLDQLR	KLLSYVDDEA	FIRDVAKVKQ	540
ENKLFKSAYL	EREYKVHINP	NSLFDVQVKR	IHEYKRQLLN	CLHIITLYNR	IKREPNRFMV	600
PRTIMIGGKA	APGYHMAKMI	IKLITAIGDV	VNHDPAGVDR	LRVIFLENYR	VSLAEKVIPA	660
ADLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	AEAEAGEENFF	IFGMRVEDVE	720
RLDQGRGNAQ	EYYDRIPELR	QIIEQLSSGF	FSPKQPDLEK	DIVNMLMHHD	RFKVFADYEE	780
YIKCQDKVSE	LYKNPREWTR	MVIRNIATSG	KFSSDRTIAQ	YAREIWGVPE	SRQRLPAPDE	840
KI						842

SEQ ID NO: 9

FEATURE
source

moltype = AA length = 364
Location/Qualifiers
1..364
mol_type = protein
organism = Mus musculus

SEQUENCE: 9

MPHPYPALTP	EQKKELSDIA	HRIVAPGKGI	LAADSTGSI	AKRLQSIGTE	NTEENRRFYR	60
QLLLTADDRV	NPCIGGVILF	HETLYQKADD	GRPPQVVIKS	KGGVVGIKVD	KGVVPLAGTN	120
GTTTTQGLDG	LSECAQYKK	DGADFAKWRC	VLKIGEHTPS	ALAIMENANV	LARYASICQ	180
NGIVPIVEPE	ILPDGDHDLK	RCCQVTEKVL	AAVYKALSDH	HVYLEGTLLK	PNMVTTPGHAC	240
TQKFSNEEIA	MATVTALRRT	VPPAVTGVTF	LSGGQSEEEA	SINLNAINKC	PLLKPWALTF	300
SYGRALQASA	LKAWGGKKEN	LKAAQEYIYK	RALANSLACQ	GKYTPSGQSG	AAASESLFIS	360
NHAY						364

SEQ ID NO: 10

FEATURE
source

moltype = AA length = 188
Location/Qualifiers
1..188
mol_type = protein
organism = Mus musculus

SEQUENCE: 10

MAPKKDVKKP	AAAPAPAPAP	APAPAKPKKE	KIDLSAIIKE	FSKEQQEDFK	EAFLLFDRTG	60
ECKITLSQVG	DVLRALGTNP	TNAEVKKVLG	NPSNEEMNAK	KIEFEQFLPM	MQAISNNKDQ	120
GGYEDFVEGL	RVPFKEGNGT	VMGAELRHVL	ATLGEKMKKE	EVEALLAGQE	DSNGCINYE	180
FVKHIMSV						188

SEQ ID NO: 11

FEATURE
source

moltype = AA length = 333
Location/Qualifiers
1..333
mol_type = protein
organism = Mus musculus

SEQUENCE: 11

MVKVGVNGFG	RIGRLVTRAA	ICSGKVEIVA	INDPFIDLNY	MVYMFQYDST	HGKFNGTVKA	60
ENGLKLVINGK	PITIFQERDP	TNIKWGEAGA	EYVVESTGVF	TMEKAGAGHL	KGGAKRVIIS	120
APSADAPMFV	MGVNHEKYDN	SLKIVSNASC	TTNCLAPLAK	VIHDNFGIVE	GLMTTVHAIT	180
ATQKTVDGPG	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCRLLEK	AKYDDIKKVV	KQASEGPKLG	ILGYTEDQV	SCDFNSNSHS	STFDAGAGIA	300
LNDNFVKLIS	WYDNEYGYSN	RVVDLMAYMA	SKE			333

SEQ ID NO: 12

FEATURE
source

moltype = AA length = 284
Location/Qualifiers
1..284
mol_type = protein
organism = Mus musculus

SEQUENCE: 12

MDAIKKKMQM	LKLDKENALD	RAEQAEADKK	AAEDRSKQLE	DELVSQKKL	KGTEDELDKY	60
SEALKDAQEK	LELAEEKATD	AEADVASLNR	RIQLVEEELD	RAQERLATAL	QKLEEAEEKAA	120
DESERGMKVI	ESRAQKDEEK	MEIQEIQLKE	AKHIAEDADR	KYEEVARKLV	IIESDLERAE	180
ERAELSEGKC	AELEEELKTV	TNNLKSLEAQ	AEKYSQKEDK	YEEBIKVLSD	KLKEAETRAE	240
FAERSVTKLE	KSIDDLEDEL	YAQKLKYKAI	SEELDHALND	MTSI		284

SEQ ID NO: 13

FEATURE
source

moltype = AA length = 1136
Location/Qualifiers
1..1136
mol_type = protein
organism = Mus musculus

-continued

SEQUENCE: 13

MPEAKPAKK	ASKGKDAPKE	APAKQTPEEP	PKEAPPEDQS	PTAEPTGIF	LKKPDSVSVE	60
TGKDAVILAK	VNGKELPGKP	TIKWFKGKWQ	ELGSKSGARF	IFKESHSTST	NVYTVELHIG	120
KVVLGDRGDY	RLEIKAKDVC	DSCSFNVDVE	APRQDSSGQS	LESFKRSGDG	KSEDAGELDF	180
SGLKKREVV	EEEEKKKKDD	DDLGIPEIWI	ELLKGAKKSE	YEKIAFQYGI	TDLRGMKLRL	240
KKAKVEVKKS	AAFTKKLDPA	YQVDRGNKIK	LVVEISDPDL	PLKWFKNQGE	IKPSSKYVFE	300
NVGKKRILTI	NKCTLADDA	YEVAVQDEK	FTELFVKEPP	VLIVTPLEDQ	QVFGVDRVEM	360
SVEVSEEGAQ	VMWMKDGVM	TREDSYKARY	RPKKDGKRHI	LIYSDVAQED	GGRYQVITNG	420
GQCEAELIVE	EKQLEVLQDI	ADLTVKAAEQ	AVFKCEVSDE	KVTGKWYKNG	VEVRPSKRIT	480
ISHVGRFHKL	VIDDVRPEDE	GDYTFVPDGY	ALSLSAKLNF	LEIKVEYVVK	QEPKIHLDL	540
SGKTSDNSIV	VVAGNKLRLD	VAITGEPPT	ATWLRGDEVF	TATEGRTHIE	QRPDCSSFVI	600
ESAERSDEGR	YTIKVTNPVG	EDVASIFLRV	VDVDPPEAV	RVTSGEDWA	ILVWEPKYD	660
GQQPVTGYLM	ERKKKGSRW	MKINFEVFTD	TTYESTKMIE	GVLYEMRVFA	VNAIGVSQPS	720
MNTKPFMPIA	PTSAPQHLTV	EDVTDTTTTL	KWRPPDRIGA	GGIDGYLVEY	CLEGSEEWVP	780
ANKEPVERCG	FTVKDLPTGA	RILFRVVGVN	IAGRSEPATL	LQPVITREIV	EQPKIRLPRH	840
LRQTYIRKVG	EALNLVIPPQ	GKPRPQVVT	KGGAPLDTSR	VNVRTSDFDT	VFFVRQAARS	900
DSGEYELSVQ	IENMKDTATI	RIRVVEKAGP	AENVVMKEVW	GTNALVEWQP	PKDDGNSEIT	960
GYFVQKADK	TMWFWNYEH	NRHTSCTVSD	LIVGNEYFR	IFSENICGLS	DSPGVSKNTA	1020
RLKGTGITLK	PLEYKEVDEH	TAPKFLTPLM	DRVVAGYTA	ALNCAVRGHP	KPKVVMKKNK	1080
MEIHEDPKFL	ITNYQGILTL	NIRRPSPFDA	GTYSCLAPNE	LGEALAECKL	DVRVPQ	1136

SEQ ID NO: 14

moltype = AA length = 1270

FEATURE

Location/Qualifiers

source

1..1270

mol_type = protein

organism = Mus musculus

SEQUENCE: 14

MPEGKKPV	AFNKKPRSAE	VTAGSAVFE	AETERSGMV	RWQRDGSIT	ANDKYGLAAE	60
GKRHTLTVRD	ASPDQGSYA	VIAAGSKVKF	DLKVTEPAPP	EKAESVAPG	APEEVPAPAT	120
ELEESVSSPE	GSVSVTQDS	AAEHQGAPDD	PIGLFLMRPQ	DGEVTVGGS	VFSAVAGAS	180
LLKPPVVKWF	KGKWDLSK	VGQHLQLHDS	YDRASKVYLF	ELHITDAQTT	SAGGYRCEVS	240
TKDKFDSCNF	NLTVEHAIGS	GDLRLRSAPR	RTSLAGAGRR	TSDSHEDAGT	PDFSSLLKKR	300
DSFRDLSKLE	APAEEDVWEI	LRQAPPSEYE	RIAPQHGVEA	CHRLPKRLKG	MKQDEKKSTA	360
FQKLEPAYQ	VNKGKIRLT	VELADPDAEV	KWLKNGQEIQ	MSGSKYIFES	VGAKRTLITIS	420
QCSLADDAAY	QCVVGEKCS	TELFVKEPPV	LITRSLEDQL	VMVGQVFEFE	CEVSEEGAQV	480
KWLKDGVELT	REETFKYRFP	KDGRKHHLLI	NEATLEDAGH	YAVRTSGGQS	LAEILIVQEKK	540
LEVYQSIADL	AVGAKDQAVF	KCEVSDENVR	GVWLKNGKEL	VPDNRIKVS	IGRVHKLITD	600
DVTPADEADY	SFVPEGFACN	LSAKLHFMEV	KIDFVPRQEP	PKIHLDPCGS	TPDTIVVVTG	660
NKLRDLVPIS	GDPAPTVMWQ	KTVTQGGKAS	AGPHDPAPED	AGADEEWVFD	KKLLCETEGR	720
VRVETTKDRS	VFTVEGAKE	DEGVYTVTVK	NPVGEDQVNL	TVKVIDVDA	PAAPKISNVG	780
EDSCTVQWEP	PAYDGGQPV	GYILERKKKK	SYRWMRLNFD	LLRELSHEAR	RMIEGVAYEM	840
RVYAVNAVGM	SRSPASQPF	MPITGPPGEP	HLAVEDVSDT	TVSLKWRPPE	RVGAGGLDGY	900
SVEYQEGGCS	EWTPALQGLT	ERRSMLVKDL	PTGARLLFRV	RAHNVAAGPG	PIVTKPEPVT	960
QEILQRPRLO	LPRHLRQTIQ	KKVGEPVNLL	IPFQGGKPRPQ	VWTWKEGQPL	AGEEVSIRNS	1020
PTDTILFLRA	ARRTHSGTYQ	VTVRIENMED	KATLILQIVD	KPSPQDRI	VETWGFNVAL	1080
EWKPPQDDGN	TEIWGYTVQK	ADKKTMEWFT	VLEHYRRTHC	VVSELIINGG	YYFRVFSHNM	1140
VGSSDKAAAT	KEPVFIPRPG	ITYEPPKYKA	LDFSEAPSFT	QPLANRSIIA	GYNAILCCAV	1200
RGSPPKISW	FKNGLDLGED	ARFRMFCKQG	VLTLTIRKPC	PYDGGVYVCR	ATNLQGEAQC	1260
ECRLEVRVPQ						1270

SEQ ID NO: 15

moltype = AA length = 1463

FEATURE

Location/Qualifiers

source

1..1463

mol_type = protein

organism = Mus musculus

SEQUENCE: 15

MSLVAVPFYQ	KRHKHFDQSY	RNIQTRYLLD	QYALKKQATT	QSSSQRSLTE	RSSSKRASSQ	60
SSAGAMTCRL	CAKRVASASE	EEVENENRYR	SQAASYGEAK	RQRFLSELAQ	LEENVHLARS	120
QARDKLDKYF	MEQTVEDNLA	WERHSFEDRM	SRAPEILVRL	RSHTIWERM	VRLCFTVQGF	180
PTPVVQWYKD	GSLICQAGEP	GKYRIESRYG	VHTLEINRAN	FEDTATYSAV	ATNSHGQVST	240
NAAVVRRYR	GEIEPFHSVG	LPIGLPLSSV	IPYTHFDVQF	LEKFGVTFRR	EGETVTLKCT	300
LLVTPDLKRV	QPRAEWYRDD	VLLKESKWK	MFFGEGQASL	SFSLNKKDDE	GLYTLRIVSR	360
GGVSDHSAPM	FVRDADPLVT	GAPGAPMDLQ	CHDANRDYVI	VTWKPPNTTT	ESPVIGYFID	420
KCEVGTNNWV	QCNDAPVKIC	KYPVTGLFEG	RSYVFRVRAV	NNAGISRPSR	ISDAVAALDP	480
VDLRLRLQAIH	LEGEKEIVYI	QDDLEGDVQI	PGPPTNVQAS	EVSRYNVVLS	WDPPSPRGKD	540
PLMYFIEKSA	VGSGSWQVRN	AQTAVRSAPR	AVFDLAEGKS	YVFRVLSANK	HGLSDPSEIT	600
PPIQAQDMIV	VPSAPGRVLA	SRNTKTSVVV	QWDRPKHEED	LLGYVDDCCV	AGTNMWEPEN	660
HKPIGYNRV	VHGLTTGEQY	IFRVKAVNAV	GTSENSQESE	VIKQAAALT	PSHPYGITLL	720
NCDGHSMTLG	WKVPKFSGGG	AIIGYYLDR	EVHHKNWHEV	NSSPVKERIL	TVEGLTEGSL	780
YEFKIAATNL	AKIGQSPDPS	EHFKCEAWTD	PEPGPAYDLT	FCEVRDTSLV	ILWKAPVYSG	840
SSPVSGYFVD	FKGSDSGEWK	TTSEAAATPNR	YLVKCDLQOG	KTYVFRIRAV	NASGPGKPSD	900
TSEPVLVEAR	PGTKEISAGV	DEBGNILGFI	DCQEMTDASQ	FTWCKAYEEI	ADEERFEVHT	960
EGDHSKLYFK	NPDKEDLGT	SVSVSDTDGV	SSSFVLDEEE	LERLMALSNE	IKNPTIPLKS	1020
ELAYEIFDKG	QVRFWLQAEH	LSPDASFRFI	INDREVSDSD	THRIKCDRST	GMIEMVMDRF	1080
TIENEGTYTV	QIQDGKAKNQ	SSLVIGDAPF	KAVLEEAEPQ	RKEFLRKQGP	HFAEYLHWDV	1140
TECEVRLVC	KVANTKRETV	FKWLKDDVLY	ETETPPDLEK	GVCELLIPKL	SKKDHEGYKA	1200

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TLKDDRGGQDV	SVLEVGKKVY	EDMILAMSRV	CGASASPLKV	LCTPEGIRLQ	CFMKYFTEEM	1260
KVSWYHKEAK	ISSSEHMRIG	GSEEMAWLQI	CEPTEKDKGK	YTFEIPDGKD	SHQRSLDLSG	1320
QAFDEAYAEF	QQLKAAFAE	KNRGKVIIGL	PDVVTIMEGK	TLNLTCTVFG	NPDPVVVWFK	1380
NDKDIELSEH	FLVKMEQSKY	VSLTIQGVTA	EDSGKYSINV	KNKYGGEEKID	VTVSVYKHGE	1440
KIPDISPPQQ	AKPKLIPAST	SSD				1463

SEQ ID NO: 16 moltype = AA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 16

MKWVTFLLLL	FVSGSAFSRG	VFRREAHKSE	IAHRYNDLGE	QHFKGLVLIA	FSQYLQKCSY	60
DEHAKLVQEV	TDFAKTCVAD	ESAANCDKSL	HTLFGDKLCA	IPNLRENYGE	LADCCTKQEP	120
ERNECFLOHK	DDNPSPLPFE	RPEAEAMCTS	FKENPTTFMG	HYLHEVARRH	PYFYAPELLY	180
YAEQYNEILT	QCCAEADKES	CLTPKLDGVK	EKALVSSVRQ	RMKCSSMQKF	GERAFKAWAV	240
ARLSQTFPNA	DFAEITKLAT	DLTKVNKECC	HGDLLECAAD	RAELAKYMCE	NQATISSKLQ	300
TCCDKPPLLK	AHCLSEVEHD	TMPADLPPIA	ADFVEDQEV	KNYAEAKDVF	LGTFLYEYSR	360
RHPDYSVSL	LRLAKKYEAT	LEKCCAEANP	PACYGTVLAE	FQPLVEEPKN	LVKTNCDLYE	420
KLGEYGFQNA	ILVRYTQKAP	QVSTPTLVEA	ARNLGRVGTK	CCTLPEDQRL	PCVEDYLSAI	480
LNRCVLLHEK	TPVSEHVTKC	CSGSLVERRP	CFSALTVDET	YVPKEFKAET	FTFHSDICTL	540
PEKEQIKKQ	TALAEVLKHK	PKATAEQKLT	VMDDFAQFLD	TCCAADKDT	CFSTEGPNLV	600
TRCKDALA						608

SEQ ID NO: 17 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 17

MAMQKIFARE	ILDSRGNTPT	EVDLHTAKGR	FRAAVPSGAS	TGIYEALCLR	DGDKARYLGK	60
GVLKAVEHIN	KTLGPALLEK	KLSVVDQEKV	DKFMIELDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVPL	YRHIADLAGN	PDLVLPVPAP	NVINGGSHAG	NKLAMQEFMI	LPVGASSFKE	180
AMRIGAEVYH	HLKGVIKAKY	KDATNVGDE	GGFAPNILEN	NEALELLKTA	IQAAGYDPKV	240
VIGMDVAASE	FYRNGKYDLD	FKSPDDPARH	ISGEKLGEY	KNFIQNYPPV	SIEDPPDQDD	300
WATWTSFLSG	VDIQIVGDDL	TVTNPKRIAQ	AVEKKACNCL	LLKVNQIGSV	TESIQACKLA	360
QSNQGWGMVS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEALGDK	420
AVFAGRKFRN	PKAK					434

SEQ ID NO: 18 moltype = AA length = 169
 FEATURE Location/Qualifiers
 source 1..169
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 18

MAPKKAKRRA	GAEGSSNVFS	MFDQTQIQEF	KEAFTVIDQN	RDGIIDKEDL	RDTFAAMGRL	60
NVKNEELDAM	MKEASGPINF	TVFLTMFGKE	LKGADPEDVI	TGAFKVLDPE	GKGTIKKQFL	120
EELLTTQCDR	FSQEEIKNMW	AAPFPDVGGN	VDYKNICYVI	THGDAKDQE		169

SEQ ID NO: 19 moltype = AA length = 574
 FEATURE Location/Qualifiers
 source 1..574
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 19

MSVQENELPQ	QLWPWFKFSQ	KDLAKSALSG	APGGPAGYLR	RASVAQLTQE	LGTAFFQQQQQ	60
LPAAMADTFL	EHLCLLDIDS	EPVAARSTSI	IATIGPASRS	VDRLEKEMIK	GMNIARLNFS	120
HGSHEYHAES	IANIREAAES	FATSPLSYRP	VAIALDTKGP	EIRTVGLQGG	PESEVEIVKG	180
SQVLTVVDPK	FRTRGDAKT	VWDYHNITQV	VAVGGRIYID	DGLISLVVRK	IGPEGLVTEV	240
EHGGFLGNRK	GVNLPNAEVD	LPGLSEQDLL	DLRFGVEHYV	DIIFASFVRK	ASDVAVVRDA	300
LGPEGRGIKI	ISKIENHEGV	KKFDEILEVS	DGIMMARGDL	GIEIPAEKVF	LAQKMMIGRC	360
NLAGKPVVCA	TQMLESMITK	ARPTRAETSD	VANAVLDGAD	CIMLSGETAK	GSFPVEAVKM	420
QHAIAREAEA	AVYHRQLFEE	LRRAAPLSRD	PTEVTAIGAV	EASFKCCAAA	IIVLTKTGRS	480
AQLLSRYRPR	AAVIAVTRSA	QAARQVHLSR	GVFPLLYREP	PEAVWADDVD	RRVQFGIESG	540
KLRGFLRVGD	LVIIVTGWPR	GSYTNIMRV	LTIS			574

SEQ ID NO: 20 moltype = AA length = 260
 FEATURE Location/Qualifiers
 source 1..260
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 20

MAKEWGYASH	NGPDHWHELY	PIAKGDNQSP	IELHTKDIKH	DPSLQPWSAS	YDPGSAKTIL	60
NNGKTCRVVF	DDTYDRSMRL	GGPLSGPYRL	RQFHLHWGSS	DDHGSEHTVD	GVKYAAELHL	120
VHWNPKYNTF	GEALKQPDGI	AVVGIFLKIG	REKGEFQILL	DALDKIKTKG	KEAPFTHFDP	180
SCLFPACRDY	WTYHGSFTTP	PCEECIVWLL	LKEPMTVSSD	QMAKLRSLFS	SAENEPVPL	240

-continued

VGNWRPPQPV KGRVVRASFK 260

SEQ ID NO: 21 moltype = AA length = 332
 FEATURE Location/Qualifiers
 source 1..332
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 21

MATLKDQLIV	NLLKEEQAPQ	NKITVVGVGA	VGMACAISIL	MKDLADELAL	VDVMEDKLKG	60
EMMDLQHGSL	FLKTPKIVSS	KDYCVTANSK	LVIIITAGARQ	QEGESRLNLV	QRNVNIFKFI	120
IPNIVKYSPL	CKLLIVSNPV	DILTYVAVKI	SGFPKNRVIG	SGCNLDSARF	RYLMGERLGV	180
HALSCHGWVL	GEHGDSSVPV	WSGVNVAGVS	LKSLNPELGT	DADKEQWKEV	HKQVVDSEAY	240
VIKLKGYTSW	AIGLSVADLA	ESIMKNLRRV	HPISTMIKGL	YGINEDVFLS	VPCILGQNGI	300
SDVVVKVTLTP	EEEARLKKA	DTLWGIQKEL	QF			332

SEQ ID NO: 22 moltype = AA length = 1942
 FEATURE Location/Qualifiers
 source 1..1942
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 22

MSSDAEMAVF	GEAAPYLRSK	EKERIEAQNR	PFDAKTSVFV	AEPKESFVKG	TIQSKDAGKV	60
TVKTEAGATL	TVKEDQIPPM	NPPKYDKIED	MAMMTHLHEP	AVLYNLKERY	AAWMIYTYSG	120
LPCVTVPYK	WLPVYNPEVV	AAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGDKKKEEA	TSGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFQKF	IRIHFGTTGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QITSNKKPEL	300
IEMLLITNTP	YDYPFVSQGE	ISVASIDDQE	ELMATDSAID	ILGFTNDEKV	SIYKLTGAVM	360
HYGNMKFKQK	QREEQAEPDG	TEVADKAAYL	QGLNSADLLK	ALCYPRVKVG	NEYVTGQQT	420
EQVTNAVVAL	AKAMYEKMFL	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNFSLQELC	480
INFTEKLLQ	FFNHMFVLE	QEEYKKEGIE	WTFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	FKNKLYEQHL	GKSANFQKPK	VVKGAEAHF	SLIHYAGTVD	YNTGWLDKN	600
KDPLNETVVG	LYQSSSVKTL	AYLFSGAQTA	EAEASSGGAA	KKGAKKKGSS	FQTVSALFRE	660
NLNKMLTNLR	STPHFVRCI	IPNETKTPGA	MEHELVLHQL	RCNGVLEGIR	ICRKGFPSPRI	720
LYADFKQRYK	VLNASAIPEG	QYIDSKKASE	KLLGSIDIDH	TQYKFGHTKV	FFKAGLLGLL	780
EEMRDDKLAQ	LITRTQAMCR	GFLARVEYQK	MVERRESIFC	IQYNIAPFMN	VKHWPWMKLF	840
FKIKPLLKSA	ETEKEMATMK	EEFQKTKDDL	AKSEAKRKEL	EEMVSLLEKE	KNDLQLQVQA	900
EAEGLADAEE	RCDQLIKTKI	QLEAKIKEVT	ERADEEEIN	AELTAKKRKL	EDECSELKKD	960
IDDLLETLAK	VEKEKHATEN	KVKNLTEEMA	GLDETIAKLT	KEKKALQEAH	QQTLDLQAE	1020
EDKVNLTAK	KIKLEQQVDD	LEGSLEQEKK	LRMDLERAKR	KLEGDLKLAQ	ESIMDIENEK	1080
QQLDERLKKK	EFEMSLQSHL	IEDEQAIGIQ	LQKKIKELQA	RIEELLEEIE	AERASRAKAE	1140
KQRSDLREL	EEISERLEE	GGATSAQIEM	NKKREAEFQK	MRRDLLEATL	QHEATAATLR	1200
KKHADSVDEL	GEQIDNLRV	KQKLEKEKSE	MKMEIDDLAS	NVETVSKAKG	NLEKMCRTLE	1260
DQVSELKSE	EEQQLINDL	TTQGRGLQTE	SGEFSRQLDE	KEALVSQLSR	GKQAFQQQIE	1320
ELKRQLEEEV	KAKNALAHAL	QSSRHDCDLL	REQYEEEQES	KAEALQALSK	ANSEVAQWRT	1380
KYETDAIQT	EELEEAQKKL	AQRLQAAEEH	VEAVNAKCA	LEKTKQRLQN	EVEDLMLDVE	1440
RTNAACAALD	KKQNFDPKIL	AEWKQKYEET	HAELEASQKE	ARSLGTELFK	MKNAYEESLD	1500
QLETILKREN	NLQQEISDLT	EQIAEGGKRI	HELEKIKKQV	EQEKCELQAA	LEEAEASLEH	1560
EEGKILRIQL	ELNQVKSEID	RKIAEKDEEI	DQLKRNRHVR	VESMQSTLDA	EIRSRNDAIR	1620
IKKKMEGDLN	EMETQLNHSN	RMAAEALRNY	RNTQGILKDT	QLHLDDALRG	QEDLKEQLAM	1680
VERRANLLQA	EIEELRATLE	QTERSRIKAE	QELLDASERV	QLLHTQNTSL	INTKKKLETD	1740
ISQIQGEMED	IVQEARNAEE	KAKKAITDAA	MAAEELKKEQ	DTSAHLERMK	KNMEQTVKDL	1800
QLRLDEAEAF	ALQVKKKIQI	KLEARVRELE	GEVESEQKRN	AEAVKGLRKH	ERRVKELTYQ	1860
TEEDRKNIIL	LQDLVDKLQA	KVKSQYKQAE	EABEQSNTNL	SKFRKIQHEL	EEAEERADIA	1920
ESQVKNLRVK	SREVTTKIIS	EE				1942

SEQ ID NO: 23 moltype = AA length = 1124
 FEATURE Location/Qualifiers
 source 1..1124
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 23

MPEPTKKEEN	EVSAPAPPPPE	EPNKEKEAGT	APAKEWSLGE	SPAGGEEQDK	QNANSQLSTL	60
FVEKPTQTSV	KVGANITFVA	KVRAEDLLRK	PTVKWFKGKW	MDLASKAGKH	LQLKETFERQ	120
TRVYTFEMQI	IKAKENYAGN	YRCEVTYKDK	FDSCSFDLEV	HESTGTTFNI	DIRSAFKRSG	180
EGQEDAGELD	FSGLLKRRREV	KQQEEEPEID	VWELLKNANP	NEYEKIAFQY	GITDLRGLMK	240
RLKRMRRVKE	KSAFAKILDL	PAYQVDKGGK	VRFVVELADP	KLEVWKFNG	QEIRPSTKYI	300
FEHKGNERNI	FINNCSLTDD	SEYYVTAGDE	KCSTELFVRE	PPIMVTKQLE	DMNAYCGERV	360
EMEVEVSEDD	ANVWFKNGE	EIFPGPKSRY	KIKVEGKKHT	LIIEGATKAD	SAEYSAMTGT	420
GQSSAKLSVD	LRPLKITPTL	TDQTVKLKKE	VCLKCEISEN	VPKGWTKNGL	PVQEGERLKV	480
VHKGRILKLV	TANALIEDEG	EYVFTPDAYN	VPLSAKVHVI	DPPIKILDGL	DADNTVTVIA	540
GSKLRLEIPV	GEPPPKAIW	SRADKAIMEG	SGRIRAESYP	DSSTLVIDVA	ERDDSGVYNI	600
NLKNEAGEAH	ASIKIKVVDI	PDPVPAPNVT	EVGDDWCIMN	WEPPAYDGGS	PILGYFIERK	660
KKQSSRWMLR	NFDLCKETTF	EPKKMIEGVA	YEVRIFAVNA	IGISKPSMPS	KPFVPLAVTS	720
PPTLLAVDSV	TDSSVTMKWR	PPDQIGAAGL	DGYVLEYCFE	GTEDWIVANT	DLIDKTKFTI	780
NGLPDIAKIF	VRVKAINAAG	ASBPKYYSQP	ILVKEIIEPP	KIRIPRHLKQ	TYIRRVGEAV	840
NLVIPFQKGP	RPELTWKKDG	EEIDKNQINI	RNSETDTIIF	IRKAERSHSG	KYDLQVKVDK	900

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YVENASIDIQ	IVDRPGPPQT	VTIEDVWGEN	VALTWTPPKD	DGNAAITGYT	IQKADKKSME	960
WFTVIEHYHR	TNATITELVI	GNEYYFRVFA	ENMCGLSEDA	TMTKESAVIA	KDGKIYKNPV	1020
YEDFNFTTEAP	MFTQPLVNTY	AIAGYNATLN	CSVGRGNPKPK	ITWMKNKVAI	VDDPRYRMFS	1080
NQGVCTLEIR	KPSPYDGGTY	CCKAVNDLGT	VEIECKLEVK	VVAQ		1124

SEQ ID NO: 24 moltype = AA length = 562
 FEATURE Location/Qualifiers
 source 1..562
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 24

MVKIVTVKTQ	AYPDQKPGTS	GLRKRKVFQ	SNANYAENFI	QSIVSTVEPA	LRQEATLVVG	60
GDGRFYMTAA	IQLIVRIAAA	NGIGRLVIGQ	NGILSTPAVS	CIIRKIKAI	GIILTASHNP	120
GGPNGDFGIK	FNISNGGPAP	EAITDKIFQI	SKTIEEYAI	PDLKVDLGLV	GKQQFDLENK	180
FKPFTVEIVD	SVEAYATMLR	NIPDFNALKE	LLSGPNRLKI	RIDAMHGVVG	PYVKKILCEE	240
LGAPANSAVN	CVPLEDFGGH	HPDPNLTYAA	DLVETMKSGE	HDFGAAPDGD	GDRNMILGKH	300
GGFVNPSDSV	AVIAANIFSI	PYFQQTGVRG	FARSMPTSGA	LDRVANATKI	ALYETPTGWK	360
FFGNLMDASK	LSLCGEESFG	TGSDHIREKD	GLWAVLAWLS	ILATRKQSVE	DILKDHQKQF	420
GRNFPTRYDY	EEVEAEGANK	MMKDLEALML	DRSFVGKQFS	ANDKVYVEK	ADNFEYSDPV	480
DGSISKNGQL	RLIFADGSRI	IFRLSGTGSA	GATIRLYIDS	YEKDVAKINQ	DPQVMLAPLI	540
SIALKVSQLQ	ERTGRTAPT	IT				562

SEQ ID NO: 25 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 25

MDAIKKKMQM	LKLDKENAID	RAEQAEADKK	QAEDRCKQLE	EEQQALQKKL	KGTEDEVEKY	60
SESVKDAQEK	LEQAEKKATD	AEADVASLNR	RIQLVEEELD	RAQERLATAT	QKLEEAEEKAA	120
DESERGMKVI	ENRAMKDEEK	MELQEMQLKE	AKHIAEDSDR	KYEEVARKLV	ILEGELERSE	180
ERAEVAESKC	GDLEELKIV	TNNLSLEAQ	ADKYSTKEDK	YEEBIKLLEE	KLKEAETRAE	240
FAERSVAKLE	KTIDDLDEV	YAQMKYKAI	SEELDNALND	ITSL		284

SEQ ID NO: 26 moltype = AA length = 249
 FEATURE Location/Qualifiers
 source 1..249
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 26

MAPTRKFFVG	GNWKMNGRKK	CLGELICTLN	AANVPAGTEV	VCAPPTAYID	FARQKLDPKI	60
AVAAQNCYKV	TNGAFTGEIS	PGMIKDLGAT	WVVLGHSERR	HVFGESDELI	GQKVSHALAE	120
GLGVIAICIG	KLDEREAGIT	EKVVFQTKV	IADNVKDWK	VVLAYEPVWA	IGTGKTATPQ	180
QAQEVHEKLR	GWLKSNVNDG	VAQSTRIIYG	GSVTGATCKE	LASQPDVDGF	LVGGASLKPE	240
FVDIINAKQ						249

SEQ ID NO: 27 moltype = AA length = 253
 FEATURE Location/Qualifiers
 source 1..253
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 27

MTTHRLVMVR	HGESLWNQEN	RFCGWFDAEL	SEKGAEAAKR	GATAIKDAKI	EPDICYTSVL	60
KRAIRTLWTI	LDVTDQMWVP	VVRTWRLNER	HYGGLTGLNK	AETAAKHGEE	QVKIWRSPD	120
TPPPPMDEKH	NYTTSISKDR	RYAGLKPEEL	PTCESLKDTI	ARALPFWNNE	IAPKIKAGQR	180
VLIAAHGNSL	RGIVKHLEGM	SDQAIMELNL	PTGIPIVYEL	DQNLKPTKPM	RFLGDEETVR	240
KAMEAVAQGG	KAK					253

SEQ ID NO: 28 moltype = AA length = 472
 FEATURE Location/Qualifiers
 source 1..472
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 28

MKALLLLCCF	LASLLLSGQA	EVEDASEEAP	LRDRSHIDKT	LMLNEDKPAD	DYSAVLQRLR	60
KIYHTSIKPL	BQSYKYNELR	QHEITDGEIT	SKPMVLFLGP	WSVGKSTMIN	YLLGLEDRY	120
QLYTGAEPPT	SEPTVLMHGP	KLKTIEGIVM	AADSARSFSP	LEKFGQNPLE	KLIGIEVPHK	180
LLERVTFVDT	PGIENRKQ	ERGYPFNDVC	QWFIDRADLI	FVVFDPTKLD	VGLELEMLFR	240
QLKGRESQIR	IILNKADNLA	TQMLMRVYGA	LFWSLAPLIN	VTEPPRVYVS	SFWPQDYKPD	300
THRELFLKEE	ISLLEDLNQV	IENRLENKIA	FIRQHAIRVR	IHALLVDRYL	QTYKDKMTFF	360
SDGELVFKDI	VEDPDKFYIF	KTILAKTNVS	KFDLPNREAY	KDFFGINPIS	NFKLLSQQCS	420
YMGCGFLEKI	ERAITQELPS	LLGSIGLGKN	PGAPNCDKTG	CGETPKNRYK	KH	472

SEQ ID NO: 29 moltype = AA length = 417
 FEATURE Location/Qualifiers
 source 1..417

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mol_type = protein
organism = Mus musculus

SEQUENCE: 29

MSLSNKLTLTD	KLDVKGKRVV	MRVDFNVPMK	NNQITNNQRI	KAAPPSIKFC	LDNGAKSVVL	60
MSHLGRPDGV	PMPDKYSLEP	VAAELKSLLG	KDVLFLKDCV	GPEVENACAN	PAAGTVILLE	120
NLRFHVHEEG	KGKASGNKV	KAEPKIDAF	RASLSKLGDV	YVNDAFGTAH	RAHSSMVGVN	180
LPQKAGGFML	KKELNYFAKA	LESPERFPLA	ILGGAKVADK	IQLINNMLDK	VNEMIIGGGM	240
AFTFLKVLNN	MEIGTSLYDE	EGAKIVKDLN	SKAEKNGVKI	TLPVDFVTAD	KFDENAKTGQ	300
ATVASGIPAG	WMGLDCGTES	SKKYAEAVGR	AKQIVWNGPV	GVFEWEAFAR	GTKSLMDEVV	360
KATSRGCITI	IGGGDTATCC	AKWNTEDEVK	HVSTGGGASL	ELLEKGVLPG	VDALSNV	417

SEQ ID NO: 30

moltype = AA length = 4691

FEATURE

Location/Qualifiers

source

1..4691

mol_type = protein

organism = Mus musculus

SEQUENCE: 30

MVAGMLMPLD	RLRAIYEVLF	REGVMVAKKD	RRPRSLHPHV	PGVTNLQVMR	AMASLKARGL	60
VRETFAWCHF	YWYLTNEGID	HLRQYLHLPP	EIVPASLQRV	RRPVAMVIPA	RRRSPHVQTM	120
QGGLGCPKPK	GPLPAEDPAR	EERQVYRRKE	REEGAPETPV	VSATTVTGTLA	RPGPPEPAPAT	180
DERDRVQKKT	FTKWNVNHLI	KHWRAEAQRH	ISDLYEDLRD	GHNLISSLEV	LSGDSLPREK	240
GRMRPHKLQN	VQIALDYLKH	RQVKLVNIRN	DDIADGNPKL	TLGLIWTIIL	HFQISDIQVS	300
GQSEDMTAK	KLLLSQRMV	EGYQGLRCDN	FTTSWRDGR	FNAIHRHKP	MLIDMNKVYR	360
QTNLENLDQA	FSVAERDLGV	TRLDDPEDVD	VQPDEKSI	TYVSSLYDAM	PRVPGAQDGV	420
RANELQLRWQ	EYRLVLVLLL	QWIRHHTAAF	EERKPPSSFE	EIELWCQFL	KFKETELPAK	480
EADKNRSKVI	YQSLEGAVQA	GQLKIPPGYH	PLDVEKEWVK	LHVAILEREK	QLRSEFERLE	540
CLQRIVSKLG	MEAGLCCEQL	NQADALLQSD	IRLLASGKVA	QRAGEVERDL	DKADGMIRLL	600
FNDVQTLKDG	RHPQGEQMYR	RVYRLHERLV	AIRTEYNLRL	KAGVGAPVTQ	VTLQSTQRRP	660
ELEDSTLRYL	QDLLAWVEEN	QRRIDSAEWG	VDLPSVEAQL	GSHRGMHQSI	EEFRAKIERA	720
RNDESQLSPA	TRGAYRDCLG	RDLQYAKLL	NSSKARLSL	ESLHGFVAAA	TKELMWLNEK	780
EEEEVGFDWS	DRNTNMAAKK	ESYSALMREL	EMKEKKIKEI	QNTGDRLLRE	DHPARPTVES	840
FQAALQTQWS	WMLQLCCCTE	AHLKENTAYF	QFFSDVREAE	EQLQKLQETL	RRKYSCDRTI	900
TVTRLEDLLQ	DAQDEKEQLN	EYKGHLSGLA	KRAKAIIVLK	PRNPAHPVRG	HVPLIAVCYD	960
KQVEVTVHHG	DQCVLVGPAQ	PSHWKVLSGS	SSEAAVPSVC	FLVPPPNQEA	QEAVALLEAQ	1020
HQALVTLWHQ	LHVDMKSLLA	WQSLSRDIQL	IRSWSLVTFR	TLKPPEQRQA	LRNLELHYQA	1080
FLRDSQDAGH	FGPEDRLVAE	REYGCSSRHY	QQLQSLQEQG	EQEESRCQRC	ISELKDIRLQ	1140
LEACETRTVH	RRLPLDKLDP	ARECAQRIAE	QQAQAQAEV	LGKGVARLSA	EAEKVLALPE	1200
PSPAAPTILRS	ELELTGKLE	QVRSLSAIYL	EKLKTISLVI	RSTQGAEEVL	KTHEEQLKEA	1260
QAVPATLQEL	EATKASLKKL	RAQAEAAQPV	FNTLRDELRG	AQEVGERLQQ	RHGERDVEVE	1320
RWRERVTLQL	RWRQAVLQET	DVRQRELEQL	GRQLRYRES	ADPLSAWLQD	AKRRQEQIQA	1380
VPIANCQAAR	EQLRQEKALL	EEIERHGEKV	EECQKFAKQY	INAIKDYELQ	LITYKAQLEP	1440
VASPAKKPKV	QSGSESVIQE	YVDLTRYSE	LTTLTSQYIK	FISETLRME	EEERLAEQQR	1500
AEERERLAEV	EAALEKQVQL	AEAAQAQAKA	AELEAQELQR	RMQEVARRE	EAAVDAQQQK	1560
RSIQEELQHL	QSSSEAEIQA	KAQQVEAAER	SRMRIEEBIR	VVRLQLETTE	RQRGGAEGEL	1620
QALRAAEAEA	EAQKRQAQEE	AERLRRQVQD	ESQKRQAEAE	ELALRVKAEA	EAAREKQAL	1680
QALDELRLQA	EEAEERRLLEE	AERARQVQV	ALETAQRSAE	VELQSKRAS	AEKTAQLERT	1740
LQEEHVTVAQ	LREEAERRAQ	QQAEEAERARE	EAERELERWQ	LKANEALRLR	LQAEEVAQK	1800
SLAQADAQEK	KEEAERBARR	RGKAEQAVR	QRELAQELE	KQRLAEBGTA	QORLAEEQEL	1860
ILRLAETEGQ	EQQRQLVLEE	LARLQHEATA	ATQKRQLEEA	ELAKVRAEME	VLLASKARAE	1920
EESRSTSEKS	KQRLAEAEAGR	FRELAEEAAR	LRALAEAAKR	QRQLAEDDAA	RQRAEAERVL	1980
TEKLAASISA	TRLKTEABIA	LKEKEAENER	LRRLAEDEAF	QRRRLQEQAA	LHKADIEERL	2040
AQLRKASESE	LERKQVQVEE	TLRQRRQVEE	EIMALKVSFE	KAAAGKAELE	LELGRIRNSA	2100
EDTMRSEKEA	ELEAARQRL	AAEEEQRRRE	AEERVQRSLA	AEEEAARQK	VALEEVEERL	2160
AKVEEARRLR	ERAEQESARQ	LQLAQEAAQK	RLQAEEKAHA	FVVQREEL	QQTLLQEQNM	2220
LDRLRSEAEA	ARRAAEEAEE	AREQAEREAA	QSRKQVEEAE	RLKQSAEEQA	QAQAQAQAAA	2280
EKLKKEAEQE	AARRAQAEQA	ALKQKQADA	EMEKHKKFAE	QTLRQKAQVE	QELTTLRLQL	2340
EETDHQKSL	DEELQRLKAE	VTEAARQRSQ	VEEELFSVRV	QMEELGKLKA	RIEAEENRALI	2400
LRDKDNTQRF	LEEEAEKMKQ	VAEEAARLSV	AAQEAARLRQ	LAEDDLAQQR	ALAEKMLKEK	2460
MQAVQEATRL	KAAEALLQQQ	KELAQEQARR	LOEDKEQMAQ	QLVEETQGFQ	RTLEAERQK	2520
LEMSABAEERL	KLRMAEMSRA	QARAEEAQR	FRKQAEETGE	KLHRTTELATQ	EKVTLVQTLE	2580
IQRQSQDHD	ERLEAEIAEL	EREKEKLQEE	AKLLQLKSEE	MQTVQEQEIL	QETQALQKSF	2640
LSEKDSLLQR	ERFIEQEKAK	LEQLFQDEVA	KAKQLREEQK	RQQQMEQEK	QELMASMEEA	2700
RRRQREAEEG	VRRQEQELQH	LEQQRQQQEK	LLAEENQRLR	ERLQRLREEH	RAALAHSEIA	2760
TTQAASTKAL	PNGRDAPDGP	SVEAPEPYTF	EGLRQKVPAP	QLQEAGILSQ	EELQRLAQGH	2820
TTVAELTQRE	DYRYLYLGRS	SIAGLLLKPT	NEKLSVYTAL	QRQLLSPGTA	LILLEAQAA	2880
GFLLDPVNRN	RLTVNEAVKE	GVVGPELHHK	LLSAERAVTG	YKDPYTGEQI	SLFQAMKKDL	2940
IVRDHGVRL	EAQIATGGII	DPVHSHRVPV	DVAYKRGYFD	EEMNRILSDP	SDDTKGFFDP	3000
NTHENLTYLQ	LLERCVEDPE	TGLRLPLPLD	KAAKGGLVY	TDTEARDVFE	KATVSAPFGK	3060
FQGRVTIWE	IINSEYFTAE	QRRDLLQQFR	TGHITVEKII	KIVITVVEEH	ERKQQLCFEG	3120
LRALVPAEEL	LDGVIISHEL	YQQLQRGERS	VREVAEADSV	RQALRGNTVI	AGVWLEEAGQ	3180
KLISYEALKK	LLDQPEHAVA	LLEAQAGTGH	IIDPATSAARL	TVDEAVRAGL	VGPELLEKLL	3240
SAEKAVTGYR	DPYSGQSVSL	FQALKKGLIP	REQGLRLLLD	QLSTGGIVDP	SKSHRVLPLDV	3300
AYARGYLDKE	TNRALTSPRD	DARVYHDPST	QEPVTYSQLQ	QRCRSDQLTG	LSLLPLSEKA	3360
VRARQEEVYS	ELQARETLEQ	AKVEVPVGSF	KGRAMTVWEL	ISSEYFTEEQ	RQELLRQFRT	3420
GKVTVEKVIK	IVITIVEEVE	TRQERLSFS	GLRAPVPASE	LDDAKILSRA	QPDQLKDGT	3480
SVKELSEVGS	VRTLLQSGSG	LAGIYLEDSE	EKVTIYEAMR	RGLLRPSTAT	LLLEAQAAATG	3540

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FLVDPVRNQR LYVHEAVKAG VVGPELHEKL LSAEKAVTGY KDPYSGNTIS LFQAMKKGLV 3600
LRDHAIRLLE AQVATGGIID PVHSHRLPVD VAYQRGYFDE EMNRVLADPS DDTKGFFDPN 3660
THENLTYLQL LERCVEDPET GLRLLPLKGA EKTEVVETQ VYTEEETRRR FEETQIDIPG 3720
GGSHGGSSMS LWEVMQSNMI PEDQARLMA DPOAGRVTKE RMIIIIIEII EKTEIIRQQN 3780
LASYDYVRRR LTAEDLYEAR IISLETYNLF REGTKNLREV LEMESAWRYL YGTGAVAGVY 3840
LPGSRQTLTI YQALKKGLLS AEVARLLLEA QAATGFLLDP VKGERLTVDE AVRKGLVGPE 3900
LHDLRLSAER AVTYGRDPYT EQTISLFQAM KKELIPAEAA LRLLDAQLAT GGIVDPRLGF 3960
HLPLEVAYQR GYLNKTDTHDQ LSEPSEVRSY VDPSTDERLS YTQLLKRCCR DDPGQMLLL 4020
LSDARKLTFR GLRKQITVEE LVRSQVMDEA TALQLQEGLT SIEEVTKNLQ KFLEGTSCIA 4080
GVFVDATKER LSVYQAMKKG IIRPGTAFEL LEAQATGYV IDPIKGLKLT VEEAVRMGIV 4140
GPEFKDKLLS AERAVTGYKD PYSGKLISLF QAMKKGLILK DHGIRLLEAQ IATGGIIDPE 4200
ESHRLPVEVA YKRGFLDEEM NEILTDPDSD TKGFFDPNTE ENLTYLQIME RCITDPQTGL 4260
CLLPLKEKKR ERKTSKSSV RKRRVVIUDP ETGKEMSVYE AYRKLIDHQ TYLELSEQEC 4320
EWEEITISS DGUVKSMIID RRSGRQYDID DAITKNLIDR SALDQYRAGT LSITEFADML 4380
SGNAGGFRGR SSSVSGSSSY PISSAGPRTQ LASWSDPTEE TGPVAGILDT ETLEKVSITE 4440
AMHRLNVDNI TQQRLLEQA CTGGIIDPST GERFPVTEAV NKGLVDKIMV DRINLAQKAF 4500
CGFEDPRTKT KMSAAQALKK GWLYYEAGQR FLEVQYLTGG LIEPDTTPGRV SLDEALQRTG 4560
VDARTAQKLR DVSAYSXYLT CPKTKLKISY KDALDRSMVE BGTGLRLLEA AAQSSKGYYS 4620
PYSVSGSGST DSRTSGSRTG SRAGSRRGSF DATSGGSFMT FSSSSYSSSG YGRRYASGPS 4680
ASLGGPESAV A 4691

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SEQ ID NO: 31      moltype = AA length = 894
FEATURE           Location/Qualifiers
source            1..894
                  mol_type = protein
                  organism = Mus musculus

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SEQUENCE: 31
MNQIEPGVQY NYVYDEDEYM IQEEEWDRDL LLDPAWEKQQ RKTFTAWCNS HLRKAGTQIE 60
NIEEDFRNGL KMLLLEVIS GERLPKPDRG KMRPHKIANV NKALDYIASK GVKLVSIGAE 120
ETVDGNVMT LGMIWTIILR FAIQDISVEE TSAKEGLLLW CQRKTAPYRN VNIQNFTSW 180
KDGLGLCALI HRHRPDLIDY SKLNKDDPIG NINLAMEIAE KHLDIPKMLD AEDIVNTPKP 240
DERAIMTYVS CFYHAFAGAE QAETAANRIC KVLAVNQENE RLMEEYERLA SELLEWIRRT 300
IPWLENRTPE KTMQAMQKKL EDPRDYRRKH KPPKVQEKCO LEINFNTLQT KLRISNRPAF 360
MPSEGMVSD IAGAWQRLQ AEKGYEELW NEIRRLERLE HLAEKFRQKA SHETWAYGK 420
EQILLQKDYE SASLTVRAL LRKHEAFESD LAAHQDRVEQ IAAIAQELNE LDYHDAVNVN 480
DRCQKICDQW DRLGTLTQKR REALERTEKL LETIDQLHLE FAKRAAPFNN WMEGAMEDLQ 540
DMFIVHSIEE IQSLITAEHQ FKATLPEADG ERQSILAIQN EVEKVIQSYS IRISSSNPYS 600
TVTMDLRNKK WDKVKQLVPV RDQSLQEELA RQHANERLRR QFAAQANAIG PWIQNMKEEI 660
ARSSIQITGA LEDQMNQLKQ YEHNINYNKN NIDKLEGDHQ LIQEALVFDN KHTNYTMEHI 720
RVGWELLTET IARTINEVET QILTRDAKGI TQEQMNEFRA SFNHFDRRKN GLMDHEDFRA 780
CLISMGYDLG EAEFARIMTL VDPNGQGTVT FQSFIDFMTR ETADTDTAEQ VIASFRILAS 840
DKPYILAEEL RRELPPDQAO YCTKRMPPYS GPGSVPGALD YTAFSSALYG ESDL 894

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SEQ ID NO: 32      moltype = AA length = 780
FEATURE           Location/Qualifiers
source            1..780
                  mol_type = protein
                  organism = Mus musculus

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SEQUENCE: 32
MTHEEHHAAK TLGIGKAIIV LTSGGDAQGM NAAVRAVVRV GIFTGARVFF VHEGYQGLVD 60
GGEHIREATW ESVMMLQLG GTVIGSARCK DFREREGRRL AAHNLVKRGI TNLGVIGGDG 120
SLTGADTFRS EWSDLLNDLQ KDGGITAEAA TKSSYLNIVG LVGSIDNDFC GTDMTIGTDS 180
ALHRIVEIVD AITTTAQSHQ RTFVLEVMMGR HCGYALVTS LSCGADWVFI PECPPDDWE 240
EHLCRRLSET RTRGSRLNII IVAEGAIDKN GKPIITSEDIK NLVVKRLGYD TRVTVLGHVQ 300
RGGTSPAFDR ILGSRMGVEA VMALLEGTPD TPACVVSLSG NQAVRLPLME CVQVTKDVTK 360
AMDEKRFDEA IKLRGRSPMN NWEVYKLLAH VRPPVSKGGL HTVAVMNVGA PAAGMNAVR 420
STVRIGLIQG NRVLVVHDGF EGLAKGQIEE AGWSYVGGWT GQGGSKLGTK RTLPKKNLEQ 480
ISANITKFNI QGLVIIGGFE AYTGGLELME GRKQFDELICI PFVVIPATVS NNVPGSDFSI 540
GADTALNTIC TTCDRIKQSA AGTKRRVFII ETMGGYCGYL ATMAGLAAGA DAAYIFEPPF 600
TIRDLQVNVV HLVQKMKTTV KRGVLVRNEK CNENYTTDFI FNLYSEEGKG IPDSRKNVLG 660
HMQQGGSPPT FDRNFATKMG AKAMNWMGSK IKESYRNGRI FANTPDSGCV LGMRKRALVF 720
QPVTLEKQDT DFEHRIPKEQ WWLKLRLPILK ILAKYEIDLD TSDHAHLEHI SRKRSGEAAV 780

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SEQ ID NO: 33      moltype = AA length = 338
FEATURE           Location/Qualifiers
source            1..338
                  mol_type = protein
                  organism = Mus musculus

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SEQUENCE: 33
MLSALARPAG AALRRSFSTS AQNNAKVAVL GASGGIGQPL SLLKNSPLV SRLTYDIAH 60
TPGVAADLSH IETRAVVKGY LGPEQLPDCL KGCDVVVIPA GVPRKPGMTR DDLENTNATI 120
VATLTAACAQ HCPEAMVCII ANPVNSTIPI TAEVFKKHGV YNPKNIPGVT TLDIVRANTF 180
VAELKGLDPA RVNVVPIGGH AGKTIIPILS QCTPKVDFPQ DQLATLTGRI QEAGTEVVKA 240
KAGAGSATLS MAYAGARFVF SLVDAMNGKE GVVECSFVQS KETECTYFST PLLLGKKGLE 300
KNLGIGKITP FEEKMIAEAI PELKASIKKG EDFVKNMK 338

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SEQ ID NO: 34 moltype = AA length = 154
 FEATURE Location/Qualifiers
 source 1..154
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 34
 MGLSDGEWQL VLNWVGKVEA DLAGHGQEV L IGLFKTHPET LDKFDKFKNL KSEEDMKGSE 60
 DLKKHGCTVL TALGTILKKK GQHAABIQPL AQSHATKHKI PVKYLEFISE IIIEVLKKRH 120
 SGDFGADAQG AMSKALELFR NDIAAKYKEL GFQG 154

SEQ ID NO: 35 moltype = AA length = 147
 FEATURE Location/Qualifiers
 source 1..147
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 35
 MVHLTDAEKA AVSCLWGKVN SDEVGGEALG RLLVVPWTQ RYFDSFGDLS SASAIMGNAK 60
 VKAHGKKVIT AFNDGLNLHD SLKGTFFASLS ELHCDKLHVD PENFRLLGNM IVIVLGHHLG 120
 KDFTPAQAQA PQKVVAGVAT ALAHKYH 147

SEQ ID NO: 36 moltype = AA length = 2726
 FEATURE Location/Qualifiers
 source 1..2726
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 36
 MMNNSNYSDA SGLGLVDEAD EMPSTEKDLA EDAPWKIKQQ NTFTRWCNEH LKCVGKRLTD 60
 LQRDLSDGLR LIALLEVLSQL KRMVYKHFHR PNFRQMKLEN VSVALEFLER EHIKLVSDS 120
 KAIVDGNLKL ILGLIWLIL HYSISMPMWE DEDDEDARKQ TPKQRLLGWI QNKVPQLPIT 180
 NFNRDWQDGK ALGALVDNCA PGLCPDWEAW DPNQPVQNAE EAMQQADDWL GVPQVIAPEE 240
 IVDPNVDEHS VMTYLSQFPK AKLKPGAPVR SKQLNPKKAI AYGPIEPQG NTVLQPAHFT 300
 VQTVDAVGVE VLVYIEDPEG HTEEAKVVPN NDKDRTYAVS YVPKVAGLHK VTVLFAGQNI 360
 ERSPEFVNVG MALGDANKVS ARGPGLEPVG NVANKPTYFD IYTAGAGTGD VAVVIVDPQG 420
 RRDTVEVALE DKGDNTRFCT YRPVMEGPHT VHVAFAGAPI TRSPFPVHVA EACNPNACRA 480
 SGRGLQPKGV RVKEVADFKV FTKGAGSGEL KVTVKGPKGK EEPVKVREAG DGVFECEYYP 540
 VVPKGVVVTI TWGGYAIPRS PFEVQVSPEA GAQKVRWAGP GLETGQVQKS ADFVVEAIGT 600
 EVGTLGFSIE GPSQAKIECD DKGDGSCDVR YWPTPEGEYA VHVICDDEDI RDSPFIAHIQ 660
 PAPPDCFPDK VKAFPGPLEP TGCIVDRPAE FTIDARAAGK GDLKLYAQDA DGCPIDIKVI 720
 PNGDGTFRCS YVPTKPIKHT IIVSWGGINV PKSPFRVNVG EGSHPERVKV YGPGVEKTGL 780
 KANEPTYFTV DCSEAGQGDV SIGIKCAPGV VGPVEADIDF DIIKNDNDTF TVKYTPPGAG 840
 HYTIMVLFAV QEIPASPFHI KVDPSHDASK VKAEGPGLSR TGVEVGKPTH FTVLTGKAGK 900
 AKLDVHFAGA AKGEAVRDFE IIDNHDSYV VKYTAVQQGN MAVTVTYGGD PVPKSPFVVN 960
 VAPPLDLSKV KVGGLNSKVA VGEQQAQFVN TRGAGGQGLQ DVRMTSPSRR PIPCKLEPGG 1020
 GAEQAQVRVM PPEEGPYKVD ITYDGHVPVG SPFAVEGVLP PDPSKVCAYG PGLKGGVLVT 1080
 PAPFSIDTKG AGTGGLGLTV EGPCEAKIEC QDNGDGSCAV SYLPTPEGEY TINILFAEAE 1140
 IPGSPFKATI QPVDGPKSP FRVGVTEGCD ASGPGLERGK AGEAATFTVD CSEAGEAELT IEILSDAGVK 1200
 AEVLIQNNAD GTYHITYSPA FPGTYTITIK YGGHPPIKFP TRVHVQPAVD TSGIKVSGPG 1260
 VEPHGVLEVL TTEFTVDARS LTATGGMHVT ARVLNPSGAK TDTYVTNDGD GTYRVQYTAY 1320
 EEGVHLVEVL YDEVAVPKSP FRVGVTEGCD PTRVRAFGPG LEGGLVNKAN RFTVETRAG 1380
 TGGLGLAIEG PSEAKMSCKD NKDGSCCTVEY IPFTPGDYDV NITFGGQPIP GSPFRVPVKD 1440
 VVDPGVKVCS GPGLGTGVRA RVPQTFTVDC SQAGRAPLQV AVLGPTGVAE PVEVRDNGDG 1500
 THTVHTPAT DGPYTVAVKY ADQEVPRSPF KIKVLPSHDA SKVRASGPGI NASGIPASLP 1560
 VEFTIDARDA GQGLLTQVIL DPEGKPKKAN IRDNGDGYT VSYLPDMSGR YTITIKYGGD 1620
 EIPYSPFRIH ALPTGDASKD LVTVSIHGHI LGACLGPRIG IGEETVITVD AKAAGKGVK 1680
 CTVSTPDGAE LDVDVVENHD GTFDIYYTAP EPGKYVITIR FGGEHIPNSP FHVLA CDPLP 1740
 HVEEPAEMLQ MRQPYAPLRP GTCPTHWATE EPVVPVEPLE SMLRPFNLVI PFTVQKGELT 1800
 GEVRMPSGKT ARPNITDNKD GTITVRYAPT EKGLHQMGIK YDGNHIPSIP LQFYVDAINS 1860
 RHVSAYGPGL SHGMVNKPAT FTIVTKDAGE GGLSLAVEGP SKAETCKDN KDGTCTVSYL 1920
 PTAPGDYSII VRFDDKHIPG SPFTAKITGD DSMRTSQLNV GTSTDVSLKI TEGDLSQLTA 1980
 SIRAPSGNEE PCLLKRLPNR HIGISFPTKE VGEHVVSVRK SGKHVTNSPF KILVGPSEIG 2040
 DASKVRVWKG GLSEGQTFQV AEFIVDTRNA GYGGLGLSIE GPSKVDINCE DMEDGTCKVT 2100
 YCPTPEPTYI INIKFADKHV PGSPFTVKVT GEGRMKESIT RRRQAPSIAT IGSTCDLNLK 2160
 IPGNWFQMVY AQERLTFTFT RSSHTYTRTE RTEISKTRGG ETKREVRVEE STQVGGDPFP 2220
 AVFGDFLGRE LGSFSGSIR QQEGEASSQD MTAQVTSFSG KTEAAEIVEG EDSAYSVRFV 2280
 PQEMGPHTVT VKYRGQHVPG SPFQFTVGPL GEGGAHKVRA GGTGLERGVA GVPAAEFSIW 2340
 REAGAGGLSI AVEGFSKAEI AFEDRKDGSC GVSYYVQEPG DYEVSIFKND EHIPDSPFVV 2400
 PVASLSDDAR RLTVTSLQET GLKVNQPSAF AVQLNGARGV IDARVHTPSG AVEECYVSEL 2460
 DSDKHTIRFI PHENGVSID VPFNGAHIPG SPFKIRVGEQ SQAGDPGLVS AYGPGLEGGT 2520
 TGVSESEFIV TQNGSGALS VTIDGPSKVQ LDCRECPEGH VVYTPMAPG NYLIAIKYGG 2580
 PQHIVGSPFK AKVTGPRLSG GSHLHETSTV LVETVTKSSS SRGASYSIP KFSSDASKVV 2640
 TRPGLSQAF VGQKNSFTVD CSKAGTNMMH VGVHGPKTFC BEVVVKHGMN RVNVNYYTVK 2700
 EKGDIYILVK WGESVPGSP FKNVNP 2726

SEQ ID NO: 37 moltype = AA length = 780
 FEATURE Location/Qualifiers
 source 1..780

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mol_type = protein
organism = Mus musculus

SEQUENCE: 37
MAPYSLLVTR LQKALGVRQY HVASVLCQRA KVAMSHFEPS EYIRYDLEK NINIVRKRLN 60
RPLTLSEKIV YGHLDDPANQ EIERGKTYLR LRPDRVAMQD ATAQMAMLQF ISSGLPKVAV 120
PSTIHCDHLI EAQVGGEKDL RRAKDINQEV YNFLATAGAK YGVGFWRPGS GIIHQIILEN 180
YAYPGVLLIG TDSHTPNNGG LGGICIGVGG ADAVDVMAGI PWELKCPKVI GVKLTGSLSG 240
WTSPKDVILK VAGILTVKGG TGAIVEYHGP GVDSISCTGM ATICNMGAIE GATTSVFPYN 300
HRMKKYLSTK GRTDIANLAE EFKDHLVPDP GCQYDQVIEI NLNELKPHIN GPFTPDLAHP 360
VADVGTVAEK EGWPLDIRVG LIGSCTNSSY EDMGRSAVA KQALAHGLKC KSQFTITPGS 420
EQIRATIERD GYAQILRDVG GIVLANACGP CIGQWDRKDI KKGEKNTIVT SYNRNFTGRN 480
DANPETHAFV TSPEIVTALA IAGTLKFNP E TDFLTGKDGK KFKLEAPDAD ELPRSDFDPG 540
QDTYQHPPKD SSGQRVDVSP TSQRLQLLEP FDKWDGKDLE DLQILIKVKG KCTTDHISAA 600
GPWLKFRGHL DNISNNLLIG AINIENKAN SVRNAVTQEF GPVPDTARYY KKHGIRWVVI 660
GDENYGEISS REHAALPRH LGGRAITKS FARIHETNLK KQGLLPLTFA DPSDYNKIHP 720
VDKLTIQGLK DPAFGPKLKC VIKHPNGTQE TILLNHTFNE TQIEWFRAGS ALNRMKELQQ 780

SEQ ID NO: 38 moltype = AA length = 110
FEATURE Location/Qualifiers
source 1..110
mol_type = protein
organism = Mus musculus

SEQUENCE: 38
MSMTDVLSAE DIKKAIGAGA AADSFHKKF FQMVGLKKIN PDEVKKVFHI LDKDKSGFIE 60
EDELGSILKG FSSDARDLSA KETKTLAAG DKDGDGKIGV EEFSTLVAES 110

SEQ ID NO: 39 moltype = AA length = 142
FEATURE Location/Qualifiers
source 1..142
mol_type = protein
organism = Mus musculus

SEQUENCE: 39
MVLSGEDKSN IKAAGWKIGG HGAEYGAEL ERMFASFPTT KTYFPHFDVS HGSAQVKGHG 60
KKVADALASA AGHLDDLPGA LSLSDLHAH KLRVDPVNFK LLSHCLLVTL ASHHPADFTP 120
AVHASLDKFL ASVSTVLTSK YR 142

SEQ ID NO: 40 moltype = AA length = 142
FEATURE Location/Qualifiers
source 1..142
mol_type = protein
organism = Mus musculus

SEQUENCE: 40
MSLMKNERAI IMSMWEKMAA QAEPIGTETL ERLFCSYPQT KTYFPHFDLH HGSQQLRAHG 60
FKIMTAVGDA VKSIDNLSSA LTKLSELHAY ILRVDPVNFK LLSHCLLVTM AARFPADFTP 120
EVHEAWDKFM SILSSILTEK YR 142

SEQ ID NO: 41 moltype = AA length = 646
FEATURE Location/Qualifiers
source 1..646
mol_type = protein
organism = Mus musculus

SEQUENCE: 41
MSKGPVAVGID LGTTYSCVGV FQHGKVEIIA NDQGNRTTPS YVAFDTERL IGDAAKNQVA 60
MNPTNTVPDA KRLIGRRFDD AVVQSDMKHW PFMVVNDAGR PKVQVEYKGE TKSFPYEEVS 120
SMVLTKMKEI AEAYLGKTVT NAVVTVPAYF NDSQRQATKD AGTIAGLNVL RIINEPTAAA 180
IAYGLDKKVG AERNVLIFDL GGGTFDVSIL TIEDGIFEVK STAGDTHLGG EDFDNRMVNH 240
FIAEFKRKKH KDISENKRAV RRLRTACERA KRTLSSSTQA SIEIDSLYEG IDFYTSITRA 300
RFEELNADLF RGTLDPVKA LRDALDKSQ IHDIVLVGGS TRIPKIQKLL QDFFNGKELN 360
KSINPDEAVA YGAAVQAAIL SGDKSENVQD LLLLDVTPLS LGIETAGGVM TVLIKRNNTI 420
PTKQTQTFIT YSDNQPGVLI QVYEGERAMT KDNLLGKFE LTGIPPAPRG VPQIEVTFDI 480
DANGILNVSA VDKSTGKENK ITITNDKGR LSKEDIERMVQ EAEKYKAED EQRDKVSSKN 540
SLESYAFNMK ATVEDEKLQG KINDEDKQKI LDKCNEIISW LDKNQTAKE EFEHQQKELE 600
KVCNPIITKL YQSAGGMPGG MPGGFPGGA PPSGGASSGP TIEEVD 646

SEQ ID NO: 42 moltype = AA length = 182
FEATURE Location/Qualifiers
source 1..182
mol_type = protein
organism = Mus musculus

SEQUENCE: 42
MGDEEKRRNA ITARRQHLKS VMLQIAATEL EKEESRRESE KENYLSEHCP PLHIPGSMSE 60
VQELCKQLHA KIDVAEEKY DMEVKVQKSS KELEDNMNQL FDLRGKFKRP PLRRVRMSAD 120
AMLKALLGSK HKVCMDLRAN LKQVKKEDTE KERDLRDVGD WRKNIEEKSG MEGRKKMFES 180
ES 182

SEQ ID NO: 43 moltype = AA length = 405

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FEATURE source	Location/Qualifiers 1..405 mol_type = protein organism = Mus musculus						
SEQUENCE: 43							
MRATDRMGAR	AVSELRLALL	FVLVLGTPRL	GVQGEDGLDF	PEYDGVDRVI	NVNAKNYKNV	60	
FKKYEVLLALL	YHEPPEDDKA	SQRQFEMEEL	ILELAAQVLE	DKGVGFGFLVD	SEKDAAVAKK	120	
LGLTEEDSVY	VFKGDEVIEY	DGEFSADTLV	EFLLDLVEDP	VELIEGEREL	QAFENIEDEI	180	
KLIGYFKSKD	SEHYKAYEDA	AEEFHPYIPF	FATPDSKVAK	KLTCLKNEID	FYEAFMEEP	240	
TIPDKPNSEE	EIVSFVEEHR	RSTLRKLKPE	SMYETWEDDL	DGIHIVAFAE	EADPDGYEFL	300	
ETLKAVAQDN	TENPDLSIIW	IDPDDFPLLV	PYWEKTFPID	LSAPQIGVVN	VTDADSIWME	360	
MDNEEDLPSA	DELEDWLEDV	LEGEINTEDD	DDDDDDDDDD	DDDD		405	
SEQ ID NO: 44							
moltype = AA length = 8886							
FEATURE source	Location/Qualifiers 1..8886 mol_type = protein organism = Mus musculus						
SEQUENCE: 44							
MDHSFSGAPR	FLTRPKAFVV	SVGKDATALSC	QIVGNPTPHV	SWEKDRQPVE	AGARFRLAQD	60	
GDVYRLTILD	LALGDSGGQV	CRARNAIGEA	FAAVGLRVDS	EGTCAEQAPH	FLLRPTSIRV	120	
REGADATFRC	RVGGSQPQAV	SWSKDGRRLG	PPDAPHVRVE	EHGESSALRI	RSARPRDGGT	180	
YEVRAENPLG	SASAAAALVV	DSDAEAVAGP	GTSTATLLAH	LQQRREAMRA	EGIPSPSPA	240	
GTRTCTVTEG	KHARLSCFVT	GEPKPEVTWK	KDGQLVTEGR	RHVVEDEQE	NFVLKILFCK	300	
QSDRGLYTCT	ASNLVGGQYIS	SVLVVVREPT	VPFKKRLQDL	EVREKESATF	QCEVAQPATE	360	
AAWFKEETRL	WASAKYDIEE	EGTERRLTVR	NVSADDDAVY	ICETTEGSRT	VAELSVQGNL	420	
TRKLPKRTAV	RTGDTAIFWV	ELAVPEGPVR	WLRNQEEMVA	GGRVAITAEG	TCHTLTIFQC	480	
TLEDMGEVAF	VSGGCRRTTQ	FCVSAAPRRP	LYPPADPVVK	AKTESSVTL S	WSAPPHGDRP	540	
VTIDGYVLEK	RKLGAAYAWS	CHEAGWLATT	EFTITGVAAE	GDFQFRVSAI	NHFGQGPYLE	600	
PPGTMHLVPM	LAVKTPLKAV	EAVEGGEVTF	SVDLTVAASS	EWFLDGEALK	ASSIYVIRCD	660	
RTRHMLTIRE	VPARLHGAQL	KFVANGIETS	IQMVVRAALG	LPSSKLPAAS	AREVLAQLHE	720	
EAQLLAELSD	QAAAVTWLKD	GRELSLGPKY	EMQVSAGRRA	LLVRDVAQDD	AGLYECVSRG	780	
SRTAYQLLVQ	DITDGYRDWG	PAGPQKHMCK	CAGAKIARYL	GSSCYRFLQY	DKGVWHWLEA	840	
ALDTRQKGKT	SSCSLHEKPK	LVPKAGQQAQ	SEVKAEAGNS	ATLSCEVTQA	QTEVTWFKDG	900	
KKLSSSSSKVR	MEASGCSRRL	VVQAGKADA	GEYSCEAGGQ	KLSFRLDVAE	PKLVFAKGQQ	960	
AHSEVKAEAG	ASATLSCEVA	QAQTEVTWFK	DGKKLSSSKVR	VRMEASGCSR	RHLVVQAGKA	1020	
DAGEYSCEAG	GQKLSFRLDV	AEPKMVFAKE	QQAARSEVKAE	AGASATLSCE	VAQAQTEVTW	1080	
FKDGKKLSSS	SKVRMEASGC	SRRLVVQAGK	KADAGEYSCE	AGGQKLSFRL	DVTEPKLVFA	1140	
KQQQARSEVK	AEVGNSATLS	CEVAQAQTEV	TWFKDGKKLS	SSSKVRMEAS	GCSRRLVVQ	1200	
AGKADAGEYS	CEAGGQKL SF	RLDVAEPKLV	FAKEBQQAAR	VKAEAGNSAT	LSCEVAQAQT	1260	
EVTWFKDGKK	LSSSSKVRME	ASGCSRRLVV	QQAGKADAGE	YSCEAGGQKL	SFHLDVTEPK	1320	
LVFAKEQQAH	SEVKAEAGAS	ATLSCEVAQA	QTEVTWFKDG	KKLSSSSKVR	MEASGCSRRL	1380	
VVQAGKADA	GEYSCEABGQ	KLSFRLDVAE	PKLVFAKEBQ	ARSEVKAEAG	ASATLSCEVA	1440	
QAQTEVTWFK	DGKKLSSSSK	VRVEASGCSR	RLVVQAGKGA	DAGEYSCEAG	GQKLSFRLDV	1500	
AEPKLVFAKE	QAQNSAEVKA	AGASATLSCE	VAQAQTEVTW	FKDGKKLSSS	SKVRVEASGC	1560	
SRRLVVQAG	KADAGEYSCE	AGGQKLSFRL	DVAEPKLAF	KEBQAHSSEV	AEAGASATLS	1620	
CEVAQAQTEV	TWFKDGKKLS	SSSKVRVEAS	GCSRRLVVQ	AGKADAGEYS	CEAGGQKL SF	1680	
RLDVAEPKLA	FAKEQQAHSE	VKAEAGASAT	LSCEVAQAQT	EVTWFKDGKK	LSSSSKVRVE	1740	
ASGCSRRLVV	QQAGKADAGE	YSCEAGGQKL	SFRLDVAEPK	LAFAKEQQAH	SEVKAEAGAS	1800	
ATLSCEVAQA	QTEVTWFKDG	KKLSSSSSKVR	VEASGCSRRL	VVQAGKADA	GEYSCEAGGQ	1860	
KLFFRLDVAE	PKLMFAKEBQ	AHSEVKAEAG	ASATLSCEVA	QAQTEVTWFK	DGKKLSSSSK	1920	
VRVEASGCSR	RLVVQAGKGA	DAGEYSCEAG	GQKLSFRLDV	AEPKLVFAKE	QQAHSSEVKA	1980	
AGASATLSCE	VAQAQTEVTW	FKDGKKLSSS	SKVRVEASGC	SRRLVVQVQG	KADAGEYSCE	2040	
ARGQKLSFRL	DVADTRLMPA	KEBQAQTEVK	AEAGNSATLS	CEVAQAQTEV	TWFKDGKKLS	2100	
SSSKVRVEAS	GCSRRLVVQ	AGKADAGEYS	CEAGGQKL SF	RLDVAEASQ	IPERPSSREP	2160	
LVVKEHETII	LTATIAAPSV	AAVTWLKDG	EIRRSKRHEA	TSLGDTHTLT	VRGAQVLD	2220	
IYSCRVGKEG	QDFPVQVEEV	AAKFSKPLEP	VEGELGGTVM	LVCELSPEQA	EVVWRCGNTQ	2280	
LRPGKRFQMT	SEGPRRLTV	SGLREDDABE	YVCSRDRTD	SARLTVKVP	VVKFTSGLSA	2340	
MVAEEGQEAT	FQCVVSPSDA	GVTWYKDMQ	LQPSEKFMV	ESGASRSLTI	LGTLLEDAGQ	2400	
VTVEAGASS	SAALRVREAP	VLFPKKLEPQ	TVEERTSVTL	EVELTRPWE	VKWTRNAAVL	2460	
TPSENVEIRA	EGARHCLVLR	SVGFADRGFF	GCETPDDKTQ	AKLNVEMRQV	RLVRGLQVE	2520	
AKEQGTASMD	VELSHAEVEG	SWTRDGLRLQ	PGPKCHLAVQ	GPVHILTLA	LQPQDSGLVA	2580	
FRAEGVHTSA	RLIVTELFSV	FTRVLQDVVA	TQKEKVTLEC	ELSRPVDVRW	LKDGVELBRAG	2640	
KAIGIVAQGT	CRSLVIYRCE	TGDQGVVCD	ALDAQTSASL	RVQGRYTYTL	FRRVLAEDAG	2700	
EVKFVAENAE	SRAHLRVKEL	PVTLLRPLRD	KIAMEKHGRV	LECCQVSRASA	QVRWFKGGE	2760	
LQSGPKYEVV	SDGLYRKLVI	NDVQPEDED	TYCDAGNVKT	SAQFFVEEQS	ITIVRGLKDM	2820	
TVMEPAPAWF	ECETSIPSVR	PPKWLLGKT	V LQAGNVGLE	QDGTVHRLTL	HKTCTSTMTGP	2880	
VHFTIGKRS	SAQLVVDIP	VVLTRPLEPK	AGRELQSVVL	SCDFRAPKA	VQWYKDDTLP	2940	
SPSEKFKMAL	EGQMAELRIL	RLTPADAGV	RCQAGSAQSS	AEDTVVEAREV	TVIQPLQDAE	3000	
AMEEGRVCF	CELSHKDEDI	EWLNGTPLY	NDSFHEISHE	GCLHTLVLS	VRQADTGTVC	3060	
ATSPKVSVA	RLVVGKFPV	FLKALDDVSA	EERGTLTLQ	EVSDPEARV	WRKDGVELGP	3120	
SDKYDFLHKA	GARGLTVDHL	SHEDAGLYTC	QVGSKETQSK	VSVHDLHVGI	TKRLKTEVL	3180	
EGESCSFECV	LSHESPDPA	VWTVGKTVG	GSGHFHAVRQ	GRKYTLTVKD	AALSDAGEVV	3240	
FSVLGLTSA	SLIIRERPVD	ITKPLEDQRT	TLGEDVMVLS	ELSRAGTSVR	WLKDGKAIRK	3300	
SQKYDLLSEG	TRAVLVVRKA	SLKDSGEYTC	ETEAASKSTAK	LCVBEKANRF	TEELADLQVE	3360	
EKGRAVFTCK	TEHPASVVTW	RKGLLEBRAS	GKHVPSQEBL	TLKLTINALE	RTDSDTYTCD	3420	

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IGQARTQARL	LVHGQKVRVI	EDLEDTAVQE	GSSAKFCCRI	APADYGPVHW	FLDKTPLHSN	3480
ELNEITVQPG	GYHVLTLRQL	ALKDSGTVYF	EAGDQRTSAA	LRVTEKPSIF	SRPLTDVTVT	3540
EGEDLTLVCE	TTTVDSSVRW	TKDGKTLRPS	ARCQLSHEGC	QAQLLITATT	PQDGGRYKCE	3600
IGGASSSSIV	RVHALPVRFR	ESLKDVEVPE	GKAATLRCVL	SSVAAPVEWR	HGDDVLKSSN	3660
KYSLRQEGAV	LELVIRDLQP	QDSGQYSCSF	GQDQTSATLT	VKTSSAQFVG	KLRNKEATEG	3720
TTVTLRCELT	KEAPVEWKKG	TETLRNGDKY	SLKQDGAVCE	LQICSLLVAD	AGEYSCVCGQ	3780
EKTSATLTVK	ALLVHFVRR	RSKEATEGDT	TTLQCELSKA	APVEWRKGTE	TLRDGDRYSL	3840
KQDGAVCELO	IRSLTIADAG	EYLCTCGQEK	TSATLTVRAL	PAKFKDSLKN	EEATEGTTAT	3900
LSCELSKAAP	VTWKKGPKTL	QSGDKYVLRQ	DGAVCGLQIH	GLTMADAGEY	SCVCGQEKTS	3960
ATLTVRGLPA	KFIEDLRSGE	LRDGATAILR	CELSKAAPVE	WRKGSETLKD	GDRYTLRQDG	4020
AVCELQIRGL	AVVDGTGYSS	LPTKFTEGLR	NEEATEGTMA	TLRCQMSKAA	PVEWRKGSET	4080
LRDGDYRSLR	QDGAMCELQI	RGLTIEDSGE	YTCVCGQEK	SATLSVKALP	SRFIEDLRSQ	4140
EATEGTMATL	RCQMSKTAPE	EWKKGSETLR	DGGYSLRQD	GPVCELQICD	LAVEDAGEYS	4200
CVCQGQEKTS	TLSEIKALPPR	FIEDLRSQEA	TEGTMATLRC	QMSKAAPVEW	RKGSETLGDG	4260
GRYSLRQNGA	VCELIQHDLA	VEDTGEYSCV	CQGEKTSATL	NVKALPPRFI	EDLRSQEATE	4320
GTMATLRCQM	SKAAPVEWRK	GSETLRDGG	YSLRQDGAVC	ELQIHDLDVE	DAGQYSCVCG	4380
QEKTSAVLTV	DALPPKFTEG	LKKEEATEGT	MVTLRCQMSK	EATVEWRKGA	KTLSDDGGYRS	4440
LRQDGAMCEL	QICGLAVEDA	GEYSCVCGQE	KTSATLSVKA	LPPRFIEDLR	SQEATEGTMA	4500
TLRCQMSKAA	PVEWRKGSSE	LRDGDYRSLR	QDGAVCELQI	RDLAVEDAGE	YLCVCGQEK	4560
SATLSVKALP	PRFIEDLRSQ	EATEGTMATL	RCQMSKAAPV	EWKKGSKTLR	DGDYRSLRQD	4620
GAMCELQICD	LAVEDTGDYS	CVCQGQEKTS	TLSEIKALPPR	FIEDLRSQEA	REGTVATLRC	4680
RMSKAAPVEW	RKGSSETLKD	DRYSLRQEGN	LCELIQIRDLA	VEDTEYSCV	CQGEKTSATL	4740
SVKALPAKFI	EDLRSQEAP	SSTVTLRCKL	SKKASVVWKK	GSETLRNGAR	YSLRQDGAVC	4800
ELIIRDLTVE	DTGEYSCVCG	QERTSATLSI	MAPQVVFQOP	LQNLQAEESG	MASLRCLESV	4860
PNAAMVWSKG	GLELQGDTRR	EARQQGCVAE	LLRLDLRRED	AGEYSCVCGS	QTTSATLMVT	4920
AAPVRFLREL	QAQDVDEGAT	ARLRCLESLR	AVSVEWRKGS	LQLFPCKAKYQ	MVQEGTTAEL	4980
LVHVGQEDDA	GEYTCDAAGT	QSTARLSVRA	PKPKFKTDLO	STEQEAGGTA	RLCCQLSEAE	5040
PQTPVQWLKE	GVELHVGSKY	EMRRQGAVCE	LLIHGLEAKD	TGEYACLVGG	QKTLASLRVK	5100
EPEVTIVRGL	VDMEVQADE	VEFTCKVSQA	GATDVQWHLQ	GLPLQSNEVT	EVAVLADGCT	5160
HVLQLKGVT	EDAGTVSFHV	GLSSSSAQLT	VRVPEVTVLE	PLKDVQLSEG	QDAHFRCLRS	5220
RASGQEARWA	LGGVPLQCN	MNDITVEQGT	LYLLTLHKVT	LEDAGTITLQ	VGSCSSEAQL	5280
KVTAKNTVLR	GLENVDAL	GEALFECQLS	QPEVAANTWL	LDDEPVHTSE	KVEVVYFENG	5340
LRHLLLLKNL	KPQDSCRVT	LAGDVVTSFA	LTVGRWRLEV	LEPHDASVK	AGMQVRFCTI	5400
LSEAVPVGEA	TDVINGAIIQ	PDDTDVTTVT	DGSHHALTSL	NAQPQHAGEV	TFAARDAVAS	5460
ARLSVLALPD	PPEDAEEVGR	SDHSVTLVSW	APMSDGGGGL	CGYRVEMKEA	STGQWQLCHD	5520
LVPGPECVVD	DLVPGKTYRF	RVAAVGPAGA	GEPVHLPMQV	KIAPAPAPAP	APAPAPAPET	5580
RQAVVGEDIC	ELDVAADGG	EVVWHKGT	IQPGGHFEVL	SRGQRQMLVI	KGFRTEDDQE	5640
YRCGPIQGLP	SSGAATFN	MTSGSGDEVP	AQPSLPPEAA	QEGDLHLWE	ALARKRRMSR	5700
EPTLDSISLE	PEEDSRVQL	RQEAETAPD	LSEGYSTADE	LARTGEADLS	HTSSDDESRA	5760
GTPLSVTYLK	KAGGFGISPL	ASKHEAQVTT	SVKPKQKQEP	VVPTCPPPGD	LSAADLMDPS	5820
LDKAAVKIQA	AFKGYKVRKE	MKQQEGPVFS	RTFGDTEAQV	GDVLRLECVL	ATKTDMRACW	5880
LKDGIELTDG	RHYHIDLQKD	GTCSLLVTLG	APTDSGRYTC	QVSTKSGRVS	HSACVVVSGT	5940
ESEAESSSGG	BLDDAFRRAA	RRLHRLFRTK	SPAELSEEL	FLSADEGPGE	PEEPADWQTY	6000
REDENFVCIR	FESLAEARQA	VTCFRNMFAT	MGIGVEISLG	EQGPRGVEMR	IGKVAPTVP	6060
AVPLAKTPGL	QTSDAAPVFL	TELQNDVQD	GYPMSFDCV	TGQVPVPSVR	FKDGKLEED	6120
ALGEPRIQGH	FIVWEGAPGA	RMPWKGNHNR	VFLFRNYLVI	CKPRDRSRTD	TFSYVFRNMM	6180
TETSSYFSAQ	GYLSSREQEG	TESDEGQLPQ	VLEELKDLQV	APGTRLAKFQ	LKVKGYPAK	6240
LYWFKDQGPL	TTSDHIRMTD	KKTLLHTLEIV	SVTREDSSGQY	AAYSINAVGA	AYSSARLLVR	6300
GPSEPEEKPA	SDVHERLVPP	RILEKFTPKK	VKRGSSITFS	VKVEGHPAPS	VHNLKEEAKE	6360
GVLWIGPDP	GYTMASSSKQ	HSLVLLDVGR	QHOGTYTCTIA	TNPAGQALCS	ASLHISGLAK	6420
EEBQERVKEA	LISFLQGT	QAVSAQMS	AGFADLVGQS	KGESLVAEAA	HSLSLAEV	6480
TEEFLLQLTS	QITEMVSAKI	SQAKLQVPGG	DSDEETKTPS	ASPRHGRSRP	SSSVQESSSE	6540
SEDGDSRGEI	FDIYVVTADY	LPLGAEQDAI	ILREGQYVEV	LDSAHPLRWL	VRTKPTKSSP	6600
SRQGWVSPAY	LDKRLKLSPE	WGTEAPEFP	GEAVSEDEYR	TRLSSVQEL	LSSEQAFVGE	6660
LQFLESHMK	HLERSPRVPA	AVASQKTVIF	RNVQDISHFH	SSFLKELQGC	GTDDVAMCF	6720
IKNQEAPEKY	LEFLVGRVQA	ESVVVSTPVQ	EFYKKAET	LSAKDPTQPP	PPPLQHYLEQ	6780
PVERVQKYQA	LLKELIRNKA	RNRQNCALLE	QAYAVVSALP	QRAENKLHVS	LMENYPTGLE	6840
ALGEPRIQGH	FIVWEGAPGA	RMPWKGNHNR	VFLFRNYLVI	CKPRDRSRTD	TFSYVFRNMM	6900
KLSSIDLNDQ	VEGDDRAFEV	WHEREDSVRK	YLLQARTVII	KNSWVKEICG	IQORLAQPVW	6960
RPPEFEEELA	QCTAELGETV	KLACRVGTGP	KPIVSWYKDG	KPVEVDPHHI	LIEDPDGSC	7020
LILDNLGTGD	SGQYMCFAAS	AGNASTLGLK	ILVQVPPRFV	NKVRATPFVE	GEDAQITCTV	7080
EGAPYPIQIRW	YKDGTLAPG	NRVRLNNEPR	SGVLVLVIAQ	ASKEDLGHYE	CELVNRLGST	7140
RGGGELYMQS	PALRARQHH	REQIVAAVED	TSVEGSAHSA	QDQADQQAAS	VLWRLGSEA	7200
LGFSPGDLN	TRQSEPPAFE	EASQIPGAA	SGTPEVSPQG	THKGLEQETT	SSGSQGWTV	7260
IRVEGTAWPG	AGTGQLLDDV	HSQVIMETTQ	RTYVCQAPDT	GVTRAPSMQV	TIEDVQVQVG	7320
DMAQFDVAVI	GHPPPIVITWY	KGSTQLTSSA	RLSQRQDGT	YSLVLTAVAP	HDAGVYTCVA	7380
NNAGGQVLCK	AKLLVHGGDK	LDAENQVYRR	KLHSFYDVQE	EIGRGVFGFV	KRVQHKGNKM	7440
FCAAKFIPLR	SKTRAQAYG	RDILATLGH	LVTGLLDQFE	TRKTLILILE	LCSSEELLDR	7500
LFKKGVVTEA	EVKVIYQQLV	EGHLHLHSHG	ILHLDIKPPN	ILMVHPARE	IKICDFGFAQ	7560
KITPSEPQYS	KYGSPEFVSP	EIIEQNPFVSE	GSDIWMAGVI	SYLSLTCS	FAGESDRATL	7620
LNVLGRVSW	SSPTAAHLE	DAKDFIKATL	QRTPRARPST	SQCLAHWPFL	KSPAEAAHF	7680
INTKQLKFL	ARSRWQSLM	SYKSILVMRS	IPELLQGGPD	SPSLGVARHL	RGEASGASS	7740
SSSSDNELAP	FARAKSLPPS	PVTHSPLHHP	RGFLRPSASL	PEETEASMP	ADAAPASPO	7800
SAGPPASPGC	VPRHSVISSL	FYQQAGEGAE	RGNKTSAGAK	HPARRRHLLK	GGYIARALPG	7860
LRPELMEYSL	LEEEAAREEQ	ASLMTKTPSF	ETALRLPSSS	VREVPGRSHS	LDNPPVTGTP	7920
SPEACKEQLL	PPPSTGLTHE	TTAKDRGHE	GFLQESVPPF	PMSGDSRPGK	QEGSSQDSCR	7980

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GKPASSCHSE	LGSGSQEGCG	PPSSQSLGSL	PPQSLKKELS	TSCGPLFSEQ	PQAAPFPTQV	8040
SPLLGSEKEP	QDGLSLEGPV	PVPSSSPGSA	SQVDASLDTE	GLSEAGDTC	FTPPPQRQPE	8100
QATTFRKFSLE	SRGGYAGVAG	YGTFAFGGDA	GGMLGQGGLW	ARMAWAVSQS	SEEQDEAATE	8160
SPQPLESLGP	IAEASGVPLR	TSPSLTPWEE	VEQVSLVQIR	DLSGDAEAD	TISLDISEVD	8220
PAYLNLSDLY	DIKYLPPFEF	IFRRVPKPIE	QPESPGSETE	AGQGLADFLE	EAAWPWPGE	8280
GLRAGLEITE	EPPEPGDLEA	LLGEAAVGRK	RKWSPSRGLF	QFPGRCLSGE	EPVELGLRQR	8340
VKASMAHISR	ILKGRPEGPE	REGPPRKKAG	LASFRLSGLK	GRDQELSDA	VVLGQSVTLA	8400
CQVLAQPTAQ	ATWSKDGVL	ESSGHLISS	TLKNFQLLTI	LVVKEEDLGT	YTCCVSNPLG	8460
TAVTTGVLRL	AERPSSSRP	EVGELYKDAV	LLVWKPVESC	GPVTYIVQCC	IEGGSWTTLA	8520
SDISDCCYL	GKLSRGGMYI	FRTACVSKAG	MGPYSSPSEQ	VLLGGPNHLA	SEESSRGRP	8580
AQLLPSTKTF	AFQMQIRRR	FSVVRQCRES	ASGRALAAKI	VPYQPEDKTA	VLREYEALKR	8640
LHHPLAQLH	AAYLSRPHLV	LILELCSGPE	LLPSLAERES	YSESDVKDYL	WQMLSATQYL	8700
HAQHILHLDL	RSENMVTEY	NLLKVIDLGN	AQSLDQEKVP	APENFKDYLE	TMAPELLEGQ	8760
GAVPQTDIWA	IGVTAFIMLS	GEYPESSEGT	RDLQKGLRKG	LIRLSRCYAG	LSGGAVAFLLQ	8820
SSLCAQPWGR	PCASTCQCQ	WLTEEGTGS	RPTPVTFPTV	RLRAVFRERE	KRRALLYKKH	8880
NLAQVR						8886

SEQ ID NO: 45 moltype = AA length = 1532

FEATURE Location/Qualifiers

source 1..1532

mol_type = protein

organism = Mus musculus

SEQUENCE: 45

MEHSKQIRIL	LLNEMEKLEK	TLFRLEQGF	LQFRLGPTLQ	GKAVTVYTN	PLPGEAFNRE	60
KFRSLDWENP	TEREDDSDKY	CKLHLQQSGS	FQYFPLQONE	KSGGGYIVVD	PILRIGVDNH	120
VLPPLDCVTLQ	TFLAKCLGPF	DEWESRLRVA	KESGYNMIHF	TPLQTLGLSR	SCYSLADQLE	180
LNPDFSRRPK	RYTWSDVQGL	VEKLKREWNI	LCITDVVYNH	TATNSKWILE	HPESAYNLVN	240
SPHLKPWWVL	DRALWHFSCD	VADGKYREKG	VPALIENDQH	MNCIRKIWE	DIFPRIQLWE	300
FFQVDVHKAV	EQFRRLLSQE	NRRVTKSEPK	EHLKIIQDPE	YRRRGCAVDM	DTALATFIPH	360
DNGPAAIEEC	CNWFRKRLEE	LNSEKHHLTS	CHQEQAIVNCL	LGNVFIYERLA	GHGPKLGPVT	420
RKYPLVTRYF	TFPFGEMALS	AEALIHLPD	KACFLMAHNG	WVMGDDPLRN	FAEPGSDVYL	480
RRELICWGDS	VKLRYGNKPE	DCPYLWAHMK	KYTEITATHF	QGVRLDNCHS	TPLHVAEYML	540
DAARKLQPNL	YVVAELFTGS	EELDNIFVTR	LGISSLIREA	MSAYNSHEEG	RLVRYGGE	600
VGSFVQPCLR	PLMPAIAHAL	FMDITHDNEC	PIVHRSAYDA	LPSTTVVSMA	CCASGSTRGY	660
DELVPHQISV	VAERFYTKW	NPASPADTG	DVNVHSGIIA	ARCAINRLHQ	ELGAKGFIQV	720
YVDQVDEIV	AVTRHSPSIH	QSVVAVSRTA	FRNPKTSFYS	KEVPQMCIPG	KIEEVVLEAR	780
TIERNTKPKY	KDENSINGMP	NMTVELREHI	QLHESKIVRQ	AGVATKGPNE	YIQEIEFENL	840
SPGSVIFPRV	SLDPHAQVAV	GILRNHLTQF	SSHFKSGSLA	VDNADPILKI	PFASIASKLT	900
LAEHLQVLYR	CESEEQEDGG	GCDYIPNWSS	LKYAGLQGLM	SVLAIEIRPKN	DLGHPFCENL	960
RSGDWMIDIV	SGRLISRSGS	IAEVGKWLQA	MFFYLKQIPR	YLIPCYFDAI	LIGAYTTLLD	1020
VAWKQMSSFV	QNGSTFVKHL	SLGVSQMGCV	GKCPCLPLLS	PSLLDVPCRL	NEITKEKEQC	1080
CASLAAGLPH	FSSGLFRWC	RDFTIALRGM	LLVTGRYLEA	RNIILAFAS	LRHGLIPNLL	1140
GEGTYARYNC	RDVAVWWLQ	IQDYCRTVPN	GLDILKCPVS	RMYPDDSDAP	LPAGTLDQPL	1200
FEVIQEAMQR	HMQGIQFRER	NAGPQIDRNM	KDEGFNITAG	IDDEETGFVYG	GNRFNCGTWM	1260
DKMGESDRAR	NRGIPATPRD	GSAVEIVGLC	KSARVWLLEL	SKKNIFFPYHE	VRVCRHGKVV	1320
AVSYDEWNRK	IQNNFEKLFH	VSEDPSPDNE	KHPNLVHKRG	IYKDSYGASS	PWCYDQLRPN	1380
FTIAMVVAPE	LFTAACKAWKA	LEIAEKKLLG	PLGMKTLDPD	DMVYCGVYDN	ALDNDNYNLA	1440
RGFNYHQGPE	WLWPIGYFLR	AKLYFSKMMG	PETAAKTVFL	VKNVLSRHYV	HLERSPWKGL	1500
PELTNENGQY	CPFSCETQAW	SMVVLETLTY	DL			1532

SEQ ID NO: 46 moltype = AA length = 463

FEATURE Location/Qualifiers

source 1..463

mol_type = protein

organism = Mus musculus

SEQUENCE: 46

MGKEKTHINI	VVIGHVDSGK	STTTGHLIYK	CGGIDKRTIE	KFEKEAAEMG	KGSFKYAWVL	60
DKLKAERERG	ITIDISLWKF	ETTKYYITII	DAPGHRDFIK	NMITGTSQAD	CAVLIVAAGV	120
GEFEAGISKN	GQTREHALLA	YTLGVKQLIV	GVNKMDDSTEP	AYSEKRYDEI	VKEVSAYIKK	180
IGYNPATVPF	VPISGWHGDN	MLEPSPNMPW	FKGWKVERKE	GNASGVSLLE	ALDTILPPTR	240
PTDKPLRLPL	QDVNFEGIGI	TVPVGRVETG	ILRPGMVVTF	APVNIITTEVK	SVEMHHEALS	300
EALPGDNVGF	NVKNVSVKDI	RRGNVCGDSK	ADPPQEAQF	TSQVILNHP	GQISAGYSPV	360
IDCHTAHIA	CFELKEKID	RRSGKKLEDN	PKSLKSGDAA	IVEMVPGKPM	CVESFSQYPP	420
LGRFAVRDMR	QTVAVGVIKN	VEKSGGAGK	VTKSAQKAQK	AGK		463

SEQ ID NO: 47 moltype = AA length = 413

FEATURE Location/Qualifiers

source 1..413

mol_type = protein

organism = Mus musculus

SEQUENCE: 47

MAPPVFAQV	PQAPPVLVFK	LTADFRDDPD	PRKVNLGVA	YRTDESQFW	LPVVRKVEQK	60
IANDNSLNHE	YLPILGLAEF	RSCASRLVLG	DNSLAIRENR	VGGVQSLGGT	GALRIGADFL	120
GRWYNGTDNK	NTPIYVSSPT	WENHNNAVFS	AGFKDIRPYC	YWDAAEKRLD	LQGFLLNDLEN	180
APFESIFVLH	ACAHNPOTGT	PTPEQWKQIA	AVMQRRFLFP	FFDSAYQGFA	SGDLEKDAWA	240
IRYFVSEGEF	LPCAQSFSKN	FGLYNERVGN	LTVVGKESDS	VLRVLSQMEK	IVRITWSNPP	300

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AQCARIVAAT	LSDPELFKEW	KGNVKTMA DR	ILTM RSELRA	RLEALKTPGT	WSHITEQIGM	360
FSFTGLNPKQ	VEYLVNEKHI	YLLPSGRINM	CGLTTKNLDY	VATSIHEAVT	KIQ	413

SEQ ID NO: 48	moltype = AA	length = 194
FEATURE	Location/Qualifiers	
source	1..194	
	mol_type =	protein
	organism =	Mus musculus

SEQUENCE: 48						
MEEKLKKAKI	IFVVGGPGSG	KGTQCEKIVQ	KYGYTHLSTG	DLLRAEVSSG	SERGKKLSAI	60
MEKGELVPLD	TVLDMLRDAM	LAKVDSSNGF	LIDGYPREVK	QGEEFEQKIG	QPTLLLYVDA	120
GAETMTQRL	LRGETSGRVD	DNEETIKKRL	ETYYNATEPV	ISFYDKRGIV	RKVNAEGTVD	180
TVFSEVCTYL	DSLK					194

SEQ ID NO: 49	moltype = AA	length = 558
FEATURE	Location/Qualifiers	
source	1..558	
	mol_type =	protein
	organism =	Mus musculus

SEQUENCE: 49						
MAALTRNPQF	QKLEWHRAN	SANLKLRELF	EADPERFNNF	SLNLNTNHHG	ILVDYSKNLV	60
NKEVMQMLVE	LAKSRGVEAA	RDNMFSGSKI	NYTENRAVLH	VALRNRSTNP	IKVDGKDVM	120
EVNRVLDKMK	SFCQVRVSGD	WKGYTGSIT	DIINIGIGGS	DLGPLMVTEA	LKPYSKGGPR	180
VWFVSNIDGT	HIAKTLASLS	PETSLPIIAS	KTFTTQETIT	NAETAKWEFL	EAAKDPSAVA	240
KHFVALSTNT	AKVKEFGIDP	QNMFEFWDWV	GGRYSLWSAI	GLSIALHVGF	DHFEQLLSGA	300
HWMDQHFLKT	PLEKNAPVLL	ALLGIWYINC	YGCETHALLP	YDQYMRFAA	YFQQGDMESN	360
GKYITKSGAR	VDHQTGPVW	GEPGTNGQHA	FYQLIHQGTK	MIPCDFLIPV	QTQHPIRKGL	420
HKILLANFL	AQTEALMGK	LPEEARKELQ	AAGKSPEDLE	KLLPHKVFE	NRPTNSIVFT	480
KLTPFILGAL	IAMEYHKIFV	QGIMWDINSF	DQWGVELGKQ	LAKKIEPELE	GSSAVTSHDS	540
STNGLISFIK	QQRDTKLE					558

SEQ ID NO: 50	moltype = AA	length = 469
FEATURE	Location/Qualifiers	
source	1..469	
	mol_type =	protein
	organism =	Mus musculus

SEQUENCE: 50						
MSQAYSSSQ	VSSYRRTFGG	APGFSLGSPL	SSPVFPFRAGF	GTKGSSSSMT	SRVYQVSR	60
GGAGGLGSL	SSRLGTTRAP	SYGAGELLDF	SLADAVNQEF	LATRTNEKVE	LQELNDRFAN	120
YIEKVRFLQ	QNAALAAEVN	RLKGREPTRV	AELYEEEMRE	LRRQVEVLTN	QRRVVDVERD	180
NLIDDLQRLK	AKLQEEIQLR	EEAENNLAAF	RADVDAATLA	RIDLERRIES	LNEEIAFLKK	240
VHEEIRELQ	AQLQEQVQV	EMDMSKPDLT	AALRDIRAQY	ETIAAKNISE	ABEWYKSKVS	300
DLTQAANKIN	DALRQAKQEM	MEYRHQIQSY	TCEIDALKGT	NDSLMLQMR	LEDRFASEAN	360
GYQDNIAELE	EEIRHLKDEM	ARHLREYQDL	LNVMKALDVE	IATYRKLEGG	EESRINLPIQ	420
TFSALNFRET	SPEQRGSEVH	TKKTVMIKTI	ETRDGEVVSE	ATQQQHEVL		469

SEQ ID NO: 51	moltype = AA	length = 419
FEATURE	Location/Qualifiers	
source	1..419	
	mol_type =	protein
	organism =	Mus musculus

SEQUENCE: 51						
MASAFSKLLT	GRNASLLFTT	LGTSALTTGY	LLNRQKVSAD	AREQHKLFPP	SADYPLDRKH	60
NNCMAECLTP	TIYAKLRNKM	TPSGYTLDDQ	IQTGVDPNGH	PFIKTVMGVA	GDEESYEVFA	120
DLFDPVIKLR	HNGYDPRVMK	HPTDLDDASKI	THGQFDERYV	LSSRVTRGRS	IRGLSLPPAC	180
SRAERREVEN	VAITALEGLK	GDLAGRYYKL	SEMTEQDQQR	LIDDHFLFDK	PVSPLLTCAG	240
MARDWPDARG	IWHNYDKTFL	IWINEEDHTR	VISMKEKGNM	KRVFERPCRG	LKEVERLIQE	300
RGWEFMWNER	LGYILTCPSN	LGTGLRAGVH	VRIPKLSKDP	RFSKILENLR	LQKRGTGGVD	360
TAAADVVDI	SNIDRIGRSE	VELVQIVIDG	VNYLVDCEKK	LERGQDIKVP	PPLPQFGRK	419

SEQ ID NO: 52	moltype = AA	length = 3284
FEATURE	Location/Qualifiers	
source	1..3284	
	mol_type =	protein
	organism =	Mus musculus

SEQUENCE: 52						
MRKHRHPLV	AVFSLLLSGI	ATTHAQHGD	VKNGAAADIV	FLVDSSWSAG	KDRFLLVQEF	60
LSDVVESLAV	GDNDFFHFAV	RLNGNPHTEF	LLNTYHSKQE	VLSHIVNMSY	IGGSNQTGKG	120
LEYVIHSHLT	EASGSRAADG	VPQVIVLTD	QSEDGFALP	SAEKLSADVN	VFAVGVEGAD	180
ERALEGEVASE	PLSMHVFNLE	NVTSLHGLVG	NLVSCIHSVV	NPERAGDKES	LKDITAQDSA	240
DIIFLIDGSG	MTGNANFDVI	RDFLVNVLER	LSVGNQVQVQ	GVVQYSEPI	TMFSLNSYPS	300
KAAVLDAVK	LSLVGESAN	IGQALDFVVE	NHFTRAGGSR	VEEGVPQVLV	LISAGPSSDE	360
IRDSVVALKQ	ASVFSFGLGA	QAASRAELQH	IATDDSLVFT	VPEFRSFGDL	QEQLIPYLVG	420
VAQRHIVLQ	PAIVTQVMEV	NKRDIVFLVD	GSSSLGPSNF	NAIRDFVTRV	IQRLEIGQDL	480
VQVSAQYAD	YTKPEFYLSN	YTNKRDAITA	VRKMRALNGS	ALYTGSSLDF	VRNNLFTSSA	540
GHRAABGVPK	LLVLITGGKS	LDEVSQPAQE	LKRGSIMALA	VGSKAAEDE	LKEIAPDSSL	600

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VFIPAEFRPA	PLQNMPLPSLM	APLRTLGTGT	EVHVNKRDII	PLLDGSNDVG	KNNFPYVRDF	660
VTNLVNSLDV	GSDNIRVLGV	QFSDTPVTEF	SLDTYQTKSE	LLAHLRRLQL	KGGSGNLNAGS	720
ALSYIHANHF	TEAGSRTRE	HVPQLLLLLM	AGPSEDAYLQ	AANALVRSVG	LTFVCVTNRA	780
DKAELEHIAF	NPSLVLYLMD	FRSLPSLPQQ	LIQPLTTYVS	GGVQEVPLSQ	PESKRDLFL	840
FDGSVNVLGQ	PPAVRDFLYR	IIELDVKPD	GTRVAIAQFS	DDVRLESRFS	EHQTKAEILN	900
LVKKMKLKTG	KALNLGYALD	YALRNIFVRS	AGSRIEDNVQ	QFLVLLVAGR	SSDAVAGPAS	960
SLKQRGVVVF	IFQAKNANPS	ELEQIVLSPA	FILAAESLPK	IGDLQSQIVS	LLKAEQSGSP	1020
VSGEKDVVFL	IDGSEGVRS	FPLLKDFVQR	VVESLDVGP	RVRVALVQYS	DRTRPEFYLN	1080
SHMDQGVIS	AIRRLTLGG	PTPNTGALE	FVLNRNLTSS	TGSRIAEGVP	QLLIVLTAEP	1140
SGDDVRGSPV	VLKQGGAVPI	GIGIGNADIS	EMQTISFIPD	FAVAIPTFRE	LGTIQQVISE	1200
RVIQLNREEL	SSLKPILTPS	TGAGVGSKKD	VVFLIDGSRN	AGPEFYIIRT	LIERIVEYLD	1260
IGFDTTRVAV	IQFSEDSKME	FPLNAHFSKD	EVQNAVRRLR	PKGGSQVYIG	NALEYVLKNI	1320
FORPLGSRIE	EGVPQFLVLI	SSGKSDEVD	DSAVELKQFG	VAPLTIARHT	DQEELVKISL	1380
SPEYVYSVST	FRELPRLEQK	LLTPITTLTS	QQIHQILAST	RYPPSVVESD	AADIVFLIDS	1440
SDAVKPDGIA	HIRDFVSRIV	RRLNIGPSKV	RIGVVQFSND	VPPEFYLKTH	KSQSSVLEAI	1500
RRLRFKGGSP	LNDGKRNPS	ARNLFVKSAG	SRIEDGVQPH	LVLPLGKGSQ	DDVARHAQVI	1560
SSSGIVSLGI	GDRNIDRTDL	QITINDPRLV	FTVREFRELP	NIEERVMLSF	GPSGATPQPP	1620
GVDLPSPSRP	EKKKADIVFL	LDGGINFRRD	SFQEVLRPAS	EIVDTVYEDG	DSIRVGLVQY	1680
NSDPTDEFPL	DFGTKKRQII	DLANKVVKYK	GRHANTRVGI	EHLRNHFVP	EAGSRDLERV	1740
PQIAFVITGG	KSVEDAQDVS	LALTQKGVKV	FAVGVRNIDS	EEVGKIASNS	ATAFRVGSVQ	1800
ELSELSETVL	ETLHDAMHET	LCPGVTDVSK	ACNLEVILGF	DGSRDQNVFV	SQKGLESKVD	1860
IILNRSIQQ	RISCSGNQLP	TVRVSVMANT	PSGPEAEADF	AEYQPELFEK	FRNMRSQRPY	1920
VLTADTLKLY	QNKFRQSSPD	TVKVVIHFTD	GADGDMADLY	RASEELRQAG	AQALILVGLE	1980
RVANLERLMH	LEFGRGMYD	RPLRLNLLDL	DYELAEQLDN	IAEKACCGVP	CKCSGERGDR	2040
GPIGSIGPKG	ISGEDGYRGI	PGDEGGPGER	PGPGVNGTQG	FQGPCGQRGV	KGSRGFPGEK	2100
GELGEIGLDG	LDGEEGDKGL	PGSSGEGKSP	GRRGDKGPKG	DKGERGDVGI	RGDPGDSGRD	2160
SQQRGPKGKT	GDIGPMGLPG	RDGIPGSPGD	PGKDGGSGRR	GPAGAKGNRG	GPQPGFEFE	2220
QQTRGSGQGP	GPIGPPGLIG	EQGIPGPRGG	GGTAGAPGER	GRTGPLGRKG	EPGEPGPKGS	2280
IGNRGPRGET	GDDGRDGVGS	EGRRGKKGER	GPPGYPGPKG	TPGEPGADGP	PGPKGIRGRR	2340
GNSSGPPGATG	QKGDPPGYPG	SGHKGNGRDS	VDQCALIQSI	RDKCPCCYGP	LECPVPFTEL	2400
AFALDTSEGV	RLDTFSRMR	VLLGIVGDLT	IAESNCPRGA	RVAVVTYNN	VTTETIRFADS	2460
KKKSALLDSI	QNLQVALTSK	QQSLETAMSF	VARNTFKRVR	SGFLMRKVAV	FFSNKPTRAS	2520
PQLREAVLKL	SDAGITPLFL	TSQEDRQLIN	ALQINNTAVG	HALVLPARRD	LTDFLKNVLT	2580
CHVCLDICNI	DPSCGFGSWR	PSFRDRRAAG	SDVDIDLAFI	LDSEATTLF	QFNEMKKYIG	2640
YVIRQLDLSP	DPKASQHFAR	VAVVQQSTYE	SVDNASVPPV	KVEFSLTDYG	AKEKLLDPLS	2700
RRMTQLQGT	GLGNAIYITI	ENIFESAPNP	RDLKIMVLM	TGDMQRQOLE	EAQRAILQAK	2760
CKGYFFVVLG	IGRKVNIKEV	YSFASEPNDV	FFKFVDKSTE	LNEEPLMRFG	RLLPSFVSSE	2820
NAFYLPDDL	KQCDWFQGDQ	PVKNGVKFGH	KQINPPHTAN	SSLTSKVVT	MKPVTTTKPT	2880
ATVNLPPAKP	APAPAPAPAP	VLAQPDPAKP	AQAPAPAPAP	ASAKLVPPQP	VHVQPAPAPQ	2940
ASVVRPAPAP	APQPAPAPAP	VPAKPAVPAQ	PAPPQAPAAK	PVPAPAPVPA	QPAPPQAPAA	3000
KVPAPAPAVP	AQPAAPAPMP	AQPVLTKSAA	VKPASANKPV	AAKPVATNTA	TATARPALAA	3060
KPAAAKPAAT	RPLAAAIRPV	ATKPEAPRQQ	AKPAATKPAT	TKPLARVSRE	VQVSEVTENS	3120
ARLHWERPEP	KLSTCFYDLV	TSAMDQSLVL	RQNLTVTDRV	IGGLLAGQLY	HVVVVSYLQS	3180
QVRAIYQGSF	NTKKTQPPPL	QAAHRASSST	INLMVNTPEL	FLTKTDICKL	SRDAGTCVDF	3240
KLLWHYDLES	KSKRFWYGG	CGGNENRPHS	QEECEKMCSP	DLLV		3284

SEQ ID NO: 53 moltype = AA length = 745
FEATURE Location/Qualifiers
source 1..745
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 53

MPLFKLTGGQ	KQIDDAMRSF	AEKVFASEVK	DEGGRHEISP	FDVDEICPIS	LHEMQAHIFH	60
MENLSMDGRR	KRRFQGRKT	NLSIPQSETS	STKLSHIEEF	ISSSPTYESV	PDFQRVQITG	120
DYASGVTVED	FEVVCCKGLY	ALCIREKYM	KSFQRFPKTP	SKYLRNIDGE	ALVGNESFYF	180
VFTPPPKKGE	DPFRTEFLPA	NLGYHLKMK	GVIYIYPDEA	AANRDEPKPY	PYPNLDDFLD	240
DMNFFLLALIA	QGPVKTYAHR	RLKFLSSKFQ	VHQMNLNEMDE	LKELKNNPHR	DFYNCRKVD	300
HIHAAACMNQ	KHLLRFIFKKS	YHIDADRUVV	STKEKSLTLK	ELFAKLNMHP	YDLTVDSLVD	360
HAGRQTFQRF	DKFNDKYNPV	GASELRDLYL	KTDNYINGEY	FATIIKEVGA	DLVEAKYQHA	420
EPRLSIYGRS	PDEWNKLSSW	FVCNRIYCPN	MTWMIQVPR	YDVFRSKNFL	PHFGKMLENI	480
FLPVFEATIN	PQAHFDLSVF	LKHITGFDVS	DDESKHSGHM	FSSKSPKPEE	WTMENNPSYT	540
YYAYMYANI	TVLNSLRKER	GMNTFLFRPH	CGEAGALTHL	MTAFMIADNI	SHGLNLKKSP	600
VLQYLFLLAQ	IPIAMSLSN	NSLFLEYAKN	PFLDFLQKGL	MISLSTDDPM	QFHFTKEPLM	660
EEYAIQAQVF	KLSTCDMCEV	ARNSVLQCGI	SHEEKAKFLG	NNYLEEGFVG	NDIRRTNVAQ	720
IRMAIRYETW	CYELNLIAEG	LKATE				745

SEQ ID NO: 54 moltype = AA length = 379
FEATURE Location/Qualifiers
source 1..379
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 54

FGNTHNKYKL	NYKSEEEYPD	LSKHNNHMAK	VLTPDLYKKL	RDKETPSGFT	LDDVIQTGVD	60
NPGHPFIMTV	GCVAGDEESY	TVFKDLFDPI	IQDRHGGFKP	TDKHKTDLNL	ENLKGGDDL	120
PHYVLSSRV	TGRSIKGYTL	PPHCSRGER	AVEKLSVEAL	NSLTGEFKGK	YYPLKSMTEQ	180
EQQQLIDHDF	LFDKPVSPLL	LASGMARDWP	DARGIWHNDN	KSFLVWVNEE	DHLRVISMEK	240

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GNMKEVFR	FCVGLQKIEE	IFKKAGHPFM	WNEHLGYVLT	CPSNLGTGLR	GGVHVKLAHL	300
SKHPKFEEIL	TRLRLQKRGT	GGVDTAAVGS	VFDISNADRL	GSSEVEQVQL	VVDGVKLMVE	360
MEKKLEKQGS	IDDMIPAQK					379

SEQ ID NO: 55 moltype = AA length = 900
 FEATURE Location/Qualifiers
 source 1..900
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 55

MMVMQPEGL	GAREGPFAGG	GGGGEYMEQE	EDWDRDLLLD	PAWEKQQRKT	FTAWCNSHLR	60
KAGTQIENIE	EDFRNGLKLM	LLLEVISGER	LPRPDKGKMR	FKHIANVNKA	LDFIASKGVK	120
LVSIGAEIEV	DGNLKM TLGM	IWTIILRF AI	QDISVEETSA	KEGLLLWCQR	KTAPYRNVNV	180
QNFHTSWKDG	LALCALIHRH	RPDLIDYAKL	RKDDPIGNLN	TAFEVAEKYL	DIPKMLDAED	240
IVNTPKPDEK	AIMTYVSCFY	HAFAGAEQAE	TAANRICKVL	AVNQENEKLM	EEYEKLASEL	300
LEWIRRTVPW	LENRVGEP SM	SAMQRKLEDF	RDYRRLHKPP	RVQEKQCQLEI	NFNTLQTKLR	360
LSHRPAPFMS	EGKLVS DIAN	AWRGLEQVEK	GYEDWLLSEI	RRLQRLQHLA	EKFQQKASLH	420
ETWTRGKEDM	LTQRDYESAS	LQEVRLALLR	HEAFESDLAA	HQDRVEHIAA	LAQELNELDY	480
HEAASVNSRC	QAICDQWN DL	GTLTQKR RDA	LERMEKLLET	IDQLQLEFAR	RAAPFNNWLD	540
GAIEDLQDVW	LVHSVEETQS	LLTAHDQFKA	TLPEADRERG	AILGIQGEIQ	KICQTYGLRP	600
SSTNPYINLS	PDINTKWD T	VRKLVPSRDQ	TLQEELARQQ	MNERLRRQFA	AQANAVGPWI	660
QAKVEEVGRL	AAGLAGSLEE	QMAGLRQQEQ	NIINYKSNID	RLEGDHQLLQ	ESLVFDNKHT	720
VYSMEHIRVG	WEQLLTSIAR	TINEVENQVL	TRDAKGLSQE	QLNEFRASFN	HFDRKRNGMM	780
EPDDFRACLI	SMYDLGVEVF	ARIMTMVDPN	AAGVVTQAF	IDFMTRETAE	TDTAEQVVAS	840
FKILAGDKNY	ITPEELRREL	PABQAEYCIH	RMAPYKGAGA	PAGALDVYAF	SSALYGESDL	900

SEQ ID NO: 56 moltype = AA length = 842
 FEATURE Location/Qualifiers
 source 1..842
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 56

SRPLSDQEKR	KQISVRGLAG	VENVTELKKN	FNRHLHFTLV	KDRNVATPRD	YYFALAHTVR	60
DHLVGRWIRT	QOHYYEKDPK	RIYYLSLEFY	MGRTLQNTMV	NLALENACDE	ATYQLGLDME	120
ELEEIEEDAG	LGNGGLGRLA	ACFLDSMATL	GLAAYGYGIR	YEFGIFNQKI	CGGWQMEEAD	180
DWLRYGNPWE	KARPEFTLPV	HFYGRVEHTS	QGAKWVD TQV	VLAMPYDTPV	PGYRNNVVNT	240
MRLWSAKAFN	DFNLKDFNVG	GYIQAVLDRN	LAENISRVLV	PNDNFFEGKE	LRLKQYFVV	300
AATLQDIIR	FKSSKFGCRD	PVRTNFDAPP	DKVAIQ LNDT	HPSLAIPELM	RVLVDLERLD	360
WDKAWEVTVK	TCAYTNHTVL	PEALERWPVH	LLETLLPRHL	QIIYEINQRF	LNRVAAAFPG	420
DVDRLRRMSL	VEEGAVKRIN	MAHLIC IAGSH	AVNGVARIHS	EILKKTIFKD	FYELEPHKFK	480
NKTNGITPRR	WLVL CNPGLA	EIIAERIGEE	YISDL DQLRK	LLSYVDDEAF	IRDVAKVKQE	540
NKLKFAAYLE	REYKVHINPN	SLPDVQVKRI	HEYKRQLLNC	LHVITLYNRI	KKEPNKFVVP	600
RTVMIGGKAA	PGYHMAKMII	KLITAIGD VV	NHDPVVGDR L	RVIFLENYRV	SLAEKVIPAA	660
DLSEQISTAG	TEASGTGNMK	FMLNGALTIG	TMDGANVEMA	EEAGEENFFI	FGMRVEDVDR	720
LDQRGYNAQE	YYDRIPELRQ	IIEQLSSGFF	SPKQPD LFKD	IVNMLMHDR	FKVFADYEEY	780
VKCQERVSAL	YKNPREWTRM	VIRNIATSGK	FSSDR TIAQY	AREIWGVEPS	RQRLPAPDEK	840
IP						842

SEQ ID NO: 57 moltype = AA length = 362
 FEATURE Location/Qualifiers
 source 1..362
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 57

HSHPALTPEQ	KKELSDIAHR	IVAPGKGILA	ADESTGSI AK	RLQSIGTENT	EENRRFYRQL	60
LLTADDRVNP	CIGGVILFHE	TLYQKADDGR	PFPQVIKSKG	GVVGIKVDKG	VVPLAGTNGE	120
TTTQGLDGLS	ERCAQYK KDG	ADPAKWRCVL	KIGEHTPSAL	AIMENANVLA	RYASICQQNG	180
IVPIVEPEIL	PDGDHDLKRC	QYVTEKVLAA	VYKALSDHHI	YLEGTL LKPN	MVTPGHACTQ	240
KYSHEEIAMA	TVTALRR TVP	PAVTGVTFLS	GGQSEEEASI	NLNAINKCPL	LKPWALTFSY	300
GRALQASALK	AWGGKKENLK	AAQEYVVKRA	LANS LACQ GK	YTPSGQAGAA	ASESLFISNH	360
AY						362

SEQ ID NO: 58 moltype = AA length = 191
 FEATURE Location/Qualifiers
 source 1..191
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 58

APKDKVKKPA	AAAAPAPAPA	PAPAPAPAKP	KEEKIDLSAI	KIEFSKEQOD	EFKEAPLLYD	60
RTGDSKITLS	QVGDVLRALG	TNPTNAEVKK	VLGNPSNEEM	NAKKIEFEQF	LPMLQAI SNN	120
KDQGTIEDFV	BGLRVFDKEG	NGTVMGAE LR	HVLATLG EKM	KEEEVEALMA	GQEDSNGCIN	180
YEA FVKHIMS	I					191

SEQ ID NO: 59 moltype = AA length = 333
 FEATURE Location/Qualifiers
 source 1..333

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mol_type = protein
organism = Oryctolagus cuniculus

SEQUENCE: 59
MVKGVNGVGF RIGRLVTRAA FNSGKVDVVA INDPPIDLHY MVYMFQYDST HGKFHGTVKA 60
ENGKLVINGK AITIFQERDP ANIKWGDAGA EYVVESTGVF TTMEKAGAHL KGGAKRVIIS 120
APSADAPMFV MGVNHEKYDN SLKIVSNASC TTNCCLAPLAK VIHDFHGIVE GLMTTVHAIT 180
ATQKTVDGPG GKLRWDGRGA AQNIIPASTG AAKAVGKVIP ELNGKLTGMA FRVPTPNVSV 240
VDLTCRLEKA AKYDDIKKVV KQASEGPLKG ILGYTEDQVV SCDFNSATHS STFDAGAGIA 300
LNDHFPVKLIS WYDNEFGYSN RVVDLMVHMA SKE 333

SEQ ID NO: 60      moltype = AA  length = 284
FEATURE           Location/Qualifiers
source            1..284
                  mol_type = protein
                  organism = Oryctolagus cuniculus

SEQUENCE: 60
MDAIKKKMQM LKLDKENALD RAEQAEADKK AAEDRSKQLE DELVSLQKKL KGTEDELDKY 60
SEALKDAQEK LELAEEKATD AEADVASLNR RIQLVEEELD RAQERLATAL QKLEEAEEKAA 120
DESERGMKVI ESRAQKDEEK MEIQEIQLKE AKHIAEDADR KYEEVARKLV IIESDLERA 180
ERAELSEGKC AELEELKTV TNNLSLEAQ AEKYSQKEDK YEEBIKVLSD KLKEAETRAE 240
FAERSVTKLE KSIDDLDEL YAQKLKYKAI SEELDHALND MTSI 284

SEQ ID NO: 61      moltype = AA  length = 1134
FEATURE           Location/Qualifiers
source            1..1134
                  mol_type = protein
                  organism = Oryctolagus cuniculus

SEQUENCE: 61
EAKPAARKAP KGKDAPKEEP PKEAAAEPPK EAPPEDQSPT AEEPTGIFLK KPDSVSVETG 60
KDAVIMAKLN GKELPAKFTI KWFPGKWLEL GSKSGARFSF KETHDAASNV YSLELHIGKV 120
VLGDRGDYRL EVKAKDVCDL CSFNIDVEAP RQDSAGQSL ESKRAGDAKP DGAGELDFSG 180
LLKKREVVEE EKKKKKDDDD LGIPPEIWEL LKGAKKSEYE RIAFQYGITD LRGMLKRLKK 240
AKVEVKKSAA FTKKLDPAYQ VDRGNKVKL V EISDPDLPL KWFKNQGEIK PSKYVFN 300
GKKRILITNK CTLADDAAYE VVNDKCKFT ELFVKEPPVL IVTPLEDQOV FVGDRVEMSV 360
EVSEEGAQVM WMKDGVELTR EGSFKSNYRF KKDGRHILI YSEVNLEDRG HYQVMTNGGQ 420
CEAELIVEEK QLEVLQDIAD LTVKASEQAM FKCEVSDEKV TGKWKNGVE VRPSKRITIS 480
HVGRFHKLVI DDVRPEDEGD YTFVPDGYAL SLSAKLNFLE IKVEYVPKQE PPKIHLDCSG 540
KTSDNAIVVV AGNKLRLDVS ITGEPPPvat WMKGDEVFTV TEGRVRIEQR PDSSSFVIES 600
AERGDEGRYT IKVTNPAGED VATVMLRVVD VPDPEAVRV TSVGEDWAIL VWDPPKYDGG 660
QPVVTGYLLER KKKGSQRWML NFEVFTETT YESTKMEIGI LYEMRVFAVN AIGVSQPSMN 720
TKPFMPIAPT SAPQHLTVED VTDTTTTLKW RPPDRIGAGG IDGYLVEYCV EGSDEWIPAN 780
KEPVERCGFT VKGLPTGAKI LFRVIGVNIA GRSEPATLVQ PVTIREIVEQ PKIRLPRHLR 840
QTYVRKVGHE INLVIPOGK PRPQVWTKG GAPVDTSRVH VRTSDFDTVF FVREAARSDS 900
GEYELSVQIE NMKDTATVRI RVVEKAGPAE NVMVKEVWGT NALVEWQPPK DNGNSEITGY 960
LVQKADKKTW EWFVYEHNR HTSCTVSDLI VGNEYFYFRVY TENVCGLSDS PGVSKNTAHI 1020
LKTGITFKFP EYKEDHFRSP PKFLTPLIDR VVVGAYAAAL NCAVRGHPKP KVVMMKNKVE 1080
IHEDPKFLMT NHQGVLTINI RRPSPFDAGT YSCRAFNELG EALAECLRDV RVPQ 1134

SEQ ID NO: 62      moltype = AA  length = 1415
FEATURE           Location/Qualifiers
source            1..1415
                  mol_type = protein
                  organism = Oryctolagus cuniculus

SEQUENCE: 62
MSLVAVPFYQ KRHKHFDQSY RNIQTKYLM D EYTAKRKAST QMSTHKSAST QMATHKSALQ 60
KASSQTSSGG TTCRLCAKRL SAAEEEEQES KNRYRSLVAS YGEAKRQRFL SELAQLEESV 120
HLARSQARDV VDKYALQOMV EDKMAWERHV FEERLSRAPE ILVQLRSHTV WEKMSVKLCF 180
TVQGFPTPVV QWYKDGLIC QAGEPGKYRI DSKYGVHTLE ISRASFDATA TYSAVATNAH 240
GOVSTNAAVV VRRYRGDEEP FYSVGLPIGY VLVKESKWTR MFFGEGQASL SFSHLNKDDE 300
GLYTLRIVSR GGVSDHSAPL FVRDADPLVT GAPGAPMDLQ CHDANRDYVI VTWKPPNTTT 360
ESPVIGYFVD RCEVGTNNWV QCNDAPVKIC KYPVTGLYEG RSYIFRVRAV NSAGISRPSR 420
VSDAVAALDP ADLRLRLQAIH LEGEKEVVYI QDDLEGDVQV PGPPTNVHAS EISRNYYVLS 480
WEPPVPRGKE PLMYFIEKSV VSGSGWQVRN AQTAVRSPRY AVFDLAEKGP YVFRVLSANK 540
MLWKAPVYSG SSPVSGYFVD FKEEDSGEWS TVNEAATPNR YLKVSDLHQG RTYVFRVRV 840
NASGVGKPSD VSEPVLEEAR PGTKEISAGV DEEGNIYLG FDCQEMTDASH FTWCKSYEEI 900
ADPERFHVET KGDHSLKYFK NPDKADLGT Y SVSVSDTDGV SSSFVLDEEE LERLLALSNE 960
IKNPTIPLKS ELAYEIPDKG QVRFWLQTEP LSPDASFREV INDREVSDSE THRMKCDKAS 1020
GIEMVMMDRF TIENEGTYTV QIHDGKAKNQ SSLVLIGDAF RKVLEEAEFQ RKEFLRKQGP 1080
HFAEVLHWDV TEECEVRLIC KVANTKKE TV FKWLKDDVLY ETETPPDLEK GVCCELLIPKL 1140
SKKDHGEYKA TLKDERGQDA SILEIAGKVY EDMILAMSRV CGMSASPLKI LCTPEGIRLQ 1200
CFMKYFTEEM RVNWHYKDAK ISSSEHMRIG GSEDMAWLQI CEPTKDKGK YTFEIDGDK 1260
NHQRLDLDSG QAFDEAFAEF QRLKAAAFAP KNRGKVI GGL PDVVTIMEGK TLNLTCTVFG 1320

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NPDPEVVWFK	NDQDIELSEH	FSVKVEQAKY	VSLTIKGVTS	EDSGKYSINV	KNKYGGEKID	1380
VTVSVYKHGE	AIPEVAPPQQ	AKPKLIPASA	SATAP			1415

SEQ ID NO: 63	moltype = AA length = 433
FEATURE	Location/Qualifiers
source	1..433
	mol_type = protein
	organism = <i>Oryctolagus cuniculus</i>

SEQUENCE: 63						
AMQKIFAREI	LDSRGNPTVE	VDLHTAKGRF	RAAVPSGAST	GIYEALRLD	GDKSRYLGKG	60
VLKAVEHINK	TLGPPALLEK	LSVVDQEKVD	KFMIELDGTE	NKSKFGANAI	LGVSLAVCKA	120
GAAEKGVPLY	RHIADLAGNH	DLVLPVPAPN	VINGGSHAGN	KLAMQEFMIL	PVGASSFREA	180
MRIGAEVYHH	LKGVIAKAYG	KDATNVGDEG	GFAPNILENN	EALLLKTAI	QAAGYDPKVV	240
IGMDVAASEF	HRNGKYDLDF	KSPDDPARHI	TGQKLGEYK	SFIKNYPVVS	IEDPFDQDDW	300
ATWTSFSLGV	DIQVGGDLT	VTNPKRIAQA	VEKKACNCLL	LKVNQIGSVT	ESIQAACKLAQ	360
SNGWGVVMSH	RSGETEDTFI	ADLVVGLCTG	QIKTGAPCRS	ERLAKYNQLM	RIEALGDKA	420
VFAGRKFRNP	KAK					433

SEQ ID NO: 64	moltype = AA length = 169
FEATURE	Location/Qualifiers
source	1..169
	mol_type = protein
	organism = <i>Oryctolagus cuniculus</i>

SEQUENCE: 64						
APKKAKRRRA	ABGGSSNVFS	MFDQTQIQEF	KEAFTVIDQN	RDGIIDKEDL	RDTFAMGRLL	60
NVKNEELDAM	MKEASGPINF	TVFLTMPGEEK	LKGADPEDVI	TGAPKVLDP	GKGTIKKQFL	120
EELLTTQCDR	FSQEEIKNMW	AAFPPDVGGN	VDYKNICYVI	THGDAKDQE		169

SEQ ID NO: 65	moltype = AA length = 1273
FEATURE	Location/Qualifiers
source	1..1273
	mol_type = protein
	organism = <i>Oryctolagus cuniculus</i>

SEQUENCE: 65						
IFSISDNAYQ	FMLTDRENQS	ILITGESGAG	KTVNTRKVIQ	YFATIAITGD	KKKEEPTPGK	60
MQGTLEDQII	SANPLLEAFG	NAKTVRNDNS	SFRGKFIRIH	FGATGKLASA	DIETYLLEKS	120
RVTFQLKAER	SYHIFYQIMS	NKKPELIELM	LITTNPYDFA	FVSQGEITVP	SIDDQEELMA	180
TDSAVDILGF	TSDEKVAIYK	LTGAVMHHYGN	LKFKQKQREE	QAEPDGTEVA	DKAAYLTSLN	240
SADLLKALCY	PRVKVGNFYV	TKGQTVQQVY	NAVGAALAKAI	YEKMFLLWMT	RINQQLDTKQ	300
PRQYFIGVLD	IAGFEIIFDN	SLEQLCINFT	NEKLQOFFNH	HMFVLEQEEY	KKEGIEWEFI	360
DFGMDLAACI	ELIEKPMGIF	SILEEECMFP	KATDTSFKNK	LYEQHLGKSN	NFQKPKPAKG	420
KAAAHFSLVH	YAGTVDYNIT	GWLDKNKDPL	NETVVGLYQK	SGMKTALFLF	SGAQTGEEGG	480
GGGKKGKKKK	GSSFQTVSAL	FRENLNKLMT	NLRSTHPHFV	RCLIPNETKT	PGAMEHELV	540
HQLRCNGVLE	GIRICRKGFP	SRILYADFKQ	RYKVLNASAI	PEGQFIDSCK	ASEKLLGSID	600
VDHTQYKFGH	TKVFFKAGLL	GLLEEMRDEK	LAQLITRTQA	ICRGYLTRVE	FKMMERRES	660
IFCIQYNIRA	FMNVKHWFPW	KLYFKIKPLL	KSAETEKEMA	NMKEEFKTK	EELAKSEAKR	720
KELEEKMAVL	MQEKNDLQLQ	VQAEADGLAD	AEERCDQLIK	TKIQLEAKIK	EVTERAEDEE	780
ETNAELTAKR	RKLDECESEL	KKDIDDELEL	LAKVEKEKHA	TENKVNLT	EMAGLDENIA	840
KLTKEKKALQ	EAHQQTLDL	QAEEDKVNTL	TKAKTKLEQQ	VDDLEGSLEQ	EKKLRMDLER	900
AKRKLEGLDK	LAQESTMDIE	NDKQQLDEKL	KKKEFEMTNL	QSKIEDEQAL	AMQLQKKIKE	960
LQARIEELEEE	EIEAERASRA	KAKQKRSDDL	RELEIEISERL	EEAGGATSAQ	IEMNKKREAE	1020
FQKMRDLREE	ATLQHEATAA	ALRKKHADSV	AELGEQIDNL	QRVKQKLEKE	KSELKMEIDD	1080
LAGNMETVSK	AKGNLEKMC	TLEDQLSEVK	TKEEEHQRLI	NELSAQKARL	HTESGEFSRQ	1140
LDEKDAMVSQ	LSRGKQAFQT	QIEELKRQLE	EETKAKSALA	HALQSSRHDC	DLLREQYEEE	1200
QEAQALQRA	MSKANSEVSQ	WRTKYETDAI	QRTEELEEA	KKLAQRLQDA	EEHVEAVNSK	1260
CASLEKTKQR	LQN					1273

SEQ ID NO: 66	moltype = AA length = 1161
FEATURE	Location/Qualifiers
source	1..1161
	mol_type = protein
	organism = <i>Oryctolagus cuniculus</i>

SEQUENCE: 66						
IFYQIMSNKK	PELIEMLLIT	TNPYDFAFVS	QGEITVPSID	DQEELMATDS	AVDILGFTSD	60
EKVAIYKLTG	AVMHYGNLKF	KQKQREEQAE	PDGTEVADKA	AYLTSLSNAD	LLKALCYPRV	120
KVGNIEYVTKG	QTVQOVYNAV	GALAKAIYEK	MFLWMVTRIN	QQLDTKQPRQ	YFIGVLDIAG	180
FEIFDFNSLE	QLCINFTEK	LQQFFNNHMF	VLEQEEYKKE	GIEWEFIDFG	MDLAACIELI	240
EKPMGIFSL	EEECMFPKAT	DTSEFKNKLYE	QHLGKSNFQ	KPKPAKGKAE	AHFSLVHYAG	300
TVDYNITGWL	DKNKDPLNET	VVGLYQKSGM	KTALAPLFGA	QTGEEGGGGG	KGGGKKKGSS	360
FQTVSALFRE	NLNKLMTNLR	STPHFVRCL	IPNETKTPGA	MEHELVLHQL	RCNGVLEGIR	420
ICRKGFPSPR	LYADFKQRYQ	VLNASAIPEG	QFIDSKKASE	KLLGSDVDH	TQYKFGHTKV	480
FFKAGLLGLL	EEMRDEKLAQ	LITRTQAICR	GYLTRVEFKK	MMERRESIFC	IQYNIRAFMN	540
VKHWPWMKLY	FKIKPLLSA	ETEKEMANMK	EEFEKTEEL	AKSEAKRKEL	EEKMVALMQE	600
KNDLQLQVQA	EADGLADAE	RCDQLIKTKI	QLEAKIKEVT	ERADEEEIN	AELTAKKRKL	660
EDECESELKID	IDDLLETLAK	VEKEKHATEN	KVKNLTEEMA	GLDENIAKLT	KEKKALQEAH	720
QQTLDLQAE	BDKVNTLTKA	KTLEQQVDD	LEGSLEQEK	LRMDLERAKR	KLEGDLKLAQ	780

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ESTMDIENDK	QQLDEKLKKK	EFEMTNLQSK	IEDEQALAMQ	LQKKIKELQA	RIELEEIEE	840
AERASRAKAE	KQRSDLSREL	EEISERLEEA	GGATSAQIEM	NKKREAEFQK	MRRDLEEATL	900
QHEATAAALR	KKHADSVael	GEQIDNLQRV	KQKLEKEKSE	LKMEIDDLAQ	NMETVSKAKG	960
NLEKMCRTL	DQLSEVKTKE	EEHQRLINEL	SAQKARLHTE	SGEFSRQLDE	KDAMVSQLSR	1020
GKQFTQQIE	ELKRQLEET	KAKSALAHAL	QSSRHDCDLL	REQYEEQEA	KALQRAMSK	1080
ANSEVSQWRT	KYETDAIQRT	EELEEAKKL	AQRLQDAEEH	VEAVNSKAS	LEKTKQRLQN	1140
EVEDLMIDVE	RSNAACAALD	K				1161

SEQ ID NO: 67 moltype = AA length = 667
 FEATURE Location/Qualifiers
 source 1..667
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 67

TLDDLQAEED	KVNTLTAKT	KLEQQVDDLE	GSLEQEKKL	MDLERAKRKL	EGDLKLAQES	60
TMDIENDKQ	LDEKLKKKEF	EMTNLQSKIE	DEQALAMQL	KKIKELQARI	EELEEIEAE	120
RASRAKAEKQ	RSDLSRELEE	ISERLEEAGG	ATSAQIEMNK	KREAEFQKMR	RDLEEATLQH	180
EATAAALRKK	HADSVaelGE	QIDNLQRVKQ	KLEKEKSELK	MEIDDLAGNM	ETVSKAKGNL	240
EKMCRITLEDQ	LSEVKTKEEE	HQRLINELSA	QKARLHTEG	EFSRQLDEKD	AMVSQLSRGK	300
QFTQQIEEL	KRQLEETKA	KSALAHALQS	SRHDCDLLRE	QYEEQEA	ELQRAMSKAN	360
SEVSQWRTKY	ETDAIQRT	LEEAKKLAQ	RLQDAEEHVE	AVNSKASLE	KTQRLQNEV	420
EDLMIDVERS	NAACAALDKK	QRNFDKVLAE	WKHKYEETQA	ELEASQKESR	SLSTEVFKVK	480
NAYEESLDHL	ETLKREKNL	QQEISDLTEQ	IAESGKHIHE	LEKVKKQIDQ	EKSELQAALE	540
EAEGLSEHEE	GKILRIQLEL	NQVKSIEDRK	IAEKDEEIDQ	LKRNLHVRVE	SMQSTLDAEI	600
RSRNDALRIK	KKMEGDLNEM	EIQLNHANRQ	AAEAIKNLNR	TQGILKDTQL	HLDDAVRGQD	660
DLKEQLA						667

SEQ ID NO: 68 moltype = AA length = 830
 FEATURE Location/Qualifiers
 source 1..830
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 68

LAQESTMDIE	NDKQQLDEKL	KKKEFEMTNL	QSKIEDEQAL	AMQLQKKIKE	LQARIEELE	60
EIEAERASRA	KAQKQSDLS	RELEEISERL	EEAGGATSAQ	IEMNKKREAE	FQKMRDLEE	120
ATLQHEATAA	ALRKHADSV	AELGEQIDNL	QRVQKLEKE	KSELKMEIDD	LAGNMETVSK	180
AKGNLEKMC	R	TKEEEHQRLI	NELSAQKARL	HTESGEFSRQ	LDEKDAMVSQ	240
LSRGKQFTQ	QIEELKRQLE	EETKAKSALA	HALQSSRHDC	DLLREQYEEE	QEAQELQRA	300
MSKANSEVSQ	WRTKYETDAI	QRTTEEEELAK	KKLAQRLQDA	EEHVEAVNSK	CASLEKTKQR	360
LQNEVEDLMI	DVERSNAAAC	ALDKKQRNFD	KVLAEWKHKY	EETQAELEAS	QKESRSLSTE	420
VFKVKNAAYE	SLDHLETLEK	ENKNLQQEIS	DLTEQIAESG	KHIHELEKVK	KQIDQEKSEL	480
QAALAEAGS	LEHEEGKILR	IQLELNQVKS	EIDRKIAEKD	EEIDQLKRNH	LRVVESMQST	540
LDAEIRSRND	ALRIKKMEG	DLNEMETQLN	HANRQAAEAI	KNLNRNTQGIL	KDTQLHLDDA	600
VRGQDDLKEQ	LAMVERRANL	MQAIEELRA	SLEQTTERSRR	VAEQELLDAS	ERVQLLHTQN	660
TSLINTKKKL	ETDISQIQGE	MEDIVQEARN	AEEKAKKAIT	DAAMMAEELK	KEQDTSAHLE	720
RMKKNMEQTV	KDLQHRLEDEA	EQALAKGGKK	QIQKLEARVR	ELENEVESEQ	KRNEVAVKGL	780
RKHERRVKEL	TYQTEEDRKN	VLRQLDLVDK	LQSKVKAYKR	QAEEAEEQSN		830

SEQ ID NO: 69 moltype = AA length = 1218
 FEATURE Location/Qualifiers
 source 1..1218
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 69

VFVADPKESF	VKATVQSR	GKVTAKTEAG	ATVTVKEDQV	FPMNPPKYDK	IEDMAMMTHL	60
HEPAVLNLK	ERYAAWMIYT	YSGLFCVTVN	PYKWLVPVNA	EVVTAYRGKK	RQEAPPHIFS	120
ISDNAYQFML	TDRENQSILI	TGESGAGKTV	NTKRVIQYFA	TIATVGDKKK	EEATSGKMQG	180
TLEDQIISAN	PLLEAFGNAK	TVRNDNSSRF	GKFIIRIHFGT	TGKLASADIE	TYLLEKSRVT	240
FQLKAERSYH	IFYQIMSNNK	PDLIEMLLIT	TNPYDYAFVS	QGEITVPSID	DQEELMATDS	300
AIDILGFTSD	ERSVIYKLTG	AVMHYGNMKF	KQKQREEQAE	PDGTEVADKA	AYLQSLNSAD	360
LLKALCYPVR	KVGNEYVTKG	QTVQQVYNVAV	GALAKAVYEK	MFLWMVTRIN	QQLDTPQPRQ	420
YFIGVLDIAG	FEIFDFNSLE	QLCINFTNEK	LQQFFNHMMF	VLEQEEYKKE	GIEWEFIDFG	480
MDLAACIELI	EKPMGIFSIL	EEECMPFKAT	DTSFKNKLYE	QHLGKSNFQ	KPKPAKGKVE	540
AHFSVLVHYAG	TVDYNTGWL	DKNKDPLNET	VVGLYQKSAM	KTALFLFTGT	AAAEAEAGGK	600
KGGKKKGSSF	QTVSALFREN	LNLKMTNLR	THPHFVRCII	PNETKTPGAM	EHVLVHLQLR	660
CNGVLEGIRI	CRKGFP	SRIL	YADFKQRYKV	LNASAIPEGQ	FIDSKKASEK	720
QYKFGHTKVF	FKAGLLGLLE	EMRDDKLAQL	ITRTQAMCRG	FLARVEYKMK	VERRESIFCI	780
QYNIRAFMNV	KHWPMKSLYF	KIKPLKSAE	TEKEMANMKE	EFEKTESLA	KAEAKRKELE	840
EKMVALMQEK	NDLQVQAE	ADSLADAEER	CDQLIKTKIQ	LEAKIKEVTE	RAEDDEEINA	900
ELTAKKRKLE	DECSLEKDI	DDLETLAKV	EKEKHATENK	VKNLTEEMAG	LDETIAKLTK	960
EKKALQEAHQ	QTLDDLQAE	DKVNTLTAKK	TKLEQQVDDL	EGSLEQEKKI	RMDLERAKRK	1020
LEGDLKLAQE	STMDIENDKQ	QLDEKLKKKE	FEMSNLQSKI	EDEQALAMQL	QKKIKELQAR	1080
IELEEIEEA	ERASRAKAEK	QSDLSRELE	EISERLEEAG	GATSAQIEMN	KKREAEFQKM	1140
RDLLEEATLQ	HEATAATLRK	KHADSVaelG	EQIDNLQRVK	KQKLEKEKSEL	KMEIDDLASN	1200
METVSKAKGN	LEKMCRTL					1218

-continued

SEQ ID NO: 70 moltype = AA length = 1574
FEATURE Location/Qualifiers
source 1..1574
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 70

TLVSDVDYRN	YLHQWTCMPD	QNDVIQAKKA	YELQSDAVYK	ADLEWLRLGIG	WMPSDSVSVN	60
HAKHAADIFS	EKKYRTKIET	LNFTPVDDRV	DYVTAKQSGE	ILDDIKYRKD	WNSIKSKYTL	120
TETPLLHTVQ	EAARILDQYL	YKESWEKQKA	TGYILPPDAV	PFVHAHHSSD	VQSELKYKAE	180
HVKQRGHVYG	VPTMRDDPKL	VWFEHAGQIQ	NDRLYKEDYH	KTKAKINIPA	DMVSVVAAKE	240
GQNLVSDIDY	RNYLHQWTC	PDQNDVIQAR	KAYDLQSDNV	YKADLEWLRLG	IGWIPLDSDV	300
HVRVTKNQEM	VNQIKYKDA	LDNYPNFTSV	VDPPAIVLAK	INAVNQSDVK	YKETFNKAKG	360
KYTFSPDTPH	ISHSKDMGKL	YSTILYKGAW	EGTKAYGYTL	DERYIPIVGA	KHADFDVNSL	420
KYKETYEKQK	GHYLAGKVIS	EFPGVVHCLD	FQKMRSVLNY	RRDYENTKAN	VHIPNDMMNH	480
VLAACKQYIL	SDLEYRHYFH	QWTSLPDEPK	VLLARNAQEI	LSDNVYKDDL	NWLKGIGCYV	540
WDTPQILHAK	KSYDLQSQLQ	YTAAGKENLQ	NYNLVTDTP	YVTALQSGIN	ASEVKYKENY	600
HQIKDKYTTV	LETVDYDRTR	NLKDLYSSNL	YKEAWNKVKA	TSYILPTSTL	SLTHAKNQKH	660
LASHIKYREE	YEKFKALYTL	PKSVEDDPNT	ARCLRVGKLN	IDRLYRSVYE	KNKMKIHIVP	720
DMVEMVTAKD	KYKSVYEHK	RLHLHEWICH	PDLQVNSHVR	KVTDQISDIV	YKDDLNLWLG	780
IGCYVWDTP	ILHAKHAYDL	RNDIKYKAHM	MKTRNDYKLV	TDTPVYVQAV	KSGKQLSDAV	840
YHHDVHVSVR	GKVAPTTKTV	DLERALHAYK	LQSENLYKTS	LRTLPTGYRL	PGDTPHFKHN	900
KDIRVMSSVF	KYKSVYEHK	ANGYTLGPND	VPFVNVRRVN	KVTSERLYRE	LYHKLKDKIH	960
TTPDTPPEIRQ	VKKTQEAUSE	LIYKSDFFKM	QGHMISLPYT	PHVLHCRYVG	DITSIDIKYE	1020
DLQILKGLGC	FLYDTPDMVR	SRHLRLKLSN	YLYTDNARKM	RDYKVVLDLT	PEYRKVQELK	1080
THLSELVYRA	SGRKQKSIFT	SVPDTPDLLR	AKRGQKLQSQ	YLYVELATKE	RPHHAGNQ	1140
TALKHARDVK	DMVSEKKYKV	QYKMKDKYT	PIPDTPILIR	AKNAYWNASD	LRYKETFOKT	1200
KGKYHTVKDA	LDIVYHRKVT	DDISKVKYKE	NYMRHLGIWR	SIPDRPEPHF	HRAVTDVSD	1260
IKYKEDLTWL	KGIGCYAYDT	PDPDLAEKNK	TLYSKYKYKE	VFERTKSDFK	YVADCPINRH	1320
FKYATQLMNE	KKYRADYEQ	KDKYQLVVDE	PRHLLAKTAG	DQISQIKYRK	QYEKSKDKFT	1380
SVVDTPPEHLR	TTKVNKQISD	ILYKLEYNKA	KPRGYTTIHD	TPMLLHVVRK	KDEVSDLYKY	1440
EYVQRNKSNC	TIKPDVAVHIK	AAKDAYKVNT	NLDYKKQYEA	NKAHWRWTPD	RPDFIQAAKS	1500
SLQQSDFEYK	LDREFLKGCK	LSVTDDKDMV	LALKNSLIES	DLKYKEKHVK	ERGSCHAVPD	1560
TPQILLAKTV	SNLV					1574

SEQ ID NO: 71 moltype = AA length = 1500
FEATURE Location/Qualifiers
source 1..1500
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 71

EHTRNVNQIQ	SDNVYKDEYN	NFFKGIGWIP	IGSLEVEKVK	KAGDALNEKK	YRQHPDTIKF	60
TSVPDSMGV	LAQHNTQQLS	DLNYKVEGEK	VKHXYTMDPE	VPQFIQARVN	AFNLSDAHYK	120
ADWKKTLAKG	YDLRDPDAIP	VAAKSSRNIA	SDCKYKEAYE	KAKGKQIGFL	SLQDDPKLVH	180
YMNVAKIQSD	REYRKGYES	KTRYHTPLDM	LSVTAAKKSQ	EVATNANYKQ	PFHNYTLPLD	240
ALPVEHSRNA	MQIQSDNLYK	SDFTGWMKGI	GWIPQISLEV	EKAKKAGEIL	SEKRYRQHP	300
KLKFTYAMDT	MEQALNKANK	LNMDKRLYTK	KWDKDKTTIH	VMPDTPDILL	SRVNQITMSD	360
KLYRAGWEEE	KKKGYDLRPD	AISIKAAKAS	RDIASDYKYK	QAYEQAKGKH	IGFRSLEDDP	420
KLVHFMQVAK	MQSDREYKKA	YEKSKTSFHT	PVDMLSVVA	KKSQEVATNA	NYRNVITHTN	480
MLPDAMGFEL	AKNMQIQSD	NQYKADYADF	MKGIGWPLG	SLEAEKNKKA	MEIIEKKYR	540
QHPDTLKYST	LMDSMNVMVA	QNNAKIMNEH	LYQAWAEADK	TKVHIMPDIP	QIVLAKANAI	600
NMSDKLYKLS	MEEAKKGGYD	LRTDAIPIKA	AKASRDIASD	YKYKYNYEKQ	KGKMGVGRSL	660
EDDPKLVHSM	QVAKMQSDRE	YKKNYEKTKT	SYHTPADMLS	VTAAKDAQAN	ITNTNYKHLI	720
HKYILLPDAM	SLELSRNMNR	IQSDNEYKQD	YNEWYKGLGW	SPAGSLDVEK	AKKATEYASD	780
QKYRQHPSNF	QFKKLTDSDM	MVLAKQNAHT	MNKHLYTIDW	NKDKTKIHMV	PDTPDILQAK	840
QNQTLFSQKF	YKLGWEEALK	KGYDLVPDAI	SVQLAKASRD	IASDFKYKQG	YRKQLGHHVG	900
FRSLQDDPKL	VLSMNVAKMQ	SEREYKDFE	KWKTKFSSPV	DMLGVVLAKK	CQALVSDVDY	960
KNYLHQWTC	PDQNDVIHAK	KAYELQSENM	YKSDLEWLKG	IGWSPLGSLE	AEKNKRASEI	1020
LSEKKYRQPP	DTNKFTSIPD	AMDIVLAKTN	AKNRSRLYR	EAWDKDKTQV	HIMPDPDII	1080
LAKANLINTS	DKLYRMGYEE	LKKKGVDLPV	DAIPIKAAKA	SREIASEYKY	KEGFRKQLGH	1140
HIGARNIKDD	PKMMMSMHVA	KIQSDREYK	DFEKKWTKFS	SPVDMLGVVL	AKKCQTLVSD	1200
IDYKNYLHWQ	TCLPDQNDVT	HARHAYDLQS	DNLYKSDLQW	LKGIGWIPTG	SIEDEKNKRA	1260
TQILSDHVR	QHPDKFKFSI	LMDSIPMVA	KNNATNMNHR	LYTEAWDKDK	TTVHIMPDP	1320
EVLAKQNGV	NCSKLYKLG	LEETKRKGVD	MRVDAIPIKA	AKASRDIASE	FKYKEGYRQK	1380
LGHHIGARAI	HDDPKMMWSM	HVAKIQSDRE	YKDFEKKWKT	KFSSPVDMLG	VVLAKKCQAL	1440
VSDVDYKNYL	HQWTCPLPDQ	DVMHARQAYD	LQSDNLYKSD	LQWLRLGIGW	PIGSLDVEK	1500

SEQ ID NO: 72 moltype = AA length = 1400
FEATURE Location/Qualifiers
source 1..1400
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 72

DVTHARHAYD	LQSDNLYKSD	LQWLKIGIGWI	PTGSIEDEKN	KRATQILSDH	VYRQHPDKFK	60
FSILMDSIPM	VLAKNATNMN	NHRLYTEAWD	KDKTTVHIMP	DTPEVLLAKQ	NQVNCSEKLY	120
KLGLEETKPK	GYDMRVDAIP	IKAAKASRDI	ASEFKYKEGY	RKQLGHGIGA	RAIHDDPKMM	180
WSMHVAKIQS	DREYKDFEKK	WKTKFSSPVD	MLGVVLAKKC	QALVSDVDYK	NYLHQWTCPL	240

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DQNDVMHARQ	AYDLQSDNLY	KSDLQWLRLGI	GWVPIGSLDV	EKCKKATEIL	SDKIYRQPPD	300
RFKFTSVTDS	LEQVLAKNNA	INMNKRLYTE	AWDKDKTQIH	IMPDTPPEIML	ARMNKINYSE	360
SLYKLANEEA	KKKGYDLRSD	AIPIVAAKAS	RDIASDYKYK	DGYRKQLGHH	IGARDIKDDP	420
KMMWMSMHVAK	IQSDREYKDK	FEKWKTKFSS	PVDMLGVVLA	KKQCTLVSDI	DYKNYLHQWT	480
CLPDQNDVMH	ARQAYDLQSD	NIYKSDLQWL	RGIGWVPIGS	LEVAKCKRAA	EILSDNIYRQ	540
PPDKLKFTSV	PDSLEQVLAK	NNAINMNKRL	YTEAWDKDKT	QIHIMPDTPPE	IMLARQNKIN	600
YESLYKLAN	EEAKKKGYDL	RSDAIPIVAA	KASRDIASDY	KYKDGYYRQL	GHHIGARDIK	660
DDPKMMWMSH	VAKIQSDREY	KKDFEKKWTK	FSSPVDMGLV	VLAKKCCQTLV	SDIDYKNYLH	720
QWTCLPDQND	VMHARQAYDL	QSDNIYKSDL	QWLRGIGWVP	IGSVDLVKCK	RAAEILSDNI	780
YRQSPDKLKF	TSVTDLSLEQV	LAKNNAINMN	KRLYTEAWDK	DKTQIHMPDP	TPEIMLARQN	840
KMNYSENFYR	QAMEEAKKEG	YDLRSDAIP	VAAKASRDIA	SDYKYKEAYR	KQLGHHIGAR	900
AIHDDPKMMW	SLHIAKVQSD	REYKKEFEKY	KTRYSSPVDM	LGIVLAKKCC	TLVSDVDYKN	960
LLHEWTCLPD	QNDVIQARKA	YDLQSDNLYK	SDLEWLRGIG	WVPIGSLEVV	KAKRATEILS	1020
DNLYRTRPDT	LKFTSVTDSL	EQVLAKNNAI	NMNKRLYTEA	WENDKKTIVH	MPDTPPEIMLA	1080
KLNKINYSK	LYKLAEESK	KEGYDLRVDA	IPIQAAKASR	DIASDYKYKE	GYRKQLGHHI	1140
GARNIKDDPK	MMWMSMHVAK	QSDREYKDKF	EKWKTKFSSP	VDMLGVVLAK	KCQILVSDID	1200
YKHPLHEWIC	LPDQNDVIQA	RKAYDLQSDA	VYKADLEWLR	GIGWVPIGSV	EVEKVKRAGE	1260
ILSDRKYRQP	ADKLKFTSIT	DTPEIVLAKS	NALTMKSHLY	TEAWDADKTS	IHVMPDTPPEI	1320
LLAKSNSANI	SQKLYTKGWD	ESKMKDYDLK	ADAIPIKSAK	ASRDIASDYK	YKEAYEKQKG	1380
HHIGAQSIED	DPKIMCAIHA					1400

SEQ ID NO: 73 moltype = AA length = 1500
 FEATURE Location/Qualifiers
 source 1..1500
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 73

DKRETSRPNW	AQVSANVPIT	SCSVEKLIIEG	HEYQPRICAE	NKYGVGDPVF	TEPAIAKNPY	60
DPPGRCDPPV	ISNVTKDHMT	VSWKPPADDG	GSPITGYLLE	KRETHAVNWT	KVNRKPVIER	120
TIKATGLQEG	TEYEFRTVAI	NKAGPGKPSD	ASKAVYAQDP	LYPPGPPAFP	KVYDTRSSV	180
SLSWGKPAYD	GGSPIIGYLV	EVKRADTDNW	VRCLNPQKLQ	KTRFEVTGLM	ENTHEYQFRVY	240
AVNKVGYS DP	SDVPDKHCPK	DILIPPEGEL	DADLRKTLIL	RAGVTMRLYV	PVKGRPPPKI	300
TWSKPNVNL	ERIGLDIKST	DPDTFLRCEN	VNKYDAGKYI	LTLNESCCKK	EYTIIVKVLD	360
TPGPPVNVTV	KEISRDSAYI	TWDPPIVDGG	SPIINYVVEK	RDAERKSWST	VTTECSKTSF	420
RVSNLLEEGKS	YFFRVFAENE	YGIGDPGETR	DAVKASETPG	PVVDLKVLTV	TKSSCNIGWK	480
KPRSDGGSRI	TGYVVDPLTE	ENKQWRVMKS	LSLQYSTKDL	NEGKQYTFRV	SAENENEGGT	540
PSEITVVAKD	DVVAPDLDLK	DLPDLCYLAK	ENSNFRLKIP	IKGKPAPSVT	WKKGEDPLAT	600
DTRVSVESSA	VNTTLVVYDC	QKSDAGKYTI	TLKNVAGTKE	GTLSIKVVVG	PGIPTGPIKF	660
DEVTAABITL	KWGPPKDDGG	SEITNYILEK	RDSVNNKWVT	CASAVQKTF	RVTRLHEGME	720
YTFRVSAENK	YGVGEGGLKE	PIVAKHPFDV	PDAPPPPNIV	DVRHDSVSLT	WTDPRKTGGS	780
PITGYHIEFK	ERNSSLWKRA	NKTPIRMKDF	KVTGLTEGLE	YEFRVMAINL	AGVGKPSLPS	840
EPVVALDPI	PPGKPEVIN	TRNSVTLIWT	EPKYDGGHKL	TGYIVEKRDL	PSKTWMKANH	900
INVPDCAFTV	TDLVEGGKYE	FRIRAKNTAG	AISAPSESTG	TIICKDEYEA	PTIVLDPTIK	960
DGLTIKAGDT	IVLNAISILG	KPLPKSSWSK	AGKDIRPSDI	TQITSTPTSS	MLTVKYASRK	1020
DAGEYITAT	NPPGTKEEHV	RVTVLDVPGP	PGPIEISNV	AEKATLTWTP	PLEDGGSPIK	1080
SVVLEKRETS	RLLTWTVVAED	IQSCRHVVTK	LIQGNEYLFR	VSAVNHYGKG	EPVQSEPVKM	1140
VDRFGPPGPP	GKPEVSNVTK	NTATVSWKRP	TDDGGSEITG	YYVERREKKG	LRWVRATKTP	1200
VSDLRCKVTG	LQEGNTYEF	VSAENRAGIG	PPSDASNYVL	MKDVAYPGPG	PSNARVTDIT	1260
KKSASGLAWK	PHYDGGLEIT	GYVVEHQVKG	DETWVKDTTG	PALRITTEFV	PDLTHTKEKYN	1320
FRISAINDAG	VEGPAVIPDV	EIVEREMAPD	FELDAELRRT	LVVRAGLSIR	IFVPIKGRPA	1380
PEVTWTKDDI	NLKTRANIEN	TESFTLLIIP	ECNRYDTGKF	VMTIENPAGK	KSGFVNVRVL	1440
DTPGPVLNLR	PTDITKDSVT	LHWDLPLIDG	GSRIITNYIVE	KREATRKSYS	TVTTKCHKCT	1500

SEQ ID NO: 74 moltype = AA length = 338
 FEATURE Location/Qualifiers
 source 1..338
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 74

DKETPSGFTL	DDVIQTGVND	PGHPFIMTVG	CVAGDEESYT	VFKDLFDPPI	QDRHGGFKPT	60
DKHKTDLNHE	NLKGDDLDLP	HYVLSRVRT	GRSIKGYTLP	PHCSRGERRA	VEKLSVEALN	120
SLTGEFGKGY	YPLKSMTEQE	QQQLIDHFL	FDKPVSPLLL	ASGMARDWPD	ARGIWHNDNK	180
SFLVWVNEED	HLRVISMKEG	GNMKEVFRFF	CVGLQKIEEI	FKKAGHPFMW	NEHLGYVLTC	240
PSNLGTGLRG	GVRVHKLALH	KHPKFEEILT	RLRLQKRGTG	GVDTAAVGSV	FDISNADRLG	300
SSEVEQVQLV	VDGVKLMVEM	EKKLEKGQSI	DDMIPAQK			338

SEQ ID NO: 75 moltype = AA length = 808
 FEATURE Location/Qualifiers
 source 1..808
 mol_type = protein
 organism = *Oryctolagus cuniculus*

SEQUENCE: 75

GGGGGGEYME	QEEWDWRDL	LDPAAWEKQQR	KTFTAWCNSH	LRKAGTQIEN	IEEDFRNGLK	60
LMLLLEVISG	ERLPRPDGK	MRFHKIANVN	KALDFIASKG	VKLVSIGAE	IVDGNLKMIL	120
GMIWTIILRF	AIQDISVEET	SAKEGLLLWC	QRKTAPYRNV	NVQNFTSWK	DGLALCALIH	180
RHRPDLIDYA	KLRKDDPIGN	LNTAFEVAEK	YLDIPKMLDA	EDIVNTPKPD	EKAIMTYVSC	240

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FYHAFAGAEQ	AETAANRICK	VLAVNQENЕК	LMEEYEKLAS	ELLEWIRRTV	PWLENRVGEP	300
SMSAMQKLE	DFRDYRRLHK	PPRVQEKQCL	EINFNTLQTK	LRLSHRPAPM	PSEGKLVSDI	360
ANAWRGLEQV	EKGYEDWLLS	EIRRLQRLQH	LAЕКFQQKAS	LHETWTRGKE	DMLTQRDYES	420
ASLQEVALL	RRHEAFESDL	AAHQDRVEHI	AALAQELNEL	DYHEAASVNS	RCQAICDQWD	480
NLGTLTQKRR	DALERMEKLL	ETIDQLQLEF	ARRAAPFNW	LDGAIEDLQD	VVLVHVSVEET	540
QSLTAAHDQF	KATLPEADRE	RGAILGIQGE	IQKICQTYGL	RPSSNTPYIN	LSPQDINTKW	600
DTVRKLVPSP	QDTLQBELAR	QQMNERLRRQ	FAAQANAVGP	WIQAKVEEVG	RLAAGLAGSL	660
EEQMAGLRQQ	EQNIINYKSN	IDRLEGDHQL	LQESLVFDNK	HTVYSMEHIR	VGWEQLLTSI	720
ARTINEVENQ	VLTRDAKGLS	QEQLNEFRAS	FNHFDRKRNG	MMEPDDFRAC	LISMVYDLGEV	780
EFARIMTMVD	PNAAGVVFQ	AFIDFMTR				808

SEQ ID NO: 76 moltype = AA length = 761
 FEATURE Location/Qualifiers
 source 1..761
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 76

IYYLSLEFYM	GRTLQNTMVN	LALENACDEA	TYQLGLDMEE	LEEIEEDAGL	GNGLGRLAA	60
CPLDSMATLG	LAAYGYGIRY	EPGIFNQKIC	GGWQMEEADD	WLRVGNPWEK	ARPEFTLPVH	120
FYGRVEHTSQ	GAKWVDTPVP	LAMPYDTPVP	GYRNNVNTM	RLWSAKAPND	FNLKDFNVGG	180
YIQAVALDRNL	AENISRVLYP	NDNFFEGKEL	RLKQYFVVA	ATLQDIIRRF	KSSKFGCRDP	240
VRTNFDAPFD	KVAIQLNDTH	PSLAIPELMR	VLVDLERLDW	DKAWEVTVKT	CAYTNHTVLP	300
EALERWPVHL	LETLLPRHLQ	IIYEINQRFL	NRVAAAPPGD	VDRLRMSLV	EEGAVKRINM	360
AHLCTAGSHA	VNGVARIHSE	ILKKTIFKDF	YELEPHKFNQ	KTNGITPRRW	LVLCPNGLAE	420
IIAERIGEEY	ISDLQQLRKL	LSYVDDEAFI	RDVAKVKQEN	KLKFAAYLER	EYKVHINPNS	480
LPDVQVKRIH	EYKRQLLNCL	HVITLYNRIK	KEPNKFVVP	TVMIGGKAAP	GYHMAKMIK	540
LITAGDVVN	HDPVVGDRLR	VIFLENYRVS	LAЕКVIPAAD	LSEQISTAGT	EASGTGNMKF	600
MLNGALTIGT	MDGANVEMAE	EAGEENFFIF	GMRVEDVDRL	DQGYNAQEY	YDRIPELRQI	660
IEQLSSGFPS	PKQPDLFKDI	VNMLMHDRF	KVFADYEEYV	KCQERVSALY	KNPREWTRMV	720
IRNIATSGKF	SSDRITIAQYA	RETWGVEPSR	QRLPAPDEKI	P		761

SEQ ID NO: 77 moltype = AA length = 351
 FEATURE Location/Qualifiers
 source 1..351
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 77

SHPALTPEQK	KELSDIAHRI	VAPGKGILAA	DESTGSIAGR	LQSIGTENTE	ENRRFYRQLL	60
LTADDRVNPC	IGGVILFHET	LYQKADDGRP	FPQVIKSKGG	VVGKVDKGV	VPLAGTNGET	120
TTQGLDGLSE	RCAQYKKGDA	DFAKWRCVLK	IGEHTPSALA	IMENANVLAR	YASICQNGI	180
VPIVEPEILP	DGDHDLKRCQ	YVTEKVLAAV	YKALSDHHIY	LEGTLLKPNM	VTPGHACTQK	240
YSHEETAMAT	VTALRRTVPP	AVTGVTFLSG	GQSEEEASIN	LNAINKCPLL	KPWALTFSYG	300
RALQASALKA	WGGKKENLKA	AQEYVKRAL	ANSLACQGY	TPSGQAGAAA	S	351

SEQ ID NO: 78 moltype = AA length = 584
 FEATURE Location/Qualifiers
 source 1..584
 mol_type = protein
 organism = Oryctolagus cuniculus

SEQUENCE: 78

EAHKSEIAHR	FNDVGEEHFI	GLVLITFSQY	LQKCPYEEHA	KLVKEVTDLA	KACVADESAA	60
NCDKSLHDIF	GDKICALPSL	RDITYGDVAD	CEKKEPERNE	CFLHHKDDKP	DLPPFARPEA	120
DVLCKAFHDD	EKAFFGHYLY	EVARRHPYFY	APELLYYAK	YKAILTECCE	AADKGACLTP	180
KLDALEGKSL	ISAAQERLRC	ASIQKPGDRA	YKAWALVRLS	QRFPKADFTD	ISKIVTDLTK	240
VHKECCHGDL	LECADDRADL	AKYMCHEQET	ISSHLKECCD	KPILEKAHCI	YGLHNDETPA	300
GLPAVAEEFV	EDKDVCKNYE	EAKDLFLGKF	LYEYSRRHPD	YSVVLLLRG	KAYEATLKKC	360
CATDDPHACY	AKVLDEFQPL	VDEPKNLVKQ	NCELYEQLGD	YNFQNALVR	YTKKVPQVST	420
PTLVEISRS	L	GKVGSKCKKH	PEAERLPCVE	DYLSVVLNRL	CVLHEKTPVS	480
LVDRRPCFSA	LGPDETYVPK	EFNAETFTFH	ADICTLPETE	RKIKKQATALV	ELVKHKPHAT	540
NDQLKTVVGE	FTALLDKCCS	AEDKEACFAV	EGPKLVESSE	ATLG		584

SEQ ID NO: 79 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 79

MPFGNTHNKH	KLNFKAEEFY	PDLSKHNNHM	AKALTLEIYK	KLRDKETPSG	FTLDDVIQTG	60
VDNPGHPFIM	TVGCVAGDEE	SYTVFKDLFD	PIIQDRHGGF	KPTDKHKTDL	NHENLKGDD	120
LDPNVVLSR	VRTGRSIRGY	ALPPHCSRGE	RRAVEKLSVE	ALNSLTGEFK	GKYYPKLSMT	180
EQEQQLLIDD	HFLDPKVPSP	LLLASGMARD	WPDARGIWHN	DNKSFLVWVN	EEDHLRVISM	240
EKGGMKEVF	RRFCVGLQKI	EEIFKKAGHP	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLA	300
HLSKHPKFEE	ILTRLRLQKR	GTGGVDTAIV	GSVFDVSNAD	RLGSSEVEQV	QLVVDGVKLM	360
VEMEKKLEKG	QSIDDMIPAQ	K				381

SEQ ID NO: 80 moltype = AA length = 901

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FEATURE
source Location/Qualifiers
1..901
mol_type = protein
organism = Bos taurus

SEQUENCE: 80

MMMLVQPEGL	GTGEGPFFAGG	RGGGEYMEQE	EDWDRDLLLD	PAWEKQQRKT	FTAWCNSHLR	60
KAGTQIENIE	EDFRNGLKLM	LLLEVISGER	LPRPDKGKMR	PHKIANVNKA	LDPIASKGVK	120
LVSIGAEIEV	DGNLKMFTGM	IWTIILRFAL	QDISVEETSA	KEGLLLWCQR	KTAPYRNVNV	180
QNFHTSWKDG	LALCALIHRH	RPDLIDYAKL	RKDDPIGNLN	TAF EVAEKYL	DIPKMLDAED	240
IVNTPKPDEK	AIMTYVSCFY	HAFAGAEQAE	TAANRICKVL	AVNQENELKM	EYEKLADEL	300
LEWIRRTVPW	LENRVGEPSP	SAMQRKLEDF	RDYRRLHKPP	RVQEKCCQLEI	NFNTLTQTKLR	360
LSHRPAPFMP	EGKLVSDIAN	AWRGLEQAEK	GYEDWLLSEI	RRLQRLQHLA	EKFQKASLH	420
EAWTRGKEDM	LSQRDYETAS	LQEVRLALLR	HEAFESDLAA	HQDRVEHIAA	LAQELNELDY	480
HEAASVNSRC	QAICDQWNL	GTLTQKRRDA	LERMEKLEET	IDQLQLEFAR	RAAPFNNWLD	540
GAVEDLQDVW	LVHSVEETQS	LVTAHQDFKA	TLPEADREGR	AILGIQGEIQ	KICQTYGLRP	600
SSTNPYITIT	PDINTKWDT	VRKLVPSRDQ	MLQEELTRQQ	VNERLRRQFA	AQANAIGPWI	660
QGVVEEVGRL	AAGMAGSLEE	QMAGLRQQEQ	NIINYKSNID	RLEGDHQLLQ	ESLVFDNKHT	720
VYSMEHIRVG	WEQLLTSTAR	TINEVENQVL	TRDAKGLSQE	QLNEFRASFN	HFDRKRNGMM	780
EPDDFRACLI	SMGYDLGEVE	FARIMTMVDP	NAAGVVTQQA	FIDFMTRETA	ETDTAEQVVA	840
SPKILAGDKN	YITAEELRRE	LPAEQAEYCI	RRMAPYKGAG	APAGALDYVA	FSSALYGESD	900
L						901

SEQ ID NO: 81 moltype = AA length = 842

FEATURE
source Location/Qualifiers
1..842
mol_type = protein
organism = Bos taurus

SEQUENCE: 81

MSRPLTDQEK	RKQISVRGLA	GVENVTELKK	NFNRLHLFTL	VKDRNVATPR	DYYFALAYTV	60
RDHLVGRWIR	TQQHYEYKDP	KRIYYLSLEF	YIGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELLEEIEEDA	GLGNGGLGRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	ISGGWQMEEA	180
DDWLRYGPNW	EKARPEFTLP	VHFGYRVEHT	SQGAkWVDTO	VVLAMPYDTP	VPGYRNNVVN	240
TNRLWSAKAP	NDPNLKDENV	GGYIQAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQEFV	300
VAATLQDIIR	RFKSSKFGCL	DPVRTNFDAP	PDKVAIQLND	THPSLAIPEL	MRILVDQERL	360
EWKAWEVTV	KTCAYTNHTV	LPEALERWPV	HLIETLLPRH	LQIIYEINQR	FLNRVAAAFP	420
GDVDRLLRRS	LVEEGAVKRI	NMAHLCIAGS	HAVNGVARIH	SEILKKTIFK	DFYELEPHKF	480
QNKTNGITPR	RWLVMCNPL	AEIIAERIGE	EYIADLDQLR	KLLSYVDDDES	FIRDVAKVKQ	540
ENKLFKSAYL	EKEYKVHINP	NSLFDIQVKR	IHEYKRQLLN	CLHVITLYNR	IKKEPNKFFV	600
PRTVMIGGKA	APGYHMAKMI	IKLITAIGDV	VNHDPVVGDR	LRVIFLENYR	VSLAEKVIPA	660
ADLSEQISTA	GTEASGTGNN	KFMLNGALTI	GTMDGANVEM	ABEAGEENFF	IFGMRVEDVE	720
RLDQKGYNQA	EYYDRIPELR	HVIDQLSSGF	FSPKQPDLPK	DIVNMLMHHD	RFKVFADYEE	780
YIKCQERVSA	LYKNPREWTR	MVIRNIATSG	KFSSDRTIAQ	YAREIWGVPE	TRQRMPPADE	840
KI						842

SEQ ID NO: 82 moltype = AA length = 364

FEATURE
source Location/Qualifiers
1..364
mol_type = protein
organism = Bos taurus

SEQUENCE: 82

MPHQYPALTP	EQKKELCDIA	HRIVAPGKGI	LAADSTGSI	AKRLQSIGTE	NTEENRRFYR	60
QLLLTADDRV	NPCIGGVILF	HETLYQKADD	GRPPQVIKA	KGGVVGKVD	KGVPVLAGTN	120
GETTTQGLDG	LSERCAQYKK	DGADFAKWRC	VLKIGEHTPS	SLAIMENANV	LARYASICQQ	180
NGIVPIVEPE	ILPDGDHDLK	RCQYVTEKVL	AAVYKALSDH	HIYLEGTLLK	PNMVTTPGHAC	240
TQKYSHEEIA	MATVTALRRT	VPPAVPGITF	LSGGQSEEEA	SINLNAINKC	PLLKPWALTF	300
SYGRALQASA	LKAWGGKKEN	LKAAQEYVK	RALANSLACQ	GKYTPSGKAG	AAASESLFIS	360
NHAY						364

SEQ ID NO: 83 moltype = AA length = 192

FEATURE
source Location/Qualifiers
1..192
mol_type = protein
organism = Bos taurus

SEQUENCE: 83

MAPKKDVKKP	AAAAAPAPAP	APAPAPAPAP	PKEEKIDLSA	IKIEFSKQQQ	DEFKEAFLLF	60
DRTGCEKITL	SQVGDVLRAL	GTNPTNAEVK	KVLGNPSNEE	MNAKIEFEQ	FLPMLQAISN	120
NKDQGTIEDF	VEGLRVFDKE	NGTVMGAEL	RHVLATLGEK	MKEEEVEALM	AGQEDSNGCI	180
NYEAFVKHIM	SN					192

SEQ ID NO: 84 moltype = AA length = 333

FEATURE
source Location/Qualifiers
1..333
mol_type = protein
organism = Bos taurus

SEQUENCE: 84

MVKVGVNGFG	RIGRLVTRAA	FNSGKVDIVA	INDPFDLHY	MVYMFQYDST	HGKFNQTVKA	60
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ENGKLVINGK	AITIFQERDP	ANIKWGDAGA	EYVVESTGVF	TMEKAGAH	KGGAKRVIIS	120
APSADAPMFV	MGVNHEKYN	TLKIVSNASC	TTNCLAPLAK	VIHDHFGIVE	GLMTTVHAIT	180
ATQKTVDGSP	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCRLEKP	AKYDEIKKVV	KQASEGPLKG	ILGYTEDQVV	SCDPNSDTHS	STPDAGAGIA	300
LNDHFPVKLS	WYDNEFGYSN	RVVDLMVHMA	SKE			333

SEQ ID NO: 85 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 85

MDAIKKMQM	LKLDKENALD	RAEQAEADKK	AAEDRSKQLE	DELVSLLQKKL	KATEDELDKY	60
SEALKDAQEK	LELAEKKATD	AEADVASLNR	RIQLVEEELD	RAQERLATAL	QKLEEAEEKAA	120
DESERGMKVI	ESRAQKBEEK	MEIQEIQLKE	AKHIAEDADR	KYEEVARKLV	IIESDLERAE	180
ERAELSEGKC	AELBEEKLTV	TNNLKSLEAQ	AEKYSQKEDK	YEEBIKVLSL	KLKEAETRAE	240
FAERSVTKLE	KSIDDLDEL	YAQKLKYKAI	SEELDHALND	MTSI		284

SEQ ID NO: 86 moltype = AA length = 1141
 FEATURE Location/Qualifiers
 source 1..1141
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 86

MPEAKPAACK	APAGKDAAK	PAPKEATPQE	APAAPTTEAP	PEDQSPTAEE	PTGVFLKKPD	60
SVSVETGKDT	VIVAKLNGKE	LPAKPAVKWF	KGKWLLELGS	SGARFSPKES	HDAASNVTYI	120
ELHITKVVVLG	DRGDYRIEVK	AKDFCDSCAF	NIDVEAPRQD	SAGQSLESFK	RSGEAKSDTA	180
GELDFSGLLK	KRQVVEEEKK	KKKDDDDLGI	PPEIWELLKG	AKKSEYERIA	FQYGITDLRG	240
MLKRLKKAIV	EVKKSAAFTK	KLDPAYQVDR	GNKIKLVVEI	SDPDLPLKWF	KNGQEIKPSS	300
KTVFENVGK	RILTINKCTL	ADDAAYEVVV	KDEKCFTELF	VKEPPVLIVT	PLEDQVVFVG	360
DRVEMAVEVS	EEGAQVMMLK	DGVELTREDS	FKARYRLKKD	GKRHLIYSE	VTMEDKGHYQ	420
VMTNGGQCEA	ELIVEEKQLE	VLQDIADLT	KASEQAVPKC	EVSDEKVTGK	WYKNGVEVRP	480
SKRITISHTG	RFHKLVIDDV	RPEDDEGDTF	VPDGYALSLS	AKLKFLLEIKV	EYVPKQEPPE	540
IHLDCSGQTS	ENAIVVVAGN	KLRMDVSITG	EPRPVATWMK	EDEVFTGTGEG	RVRIEQRGDI	600
SSFVIESAER	GDEGRYTIKV	TNPVGEDVAS	ILLKVVDVPD	PPEAVRVTSV	GEDWAILVWE	660
PKYDGGRPV	TGYLLERKKK	GSQRWMKLN	EVFTETTYES	TNMIIEGILYE	MRVFAVNVIG	720
VSQPSMNTKP	FMPIAPTSEP	QHLTVEDVTD	TTTTLKWRPP	DKIGAGGIDG	YLIEYCVESG	780
DEWIPANTEP	VERCGFTVKI	LPTGAKITFR	VVGVNIAGRS	QPATLAQPVT	IREIVEQPKI	840
RLPRHLRQTY	VRKVGHEHNL	VIPFQKPRP	QVWTKGGAP	IDTSHVHVRT	SDFTVFFVR	900
QAARSDSGEY	DLTVQIENMK	DTATVCIRVV	EKAGPPQNVN	VKEVWGTNAL	VEWQPPKDDG	960
NSEITGYFVQ	KADKKTMEWF	TVYERNRHTS	CTVSDLMIGN	EYYFRVYSEN	VCGLSDLPGV	1020
SKNTARIVKT	GMTLKLPEYK	EHDFTTPPKF	LTPLPDRVVV	AGYAAALNCA	VRGHPKPKVV	1080
WMKNKMEIRE	DPKFLMTNQ	GVLTLNIRRP	SPFDSGTYSC	RAVNELGEAL	AECKLEVVRP	1140
Q						1141

SEQ ID NO: 87 moltype = AA length = 1495
 FEATURE Location/Qualifiers
 source 1..1495
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 87

MSLVAVPFYQ	KRHKHFDQSY	RNIQTRYLLD	EYSAKKKRAS	AQSSLQRSVT	RKSSSQRESS	60
RAALGESTLG	ETTLGETSLG	DTTLGRTTCR	LCAKRMSSAL	EEEAQERKRR	YQSMVAAYGE	120
AKRERYLSEL	AQLEEDVHVA	RTHARDQLDR	LALQHAGRHG	LHHAVDDRLA	WERHSFEERI	180
SRAPEILVRL	RSHTVWERMS	VKLCTVQGF	PTPVVQWYKD	GSLICQAGEL	GKYRIDSQYK	240
VHTLEINRAD	FDDTATYSAV	ATNVHGQVST	NAAVVVRFR	GDEEPPFHSVG	LPIGLPLSSV	300
IPYTHFDVQF	IEKPGVTFRR	EGETLTLKCT	LLVTPDLKRV	QPRAEWYRDD	VLLKESKWT	360
MFFGEGQASL	SFSLNKDDE	GLYTLRIVSR	GGITDHSAPL	FVRDADPLVT	GAPGAPMDVH	420
CHDANRDYVI	VTWKPPNATK	ESPVIGYFID	RCEVGTDNWI	QCNDAPVKIC	KYPVTGLFEG	480
RSYIFRVRAV	NSAGISRPSR	VSEAVAALDP	VDLRRLLQAVH	LEGEKDIVVY	QEELEGEVQI	540
PGPPTNVHAS	ETSTRTYVLS	WDPEPRGKE	PLMYFIEKSM	VGSGSWQRVN	AQTAVRSPRY	600
AVFDLAEGKP	YVFRVLSANK	HGLSDPSEIT	APIQAQDTIV	VPSAPGRVLA	SRNTRTSVVV	660
QWDRPKHEED	LLGYVDCSV	AGSNVWPCN	HKPIGYNRFV	VHGLTTGEQY	IFRVKAVNAV	720
GTSSENSQESD	VIKQAAALTV	PSHPYGITLL	NCDGHSMILG	WKVPKFSGGS	PILGYVVDKR	780
EAHHKNWHEV	NSSPLKERIL	TVEGLTEGSL	YEFKIAAANM	AGIGQPSDPS	ELFKCEAWTM	840
PEPGPAYDLT	FCEVRDTSLV	LLWKPPVYSG	SSPVSGYFVD	YKEEDSGEWL	TVHETATPHR	900
YLKVCIDLHQ	KTYVFRVRAV	NASGVGRPSD	TSEPVLEVAR	PGTKEVSAGV	DEEGNVYLSF	960
DCPEVTDASH	FTWCKSYEDI	ADDDRPFKVT	AGDHSKLYFK	NLDKEDLGT	SVSVSDTDGV	1020
SSSFVLDEEE	LRLKALSNE	IKNPTIPLKS	ELAYEIFDKG	QVRFWLQAEH	LSPDSSYRFI	1080
INDREVADSE	THRIKCDKST	GIEMVMDFR	TIDNEGTYTV	QIQDGKAKSQ	SSSLVLIGDAF	1140
KAVLKEAEFQ	KRFLRKQGP	HFAEYLHWDV	TEECEVRLVC	KVANTKKETV	FKWLKDDVLY	1200
ETEKMPDLEK	GVCELLIPKL	SKKDHGEYKA	TLKDDRGQDV	STLEIAGKVY	EDMILAMSRV	1260
CGASASPLKI	LCTPEGRLQ	CFMKYFTEEM	KVNWYHKDAK	ITSSEHMRI	GNEQMAWLQI	1320
CEPTEKDKGK	YTFEILDGKK	NHQRSLDLSG	QAFDEAFAEF	QQLKAAAFAE	KNRGKVIIGL	1380
PDVVTIMEGK	TLNLTCTVFG	NPDPPEVWFK	NDKDIELSDH	FSVKVEQGGY	VSMTIKGVT	1440
EDSGKYSIHV	KNKYGGEKID	VTVSVMYKHGE	EIPDVVLPQQ	AKPKLLPAA	PSPAH	1495

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SEQ ID NO: 88 moltype = AA length = 434
FEATURE Location/Qualifiers
source 1..434
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 88

MAMQKIFARE	ILDSRGNPTV	EVDLHTAKGR	FRAAVPSGAS	TGIYEALRLR	DGDKSRYLGK	60
GVLKAVEHIN	KTLGPALLEK	KLSVVDQEKV	DKFMIELDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVPL	YRHIADLAGN	PELILPVPAF	NVINGGSHAG	NKLAMQEFMI	LPVGASSFRE	180
AMRIGAEVYH	HLKGVIKAKY	GKDATNVGDE	GGFAPNILEN	NEALELLKTA	IQAAGYPDKV	240
VIGMDVAASE	FYRNGKYDLD	FKSPDDPARH	ISGEKLGELY	KNFIKNYPVV	SIEDPFDQDD	300
WATWTSFSLG	VNIQIVGDDL	TVTNPKRIAQ	AVEKKACNCL	LLKVNQIGSV	TESIQACKLA	360
QSNQWGVMS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEEALGDK	420
AVFAGRKFFN	PKAK					434

SEQ ID NO: 89 moltype = AA length = 170
FEATURE Location/Qualifiers
source 1..170
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 89

MAPKKAKRRA	AAEGGSSSVF	SMFDQTQIQE	FKEAFTVIDQ	NRDGIIDKED	LRDTFAAMGR	60
LVNKNEELDA	MMKEASGPIN	FTVFLNMFGE	KLKGADPEDV	ITGAFKVLDP	EGKGTIKKKF	120
LEELLTTQCD	RFSQBEIKNM	WAAFPDPVGG	NVDYKNICYV	ITHGDAKDQE		170

SEQ ID NO: 90 moltype = AA length = 1938
FEATURE Location/Qualifiers
source 1..1938
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 90

MSSDQEMAVF	GEAAPYLRSK	EKERIEAQNK	PFDAKTSVFV	ADPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQVFP	NPPKFDKIED	MAMMTHLHEP	AVLYNLKERY	AAMMIYTSYG	120
LFCVTVNPYK	WLPVYNAEVV	TAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKKEEP	TSGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFQKF	IRIHFGTTGK	LASADIETVL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITNTP	YDYAYVSQGE	ITVPSIDDQE	ELMATDSAIE	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFQKQ	QREEQAEPDG	TEVADKAAYL	QGLNSADLLK	ALCYPRVKVG	NEFVTGQQT	420
EQVYNAVVAL	AKAYYDKMFL	WMVARINQQL	DTKQPRQYFI	GVLDIAGFEI	PDFNSLEQLC	480
INFTEKLLQ	FFNHMHFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATMS	FKNKLYEQHL	GKSNNFQPKP	PAKGKAAEHF	SLIHYAGTVD	YNTGWLDKN	600
KDPLNETVVG	LYQKSSVGAT	ALLFSGPASG	EAEFGPKKGG	KKKGSSTQTV	SALFRENLNK	660
LMTNLRSTHP	HFVRCIIPNE	TKTPGAMEHE	LVLHQLRCNG	VLEGIRICRK	GFPSRILYAD	720
FKQRYKVLNA	SAIPEGQFID	SKKASEKLLA	SIDVDHTQYK	FGHTKVFFKA	GLLGLEEMR	780
DEKLAQLITR	TQARCRGFLA	RVEYQKMVER	RESIFCIQYN	VRAFNMVKNH	PWMKLYFKIK	840
PLLKSAETEK	EMANMKEEFE	KTKEELAKSE	AKRKELEEM	VTLTQEKNDL	QLQVQSEADA	900
LADAEERCQD	LIKTKIQLEA	KIKEVTERAE	DEEINAELT	AKRKLEDEC	SELKDDIDL	960
ELTLAKVEKE	KHATENKVEK	LTEEMAGLDE	TIAKLTKEKK	ALQEAHQOTL	DDLQAEEDKV	1020
NTLTAKAKTKL	EQQVDDLEGS	LEQEKKLKRD	LERAKRKLEG	DLKLAQESTM	DIENDKQQLD	1080
EKLKKKEFEM	SNLQSKIIDE	QALAMQLQKK	IKELQARIEE	LEEEIEAERA	SRAKAEKQRS	1140
DLRLELEEIS	ERLEEAGGAT	SAQIEMNNKR	EAEFQKMRD	LEEATLQHEA	TAAALRKKHA	1200
DSVAELGEQI	DNLQRVQKQL	EKEKSEMME	IDDLASNMET	VSKAKGNLEK	MCRALEDQLS	1260
ELKTKEDEQQ	RLINDLTQAR	ARLQTESGEF	SRQLDEKDAL	VSQLSRGKQA	FTQQIEELKR	1320
QLEEEIKAKS	ALAHALQSAR	HDCDLLREQY	EEEQEGKAE	GRAMSKANSE	VAQWRTKYET	1380
DAIQRTTEEL	EAKKKLAQRL	QDAEEHVEAV	NAKCASLEKT	KQRLQNEVED	LMIDVERTNA	1440
ACAALDKKQR	NFDKILSEWK	QKYEETHAEL	EASQKESRSL	STELFKIKNA	YEESLDQLET	1500
LKRENKLLQ	EISDLTEQIA	EGGKRIHELE	KVKKQVEQEK	SEIQAALEEA	EASLEHEEGK	1560
ILRIQLELNQ	VKSEIDRKIA	EKDEEIDQLK	RNHIRIVESM	QSTLDAEIRS	RNDAIRLKKK	1620
MEGDLNEMEI	QLNHANRMAA	EALKNYRSTQ	AILKDTQIHL	DDALRGQEDL	KEQLAMVERR	1680
ANLLQAEIEE	LRATLEQTER	SRKIAEQELL	DASERVQLLH	TQNTSLINTK	KKLETDTQI	1740
QGEMEDIIQE	ARNAEEKAKK	AITDAAMMAE	ELKKEQDTS	HLERMKKNLE	QTVKDLQHRL	1800
DEAEQLALKG	GKKQIQKLEA	RVRELEGEVE	SEQKRNVEAV	KGLRKHHERR	KELTYQTEED	1860
RKNILRLQDL	VDKLQAKVKS	YKRQAEAEAE	QSNVNLSKFR	KLQHELEBAE	ERADIAESQV	1920
NKLRVKSREV	HTKIIISEE					1938

SEQ ID NO: 91 moltype = AA length = 7100
FEATURE Location/Qualifiers
source 1..7100
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 91

MPNFQTRTRF	YETTTTTRSD	YEQSETSRPA	LAQPVPEKPV	ERKRVRIRKV	DPSKFMTPIYI	60
AHSQKMQNLF	STNKYKENYE	KAKGKPYAIT	TDTPELRRIK	KVQDQLSEVK	YRVDGDAVKT	120
ICHVDEKAKD	IEHAKKYSQQ	VSKVLYQKNW	EDTKDKYLLP	PDAPELVQAI	KNTAMFSKKL	180
YTDEWEADKT	MFYPYNDSP	LRRVQAQAQA	LSDIAYKKKL	AEQQTQFTSL	PDPPIEFKAK	240

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KVTNQVSKQK	YKEDYENKVK	GKWSETPCFE	IATARMNSNN	ISTKKYQEDF	EHMKDQIYFM	300
QTETPEYKVN	KQAGVAASKV	KYKQDYEKKN	GKADYNVLP	SENPLLRQLK	AAGDALSDKL	360
YKENYEKTKA	KSINYCETPK	FKLDTVLHNF	SSDTKYKDSY	LKDILGHYVG	SYEDPYHTHC	420
MRVSAQNSDK	NYKAEYEEDR	GKGFPPQTIT	QEYEAIKKLD	QCKDHTYKVH	PKTKTPTQVT	480
DSPVLMQAQV	NSKQLSDLNY	KAKHENEKFK	CHIPDTPAF	IQHKLNAYNL	SDNIYKHDWE	540
KTKAKKFDIK	VDAIPLLAOK	ANTKIASDVM	YKKDYEKSKG	KMIGALSIND	DPKMLHSLKT	600
AKNQSDRLYK	ENYKTKAKS	MNYCETPKYQ	LDTLKKNFSE	AKYKDSYVQN	VLGHYIGSFE	660
DPYQVHCLKI	SAQNSDKNYK	AEYEEDKGKC	YFPQTITQEY	EAIKKLDQCK	DHTYKVHPDK	720
TKFTAVTDTF	VLLQAQLNTK	QLSDLNYKAK	HEGEKFKCHI	PADAPQPIQH	RVNAYNLSDN	780
IYKHDWEKTK	AKKFDIKVDA	IPLLAOKANT	KIASDVMYKK	DYEKSKGKMI	GALSINDDPK	840
MLHSLKTAKN	QSDHEYRKDY	EKSCTIYTAP	LDMLPLTHAK	KSQAIASDVD	YKHLHNYSY	900
PPDSVNVDLA	KKAYALQSDV	EYKADYNSWM	KGCGWMPFGS	LEMEKAKRAS	DILNEKKYRQ	960
HPDTLKFTSI	EDAPIIVQSK	INQAQRSDVA	YKAKGEEVIH	KYSLPADLPQ	FIQAKVNAYN	1020
ISENLYKADL	KDLSKKGYDL	RIDAIPKAA	KAARQAASDV	QYKKDYEKAK	GKMGVQFQSLQ	1080
DDPKLVHYMN	VAKIQSDREY	KKAYEKTCTR	HNTPHDMVNI	VAAKKAQDVA	SNVNYKHS LH	1140
HTTYLPDAMD	LELSKNNMHI	QSDNVYKEDY	NNWMKGIGWI	PIGSLEVEKV	KKAGDALNEK	1200
KYRQHPDTLK	FTSIVDSPVM	VQAKQNTQQV	SDILYKAKGE	DVKHKYTMSP	DLPQFLQAKC	1260
NAYNLSDVY	KRDWHDLIAM	GTNVLGDAIP	ITAAKASRNI	ASDYKYKEAY	EKAKGKQVGF	1320
RSLODDPKLV	HRMNVAKLQS	DREYKKNYEN	TKTSYHTPGD	MVSITAAKMA	QDVATNVNYYK	1380
QPIHHYTYLP	DALSLEHIRN	VNQIQSDNVY	KDEYNHFFKG	MGWIPIGSLE	VEKVKKAGDA	1440
LNEKKYRQHP	DTIKFTSVDP	SMGMVLAQHN	TKQLSDLNYK	VEGEKVHKHY	TMDPDVPPFI	1500
QAKVNAYNMS	DSHYKADWKK	TLAKGYDLRP	DAIPIVAAS	SRNIASDFKY	KEAYEKTGKG	1560
QIGFRSLQDD	PKLVHFMNVA	KMQSDREYK	GYEASKTKYH	TPLDMLSVTA	AKKSQEVATN	1620
TNYKQPFHHY	TLLPDALNVE	HSRNAMQIQS	DNLYKSDFTN	WMKGIGWLPL	ESLEVEKAKK	1680
AGEILSEKKY	RQHPEKLFKT	YAMDTMEQAL	NKSNKLIMDK	RLYTEKWNKD	KTTIHVMPDT	1740
PDILLSRVNQ	ITMSDKLYKA	GWEEKKKGY	DLRPAISIK	AARASRDIA	DYKYKQAYEQ	1800
AKGKQIGFRS	LEDDEPKLVH	MQVAKMQSDR	EYKAYEKS	TSFQTPVDML	SVVAAKKSQE	1860
VATNANYRNV	IHTYNMLPDA	MSLELAKNMM	QIQSDNQYKA	DYADFMKGIG	WLPLGSLEAE	1920
KNKKAMEIIS	EKKYRQHPDT	LKYSTLMDSM	NMVLAKNNAK	IMNEHLYKQA	WEADKTKVHI	1980
MPDIPQIILA	KANAINISDK	LYKLSLEEAK	KKGYDLRTDA	IPIKAAKASR	DIASDYKYKH	2040
SYEKEGKQMV	GFRSLEDDEPK	LHVSMQVAKM	QSDREYKKNY	EKTKTSYHTP	ADMLSVTAAK	2100
DAQANITNTN	YKHLIHKYIL	LPDAMNIQLS	KNMNRIQSDN	EYKQDYNEWY	KGLGWSPAGS	2160
LEVEKAKKAT	EYASDQKYRQ	HPSNFQFTKL	NDSDMDVLAK	QNAHTMKNYL	YTVDNKDKT	2220
KIHVMPDTPD	ILQAKQKQTM	YSQKLYKLGW	EEALKKGYDL	PVDAISVQAA	KASRDIVSDF	2280
KYKQGYRKQI	GHHIGFRSLQ	DDPKLVLSMN	VAKMQSREY	KKDFEKWKTK	FSSPVDMLGV	2340
VLAKKQCALV	SDVDYKKNYLH	GWTCCLPDQND	VIHAKKAYDL	QSENLYKSDL	EWLKGIGWSP	2400
LGSLEAEKKN	TSIIPDAMDIV	YRQPPDRNKF	TSIPDAMDIV	LAKTNAKNRS	DILYREAWDK	2460
DKTQIHIMPD	TPDIILAKAN	LINTSDKFYR	MGYEELRKKG	YDLPVDAIPI	KAAKASREIA	2520
SEYKYKEGFR	MQLGHHIGAR	NIKDDPKMMW	SMHVAKIQSD	REYKKDFEKK	KTKFSSPVDM	2580
LGVLAKKQ	TLVSDIDYKN	YLHQWTCCLPD	QSDVIHARRA	YDLQSDNLYK	SDLQWLGRIG	2640
WLPSGSLDE	KNKRASQILS	DHVRQHPDQ	FKFSSLMDSI	PMVLAKNNAI	TMNHRLYTEA	2700
WDKDKTTHVI	MPDTPEVLLA	KQKQINYSK	LYKLGLDEAK	RKGVDMRIDA	IPIRAAKASR	2760
DIASEFKYKE	GYYRQLGHHI	GARAIHDDPK	MMWSMHVAKI	QSDREYKDKF	EKWKTKFSSP	2820
VDMLGVVLAK	KCQTLVSDID	YKNYLHQWTC	LPDQSDVIHA	RQAYDLQSDN	VYKSDLQWLR	2880
GIGWVPIGSL	DVEKSKRASE	ILSDKLYRQP	PKFKFTSVT	DSLEQVLAKN	NAINMKNRLY	2940
TEAWDKDKTQ	HVIMPDTEI	TLARTNKVNY	SENLYKLAHE	EAKKKGYDLR	SDAIPIVAOK	3000
ASRDIISDYK	YKGYRKQLG	HHIGARDIKD	DPKMMWSMHV	AKIQSDREYK	KDFEKWKTKF	3060
SSPVDMLGVV	LAKKQCTLVS	DIDYKNYLHQ	WTCLPDQSDV	IHARQAYDLQ	SDNVYKSDLQ	3120
WLRGIGWVPI	GKSDVVKAKR	AAEILSDNLY	RQPPDTLKFT	SVDPDSLEQVL	AKNNAINMKN	3180
RLYTEAWDKD	KTQIHIMPD	PEIMLARQNK	LNYSSETLYK	ANEEAKKGY	DLRSDAIPIV	3240
AAKASRDIIS	DYKYKGYRK	QLGHHIGARD	IKDDPKMMWS	MHVAKIQSDR	EYKDKFEKWK	3300
TKFSSPVDML	GVVLAKKQ	LVSDIDYKNY	LHQWTCCLPDQ	SDVIHARRAY	DLQSDNLYKS	3360
DLQWLGRIGW	VPIGSVDVVK	CKRAAEILSD	NLYRQRPDTL	KFTSVDPDSLE	QVLAKNNAIN	3420
MNKLRYTEAW	DKDKTQIHMM	PDTPEITLAR	QNKINYSNL	YRQAMEEAKK	EGYDLRSDAI	3480
PIVAAKASRE	IASDYKYKEA	YRKQLGHHIG	ARAIHDDPKM	MWSVHVAKMQ	SDREYKDKFE	3540
KYKTRFSSPV	DMLGIVLAKK	CQTLVSDVDY	KHPLHEWTCL	PDQNDIIHAR	KAYDLQSDNL	3600
YKSDLEWLKG	IGWVPIGSVE	VLKAKRAGEI	LSDNIRYRQP	DTLKFTSVTD	SPEQVLAKNN	3660
AINMKNRLYT	BAWDNDKKT	HVMPDTPEIM	LAKLNRINYS	DKLYKLALKE	SRKEGYDLRL	3720
DAIPIQAAKA	SREIASDYKY	KEGYRKQLGH	HIGARNIKDD	PKMMWSIHAG	KLQSDLEYKK	3780
DPEKWKTKFS	SPVDMMLGVV	AKKQCILVSD	IDYKNLLHEW	TCLPDQNDII	QARKAYDLQS	3840
DAIYKADLEW	LRGIGWVPI	SVGVEHAKRA	GEILSERKYR	QPAVQLKFTS	ITDTPETVLA	3900
KNNALNVSKH	LYTEAWDADK	TSIHVMPDTP	EILLAKSNSA	NISHKLYTKG	WDESKMKDYD	3960
LRADAIISKS	AKASRDIASD	YKYKEAYEKH	KGHHIGARSV	EDDPRIMCAM	NAGRIQSERE	4020
YKKEPQKWK	KFSSPVDMLG	ILLAKKQCTL	VSDIDYRNYL	HHWTCLPDQN	DIIQARKAYD	4080
LQSDNVYKSD	LEWLGRIGWM	PEGSVEMNRV	KNAQDLINER	RYRTRPEALT	FTSIVDTPEV	4140
VLAKANSLLQM	SEKLYQEAWN	KDKTNISIPS	DTPLILQAQI	NAMQISNKLY	QKDWDKAKQK	4200
GYDLRADAIE	IKHAKASREI	ASEYKYKEGY	RKQLGHHVGF	RTLQDDPKLV	WSIHAAKIQS	4260
DREYRKAYEK	SGKIHNTPLD	MMSIVQAKKC	QVLVSDIDYR	NYLHQWICLP	DQNDVIQAKK	4320
AYELQSDAIY	KADLEWLRLGI	GWMEGSPEV	LRVKNAQHIF	RDSVYRTPV	KLKYTSIVDT	4380
PEVVLAKSNA	ENISIPKYRE	VWDKDKTSIH	IMPDTPEINL	ARTNALNVSN	KLYREGWDEM	4440
KMSCDVRLDA	PIQAQAKASR	EIASDYKYKL	DHEKQKGHYV	GTRTARDNKK	IRWALIAKGI	4500
QNEREYRLHW	AKWKSFKQSP	PDMLSIEHSK	NSQTLVSDID	YRHYLHQWTC	MPDQNDVIQA	4560
RKAYDLQSDA	IYKADLEWLR	GIGWMPADSV	SVNHAKHMAD	IFNEKKYRTK	IETLSFTPVD	4620
DRVDYVTAKH	SGEILNDIKY	RKDWNDTKSK	YTLTETPQLH	TAQEAARILD	QYLYKESWEK	4680
KATGYILP	DAVPFVHAHH	SGDVQSELKY	KABHVQKQGH	YVGVPTMRDD	PKLVWFEHAG	4740
QIQNDRLYKE	NYHKTAKAIIH	IPDPMVSVLA	AKEGQALASD	IDYRNYLHQW	ICHPDQNDVI	4800

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QARKAYDLQS	DNIYKADLEW	LRGIGWIPLD	SVDHVRVTRN	QEMMNQIKYK	KDALANYPNF	4860
TSVVDPPPEIV	LAKINSNVQS	DVKYKETFNK	LIKGYIFSP	DTPYITHSKD	MEKLYSTILY	4920
KRAWDGTKAY	GYTLDERYIP	IVGAKHADFV	NSELKYKETY	EKLKGHYLAG	KEISEFPNVV	4980
HCLDFQKMRS	LLNRYRRHPD	TKANVHIPQD	MMNHVLAKRC	QYILSDLEYR	HYFHWTSLSL	5040
EEPNVVRVRH	AQEILSDNVY	KDDLNLWLKGI	GCYVWDTPQI	LHAKKSYDLQ	SQILYTAAGK	5100
ENLNKNYLVLT	DTPLYVTALQ	SGINASEVKY	KENYHQTKDK	YTTVLETVDY	DRINKLKDLE	5160
SSNLKYKAWD	KVKATSYILP	PNTVSLTHAK	NQKYMASHIK	YREEYEKFKK	LYTLPRSVED	5220
DPNTARCLRV	GKFNIDRLYR	SVYEKNKMKI	HIVPDMVEMV	TAKDSQKKVS	EIDYRLHLHE	5280
WICHDPDLQVN	SHVRKVTDQI	SDIVYKDDL	WLKGIGCYVW	DTPEILHAKH	AYDLRNDIKY	5340
KAHVLKTRNN	YKLVTDTPVY	VQAVKSGKQL	SDAVYHYDYV	HSIRGRVAPT	TKTVDLDRAL	5400
HAYKLQSENL	YRKAGLHALP	TGYRLPVDTP	HPKHTKDTRY	MSSYFKYKEV	YEHMKAYGYT	5460
LGPNDVPPFN	VRRVNNITSE	RLYRQLYHKL	KDKIHTTPTD	PEIRQVKKQT	EAVSELIYKS	5520
DPFKMQGHMI	SLPYTPQVLH	CRYVGDITS	IKYKEDLQVL	RGMGCFLYDT	PDMVRSRHLR	5580
KLWSHYLYTD	KARKMRDKYK	VVLDTPEYRK	VQELKTHLSE	LVIYRAAGRKQ	KSIFTSVPDT	5640
PDLTRAKRGQ	KLQSQYLYVE	LATKERPHHH	AGNQTALKH	ARHVKDMVSE	NKYKIYQEKM	5700
KDKYTPVPDT	PILIRAKRAY	WNASDLRYKE	TPQKTGKYH	TVKDALDIVY	HRTVTDHISK	5760
IKYKENYMSQ	LGIWRSIPDR	PEHFHHRVAVT	DAVSDVKYKQ	DLTWLKGIGC	YAYDTPDFTL	5820
AEKNKTLYSK	YKYKEVFERT	KSNFKYVADC	PINRHFKEPAT	QLMNEKKYRA	DYBQRKDKYH	5880
LVVDEPRHLL	AKIAGDQISQ	IKYRKNYEET	KDKFTSIVDT	PEHLRTTKVN	KQISDILYKL	5940
EYNKAKPRGY	TTIHDTMPLL	HVRKVKDEVS	DLKYKEVYQR	TKSNCTIEPD	AVHIKAAKDA	6000
YKVNNTLDYK	KKYEATKAYW	KWTPDRPDFI	QAAKASTLQOS	DFEYKLDREY	LKGCKLSVTD	6060
DKDMVLALKN	SIIESDLKYK	EKHVKERGSC	HAVPDTPQIL	LAKTVSSLVS	ENKYKSYVKK	6120
HLAQGSYTTL	PETRDTHIVK	EVTKNVSDTN	YKKKFVKEKG	KSNYSIMLEP	PDVKHAMDVA	6180
KKQSNVAYKK	DAKENLHYTT	VADRPDIKKA	TQAAKQASEV	EYRAKHKKEG	SHGLSMLGRP	6240
DIEMAKKAAK	LSSQVQYREN	FNKEKGKTPK	YNPKDSQLYK	VMKDANTLAS	EVYKQADLKK	6300
LHKPVTDMKE	SLIMNHVLTN	SHLASSYQYK	KNYEKSNGHY	HTIPDNLEQL	HLKEATELQS	6360
IVKYKEKYK	ERKPMPLDPE	TPYITAKES	QQMQSGKEYR	KDYBESIKGR	NLTGLEVTPTA	6420
LLHVKYATKI	ASEKEYRKDL	EESIRGKGLS	EMEDTPDMLR	AKNATQILNE	KEYKRDELE	6480
VKGRGLNAMA	NETPDFLRAR	NATDIASQIK	YKQSAEMEKA	NFTSVVDTP	IIHAQQVKNL	6540
SSQKKYKEDA	EKCMSYETV	LDPTEMQVRV	ENQKNFSLQ	YQYDLNKSNG	KITTVQDTP	6600
ILRVKENQKN	FSSVLYKEDV	SPGTAIGKTP	EMMRVKQTD	HISSVYKKEV	IGQGTPIPD	6660
PEVKRVKQQT	KHISVVMYKE	NLGTGIPTPV	TPEIERVKRN	QENFSSNLSL	KHEVTSKSI	6720
LTKYNSLKL	PHTPLPPVLY	KENLGKGTPT	PITPEMERVK	RNQNENFSSV	YKENMGKGT	6780
LPVTPEMERI	KHNQENISSV	LYKENMRKAT	PTPVTPEMER	AKRNQENISS	VLYSDSFRKQ	6840
IQGKAAAYVLD	TPEMRRVRET	QRHISTVKYH	EDFEKHKGCF	TPVVTDPITE	RVKKNQDQFS	6900
DICYRGIQRK	VVEMEQRKND	QDQETITGLR	WVRTNPGSVF	DYDPAEDNIQ	SRLHMINAQ	6960
AQRRSREQSR	SASGLSISGG	EEKSEHSEAA	HLSTYSDDGV	FFSAASTGYK	HARTTELPPQ	7020
RSSSVATQQT	TVSSIPSHPS	TAGKIFRAMY	DYMAADADEV	SFKDGDIAVN	VQAIDEGWMY	7080
GTVQRTGRTG	MLPANYVEAI					7100

SEQ ID NO: 92 moltype = AA length = 1935
 FEATURE Location/Qualifiers
 source 1..1935
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 92

MSSDQEMAIF	GEAAPYLRSK	EKERIEAQNK	PFDAKTSVFV	VDPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQVFP	NPPKYDKIED	MAMMTHLHEP	AVLYNLKERY	AAMWIYTYSG	120
LFCVTVPYK	WLPVYNAEV	TAYRGKKRQE	APPHIFSIIS	NAYQFMTLGE	LDVSIIGESGA	180
GKTVNTKRV	QYFATIAVGT	EKKKEEPTSG	KMQGTLEDQI	ISANPLLEAF	GNAKTVRNDN	240
SSRFKGFIRI	HFGTTGLKAS	ADIETYLLEK	SRVTFQLKAE	RSYHIFYQIM	SNKKPELIEM	300
LLITTPYDVI	AYVSQGETTV	PSIDDDQELI	ATDSAIDILG	FTSDERSIY	KLTVGAVMHY	360
NLKFKQKQRE	EQAEPDGTEV	ADKAAYLQSL	NSADLLKALC	YPRVKVGNF	VTGQQTVOQV	420
YNAVGAALAK	VYEMFVLMV	TRINQQLDTK	QPRQYFIGVL	DIAGFEIFDF	NSLEQLCINF	480
TNEKLQQFFN	HHMFVLEQEE	YKKEGIEWEF	IDFGMDLAAC	IELIEKPMGI	FSILEEECMF	540
PKATDTSFKN	KLYEQHLGKS	NNFQKPKPAK	GKAEAHFSLV	HYAGTVDYNI	AGWLDKNKDP	600
LNETHVGLYQ	KSAMKTLAFL	PAGGSAEEG	GGGKKGKKK	GSSPQTVSAL	FRENLNKLMT	660
NLRSTHPHFV	RCIIPNETKT	PGAMEHELVL	HQLRCNGVLE	GIRICRKGF	SRILYADFKQ	720
RYKVLNASAI	PEGQFIDSKK	ASEKLLGSID	IDHTQYKFGH	TKVFFKAGLL	GILEEMRDEK	780
LAQLITRTQA	ICRGFLMRVE	FRKMERRRES	IFCIQYNVRA	FMNVKHWPM	KLYFKIKPL	840
KSAETEKEMA	NMKEEFKTK	EELAKSEAKR	KELEEKMTVL	MQEKNDLQLQ	VQSEADGLK	900
AEERCQDLIK	TKIQLEAKIK	ELTERAEDEE	EINAELTAKK	RKLEDECSEL	KKDIDDEL	960
LAKVEKEKHA	TENKVNKLTE	EMAGLDENIA	KLTKKKALQ	BAHQQTLDL	QAEEDKVNTL	1020
TKSKTKLEQQ	VEDELSLEQ	EKKLRMDLER	AKRLEGLDL	LAQESTMDIE	NDKQQLDEKL	1080
KKKEFEMSNI	QSKIEDQAL	AMQLQKKIKE	LQARIEELE	EIEAERASRA	KAEKLRSDLS	1140
RELEIEISERL	EEAGGATSAQ	IEMNKKREAE	FQKMRRLLEE	ATLQHEATAA	ALRKKHADS	1200
AELGEQIDNL	QRVKQKLEKE	KSEMMEIDD	LASNMETVSK	AKGNLEKMC	TLEDQLSEVK	1260
SKEEHEQRLL	NELSVQKAHL	HTSEGEFSRQ	LEEKDALVSQ	LSRGKQAF	QIEELKRQLE	1320
EETKAKSALA	HALQSARHDC	DLLEQYEEE	QEGKAEQRA	LSKANSEVAQ	WRTKYETDAI	1380
QRTTELEEA	KKLAQRQLDA	EEHVEVNAK	CASLEKTKQ	LQNEVEDLML	DVERNAACA	1440
ALDKKQRNFD	KLVSEWKQY	EETQADLEAS	QKESRSLSTE	LKVKKNAYEE	SLDQLETLKR	1500
ENKNLQQEIS	DLTEQIAEGG	KHIHELEKVK	KQIEQEKGE	QAALKEEAS	LEHEEGKILR	1560
IQLELNQVKS	BIDRKIAEKD	EEIDQLKRNH	LRVVESMQST	LDABIRSRND	ALRIKKKMEG	1620
DLNEMEIQLN	HANRQATEAI	KNLRNTQGV	KDTQLHLDDA	IRGQDDLKEQ	LAMVERRANL	1680
MQAIEELRA	SLEQTERSRR	VABQELLDAS	ERVQLLHTQN	TSINTKKKL	ETDITIQGE	1740
MEDIVQEAR	ABEAKAKAIT	DAAMMAELK	KEQDTSAHLE	RMKNMEQTV	KDLQHLRDEA	1800

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EQLALKGGKK	QIQKLEARVR	ELENEVEVEQ	KRNAEAVKGL	RKHERRVKEL	TYQTEEDRKN	1860
VLRLQDLVDK	LQSKVKAYKR	QABEEAEQSN	VNLAKFRKLQ	HELEEAEEERA	DIAESQVNKL	1920
RVKSREVTHTK	VISEE					1935

SEQ ID NO: 93 moltype = AA length = 607
 FEATURE Location/Qualifiers
 source 1..607
 mol_type = protein
 organism = Bos taurus

SEQUENCE: 93

MKWVTFISLL	LLFSSAYSRG	VFRDTHKSE	IAHRFKDLGE	EHFKGLVLIA	FSQYLQCCPF	60
DEHVKLVLNE	TEFAKTCVAD	ESHAGCEKSL	HTLFGDELCK	VASLRETYGD	MADCCCKQEP	120
ERNECFLSHK	DDSPDLPLK	PDPNTLCDEF	KADEKKFWGK	YLYETARRHP	YFYAPELLYY	180
ANKYNGVQFE	CCQAEDKGAC	LLPKIETMRE	KVLASSARQR	LRCASIQKFG	ERALKAWSVA	240
RLSQKFPKAE	FVEVTKLVTD	LTKVHKECCH	GDLLCACCDD	ADLAKYICDN	QDTISSKLKE	300
CCDKPLLEKS	HCIAEVEKDA	IPENLPPLTA	DFAEDKDVC	NYQEAQDAFL	GSFLYEYSRR	360
HPEYAVSVLL	RLAKEYEATL	EECCAADDPH	ACYSTVFDKL	KHLVDEPQNL	IKQNCDDQFEK	420
LGEYGFQNAL	IVRYTRKVPQ	VSTPTLVEVS	RSLGKVGTRC	CTKPESERMP	CTEDYLSLIL	480
NRLCVLHEKT	PVSEKVTKCC	TESLVNRRPC	FSALTPDETY	VPKAFDEKLF	TFHADICTLP	540
DTEKQIKKQT	ALVELLKHKP	KATEEQLKTV	MENFVAFVDK	CCAADDKEAC	FAVEGPKLVV	600
STQTALA						607

SEQ ID NO: 94 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 94

MPFGNTHNKH	KLNFKAEEY	PDLSKHNNHM	AKALTLEIYK	KLRDKETPSG	FTLDDVIQTG	60
VDNPGHPFIM	TVGCVAGDEE	SYTVFKDLFD	PIIQDRHGGF	KPTDKHKTDL	NHENLKGDD	120
LDPHYVLSRR	VRTGRSIKGY	ALPPHCSRGE	RRAVEKLSVE	ALNSLTGEFK	GKYYPLKSMT	180
EQEQQQQLIDD	HFLFDKVPSP	LLLASGMARD	WPDARGIWHN	DNKSFLVWVN	EEDHLRVISM	240
EKGGMNKEVF	RRFCVGLQKI	EETFKKAGHP	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLA	300
HLSKHPKFEE	ILTRLRLQKR	GTGGVDTAAV	GSVFDVSNAD	RLGSSEVEQV	QLVVDGVKLM	360
VEMEKKLEKG	QSIDDMIPAQ	K				381

SEQ ID NO: 95 moltype = AA length = 897
 FEATURE Location/Qualifiers
 source 1..897
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 95

MMMVLPQEG	GTGEGPFAAG	GRGGGEYMEQ	EEDWDRDLLL	DPAWEKQQRK	TFTAWCNSHL	60
RKAGTQIENI	EEDFRNGLKL	MLLLEVISGE	RLPRPDKGKM	RFHKIANVKN	ALDFIASKGV	120
KLVSIGAEIE	VDGNLKMTLG	MIWTIILRFA	IQDISVEETS	AKEGLLLWCQ	RKTAPYRNVN	180
VQNFTSWKD	GLALCALIHR	HRPDLIDYAK	LRKDDPIGNL	NTAFEVAEKY	LDIPKMLDAE	240
DIVNTPKPDE	KAIMTYVSCF	YHAFAGAEQA	ETAANRICKV	LAVNQENELK	MEEYEKLASE	300
LLEWIRRTVP	WLENRVGEPS	MSAMQRKLED	FRDYRRLHKP	PRVQEKCOLE	INFNTLQTKL	360
RLSHRPAPMP	SEGKLVSDIA	NAWRGLEQAE	KGYEDWLLSE	IRRLQRLQHL	AEKFQQKASL	420
HEAWTRGKEE	MLSQDYETA	SLQEVRLALL	RHEAFESDLA	AHQDRVEHIA	ALAQELNELD	480
YHEAASVNSR	CQATCDQWDN	LGLTLQKRRD	ALERMEKLEL	TIDQLQLEFA	RRAAPFNNWL	540
DGAVEDLQDV	WLHVSVETQ	SLVTAHDQFK	ATLPEADRER	GAILGIQGEI	QKICQTYGLR	600
PSSTNPYIAL	TPQDINTKWD	TVRKLVPSPD	QMLQEEELTRQ	QMNERLRRQF	AAQANAIGPW	660
IQQKVEEGR	LAAGMAGSLE	EQMAGLRQQE	QNIINYKSN	DRLEGDHQLL	QESLVFDNKH	720
TVYSMEHIRV	GWEQLLTSIA	RTINEVENQV	LTRDAKGLSQ	EQLNEFRASF	NHFDRKRNGM	780
MEPDDFRACL	ISMGYDLGEV	EFARIMTMVD	PNAAGVVTFQ	AFIDFMTRET	AETDTAEQVV	840
ASFKILAGK	NYITAEELRR	ELPAEQARRA	PDRGAGAPAG	ALDYVAFSSA	LYGESDL	897

SEQ ID NO: 96 moltype = AA length = 842
 FEATURE Location/Qualifiers
 source 1..842
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 96

MSRPLTDQEK	RKQISVRGLA	GVENVTELKK	NFNRHLHFTL	VKDRNVATPR	DYYFALAYTV	60
RDHLVGRWIR	TQQHYEKD	KRIYYLSLEF	YIGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELEEIEEDA	GLGNGGLGRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	ISGGWQMEEA	180
DDWLRYGNPW	EKARPEFTLP	VHFYGRVEHT	SQGAQWVDQ	VVLAMPYDTP	VPGYRNNVVN	240
TMRLWSAKAP	NDFNLKDFNV	GGYIQAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQYFV	300
VAATLQDIIR	RFKSSKFGCL	DPVRTNFDAF	PDKVAIQLND	THPSLAIPEL	MRILVDQERL	360
EWEKAWETV	KTCAYTNHTV	LPEALERWPV	HLIETLLPRH	LQIYEINQR	FLNRVAAAF	420
GDVDRLRMS	LVEEGAVKRI	NMAHLCIAGS	HAVNGVARIH	SEILKKTIFK	DFYELEPHKF	480
QNKTNGITPR	RWLVMCNPL	AEVIAERIGE	EYIADLDQLR	KLLSYVDD	FIRDVAKVQ	540
ENKLFKFSAYL	EKEYKVHINP	NSLFDIQVKR	IHEYKRQLLN	CLHVITLYNR	IKKEPNKFFV	600
PRTVMIGGKA	APGYHMAKMI	IRLITAIGDV	VNHDPVVGDR	LRVIFLENYR	VSLAEKVIPA	660
ADLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	ABEAGEENFF	IFGMRVEDVE	720

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RLDQKGYNQ	EYYDRIPELR	HIIDQLSSGF	FSPKQPDLEK	DIVNMLMHH	RKFVFADYEE	780
YVKCQERVSA	LYKNPREWTR	MVIRNIATSG	KFSSDRTIAQ	YAREIWGVPE	TRQRMPPAPDE	840
KI						842

SEQ ID NO: 97 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 97

MPHQYPALTP	EQKKELCDIA	HRIVAPGKGI	LAADSTSGSI	AKRLQSIGTE	NTEENRRFYR	60
QLLLTADDRV	NPCIGGVILF	HETLYQKADD	GRPPQVVIKA	KGGVVGKVD	KGVVPLAGTN	120
GETTTQGLDG	LSERCAQYKK	DGADFAKWRC	VLKIGEHTPS	SLAIMENANV	LARYASICQQ	180
NGIVPIVEPE	ILPDGDHDLK	RCQYVTEKVL	AAVYKALSDH	HIYLEGTLK	PNMVTPGHAC	240
TQKYSHEEIA	MATVTALRRT	VPPAVPGITF	LSGGQSEEEA	SINLNAINKC	PLLPWALT	300
SYGRALQASA	LKAWGGKKEN	LKAAQEYEVK	RALANSLACQ	GKYTPSGKAG	AAASESLFIS	360
NHAY						364

SEQ ID NO: 98 moltype = AA length = 197
 FEATURE Location/Qualifiers
 source 1..197
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 98

ILLNKMAPPK	DVKKPAAAAA	PAPAPAPAPA	PAPAPPKEEK	IDLSAIKIEF	SKQQQDEFKE	60
AFLLYDRTGD	GKITLSQVGD	VLRALGTNPT	NAEVKKVMGN	PSNEEMNAKK	IEFEQFLPML	120
QAISNNKDGQ	TYEDFVEGLR	VFDKEGNGTV	MGAELRHVLA	TLGEKMKEEE	VEALMAGQED	180
SNGCINYEAF	VKHIMSN					197

SEQ ID NO: 99 moltype = AA length = 322
 FEATURE Location/Qualifiers
 source 1..322
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 99

IGRLVTRAAF	NTGKVDIVAI	NDPFIDLHYM	VYMFQYDSTH	GKPHGTVKAE	NGKLIVINGKA	60
ITIFQERDPA	NIKWDAGAAE	YVVESTGVFT	TMEKAGAHLK	GGAKKVIISA	PSADAPMFVM	120
GVNHEKYNNT	LKIVSNASCT	TNCLAPLAKV	IHDHFGIVEG	LMTTVHAIAT	TQKTVDGPGS	180
KLWRDGRGAA	QNIIPASTGA	AKAVGKVIPE	LNGKLTGMAR	RVPTPNVSVV	DLTCRLEKPA	240
KYDEIKKVVQ	QASEGPLEKI	LGYTEQVVS	CDFNSDTHSS	TFDAGAGIAL	NDHFVKLISW	300
YDNEFGYSNR	VVDLMVHMAS	KE				322

SEQ ID NO: 100 moltype = AA length = 294
 FEATURE Location/Qualifiers
 source 1..294
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 100

MDAIKKMQM	LKLDKENALD	RAEQAEADKK	AAEDRSKQLE	DELVSQKKL	KATEDELDKY	60
SEALKDAQEK	LELAEEKATD	AEADVASLNR	RIQLVEEELD	RAQERLATAL	QKLEAEAKAA	120
DESERGMKVI	ESRAQKDEEK	METQEIQLKE	AKHIAEDADR	KYEEVARKLV	IIESDLERA	180
ERAELSEGQV	RQLEEQLRIM	DQTLKALMAA	EDKYSQKEDK	YEEIKVLSL	KLKEAETRAE	240
FAERSVTKLE	KSIDDLEDEL	YAQKLKYKAI	SEELDHALND	MTSIESGSCQ	RRKP	294

SEQ ID NO: 101 moltype = AA length = 1141
 FEATURE Location/Qualifiers
 source 1..1141
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 101

MPEAKPAACK	APAGKDAAAK	PAPKEAPPQE	APAAPSTEAP	PEDQSPTAAE	PTGVFLKKPD	60
SVSVENGKDT	VIVAKLNGKE	LPAKPAVKWF	KGKWLGLGSK	SGARFSFKES	HDAASNVTYV	120
ELHITKVLVG	DRGDYRIEVK	AKDFCDSCFT	NIDVESPRQD	SAGQSLESFK	RSSEAKSDTA	180
GELDFSGLLK	KQVVEEEKK	KKKDDDLGI	PPEIWELLKG	AKKSEYERIA	FQYGITDLRG	240
MLKRLKKAKV	EVKKSAAFTK	KLDPAYQVVK	GNKIKLVVEI	SDPDLPLKWF	KNGQEIKPSS	300
KYVFENVGKK	RILTINKCTL	ADDAAYEVVV	KDEKCTELF	VKVEPPVLIV	TPLEDQOVFV	360
GNRVEMAVEV	SEGAQVMWL	KDGVELTRED	SFKSRYRLKK	DGKRHILIYA	EVALEDKGYH	420
KVMTNGDQCE	AEILVEEKQL	EVLQDIADLT	VKASEQAVFK	CEVSEKVTG	KWYKNGVEVR	480
PSKRITISHT	GRIHKLVIDD	VRPEDEGDT	FVPDGYALSL	SAKLNFLGAG	LSPLPPEPPK	540
IHLDCSGQTS	ENSVVVVAGN	KLRLDVSITG	EPRPVATWMK	EDEVFTGTG	RVRIEQRGDI	600
SSFVIESAER	GDEGRYTIKV	TNPAGEDVAS	ILLKVVDPD	PPEAVRVTSV	GEDWAVLVWE	660
PPKYDGGRPV	TGYLLERKKK	GSQRWMKLN	EVFTETTYES	TNMIIEGILY	MRVFAVNAIG	720
VSQPSMNTKP	FMPIAPTSEP	QHLTVEDVTD	TTTTLKWRPP	EKIGAGGIDG	YLIEYCEVGS	780
DEWIPANTEP	VERCGFTVKN	LPTGAKITFR	VVGVNIAGRS	QPATLAQPV	IREIVEQPKI	840
RLPRHLRQTY	VKRGHEHLNL	VIPFQKPRP	QVWTKGGAP	IDTNHVHVRT	SDFDTVFFVR	900
QAARSDSGEY	BLTVQIENMK	DTATIRIRVV	EKAGPENVM	VKEVWGTNAL	VEWQPPRDDG	960

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NSEITGYLVQ	KADKKTMEWF	TVYERNRHTS	CTVSDLIMGN	EYYFRVYSEN	ICGLSDLPGV	1020
SKNTARIVKT	GMTLKLPEYK	EHDFTTPPKF	LTPLFDRVVV	AGYAAALNCA	VRGHPKPKVV	1080
WMKNKMEIRE	DPKFLMTNHQ	GVLTLNIRRP	SPFDSGTYSY	RAVNELGEAL	AECKLEVRGE	1140
P						1141

SEQ ID NO: 102 moltype = AA length = 1483
 FEATURE Location/Qualifiers
 source 1..1483
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 102

MSLVAVPFYQ	KRHKHFQDSY	RNIQTRYLLD	EYAAKKKRAS	AQSSLQRSVT	RKSSSQRESS	60
RAALGETTLG	ETTLGRITCP	LCAKRMSSAL	EEEAQERKRR	YQSMVAAYGE	AKRERYLSEL	120
AQLEEDVHVA	RTHAREELDR	LALQHAGRHG	LHHAVDDRLA	WERHWFEDRI	SRAPEILVRL	180
RSHTVVERMS	VKLCTVQGF	PTPVVQWYKD	GSLICQAGEP	GKYRIDSKYG	VHTLEINRAD	240
FDDTATYSAV	ATNVHGQVST	NAAVVVRRFR	GDEEPFHSVG	LPIGLPLSSV	IPYTHFDVQF	300
IEKFGVTFRR	EGETLTCLKT	LLVTPDLKRV	QPRAEWYRDD	VLLKESKWTK	MFFGEGQASL	360
SFSLHNKDDE	GLYTLRIVSR	GGVTDHSAFL	FVRDADPLVT	GAPGAPMDVH	CHDANRDYVI	420
VTWKPPNTTK	ESPVIGYFID	RCEVGTDNWW	QCNDAPVKIC	KYPVTGLFEG	RSYIFRVRAV	480
NSAGISRPSR	VSEAVAALDP	VDLRRLQAVH	LEGEKDIVIY	QEDLEGEVQI	PGPPTNVHAS	540
ETSRTYVVLS	WDPEPRGKE	PLMYFIEKSM	VGSGSQWRVN	AQTAVRSPRY	AVFDLAEGKP	600
YVFRVLSLNG	HGLSEPSBIT	APIQAQDTIV	VPSAPGRVLA	SRNTRTSVVV	QWDRPKHEED	660
LLGYVDCSV	AGSNVWPCN	HKPIGYNRFV	VHGLTTGEQY	IFRVKAVNAV	GTSENSQESD	720
VTKVQAALTV	PSHPYGITLL	NCDGHSMILG	WKVPKFSGGS	PILGYIIDKR	EAHHKNWHEV	780
NSSPLKERIL	KVEGLTEGSL	YEPKIAAANM	AGMGSQQTEL	FKCEAWTMEP	PGPAYDLTFC	840
EVRDTSVLVL	WKPPVYSGSS	PVSGYFVDYK	EEDSGEWLTV	NEVTTPHRYL	KVCDLHQGKT	900
YVFRVRAVNA	SGVGRPSDTS	EPVLVEARPG	TKEVSAGVDE	EGNIYLSFDC	PEVTDASHFT	960
WCKSYEDIA	DDRFKVVETAG	DHCKLYFKNL	DKEDLGTYSV	SVSDTDGVSS	SFVLDEEELE	1020
RLKALSNEIK	NPTIPLKSEL	AYEIFDKGQV	RFWLQAEHLS	PDSSYRFIIN	DREVADSETH	1080
RTKCDKSTGV	TEMVMDRFTV	DNEGTYTVQI	QDGKAKSQSS	LVLIGDAFKA	VLEEAFFQRK	1140
EFLRKQGPHE	AEYLHWDVTE	ECEVRLVCKV	ANTKKETVFK	WLKDDVLYET	EKMPPLEKGV	1200
CELLIPKLKSK	KDHGEYKATL	KDDRGQDVST	LEIAGKVYED	MILAMSRVCG	ASASPLKILC	1260
TPEGIRLQCF	MKYFTEEMKV	NWYHKDAKIA	SSEHMRIGGS	EEMAWLQICE	PTEKDKGKYT	1320
FEILDGKKNH	QRSLDLGQQA	FDEAFAEFQQ	LKAAAFAEKN	RGKVIIGLPD	VVTIMEGKTL	1380
NLTCTVFGNP	DPEVVWFKN	KDIELSDHFS	VKVEQAKYVS	MTIKGVTSED	SGKYSMHVKN	1440
KYGGKIDVT	VSVKHGEIE	PDVVLPPQAK	PKLLPAAAPS	PAH		1483

SEQ ID NO: 103 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 103

MAMQKIFARE	ILDSRGNPVT	EVDLHTAKGR	FRAAVPSGAS	TGIYEALRL	DGDKSRYLGK	60
GVLKAVEHIN	KTLGPALLEK	KLSVVDQEKV	DKFMIELDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVL	YRHLADLAGN	PELILPVPAF	NVINGGSHAG	NKLAMQEFMI	LPVGASSFRE	180
AMRIGAEVYH	HLKGVIKAKY	GKDATNVGDE	GGFAPNILEN	NEALELLKTA	IQAAGYPDKV	240
VIGMDVAASE	FYRNGKYDL	FKSPDDPARH	ISGEKLGLY	KSFINKYPPV	SIEDPFDQDD	300
WATWTSFLSG	NVNIQIVGDDL	TVTNPRIKIAQ	AVEKKACNCL	LLKVNQIGFS	TESIQACKLA	360
QSNQWGVMS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEALGDK	420
AVFAGRKFRN	PKAK					434

SEQ ID NO: 104 moltype = AA length = 170
 FEATURE Location/Qualifiers
 source 1..170
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 104

QAPKAKARRA	AAEGSSNVF	SMFDQTQIQE	FKEAFTVIDQ	NRDGIIDKED	LRDTFAAMGR	60
LNKNEELDA	MMKEASGPIN	FTVFLTMFGE	KLKGADPEDV	ITGAFKVLDP	EGKGTIKKQF	120
LEELLTTQCD	RFSQEIRNM	WAAPPPDVGG	NVDYKNICYV	ITHGDAKDQE		170

SEQ ID NO: 105 moltype = AA length = 1937
 FEATURE Location/Qualifiers
 source 1..1937
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 105

MSSDQEMAIF	GEAAPYLRSK	EKERIEAQNK	PPDAKTSVFF	ADPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQVPPM	NPPKFDKIED	MAMMTHLHEP	AVLYNLKERY	AAWMIYTYSG	120
LPCVTVNPKY	WLPVYNAEVV	AAYRGKKRQE	APPHIFSID	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKKEEP	TSQKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSRFRGFG	IRIHFGTGTG	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTMP	YDYAFVSQGE	ITVPSIDDQE	ELMATDSAIE	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFQKQ	QREQAEPDG	TEVADKAAYL	QGLNSADLLK	ALCYPRVKVG	NEFVTKGQTV	420
EQVYNVAVGAL	AKAIYDKMFL	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	PDFNSLEQLC	480

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INFTNEKLQQ	PFNNHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	FKNKLYEQHL	GKSNNFQKPK	PAKGKAAEHF	SLIHYAGTVD	YNITGWLDKN	600
KDPLNETVVG	LYQKSSVKTL	APLFSGAASA	EAEGGGAKKG	GKKKGSSFQT	VSALFRENLN	660
KLMTNLRSTH	PHFVRCIIPN	ETKTPGAMEH	ELVLHQLRCN	GVLEGIRICR	KGFPSRILYA	720
DFKQRYKVLN	ASAIPEGQFI	DSKKASEKLL	GSIDVDHTQY	KFGHTKVFFK	AGLLGLLEEM	780
RDEKLAQLIT	RTQARCRGFL	ARVEYQRMVE	RRESIFCIQY	NVRAFMMNVKH	WPWMKLYFKI	840
KPLLSAETE	KEMANMKKEE	EKTKEELAKS	EAKRKELEEK	MVTLTQEKND	LQLQVQSEAD	900
ALADAEERCD	QLIKTKIQLE	AKIKEVTERA	EDEEEINAEL	TAKKRKLEDE	CSELKKDIDD	960
LELTAKVEK	EKHATENVKV	NLTEEMAGLD	ETIAKLTKEK	KALQEAHQQT	LDDLQAEEDK	1020
VNTLTAKATK	LEQQVDDLEG	SLEQEKLLRM	DLERAKRKLE	GDLKLAQEST	MDIENDKQQL	1080
DEKLKKKEFE	MSNLQSKIED	EQALGMQLQK	KIKELQARIE	ELEEEIEAER	ASRAKAEKQR	1140
SDLRELEEI	SERLEEAGGA	TSAQIEMNKK	REAEFQKMRR	DLEEAATLQHE	ATAAALRKKQ	1200
GGVGEQLDNL	QRVQKLEKE	KSEMMEIDD	LASNMTVSK	AKGNLEKMCR	TLEDQLSELK	1260
TKEDQQRLLI	NDLTTQRRARL	QTESGEFSRQ	LDEKDALVSQ	LSRGKQAFQTQ	QIEELKRQLE	1320
EEIKAKSALA	HALQSARHDC	DLLREQYEEE	QEAKAELQRA	LSKANSEVAQ	WRTKYETDAI	1380
QRTELEEEAK	KKLQARLQDA	EEHVEAVNAK	CASLEKTKQR	LQNEVEDLMI	DVERTNAACA	1440
ALDKKQRNFD	KILSEWKQKY	EETHAELEAS	QKESRSLSTE	LFKIKNAYEE	SLDQLETLKR	1500
ENKNLQPTTE	ISDLTEQIAE	GGKRIHELEK	VKKQVEQESK	EQIAALEEAE	ASLEHEEGKI	1560
LRIQLELNQV	LEQIDQLKRA	KDEIDQLKR	NHIRVVEVMQ	STLDAEIRSR	NDAIRLKKKM	1620
EGDLNEMETQ	LNHANRMAAE	ALKNYRNTQA	ILKDTQLHLD	DALRGQEDLK	EQLAMVERRR	1680
NLLQAEIEEL	RATLEQTERS	RKIAEQELLD	ASERVQLLHT	QNTSLINTKK	KLETDITQIQ	1740
GEMEDIQIEA	RNAEEKAGGA	ITDAAMMAEE	LKKEQDTSAH	LERMKKNLEQ	TVKDLQHRLD	1800
EAEQLALKGG	KKQIQKLEAR	VRELEAEVES	EQKRNVTETK	GLRKHERRVK	ELTYQTEEDR	1860
KNILRLQDLV	DKLQAKVKS	KRQAEAEAEQ	SNVNLKSKFRK	LQHELEAEAE	RADIAESQVN	1920
KLRVKSREVV	TKIISEE					1937

SEQ ID NO: 106 moltype = AA length = 6911
 FEATURE Location/Qualifiers
 source 1..6911
 mol_type = protein
 organism = Ovis aries

SEQUENCE: 106

MADDEEYEEV	VEYYTEETVY	EEVPGETITE	FYETTTTTRS	DYEQSDTSRP	ALAKPVPEKP	60
VERKRVIRKK	VDPKFMTPY	IAHSQKMDL	FSTNKYKENY	EKAGQPYAI	TTDTPELRRI	120
KKVQDQLSEV	KYRTDGDVAK	TICHVDEKAK	DIEHAKKVSQ	QVSKVLYKQN	WEDTKDKYLL	180
PPDAPELVQA	IKNTAMPSKK	LYTEDWEADK	TLFPYPYNDSP	ELRRVAQAQK	ALSDITYKKG	240
LAEQQTQFTS	LPDPPDIEFA	KKVTNQVSKQ	KYKEDYENKV	KGKWESETPCF	EIATARMSQ	300
NISTKKYQED	FENTKDQIYF	MQTETPEYKV	NKQAGVAASK	VKYKQDYEKN	KGKADYNVLP	360
ASENPLLRQL	KAAAGDALSDK	LYKENYEKTK	AKSINYCETP	KFKLDTVLHN	FSSDTKYKDS	420
YLNKILGHYV	GSYLEDPHYTH	CMRVSAQNDS	KNYKAEYEED	RGKGFPPQTI	TQYEYAIKKL	480
DQCKDHTYKV	HPDKTKFTQV	TDSPVLVQAQ	VNSKQLSDLN	YKAKHENEKF	KCHIPPDTPA	540
FTQHKVNAYN	LSDNIIYKHDW	EKTAKKFDI	KVDAIPLLA	KANTKIASDV	MYKKDYEKSK	600
GKMGALSIN	DDPKMLHSLK	TAKNQSDRLY	KENYEKTKAK	SMNYCETPKY	QLDTLLKNFS	660
EAKYKDSYVQ	NVLGHYIGSF	EDPYREHCKM	VAAQNSDKNY	KAEYEEDKKG	CYPPQITITQE	720
YBAIKKLQDC	KDHTYKVPD	KTKFTAVTDT	PVLLQAQLNT	QQLSDLNKYA	KHEGEKFKCH	780
IPADAPQFTQ	HRVNAVNLSD	NIYKHDWEKT	KAKKFDIKVD	AIPLLAAKAN	TKIASDVMYK	840
KDYKSKGKGM	IGALSINDDP	KMLHSLKTAK	NQSDREYRKD	YEKSKTIYTA	PLDMLPLTHA	900
KKSQEIASDV	DYKHLHNNYS	YPPDSVNVDL	AKKAYALQSD	VEYKADYNSW	MKGCGWVPFG	960
SLEMEKAKRA	SIDILNEKPYR	QHPDTLKFTS	IEDAPIIVQS	KINQAQRSDV	AYKAKGEEVI	1020
HKYSLPADLP	QFIQAKVNAY	NISENMYKAD	LKDLSKKGYD	LRIIDAIPKA	AKAARQAASD	1080
VQYKKDYDEKA	KGKMGVQPSL	QDDPKLVHYM	NVAKIQSDRE	YKDYDEKTKT	RHNTPHDMVN	1140
IVAAKKAQDV	ASNVMYKHSK	HHYTYLPDAM	DLELSKNMMH	IQSDNVYKED	YNNWMKGIGW	1200
IPIGSLEVEK	VKKAGDALNE	KKYRQHPDTL	KFTSIVDSPV	MQVAKQNTQQ	VSDILYKAKG	1260
EDVHKHYTMS	PDLQFLQAK	CNAYNLSDVC	YKRDWHDLIA	KGTVNLGDAI	PITAAKASRN	1320
IASDYKYKEA	YEKAGKQVVG	FRSLQDDPKL	VHYMNVAKLQ	SDREYKKNYE	NTKTSYHTPG	1380
DMVSITAAMK	AQDVATNVNY	KQPIHHYTYL	PDALSLEHIR	NVNQIQSDKV	YKDEYNHFFK	1440
GIGWIPIGSL	EVEKVKKAGD	ALNEKKYRQH	PDTIKFTSVP	DSMGVMVAQH	NTKQLSDLNY	1500
KVEGEKVKHK	YTLDPDVPQF	IQAKVNANFM	SDSHYKADWK	KTAKGYDLR	PDAIPVAAK	1560
SSRNIAISDFK	YKEGYEKAKG	KQIGFRSLQD	DPKLVHFMNV	AKMQSDREYK	KGYEASKTKY	1620
HTPLDMLSVT	AAKKSQEVAT	NTNYKQPFHH	YTLPLDALNV	EHSRNAMQIQ	SDNLYKSDFT	1680
SWMKIGWMP	LESLEVAKAK	KAGEILSEKK	YRQHPKELKF	TYAMDTEQA	LNKSNKLIMD	1740
KRRYTEKWNK	DKTTIHVMPD	TPDILLSRVN	QITMSDKLYK	AGWEEKKKKG	YDLRPDAIPI	1800
KAARASRDIA	SDYKYQAYE	QAKGKQIGFR	SLEDDPKLVH	FMQVAKMQSD	REYKKAYEKS	1860
ADYADFMKGI	LSVPLGSLAE	EKNKKAMEII	SEKKYRQHPD	TLKYSTLMDS	MMNVLAKNNA	1920
KIMNEHLYKQ	AWEADKTKVH	IMPDIQIIL	AKANAINMSD	KLYKLSLEEA	KKKGYYDLRTD	2040
AIPIKAAKAS	RDIASDYKYK	HNYEKERGKM	VGFRSLEDDP	KLVSMSQVAK	MQSDREYKKN	2100
YEKTKTSYHT	PADMLSYTAA	KDAQANITNT	NYKHLIHKYI	LLPDAMNIIQ	SKNMNRLQSD	2160
NEYKQDYNEW	YKGLGWSPAG	SLEVEKAKRA	TEYASDQKYR	QHPSNFQFSK	LNDSDMDVLA	2220
KQNAHTMNKY	LYTVDNWKKD	TKIHVMPDTP	DILQAKQNTQ	IYSQKLYKLG	WEALKKGYD	2280
LPVDAISVQA	AKASRDIVSD	FKYKQGYRQK	IGHHIGFRSL	QDDPKLVLSM	NVAKMQSERE	2340
YKPDFEKWKT	KFSSPVDMLG	VVLAKKCCAL	VSDVDYKNYL	HGWTCLPDQN	DVIHAKKAYD	2400
LQSENLYKSD	LEWLKGLGWS	PLGSLEAEKN	KRASEIIESE	KYRQPPDRNK	FTSIPDAMDI	2460
VLAKTNAKNR	SDILYREAWD	KDKTQVHIMP	DTDPDIILAKA	NLINASDKFY	RMGYEELRKK	2520
GVDLPVDAIP	TKAAKASREI	ASEYKYKEGF	RKQLGHHIGA	RNIKDDPKMM	WSMHVAKIQS	2580
DREYKKDFEK	WTKTFSSPVD	MLGVVLAKKC	QTLVSDIDYK	NYLHQWTCPL	DQSDVIHARR	2640

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AYDLQSDNLY	KSDLQWLKGI	GWLPSPGSLED	EKNKRATQIL	SDHVYRQHPD	KFKFSSLMD5	2700
IPMVLAKNNA	ITMNRHLYTE	AWDKDKTTVH	IMPDPGVL	AKQNIYINSE	KLYKLGLDEA	2760
KRKGVDMDR	AIPIRAAKAS	RDIASEFKYK	EGYRKQLGHH	IGARAIHDDP	KMMWSMHVAK	2820
IQSDREYKKD	FEKWKTKFSS	PVDMLGVVLA	KKCQTLVSDI	DYKNYLHQWT	CLPDQSDVIH	2880
ARQAYDLQSD	NVYKSDLQWL	KGIGWVPIGS	LDVEKCKRAS	EILSDKLYRQ	PPDKFKFTSV	2940
TDSLEQVLAK	NNAINMNRKL	YTEAWDKDKT	QVHIMPDPTE	ITLARMNKVN	YSESLYKLAH	3000
EEAKKKGVDL	RSDAIPVAA	KASRDIISDY	KYKDGYSKQL	GHHIGARDIQ	DDPKMMWSMH	3060
VAKIQSDREY	KKDFEKKWTK	FSSPVDMGLV	VLAKKCQTLV	SDIDYKNYLH	QWTCLPDQSD	3120
VIHARRAYDL	QSDNVYKSDL	QWLRGIGWVP	IGSVDVVKCK	RAAEILSDNL	YRQPPDTLKF	3180
TSVPDSLEQV	LAKNNAINMN	KRLYTEAWDK	DKTQVHIMP	TPEIMLARQN	KLNYSETLYK	3240
LANEEAKKKG	YDLRSDAIP	VAAKASRDII	SDYKYKGYR	KQLGHHIGAR	DIKDDPKMMW	3300
SMHVAKIQSD	REYKDKFEKW	KTKFSSPVDM	LGVLVLAKKCQ	TLVSDIDYKN	YLHQWTCCLPD	3360
QSDVIHARQA	YDLQSDNIYK	SDLQWLRGIG	WVPIGSVDV	KCKRAAEILS	DNLYRQRPD	3420
LKFTSVPSDL	EQVLAKNNAI	NMNRKLYTEA	WDKDKTQIHM	MPDTPETILA	RQNKINYSEN	3480
LYRQAMEEAK	KEGYDLRSDA	IPVAAKASR	ETASDYKYKE	AYRKQLGHHI	GARAIHDDPK	3540
MMWSHVAKM	QSDREYKKDF	EKYKTRFSSP	VDMGLIVLAK	KCQTLVSDVD	YKHPHLEWTC	3600
LPDQNDIIHA	RKAYDLQSDN	LYKSDLEWLK	GIGWVPIGSF	DVVKAKRAGE	ILSDNIYRQR	3660
PDTLKFTSVT	DSPEQVLAKN	NALNMKRLY	TEAWDNDKKT	IHVMPDTPET	MLAKLNIRIN	3720
SKLYKLAL	ESKKEGYDLR	DLDAIPQAQ	ASREIASDYK	YKEGYRKQLG	HHIGARNIKD	3780
DPKMMWSIHA	TKLQSDREYK	KDFEKKWTKF	SSPVDMMLV	QAKKCQILVS	DIDYKNLLHE	3840
WICLPDQNDV	IQARKAYDLQ	SDAIYKADLE	WLRGIGWVPI	GSVGVENVKR	AGEILSERKY	3900
RQPAVQLKFT	SITDTPETVL	AKNNALNMK	HLYTEAWDAD	KTSIHVMPDT	PEILLAKSNS	3960
ANISHKLYTK	GWDESKMKDY	DLRADAISIK	SAKASRDIA	DYKYKEAYEK	QKGHHIGARS	4020
VEDDPKIMCA	MNAGKIQSER	EYKKEPQKWK	TKFSSPVDM	GILQAKKCQI	LVSDIDYRNY	4080
LHHWTCLPDQ	NDIIQARKAY	DLQSDAIYKA	DLEWLRGIGW	MPEGSPEVLR	VKNAQHIFRD	4140
SVYRTPVVKL	KYTSIVDTPE	VVLAKSNAEN	ISIPKYREVW	DKDKTSIHIM	PDTPEINLAR	4200
TNALNVSNKL	YREGWDEMKM	SCDVLDAIP	IQAAKASREI	ASDYKYKLDH	EKQKGHYVGT	4260
RTARDDNKIR	WALTAGKIQN	ERERYMHWA	WKSFKQSPDP	MLSIEHSKNS	QTLVSDIDYR	4320
HYLHQWTCMP	DQNDVIQARK	AYDLQSDAIY	KADLEWLRGI	GWMPADSVSV	NHAKHMGDIF	4380
NEKKYRTKIE	TLNFTPVDDR	VDYVTAKHSG	EILNDIKYRK	DWNTDKSKYT	LTETPQLHAA	4440
QEAARILDQY	LYKESWEKQK	ATGYILPPDA	VFPVHAHSG	DVQSELKYKA	EHVKQKGYHV	4500
GVPTMRDDPK	LVWFEHAGQI	QNDRLYKESY	HKTAKIHIP	PDMVSVLAAK	EGQALASDID	4560
YRTYLHQWIC	HPDQNDVIA	RKAYDLQSDN	IYKADLEWLR	GIGWMLPDSV	DHVRVTRNQE	4620
MVNQIKYKDK	ALANYPNFTS	VVDPEIVLA	KINAVNQSDV	KYKETFNKLI	KGKYSFSPDT	4680
PYITHSKDME	KLYSTILYKR	AWEGTKAYGY	TLDERYIPIV	GAKHADFVNS	ELKYKETYEK	4740
LKGHYLAGKE	ISEFPNVVHC	LDFQKMRSLL	NYRRHYEDTK	ANVHIPQDMM	NHVLAKRCCQY	4800
ILSDLEYRHY	PHQWTSLEPE	PNVVRVRAHQ	EILSDNVYKD	DLNWLKIGIG	YVWDTPOILH	4860
AKKSYDLQSQ	ILYTAAGKEN	LKNYNLVTD	PLYVTALQSG	INASEVYKYE	NYHQIKDKYT	4920
TVLETVDYDR	TRNLKDLFSS	NLYKEAWDKV	KATSYILPPN	TVSLTHAKNQ	KHMASVYKYR	4980
EYDYKFKALY	TLPRSVDKDP	NTARCLRVGK	FNIDRLYRSV	YEKNKMNHI	VPMVMEMVTA	5040
KDSQKKVSEI	DYRLHLHEWI	CHPDLQVNSH	VRKVTQDISD	IVYKDDLTWL	KGIGCYVWDT	5100
PEILHAKHAY	DLRNDIKYKA	HVLKTRNNYK	LVTDPVYVQ	AVKSGKQLSD	AVYHYDYVHS	5160
IRGRVAPTTK	YVDLDRALHA	YKLQSENLYR	KAGLHALPTG	YRLPVDTPHF	KHTKDTRYMS	5220
SYFKYKEVYE	HMKAYGYTLG	PNDVPFVNVR	RVNNITSERL	YRQLYHKLKD	KIHTTPDTPE	5280
IRQVKKTQEA	VSELIYKSD	FKMQGHMISM	PYTPHVLHCR	YVGDISDIK	YKEDLQILRG	5340
LGCFLYDTPD	MVRSRHLRLE	WSHYLYTDKA	RKMRDKYQV	LDTPEYRTVQ	ELKTHLSELV	5400
YRASGRKQKS	IFTSVPTDTP	LIRAKRGQKL	QSQYLYVELA	TKERPHHHAG	NQTTALKKHAR	5460
HVKDMVSENK	YKIQYKMKD	KYTPVPDTP	LIRAKRAYWN	ASDLRYKETP	QKTKGRYHTV	5520
KDALDIVYHR	TVTTHISKIK	YKENYMSQLG	IWRSDPDRPE	HFHHRVAVTA	VSDVYKADL	5580
TWLKGIGCYA	YDTPDFTLAE	KNKTLYSKYK	YKEVFERTKS	NFKYVADCP	NRHFKFATQL	5640
MNEKKYRADY	BQRDKYHLV	VDBPRHLLAK	IAGDQISQIR	YRKNYKTKD	KFTSIVDTPE	5700
HLRTTKVKNQ	ISDILYKLEY	NKAKPRGYTT	IHDTPMLLHV	RKVDEVSDDL	KYKEVYQRTK	5760
SNCTIEPDAV	HIKAADAYK	VNTNLDYKKQ	YEATKAYWKW	TPDRPDFIQA	AKSTLQQSDF	5820
EYKLDREYK	GCKLSVTDK	DMVLALKNSI	IESDLKYKEK	HVKERGSCHA	VPDTPQILLA	5880
KTVSSLVSEN	KYQYVKKHL	AQGSYTLTPE	TRDTIHVKEV	TKNVSDTNYK	KKFVKEKGKS	5940
NYSIMLEPPD	VKHAMDVAKK	QSNMKVAYKK	DAKENLHYTT	VADRPDIKKA	TQAAKQASEV	6000
EYRAKHREK	SHGLSMLGRP	DIEMAKKAAK	LSSQVKYREN	FNKEKGKTPK	YNPKDSQLYK	6060
VMKDANTLAS	EVKYKADLKK	LHKPVTDMKE	SLIMNHVLTN	SHLASSYQYK	KNYEKSNGHY	6120
HTIPDNLEQL	HLKEATELQS	IVKYKEKYEK	ERKPMPLDPE	TPTYITAKES	QQMQSGKEYR	6180
RDYEEISIKGR	NLTGLEVTTPA	LLHVKYATKI	ASEKEYRKDL	EESIRGKGLS	EMEDTPDMLR	6240
AKNATQILNE	KEYKRDLELE	VKGRGLNAMA	NETPDFLRAR	NATDIASQIK	YKQSAEMEKA	6300
NFTSVVDTP	IIHAQQVKNL	SSQKKYKEDA	EKCMSYYETV	LDTPEMQRVR	ENQKNFSLQ	6360
YQYDLKNSKG	KITVVQDTP	ILRVKENQKN	FSSVLYKEDV	SPGTAIGKTP	EMMRVKQTQD	6420
HISVVKYKEV	IGQGTPIPLD	PEVKRVKQTO	KHISVYMYKE	NLGTGIPTV	TPEIERVKRN	6480
QENFSSVLYK	ENLGKGTPTP	ITPEMERVKR	NQENFSSILY	KENLSKGTPL	PVTPEMERVK	6540
RNQENFSSVL	YKENVGKGT	TPVTPEMQV	KRNQENISSV	LYKENLGKAT	PTPFTPEMER	6600
VKRQENFSS	VLKENMRKA	TPTVTPEME	RAKRQENIS	SVLYSDSFRK	QIQGKAAYVL	6660
DTPEMRRVRE	TQRHISTVKY	HEDFEKHKG	FTPVVTDPI	ERVKNMQQDF	SDISYRGIQR	6720
KVVEMEQRKN	DQDQETVTGL	RVWRTNPGSV	FDYDPAEDNI	QSRSLHMINA	QAQRSSREQS	6780
RSASGLSVSG	GEEKSEHSEA	AHLSTYSDDG	VFFSATSTGY	KHARTTELPO	QRSSSVATQQ	6840
TTVSSIPSPH	STAGKIFRAM	YDYMAADADE	VSPKDGDAIV	NVQAIDEGWM	YGTVQRTGRT	6900
GMLPANYVEA	I					6911

SEQ ID NO: 107
FEATURE
source
moltype = AA length = 1948
Location/Qualifiers
1..1948

-continued

mol_type = protein
organism = *Ovis aries*

SEQUENCE: 107

MSSDQEMAIF	GEAAPYLRSK	EKERIEAQNK	PFDAKTSVFV	ADPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQVFP	NPPKFDKIED	MAMMTHLHEP	GVLYNLKERY	AAWMIYTYSG	120
LPCVTVNPYK	WLPVYNAEVV	TAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKKEEP	TSQKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFQKF	IRIHFGTTGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTNP	YDYAFVSQGE	ITVPSIDDOE	ELMATDSAIE	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFKQK	QREEQAEPDG	TEVADKAAYL	QSLNSADLLK	ALCYPRVKVG	NEFVTKGQTV	420
QQVYNAVAGL	AKAVYEKMF	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDFNSLEQLC	480
INFTEKLLQ	PFNHMHFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	PKNKLYEQHL	GKSNNFQKPK	PAKGKAEAHF	SLVHYAGTVD	YNIAGWLDKN	600
KDPLNETVVG	LYQKSAMKTL	AFLFAGGPSA	EVHVSEGGGG	KKGGKKKGSS	FQTVSALFRE	660
NLNKLMNTLR	STHPHFVRCI	IPNETKTPGA	MEHELVLHQL	RCNGVLEGIR	ICRKGFPSPRI	720
LYADFKQRYK	VLNASAIPEG	QFIDSKKASE	KLLGSIDIDH	TQYKFGHTKV	FFKAGLLGIL	780
EEMRDEKLAQ	LITRTQAICR	GFLTRVEFRK	MMERSRESIF	CIQYNVRAFM	NVKHWPWMKL	840
YFKIKPLLKS	AETEKEMANM	KEEFECTKEE	LAKSEAKRKE	LEEKMTVLMQ	EKNDLQLQVQ	900
SEADGLADAE	AECEEEHQR	IQLEAKIKEL	TERAEDEEEI	NAELTAKRKR	LEDECSSELKK	960
DIDDLLETLA	KVEKEKHATE	NKVKNLTEAM	AGLDENIAKL	TKEKKALQEA	HQQTLDDLQA	1020
EEDKVNLTAK	SKTKLEQQVD	DLEGSLEQEK	KLRMDLERAK	RKLEGLKLA	QESTMDIEND	1080
KQQLDEKLLK	KEFEMSNLQS	KIIDEQALGM	QLQKKIKELQ	ARIEELEEEI	EAERASRAKA	1140
EKQSDLSRE	LEEISERLEE	AGGATSAQIE	MNKKREAEFQ	KMRDLLEAT	LQHEATAAAL	1200
RKKHADSVAE	LSEQIDNLQR	VKQKLEEEGS	NLTVOALDII	SNFKIINRYI	YIEGNLEKMC	1260
RTLEDQLSEV	KSEEEHQR	INELSAQKAR	LHTESEGEFS	QLDEKDALVS	QLSRGQAFT	1320
QQIEELKRQL	BEETKAKSAL	AHALQSARHD	CDLLREQYEE	EQAQAEQLR	SMSKANSEVA	1380
QWRTKYETDA	IQTREELAEA	KKKLAQRLQE	AEHVEAVNA	KCASLEKTKQ	RLQNEVEDLM	1440
LDVERSNAAC	AALDKKQRF	DKVLSWKQK	YEETQAELEA	SQKESRSLST	ELFKVKNAVE	1500
ESLDQLETLK	RENKNLQOEI	SDLTEQIAEG	GKHIHELEKV	KKQIEQEKGE	LQAALAEAEA	1560
SLEHEEGKIL	RIQLELNVQK	SEIDRKIAEK	DEEIDQLKRN	HLRVVESMQS	TLDAEIRSRN	1620
DALRIKKKME	GDLNMEIQL	NHANRQATEA	TKNLRNTQGV	LKDTQLHLDD	AIRGQDDLKE	1680
QLAMVERRAN	LMQAEIEELR	ASLEQTERS	RVAEQELLLA	SERVQLLHTQ	NTSLINTKKK	1740
LETDITRVHG	SVVSGSIISK	HLTFVYISVG	CAFQAAMMAE	ELKKEQDTS	HLERMKKNLE	1800
QTVKDLQHR	DEAEQLALKG	GKKQIQKLEA	RVRELENEVE	NEQKRNAEAV	KGLRKHERRV	1860
KELTYQTEED	RKNVLRQLDL	VDKLQSKVKA	YKRQAEAEAE	QANVNLSKFR	KLQHELEAE	1920
ERADIAESQV	NKLRVKSREV	HTKVISEE				1948

SEQ ID NO: 108 moltype = AA length = 607
FEATURE Location/Qualifiers
source 1..607
 mol_type = protein
 organism = *Ovis aries*

SEQUENCE: 108

MKWVTFISLL	LLFSSAYSRG	VFRDRTHKSE	IAHRFNDLGE	ENFQGLVLIA	FSQYLQCCPF	60
DEHVKLVKEL	TEFAKTCVAD	ESHAGCDKSL	HTLPGDELCK	VATLRETYGD	MADCCCKQEP	120
ERNECFNLHK	DDSPDLPLKK	PEPDTLCAEF	KADEKKFWGK	YLYEVARRHP	YFYAPELLYY	180
ANKYNGVFQE	CCQAEDKGAC	LLPKIDAMRE	KVLASSARQR	LRCASIQKFG	ERALKAWSVA	240
RLSQKFPKAD	FTDVTKIIVD	LTKVHKECCH	GDLLCADDR	ADLAKYICDH	QDALSSKLKE	300
CCDKPVLEKS	HCIAEVDKDA	VPENLPLPLA	DFAEDKEVCK	NYQAEKDVFL	GSFLYEYSRR	360
HPEYAVSVLL	RLAKEYEATL	EDCCAKEDPH	ACYATVFDKL	KHLVDEPQNL	IKKNCELFEK	420
HGEYGFQNAL	IVRYTRKAPQ	VSTPTLVEIS	RSLGKVGTKC	CAKPESERMP	CTEDYLSLIL	480
NRLCVLHEKT	PVSEKVTKCC	TESLVNRRPC	FSDLTLDETY	VPKPFDEKFF	TFHADICTLP	540
DTEKQIKQKT	ALVELLKHKP	KATDEQLKTV	MENFVAFVDK	CCAADDKEGC	FVLEGPKLVA	600
STQAALA						607

SEQ ID NO: 109 moltype = AA length = 381
FEATURE Location/Qualifiers
source 1..381
 mol_type = protein
 organism = *Sus scrofa*

SEQUENCE: 109

MPFGNTHNKY	KLNFKAEEY	PDLSKHNNHM	AKALTLEIYK	KLRDKETPSG	FTLDDVIQTG	60
VDNPGHPFIM	TVGCVAGDEE	SYVVFKDLPD	PIIQDRHGGY	KPTDKHKTDL	NHENLKGDD	120
LDPNVVLSRR	VLTRGSIKGY	TLPPHCSRGE	RAVEKLSVE	ALNSLTGEFK	GKYYPKLSMT	180
EQEQQLLIDD	HFLDPKVPSP	LLLASGMARD	WPDARGIWHN	DNKSFLVWVN	EEDHLRVISM	240
EKGGMNKEVF	RRFCVGLQKI	EEIFKKAGHP	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLA	300
HLSKHPKFEE	ILTRLRLQKR	GTGGVDTA	AAVGSVPDVS	NADRLGSSSE	VEQVQLVVDG	360
VEMEKKLEKG	QSIDDMIPAQ	K				381

SEQ ID NO: 110 moltype = AA length = 902
FEATURE Location/Qualifiers
source 1..902
 mol_type = protein
 organism = *Sus scrofa*

SEQUENCE: 110

MMMVMQPEGL	GTREGPFAGG	SGGGGEYMEQ	EEDWDRDLLL	DPAWEKQQRK	TFTAWCNSHL	60
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RKAGTQIENI	EEDFRNGLKL	MLLLEVISGE	RLPRDPKGM	RFHKIANVVK	ALDFIASKGV	120
KLVSIGAEI	VDGNLKMILG	MIWTIILRFA	IQDISVEETS	AKEGLLLWCQ	RKTAPYRNVN	180
VQNFHTSWKD	GLALCALIHR	HRPDLIDYAK	LRKDDPIGNL	NTAFEVAEKY	LDIPKMLDAE	240
DIVNTPKPDE	KAIMTYVSCF	YHAFAGAEQA	ETAANRICKV	LAVNQENEKL	MEEYEKLASE	300
LLEWIRRTVP	WLENRVGEPS	MSAMQRKLED	FRDYRRLHKP	PRVQEKQCLE	INFNTLQTKL	360
RLSHRPAPMP	SEGKLVSDIA	NAWRGLEQVE	KGYEDWLLSE	IRRLQRLQHL	AEKFQQKASL	420
HEAWTRGKED	MLSQRDYESA	SLQEVRLALL	RHEAFESDLA	AHQDRVEHIA	ALAQELNELD	480
YHEASVNSR	CQAICDQWDN	LGTLTQKRRD	ALERMEKLE	TIDQLQLEFA	RRAAPFNWL	540
DGAVEDLQDV	WLVHVSVEETQ	SLVTAHEQFK	ATLPEADRER	GAILGIQGEI	QKICQTYGLR	600
PSSTNPYIAL	SPQDINTKWD	TVRKLVPSRD	QMLQEELTRQ	QVNERLRRQF	AAQANAIGPW	660
IQGKVEEVGR	LAAGMAGSLE	EQMAGLRQQE	QNIINYKSNI	DRLEGDHQLL	QESLVFDNKH	720
TVYSMEHIRV	GWEQLLTSIA	RTINEVENQV	LTRDAKGLSQ	EQLNEFRASF	NHFDRKRNGM	780
MEPDDFRACL	ISMGYDLGEV	EFARIMTMVD	PNAAGVVTQ	AFIDFMTRET	AETDTAEQVV	840
ASFKILAGDK	NYITAEELRR	ELPAEQAEYC	IRRMAYKGA	GAPAGALDYV	AFSSALYGES	900
DL						902

SEQ ID NO: 111 moltype = AA length = 844
 FEATURE Location/Qualifiers
 source 1..844
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 111

MAMSRPLTDQ	EKRKQISVRG	LAGVENVAEL	KKNFNRHLHF	TLVKDRNVAN	ARDYYFALAY	60
TVRDHLVGRW	IRTQQHYEYK	DPKRIYYLSL	EFYMGRTLQN	TMVNLALENA	CDEATYQLGL	120
DMEELEEMEE	DAGLNGGLG	RLAACFLDSM	ATLGLAAYGY	GIRYFEGIFN	QKITGGWQME	180
EADDWLRYGN	AWEKARPEFT	LPVHFYGRVE	HTNQGAKWVD	TQVVLPMPYD	TPVPGYRNNV	240
VNTMRLWSAK	APDNFNLKDF	NVGGYIQAVL	DRNLAENISR	VLYPNDNFFE	GKELRLKQEY	300
FVVAATLQDI	IRRFKSSKFG	CRDPVRTSFD	AFPDKVAIQL	NDTHPSLAIP	ELMRILVDEE	360
RLEWEKAWDV	TVRTCAYTNH	TVLPEALERW	PVHLMETLLP	RHLQIYEIN	QRFLNRVAAA	420
YPGDVDRLLR	MSLVEEGAVK	RINMAHLCIA	GSHAVNGVAR	IHSEILKKT	FKDFEYELPH	480
KFQNKTNGIT	PRRWLVMCNP	GLAEVIAERI	GEEYIADLDQ	LRKLLSYVDD	EAFIRDVAKV	540
KQENKLKPSA	YLREYKVHI	NPNSLFDIQV	KRIHEYKRQL	LNCLHVITLY	NRIKKEPNKF	600
VVPRTVMIGG	KAAPGYHMAK	MIKLITAIG	DVVNHDPVVG	DRLRVIFLEN	YRVSLAEKVI	660
PAADLSEQIS	TAGTEASGTG	NMKFMLNGAL	TIGTMDGANV	EMAEAEAGEEN	FFIFGMRVED	720
VERLDQKGYN	AQEYYDRIPE	LRHVIDQLSS	GFFSPKQDDL	FKDIVNMLMH	HDRFKVFADY	780
EDYIKCQERV	SALYKNPREW	TRMVIRNIAT	SGKFSSDRTI	AQYARETWGV	EPSRQRLPAP	840
DEKI						844

SEQ ID NO: 112 moltype = AA length = 368
 FEATURE Location/Qualifiers
 source 1..368
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 112

MPHQYPALTP	EQKKELSDIA	HRIVAPGKGI	LAADSTGSI	AKRLQSIGTE	NTEENRRFYR	60
QLLLTADDRV	NPCIGGVILF	HETLYQKADD	GRPPQVIKA	KGGVVGKVD	KGVPVLAGTN	120
GETTTQGLDG	LSERCAQYKK	DGADFAKWRC	VLKIGEHTPS	SLAIMENANV	LARYASICQQ	180
VGLQNGIVPI	VEPILPDGD	HDLKRCQYVT	EKVLAAVYKA	LSDHHIYLEG	TLLKPNMVT	240
GHACTQKYSH	BEIAMATVTA	LRRTPPPAVP	GITFLSGGQS	EEAEASINLA	INKCPLLKPW	300
ALTFSYGRAL	QASALKAWGG	KKENLKAAQE	EYVKRALANS	LACQKGKYTPS	GKAGAAASES	360
LFISNHAY						368

SEQ ID NO: 113 moltype = AA length = 191
 FEATURE Location/Qualifiers
 source 1..191
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 113

MAPKKDVKKP	AAAAAPAPAP	APAPAPAPAP	PKEEKIDLSA	IKSFSADQIA	EFKEAFLFLD	60
RTGECKITLS	QVGDVLRALG	TNPTNAEVKK	VLGNPSNEEM	NAKKIEFEQF	LPMLQAISSN	120
KDQGSYEDFV	BGLRVFDKEG	NGTVMGAELR	HVLATLGEKM	KEEEVEALMA	GQEDSNGCIN	180
YEAUVKHIMS	I					191

SEQ ID NO: 114 moltype = AA length = 333
 FEATURE Location/Qualifiers
 source 1..333
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 114

MVKVGNGVFG	RIGRLVTRFA	FNSGKVDIVA	INDPFIDLHY	MVYMFQYDST	HGKFHGTVKA	60
ENGKLVINGK	AITTFQERDP	ANIKWGDAGA	TYVVESTGVF	TMEKAGAH	KGAKRVIIS	120
APSADAPMFV	MGVNHEKYDN	SLKIVSNASC	TTNCLAPLAK	VIHDHFGIVE	GLMTTVHAIT	180
ATQKTVDGSP	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCLRLEK	AKYDDIKKVV	KQASEGPKLG	ILGYTEDQV	SCDFNSDTHS	STPDAGAGIA	300
LNDHFVKLIS	WYDNEFGYSN	RVVDLMVHMA	SKE			333

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SEQ ID NO: 115 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 115

MDAIKKMQM	LKLDKENALD	RAEQAEADKK	AAEDRSKRLE	DELVSQKKL	KATEDELDKY	60
SEAPKDAQEK	LELAEEKATD	AADVASLNR	RIQLVEEELD	RAQERLATAL	QKLEEAEEKAA	120
DESERGMKVI	ESRAQKDEEK	MEIQEIQLKE	AKHIAEADR	KYEEVARKLV	IIESDLERAE	180
ERAEELSEGKC	AEEELKTV	TNNLKSLEAQ	AEKYSQKEDK	YEEBIKVLS	KLKEAETRAE	240
FAERSVTKLE	KSIDDLDEL	YAQKLKYKAI	SEELDHALND	MTSI		284

SEQ ID NO: 116 moltype = AA length = 1220
 FEATURE Location/Qualifiers
 source 1..1220
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 116

MPEAKPATKK	APAGKDAAKP	APKEAPPKGA	PEEPPKEAPP	EDQSPTAEEP	TGIFLKKPDS	60
VSVENGKDAL	IVAKVNGKEL	PGKPEIKWFK	GKWLGLGSKS	GARFSFKESH	DSASNVTLE	120
LRIGKVVLGD	RGDYRLVKA	KDFCDSCAFN	IDVEAPRQDS	AGQSLESFKR	SGEGKPEGAG	180
ELDFSGLLKK	RQVVEEEKK	KKDDDDLGI	PEIWELLKGA	KKSEYEKIAF	QYGITDLRGM	240
LKRLKKAKVE	VKKSAAFTKK	LDPAYQVDQG	NKIKLVVEIS	DPDLPLKWKY	NGQEIKPSSK	300
YVFENVGKKR	ILCINKCTLA	DDAAEVAVK	DEKCFTELFV	REPPVLIVTP	LEDQQVYVGD	360
RVEMTVEVSE	EGAQVMWLKD	GVELTREDSE	KSRYRFKKDG	KRHILIYSDV	SLEDRGRYQV	420
MTNGGQCEAE	LSVEEKQLEV	LQDIADLTVK	ASEQAVFKCE	VSDEKVTGKW	YKNGVEVRPS	480
KRITISHVGR	PHKLVIDDVR	PEDEGDYTFV	PDGYALSLSA	KLNFLEIKVE	YVPKQEPPI	540
HLDCSGKTSE	NAVVVVAGNK	LRLDVSTIGE	PPPVATWMKE	DEVFSGTEGR	VRLEQRADIS	600
SFVIESAERG	DEGRYTIKVT	NPAGEDVASI	FLRVVDVDPD	PEAVRVTTVG	EDWAILVWEP	660
PKYDGGQPV	GYLLERKKKG	SQRWMLNFE	VFTETTYEST	NMIEGILYEM	RVFAVNAIGV	720
SQPSVNTKPF	MPIAPTSEPQ	HLTVEDVTD	TTTLKWRPDP	RIGAGGIDGY	LVEYCVEGSD	780
EWIPANTEPV	ERCGETVKNL	PTGARITFRV	GVGNIAGRSQ	PATLAQPVTI	REIVGEPPRQ	840
SSVLIRASPD	SLVATTLVSP	LRRVPQPYAE	PAPGIVSPEQ	PKIRLPRHLR	QTYIRKVGHE	900
LNLVIPPQKG	PRPQVVWTKG	GAPIDTSRVH	VRTSNFDTVF	FVRQAARSDS	GEYELSVQIE	960
NMKDTATVHI	RVVEKAGPPQ	NVMVKEVWGT	NALVEWQPPK	DDGNTETITG	FVQKADKKT	1020
EWFNVEYHNR	HTSCTVSDLI	MGNEYFRVY	SENICGLSDS	PGVSKNTARI	LKTGITFKLP	1080
EFKEHDFRTP	PKFLTPLPDR	VVVAGYTAAL	NCAVRGHKPK	KVVVMKNQME	IGEDPKFLMT	1140
NHGGVLTINI	RRSPFDSGT	YSCRASTSWG	RPWPSASWKS	ECRSEAPAST	QGSWLLSPDP	1200
DLPEPLSLSR	VITRFLSRTQ					1220

SEQ ID NO: 117 moltype = AA length = 869
 FEATURE Location/Qualifiers
 source 1..869
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 117

TMQHVYTKMS	LVAVPFYQKR	HRHFDQSYRN	IQTRYLLDEY	AAKRASTQS	SSQKSVTQES	60
SSQRESSSR	ASSQTLGGT	TCRLCAKRV	TALEEEEQER	KRRYHSLAA	YGEAKRERYL	120
SELAQLEEDV	HLATHARDK	LDRFALQHVD	THALRHAVGD	RLAWERHSFE	DRISRAPEIL	180
VRLRSHTVWE	RMSVKLCFTV	QGFPTPVVQW	YKDGSLICQA	GEPGKYRIES	KYGVHTLEIN	240
RADFDATATY	SAVATNVHGQ	VSTNAVVVR	RFHGDEEPPH	SVGLPIRLPL	SSVIPYTHFD	300
VQFLEKFGVT	FRREGETLTL	CTLLLVTPDL	KRVQPRAEWY	RDDVLLKESK	WTKMFFGEGQ	360
ASLSFSLHNL	DDEGLYTLRI	VSRGGISDHS	AFLFVRDADP	LVMGAPGAPM	DVHCHDANRD	420
YVIVTWKPPN	TTKESPVLYG	FIDRCEVGT	NWVQCNDAPV	KICKYPVSGL	VEGRSYTFRV	480
RAVNSAGVSR	PSRASEAVAA	LDPLDLKRLQ	GIHWEGDKDI	VIYQDDLEGE	VQIPRPPTNV	540
HASEISRTRY	VLSWDPPTPR	GKEPLMYFIE	KSVVSGSGTWQ	RENAQTAVRS	PRYAVFDLAE	600
GKSYVFRVLS	ANKHGLSDPS	EITSPIQAQD	TIVVPSAPGR	VLASRNTKTS	VVVQWDQPKH	660
GEDLLGYVVD	CCVAGSNVWE	PCNHKPIGYN	RFVVHGLTTG	EQYIFRVKAV	NAVGTSENSQ	720
ESDVIKVVQA	LTVPSHPYGI	TLLNCDGHSM	TLGWKVPKFS	GGSPIIGYVY	DKREGHHKNW	780
HEVNSSPLKE	RILTVEGLTE	GSTYEFKIAA	PTWWGSGCRR	IPVSSSSVRP	GQRLSQVPTS	840
GWVGTLGFFL	VLLSLYRRYL	LGSGEHGD				869

SEQ ID NO: 118 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 118

MAMQKIFARE	ILDSRGNPTV	EVDLHTAKGR	FRAAVPSGAS	TGIYEALRL	DGDKSRYLGK	60
GVLKAVEHIN	KTGLPALLEK	KLSVVDQEKV	DKFMIELDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVPL	YRHADLAGN	PDLLVPVPAF	NVINGGSHAG	NKLAMQEFMI	LPVGASSFRE	180
AMRIGAEVYH	HLKGVIKGKY	GKDATNVGDE	GGFAPNILEN	NEALELLKTA	IQAAGYPDKV	240
VIGMDVAASE	FYRNGKYDLD	FKSPDDPSRH	ITGEKLGLY	KSFINKYPVV	SIEDPFDQDD	300
WKTWTSFLSG	VNIQIVGDDL	TVTNLKRIAQ	AVEKKACNCL	LLKVNQIGSV	TESIQACKLA	360
QSNGWGMVVS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEEALGDK	420
AVFAGRKFERN	PKAK					434

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SEQ ID NO: 119 moltype = AA length = 169
 FEATURE Location/Qualifiers
 source 1..169
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 119
 MAPKKAKRRA AAEAGSSNVFS MFDQTQIQEF KEAFTVIDQN RDGIIDKEDL RDTFAAMGRL 60
 NVKNEELDAM MKEASGPINF TVFLTMFGKEK LKGADPEDVI TGAFKVLDPD GKGTIKKHFL 120
 EELLTTQCDR FSQEEIKNMW AAFPPDVGGN VDYKNICYVI THGDAKDQE 169

SEQ ID NO: 120 moltype = AA length = 1939
 FEATURE Location/Qualifiers
 source 1..1939
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 120
 MSSDQEMAIF GEAPYLRKS EKERIEAQNK PFDAKTSVFV AEPKESFVKG TVQSREGGKV 60
 TVKTEAGATL TVKEDQVFPN NPPKFDKIED MAMMTHLHEP AVLNLKERY AAWMIYTYSG 120
 LFCVTVNPNYK WLPVYNAEVV TAYRGKKRQE APPHIFSISD NAYQFMLTDR ENQSILITGE 180
 SGAGKTVNTK RVIQYFATIA VTGEKKKEEP TSGKMQGTLE DQIISANPLL EAFGNAKTVR 240
 NDNSSRFGEF IRIHFGTTGK LASADIETVL LEKSRVTFQL KAERSYHIFY QIMSNKKPEL 300
 IEMLLITTNP YDYAFVSQGE ITVPSIDDQE ELMATDSAIE ILGFTSDERV SIYKLTGAVM 360
 HYGNLKFQKQ QREEQAEFDG TEVADKAAYL QGLNSADLLK ALCYPRVKVG NEFVTKGQTV 420
 QQVYNAVQAL AKAVYDKMFL WMVTRINQQL DTKQPRQYFI GVLDIAGFEI FDFNSLEQLC 480
 INFTNEKLQQ PFNHHMFVLE QEEYKKEGIE WEFIDFGMDL AACIELIEKP MGIFSILEEE 540
 CMFPKATDTS FKNKLYEQHL GKSNFPQPKP PAKGKVEAHF SLIHYAGTVD YNITGWLDKN 600
 KDPLNETVVG LYQKSSVKTL AFLFTGAAGA DAEAGGKKG GKKKGSSFQT VSALFRENLN 660
 KLMTNLRSTH PHFVRCIIPN ETKTPGAMEH ELVLHQLRCN GVLEGIRICR KGFPSRILYA 720
 DFKQRYKVLN ASAIPEGQFI DSKKASEKLL GSIDIDHTQY KFGHTKVFFK AGLLGLEEM 780
 RDEKLAQLIT RTQARCRGFL ARVEYQKMEV RRESIFCIQY NIRAFMNVKH WPMWKLYFKI 840
 KPLLKSAETE KEMANMKEEF EKTKEKSLAK EAKRKELEEK MVALMQEKND LQLQVQAEAD 900
 SLADAEERCD QLIKTKIQLE AKKEVTERA EDEEINAEEL TAKKRKLEDE CSELKKDIDD 960
 LELTLAKVEK EKHATENKVK NLTEEMAGLD ETIAKLTKEK KALQEAHQOT LDDLQAEEDK 1020
 VNTLTAKAKT LEQQVDDLEG SLEQEKKLRL DLERAKRKE GDLKLAQEST MDIENDKQQL 1080
 DEKLKKKEFE MSNLQSKIED EQALAMQLQK KIKELQARIE ELEEIEAER ASRAKAEKQR 1140
 SDSLRELEEI SERLEEAGGL TSAQIEMNKK REAEFQKMRR DLEEATLQHE ATAATLRKKH 1200
 ADSVAELGEQ IDNLQVRKQK LEKEKSEMKM EIDDLASNME TVSKAKGNLE KMCRTLEDQL 1260
 SELKTKEEEQ QRLINDLTAQ RARLQTESGE YSRQLDEKDT LVSQSLRGKQ APTQQIEELK 1320
 RQLEEEIKAK SALAHVQSS RSRKVAEQEL LDASERVQLL HTQNTSLINT KKKLETDISQ 1380
 TDAIQRTEEL EEAKKKLAQR LQDAEEHVEA VNAKASLEK TKQRLQNEVE DLMIDVERSN 1440
 AACAAALDKKQ RNFDKILAEW KQKYEETHAE LEASQKESRS LSTELFKVKV AYEEESLDQLE 1500
 TLKRENKNIK QEISDLTEQI AEGGKRIHEL EKIKKQVEQE KSEIQAALAE ABASLEHEEG 1560
 KILRIQLELN QVKSEVDRKI AEKDEEIDQL KRNVHVRVVE MQSMLDAEIR SRNDAIRLKK 1620
 KMEGDLNEME IQLNHNHMA AEALNRYRNT QGILKDTQIH LDDALRSQED LKEQLAMVER 1680
 RANLLQAEIE ELRATLEQTE RSRKVAEQEL LDASERVQLL HTQNTSLINT KKKLETDISQ 1740
 IQGEMEDIQ EARNAEKAK KAITDAAMMA EELKKEQDTS AHLERMKKNL EQTVKDLQHR 1800
 LDEAEQLALK GKKQIQKLE ARVRELEGEV ESEQKRVET VKGLRKHERR VKELTYQTEE 1860
 DRKNILRLQD LVDKLQAKVK SYKRQAEAE EQSNVNLSKF RKLQHELEEA EERADIAESQ 1920
 VNKLRVKSRE VHTKIISEE 1939

SEQ ID NO: 121 moltype = AA length = 6665
 FEATURE Location/Qualifiers
 source 1..6665
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 121
 MADDEEYEEV VEYYTEETVY EEVPGETITE VYETTTTRTS GYEQSDTSQP AVAQPMPSKP 60
 VERKKVIRKK VDPKFMPTPY IAHSQKMQDL FSTNKYKKNY EKAKGKPYAI TTDTPELRRI 120
 KKVQDLSEV KYRIDGDAK TICHVDEKAK DIEHAKKVSQ QVSKVLYKQN WEDTKDKYLL 180
 PPDAPELVQA IKNTAMFSKK LYTEDWEADK SLFYYPNDSP ELRRVAQAQK ALSDIAYKKG 240
 HAEQQTQFTS LPDPPDIEFA KKVTVNQVSKQ KYKEDYENKI KGKWSSETPCF EIATARMNAK 300
 NISTRKYQED FENMKDQIYF MQTETPEYKV NKQAGTAASK VKYKEDYEKN KGKADYNVLP 360
 ASENPLLRHL KAAGDALSDK RYKENYEKTK AKSINCYETP KFKLDTVLHN FSSDTKYKDS 420
 YLKNILGHYV GSFEDPYHTH CMKVAEENS DKNYKAEYED RGKGFPPQTI TQEYEAIKKL 480
 DQCKDHTYKV HPDKTKFTQV TDSFVLVQAQ VNSQKLSDLN YKAKHENEKF KCHIPPDTPA 540
 FIQHKVNAYN LSDNVYKHDW EKTAKKFDI KVDAIPLLA KANTKNTSDV MYKKDYEKSK 600
 GKMIGALSIN DDPKMLHSLK TAKNQSDRLY KENYEKTKAK SMNYCETPKY QLDTLKNFS 660
 EAKYKDSYVQ NVLGHYIGSF EDPYQLHCMK VSAQNSDKNY KAEYEEDKGK CYFPQTITQE 720
 YEAIKKLDQC KDHTYKVPD KTKFTAVTDT PVLLQAQLNT KQLSDLNKYA KHEGEKFCN 780
 IPADAPQFTI HRVNAYNLS DNVYKHDWEKT KAKKFDIKVD AIPLLAAKAN TKNTSDVMYK 840
 KDYEKSGKM IGALSINDDP KMLHSLKTAK NQSDREYRKD YEKSKTIYTA PLDMLQVTA 900
 KKSQAIASDV DYKHLHNYS YPPDSINVDL AKKAYALQSD VEYKADYNSW MKGCGWIPFG 960
 SLEMEKAKRA SDILNEKKYR QHPETLKFTS IEDAPIVQS KINQAQRSDI AYKAKGEEII 1020
 HKYSLPADLP FIQAKVNAYN ISESMYKADL KDLSKKGYDL RTDAIPIKAA KAARQAASDV 1080
 QYKKDYEKAK GKMVGFSQSL DDPKLHYHMY VAKIQSDREY KKDYEKTKTK YNTPHDMFNV 1140

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VAAKQAQDVA	SNVNYRHS LH	HYTYLPDAMD	LELSKNMMHI	QSDNAYKEDY	NTWMKGIGWI	1200
PLGSLEVEKV	KKAGDALNEK	KYRQHPDTLK	FTSIVDSPVM	VQAKQNTROV	SDILYKAKGE	1260
DVKHKYTMGP	DLPQFLQAKC	NAYNISDV CY	KRDWHDLIAK	GNNVLGDAIP	ITAAKASRNI	1320
ASDYKYKEAY	EKAKGKHVGF	RS LQDDPKLV	HYMNVAKLQS	DREYKKNYEN	TKTSYHTPGD	1380
MVSITAAKMA	QDVATNVN YK	QPIHHYTYLP	DALSLEHTRN	VNQIQSDN VY	KDEYNHFFKG	1440
IGWIPLGSL E	VEKVYKAGDA	LNERKYRQHP	DTIKFTSVPD	SMGMVLAQHN	TKQLSDLN YK	1500
VEGEKLHKYK	TMDPDVPQFI	QAKVNAINMS	DAH YKADWKK	TLAKGYDLTP	DAIPIVAAKS	1560
SRNIASDFKY	KEAYEKAKGK	QVGFRSLQDD	PKLVHYMNVA	KIQSDREYKK	GYEASKTRYH	1620
TPLDMLSVTA	AKKSQEVATN	TNYKQPFHHY	TLLPDALNVE	HSRNAMQIQS	DNLYKSDFTN	1680
WMKGIGWIPM	ESLEVEKAKK	AGEILSEKKY	RQHPEKLKFT	YAMDTMEQAL	NKSNKLIMDK	1740
RLYTEKW NDK	KTTIHVMPDT	PDILLSRVNO	ITLSDKLYKA	GWEEBKKKG Y	DLRPDAISIK	1800
AARASRD IAS	DYKYQAYEQ	AKGKHIGFRS	LEDDPKLVHF	MQVAKMQSDR	EYKKA YEKSK	1860
TSFHTPVDML	SVVAAKKSQE	VATNANYRNV	IHTYNMLPDA	MSFELAKNMM	QLQSDNQYKA	1920
DYADPMKGIG	WLPLGSLEAE	KNKKAMEIIS	EKKYRQHPDT	LKYSTLMDSM	NMVLAKNNAK	1980
IMNEHLYKQA	WEADTKV KHI	MPDIPQIILA	KANAVNMSDK	LYKLSLEESK	KKGYDLRTDA	2040
IGIQAAKASR	DIASDYKYKH	SYEKERGMV	GFRSLEDDPK	LVHSMQVAKM	QSDREYKKS Y	2100
EKTKTSYHTP	ADMLSVTAAR	DAQANITNTN	YKHLIHKYIL	LPDAMNIQLS	KNMNHIQSDN	2160
EYKQDYNECY	KGLGWSPAGS	LEVEKAKKAT	EYASDQKYRQ	HPSNFQFQKL	NDSDMDVLAK	2220
QNAHTMKNXL	YTAHWNKDKT	KIHVMPDPTD	ILQAKQNQTL	FSQKLYKLGW	EEALKKG YDL	2280
PVDAISVQLA	KVSRDNASDF	KYKQGYRKQL	GHHIGFRNLQ	DDPKLVLSMN	VAKMQSEREY	2340
KKDFEKWKTK	FTSPVDM LGV	VLAKKCQALV	SDVDYKNYLH	QWTC LDPQND	VIQAKRSYEL	2400
QSENL YKSDL	EWLRGIGWSP	LGSLAEKKNK	RASEIISEKK	YRQPPDRNKF	TSIPDAMDIV	2460
LAKTNAKNRS	DRLYREAWDK	DKTQVHIMPD	TPDIILAKAN	LINSSDKLYR	MGYEELKKKG	2520
YDLVPDAIPI	KAASASREIA	SEYKYKEGFR	KQLGHHVGAR	NIEDDPKMMW	SMHVAKIQSD	2580
REYKDKFEKW	KTKFSSPVDM	LGVLVLA KCCQ	TLVSDIDYKN	YLHQWTC LDP	QSDVIHARRA	2640
YDLQSDNLYK	SDLQWLKGIG	WLPSPGSLEDE	KNKRATQILS	DHVYRQHPDK	FKFSSLMDSI	2700
PMVLAKNNAI	TMNHRLYTEA	WDKDKTTVHI	MPDTPPEVLLA	KQNKIN YSEK	LYKLGLGEEAK	2760
RKGYDMRLDA	IPIRAAKASR	DIASEFKYKE	GYRKQLGHHI	GARAIHDDPK	MMWSMHVAKV	2820
QSDREYKKDF	EKWTKFSSP	VDM LGVV LAK	KCQTLVSDID	YKNYLHQWTC	LPDQSDVIHA	2880
RRAYDLQSDN	MYKSDLQWLR	GIGWVPIGSL	DVEKCKRATE	ILSDKIYRQP	PDTFKFTSV P	2940
DSLEQVLAKN	NAIMNVRDAI	TEAWDKDKTQ	VHIMPDTPEI	MLARQNRLNY	SETLYKLANE	3000
EAKKKGYDLR	SDAIPIVAAK	ASRDIVSDYK	YKDG YRKQLG	HHIGARNIED	DPKMMWSMHV	3060
AKIQSDREYK	KDFEKWKTKF	SSPVDM LGLV	QAKKCCQILVS	DVDYKNYLHQ	WTC LDPQNDV	3120
IHARQAYDLQ	SDNIYKSDLQ	WLKGIGWVPI	GSVDVVKCKR	AAEILSDHIY	RQPPDKLKFT	3180
SVPDSLEQVL	AKNNAINMKN	RLYTEAWDKD	KTQIHIMPDT	PEIMLARQNK	INYESLYRQ	3240
AMEEAKKEGY	DLRSDAIPIV	AAKASRD IAS	DYKYKEAYRK	QLGHHIGARA	VHDDPKMMWS	3300
LHVAKMQSDR	EYKDKFEKYK	TRFSSPVDM L	GIVLAKKCQT	LVSDVDYKHP	LHEWTC LDPQ	3360
NDIIHARKAY	DLQSDNLYKS	DLEWMKGIGW	IPLGSVEVVK	AKRAGEILSD	NIYRQRPDTL	3420
KFTSVTDSPE	LVLAKNNALN	MNKRLYTEAW	DNDKKTIVHM	PDTPEIMLAK	LNRINYS DKL	3480
YRLALEESKK	BGCDLRVDAI	PIQAAKASRE	IASDYKYKEA	YRKQLGHHIG	ARNIEDDPKM	3540
MWSIHAAKIQ	SDREYKKDFE	KWKTRFSSPV	DMLGLVQAKK	CQILVSDIDY	KHPLHEWICL	3600
PDQNDI IQAR	KAYDLQSDAI	YKADLEWLRG	IGWVPLGSVE	VEKVKRAEIG	LSERKYRQPA	3660
DQLKFTCIDT	TPEIVLAKNN	ALTMSKHLYT	EAWDTDKTSI	HVMPDTP EIL	LAKSNSANIS	3720
QKLYTKGWDE	SKMKDYDLKA	DAISIKSAKA	SRDIASDYKY	KEAYEKQKGH	HIGAQSVEDD	3780
PKIMCAIHAG	KIQSEREYKK	DFQKWKTKFT	SPVDM LGILL	AKKCQTLVSD	IDYRNYLHHW	3840
TCLPDQNDII	QARKAYELQS	DALYKADLEW	LRGIGWMPQG	SPEVLRVKNA	QEILRDSVYR	3900
TPVVKLKYTS	IVDTPEV VLA	KSNAENISIP	KYREVWDKDK	TSVHIMPDTP	EINLARANAL	3960
NVSNKIYREB	WDEMKMSCDV	RLDAIPIQAA	KASREIASDY	KYKLDHEKQK	GHYVGTLTAR	4020
DDNKIRWALI	AGKIDFEEYK	RLHWA KWKSK	FQSPADMLS I	THSKNCQALV	SDIDYRNYLH	4080
QWTC LDPQHD	VIQARKAYDL	QSDAIYKADL	EWL RGIGWMP	SDSVSVNHAK	HAADIFSEKK	4140
YRTKIETLNF	TPVDDRDYDV	TAKQS GEILN	DIKYRKDWND	TKSKYT LTET	PQLHAAQEAA	4200
RILDQYLYKE	SWEKQDFEYK	ILPPDAVPFV	HAHHS SDVQS	ELKYKADHVK	QKGHYVGVP T	4260
MRDDPKLVWF	EHAGQIQNDR	LYKEHYHKT K	AKINIPPM DV	SVVAAKEGQI	LVSDIDYRNY	4320
LHQWTC HPDQ	NDVIQARKAY	DLQSDN VYKA	DLEWLRGIGW	IPLDSVDHVR	VTRNQEMVNO	4380
IKYKKDALAN	YPNFTSVVDP	PAVV LAKINS	VNQSDVKYKE	TFNKL LKGKY	IFSPDTPYIT	4440
HSKDMGKLYS	TILYKGAWEG	TKAYGYTLDE	RYIPIVGAKH	ADFNSELKY	KETYEKQKGH	4500
YLAGKEISEF	PNV VHC LDFQ	KMRSVLN YRR	HYEDTKANVH	IPNDMMNHVL	AKRCQYILSD	4560
LEYRHYFHQW	TSLPEEPNVI	RARNAQEILS	DNVYKDDLNW	LKGIGCYVWD	TPQILHAKKS	4620
YDLQSQIILYT	ADGKENLKNY	NMVTDTPLYV	TALQSGINAS	EVKYKENYHQ	IKDKYTTVLE	4680
TVDYDRTRNL	KDLYSSN VYK	EAWDKVKATS	YILPSNTVSL	THAKNQKHLA	SHIKYREEYE	4740
KFKALYTLPK	SVDDDPNTAR	CLRVGKENID	RLYRSVYEKN	KMKIHMVPDM	VENMTAKDSQ	4800
KKVSEIDYRL	HLHEWICH PD	LQVNSHVRKV	TDQISDIVYK	DDLNLWLGIG	CYVWDTP EIL	4860
HAKHAYDLRN	DIKQAHVKK	TRNNYKL VTD	TPLYVQAVKS	GKQLSDAVYH	YDVVHVSVRGK	4920
VAPTTKTVDL	DRALHAYKLQ	SENLYRKDGL	NSLPTGYRLP	GDTPHFKHNK	DIRYMSSYFK	4980
YKEVYEHFKA	NGYTLGPKDV	PFVHVRRVNN	VTSERLYRQL	YHKLKDKIHT	TPDTP EIRQV	5040
KKTQEA VSEL	IYKSDFFKMQ	GHMISMPYTP	HVLHCRYVGD	ITSDIKYKED	LQILRGLGCF	5100
LYDTPDMVRS	RHLRLKWSNY	LYTDNARKMR	DKYKVVLDT P	EYRKVQELKT	HLSELVYRAA	5160
GKKQKSIPTS	VPDTPDLLRA	KRGQKLQSQY	LYVELATKER	PHHAGNQTT	ALKHARDVKD	5220
MVSEKKYKVQ	YEKMKDKYTP	VPDTPILIRA	KKAYWNASDL	RYKETFQKTK	GKYHTVKDAL	5280
DIVYHRKVTD	HISKIKYKEN	YMSQLGIWRS	IPDRPEHFFH	RAVTDAVSDV	KYKEDLTWLK	5340
GIGCYAYDTP	DKTLAEKNKT	LYSKYKYKEV	FERTKSDFKY	EADCPINRHF	KFATQLMNEK	5400
KYRADYEQRK	DKYQLVVD EP	RHLLAKIAGD	QISQIKYRK N	YEKTKDKFTS	VVDTP EHLRT	5460
TKV NKQISDI	LYKLEYNKAK	PRGYTTIHDT	PMLLHV RKVK	DEVSDLYKE	VYQRTKSNCT	5520
IEPDVAHIKA	AKDAYKVNTN	LDYKKQYEAN	KAHWRWTPDR	PDFIQAAKSS	LQQSDFEYKL	5580
DREFLKGCKL	SVTTDDKDMVL	ALRNSVIESD	LKYKEKHVKE	RGTCHAVPDT	PQILLAKTYS	5640
NLVSENKYKR	HVKKHLAQGS	YTTLPETRTD	VHVKEVTKHV	SDTNYKKKFV	KERGKSNYSI	5700

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MLEPPEVKHA	MEVAKQSDV	AYKKDAKENL	HYTTVADRPD	IKKATQAAKQ	ASEVEYRAKH	5760
RKEGSHGLSM	LGRPDIEMAK	KAATLSSQVK	YRENFNKEKG	KTPKYNPKES	QLYKIMKDAN	5820
NLASEVKYKA	DLKKLHKPVT	DMKESLIMNH	VLNTSHLASS	YQYKKNYEKS	KGHYHTIPDN	5880
LEQLHLKEAT	ELQSIVKYKE	KYKEKRGKPM	LDFETPTYIT	AKESQQMQSG	KEYRKDYEES	5940
IKGRNLTGLE	VTPALLHVKY	ATKIASEKEY	RKDLEESIRG	KGLSEMEDTP	DMLRAKNATQ	6000
ILNEKEYKRD	LELEVKGKRL	NAMANETPDF	MRARNATDIA	SQIKYKQLAE	MEKANFTSVV	6060
DTPEIIHAQQ	VKNLSSQKKY	KEDAECMSY	YETVLDTPEM	QRVRENQKNF	SLQYQCDLK	6120
NSKGKITVVQ	DTPEILRVKE	NQKNFSSVLY	KEDVSPGTAI	GKTPEMMRVK	QTQDHISSVK	6180
YKETIGQGTP	IPDLPEVKRV	KETQKHISV	MYKENLGTGI	PTPVTPEIER	VKRNNQENFSS	6240
VLYKENLGKG	TPTPITPEME	RVKRNNQENF	SVLYKENMGK	GTPLPVTPEM	ERVKHNNQENI	6300
SSILYKENVG	KGTPTPTVPE	MQRVKRNNQEN	ISSVIYKENL	GKATPTPTPT	EMERVKRNNQE	6360
NFSSVLYKEN	MRKATPTPTV	PEMERAKRNNQ	ENISSVLYSD	SFRKQIQGKA	AYVLDTPEMR	6420
RVRETQRHIS	TVKYHEDFEK	HKGCFTPVVT	DPITERVKKK	TQDFSDINYR	GIQRKVVEME	6480
QKRNDQDQET	ITGLRVWRTN	PGSVFDYDPA	EDNIQSRSLH	MLNAQAQRRS	REQSRASASAL	6540
SISGGEKESE	HSEARLSTY	SDGGVFPSAT	SAAYKHAKTT	ELPQQRSSSV	ATQQTTVSSI	6600
PSHPSTAGKI	FRAMYDYMAA	DADEVSPFRDG	DAIVNVQAID	BGWMYGTVQR	TGRTGMLPAN	6660
YVEAI						6665

SEQ ID NO: 122 moltype = AA length = 1937
 FEATURE Location/Qualifiers
 source 1..1937
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 122

MSSDQEMAI	GEAAPYLRS	EKERIEAQN	PFDAKTSV	AEPKESFVK	TVQSREGGKV	60
TVKTEAGATL	TVKEDQVFP	NPPKFDKIE	MAMMTHLEP	AVLYNLKERY	AAWMIYTYSG	120
LPCVTVNPYK	WLPVYNAEV	TAYRGKKRQ	APPHIFSIS	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKKEEP	TPGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFGKF	IRIHFGTTGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITNTP	YDYAFVSGQ	ITVPSIDDQ	ELMATDSAE	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFQKQ	QREEQAEFDG	TEVADKAAYL	QGLNSADLLK	ALCYPRVKVG	NEFVTKGQTV	420
QQVYNAVVAL	AKAVYDKMFL	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNFSLQELC	480
INFTEKLLQ	FFNHHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	PKNKLYEQHL	GKSNNFQKPK	PAKGKAEAHF	SLIHVAGTVD	YNTGWLDKN	600
KDPLNETVVG	LYQKSSVKTL	AFLEAERQSS	EEGGTKKGGK	KKGSSPQTVS	ALPRENLNKL	660
MTNLRSTHPI	FVRCIIPNET	KTPGAMEHEL	VLHQLRCNGV	LEGIRICRKG	FPSRILYADF	720
KQRYKVLNAS	AIPEGQFIDS	KKASEKLLGS	IDIDHTQYKF	GHTKVFFKAG	LLGTLEEMRD	780
EKLAQLITRT	QAMCRGFLMR	VEPRKMMERR	ESIFCIQYNI	RAFMNVKHWP	WMKLYFKIKP	840
LLKSAETEKE	MANMKEEFEK	TKEDLAKSEA	KRKELEEKMV	ALMQEKNDLQ	LQVQAEADGL	900
ADAERECQDL	IKTKIQLEAK	IKEVTERAED	EEEINAELTA	KKRKLEDECS	ELKKDIDDL	960
LTLAKVEKEK	HATENKVKNL	TEEMAGLDEN	IAKLTKEKKA	LQEAHQQTLD	DLQAEEDKVN	1020
TLTKAKTKLE	QQVDDLESSL	EQEKKLRMDL	ERAKRKLEGD	LKLAQESMTD	IENDKQQLDE	1080
KLKKKEFEMS	NLQSKI EDEQ	ALAMQLQKKI	KELQARTEEL	EEETEAERAS	RAKAEQORS	1140
LSRELEEISE	RLEEAGGATS	AQIEMNKKRE	AEFQKMRRDL	BEATLQHEAT	AAALRKKHAD	1200
SVAELGEQID	NLQRVQKLE	KEKSELKMEI	DDLASNMTV	SKAKGNLEKM	CRTELDQLSE	1260
VKTKEEEHQR	LINELSAQKA	RLQTESGEFS	RQLDEKEALV	SQLSRGKQAF	TQQIEELKRQ	1320
LEEETAKAKS	LAHAVQSSRH	DCDLLREQYE	EEQEAKAELQ	RAMSKANSEV	AQWRTKYETD	1380
ATQRTHEEL	AKKLAQLRLQ	DABEHVEAVN	AKCASLEKTK	QRLQNEVEDL	MLDVERSNA	1440
CAALDKKQRN	FDKILAWEQ	KYEETQAELE	ASQKESRSL	TELPKVKVNA	EESLDQLETL	1500
KRENKNLQOE	ISDLTEQIAE	GGKHIHELEK	VKKQIEQEK	ELQAALEBAE	ASLEHEEGKI	1560
LRIQLELNVQ	KSEIDRKIAE	KDEEIDQMKR	NHIRVVESMQ	STLDAEIRSR	NDALRIKKKM	1620
EGDLNEMETI	LNHANRQATE	AIRNLRNTQG	VLKDTQLHLD	DAIRGQDDLK	EQLAMVERR	1680
NLMQAEIEEL	RASLEQTERS	RRVAEQELLD	ASERVQLLHT	QNTSLINTKK	KLETDISQIQ	1740
GEMEDIVQEA	RNAEEKAKKA	ITDAAMMAEE	LKKEQDTSAH	LERMKKNMEQ	TVKDLQHRLD	1800
EAEQLALKGG	KKQIQKLEAR	VRELENEVEN	EQKRNVAVK	GLRKHERRVK	ELTYQTEEDR	1860
KNVLRQLDLV	DKLQSKVKAY	KRQAEAEAEQ	SNVNLSKFRK	LQHELEAEAE	RADIAESQVN	1920
KLRVKSREHV	TKVISEE					1937

SEQ ID NO: 123 moltype = AA length = 607
 FEATURE Location/Qualifiers
 source 1..607
 mol_type = protein
 organism = Sus scrofa

SEQUENCE: 123

MKWVTFISLL	FLFSSAYSRG	VFRRDYKSE	IAHRFKDLGE	QYFKGLVLIA	FSQHLQQCPY	60
EEHVKLVRV	TEFAKTCVAD	ESAENCDSKI	HTLFGDKLCA	IPSLREHYGD	LADCCKEEEP	120
ERNECFLOKH	NDNPDIPLK	PDPVALCADF	QEDEQKFWGK	YLYEIAARRHP	YFYAPELLYY	180
AIYKDVFS	CCQAADKAC	LLPKIEHLRE	KVLTSAAKQR	LKCAISIQFG	ERAFKAWSLA	240
RLSQRFPKAD	PTEISKIVTD	LAKVHKECCH	GDLLECADDR	ADLAKYICEN	QDTISTKLKE	300
CCDKPLLEKS	HCIAEAKRDE	LPADLNPLEH	DFVEDKEVCK	NYKEAKHVFL	GTFLYEYSRR	360
HPDYSVSLLL	RIAKIYEATL	EDCCAKEDPP	ACYATVDFDK	QPLVDEPKNL	IKQNCLEFEK	420
LGEYGFQNAL	IVRYTKKVPQ	VSTPTLVEVA	RKGLGVGSRC	CKRPEEERLS	CAEDYLSLVL	480
NRLCVLHEKT	PVSEKVKCC	TESLVNRRPC	FSALTPDETY	KPKEFVEGTF	TFHADLCTLP	540
EDEKQIKKQT	ALVELLKHKP	HATEEQRLTV	LGNFAAFVQK	CCAAPDHEAC	FAVEGPKFVI	600
EIRGILA						607

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SEQ ID NO: 124 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Bison bison

SEQUENCE: 124

MPFGNTHNKH	KLNFKAEEY	PDLSKHNNHM	AKALTLEIYK	KLRDKETPSG	FTLDDVIQTG	60
VDNPGHPPIM	TVGCVAGDEE	SYTVFKDLFD	PIIQDRHGGF	KPTDKHKTDL	NHENLKGDD	120
LDPNVVLSSR	VRTGRSIKGY	ALPPHCSRGE	RRAVEKLSVE	ALNSLTGEFK	GKYYPLKSMT	180
EQEQQLLIDD	HFLRDKVPSP	LLLASGMARD	WPDARGIWHN	DNKSFLVWVN	EEDHLRVISM	240
EKGGMNKEVF	RRFCVGLQKI	EEIFKKAGHP	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLA	300
HLSKHPKFEE	ILTRLRLQKR	GTGGVDTAAV	GSVPDVSAD	RLGSSEVEQV	QLVVDGVKLM	360
VEMEKKLEKG	QSIDDMIPAQ	K				381

SEQ ID NO: 125 moltype = AA length = 894
 FEATURE Location/Qualifiers
 source 1..894
 mol_type = protein
 organism = Bison bison

SEQUENCE: 125

MNQIEPGVQY	NYVYEDDEYM	IQEEEWDRDL	LLDPAWEKQQ	RKTFTAWCNS	HLRKAGTQIE	60
NIEEDFRNGL	KLMLLLEVIS	GERLPKPD RG	KMRPHKIANV	NKALDYIASK	GVKLVSIGAE	120
EIVDGNVKMT	LGMWITIILR	FAIQDISVEE	TSAKEGLLLW	CQRKTAPYRN	VNIQNPHTSW	180
DKGLGLCALI	HRHRPDLIDY	SKLNKDDPIG	NINLAMEIAE	KHLDI PKMLD	AEDIVNTPKP	240
DERAMTYVVS	CFYHAFAGAE	QAETAANRIC	KVLAVNQENE	RLMEEYERLA	SELLEWIRRT	300
IPWLENRTPE	KTMQAMQKKL	EDPRDYRRKH	KPPKVQEKQ	LEINFNTLQT	KLRISNRPAF	360
MPSEGKMSVD	IAGAWQRLAQ	AEKGYEELW	NEIRRLERVE	HLAEKFRQKA	STHETWAYGK	420
EQILLQKDYE	SSTLTEVRAL	LRKHEAFESD	LAHQDRVEQ	IAAIAQELNE	LDYHDAVNVN	480
DRCQKICDQW	DRLGTLTQKR	REALERTEKL	LETIDQLHLE	FAKRAAPFNN	WMEGAMEDLQ	540
DMFIVHSIEE	IQSLITAHEQ	FKATLPEADG	ERQSILAIQN	EVEKVIQSYS	IRISSNPYS	600
TVTVEIRISK	WDKVKQLVPI	RDQSLQEELA	QHANERLRR	QFAAQANAIG	PWQNKMEEI	660
ARSSIQITGA	LEDQMNQLKQ	YEHNIIINYKN	NIDKLEGDHQ	LIQEALVPDN	KHTNYTMEHI	720
RVGWELLTIT	IARTINEVET	QILTRDAKGI	TQEQMNEFRA	SFNHFDRRKN	GLMDHEDFRA	780
CLISMGYDLG	BAEFARIMTL	VDPNGQGTVT	FQSFDIFMTR	ETADTDTAEQ	VIASFRI LAS	840
DKPYILAEEL	RRELPPDQAQ	YCIKRM PAYS	GPGSVPGALD	YTAFFSSALYG	ESDL	894

SEQ ID NO: 126 moltype = AA length = 842
 FEATURE Location/Qualifiers
 source 1..842
 mol_type = protein
 organism = Bison bison

SEQUENCE: 126

MSRPLTDQEK	RKQISVRGLA	GVENVTELKK	NFNRHLHFTL	VKDRNVATPR	DYFALAYTV	60
RDHLVGRWIR	TQOHYYEKDP	KRIYYLSLEF	YIGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELEIEEDA	GLNGGGLGRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	ISGGWQMEEA	180
DDWLRYGNFW	EKARPEFTLP	VHFYGRVEHT	SQGAKWVDTQ	VVLAMPYDTP	VPGYRNNVVN	240
TMRLWSAKAP	NDPNLKDFNV	GGYIQAVLDR	NLAENISRVL	YPNDNPFEGK	ELRLKQYFV	300
VAATLQDIIR	RFKSSKFGCL	DPVRTNFDAP	PDKVAIQLND	THPSLAIPEL	MRILVDQERL	360
EWKAWEVTV	KTCAYTNHTV	LPEALERWPV	HLIETLLPRH	LQIIEINQR	FLNRVAAAP	420
GDVDRLRMS	LVEEGAVKRI	NMAHL CIAGS	HAVNGVARIH	SEILKKTIFK	DFYELEPHKF	480
QNKTINGITPR	RWLVMCNPGL	AEIIAERIGE	EYIADLDQLR	KLLSYVDDSE	FIRDVAKVKQ	540
ENKLFESAYL	EKEYKVHINP	NSLFDIQVKR	IHEYKRQLLN	CLHVITLYNR	IKKEPNKFFV	600
PRTVMIGGKA	APGYHMAKMI	IKLITAIGDV	VNHDPVVGDR	LRVIFLENYR	VSLAEKVIPA	660
ADLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	ABEAGEENFF	IFGMRVEDVE	720
RLDQKGYNQA	EYYDRIPELR	HIIDQLSSGF	FSPKQPDLEK	DIVNMLMHHD	RFKVVFADYEE	780
YVKCQERVSA	LYKNPREWTR	MVIRNIATSG	KFSSDRTIAQ	YAREIWGVPE	TRQRMPAPDE	840
KI						842

SEQ ID NO: 127 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = Bison bison

SEQUENCE: 127

MPHQYPALTP	EQKKELCDIA	HRIVAPGKGI	LAADSTGSI	AKRLQSIGTE	NTEENRRFYR	60
QLLLTADDRV	NPCGGVILF	HETLYQKADD	GRFPQVQIKA	KGGVVGIKVD	KGVVPLAGTN	120
GETTTQGLDG	LSERCAQYKK	DGADFAKWR	VLKIGEHPTPS	SLAIMENANV	LARYASICQQ	180
NGIVPIVEPE	ILPDGDHDLK	RCQYVTEKVL	AAVYKALSDH	HIYLEGTLK	PNMVT PGHAC	240
TQKYSHEEIA	MATVTALRRT	VPPAVPGITF	LSGGQSEEEA	SINLNAINKC	PLLKPWALTF	300
SYGRALQASA	LKAWGGKKEN	LKAAQEYVK	RALANSLACQ	GKYTPSGKAG	AAASESLFIS	360
NHAY						364

SEQ ID NO: 128 moltype = AA length = 192
 FEATURE Location/Qualifiers
 source 1..192
 mol_type = protein

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                                organism = Bison bison
SEQUENCE: 128
MAPKKDVKNP AAAAAAPAPAP APAPAPAPAP PKEEKIDLSA IKIEFSKQQQ DEFKEAFLLF 60
DRTGECKITL SQVGDVLRAL GTNPTNAEVK KVLGNPSNEE MNAKKIEFEQ FLPMLQAISN 120
NKDQGTIEDF VEGLRVFDKE GNGTVMGAEI RHVLATLGEK MKEEEVEALM AGQEDSNGCI 180
NYEAFVKHIM SN 192

SEQ ID NO: 129      moltype = AA length = 333
FEATURE            Location/Qualifiers
source             1..333
                   mol_type = protein
                   organism = Bison bison
VARIANT            158
                   note = Any amino acid
VARIANT            290
                   note = Any amino acid

SEQUENCE: 129
MVKVGVNGFG HIGRLVTRAA FNSGKVDVIA INDPFIDLHY IVYMFQYDIT HGKPHGTVKA 60
ENGKLIINGK AVTIFQERDP ASIKWGDAGA EYVVESTGVF TTMEKAGAHL KGGAKSVIIS 120
APSADAPMFV LGMNHEKYNL NLKIVSNASC TTNCLAPXAK VIHDHFGIVE GLMTTFHAIT 180
VTQKTVDGGS GKLWHDGGA AQNIIIPASTG AAKAVGKVIP ELNRKLTGMA FCVPTPNVSV 240
MDLTYHLEKP AKYDEIKKV KQTSEGLEP ILGYTQDQVV SCDPNDSYX STPDAGAGIA 300
LNDHFPVKLIS WYDNEFGYSN RVVDLMVHMA SKE 333

SEQ ID NO: 130      moltype = AA length = 284
FEATURE            Location/Qualifiers
source             1..284
                   mol_type = protein
                   organism = Bison bison
VARIANT            30
                   note = Any amino acid

SEQUENCE: 130
MDAIKKKMQM LKLDKENALD RAEQAEADKX VPPHGHSQLD DELVSLQKKL KATEDELDKY 60
SEALKDAQEK LELAEEKATD AEADVASLNR RIQLVEEELD RAQERLATAL QKLEEAKEAA 120
DESERGMKVI ESRAQKDEEK MEIQEIQLKE AKHIAEDADR KYEEVARKLV IIESDLERAE 180
ERAELSEGKC AELEELKTV TNNLKSLEAQ AEKYSQKEDK YEEIKVLSL KLKEAETRAE 240
FAERSVTKLE KSIDLEDEL YAQKLKYKAI SEELDHALND MTSM 284

SEQ ID NO: 131      moltype = AA length = 1183
FEATURE            Location/Qualifiers
source             1..1183
                   mol_type = protein
                   organism = Bison bison

SEQUENCE: 131
MPEAKPAKK APAGKDAAK PAPKEATPQE APAAPTGEI DGAPAGRAEP GGSSAASRVV 60
PPRPLLRVSV PTAPGLTALP YSQPCRAPQR FQGGGLTVPL PPHSPLLPGK DTIVIVAKLNG 120
KELPAKPAVK WFKGKWLLEL SKSGARFSFK ESHDAASNVE TIELHITKVV LGDRGDYRIE 180
VAKDFCDSC AFNIDVEAPR QDSAGQSLES FKRSGEAKSD TAGELDFSGL LKKRQVVEE 240
KKKKKDDDDL GIPPEIWELL KGAKKSEYER IAFQYGITDL RGMLKRLKKA KVEVKKSAAF 300
TKKLDPAYQV DRGNKIKLVV EISDPDLPLK WFKNGQEIKP SSKYVFENVG KKRILTINKC 360
TLADDAAYEV VVKDEKCFTE LFKVEPPVLI VTPLEDQQVF VGDRVEMAVE VSEGAQVMW 420
LKDGVELTRE DSKFARYRLK KDGKRHLIIY SEVTMEDKGH YQVMTNGGQC EAEELIVEEKQ 480
LEVLDIADL TVKASEQAVF KCEVSDEKVT GKWKYNGVEV RPSKRITISH TGRFHKLVLD 540
DVRPEDEG DYTFVPDGYALS LSAKLKFLFI KVEYVPKQEP PKIHLDCSGQ TSENAIVVVA 600
GNKLKMDVSI TGEPRPVATW MKEDEVFTGT EGRVHIEQRG DISSFVIESA ERGDEGRYTI 660
KVTNPVGEDV ASILLKVVDV PDPPEAVRVT SVGEDWAVLV WEPKPYDGGR PVTGYLLERK 720
KKGSRWMKL NFEVFTETTY ESTNMIEGIL YEMRVFAVNV IGVSQPSMNT KPFMPIAPTS 780
EPRHLTVEDV TDTTTLTKWR PPDKIGAGGI DGYLIEYCVE GSDEWIPANT EPVERCGFTV 840
KNLPTGAKIT PRVVGVIAG RSQPATLAQP VTIREIVEQP KIRLPRHLRQ TYVRKVGEHL 900
NLVIPFGQPK RPQVWTKGG APIDTSHVHV RTSDFDTVFF VRQAARSDSG EYELTVQIEN 960
MKDTATVCIR VVEKAGPPQN VMVKEVWGTN ALVEWQPPK DGNSEITGYF VQKADKKTME 1020
WFTVYERNRH TSCTVSDILM GNEYFRVYS ENVCGLSDLP GVSKNATARIV KTGMTLKLPE 1080
YKEHDFRTPP KFLTPLDRV VVAGYAAALN CAVRGHPKPK VVWMKNKMEI REDPKFLMTN 1140
QQGVLTNLIR RPSPFDSGTY SCRARNELGE ALAECKLEVR VPQ 1183

SEQ ID NO: 132      moltype = AA length = 1430
FEATURE            Location/Qualifiers
source             1..1430
                   mol_type = protein
                   organism = Bison bison

SEQUENCE: 132
MSLVAVPFYQ KRHKHFDQSY RNIQTRYLLD EYSAKKKRAS AQSSLQRSVT RKSSSQRESS 60
RAALGESTLG ETTLGETSLG DTTLGRTTCR LCAKRMSSAL EEEAQERKRR YQSMVAAYGE 120
ILVRLRSHTV WERMSVKLCF TVQGFPTPVV QWYKDGSLIC QAGEPGKYRI DSKYGVHTLE 180
INRADFDDTA TYSAVATNVH GQVSTNAAVV VRRFRGDEEP FHSVGLPIGL PLSSVIPYTH 240
FDVQFIEKFG VTFRREGETL TLKCTLLVTP DLKRVQPRAE WYRDDVLLKE SKWTKMFPGE 300

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GQASLSFSHL	NKDDEGLYTL	RIVSRGGITD	HSAFLFVRDA	DPLVTGAPGA	PMDVHCHDAN	360
RDYVIVTWKP	PNTTKESPI	GYFIDRCEVG	TDNWIQCND	PVKICKYPVT	GLFEGRSYIF	420
RMRAVNSAGI	SRPSRVSEAV	AALDPVDLRR	LQAVHLEGEK	DIVIYQEELE	GEVQIPGPPT	480
NVHASETSRT	YVVLSDWDPPE	PRGKEPLMYF	IEKSMVGS	WQRVNAQTAV	RSPRYAVFDL	540
AEGKPYVFRV	LSANKHGLSD	PSEITAPIQA	QDTIVVPSAP	GRVLASRNTR	TSVVVQWDRP	600
KHEEDLLGY	VDCSVAGSNV	GEPCHHKPIG	YNRFVHGLT	TGEQYIFRVK	AVNAVGTSEN	660
SQESDVIKVG	AALTVPSHPY	GITLLNCDGH	SMILGWKVPK	FSGGSPILGY	YVDKREAHK	720
NWHEVNSSPL	KERILTV EGL	TEGSLYEFKI	AAANMAGIGQ	PSDPSSELFKC	EAWTMPPEGP	780
AYDLTFCEVR	DTSLVLLWKP	PVYSGSSPVS	GYFVDYKEED	SGEWLTVHET	ATPHRYLKVC	840
DLHQGKTYVF	RVRVNASGV	GRPSDTSEPV	LVEARPGTKE	VSAGVDEEGN	VYLSFDCPEV	900
TDASHFTWCK	SYEDIADDDR	FKVETAGDHS	KLYFKNLDKE	DLGTVSVSVS	DTDGVSSSFV	960
LDEEELERLK	ALSNEIKNPT	IPLKSELAYE	IFDKGQVRFW	LQAEHLSPDS	SYRFIINDRE	1020
VADSETHRIK	CDKSTGIIEM	VMDRFTIDNE	GTYYTVQIQDG	KAKSQSSLVL	IGDAFKAVLK	1080
EAEFQRKEFL	RKQGPFAEY	LHWVDTEECE	VRLVCKVANT	KKETVFKWLK	DDVLYETEK	1140
PDLEKGVCEL	LIPKLSKIDH	GEYKATLKDD	RQGDVSTLEI	AGKVIYEDMIL	AMSRVCGASA	1200
SPLKILCTPE	GIRLQCFMKY	FTEEMKVNWY	HKDAKITSSE	HMRIGGSEQM	AWLQICEPTE	1260
KDKGKYTFEI	LDDKKNHORS	LDLSGQAFDE	AFAEFQQLKA	AAFAEKNRGK	VIGGLPDVVT	1320
IMEGKTLNLT	CTVFGNPDPE	VVWFKNDDKI	ELSDHFSVKV	BQAKYVSMTI	KGVTSEDSGK	1380
YSIHVNKNYG	GEKIDVTVSV	YKHGEEIPDV	VLPQQAKPKL	LPAAAPSPA		1430

SEQ ID NO: 133 moltype = AA length = 440
 FEATURE Location/Qualifiers
 source 1..440
 mol_type = protein
 organism = Bison bison

SEQUENCE: 133

MRTLAMAMQ	KIFAREILDS	RGNPTVEVDL	HTAKGRFRAA	VPSGASTGIY	EALELRDGDK	60
SRYLKGVVLK	AVEHINKTLG	PALLEKKLSV	VDQEKVDKFM	IELDGTENKS	KFGANAILGV	120
SLAVCKAGAA	EKGVPYRHI	ADLAGNPELI	LPVPAPNVIN	GGSHAGNKLA	MQEFMILPVG	180
ASSFREAMI	GAEVYHLKG	VIAKAYGKDA	TNVGDEGGFA	PNILENNEAL	ELLKTAIQAA	240
GYPDKVVIOM	DVAASEFYRN	GKYDLDFKSP	DDPARHISGE	KLGLYKYNFI	KNYPVVSIED	300
PPDQDDWATW	TSPLSGVNIQ	IVGDDLTVTN	PKRIAQAVEK	KACNCLLLKV	NQIGSVTESI	360
QACKLAQSNQ	WGVVMVSHRS	ETEDTFIADL	VVGLCTGQIK	TGAPCRSERL	AKYNQLMRIE	420
EALGDKAVFA	GRKFRNP					440

SEQ ID NO: 134 moltype = AA length = 170
 FEATURE Location/Qualifiers
 source 1..170
 mol_type = protein
 organism = Bison bison

SEQUENCE: 134

MAPKKAKRRA	AAEGGSSSVF	SMFDQTQIQE	FKEAFTVIDQ	NRDGIIDKED	LRTDTFAAMGR	60
LNVKNEELDA	MMKEASGPI	FTVFLNMFG	KLKGADPEDV	ITGAFKVLDP	EGKGTIKKQF	120
LEELLTTQCD	RFSHEEIKNM	WAAFPDPVGG	NVDYKNICYV	ITHGDAKDQE		170

SEQ ID NO: 135 moltype = AA length = 1938
 FEATURE Location/Qualifiers
 source 1..1938
 mol_type = protein
 organism = Bison bison

SEQUENCE: 135

MSSDQEMAIF	GEAAPYLRSK	EKERIEAQNK	PFDAKTSVFV	ADPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQVFP	NPPKFDKIED	MAMMTHLHEP	AVLYNLKERY	AAWMIYTSYG	120
LPCVTVNPKY	WLPVYNAEVV	TAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKKEEP	TSGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFKGF	IRIHFGTTGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNNKKPEL	300
IEMLLTITNP	YDYAYVSQGE	ITVPSIDDQE	ELMATDSIAE	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFQKQ	QREBQAEPDG	TEVADKAAYL	QGLNSADLLK	ALCYPRVKVG	NEFVTKGQTV	420
EQVYNAVVAL	AKAVYDKMFL	WMVARINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNLSLEQLC	480
INFNTNEKLQ	FFNHHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDMS	FKNKLYEQHL	GKSNNFQKPK	PAKGKAEAHF	SLIHYAGTVD	YNTTGWLDKN	600
KDPLNETVVG	LYQKSSVKTL	ALLFSGPASG	EAEKGPKKGG	KKKGSSPQTV	SALFRENLNK	660
LMTNLRSTHP	HVRCIIPNE	TKTPGAMEHE	LVLHLQRCNG	VLEGIRICRK	GFPSRILYAD	720
FKQRYKVLNA	SAIPEGQFID	KKASEKLLA	SIDVDHTQYK	FGHTKVFFKA	GLGLLEEMR	780
DEKLAQLITR	TQARCRGFAL	RVEYQKMVER	RESIFCIQYN	VRAFNMVKNH	PWMKLYFKIK	840
PLLKSAETEK	EMANNKEEFE	KTKEELAKSE	AKRKELEEK	VTLTQEKNDL	QLQVQSEADA	900
LADAERECDO	LIKTKIQLEA	KIKEVTERAE	DEEEINAEIT	AKKRKLEDEC	SELKKDIDDL	960
EITLAKVEKE	KHATENKVK	LTEEMAGLDE	TIAKLTKEKK	ALQEAHQOTL	DDLQAEEDKV	1020
NTLTAKTKL	EQQVDDLEGS	LEQEKKLKMD	LERAKRKLE	DLKLAQESTM	DIENDKQQLD	1080
EKLKKKEFEM	SNLQSKIEDE	QALAMQLQKK	IKELQARIEE	LEEEIEAERA	SRAKAEKQRS	1140
DLRLEEEIS	LRLEEAGGAT	SAQIEMNNKR	EAEFQKMRD	LEEATLQHEA	TAAALRKHA	1200
DSVAELGEQI	DNLQRVKQKL	EKEKSEMKME	IDDLASNMET	VSKAKGNLEK	MCRALDQLS	1260
ELKTKEDEQQ	RLINDLTQOR	ARLQTESGEF	SRQLDEKDAL	VSQLSRGKQA	FTQQIEELKR	1320
QLEEEIKAKS	ALAHALQSAR	HDCDLLREQY	EEEQEGKAE	GRAMSKANSE	VAQWRTKYET	1380
DAIQRTTEEL	EAKKLAQRL	QDAEEHVEAV	NAKASLEKT	KQRLQNEVED	LMIDVERTNA	1440
ACAALDKKQR	NFDKILSEWK	QKYETHAE	EASQKESRSL	STELFKIKNA	YEESLDQLET	1500

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LKRENKQLQQ	EISDLTEQIA	EGGKRIHELE	KVKKQVEQEK	SEIQAALEEA	EASLEHEEGK	1560
ILRIQLELNQ	VKSEIDRKIA	EKDEEIDQLK	RNHIRVVESM	QSTLDAEIRS	RNDAIRLKKK	1620
MEGDLNEMEI	QLNHANRMAA	EALKNYRNTQ	AILKDTQIHL	DDALRGQEDL	KEQLAMVERR	1680
ANLLQAEIEE	LRATLEQTER	SRKIAEQELL	DASERVQLLH	TQNTSLINTK	KKLETDTIQI	1740
QGEMEDIIEQ	ARNAEKAKK	AITDAAMMAE	ELKKEQDTS	HLERMKKNLE	QTVKDLQHRL	1800
DEAEQLALKG	GKKQIQKLEA	RVRELEGEVE	SEQKRNVEAV	KGLRKHERRV	KELTYQTEED	1860
RKNILRLQDL	VDKLQAKVKS	YKRQAEAEAE	QSNVNLSKFR	KLQHELEBAE	ERADIAESQV	1920
NKLRVKSREV	HTKIISEE					1938

SEQ ID NO: 136 moltype = AA length = 6909
 FEATURE Location/Qualifiers
 source 1..6909
 mol_type = protein
 organism = Bison bison

SEQUENCE: 136

MADDEEYEEV	VEYYTEETVY	KEVPGETITG	FYETTTTTRS	DYEQSETSRP	ALAQPVPEKP	60
VERKRVIRKK	VDPSPKFTPY	IAHSQKMQL	FSTNKYKENY	EKAKGKPYAI	TTDTPELRRI	120
KKVQDQLSEV	KYRVGDGDAK	TICHVDEKAK	DIEHAKKVSQ	QVSKVLQKQN	WEDTKDKYLL	180
PPDAPELVQA	IKNTAMFSKK	LYTEDWEADK	TMFYPYNDSP	ELRRVAQAQK	ALSDIAYKKG	240
LAEQQTQFTS	LPDPPDIEFA	KKVTNQVSKQ	KYKEDYENKV	KGKWEETPCF	EIATARMNSN	300
NISTKKYQED	FEHMKDQIYF	MQTETPEYKV	NKQAGVAASK	VKYKQDYEKN	KGKADYNVLP	360
ASENPLLRQL	KAAGDALSDK	LYKENYEKTK	AKSINYCETP	KFKLDTVLHN	FSSDTKYKDS	420
YLKDILGHYV	GSYEDPYHTH	CMRVSAQNSD	KNYKAEYEED	RGKGFFPQTI	TQEYEAIKKL	480
DQCKDHTYKV	HPDKTKFTQV	TDSPVLMQAA	VNSKQLSDLN	YKAKHENEKF	KCHIPPDTPA	540
FIQHKLNAYN	LSDNIYKHDW	EKTKAKKFDI	KVDAIPLLA	KANTKIASDV	MYKKDYEKSK	600
GKMIGALSIN	DDPKMLHSLK	TAKNQSDRLY	KENYEKTKAK	SMNYCETPKY	QLDTLLKNFS	660
EAKYKDSYVQ	NVLGHYIGSF	EDPYQVHCLK	ISAQNSDKNY	KAEEEDKGGK	CYFPQTITQE	720
YEAIKKLDQC	KDHTYKVHPD	KTKFTAVTDT	PVLLQAQLNT	QQLSDLNKYA	KHEGEKFKCH	780
IPADAPQFIQ	HRVNAYNLS	NIYKHDWEKT	KAKKFDIKVD	AIPLLAAKAN	TKIASDVMYK	840
KDYEKSNGKM	IGALSINDDP	KMLHSLKTAK	NQSDHEYRKD	YEKSKTIYTA	PLDMLPLTHA	900
KKSQAIAASV	DYKHLHNNYS	YPPDSVNVDL	AKKAYALQSD	VEYKADYNSW	MKGCGWMPFG	960
SLEMEKAKRA	SDILNEKKYR	QHPDTLKFTS	IEDAPIIVQS	KINQAQRSDV	AYKAKGEEVI	1020
HKYSLPADLP	QFIQAKVNAY	NISENLYKAD	LKDLKSKGYD	LRIDAIPIKA	AKAARQAASD	1080
VQYKKLDYKA	KGKVMGFQSL	QDDPKLVHYM	NVAKIQSDRE	YKAYEKTCT	RHNTPHDMVN	1140
IVAAKKAQDV	ASNVMYKHS	HHYTYLPDAM	DLELSKNMMH	IRSDNVYKED	YNNWMKGIGW	1200
IPIGSLEVKE	VKKAGDALNE	KKYRQHPDPL	KFTSIVDSPV	MVQAKQNTQQ	VSDILYKAKG	1260
EDVKHKYTMS	PDLPQFLQAK	CNAYNLSKVC	YKRDWHDLIA	KGTNVLDGAI	PITAAKASRN	1320
IASDYKYKEA	YEKAKGKQVG	FRSLQDDPKL	VHYMNVAKLQ	SDREYKKNYE	NTKTSYHTPG	1380
DMVSIATAAM	AQDVATNVNY	KQPIHHYTYL	PDALSLEHIR	NVNQIQSDNV	YKDEYNHFFK	1440
GMGWIPIGSL	EVEKQSFVAT	ALNEKKYRQH	PDTIKFTSVP	DSMGMVLAQH	NTKQLSDLNY	1500
KVEGEKVKKH	YTMDDPVPQF	IQAKVNAYNM	SDSHYKADWK	KTAKGYDLR	PDAIPIVAAK	1560
SSRNIASDFK	YKEAYEKTGK	KQGFGRSLQD	DPKLHVFMMV	AKMQSDREYK	KGYEARKTKY	1620
HTPLDMLSVT	AAKSKQFVAT	NTNYKQPFHH	YTLPLDALNV	EHSRNAMQIQ	SDNLYKSDFT	1680
NWMKGIGWLP	LESLEVEKAK	KAGEILSEKK	YRQHPEKLF	TYAMDTMEQA	LNKSNNKLMD	1740
KRLYTEKWNK	DKTTIHVMPD	TPDILLSRVN	QITMSDKLYK	AGWEEEEKKG	YDLRPAIS	1800
KAARASRDIA	SDYKYKQAYE	QAKGRQIGFR	SLEDDPKLVH	FMQVAKMQSD	REYKAYEKS	1860
KTSFHTPVDM	LSVVAAKKSQ	EVATNANYRN	VIHTYNMLPD	AMSLELAKNM	MQIQSDNQYK	1920
ADYADFMKGI	GWPLGSLSEA	EKNKKAMEII	SEKKYRQHPD	TLKYSTLMDS	MMNVLAKNNA	1980
KIMNEHLYKQ	AWEADKTKVH	IMPDIQIIL	AKANAINISD	KLYKLSLEEA	KKKGYDLRTD	2040
AIPIKAAKAS	RDIASDYKYK	HSYEKERGKM	VGFRSLEDDP	KLHVSMQVAK	MQSDREYKKN	2100
YEKTKTSYHT	PADMLSVTAA	KDAQANITNT	NYKHLIHKYI	LLPDAMNIQL	SKNMNRIQSD	2160
NEYKQDYNSV	KYGLGWSPAG	SLEVEKAKKA	TEYASDQKYR	QHPNSNFQTK	LNSMDMVLVA	2220
KQNAHTMNKY	LYTIDWNKDK	TKIHVMPDTP	DILQAKQNTQ	MYSQKLYKLG	WEEALKKGYD	2280
LPVDAISVQA	AKASRDIVSD	PKYKQGYRKQ	IGHHIGFRSL	QDDPKLVLSM	NVAKMQSERE	2340
YKQDFEKWKT	KFSSPVDMLG	VVLAKKQCAL	VSDVDYKNYL	HGWTCLPDQN	DVIHAKKAYD	2400
LQSENLYKSD	LEWLKIGIWS	PLGSLEAEKN	KRASEIIESE	KYRQPPDRNK	FTSIPDAMDI	2460
VLAKTNAKNR	SDILYREAWD	KDKTQIHIMP	DTPTDILAKA	NLINTSDKFY	RMGYEELRKK	2520
GYDLVPDAIP	IKAAKASREV	ASEYKYKEGF	RMQLGHHIGA	RNIKDDPKMM	WSMHVAKIQS	2580
DREYKDFEKF	WTKFSSPVD	MLGVVLAKKC	QTLVSDIDYK	NYLHQWTCPL	DQSDVIHARR	2640
AYDLQSDNLY	KSDLQWLRGI	GWLPSPGSLD	EKNKRASQIL	SDHVYRQHPD	QFKFSSLMDS	2700
IPMVLAKNNA	ITMNHRLYTE	AWDKDKTTVH	IMPDTPEVLL	AKQNKINYS	KLYKLGLDEA	2760
KRKGYDMRID	AIPIRAAKAS	RDIASEFKYK	EGYRRQLGHH	IGARAIHDDP	KMMWSMHVAK	2820
IQSDREYKSD	FEKWKTFSS	PVDMLGVVLA	KKCQTLVSDI	DYKNYLHQWT	CLPDQSDVIH	2880
ARRAYDLQSD	NVYKSDLQWL	RIGYGVPIGS	LDVEKSKRAS	EILSDKLYRQ	PPDKFKFTSV	2940
TDSLEQLLAK	NNAINMKNRL	YTEAWDKDKT	QVHIMPDTPE	ITLARTNKVN	YSENLYKLALH	3000
EEAKKKGVDL	RSDAIPVAA	KASRDIIISD	KYKQGYRKQL	GHHIGARDIK	DDPKMMWSMH	3060
VAKIQSDREY	KKDFEKWTK	FSSPVDMLGV	VLAKKCQTLV	SDIDYKNYLH	QWTCLPDQSD	3120
VIHARRAYDL	QSDNVYKSDL	QWLRGIGWVP	IESVDVVKCK	RAAEILSDNL	YRQPPDTLKF	3180
TSVPDSLEQV	LAKNNAINNM	KRLYTEAWDK	DKTQIHIMP	TPEIMLARQN	KLNYSETLYK	3240
LANEEAKKKG	YDLRSDAIP	VAAKASRDII	SDYKYKGYR	KQLGHHIGAR	DIKDDPKMMW	3300
SMHVAKIQSD	REYKDFEKVH	KTKFSSPVDM	LGVLAKKCCQ	TLVSDIDYRN	YLHQWTCPLD	3360
QSDVIHARQA	YDLQSDNIYK	SDLQWLRGIG	WVPIGSVDV	KCKRAAEILS	DNLRYRQRPDT	3420
LKFTSVPSDL	BQVLAKNNAI	NMKNRLYTEA	WDKDKTQIHM	MPDTPETLA	RQNKINYSN	3480
LYRQAMEEAK	KEGYDLRSDA	IPIVAAKASR	EIASDYKYKE	AYRKQLGHHI	GARAIHDDPK	3540
MMWSVHVAKM	QSDREYKDF	EKYKTRFSSP	VDMLGIVLAK	KCQTLVSDVD	YKPLHEWTC	3600
LPDQNDIIHA	RKAYDLQSDN	LYKSDLEWLK	GIGWVPIGSV	EVLKAKRAGE	ILSDNIYRQR	3660

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PDTLKFTSVT	DSPEQVLAKN	NAINNMNRLY	TEAWNDKKT	IHVMPDTPET	MLAKLNRINY	3720
SDKLYKLAL	ESRKEGYDLR	LDAIPIQAAK	ASREIASDYK	YKEGYRKQLG	HHIGARNIKD	3780
DPKMMWSIHA	GKIQSDLEYK	KDFEKWKTKF	SSPVDMMGLV	QAKKCQILVS	DIDYKNLLHE	3840
WTCLPDQNDI	IQARKAYDLQ	SDAIYKADLE	WLRGIGWVPI	DSVGVEHAKR	AGEILSERKY	3900
RQPAVQLKFT	SITDTPETVL	AKNNALNVSK	HLYTEAWDAD	KTSIHVMPDT	PEILLAKSNS	3960
ANISHKLYTK	GWDESKMKDY	DLRADAIK	SAKASRDIA	DYKYKEAYEK	HKGHHIGARS	4020
VEDDPRIMCA	MNAGRIQSER	EYKKEPQKWK	TKFSSPVDML	GILLAKKCQT	LVSDIDYRNY	4080
LHHWTCPLDQ	NDIIQARKAY	DLQSDAIYKA	DLEWLRGIGW	MPEGSPEVLR	VKNAQHIFRD	4140
SVYRTPVVKL	KYTSIVDTPE	VVLAKSNAEN	ISIPKYREVW	DKDKTSIHIM	PDTPEINLAR	4200
TNALNVSANKL	YREGWDEMKN	SCDVRDLAIP	IQAAKASREI	ASDYKYKLDH	EKQKGHYVGT	4260
RTARDNKKIR	WALIAGKIQN	EREYRLHWAK	WKSQFQSPPD	MLSIEHSKNS	QTLVSDIDYR	4320
HYLHQWTCMP	DQNDVIQARK	AYDLQSDAIY	KADLEWLRGI	GWMPADSVSV	NHAKHMADIF	4380
NEKKYRTKIE	TLSTPTVDDR	VDYVTAHSG	EILNDIKYRK	DWNDTKSXYT	LTETPQLHTA	4440
QEAARILDQY	LYKESWEKQK	ATGYILPPDA	VPFVHAHSG	DVQSELKYKA	EHVKQKGHYV	4500
GVPTMRDDPK	LVWFEHAGQI	QNDRLYKENY	HKTAKIHIP	PDMVSVLAAK	EGQALASDID	4560
YRNYLHQWIC	HPDQNDVIQA	RKAYDLQSDN	IYKADLEWLR	GIGWIPLDSV	DHVRVTRNQE	4620
MMNQIKYKDK	ALANYPNFTS	VVDPPETVLA	KINSVNSQSDV	KYKETFNKL	KGKYLFPDPT	4680
PYITHSKDME	KLYSTILYKR	AWDGTKAYGY	TLDERYIPIV	GAKHADPVNS	ELKYKETYEK	4740
LKGHYLAGKE	ISEFPNVVVC	LDFQKMRSL	NYRRHYEDTK	ANVHIPQDMM	NHVLAKRCQY	4800
ILSDLEYRHY	PHQWTSLPPE	PNVVRVRHAQ	EILSDNVYKD	DLNWLKGIGC	YVWDTQPILH	4860
AKKSYDLQSQ	ILYTAAGKEN	LKNYNLVTDT	PLYVTALQSG	INASEVKYKE	NYHQIKDKYT	4920
TVLETPVDYDR	IKNLKDLFSS	NLYKEAWDKV	KATSYILPPN	TVSLTHAKNQ	KYMASHIKYR	4980
EEYEKFKALY	TLPRSVEDDP	NTARCLRVGK	FNIDRLYRSV	YEKNMKIHI	VPDMVEMVTA	5040
KDSQKQVSEI	DYRLHLHEWI	CHPDQLQVNSH	VRKVTDAQSD	IVYKDDLTWL	KGIGCVVWDT	5100
PEILHAKHAY	DLRNDIKYKA	HVLKTRNNYK	LVTDTPVVYQ	AVKSGKQLSD	AVYHYDYVHS	5160
IRGRVAPTTK	TVDDLDRALHA	YKLQSENLYR	KAGLHALPTG	YRLPVDTPHF	KHTKDTRYMS	5220
SYFKYKEVYE	HMKAYGYTLG	PNDVPFVNV	RVNNITSERL	YRQLYHKLKD	KIHTTPDTP	5280
IRQVKKTQEA	VSELYTKSDF	FKMQGHMISL	PYTPQVLHCR	YVGDITSDIK	YKEDLQVLRG	5340
MGCFLYDTPD	MVRSRHLRKL	WSHYLYTDKA	RKMRDKYKVV	LDTPEYRKVQ	ELKTHLSELV	5400
YRAAGRKQKS	IFTSVPTDTP	LTRAKRGQKL	QSQYLYVELA	TKERPHHAG	NQTTALKKHAR	5460
HVKDMVSENK	YKIQYEKMKD	KYTPVPDTP	LIRAKRAYWN	ASDLRYKET	QKTGKYHTV	5520
KDALDIVYHR	TVTDHVSNIK	YKENYMSQLG	IWRSDPRPE	HFHHRVTD	VSDVKYKQDL	5580
TLWLKIGCYA	YDTPDFTLAE	KNKTLYSKYK	YKEVFERTKS	NFKYVADCP	NRHFKFATQL	5640
MNEKKYRADY	BQRDKYHLV	VDEPRHLLAK	IAGDQISQIK	YRKNYEETKD	KFTSIVDTPE	5700
HLRTTKVKNQ	ISDILYKLEY	NKAKPRGYTT	IHDTPMLLHV	RKVKDEVSDL	KYKEVYQRTK	5760
SNCTIEPDPAV	HIKAAKDAYK	VNTNLDYKKK	YEATKAYWKW	TPDRPDPIQA	AKSTLQQSDF	5820
EYKLDREYLYK	GCKLSVTDDK	DMVLAKNSI	IESDLKYKEK	HVKERGSCHA	VPDTPQILLA	5880
KTVSSLVSEN	KYKSYYVKKHL	AQGSYTTLPE	TRDTIHVKEV	TKNVSDTNYK	KKPVKEKGKS	5940
NYSIMLEPPD	VKHAMDVAKK	QSNVAYKKDA	KENLHYTTVA	DRPDIKKATQ	AAQQAQSEVEY	6000
RAKHRKEGSH	GLSMLGRPDI	EMAKKAAKLS	SQVQYRENFN	KEKGKTPKYN	PKDSQLYKVM	6060
KDANTLASEV	KYKADLKKLH	KPVTDMKESL	IMNHVLTNSH	LASSYQYKKN	YEKSKGHYHT	6120
IPDNLEQLHL	KEATELQSI	KYKEKYEKER	GKPMDFETP	TYITAKESQ	MQSGKEKRYD	6180
YEESIKGRNL	TGLEVTYPALL	HVKYATKIAS	EKEYRKDLLE	SIRGKGLSEM	EDTPDMLRAK	6240
NATQILNEKE	YKRDLELEV	GRGLNAMANE	TPDFLRARNA	TDIASQIKYK	QSAEMEKANF	6300
TSVVDTPETI	HAQQVKNLSS	QKKYKEDAEK	CMSYYETVLD	TPEMQVRREN	QKNFSLQYQ	6360
YDLKNSKGKI	TKVQDTPETI	RVKENQKNFS	SVLYKEDVSP	GTAIGKTPEM	MRVKQTQDHI	6420
SSVKYKEVIG	QGTPIPDLP	VKRVKQTQKH	ISSVMYKENL	GTGIPTPVTP	EIERVKRNQE	6480
NFSSVLYKEN	LGKGTPTPT	PEMERVKRNQ	ENFSSVLYKE	NMGKGTPLPV	TPEMERIKHN	6540
QENISSVLYK	QENISSVLYK	QENISSVLYK	QENISSVLYK	QENISSVLYK	QENISSVLYK	6600
RQENFSSVL	YKENMRKATP	TPVTPEMERA	KRNQENISSV	LYSDSFRKQI	QGKAAYVLD	6660
PEMRVRRETQ	RHISTVYKHE	DFEKHKGCF	PVVTDPITER	VKKNQDQFSD	ICYRGIQRKV	6720
VEMEQKRNQ	DQETITGLRV	WRTPNPGSVF	YDPAEDNIQS	RSLHMINAQ	QRRSRQSR	6780
ASGLSISGGE	EKSEHSEAAH	LSTYSGGGVF	FSAASTGYKH	ARTTELPQQR	SSSVATQQT	6840
VSSIPSHPT	AGKIFRAMYD	YMAADADEVS	FKDGDIVNV	QAIDEGWYMG	TVQRTGRTGM	6900
LPANYVEAI						6909

SEQ ID NO: 137 moltype = AA length = 1938
 FEATURE Location/Qualifiers
 source 1..1938
 mol_type = protein
 organism = Bison bison

SEQUENCE: 137

MSDQEMAI	GEAAPYLRSK	EKERIEAQN	PFDAKTSVFV	VDPKESFVKA	TVQSREGGKV	60
TAKTEAGATV	TVKEDQIFMP	NPPKYDKIED	MAMMTHLHEP	AVLYNLKERY	AAMMIITYSG	120
LFCVTVNPKY	WLPVNAEVV	YAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIIQYFATIA	VTGEKKKEEP	TSQKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFKGF	IRIHFGTTGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTPN	YDYAFVSGQE	ITVPSIDQDE	ELIATDSAD	ILGFTSDERV	SIYKLTGAVM	360
HYGNLKFQKQ	QREEQAEPDG	TEVADKAAYL	QSLNSADLLK	ALCYPRVKVG	NEFVTKGQTV	420
QQVYNVAGAL	AKAVYEMKFL	WMVTRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNFSLQELC	480
INFNTNEKLQ	FFNHHMVFLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSLILEE	540
CMFPKATDTS	FKNKLYEQHL	GKSNFPQKPK	PAKGKAEAHF	SLVHYAGTVD	YNIAGWLDKN	600
KDPLNETVVG	LYQKSAMKTL	AFLFAGGPSA	EEGGGKKGK	KKKGSSFTQV	SALFRENLNK	660
LMTNLRSTHP	HFVRCIIPNE	TKTPGAMEHE	LVLHQLRCNG	VLEGIRICRK	GFPRIILYAD	720
FKQRYKVLNA	SAIPEGQFID	SKKASEKLLG	SIDIDHTQYK	FGHTKVFFKA	GLLGILEEMR	780
DEKLAQLITR	TQAICRGFLM	RVEFRKMMER	RESIFCIQYN	VRAFMMNVKH	PWMKLYFKIK	840

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PLLSAETEK	EMANMKEEFE	KTKEELAKSE	AKRKELEEKM	VTLMQEKNDL	QLQVQSEADG	900
LADAERCDQ	LIKTKIQLEA	KIKELTERAE	DEEEINAELT	AKKRKLEDEC	SELKKDIDDL	960
ELTLAKVEKE	KHATENKVKN	LTEEMAGLDE	NIAKLTKEKK	ALQEAHQOTL	DDLQAEDKV	1020
NTLTSSKTKL	EQQVDDLEGS	LEQEKLRMD	LERAKRKLEB	DLKLAQESTM	DIENDKQQLD	1080
EKLKKKEFEM	SNLQSKIEDE	QALAMQLQKK	IKELQARIEE	LEEEIEAERA	SRAKAEKQRS	1140
DLSRLEEEIS	ERLEEAGGAT	SAQIEMNKKR	EAEFQKMRRD	LEEATLQHEA	TAAALRKHA	1200
DSVALGEGQI	DNLQRVKQKL	EKEKSEMME	IDDLASNMT	VSKAKGNLEK	MCRTELEDQLS	1260
EVKSKEEEHQ	RLINELSAQK	ARLHAESGEF	SRQLEEKDAL	VSQLSRGKQA	FTQQIEELKR	1320
QLEETKAKS	ALAHALQSAR	HDCDLLREQY	EEEQEGKAE	QRALSKANSE	VAQWRTKYET	1380
DAIQRTEBLE	EAKKKLAQRL	QDAEEHVEAV	NAKCASLEKT	KQRLQNEVED	LMLDVERSNA	1440
ACAALDKKQR	NFDKVLSEWK	QKYEETQAE	EASQKESRSL	STELFKVKNA	CEESLDQLET	1500
LKRENKNLQ	EISDLTEQIA	EGGKHIHELE	KVKKQIEQEK	GELQAALAEA	EASLEHEEGK	1560
ILRIQLELNQ	VKSEIDRKIA	EKDEEIDQLK	RNHLRVMEST	QSTLDAEIRS	RNDALRIKKK	1620
MEGDLNEMEI	QLNHANRQAT	EAIKNLRNTQ	GVLKDTQLHL	DDAIRGQDDL	KEQLAMVERR	1680
ANLMQAEIEE	LRASLEQTER	SRRAEQELL	DASERVQLLH	TQNTSLINTK	KKLETDTIQI	1740
QGMEDILQE	ARNAEKAKK	AITDAAMMAE	ELKKEQDTS	HLERMKNME	QTVKDLQHRL	1800
DEAEQLALKG	GKKQIQKLEA	RVRELENEVE	NEQKRNAEAV	KGLRKHERRV	KELTYQTEEY	1860
RKNVLRQLDL	VDKLQSKVKA	YKRAEEAE	QSNVNLAKFR	KLQHELEAE	ERADIAESQV	1920
NKLRVKSREV	HTKVISSE					1938

SEQ ID NO: 138 moltype = AA length = 607
 FEATURE Location/Qualifiers
 source 1..607
 mol_type = protein
 organism = Bison bison

SEQUENCE: 138

MKWVTFISLL	LLFSSAYSRG	VFRDRTHKSE	IAHRFKDLGE	EHFKGLVLIA	FSQYLQCCPF	60
DEHVKLVNEL	TEFAKTCVAD	ESHAGCEKSL	HTLFGDELCK	VASLRETYGN	MADCCCKQEP	120
ERNECFLSHK	DDSPDLPLK	PDPNTLCDEF	KADEKKFWGK	YLYEIAARRHP	YFYAPELLYY	180
ANKYNGVFQE	CCQAEKGCAC	LLPKIETMRE	KVLTSSARQR	LRCASIQKFG	ERALKAWSVA	240
RLSQKFPKAE	PVEVTKLVTD	LTKVHKECCH	GDLLECADDR	ADLAKYICDN	QDTISSKLKE	300
CCDKPLLEKS	HCIAEVEKDA	VPENLPPLTA	DFAEDKDVC	NYQEAQDAFL	GSFLYEYSRR	360
HPEYAVSVLL	RLAKEYEATL	EECCAKDDPH	ACYSTVFDKL	KHLVDEPNQL	IKQNCQDFEK	420
LGEYGFQNAL	IVRYTRKVPQ	VSTPTLVEVS	RLGKVGTRC	CTKPESEMRP	CTEDYLSLIL	480
NRLCVLHEKT	PVSEKVTKCC	TESLVNRRPC	FSALTPDETY	VPKAFDEKLF	TFHADICTLP	540
DTEKQIKKQT	ALVELLKHKP	KATEEQKLT	MENFVAFVDK	CCAADDKEAC	FAVEGPKLVV	600
STQTALA						607

SEQ ID NO: 139 moltype = AA length = 667
 FEATURE Location/Qualifiers
 source 1..667
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 139

MLDAEDIVNT	PKPDEKAIMT	YVSCFYHAF	GAEQAETAAN	RICKVLAVNQ	ENEKLMEEYE	60
KLASELLEWI	RRTVPWLENR	VGEPSMSAMQ	RKLEDFRDYR	RLHKPPRVQE	KCQLEINFNT	120
LQTKLRLSHR	PAPMPSEGKL	VSDIAKAWRG	LEQVEKGYED	WLLSEIRRLQ	RLQHLAEKFQ	180
QKASLHEAWT	RGKEEMLSQ	DYETASLQEV	RALLRRHEAF	ESDLAAHQDR	VEHIAALAE	240
LNELDYHEAA	SVNSRCAIC	DQWDNLGTLT	QKRRDALERM	EKLEETIDQL	QLEFARRAAP	300
FNNWLDGAVE	DLQDVWLVS	VEETQSLVTA	HDQFKATLPE	ADRERGAILG	IQGEIQKICQ	360
TYGLRPSSTN	PYIALTPQDI	NTKWDTVRKL	VPSRDQMLQE	ELTRQOVNER	LRRQFAAQAN	420
ATGPWIQGVK	EEVGRLAAGM	AGSLEEOMAG	LRQQEQNIIN	YKSNIDRLEG	DHQLLQESLV	480
FDNKHTVYSM	EHIRVGWEQL	LTSIARTINE	VENQVLTRDA	KGLSQEQQLNE	FRASPNHFDR	540
KRNGMMEPDD	FRACLISMGY	DLGEVEFARI	MTMVDPNAA	GVTFQAFIDF	MTRETAETDT	600
AEQVVASPKI	LAGDKNYITA	EELRRELPAE	QAEYCIRMA	PYKAGAGAPAG	ALDYVAFSSA	660
LYGESDL						667

SEQ ID NO: 140 moltype = AA length = 792
 FEATURE Location/Qualifiers
 source 1..792
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 140

MSRPLTDQEK	RKQISVRGLA	GVENVTELKK	NFNRLHFTL	VKDRNVATPR	DYFALAYTV	60
RDHLVGRWIR	TQQHYEYKDP	KRIYYLSLEF	YMGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELEIEEEDA	GLGNGGLGRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	ISGGWQMEEA	180
DDWLRYGPNW	EKARPEFTLP	VHFYGRVEHT	SQGAQWVDTO	VVLAMPYDTP	VPGYRNNVNV	240
TMRLWSAKAP	NDNFNLKDFNV	GGYIQAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQEFV	300
VAATLQDIIR	RFKSSKFGCL	DPVAIQNLDT	HPSLAPELM	RILVDQERLE	WEKAEVTVK	360
TCAYTNHTVL	PEALERWPFVH	LIETLLPRHL	QIIYEINQRF	LNRVAAAPFG	DVDRLRRMSL	420
VEEGAVKRIN	MAHLCIAGSH	AVNGVARIHS	EILKKTMTSK	SWSLINSRIR	PMGSPLAGAGW	480
LCRIGEEYIA	DLQRLKLLS	YVDESFIIRD	VAKVKQENKL	KFSAYLEKEY	KVHINPNSLF	540
DIQVKRIHEY	KRQLLNLCHA	APGYHMAKMI	IKLITAIGDV	VNHDPVVGDR	LRVIFLENYR	600
VSLAEKVIPA	ADLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	ABEAGEENFF	660
IFGMRVEDVE	RLDQRGYNAQ	EYDRIPELR	HVIDQLSSGF	FSPKQPDLFK	DIVNMLMHHD	720
RFKVFADYEE	YVKCQERVSA	LYKNPREWTR	MVIRNIATSG	KFSSDRITIAQ	YAREIWGVPE	780

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NRQMPAPDE KI 792

SEQ ID NO: 141 moltype = AA length = 412
 FEATURE Location/Qualifiers
 source 1..412
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 141

MAQRKPEGS	FHMTCLSMAL	AYSFRPDANI	QPHPOLGNTQ	KQTELGKELT	GTMPHQYPAL	60
TPEQKLEL	CDIAHRIVAPGK	GILAADESTG	SIAKRLQSIG	TENTENRRF	YRQLLLTADD	120
RVNPCIGGVI	LFHETLYQKA	DDGRPPFPQVI	KAKGGVVDKG	VVPLAGTNGE	TTTQGLDGLS	180
ERCAQYKKDG	ADFAKWRCVL	KIGEHTPSSL	AIMENANVLA	RYASICQQNG	IVPIVEPEIL	240
PDGDHDLKRC	QYVTEKVLAA	VYKALSDHHI	YLEGTLKPN	MVTPGHACTQ	KYSHEETAMA	300
TVTALRRTPV	PAVPGITFLS	GGQSEEEASI	NLNAINKCPL	LKPWALTFSY	GRALQASALK	360
AWGGKKENLK	AAQEYVKRA	LANSLACQCK	YTPSGKAGAA	ASESLFISNH	AY	412

SEQ ID NO: 142 moltype = AA length = 167
 FEATURE Location/Qualifiers
 source 1..167
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 142

PAPAPPKKEK	IDLSAIKIEF	SKQQQDEFKE	AFLLFDRTGE	CKITLNQVGD	VLRALGTNPT	60
NAEVKKVLGN	PSNEEMNAKK	IEFEQFLPML	QAISNNKDQG	TYEDFVEGLR	VFDKEGNGTV	120
MGAELRHVLA	TLGKMKKEE	VEALMAGQED	SNGCINYEAF	VKHIMSN		167

SEQ ID NO: 143 moltype = AA length = 206
 FEATURE Location/Qualifiers
 source 1..206
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 143

MFVMGVNHEK	YNNTLKIVSN	ASCTTNCLAP	LAKVIHDHFG	IVEGLMTTVH	AITATQKTVD	60
GPSGKLWRDG	RGAAQNIIPA	STGAAKAVGK	VIPELNGKLT	GMAFRVPTPN	VSVVDLTCRL	120
EKPAKYDEIK	KVVVKQASEGP	LKGILGYTED	QVVSDFNSD	THSSTFDAGA	GIALNDHFVK	180
LISWYDNEFG	YSNRVVDLMV	HMASKE				206

SEQ ID NO: 144 moltype = AA length = 428
 FEATURE Location/Qualifiers
 source 1..428
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 144

MRTLAMAMQ	KIFAREILDS	RGNPTVEVDL	HTAKGRFRFA	VPSEGASTGIY	EALRLDGDGK	60
SRYLKGVLK	AVEHINKTLG	PALLEKASLG	RLYLLPPAKF	GANAILGVSL	AVCKAGAAEK	120
GVPLYRHIAD	LAGNPELILP	VPAFNIVINGG	SHAGNKLAMQ	EFMILPVGAS	SFKEAMRIGA	180
EVYHHLKQVI	KAKYKDATN	VGDEGGFAPN	ILENNEALEL	LKTAIQAAAGY	PDKVIGMDV	240
AASEFYRNKG	YDLDFKSPDD	PARHISGEKL	GELYKNFIKN	YPVVSIEDPF	DQDDWATWTS	300
FLSGVNIQIV	GDDLTVTNPK	RIAQAVEKKA	CNCLLLKVNQ	IGSVTESIQA	CKLAQSNQWG	360
VMVSHRSGET	EDTFIADLVV	GLCTGQIKTG	APCRSERLAK	YNQLMRIEEA	LGDKAVFAGR	420
KFRNPKAK						428

SEQ ID NO: 145 moltype = AA length = 181
 FEATURE Location/Qualifiers
 source 1..181
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 145

MPDQTQIQEF	KEAPKKAKRR	AAAEKGSSSV	FSMPDQTQIQ	EFKEAFTVID	QNRDGIIDKE	60
DLRDTFAAMG	RLNVKNEELD	AMMKEASGPI	NFTVPLNMFQ	EKLKGADPED	VITGAFKVL	120
PEGKGTIKKQ	FLEELLTTQC	DRFSQEEMRN	MWAAFPDPVG	GNVDYKNICY	VITHGDAKDQ	180
E						181

SEQ ID NO: 146 moltype = AA length = 340
 FEATURE Location/Qualifiers
 source 1..340
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 146

MCEEDSTAL	VCDNGSLCK	AGFAGDDAPR	AVFPSIVGRP	RHQGMVMGMG	QKDSYVGDEA	60
QSKRGILLTK	PIEHGIIITN	WDDMEKIMFE	TFNVPMYVYA	IQAVLSLYAS	GRTTGIVLDS	120
GDGVTHNVPI	YEGYALPHAI	MRLDLAGRDL	TDYLMKILTE	RGYSFVTAAE	REIVRDIKEK	180
LCYVALDFEN	EMATAASSSS	LEKSYELPDG	QVITIGNERF	RCPETLFPQS	FIGMESAGIH	240
ETTYNSIMKC	DIDIRKDLIA	NNVLSGGTTM	YPGIADRMQK	EITALAPSTM	KIKIIAPPER	300
KYSVWIGGSI	LASLSTFQQM	WISKQYDEA	GPSIVHRKCF			340

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SEQ ID NO: 147 moltype = AA length = 605
 FEATURE Location/Qualifiers
 source 1..605
 mol_type = protein
 organism = Cervus elaphus

SEQUENCE: 147

MKWVTFISLL	LLFSAYSARG	VFRDTHKSE	IAHRFKDLGE	DNFQGLVLIA	FSQYLQQCPF	60
DEHVKLVKEL	TDFAKECVAD	ESHAGCEKSL	HTLFGDELCK	VASLRETYGD	MADCCEKQEP	120
ERNECFLKHK	DDNPDLPKLL	PDPDTLCAEF	KADEKKFWGK	YLYEVARRHP	YFYAPELLYY	180
ANKYNGVFQE	CCQADDKAC	LLPKIDTMRE	KVLTSSARQR	LKCASIQKFG	ERALKAWTVA	240
LLSQKFPKAD	FAEVTKIIVT	LTKVHKECCH	GDLLECADDR	ADLAKYICDN	QDTLSSKLKE	300
CCDKPVLEKS	HCIAEIEKDA	VPENLPPLTA	DFAEDKEVCK	NYQEAQDVFL	GTFLYEYARR	360
HPEYAVSVLL	RVAKEYEATL	EDCCAKDDPH	ACYATVFDKL	QHLVDEPNQL	IKQNCDLFEK	420
VGEYGFQNAL	IVRYTKKAPQ	VSTPTLVEVS	RNLGRVGTCK	CTKPESERMS	CTEDYLSLVL	480
NRLCVLHEKT	PVSEKVTKCC	TESLVNRRPC	FSALTLDETY	VPKAFDEKLF	TFHADICTVP	540
DAEKQIKKQT	ALVELLKHKP	KATDEQLKTV	MENFVAFVDK	CCAAVDKEAC	FAVEPTLRVR	600
GKKEK						605

SEQ ID NO: 148 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 148

MPFSSTHNKH	KLKFSAAEEF	PDLSKHNNHM	AKVLTPELYK	RLRDKETPSG	FTLDDVIQTG	60
VDNPGHPPIM	TVGCVAGDEE	SYEVFKDLFD	PVIQDRHGGY	KPTDKHRTDL	NHENLKGDD	120
LDPKVVLSRR	VRTGRSIKGY	SLPPHCSRGE	RRAVEKLSVE	ALNSLEGEFK	GRYYPLKAMT	180
EQEQQLIDD	HFLFDKPVSP	LLLASGMARD	WPDARGIWHN	DNKTFLVWVN	EEDHLRVISM	240
EKGGMKEVF	RRFCVGLKKI	EEIFKKAGHP	FMWTEHLGYI	LTCPSNLGTG	LRGGVHVKLP	300
KLSQHPKFEE	ILHRLRLQKR	GTGGVDTAHV	GAVFDISNAD	RLGFSEVEQV	QMVVDGVKLM	360
VEMEKKLEQN	QPIDDMIPAQ	K				381

SEQ ID NO: 149 moltype = AA length = 897
 FEATURE Location/Qualifiers
 source 1..897
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 149

MNSMNQIETN	MQYTYNYEED	EYMTQEEEDW	RDLLDPAPAE	KQQRKTFTAW	CNSHLRKAGT	60
QIENIEEDFR	NGLKMLLLE	VISGERLPKP	DRGKMRFHKI	ANVNKALDYI	ASKGVKLVSI	120
GAEIIVDGNV	KMTLGMWITI	ILRFAIQDIS	VEETSAKEGL	LLWCQRKTAP	YRNVNIQNPH	180
LSWKDGLGLC	ALIHRRHPDL	IDYSKLNKDD	PIGNINLAME	IAEKHLDIPK	MLDAEDIVNT	240
PKPDERAIMT	YVSCFYHAF	GAEQAETAAN	RICKVLAVNQ	ENERLMEEYE	RLASELLEWI	300
RRTIPWLENR	TPEKTMQAMQ	KKLEDPRDYR	RKHKKPKVQE	KCQLEINFNT	LQTKLRISNR	360
PAFMPSEGMK	VSDIAGAWQR	LEQAEKGYYE	WLLNEIRRL	RLEHLAEKFR	QKASTHEQWA	420
YGKEQILLQK	DYESASLTFV	RAMLRKHEAF	ESDLAAHQDR	VEQIAAIAQE	LNELDYHDAA	480
SVNDRQCQIC	DQWDSLGLTL	QKRREALERT	EKLEITIDQL	HLEFAKRAAP	FNNWMEGAME	540
DLQDMFIVHS	IEETQSLISA	HDQFKATLPE	ADGERQAALS	IQNEVEKVIQ	SYSMRISASN	600
PYSTVTVEEI	RTKWEKVKQL	VPQRDQSLQE	ELARQHANER	LRRQFAAQAN	VIGPWIQTKM	660
EEIARSSIEM	TGPLEDQMNQ	LKQYEQNIIN	YKHNDKLEG	DHQLIQEALV	FDNKHTNYTM	720
EHIRVGWELL	LTTIARTINE	VETQILTRDA	KGITQEQMND	FRASFNHEDR	RKNGLMDHDD	780
FRACLISMGY	DLGEAEFARI	MSLVDPNGQG	TVTQFSFIDF	MTRETADTDT	AEQVIASFRI	840
LASDKPYILA	DELRRRELPE	QAQYCIKRMP	QYTGPGSVPG	ALDYTSFSSA	LYGESDL	897

SEQ ID NO: 150 moltype = AA length = 843
 FEATURE Location/Qualifiers
 source 1..843
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 150

MAAPLSDGER	RKQISVRGIA	GLGDVAEVRK	SFNRHLHFTL	VKDRNVATPR	DYFFALAHTV	60
RDHLVGRWIR	TQOHYYERDP	KRIYYLSLEF	YMGRTLQNTM	VNLGLQNACD	EAIYQLGLDL	120
EELIEIEEDA	GLNGGGLRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	IIDGWQVEEA	180
DDWLRYGNPW	EKARPEYMLV	VHFYGRVEHS	PEGVKWVDTP	VVLAMPYDTP	VPGYKNNTVN	240
TMRLWSAKAP	NDFNLQEFNV	GDYIEAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQYFV	300
VAAATLQDIIR	RFKSSKFGCQ	DPVRTSFETF	PDKVAIQLND	THPALSIPEL	MRILVDVEKV	360
DWDKAWETIK	QTCAYTNHTV	LPEALERWPV	SMFEKLLPRH	LEIIYALNQM	HLDRVAALYP	420
GDVRLRLRMS	VIEEGDCKRI	NMAHLCVIGS	HAVNGVARIH	SDIVKNTVFK	DFYELEPEKF	480
QNKTNGITPR	RWLLLCNPGL	ADVIAEKIGE	DFVTDLSQLK	KLLDFINNET	FIRDVAKVKQ	540
ENKLKFSAYL	EERYKVKINP	SSMFDIQVKR	IHEYKRQLLN	CLHAITLYNR	IRSDPSKSFV	600
PRTIMIGGKA	APGYHMAKMI	IKLITIGEV	VNNDPYVGDK	LKVIFLENYR	VSLAEKVIPA	660
ADLSQQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	AEAEAGEENLF	IFGMRVEEVE	720
ALDRRGYNAR	EYYDRIPELR	QAIDQISSGF	FSPRDPGCFR	DVVNMLMHHD	RFKVFADYEA	780
YIKCQGQVDQ	LFMDPREWTK	KVIRNIACSG	KFSSDRITITE	YAREIWGVPE	SATKIPPPNL	840
PRD						843

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SEQ ID NO: 151 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 151

MTHQYPALTA	BQKKELSDIA	LRIVAPGKGI	LAADESVGSM	AKRLNQIGVE	NTEENRRLYR	60
QILFSADSRV	KKCIGGVIFF	HETMYQKADD	GTPFVQMIKD	KGIVVGIKVD	KGVVPLAGTD	120
GETTTQGLDG	LSERCAQYKK	DGADFAKWRC	VLKISDNTPS	ALAIMENANV	LARYASICQQ	180
NGIVPIVEPE	ILPDGDHDLK	RCQYVTEKVL	AAVYKALSDH	HVYLEGTLK	PNMVTPGHSC	240
PTKYSPEEIA	MATVTALRRT	VPPAVPGVTF	LSGGQSEEEA	SINLNAINTC	PLVRPWALTF	300
SYGRALQASA	LSAWRGQRDN	ANAATEEFVK	RAEVNGLAAL	GKYEGSGDDS	GAAGQSLYVA	360
NHAY						364

SEQ ID NO: 152 moltype = AA length = 192
 FEATURE Location/Qualifiers
 source 1..192
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 152

MAPKKDVKKP	AAAAAPAPAP	APAPAPAPAK	PKEPAIDLKS	IKIEFSKEQQ	DDFKEAFLLF	60
DRTGDAKITL	SQVGDIVRAL	GQNPTNAEIN	KILGNPSKEE	MNAKKITFEE	FLPMLQAAAN	120
NKDQGTFFED	VEGLRVFDKE	GNGTVMGAEI	RHVLATLGEK	MTEEEVEELM	KGQEDSNGCI	180
NYEAFVKHIM	SV					192

SEQ ID NO: 153 moltype = AA length = 333
 FEATURE Location/Qualifiers
 source 1..333
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 153

MVKVGVNGFG	RIGRLVTRAA	VLSGKVQVVA	INDPFIDLNY	MVYMFKYDST	HGHFKGTVKA	60
ENGKLVINGH	AITIFQERDP	SNIKWADAGA	EYVVESTGVF	TTEKAGAGHL	KGGAKRVIIS	120
APSADAPMFV	MGVNHEKYDK	SLKIVSNASC	TTNCLAPLAK	VIHDNFGIVE	GLMTTVHAIT	180
ATQKTVDGPG	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCRLKEP	AKYDDIKRVV	KAAADGPLKG	ILGYTEDQVV	SCDFNGSDSH	STFDAGAGIA	300
LNDHFVKLV	WYDNEFGYSN	RVVDLMVHMA	SKE			333

SEQ ID NO: 154 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 154

MDAIKKMQM	LKLDKENALD	RAEQAEADKK	AAEERSKQLE	DELVALQKKL	KGTEDELDKY	60
SESLKDAQEK	LELADKKATD	AESEVASLNR	RIQLVEEELD	RAQERLATAL	QKLEEAEEKAA	120
DESERGMKVI	ENRAQKDEEK	MEIQEIQLKE	AKHIAEEADR	KYEEVARKLV	IIEGDLERAE	180
ERAELSESKC	AELSEELKTV	TNNLKSLEAQ	AEKYSQKEDK	YEEIKVLTD	KLKEAETRAE	240
FAERSVTKLE	KSIDDELEDEL	YAQKLKYKAI	SEELDHALND	MTSI		284

SEQ ID NO: 155 moltype = AA length = 1132
 FEATURE Location/Qualifiers
 source 1..1132
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 155

MPEPSKAAPK	KEAKKKEEKK	EEKKEAPPPQ	EHKDEAPDDV	HPPETPDPEG	LFLSKPQNMV	60
VESGRDVTVS	ARVAGAALPC	APAVKWFKGK	WAEELGDKSAR	CRLRHSVDDD	KVHTFELTIT	120
KVAMGDRGDY	RCEVTAKEQK	DSCSFSIDVE	APRSSEGNVL	QAFKRTGEGK	DDTAGELDFS	180
GLLKKREVQV	EEKKKKKDED	DQPPPEIWEI	LKGVTKKSEY	ERIAFYQGIT	DLRGMLKRLK	240
KVHVEPKKSE	AFIRKLDPAY	QVDKGNKIKL	VVELSDPDLP	LKWYKNGQLL	KPSTKYVFEN	300
VGLKRILTIH	KCSLADDAAY	ECRVNDEKCF	TEVFVKEPPV	TVVRGLEDQ	VVVGDRVVLE	360
AEVSEGAQV	MWLDGDVDT	RDDAFKYRFP	KDGKKHFLII	NEAELSDSAH	YKIMTNGGES	420
EAELSVEEKQ	LEVLDQMADL	TVKASEQAVF	KCEVSDEKVT	GRWFRNGVEV	KPSKRIHISH	480
NGRFHKLVID	DVRPEDEGDY	TFIPDGYALS	LSAKLNFLEI	KVEYVPKQEP	PKIHLDCSGK	540
AAENTIVVVA	GNKVRLDVPI	SGEPAPTIVT	WRGDLFTAT	EGRVHIDSQA	DLSSFVIESA	600
ERSDEGRYCI	TVTNPVGEDS	ATLHVRVVDV	PDPPQSVRVT	SVGEDWAVLS	WEAPPFDGGM	660
PITGYLMERK	KKGSMRWMLK	NFEVFPDPTY	ESTKMIEGVF	YEMRVFAVNA	IGVSQPSLNT	720
QPFMPIAPTS	EPTHVLEDDV	TDTTATIKWR	PPERIGAGGV	DGYLVEWCRE	GSNEWVAANT	780
ELVERCGLTA	RGLPTGERLL	FRVISVMAG	KSPATMAQOP	VTIREIVERP	KIRLPRHLRQ	840
TYIRRVGEQV	NLVIPIFQKPK	RPOQVTSREG	GALPAEVQTR	TSVDVSVFFI	RSARPLSGN	900
YEMRVRIDNM	EDCATLRLRV	VERPGPPQAV	RVMEVWGSNA	LLQWEPPKDD	GNAEISGYTV	960
QKADTRTMEW	FTVLEHSRPT	RCTVSELVMG	NEYRFRVYSE	NVCGTSQEP	TSHTNTARIAK	1020
EGLTLKMPVP	KERDLRAAPQ	FLTPLVDRSV	VAGYTVTLNC	AVRGHPKPKV	TWLKNSVEIG	1080
ADPKFLSRHG	LGVLSLLIRR	PGPFDGGTYG	CRAVNEMGEA	TTECRLDVRV	PQ	1132

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SEQ ID NO: 156 moltype = AA length = 1450
 FEATURE Location/Qualifiers
 source 1..1450
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 156

MSSVAVPFYQ	RRHKHFDQSY	RNIQTRYVLE	EYAARKAASR	QAAHYESTGL	GKTTCRLCAR	60
RARSLAHEAM	QESRKRTHEQ	KSHASDEKRI	KFASLSSLE	REIHMARHHA	REQLDRLAIQ	120
RMVEENMALE	RHVVEEKISR	APEILVRLRS	HTVWEKMSVR	LCFTVQGFP	PPVQWYKNEE	180
LITPASDPAK	YSVENKYGVH	VLHINRADFD	DSATYSAVAT	NIHQASTNC	AVVRRFRRES	240
EEPHPAGIMP	PHLPLSYDVC	FTHFDVQFLE	KFGVTFATEG	ETLTLKCSVL	VTPELKRLRP	300
RAEWYRDDVL	IKDSKWTKLY	FGEGQAALSF	THLNKDDDEGL	YTLRMVTKGG	VNECSAFLFV	360
RDADALIAGA	PGAPMDVKCH	DANRDYVIVT	WKPPNTTSQN	PVIGYFVDKC	EVGLENWVQC	420
NDAPVKICKY	PVTGLYEGRS	YIFRVRAVNS	AGISRPSRVS	EPVAALDPVD	LERTQTVHVD	480
EGRKIVISKD	DLEGDIQIPG	PPTNVHASEI	SKTYVVLSDW	PPVPRGREPL	TYFIEKSMVG	540
SGSQWRVNAQ	VAVKSPRYAV	FDLAEGKPYV	FRVLSANKHG	ISDPSEITEP	IQPDIVVVP	600
SAPGRVVATR	NTKTSVVVQW	DKPKHEENLY	GYIDYSVVG	SNQWEPANHK	PINYNRFVVH	660
GLETGEQYIF	RVKAVNAVGF	SENSQSESEI	KVQAALTCPS	YPHGITLLNC	DGHSMTLGWK	720
APKYSGGSPI	LGYYIDKREA	NHKNWHEVNS	SVISRTIYTV	EDLTEDAFYE	FKIAAANVVG	780
IGHPSDPSEH	FKCKAWTMPE	PGPAYDLTVC	EVRNTSLVLL	WKAPVYEGKS	PITGYLVDYK	840
EVDTEWDITA	NEKPTSHRYF	KVTDLHQGHT	YVFKVRAVND	AGVGKSSEIS	EPVFVEASPG	900
TKEIFSGVDE	BGNYYLGFEC	KEATDASHFL	WGKSYEEIED	SDKFKIETKG	DHCKLYFKHP	960
DKSDLGTYCI	SVSDTDGVSS	SFVLDEEELE	RLMTLSNEIK	NPTIPLKSEL	AYEVLDKGEV	1020
RFWIAEASLS	PNSTYRFVIN	DKEVENGDRH	KISCDHSNGI	IEMVMDKFTI	DNEGTYTVQI	1080
QDGKAKNQSS	LVLIGDAFKA	ILAESELQRK	EPLRKQGPHE	SEFLYWEVTE	ECEVLLACKI	1140
ANTKKETVPK	WYRNGSGIDV	DEAPDLQKGE	CHLTPVKLSR	KDEGVYKATL	SDDRGHVDVST	1200
LELSGKVYND	IILALSRVSG	KTASPLKILC	TEEGIRLQCF	LKYYNEMMKV	TWSHRESKIS	1260
SGEKMKIGGG	EDVAWLQITE	PTFKDKGNVT	FEIFSDKESF	KRTLDSLQQA	FDDALTEFQR	1320
LKAAAFPAEK	RGKVIIGLDP	VVTIMDGKTL	NLTCTVFGNP	DPEVVWFQND	KALELNEHYL	1380
VSLEQGYKAS	LTIKGVTSED	SGKYSIYVKN	KYGGETVDVT	VSVYRHGEKI	PEVNGQGLAK	1440
PRLIPSSST						1450

SEQ ID NO: 157 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 157

MSIQKIHAARE	ILDSRGEPTV	EVDLHTAKGH	FRAAVPSGAS	TGIHEALEPR	DGDKKRFLGK	60
GVLKAVEHIN	KTIGFALIEK	KISVVEQEKI	DKVMIEMDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVP	YRHIADLAGN	TELILPVPAP	NVINGGSHAG	NKLAMQEFMV	LPVGAASFHD	180
AMRVGAEVH	SLKGVIAKAY	GKDATNVGGE	GGFAPNILDN	HEALELLKAA	IAQAGYTDKV	240
VIGMDVAASE	FCRDGRVHLD	FKSPPHTKRY	ITGEQLGEIY	RGFIKDYPPV	SIEDPPDQDD	300
WEAWKRFPFH	VDIQVVGDDL	TVTNPKRIRH	GAEQHACNCL	LLKVNQIGSV	TESIQACKLA	360
QSHGWGMVVS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEALGDK	420
AKFAGRKFRN	PKAK					434

SEQ ID NO: 158 moltype = AA length = 168
 FEATURE Location/Qualifiers
 source 1..168
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 158

MAPKKAKRRA	AEGSSNVFSM	FDQTQIQEFK	EAFITVIDQNR	DGIIDKDDL	ETFAAMGRIN	60
VKNEELDAMI	KEASGPINFT	VFLTMFGEKL	KGADPEDVIM	GAFKVLDPDG	KGSIKKSFL	120
ELLTTQCDRF	TPPEIKNMWA	AFPPDVAGNV	DYKNICYVIT	HGEDKEGE		168

SEQ ID NO: 159 moltype = AA length = 1940
 FEATURE Location/Qualifiers
 source 1..1940
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 159

MATDADMAIF	GEAAPYLRSK	EKERIEAQNK	PFDAKSSVVF	VHAKESYVKS	TIQSKESGKV	60
TVKTEGGETL	TVKEDQIFSM	NPPKYDKIED	MAMMTHLHEP	AVLYNLKERY	AAWMIYTSYG	120
LPCVTVNPKY	WLPVYNPEVV	LAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	ASGDKKKEEQ	PAGKMQGTLE	DQIISANPLL	EAFNGAKTVR	240
NDNSSRFGFK	IRIHFGATGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTNP	YDYQYVSQGE	ITVPSINDQE	ELMATDSALD	ILGFTPDEKT	AIYKLTGAVM	360
HYGNLKFQYK	QREEQAEPPG	TEVADKAAYL	MGLNSADLLK	ALCYPRVKVG	NEYVTGQQT	420
QQVNSVGLAL	AKSVFEKMLF	WMVVRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNLSLEQLC	480
INFTNEKLQQ	FFNHMHVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	FKNKLYDQHL	GKSNFPQPKP	PGKGAEAEHF	SLVHYAGTVD	YNTGWLEKN	600
KDPLNETVVG	LYQKSSSLKTL	ALLFASVGGG	EAESGAGGKK	GGKKKGSSSQ	TVSALPRENL	660
NKLMSNLRST	HPHFVRCLIP	NETKTPGAME	HELVHLQLRC	NGVLEGRIC	RKGFPIRILY	720
ADFKQRYKVL	NASAIPEGQF	IDSKKASEKL	LGSIDVDHTQ	YKFGHTKVFF	KAGLLGLEEE	780

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MRDEKLAQLI	TRTQARCRGF	LMRVEFKMM	ERRESIFCIQ	YNVRAFMNVK	HWPWMKLFFK	840
IKPLKSAES	EKEMANMKEE	FEKTKEELAK	SEAKRKELEE	KMVSLQEK	DLQLQVQAEA	900
DGLADAEERC	DQLIKTKIQL	EAKIKELTER	AEDDEEMNAE	LTAKKRKLED	ECSELKKDID	960
DLELTLAKVE	KEKHATENKV	KNLTEEMAAL	DETIAKLTKE	KKALQEAHQ	TLDDLQAEED	1020
KVNTLTAKT	KLEQQVDDLE	GSLEQEKKL	MDLERAKRL	EGDLKMTQES	TMDLENDKQ	1080
LDEKLKKDF	EISQIQSKIE	DEQALGMQLQ	KKIKELQARI	EELEEEIEAE	RTSRKAKEKH	1140
RADLSRELEE	ISERLEEAGG	ATAAQIDMKN	KREAEFQKMR	RDLEEATLQH	EATAAALRKK	1200
HADSTADVGE	QIDNLQVRVKQ	KLEKEKSELK	MEIDDLASNM	ESVSKAKANL	EKMCRSLEDQ	1260
LSEIKTKEEE	QORTINDISA	QKARLQTESG	EYSRQVEEKD	ALISQLSRGK	QAFTQQIEEL	1320
KRHLEEEIKA	KKCPAHALQS	ARHDCDLLRE	QYEEEQEAKG	ELQRALSKAN	SEVAQWRTKY	1380
ETDAIQRTTEE	LEEAKKKLAQ	RLQDAEEHVE	AVNSKCASEL	KTKQRLQNEV	EDLMIDVERS	1440
NAACAALDKK	QKNFDKILSE	WKQKYETQA	ELEASQKESR	SLSTELFKMK	NAYEESLDHL	1500
ETLKRENKNL	QQEISDLTEQ	IAEGGKAIHE	LEKVKKQIEQ	EKSELQTALE	EAEASLEHEE	1560
GKILRVQLEL	NQVKSIDIRK	IAEKDEEIDQ	LKRNHLRVVD	SMQSTLDAEI	RSRNEALRLK	1620
KKMEGDLNEI	EIQLSHANRQ	AAEAQKNLRN	TQGVLDKTDI	HLDDALRSQE	DLKEQVAMVE	1680
RRANLLQAEI	EELRAALEQT	ERSRKVAEQE	LLDASERVQL	LHTQNTSLIN	TKKKLESDIS	1740
QIQSEMEDIT	QEARNAEKA	KKAITDAAMM	AEELKKEQDT	SAHLERMKKN	LDQTVKDLQH	1800
RLDEAEQLAL	KGGKKQIQKL	EARVRELEGE	VDAEQKRSAE	AVKGVRYKER	RVKELTYQSE	1860
EDRKNVLRQ	DLVDKQMKV	KSYPKQAEAA	EELSNVNLK	FRKIQHELEE	AERADIAES	1920
QVNLRAKSR	EIGKKAESSE					1940

SEQ ID NO: 160 moltype = AA length = 1940
 FEATURE Location/Qualifiers
 source 1..1940
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 160

MATDADMAIF	GEAAPYLRSK	EKERIEAQNK	PFDAKSSVFV	VHAKESYVKS	TIQSKESGKV	60
TVKTEGGETL	TVKEDQIFSM	NPPKYDKIED	MAMMTHLHEP	AVLYNLKERY	AAWMIYTYSG	120
LPCVTVNPYK	WLPVYNPEVV	LAYRGKKRQE	APPHIFSISD	NAYQFMLTDR	ENQSILITGE	180
SGAGKTIVNK	RVIQVFATIA	ASGDKKKEEQ	PAGKMQGTLE	DQIISANPLL	EAFGNKATVR	240
NDNSSRFGKF	IRIHFGATGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IEMLLITTMP	YDYQYVSQGE	ITVPSINDQE	ELMATDSAD	ILGFTPDEKT	AIYKLTGAVM	360
HYGNLKFKQK	QREBQAEEDG	TEVADKAAYL	MGLNSADLLK	ALCYPRVKVG	NEVVTGQTV	420
QQVYNSVGAL	AKSVFEKMPF	WMVVRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNLSLEQLC	480
INFTEKLLQ	FFNHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	PKNKLYDQHL	GKSNNFQKPK	PGKGKAEAHF	SLVHYAGTVD	YNTGWLEKN	600
KDPLNETVVG	LYQKSSSLKTL	ALLFASVGGG	EAESGAGGKK	GGKKKGSSFQ	TVSALFRENL	660
NKLMSNLRST	HPHFVRCLIP	NETKTPGAME	HELVLHQLRC	NGVLEGIRIC	RKGFPSPRILY	720
ADFKQRYKVL	NASAIPEGQF	IDSKKASEKL	LGSIDVDHTQ	YKFGHTKVFF	KAGLLGLLEE	780
MRDEKLAQLI	TRTQARCRGF	LMRVEFKMM	ERRESIFCIQ	YNVRAFMNVK	HWPWMKLFFK	840
IKPLKSAES	EKEMANMKEE	FEKTKEELAK	SEAKRKELEE	KMVSLQEK	DLQLQVQAEA	900
DGLADAEERC	DQLIKTKIQL	EAKIKELTER	AEDDEEMNAE	LTAKKRKLED	ECSELKKDID	960
DLELTLAKVE	KEKHATENKV	KNLTEEMAAL	DETIAKLTKE	KKALQEAHQ	TLDDLQAEED	1020
KVNTLTAKT	KLEQQVDDLE	GSLEQEKKL	MDLERAKRL	EGDLKMTQES	TMDLENDKQ	1080
LDEKLKKDF	EISQIQSKIE	DEQALGMQLQ	KKIKELQARI	EELEEEIEAE	RTSRKAKEKH	1140
RADLSRELEE	ISERLEEAGG	ATAAQIDMKN	KREAEFQKMR	RDLEEATLQH	EATAAALRKK	1200
HADSTAELE	QIDNLQVRVKQ	KLEKEKSELK	MEIDDLASNM	ESVSKAKANL	EKMCRSLEDQ	1260
LSEIKTKEEE	QORTINDISA	QKARLQTESG	EYSRQVEEKD	ALISQLSRGK	QAFTQQIEEL	1320
KRHLEEEIKA	KNALAHALQS	ARHDCDLLRE	QYEEEQEAKG	ELQRALSKAN	SEVAQWRTKY	1380
ETDAIQRTTEE	LEEAKKKLAQ	RLQDAEEHVE	AVNSKCASEL	KTKQRLQNEV	EDLMIDVERS	1440
NAACAALDKK	QKNFDKILSE	WKQKYETQA	ELEASQKESR	SLSTELFKMK	NAYEESLDHL	1500
ETLKRENKNL	QQEISDLTEQ	IAEGGKAIHE	LEKVKKQIEQ	EKSELQAAL	EAEASLEHEE	1560
GKILRVQLEL	NQVKSIDIRK	IAEKDEEIDQ	LKRNHLRVVD	SMQSTLDAEI	RSRNEALRLK	1620
KKMEGDLNEI	EIQLSHANRQ	AAEAQKNLRN	TQGVLDKTDI	HLDDALRSQE	DLKEQVAMVE	1680
RRANLLQAEI	EELRAALEQT	ERSRKVAEQE	LLDASERVQL	LHTQNTSLIN	TKKKLESDIS	1740
QIQSEMEDIT	QEARNAEKA	KKAITDAAMM	AEELKKEQDT	SAHLERMKKN	LDQTVKDLQH	1800
RLDEAEQLAL	KGGKKQIQKL	EARVRELEGE	VDAEQKRSAE	AVKGVRYKER	RVKELTYQSE	1860
EDRKNVLRQ	DLVDKQMKV	KSYPKQAEAA	EELSNVNLK	FRKIQHELEE	AERADIAES	1920
QVNLRAKSR	EIGKKAESSE					1940

SEQ ID NO: 161 moltype = AA length = 615
 FEATURE Location/Qualifiers
 source 1..615
 mol_type = protein
 organism = Gallus gallus

SEQUENCE: 161

MKWTVLISFI	PLFSSATSRN	LQRFARDAEH	KSEIAHRYND	LKEETFKAVA	MITFAQYLQR	60
CSYEGLSKLV	KDVVDLAQKC	VANEDAPECS	KPLPSIILDE	ICQVEKLKRS	YGAMADCCSK	120
ADPERNECFI	SFKVSQPDFV	QPYQRPASDV	ICQEQDNRV	SFLGHFIYSV	ARRHPFLYAP	180
AILSFADVFE	HALGSCCKES	DVGACLDTKE	IVMREKAKGV	SVKQQYFCGI	LKQFGDRVFP	240
ARQLIYLSQK	YPKAPFSEVS	KFVHDSIGVH	KECCGDMVE	CMDDMARMS	NLCSQQDVFS	300
GKIKDCCCEP	IVERSQCIME	AEFDEKPADL	PSLVEKYIED	KEVCKSPFAG	HDAFMAEFVY	360
EYSRRHPEFS	FLIMIRIAKG	YESLLEKCK	TDNPAECYAN	AQEQLNQHIK	ETQDVVKTNC	420
DLHHDHGEAD	IKKSILIRYT	KKMPQVPTDL	LLETGKKMTT	IGTKCCQLGE	DRRMACESEY	480
LSIVIHDTCT	KQETTPINDN	VSQCCSQLYA	NRPRCPTAMG	VDTKYVPPFP	NPDMFSFDEK	540

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LCSAPAEERE VGQMKLLINL IKRKPQMTEE QIKTIADGFT AMVDKCKQS DINTCFGEEG 600
ANLIVQSRAT LGIGA 615

SEQ ID NO: 162 moltype = AA length = 380
FEATURE Location/Qualifiers
source 1..380
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 162
MPFSNSHLL KMKHSADDEF PDLSAHNHMH AKVLTLDLYK KLRDRQTSSG FTLDDVIQTG 60
VDNPGHPPIM TVGCVAGDEE SYEVFKELFD PVIEDRHGGY KPTDEHKTDL NADNLQGGDD 120
LDPNVVLSR VRTGRSIRGF CFPHCSSRGER RAIEKLSVEA LGS LGGDLKG KYALRNMTD 180
AEQQQLIDDH VRTGKPVSP LLSGMDARDW PDARGIWHND NKTFLVWINE EDHLRVISMQ 240
KGSNMKEVFT RFCTGLTQIE TLFKSKNYEF MWNPHLGYIL TCPSNLGTGL RAGVHIKLPN 300
LGKHEKFGEV LKRLRLQKRG TGGVDTAAVG GVFDVSNADR LGFSEVELVQ MVVDGVKLLI 360
EMEKRLKQKQ SIDDLMPAQK 380

SEQ ID NO: 163 moltype = AA length = 897
FEATURE Location/Qualifiers
source 1..897
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 163
MNSMNQIETN MQYTYNYEED EYMTQEEWD RDLLEDPAWE KQQRKTFTAW CNSHLRKAGT 60
QIENIEEDFR NGLKMLLLE VISGERLPKP DRGKMRFHKI ANVNKALDYI ASKGVKLVS I 120
GAEIIVDGNV KMTLGMWITI ILRFAIQDIS VEETSAKEGL LLWCQRKTAP YRNVNIQNPH 180
LSWKDGLGLC ALIHRHRPDL IDYSKLNKDD PIGNINLAME IAEKHLDIPK MLDAEDIVNT 240
PKPDERAINT YVSCFYHAF GAEQAETAAN RICKVLAVNQ ENERLMEEYE RLASELLEWI 300
RRTIPWLENR TPEKTMQAMQ KKLEDFRDYR RKHKPPKVQE KCQLEINFNT LQTKLRISNR 360
PAPMPSEGKM VSDIAGAWQR LEQAEKGYYE WLLNEIRRL RLEHLAEKFR QKASTHEQWA 420
YGKEQILLQK DYESASLTV RANLRKHEAF ESDLAHQDR VEQIAAIAQE LNELDYHDAA 480
SVNDRQCQIC DQWDSLGLT QKRREALERT EKLETTIDQL HLEFAKRAAP FNNWMEGAME 540
DLQDMFIVHS IEEIQSLISA HDQFKATLPE ADGERQAILS IQNEVEKVIQ SYSMRISASN 600
PYSTVTVEEI RTKWEKVQL VPQRDQSLQE ELARQHANER LRRQFAAQAN VIGPWIQTKM 660
EEIARSSIEM TGPLDQMNQ LKQYEQNIIN YKHNDKLEG DHQLIQEALV FDNKHTNYTM 720
EHIRVGWELL LTTIARTINE VETQILTRDA KGITQEQMND FRASFNHEDR RKNGLMDHDD 780
FRACILSMGY DLGEAEFARI MSLVDPNGQG TVTFQSFIDF MTRETADTDT AEQVIASFRI 840
LASDKPYILA DELRRELPE QAQYCIKRM QYTPGGSVPG ALDYTSFSSA LYGESDL 897

SEQ ID NO: 164 moltype = AA length = 839
FEATURE Location/Qualifiers
source 1..839
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 164
MAAPLSDGER RKQISVRGIA GLGDVAEVRK SFNRHLHFTL VKDRNVATPR DYFFALAHTV 60
RDHLVGRWIR TQQHYERDP KRIYYLSLEF YMGRTLQNTM VNLGLQNACD EAIYQLGLDL 120
EELEIEEDA GLNGGLGRL AACFLDSMAT LGLAAYGYI RYFPGIFNQK IVDGWQVEEA 180
DDWLRYGNFW EKARPEYMLP VHFYGRVEHS PEGVKWVDTP VVLAMPYDTP VPGYKNNTVN 240
TMRLWSAKAP NDFNLQEFNV GDYIEAVLDR NLAENISRVL YPNDNPFEGK ELRLKQYEFV 300
VAATLQDIIR RFKSSKFGCQ DPVTSFETP PDKVAIQLND THPALSIPEL MRILVDVEKV 360
DWDKAWIITK QTCAYTNHTV LPEALERWPV SMFEKLLPRH LEIYALNQM HLDRAVALYP 420
GDIDRLRRMS VIEEGDCKRI NMAHLCVIGS HAVNGVARIH SDIVKNTVFK DFYELDPEKF 480
QNKTNGITPR RWLLCNFGL ADVIAEKIGE DFTVTLSQLK KLDFIDDET FIRDVAKVKQ 540
ENKLKFSAYL EEHYKVKNP SSMFDIQVKR IHEYKRQLLN CLHAITLYNR KSDPSKSFVP 600
RTIMIGGKAA PGYHMAKMI KLITSIGEVV NNDPYVGDKL KVIFLENYRV SLAEKVIPAA 660
DLSQQISTAG TEASGTGNMK FMLNGALTIG TMDGANVEMA EEAGEENLFI FGMRVVEEVA 720
LDRQGYNARE YYDRIPELRQ AIDQISSGFF SPRDPGCFRD VVNMLMHDR FKVPADYEAY 780
IKCQGQVDQL FMDPREWTKK VIRNIACSGK FSSDRITIEY AREIWGVEPS GDVFPAAWA 839

SEQ ID NO: 165 moltype = AA length = 364
FEATURE Location/Qualifiers
source 1..364
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 165
MTHQYPALTA EQKKELSDIA LRIVAPGKI LAADESVGSM AKRLNQIGVE NTEENRRLYR 60
QLLFSADSRV KKCIGGVIF HETMYQKADD GTPFVQMIKD KGIVVGKVD KGVVPLAGTD 120
GETTTQGLDG LSERCAQYKK DGADFAKWRV VLKISDNTPS ALAIMENANV LARYASICQQ 180
NGIVPIVEPE ILPDGDHDLK RCQYVTEKVL AAVYKALSDH HVYLEGTLK PNMVTPGHSC 240
PTKYSPEETA MATVPAVGVT LSGGQSEEEA SINLNAINTC PLVRPWALTF 300
SYGRALQASA LSAWRGQRDN ANAATEEFVK RAEVNGLAAL GKYEGSGDDS GAAGQSLYVA 360
NHAY 364

SEQ ID NO: 166 moltype = AA length = 192
FEATURE Location/Qualifiers

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source 1..192
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 166
MAPKKDVKKP AAAAAAPAPAP APAPAPAPAK PKEPAIDLKS IKIEFSKEQQ DDFKEAFLLF 60
DKTGDAKITL SQVGDIVRAL GQNPTNAEMN KILGNPSKEE MNAKKITFEE FLPMLQAAAN 120
NKDQGTYEDF VEGLRVFDKE GNGTVMGAEL RHVLATLGEK MTEEEVEELM KGQEDSNGCI 180
NYEAFVKHIM SV 192

SEQ ID NO: 167 moltype = AA length = 333
FEATURE Location/Qualifiers
source 1..333
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 167
MVKVGVGNGFG RIGRLVTRAA VLSGKVQVVA INDPPIDLNY MVYMFKYDST HGHFKGTVKA 60
ENGKLVINGH AITIFQERDP SNIKWADAGA EYVVESTGVF TTMEKAG AHL KGGAKRVIIS 120
APSADAPMFV MGVNHEKYDK SLKIVSNASC TTNCLAPLAK VIHDFNGIVE GLMTTVHAIT 180
ATQKTVDGFS GKLRWDGRGA AQNIIPASTG AAKAVGKVIP ELNGKLTGMA FRVPTPNVSV 240
VDLTCRLEKP AKYDDIKRVV KAAADGPLKG ILGYTEDQVV SCDPNGDSHS STPDAGAGIA 300
LNDHVFVKLV WYDNEFGYSN RVVDLMVHMA SKE 333

SEQ ID NO: 168 moltype = AA length = 284
FEATURE Location/Qualifiers
source 1..284
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 168
MEAIKKKMQM LKLDKENALD RAEQAEAEQK QAEERSKQLE DELAAMQKKL KGTEDELDKY 60
SEALKDAQEK LELAEKKAAD AEAEVASLNR RIQLVEEELD RAQERLATAT QKLEEAEEKA 120
DESERGMKVI ENRALKDDEK MELQEIQLKE AKHIAEEADR KYEEVARKLV IIEGDLERE 180
ERAEALAEK SCLEEEELKNV TNNLSLEAQ AEKYSQKEDK YEEBIKILTD KLKEAETRAE 240
FAERSVAKLE KTIDDLEDEL YAQKLKYKAI SEELDHALND MTSI 284

SEQ ID NO: 169 moltype = AA length = 968
FEATURE Location/Qualifiers
source 1..968
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 169
MCLQLFVFSE LPPPFSLKFN KELTFVYCLH KALMVECWCH SICSSPTLAD QIREVVGLSE 60
LVSSGKCILE RIKHWLIIDI FLFLCISVWS IGESPPPEETE KHDSQSRSTL FIEKPQSGSV 120
SVGGNITFIA KVEAKDLLRK PNWKWFKGKW MDLASKAGKH LQLKESFERH TKIHTFEMHI 180
IQAKENYAGN YRCEVSYKDK FDCSFDLEV TESSQAAPSI DIRSAFKRSG DGQDDAGELD 240
FSGLLKRREI KQEEEEPEVD VWELLKNANP SEYEKIAFOY GITDLRGLMK RLKRMRRREV 300
KSAAFSGKLD PAYQVDKGGK VRFMVELADP TVELKWKYNG QEIRPSAKYI FEHKGQRIL 360
FINNCTMTDD ARYYVTAGDE KCSTELFVRD PPILVTKGLE DTNTYVGERV ELSCEVS EDD 420
ANVKWFKNV ELTNEPKSRY RIKVEGKKHT LIIIEAAKND NATYSVMTTG GQSEAKLSVD 480
LRPLKISLAL EDQTVRLQOE IHLKCEISEN VEGKWKYNGQ LIEASDRVKL YHKGRIHRFI 540
IASAAVDDEG EYMFVPDAYN INIPCKVHV DPPLKHL DGL GENNTVTVVA GTKLRL EIP 600
TGEPTPKVMW SKGDKWITDS GRIRAEYSD SSCLVIDTAE REDSGPFRIT LKNEAGEDSA 660
LNIKVVDVP DPPQAPNVE VGEDWCVMTW EPPANDGGSP ILGYFIERKK QSSRWMLRN 720
FELCKETTPE PKKMIEGVAY EVRVFAVNAI GTSKPSMP SK SFVPLAVTSP PTLLAVDSVT 780
DTSVTMKWRP PDHIGAAGLD GYVVEYCFEG TDEWIVANPE LTDKTKFTIN GLPTGSKILV 840
RVKAVNAAGE SEPRYHPQPI LVKEVIEPPK IRLPRHLKQT YTRRVGETVN LVIPFQVRTK 900
NTDRKTLNTH PSYATASYLL FLYVNLGYWT AFVNILHPEG KSHDKASGII HSYLGTKIGH 960
YHMSQSN 968

SEQ ID NO: 170 moltype = AA length = 1309
FEATURE Location/Qualifiers
source 1..1309
mol_type = protein
organism = Meleagris gallopavo

SEQUENCE: 170
MNVKGEISAY ASLVVKRYKG EIDASHLHAG TFSMPLSLGV TPHGFASRFE IKFIDKFEVM 60
FGREGETMSL GCTVVISDPI KRFPKDIEWY RINGVLITPSK WVQMHWTGER ASLTLTHANK 120
EDEGLYTLRV VMGEYEEYS GYVFVRDADA EIPGAPGAPL DVQCLDANKD YVIVSWKQPA 180
VDGNNPILGY FIDRCEVGTI HWTQCNETPV KFAFPFVTGL IEGRSYIFRV RAVNRAGVSL 240
PSRVSEAVAA LDPADRARLR SHPSAPWTGQ IIVTEEEP AE GVIPGPPTDL QVIEATKNYV 300
VLSWKPPGQR GHEGIMYFVE KCIAGTEDWQ RVNTEIPVKS PRFAVFDLSE GKSYPFRVRC 360
CNSAGIGEPS EATEATVYDD KLDIPKPPGR IMPTRNTDTS VVVWTWTEPPD AKELVGYYIE 420
SSVVGSGHWE PCNNNPVKGT RFICHGLVNG EKYIFRVRAV NAAGLSEFSQ DSEPIEVQAA 480
IAAPSPPYDI TVLESVRDSM VLGWKQPKAT GGTEITGYV NYREVIGGVP GKWREANIKA 540
ISERAYRIEN LKENMVYQFQ VAAANLAGVG TPLSLSPKPK CEEWTIAVPG PPHDVITYTEV 600
RKDSLVLWK EPVYTGSPSI AGYYIDMKET EAKEEHWRSV NEKPLQKFL KIAGLTEGVS 660
YVFRVRATNQ AGVGRPSDIT DPVVAETRPG TKEVVVNVD DGVISLNFEC DQMSPN SKFV 720

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WSKNYEPED	DSRLDIDSKG	GSKAIFKQL	GEDDLGIYSC	VVTDTDGVSS	GYTIDEEMK	780
RLALSHERK	PFTIPLASEL	AVEILEKGEV	RFWLQAEKLS	GNAKANFVFN	DKEIFNGEKY	840
KMKVDQRTGL	VEMIMDKLED	KDEGTYTQFL	QDGKATNQSS	LVLIGDVFKK	LRDEAEFQRK	900
EWHRKQGPHE	VDQLGWEVTD	ECNVMKCKV	ANIKKETHIV	WYKDDREIMV	DEEHDFKDG	960
CTLLISEFSK	KDTGIYEVIL	KDDRGKDVSE	LKLTDATAPAD	LMNEVCRRIA	LSATDLKIQS	1020
TAEGRILYSF	VTYYVDLRLV	GWVHNDTQIR	YTDRVKTGVT	GEQIWLQINE	PTPDQKGKYT	1080
MELFDGKTGH	KKTVDLSGQA	FDEAFAEFQR	LKQAAIAEKN	RARVLGGLPD	VVTIQEGKAL	1140
NLSCTVWGDP	TPEVSWLKNE	KSFVSDANCI	LKFEAGKNVS	FTISAVSTHD	SGKYSLVVKN	1200
KYGTETSDVT	VIVITVSGQT	DTAKHTVSIS	GQSQNNITVS	KKKKATTMSK	PQQEKEKKDI	1260
KPHVDEIIAT	BETSDSKEIP	RVEKEKNTE	NQQTAKKTHG	KAAAARTVP		1309

SEQ ID NO: 171 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Meleagris gallopavo

SEQUENCE: 171

MSILKIQARE	IFDSRGNPTV	EVDLYTNKGL	FRAAVPSGAS	TGIYEALRLR	DNDKTRYLGK	60
GVSKAVEHVN	KTIAPALISK	NVNVVEQEKI	DKLMLEMDGT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVP	YRIADLAGN	PEVILPVPAP	NVINGGSHAG	NKLAMQEFMI	LPVGADNFKE	180
AMRIGAEVYH	NLKNVIKEKY	GKDATNVGDE	GGFAPNILEN	KEALELLKTA	IGKAGYSDKV	240
VIGMDVAASE	FYRDGKYDL	FKSPDDPSRY	ISPDQLADLY	KGFVKNYPLV	SIEDPPDQDD	300
WAAWKFTAS	VGIQVVGDDL	TVTNPRIAK	AVEEKSCNCL	LLKVNQIGSV	TESLQACKLA	360
QSGNGWVMVS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	LRIEELGSK	420
ARFAGRNFRN	PRIN					434

SEQ ID NO: 172 moltype = AA length = 165
 FEATURE Location/Qualifiers
 source 1..165
 mol_type = protein
 organism = Meleagris gallopavo

SEQUENCE: 172

MAPKAKKRI	EGANSNVFSM	FEQAQIQEFK	EAFITMDQNR	DGFIDKADLR	DTFAALGRLN	60
VKNEEIDEMI	KEAPGPINFT	VFLTMFGEKL	KGADPEETIL	NAFKVDFPEG	KGLKSAYIKE	120
MLMTQGERFS	QEEIDQMFPA	FPPDVSGNLD	YKNLVHVITH	GEEKD		165

SEQ ID NO: 173 moltype = AA length = 1974
 FEATURE Location/Qualifiers
 source 1..1974
 mol_type = protein
 organism = Meleagris gallopavo

SEQUENCE: 173

MSSDAEMAIF	GEAAPYLRSK	EKERIEAQR	PFDKAAACFV	VDQKQMYVK	TIQSREGGKV	60
TVKTYDDTTV	TVKDDEVFPM	NPPKYDKIED	MAMMTHLEP	AVLYNLKERY	AAWMIYTYSG	120
LPCVTVPYK	WLPVYNPEV	SGYRGKQRQ	APPHIFSISD	NAYQFMLTDR	DNQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VTGEKKDQ	PSKMQGTLED	QIISANPLLE	AFGNKTVRN	240
DNSSRFKFI	RIHFGTTGKL	ASADIETYLL	EKSRTVFQLA	SERSYHIFYQ	IMSNKKPELI	300
GEYLLITTN	PYDPTYVSQ	EITVASIDDS	EELLATDNAV	DILGFSPEDE	VGMYKLTGAV	360
MHYGNMKFKQ	KQREEQAEPD	GTEVADKAAY	LMGLNSAELL	KALCYPRVKV	GNEYVTKGQN	420
VQQVYNSVGA	LAKSVYEKMF	LWMVTRINQQ	LDTKQPRQHF	IGVLDIAGFE	IFDFNSLEQL	480
CINFNTNEKLQ	QFFNHMFVL	EQBEYKKEGI	DWFEIDFGMD	LAACIELIEK	PMGIFPSILEE	540
ECMPFKATDT	SFKNKLYDQH	LKGSNNFQPK	KPAKGKAEAH	FSLIHYAGTV	DYNITGWLEK	600
NKDPLNETVV	GLYQKSSLLK	LSFLFSNYAG	AETSDSGSGK	KGGKKKGGSF	QTVSAVFREN	660
LNKLMTNLRS	THPHFVRLCI	PNETKTPGEM	DHYLVMHQLR	CNGVLEGIRI	CRKGFPSSRL	720
YADFKQRYKI	LNASAIPEGQ	FIDSKKASEK	LLSSIDVDHN	QYRFGHTKVF	FKAGLLGLLE	780
EMRDEKLVS	ITHMQAMCRG	YLMRTEFKKM	NARRESIYII	QYNVRAFPMN	KHWPWMKLYF	840
KMKPMLKSVE	SEKEMANMKE	EFEKTKEELA	KSEAKRKELE	EKMVSLLOEK	NDLQLQVQSE	900
SENLADEAER	CEGLIKSKIQ	LEARIKELNE	RMEDEEEMNA	ELTAKKRKLE	DECSELKKDI	960
DDLELTAKV	EKEKHATENK	VKNLTEEMAA	LDENISKLTK	EKKALQEAHQ	QTLEYLQAE	1020
DKVSILTKIK	TKLEQQVDDL	EGSLEQERKQ	RMDLERAKRK	LDGDLKMSQD	SIMDLKNDKQ	1080
QTDRLKKKD	FEISHLQSKI	EDQQAQSSQL	QKKIKELQAR	TEELEEEIEA	ERANCAKTEK	1140
QRADLSRELE	EISERLEEAG	GATAAQIEMN	KKRAEFQKM	RRDLEEATLQ	HEATAAALRK	1200
KHADSTAELG	EQIDNLQRVK	QKLEKEKSEL	KMEIDDLASN	MESVSKAKSN	LEKMCRALED	1260
QYSEVRTKDD	BHMRILINDM	TQKTRLQTE	GELTRQLEEK	ESLISQLTRG	KQAITQQTEE	1320
LKRQLEENK	AKNALAHGLQ	SARHDCDLLR	EQYEELEAK	GELQRLASKA	NSEVAQWRTK	1380
YETDAIQCME	ELEDAKKKLA	QRLQDSEEQI	EAVNSKCASL	EKTKQRLQGE	VDDLMIAYER	1440
SNAACAAPDK	KQKNFDKUMA	EWKQKYQESQ	AELEAAQKES	RTLSTEIFRM	KNAYEEVLDQ	1500
VETVRRENKN	LQQEISDLTE	QLAEAGKANH	GLEKAKKQTE	QEKYDLQAA	EEAEVSLHE	1560
EGKILRVQLE	LNQVKSQVDR	RSAEKDEEIQ	QLKKNHQVRL	ESMQTMLDAE	IRSRNDALRL	1620
KKKMEGDLND	MEIQLSRANC	QIAEIQKHLK	GMQGGQLKDSQ	LHLDLALREN	DDLKEQLAVV	1680
ERRNNLMTTE	LEQMRALAEQ	TERARKVSEQ	ELTDTSERVQ	ILHSQNTSL	NTKKKLEVDI	1740
THLQNEVEDS	IQEASNAEEK	AKKAITDAAM	MAEELKKEQD	TSALHERMKR	NLEQMVKDLQ	1800
HRLDAAEQLT	LKGGKKQLQK	LEARIHELEN	ELDIEQKRG	ESLKGARKYE	RRLKELTYQS	1860
EEDKKNILRL	QNLVDKQLK	VKAYKKQAE	AEEQASTNLS	RCRKTQHELE	EAEERADIAE	1920
CQVNKLRAKS	RDTGRQKSG	SSELGDNTFL	PVIWHNMNCNI	DRPETTPQVI	LLHQ	1974

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SEQ ID NO: 174 moltype = AA length = 615
FEATURE Location/Qualifiers
source 1..615
 mol_type = protein
 organism = Meleagris gallopavo

SEQUENCE: 174

MKWVTLISFI	PLFSSATSRN	LQRVARDAEH	KSEIAHRYND	LKEETFKAVA	MITFAQYLQR	60
CSYDGLSKLV	KDVVDLAQKC	VANEDAPECS	KSLPSIILDE	ICQVEKLRDS	YGAMADCCSK	120
ADPERNECFL	SFKVSDPFI	QPYQRPASDV	ICKEYQDNRV	SLLGHFIYSV	ARRNPYMYAP	180
TILSPAADYE	HALQSCQES	DVGACLDAKE	TAMREKAKKI	SVKQQYSCGI	LKKFGDRVFO	240
AEKLVRLSQK	YPKAPFSEVA	KLVRSEKDVH	KECCEGDMVE	CMDDMAEII	NMCSKQDVFS	300
SKIKGCCCKP	IMERSQCIME	AEFDEKPADL	PSLVEKYIQD	KEVCKSFEAG	HDAFMSEFVY	360
EYSRRHPFES	TQLILRIAAG	YESLLEKCKC	TDNPAECYAN	AQDQLNQHIK	ETQDVVKTNC	420
DLLSAHGEPD	FLKSILIRYT	KKMPQVSTET	LLEIGKKMTA	IGTKCCQLSE	DRMACSEGY	480
LSIVIHDMCR	RQETTPINDN	VSHCCSDSYA	HRRPCFTAMG	VDTKYVPPPF	NPDMFSFDEK	540
LCSAPAEERE	VGQMKLLINL	IKRKPQMTTE	QIKTIADGFT	AMVDKCKCKS	DINTCFGEEG	600
ANLIVQSRAT	LGIGV					615

SEQ ID NO: 175 moltype = AA length = 381
FEATURE Location/Qualifiers
source 1..381
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 175

MPFSNSHLL	KMKHSATDEF	PDLSAHNHNM	AKVLTLDLYK	KLRDKQTPSG	FTLDDVIQTG	60
VDNPGHPFIM	TVGCVAGDEE	SYEVFKELFD	PVIEDRHGGY	KPTDQHKTDL	NAGNLQGGDD	120
LDPNVVLSSR	VRTGRSIRGF	CLPPHCSSRG	RRAIEKLSVE	ALGSLGGDLK	GKYYALRNMT	180
DAEQQLIDD	HFLFDKPVSP	LLLASGMARD	WPDARGIWHN	DNKTFLVWIN	EEDHLRVISM	240
QKGGNMKEVF	TRPCTGLTQI	ENLFKSKNHE	FMWNPHLGYI	LTCPSNLGTG	LRAGVHIKLP	300
HLGKHEKFGE	VLLKLRQKQR	GTGGVDTAAV	GGVFDVSNAD	RLGFSEVELV	QMVVDGVKLL	360
IMEKRLEKKG	QSIDDLIPAQ	K				381

SEQ ID NO: 176 moltype = AA length = 897
FEATURE Location/Qualifiers
source 1..897
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 176

MNSMNQIETN	MQYTYNYEED	EYMTQEEEDW	RDLLDPAPAE	KQQRKTFTAW	CNSHLRKAGT	60
QIENIEEDFR	NGLKMLLLE	VISGERLPKP	DRGKMRFHKI	ANVNKALDYI	ASKGVKLVI	120
GAEIIVDGNV	KMTLGMWITI	ILRFAIQDIS	VEETSAKEGL	LLWCQRKTAP	YRNVNIQNPH	180
LSWKDGLGLC	ALIHRRPDL	IDYSKLNKDD	PIGNINLAME	IAEKHLDIPK	MLDAEDIVNT	240
PKPDERAIMT	YVSCFYHAF	GAEQAETAAN	RICKVLAVNQ	ENERLMEEYE	RLASELLEWI	300
RRTIPWLENR	TPEKTMQAMQ	KKLEDPRDYR	RKHKPPKVQE	KQCLEINFNT	LQTKLRISNR	360
PAFMPSEGMK	VSDIAGAWQR	LEQAEKGYYE	WLLNEIRRL	RLEHLAEKFR	QKASTHEQWA	420
YGKEQILLQK	DYESASLTFV	RAMLKHEAF	ESDLAAHQDR	VEQIAAIAQE	LNELDYHDAA	480
SVNDRQCQIC	DQWDSLGLTL	QKRREALERT	EKLEITIDQL	HLEFAKRAAP	FNNWMEGAME	540
DLQDMFIVHS	IEETQSLISA	HDQFKATLPE	ADGERQAILS	IQNEVEKVIQ	SYSMRISASN	600
PYSTVTVEEI	RTKWEKVKQL	VPQRDQSLQE	ELARQHANER	LRRQFAAQAN	VIGPWIQTKM	660
EEIARSSSIEM	TGPLEDQMNQ	LKQYEQNIIN	YKHNDKLEG	DHQLIQEALV	FDNKHTNYTM	720
EHIRVGWELL	LTTIARTINE	VETQILTRDA	KGITQEQMND	FRASFNFHFR	RKNGLMDHDD	780
FRACILSMGY	DLGEAEFARI	MSLVDPNQGG	TVTQFSFIDF	MTRETADTDT	AEQVIASFRI	840
LASDKPYILA	DELRRLEPPE	QAQYCIKRM	PYTGPGSVPG	ALDYTSFSSA	LYGESDL	897

SEQ ID NO: 177 moltype = AA length = 843
FEATURE Location/Qualifiers
source 1..843
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 177

MAAPLSDGKEK	RKQISVRGIA	GLGDVAEVRK	SFNRHLHFTL	VKDRNVATPR	DYFFALAHTV	60
RDHLVGRWIR	TQOHYYERDP	KRIYYLSLEF	YMGRTLQNTM	VNLGLQNACD	EAIYQLGLDL	120
EELIEIEEDA	GLNGGGLRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	IVDGWQVEEA	180
DDWLRYGNPW	EKARPEYML	VHFYGRVEHT	PEGVKWVDTP	VVLAMPYDTP	VPGYKNNTVN	240
TMRLWSAKAP	NDFNLQEFNV	GDYIEAVLDR	NLAENISRLV	YPNDNFFEGK	ELRLKQYFV	300
VAATLQDIIR	RFKSSKFCGR	DPVRTSFETF	PDKVAIQLND	THPALSIPEL	MRILVDVEKV	360
DWDKAWETIK	RTCAYTNHTV	LPEALERWPV	SMFEKLLPRH	LEIIYALNQM	HLDRVAALYP	420
GDVRLRLRMS	VIEEGDCKRI	NMAHLCVIGS	HAVNGVARIH	SDIVKHSVFK	DFYELEPEKF	480
QNKTNGITPR	RWLLLCNPLG	ADVIAEKIGE	GFVTDLSQLK	KLLDFIDNEM	FIRDVAKVKQ	540
ENKLFKSAYL	EEQYVKIKNP	SSMFDVQVKR	IHEYKRLQLN	CLHAITLYNR	IRSDPSKSPV	600
PRTIMIGGKA	APGYHMAKMI	IKLITSIGEV	INNDPYVGDK	LKVIFLENYR	VSLAEKVIPA	660
ADLSQQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	AEAGEENLFF	IFGMRVEDVE	720
ALDRKGYNAL	EYYERLPELR	QAIDQISSGF	FSPRDPGCFR	DVVNMLMHHD	RFKVFDAYEA	780
YIKCQGQVDQ	LFMDPREWTK	KVIRNIACSG	KFSSDRITITE	YAQEIWGVPE	SSTKIPPPNI	840
PRD						843

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SEQ ID NO: 178 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 178

MTHQFPALSP	EQKKALSDIA	QRIVASGKGI	LAADESVGTM	GNRLQRINVE	NTEENRRAPR	60
EILFSSDTSI	NQSIGGVILF	HETLYQKDSS	GKPPPAIIE	KGIVVGIKLD	KGTSPLAGTN	120
GETTIEGLDG	LSERCAQYKK	DGVDFGKWRA	VLKITSTTPS	QLAIQENANT	LARYASICQQ	180
NGLVPIVEPE	VLPDGDHDLQ	RCQYVTEKVL	AAVYKALNDH	HVYLEGTLLK	PNMVTAGHAC	240
TKKYTPQDVA	AATVTTLRLT	VPAAVPGICF	LSGGQSEEEA	SLNLNAMNQS	SLPKPWKLTF	300
SYGRALQASA	LAAWLGKSEN	KKAAQEAFRK	RAQINSLACR	GQYVVSCKTD	AAATQSLFTA	360
SYTY						364

SEQ ID NO: 179 moltype = AA length = 192
 FEATURE Location/Qualifiers
 source 1..192
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 179

MAPKKDVKKP	AAAAAPAPAP	APAPAPAPAK	PKEPAIDLKS	IKIEFSKEQQ	DDFKEAFLLF	60
DRTGDAKITL	SQVGDIVRAL	GQNPTNAEIN	KILGNPSKEE	MNAKKITFEE	FLPMLQAAAN	120
NKDQGTIEDF	VEGLRVFDKE	GNGTVMGAEI	RHVLATLGEK	MTEEEVDELM	KGQEDSNGCI	180
NYEAFVKHIL	SV					192

SEQ ID NO: 180 moltype = AA length = 339
 FEATURE Location/Qualifiers
 source 1..339
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 180

MVKVGVNGFG	RIGRLVTRAA	ILSGKVQVVA	INDPFIDLNY	MVYMFKYDST	HGHFRGTVKA	60
ENGKLVINGN	AITIFQERDP	SNIKWADAGA	EYVVESTGVF	TMEKAGAGHL	KGGAKRVIIS	120
APSADAPMFV	MGVNHKKYDK	SLKIVSNASC	TTNCLAPLAK	VIHDNFGIVE	GLMTTVHAIT	180
ATQKTVDGPS	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCRLLEK	AKYDDIKRVV	KAAADGPLKG	ILAYTEDQVV	SCDFNGDSHS	STFDAGAGIA	300
LNDHFVKLV	WYDNEYGYSN	RVLALSTLLH	AANIFPSSR			339

SEQ ID NO: 181 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 181

MEAIKKKMQM	LKLDKENALD	RAEQAEAEQK	QAEERSKQLE	DELAAMQKKL	KGTEDELDKY	60
SEALKDAQEK	LELAEKKAAD	AEAEVASLNR	RIQLVEEELD	RAQERLATAL	QKLEEAEEKAA	120
DESERGMKVI	ENRALKDEEK	MELQEIQLKE	AKHIAEEADR	KYEEVARKLV	IIEGDLERTE	180
ERAELAESKC	SELEEEELKNV	TNNLKSLEAQ	AEKYSQKEDK	YEEIEKILTD	KLKEAETRAE	240
FAERSVAKLE	KTIDDELEDEL	YAQKLKYKAI	SEELDHALND	MTSI		284

SEQ ID NO: 182 moltype = AA length = 1247
 FEATURE Location/Qualifiers
 source 1..1247
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 182

MPEPVKKTVS	AFTKKPKTTE	VTAGSTAVFE	AETEKASVKV	KWQRAGTEIT	DSEKYVIKAE	60
GNKHSLTINN	VVKEDDVTYA	VIAGSSKVKF	ELKVKEPEKS	ETVTPAEAAP	APAASELPAP	120
PAESSQNTVE	PSAGTQPEEP	ADPIGLFVTR	PQDGEVTVGG	NITFTAKVAG	ESLLKKPSVK	180
WPKGKWMDLA	SKVGQHLQLH	DSYDRNNKVY	TFELEIIEAN	MTFAGGYRCE	VSTKDKFDSS	240
NFNLTVTNEAP	LSGDLDIRAA	FRRTTEGHDE	AGELNFSALL	KKRGDGKTES	QPDVDVWEIL	300
RKAPPSEYEK	IAFYQGITDL	RGMLKRLKRI	KKEEKKSTAF	LKKLDPAYQV	DKGQKIKLMV	360
EVANPDADVK	WLKNGQEIQV	SGSRYIFEAV	GNKRILTIIN	CSLADDAAYE	CVVGEEKSFT	420
ELFVKEPPIL	ITRPLEDQMV	MVGERVEFEC	EVSEEGATVK	WEKDGVELTR	EETFKYRFKK	480
DGKKQYLIIN	ESTKEDSGHY	TVKTNGGVSV	AELIVQEKKL	EVYQSIADLT	VKARDQAVFK	540
CEVSDENVKG	IWLKNGKEVV	PDERIKISHI	GRIHKLTIED	VTPGDEADYS	FIPQGFAYNL	600
SAKLQFLEVK	IDFVPREEPP	KIHLDCLGQS	PDTIVVVAGN	KLRLDVPISG	DPTPTVIWQK	660
VNKKTDLLRS	NDDSMTPSEN	SSDLNTDSKL	LFSEGRVRV	EKHEDHCVFI	IEGAKEDEGE	720
VYRVIVKNPV	GEDKADITVK	VIDVPDPPEA	PKISNIGEDY	CTVQWQPPKY	DGGQPVLGVI	780
LERKKKKSyr	WMRLNFLLLK	ELTYEAKRMI	EGVYVEMRIY	AVNSIGMSRP	SPASQPFMPI	840
APPSPTHTFT	VEDVSDSTVA	LKWRPPERIG	AGGLDGYIVE	YCKEGSTEWI	PALPGLTERT	900
SALIKDLVTG	DKLHFRIKAI	NLAGESGAAI	IKEPITVQEI	LQRPKIWLPR	HLRQTLVKKV	960
GETINIVIPF	QGGPRPKITW	TKDGQSLDAK	DVGIRSNNTD	TILFIRKAEI	HHSYGAYEITL	1020
QIENMTDKVA	ITIQIIDLKPG	PPQNIKLTDV	WGFNAALEWT	PPQDDGNAQI	LGTYVQKADK	1080
KTMWYTVDFD	HYRRTNVSVS	ELTMGNEYFF	RVSSENLCGL	SETAATTKNP	AYIQKTGTTY	1140
KPPCYKEHDF	SEPPKFTHPL	VNRSVIAGYN	TTLSCAVRGI	PKPKIYWKYN	KVDLSGDAKY	1200

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RMFSKQGVLT LEIRKPTPPD GGPYTCKAVN ECGEAEIECK LDVRVPQ 1247

SEQ ID NO: 183 moltype = AA length = 1451
FEATURE Location/Qualifiers
source 1..1451
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 183

MSSLAVPFYQ	RRHKHFQDSY	RNIQTRYVLE	EYAARKAASR	QAAYESSGL	GKTCRLCAR	60
RARTLAHETM	QESRKRTHEQ	KSHASEEKRI	RFASELSALE	KEIHTARHRA	REQLDRLAIQ	120
RMVEENNAME	RHAVEERISR	APEILVRLRS	HTVWEKMSVS	LCFTVQGFPT	PVVQWYKNEE	180
LITPASDPAP	YRVESKYGVH	VLHINRADFD	DSATYSAVAT	NVHGQASTNC	AVVRRFRFEG	240
EEPHPAGIMP	FHLPLSYDVC	FTHFDVQFLE	KPGVTFATEG	ETLTLKCSVL	VTPELKRLRP	300
RAEWYRDDVL	IKDSKWTCLY	FGEGQAALSF	THLNKDDEGL	YTLRMITKGG	VNECSAYLFV	360
RDADALIAGA	PGAPMDVKCH	DANRDYVIVT	WKPPNTTSQS	PVIGYFVDKC	EVGLENWVQC	420
NDAPVKICKY	PVTGLYEGRS	YIFRVRVANN	AGISRPSRVS	EPVAALDPVD	LERTQTIHVD	480
EGRKIVISKD	DLEGDIQIPG	PPTNVHASEI	SKTYVVLSDW	PPVPRGREPL	TYFIEKSMVG	540
SGSWQRVNAQ	VAVKSPRYAV	FDLAEGKPYV	FRVVSANKHG	ISDPSEITEP	IQPQDTVVVP	600
SAPGRVVATR	PVSTDYGVS	DKPKHEEDLY	GYIDYSVVG	SNHWEPSNHK	PIKYNRFVVH	660
GLETGEQYIF	RVKAVNAVGF	SENSQSESAI	KVQAALTCPS	YPHGITLLNC	DGHSMTLGWK	720
APKYSGGSPI	LGYYMDKREA	NHQNWHEVNS	SLITRTIYTV	EDLTEDAFYE	FKIAAANIVG	780
IQQSPDPSH	FKCKAWTMPE	PGPAYDLTVC	EVRNTSLVLL	WKAPVYEGKS	PITGYLVDYK	840
EVDKEDWITA	NEKPTAQRYP	KVTDLHEGHT	YVFKVRAVND	AGVGKSSEVS	EPVLVEAKPG	900
TKEIFSGVDD	BGNIYLAPEC	KEATDASHFA	WGKSYEEIED	SDKYKIESEG	DHSLKLYFKNP	960
DKSDLGTYSI	SVSDTDGVSS	SFVMDDEELE	RLMAISNEIK	NPTIPLKSEL	AYEVLEKGEV	1020
RFWIQAESLS	PDSTFRFVIN	DKEVENSDRH	KINCDRPNGI	IEMVMDRFTI	DNEGTYTVQI	1080
QDGKAKNQSS	LVLIGDAFKA	VLAESLQKRE	EFLRKQGPHE	SEPLYWEVTE	ECEVLLACKV	1140
ANTKKETVFK	WYKDGSGIDV	DEMPDLQKGE	CHLSIPRLSR	KDEGVYKATL	SDDRGRHDVST	1200
LELSGKVYDD	IILALSRVSG	KTASPLKILC	TEEGIRLQCF	LKYYNEMMKI	SWSHRESKIS	1260
SGEKMKIGGG	ENIAWLQITE	PTEKDKGNYT	FEIPGDKESF	KRTLDLSGQA	FDDAYAEFQR	1320
LKAAAFAEKN	RKGVIGGLPD	VVTIMDGKTL	NLTCTVFGNP	DPEVVWFKNP	KALELNEHYV	1380
VSLEQGGKYAS	LTIKGVTSED	SGKFSIHVKN	KYGETVDVTD	VSVYRHGEKV	PEVKPGQLAK	1440
PRPIPPSEDS	S					1451

SEQ ID NO: 184 moltype = AA length = 408
FEATURE Location/Qualifiers
source 1..408
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 184

MPERSWIPVG	TPLLRTWCTH	TKACFEQRSP	AVPPLVSTKH	WSSETTTSPV	SLEKGLSVVD	60
QEKIDNLMLE	MDGTENSKKF	GANAILGVSL	AVCKAGAAEK	DVPLYRHIAD	LAGNSDLILP	120
VPAFNVINGG	SHAGNKLAMQ	EFMILPVGAE	SFRDAMRIGA	EVYHNLKSVI	KEKYGKDATN	180
VGDEGGFAPN	ILENSEALEL	LKEAIDKAGY	TDKIVIGMDV	AASEFYRDGK	YDLDFKSPDD	240
PSRYISADEL	GDLYQSFVRD	YPVVSIEDPF	DQDDWEAWSK	FTANVGIIQV	GDDLTVTNPK	300
RIERAVEEKA	CNCLLLKYNQ	IGSVTEAIQA	CKLAQENGWG	VMVSHRSGET	EDTFIADLVV	360
GLCTGQIKTG	APCRSERLAK	YNQLMRIEEE	LGDEARFAGH	NFRNPSVL		408

SEQ ID NO: 185 moltype = AA length = 171
FEATURE Location/Qualifiers
source 1..171
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 185

MQGENHRAPK	KAKKRIEGAN	SNVFSMFEQA	QIQEFKEAFT	IMDQNRDGI	DKADLRDTFA	60
ALGRNLVKNE	EIDEMIKEAP	GPINFVFLT	MPGEKLKPAD	PEETILNAFK	VFDPEGKGLK	120
SAYIKEMLMT	QGERFSQEEV	DQMFAPPPD	VSGNLDYKNL	VHVITHGEEK	D	171

SEQ ID NO: 186 moltype = AA length = 1626
FEATURE Location/Qualifiers
source 1..1626
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 186

FSISDNAYQF	MLTDRENQSI	LITGESGAGK	TVNTRKVIQY	FATIAASGEK	KKEEQSGKMQ	60
GTLEDQIISA	NPLLEAFGNA	KTVNRDNSSR	FGKFIRIHFG	ATGKLASADI	ETVLLKESRV	120
TFQLKAERSY	HIFYQIMSNK	KPELIDMLLI	TTNPYDYHFV	SQGEITVASI	NDQEEELMATD	180
SAIDILGFPA	DEKVAIYKLT	GAVMHYGNLK	FKQKQREEQA	EPDGTVEADK	AAYLMGLNSA	240
DLLKALCYPR	VKVGNEYVTK	GQTVQQVNNS	VGALAKAVYE	KMFLWMVVRI	NQQLDTKQPR	300
QYFIVGLDIA	GFETFDNFSL	EQLCINFNTNE	KLQQFFNHHM	FVLEQEEYKK	EGIEWEFIDF	360
GMDLAACIEL	IEKPMGLFSI	LEEECMFPKA	TDTSFKNKLY	DQHLGKSNF	QKPKPAKGKV	420
EAHFSLVHYA	GTVDYNITGW	LEKNKDPLNE	TVIGLYQKSS	VKTLALLFAN	YGGAEINPQK	480
IIIKDTFKGS	SFQTVSALFR	ENLNKLMTNL	RSTHPHFVRC	IIPNETKTPG	AMEHELVLHQ	540
LRNCGVLEGI	RICKRGFPPSR	VLYADFKQRY	KVLNASAIPE	GQFIDSKKAS	EKLKLSIDVD	600
HTQYKFGHTK	VFFKAGLLGL	LEEMRDEKLA	QLITRTQARC	RGFLMRVEYQ	RMVERRESIF	660
CIQYNVRAFM	NVKHWPWMKL	FFKIKPLLSK	AESEKEMANM	KEEFKTKKEE	LAKSEAKKKE	720

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LEEKMKVLQVQ	EKNDLQQLQVQ	AEADALADAE	ERCDQLIKTK	IQLEAKIKEV	TERAEDEEEI	780
NAELTAKKRR	LEDECSSELKK	DIDDLELTLA	KVEKEKHATE	NKVKNLTEEM	AALDEETIAKL	840
TKEKKALQEA	HQQTLDLQQA	EEDKVNTLTK	AKTKLEQQVD	DLEGSLEQEK	KLRMDLERAK	900
RKLEGLDLKLS	QDSIMDLEND	KQQLDEKLKK	KDFEISQIQS	KIEDEQALGM	QLQKKIKELQ	960
ARIEELEEEI	EAERTSRAKA	EKHRADLSRE	LEEISERLEE	AGGATAAQIE	MNKKREAEFQ	1020
KMRRDLEEAT	LQHEATAAAL	RKKHADSTAE	LGEQIDNLQR	VKQKLEKEKS	ELKMEIDDLA	1080
SNMESVSKAK	ANLEKMCRTL	EDQLSEIKTK	EEHQRMIND	LTAQRARLQT	ESGEYSRQVE	1140
EKDSQISQLS	RSKQAFQTOI	EELKRHLEEE	IKAKNALAHA	LQSAHDCDL	LREQYEEEQE	1200
AKGELQRLS	KANSEVAQWR	TKYETDAIQR	TEELEEAQKK	LAQRLQDAEE	HVEAVNAKCA	1260
SLEKTKQRLQ	NEVEDLMIDV	ERSNAACAAL	DKKQKNFDKI	LAEWKQKYEE	TQAELEASQK	1320
ESRSLSTELF	KMKNAYEESL	DHLETLEKRE	KNLQQEISDL	TEQIAEGGKA	IHELEKVKKQ	1380
IEQEKSEIQA	ALEEAEASLE	HEEGKILRLQ	LELNQVKSEI	DRKIAEKDEE	IDQLKRNLHR	1440
IVESMQSTLD	AEIRSRNEAL	RLKKKMEGDL	NEMEIQLSHA	NRQAEEAQKN	LRNTQGVVKD	1500
TQIHLDLALR	TQEDLKEQVA	MVERRANLLQ	AEIEELRAAL	EQTERSARKA	EQELMDASER	1560
VQLLHTQNTS	LINTKKKLET	DIQIQSEME	DAIQEARNAE	EKAKKAITDA	SMAAEELKKE	1620
QDTSAH						1626

SEQ ID NO: 187 moltype = AA length = 565
 FEATURE Location/Qualifiers
 source 1..565
 mol_type = protein
 organism = Anas platyrhynchos

SEQUENCE: 187

MITFAQYLQR	CSYDGLSKLV	KDVVDLAQKC	VANEDAPECS	KSLPSIFLDE	ICQVEKLKRS	60
YGAMADCCAK	SDPERNECFL	SFKIPQPDFV	QPYQRPASDV	ICKEYEDNRV	SFLGHFIYSV	120
ARRNPFLYAP	TILGVAADYE	HALKSCCKES	DVGACLDGKE	TGIREKVKKI	SVKQQYSCGI	180
LKKFGDRIQ	ADKLALLSQK	YPKASFAEIS	KLIHDKVDVY	KECCGDMVE	CMDRAELVN	240
YMSCKHEAFS	SKIKDCCEKP	IVERSQCIME	AEFDEKPADL	PSLVEKVIQD	KEVCKSFEAG	300
HDAFMSEFVY	EYSRRHPEFS	TQLILRIAKG	YETLLEKCK	TDNPAECYAN	AQDQLNEHIK	360
ETQDVVTKNC	DLTTHGEAD	FLKALLIRYT	KMPQVSTET	LLEIGKKMTG	VGTKCKCKLPE	420
DRRLPCSEGY	LSMVIQDMCR	RQETSPINDN	VSHCCSDSYA	YRRPCFSAMG	VDTKYVPPPF	480
NPDVFNFDEK	LCSAPAEERA	EGQMKLLINL	IKRKPQMTTE	QIKTIADGFT	AMVDKCKCKQS	540
DIETCFGEEG	ANLEVQSRAT	LGIGA				565

SEQ ID NO: 188 moltype = AA length = 418
 FEATURE Location/Qualifiers
 source 1..418
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 188

MCCLKCQQGF	WDFCAPAVLA	RGPCFGCGAL	VTAVGSAMPF	SNSHNLLKMK	HSATDEFDPDL	60
SAHNHMAKV	LTLDLYKKLR	DKQTPSGFTL	DDVIQTGVND	PGHPFIMTVG	CVAGDEESYE	120
VPKELFDPVI	EDRHGGYKPT	DQHKTDNLNAG	NLQGGDDLDP	NYVLSSRVRT	GRSIRGFCLP	180
PHCSRGERRA	IEKLSVEALG	SLGGDLKGY	YALRNMTDAE	QQQLIDDHFL	FDKPVSPLLL	240
ASGMARDWPD	ARGVHNDNK	TFLVWINEED	HLRVISMQKG	GNMKEVFTFR	CTGLTQIENL	300
FKSKNHFMFW	NPHLGYLLTC	PSNLGTGLRA	GVHIKPLHLG	KHEKFGEVLK	KLRLQKRGTTG	360
GVDTAAVGGV	FDVSNADRLG	FSEVELVQMV	VDGVKLLIEM	EKRLEKQCSI	DDLIPAQK	418

SEQ ID NO: 189 moltype = AA length = 885
 FEATURE Location/Qualifiers
 source 1..885
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 189

RAGDKPQDEY	MTQEEEWDRD	LLLDPAWEKQ	QRKTFTAWCN	SHLRKAGTQI	ENIEEDFRNG	60
LKLMLLLEVI	SGERLPKPD	GKMRPFHKIAN	VNKALDYIAS	KGVKLVSIGA	EEIVDGNVNM	120
TLGMIWTIIL	RFATQDISVE	ETSAKEGLLL	WCQRKTAPYR	NVNIQNFHLS	WKDGLGLCAL	180
IHRHRPDLID	YSKLNKDDPI	GNINLAMEIA	EKHLDIPKML	DAEDIVNTPK	PDERAIMTYV	240
SCFYHAFAGA	EQAETAANRI	CKVLAVNQEN	ERLMEEYERL	ASELLEWIRR	TIPWLENRTP	300
EKTMQAMQKK	LEDPRDYRRK	HKPPKVQEK	QLEINFNTLQ	TKLRISNRPA	FMPSEGKMS	360
DIAGAWQRL	QAEKGYEEWL	LNELRRRLER	EHLAEKFRQK	ASTHEQWAYG	KEQILLQKDY	420
ESASLTEVRA	MLRKHEAFES	DLAAHQDRVE	QIAAIAQELN	ELDYHDAASV	NDRCKQICDQ	480
WDSLGLTLTK	RREALERTEK	LLETIDQLHL	EFAKRAAPFN	NWMEGAMEDL	QDMFIVHSIE	540
EIQLSLITAH	QFKATLPEAD	GERQAILSIQ	NEVEKVIQSY	SMRISASNPY	STVTVEEIRT	600
KWEKVQLVLP	QRDQSLQEEL	ARQHANEERL	RQFAAQANVI	GPWIQTKMEE	IARSSIEMTG	660
PLEDQMNQLK	QYEQNIINIK	HNIDKLEGDH	QLIQEALVFD	NKHTNYTMEH	IRVGWELLLT	720
TIARTINEVE	TQILTRDAKG	ITQEQMNDFF	ASFNFHFRRK	NGLMDHDDFR	ACLISMGYDL	780
GEAEFARIMS	LVDPNQGQTV	TFQSFIDFMT	RETADTDTA	QVIASFRLA	SDKPYILADE	840
LRRELPEQA	QYCIKRMPPY	TGPGSVPGAL	DYTSFSSALY	GESDL		885

SEQ ID NO: 190 moltype = AA length = 843
 FEATURE Location/Qualifiers
 source 1..843
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 190

-continued

MAAPLSDGK	RKQISVRGIA	GLGDVAEVRK	SPNRHLHFTL	VKDRNVATPR	DYFFALAHTV	60
RDHLVGRWIR	TQQHYERDP	KRIYYLSLEF	YMGRTLQNTM	VNLGLQNACD	EAIYQLGLDL	120
EELLEEIEDA	GLGNGGLGRL	AACFLDSMAT	LGLAAYGYGI	RYEFGIFNQK	IVDGWQVEEA	180
DDWLRYGNPW	EKARPEYMLP	VHFGYRVEHN	PEGVKWVDQ	VVLAMPYDTP	VPGYKNNTVN	240
TMRLWSAKAP	NDFNLQEFNV	GDYIEAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQEYFV	300
VAATLQDIIR	RFKSSKFGCR	DPVRTSFETF	PDKVAIQLND	THPALSIPEL	MRILVDVEKV	360
DWDKAWEITK	RTCAYTNHTV	LPEALERWPV	SMFEKLLPRH	LEIIYALNQM	HLDRVAALYP	420
GDVDRLRMS	VIEEGDCKRI	NMAHLCVIGS	HAVNGVARIH	SDIVKHSVFK	DFYELEPEKF	480
QNKTINGITPR	RWLLLCNPL	ADVIAEKIGE	GFVTDLSQLK	KLLDFIDNKM	FIRDVAKVQK	540
ENKLFKSAYL	EEQYKVKINP	SSMFDVQVKR	IHEYKRQLLN	CLHAITLYNR	IRSDPSKSFV	600
PRTIMIGGKA	APGYHMAKMI	IKLITSIGEV	INNDPYVGDK	LKVIFLENYR	VSLAEKVIPA	660
ADLSQQISTA	GTEASGTGNN	KFMLNGALTI	GTMDGANVEM	AEAGEENLF	IFGMRVEDVE	720
ALDRKGYNAL	EYERLPELR	QAIDQISSGF	FSPGDPGCFR	DVVNMLMHHH	RFKVFADYEA	780
YIKCQGQVDQ	LFMDPWEWTK	KVIRNIACSG	KFSSDRITITE	YAQEIWGVPE	SSTKIPPPNI	840
PRD						843

SEQ ID NO: 191 moltype = AA length = 364
 FEATURE Location/Qualifiers
 source 1..364
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 191

MTHQFPALSP	EQKKALSDIA	QRIVASGKGI	LAADSVGTM	GNRLQRINVE	NTEENRRAPR	60
EILFSSDTSI	NQSIGGVILF	HETLYQKDSS	GKPPPAIIE	KGIVVGIKLD	KGTSPLAGTN	120
GETTIEGLDG	LSERCAQYKK	DGVDGKWRRA	VLKITSTTPS	QLAIQENANT	LARYASICQQ	180
NGLVPIVEPE	VLPDGDHDLQ	RCQYVTEKVL	AAVYKALNDH	HVYLEGTLK	PNMVTAGHAC	240
TKKYTPQDVA	AATVTTLLRT	VPAAVPGICF	LSGGQSEEEA	SLNLNANMQS	SLPKPWKLTF	300
SYGRALQASA	LAAWLGKPEN	KKAAQEAPFR	RAQINSILACR	GQYVVSQKTD	AAATQSLFTA	360
SYTY						364

SEQ ID NO: 192 moltype = AA length = 333
 FEATURE Location/Qualifiers
 source 1..333
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 192

MVKVGVNGFG	RIGRLVTRAA	ILSGKVQVVA	INDPPIDLNY	MVYMFKYDST	HGHFRGTVKA	60
ENGKLVINGN	AITIFQERDP	SNIKWADAGA	EYVVESTGVF	TTMEKAGAHL	KGAKRVIIS	120
APSADAPMFV	MGVNHKEYDK	SLKIVSNASC	TTNCLAPLAK	VIHDNFGIVE	GLMTTVHAIT	180
ATQKTVDGFS	GKLWRDGRGA	AQNIIPASTG	AAKAVGKVIP	ELNGKLTGMA	FRVPTPNVSV	240
VDLTCRLEKP	AKYDDIKRVV	KAAADGPLKG	ILAYTEDQVW	SCDFNGDSHS	STPDAGAGIA	300
LNDHVFVKLV	WYDNEYGYSN	RVVDLMVHMA	SKE			333

SEQ ID NO: 193 moltype = AA length = 284
 FEATURE Location/Qualifiers
 source 1..284
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 193

MDAIKKKMQM	LKLDKENALD	RAEQAEADKK	AAEERSKQLE	DELVALQKKL	KGTEDELDKY	60
SESLKDAQEK	LELADKKATD	AESEVASLNR	RIQLVEEELD	RAQERLATAT	QKLEEAKEAA	120
DESERGMKVI	ENRAQKDEEK	MEIQEIQLKE	AKHIAEEADR	KYEEVARKLV	IIESDLERAE	180
ERAELSESKC	AEEELKKTIV	TNNLSLEAQ	AEKYSQKEDK	YEEBIKVLTG	KLKEAETRAE	240
FAERSVTKLE	KSIDDLLEDEL	YAQKLKYKAI	SEELDHALLND	MTSM		284

SEQ ID NO: 194 moltype = AA length = 1258
 FEATURE Location/Qualifiers
 source 1..1258
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 194

YPTTKPKTTE	VTAGSTAVFE	AETEKAGVKV	KWQRAGTEIT	DSEKYVIAKE	GNKHSLTINN	60
VVKEDDVTYA	VIAGSSVKVF	ELKVKEPEKS	ETVTPAEAAP	APAASELPAP	PAESSQNTTEV	120
PSAGTQPEEP	VDPIGLFVTR	PQDGEVTVGG	NITFTAKVAG	ESLLKKPSVK	WFKGKWMDLA	180
SKVGQHLQLH	DSYDRNNKVY	TFELEIIIEAN	MTFAGGYRCE	VSTKDKFDDSS	NFNLTVNEIY	240
HLCHDIVSFP	FPSKTNIFYL	LLRSTEGHDE	AGELNFSALL	KKRGDGKTES	QPDVDVWEIL	300
RKAPPSEYEK	IAFOYGITDL	RGMLKRLKRI	KKEEKKSTAF	LKKLDPAYQV	DKGQKIKLMV	360
EVANPDADV	WLKNGKEVVP	SGRYIFEAVG	NKRILITINH	SLADDAAYEC	VVGEEKSFTE	420
LFVKEPPILI	THPLEDQVMV	VGERVEFECE	VSEEGATVKW	EKDGVELTRE	ETPKYRFPKD	480
GKKQVLIINE	STKEDSGHYT	VKTNGGVSV	ELIVQEKKLE	VYQSIADLT	KARDQAVFKC	540
EVSDENVKGI	WLKNGKEVVP	DERIKISHIG	RIHKLTIEDV	TPGDEADYSF	IPQGFAYNLS	600
AKLQFLEVKI	DFVPREEPPK	IHLDCLGQSP	DTIVVVAGNK	LRLDVPISGD	PTPTVIWQKV	660
NKKTDLRLSN	DDSMTPSENS	SDLNTDSKLL	FESEGRVRVE	KHEDHCVFII	EGAEKEDEGV	720
YRVIVKNPVG	EDKADITVKV	IDVPDPPEAP	KISNIGEDYC	TVQWQPPKYD	GGQPVLGYYL	780
ERKKKKSYYW	MRLNFDLLKE	LYTEAKRMIE	GVVYEMRIYA	VNSIGMSRPS	PASQPFMPIA	840
PPSEPTHTFTV	EDVSDSTVAL	KWRPPERIGA	GGLDGYIVEY	CKEGSTEWIP	ALPGLTERTS	900

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ALIKDLVTGD	KLHPRIKAIN	LAGESGAII	KEPITVQEIL	QRPKIWLPRH	LRQTLVKKVG	960
ETINIVIPFQ	GKPRPKITWT	KDQSLDAKD	VGIRNSNTDT	ILFIRKAEHL	HSGAYEITLQ	1020
IENMTDKVAI	TIQIIDKPGP	PQNIKLTDVW	GFNAALEWTP	PQDDGNAQIL	GYTVQKADKK	1080
TMEWYTVFDH	YRRNTNCVSE	LIMGNEYFFR	VFSENLCGLS	ETAATTKNPA	YIQTGTITYK	1140
PPCYKEHDFS	EPPKFTHPLV	NRSVIAGYNT	TLSCAVRGIP	KPKIYWKYKN	MDLSGDAKYR	1200
MFSKQGVLT	EIRKPTPFDG	GFTTCKAVNE	RGEAEIECKL	DVRGNHMSIM	HNTNTAVK	1258

SEQ ID NO: 195 moltype = AA length = 1451
 FEATURE Location/Qualifiers
 source 1..1451
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 195

MSSLAVPFYQ	RRHKHFDQSY	RNIQTRYVLE	EYAARKAASR	QAAYYESSGL	GKTTTCRLCAR	60
RARTLAHEM	QESRKRTHEQ	KSHASEEKRI	RFASELSALE	KEIHTARHRA	REQLDRLAIQ	120
RMVEENMAME	RHAVEERISR	APILVRLRS	HTVWEKMSVS	LCFTVQGQPT	PPVQWYKNEE	180
LITPASDPAK	YRVESKYGVH	VLHINRADFD	DSATYSAVAT	NVHGQASTNC	AVVVRRFREG	240
EEPHFAGIMP	FHLPLSYDVC	FTHFDVQFLE	KFGVTFATEG	ETLTLKCSVL	VTPELKRLRP	300
RAEWYRDDVL	IKDSKWTCLY	FGEQQAALSF	THLNKDDDEGL	YTLRMITKGG	VNECSAYLFV	360
RDADALIAGA	PGAMPDVKCH	DANRDYVIVT	WKPPNNTTSQS	PVIGYFVDKC	EVGLENWVQC	420
NDAPVKICKY	PVTGLYEGRS	YIFRVRVAVN	AGISRPSPRS	EPVAALDPVD	LERTQTIHVD	480
EGRKIVISKD	DLEGDIQIPG	PPTNVHASEI	SKTYVVLSDW	PPVPRGREPL	TYFIEKSMVG	540
SGSWQRVNAQ	VAVKSPRYAV	FDLAEGKPYV	FRVVSANKHG	ISDPSEITEP	IQQPDTVVVP	600
SAPGRVVATR	NTKTSVVVQW	DKPKHEEDLY	GYIDYSVVVG	SNHWEPNHNK	PIKYNRFVVH	660
GLETGEQYIF	RVKAVNAVGF	SENSQSESAI	KVQAALTCPS	YPHGITLLNC	DGHSMTLGWK	720
APKYSGGSPI	LGYYMDKREA	NHQNWHVENS	SLITRTIYTV	EDLTEDAFYE	FKIAAANIVG	780
IQQPSDPSH	FKCKAWTMPE	PGPAYDLTVC	EVNRTSLVLL	WKAPVYEGKS	PITGYLVVDYK	840
EVDKEDWITA	NEKPTAQRYF	KVTDLHEGHT	YVFKVRVAVN	AGVGKSSEVS	EPVLVEAKPG	900
TKEIFSGVDD	EGNIYLAPEC	KEATDASHFA	WGKSYEEIED	SDKYKIELEG	DRSKLYFKNP	960
DKSGLGTYSI	SVSDTDGVSS	SFVMDDEELE	RLMAISNEIK	NPTIPLKSEL	AYEVLEKGEV	1020
RFWIQAESLS	PDSTFRFVIN	DKEVENSDRH	KINCDRPNGI	IEMVMDRFTI	DNEGTYTVQI	1080
QDGKAKNQSS	LVLIGDAPKA	VLAESLQQRK	EFLRKQGPFP	SEFLYWEVTE	ECEVLLACKV	1140
ANTKKETVFK	WYKDGSGIDV	DEMPDLQKGE	CHLSIPRLSR	KDEGVYKATL	SDDRGHVDST	1200
LELSGKVYDD	IILALSRVSG	KTASPLKILC	TEEGIRLQCF	LKYVNEEMKI	SWSHRESKIS	1260
SGEKMKIGGG	ENIAWLQITE	PTKDKGNYT	FEIFGDKESF	KRTLDSLQQA	FDDAYAEPQR	1320
LKAAAFAEKN	RGKVIIGLPD	VVTIMDGKTL	NLTCTVFGNP	DPEVWVFNKD	KALELNEHYV	1380
VSLEQGGYAS	LTIKGVTSED	SGKFSIHVKN	KYGGETVDVT	VSVYRHGEKV	PEVKPGQLAK	1440
PRPIPPSEGS	S					1451

SEQ ID NO: 196 moltype = AA length = 438
 FEATURE Location/Qualifiers
 source 1..438
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 196

MVFKMSILKI	HAREIFDSRG	NPTVEVDLYT	NKGLFRAAVP	SGASTGIYEA	LELRDNDKTR	60
YMGKGVSKAV	EHVNKTIAPA	LISKNVNVVE	QEKIDKLMLD	MDGSENKSKF	GANAILGVSL	120
AVCKAGAAEK	GVPLYRHIAI	LAGNPEVILP	VPAFNVINGG	SHAGNKLAMQ	EFMILPVGAD	180
SPKEAMRIGA	EVYHNLNKVI	KEYYKDATN	VGDEGGFAPN	ILENKEALEL	LKTAIGKAGY	240
SDKVVIQMDV	AASEFYRDGK	YDLDFKSPDD	PSRYISPDQL	ADLYKGFVKV	YPVVSIEDPF	300
DQDDWAAWK	FTGSVGIQV	GDDLTVTNPK	RIAKAVEEKS	CNCLLLKVNQ	IGSVTESLQA	360
CKLAQSNQGW	VMSHRSGET	EDTFIADLVV	GLCTGQIKTG	APCRSERLAK	YNQLLRIEEE	420
LGSKARFAGR	NFRNPRIN					438

SEQ ID NO: 197 moltype = AA length = 165
 FEATURE Location/Qualifiers
 source 1..165
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 197

MAPKKAKKRI	EGANSNVFSM	FEQAQIQEFK	EAFITMDQNR	DGFIDKADLR	DTFAALGRLN	60
VKNEEIDEMI	KEAPGPINFT	VFLTMFGEKL	KGADPEETIL	NAFKVFDPEG	KGLKSAYIKE	120
MLMTQGERFS	QEEVDQMAFA	FPPDVSQNL	YKNLVHVITH	GEEKD		165

SEQ ID NO: 198 moltype = AA length = 1945
 FEATURE Location/Qualifiers
 source 1..1945
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 198

MTSPDAEMAA	FGEAAPYLK	SEKERIEAQN	KPFDAKSSVF	VAHPKESFVK	GTIQSKESGK	60
VTVKTEGGET	LTVKEDQIFS	MNPPKYDKIE	DMAMMTHLHE	PAVLYNLKER	YAAWMIYTYS	120
GLFCVTVNRY	KWLPVYNFV	VLAYRGKKRQ	EAPPHIFSIS	DNAYQFMLTD	RENQSILITG	180
ESGAGKTVMT	KRIYQYFATI	AASGEKKKEE	QSGKMQGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSRFGFGF	TRIHFHGETK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IDMLLITTNP	YDYHFVSQGE	ITVASINDQE	ELMATDSAD	ILGPTADEKV	AIYKLTGAVM	360

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HYGNLKFKQK	QREEQAEPDG	TEVADKAAYL	MGLNSADLLK	ALCYPRVKVG	NEYVTKGQTV	420
QQVNNVSGAL	AKAVYEKMFL	WMVVRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNFNSLEQLC	480
INFNTNEKLQQ	FFNHHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	FKNKLYDQHL	GKSSNFQKPK	PAKGKAEAHF	SLVHYAGTVD	YNITGWLEKN	600
KDPLNETVIG	LYQKSSVKTL	ALLFANYGGA	EAEASGGGAS	GGKKGGKKKG	SSFQTVSALF	660
RENLNKLMTN	LRSTHPHFVR	CIIPNETKTP	GAMEHELVLH	QLRCNGVLEG	IRICRKGFP	720
RVLYADFKQR	YKVLNASAI	EGQFIDSKA	SEKLLGSIDV	DHTQYKFGHT	KVFFKAGLLG	780
LLEEMRDEKL	AQLITRTQAR	CRGFLMRVEY	QRMVERRESI	FCIQYNVRAF	MNVKHWPMWK	840
LFYKIKPLLK	SAESEKEMAN	MKEEFKTK	ELAKSEAKRK	ELEEKMKLV	QEKNDLQLQV	900
QAEADALADA	EERCQDLIKT	KIQLEAKIKE	VTAEDEEEE	INAELETAKR	KLEDECSSELK	960
KDIDDLLETL	AKVEKEKHAT	ENKVKNLTEE	MASLDETIK	LTKEKKALQE	AHQQTLDLQ	1020
AEEDKVNTLT	KAFTKLEQV	DDLEGSLEQE	KKLRMDLERA	KKRLEGLKL	SQDTIMDLN	1080
DKQQLDEKLK	KKDFEISQIQ	SKIEDEQALG	MQLQKKIKEL	QARIEELEE	IEAERTSRK	1140
AEKHRADLSR	ELEEISERLE	EAGGATAAQV	EMNKKREAEF	QKMRDLLEE	TLQHEATAAT	1200
LRKKHADSTA	ELGQIDNLQ	RVKQKLEKEK	SELKMEIDDL	ASNMESVSKA	KANLEKMCRT	1260
LEDQLSEIKT	KEEEHQRMIN	DLTAQRARLQ	TESGEYSRQV	EEKDSQISQL	SRSKQAFQ	1320
IEELKRHLLE	EIKAKNALAH	ALQSRHDCD	LLREQYEEEQ	EAKGELQRAL	SKANSEVAQW	1380
RTKYETDAIQ	RTEELEBAK	KLAQRLQDAE	EHVEAVNAK	ASLEKTKQRL	QNEVEDLMID	1440
VERTNAACAA	LDKKQKNFDK	ILAEWKQKYE	ETQAELEASQ	KESRSLSTEL	FKMKNAYEES	1500
LDHLETLKRE	NKNLQOEISD	LTEQIAEGGK	AIHELEKVKK	QIEQEKSEIQ	AALEEAEASL	1560
EHEEGKILRL	QLELNQVKSE	IDRKIAEKDE	EIDQLKRNHL	RIVESMQSTL	DAEIRSRNEA	1620
LRLKKKMEGD	LNEMEIQLSH	ANRQAAEAQK	NLRNTQAVLK	DTQIHLDDAL	RTQEDLKEQV	1680
AMVERRANLL	QAEIEELRAA	LEQTERSRSK	AEQELMDASE	RVQLLHTQNT	SLINTKKKLE	1740
TDIVQIQGEM	EETIQEARN	EAKAKAITD	ASMAEELKK	BQDTSALHER	MKNLDQTVK	1800
DLQHRLEDAE	QLALKGGKKQ	IQKLEARVRE	LEGDVDAEQK	RSABAVKGV	KYERRVKELT	1860
YQSEEDRKNI	LRLQDLVDKL	QMKVKSYSKRQ	ABEAEELSNV	NLSKFRKIQH	ELEEAEERAD	1920
IAESQVNLKL	AKSREFHGKK	IEEEE				1945

SEQ ID NO: 199 moltype = AA length = 1945
FEATURE Location/Qualifiers
source 1..1945
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 199

MTSPDAEMAA	FGEAAYPLRK	SEKERIEAQN	KPFDAKSSVF	VAHPKESFVK	GTIQSKESGK	60
VTVKTEGGET	LTVKEDQIFS	MNPPKYDKIE	DMAMMTHLHE	PAVLYNLKER	YAAWMIYTS	120
GLFCVTVNPN	KWLPVYNPEV	VLAYRGKKRQ	EAPPHIFSIS	DNAYQFMLTD	RENQSILITG	180
ESGAGKTVM	KRVIQYFATI	AASGEKKKEE	QSGKMGGTLE	DQIISANPLL	EAFGNAKTVR	240
NDNSSRFGFK	TRIHFGATGK	LASADIETYL	LEKSRVTFQL	KAERSYHIFY	QIMSNKKPEL	300
IDMLLITNTP	YDYHFVSQGE	ITVASINDQE	ELMATDSAID	ILGPTADEKV	AIYKLTGAVM	360
HYGNLKFKQK	QREEQAEPDG	TEVADKAAYL	MGLNSADLLK	ALCYPRVKVG	NEYVTKGQTV	420
QQVNNVSGAL	AKAVYEKMFL	WMVVRINQQL	DTKQPRQYFI	GVLDIAGFEI	FDNFNSLEQLC	480
INFNTNEKLQQ	FFNHHMFVLE	QEEYKKEGIE	WEFIDFGMDL	AACIELIEKP	MGIFSILEEE	540
CMFPKATDTS	FKNKLYDQHL	GKSSNFQKPK	PAKGKAEAHF	SLVHYAGTVD	YNITGWLEKN	600
KDPLNETVIG	LYQKSSVKTL	ALLFANYGGA	EAEASGGGAS	GGKKGGKKKG	SSFQTVSALF	660
RENLNKLMTN	LRSTHPHFVR	CIIPNETKTP	GAMEHELVLH	QLRCNGVLEG	IRICRKGFP	720
RVLYADFKQR	YKVLNASAI	EGQFIDSKA	SEKLLGSIDV	DHTQYKFGHT	KVFFKAGLLG	780
LLEEMRDEKL	AQLITRTQAR	CRGFLMRVEY	QRMVERRESI	FCIQYNVRAF	MNVKHWPMWK	840
LFYKIKPLLK	SAESEKEMAN	MKEEFKTK	ELAKSEAKRK	ELEEKMKLV	QEKNDLQLQV	900
QAEADALADA	EERCQDLIKT	KIQLEAKIKE	VTAEDEEEE	INAELETAKR	KLEDECSSELK	960
KDIDDLLETL	AKVEKEKHAT	ENKVKNLTEE	MASLDETIK	LTKEKKALQE	AHQQTLDLQ	1020
AEEDKVNTLT	KAFTKLEQV	DDLEGSLEQE	KKLRMDLERA	KKRLEGLKL	SQDTIMDLN	1080
DKQQLDEKLK	KKDFEISQIQ	SKIEDEQALG	MQLQKKIKEL	QARIEELEE	IEAERTSRK	1140
AEKHRADLSR	ELEEISERLE	EAGGATAAQV	EMNKKREAEF	QKMRDLLEE	TLQHEATAAT	1200
LRKKHADSTA	ELGQIDNLQ	RVKQKLEKEK	SELKMEIDDL	ASNMESVSKA	KANLEKMCRT	1260
LEDQLSEIKT	KEEEHQRMIN	DLTAQRARLQ	TESGEYSRQV	EEKDSQISQL	SRSKQAFQ	1320
IEELKRHLLE	EIKAKNALAH	ALQSRHDCD	LLREQYEEEQ	EAKGELQRAL	SKANSEVAQW	1380
RTKYETDAIQ	RTEELEBAK	KLAQRLQDAE	EHVEAVNAK	ASLEKTKQRL	QNEVEDLMID	1440
VERTNAACAA	LDKKQKNFDK	ILAEWKQKYE	ETQAELEASQ	KESRSLSTEL	FKMKNAYEES	1500
LDHLETLKRE	NKNLQOEISD	LTEQIAEGGK	AIHELEKVKK	QIEQEKSEIQ	AALEEAEASL	1560
EHEEGKILRL	QLELNQVKSE	IDRKIAEKDE	EIDQLKRNHL	RIVESMQSTL	DAEIRSRNEA	1620
LRLKKKMEGD	LNEMEIQLSH	ANRQAAEAQK	NLRNTQAVLK	DTQIHLDDAL	RTQEDLKEQV	1680
AMVERRANLL	QAEIEELRAA	LEQTERSRSK	AEQELMDASE	RVQLLHTQNT	SLINTKKKLE	1740
TDIVQIQGEM	EETIQEARN	EAKAKAITD	ASMAEELKK	BQDTSALHER	MKNLDQTVK	1800
DLQHRLEDAE	QLALKGGKKQ	IQKLEARVRE	LEGDVDAEQK	RSABAVKGV	KYERRVKELT	1860
YQSEEDRKNI	LRLQDLVDKL	QMKVKSYSKRQ	ABEAEELSNV	NLSKFRKIQH	ELEEAEERAD	1920
IAESQVNLKL	AKSREFHGKK	IEEEE				1945

SEQ ID NO: 200 moltype = AA length = 615
FEATURE Location/Qualifiers
source 1..615
 mol_type = protein
 organism = Cairina sp.

SEQUENCE: 200

MKWTLSIFI	PLFSSATSRN	LQVRADADH	KSEIAHRYND	LKEETFKAVA	MITFAQYLQR	60
CSYDGLSKLV	KDVVDLAQKC	VANEDAPECS	KSLPSIFLDE	ICQVEKLKRS	YGAMADCCAK	120

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SDPERNECF	SFKIPQPDFV	QPYQRPASDV	ICKEYEDNRV	SFLGHFIYSV	ARRNPFLYAP	180
TILGVAADYE	HALKSCCKES	DVGACLDGKE	TAIREKVKKI	SVKQQYSCGI	LQKFGDRIFQ	240
ADKLAFLSQK	YPKASFTEIS	KLIHDVKDVY	KECCEGDMVE	CMDDRAELVN	YICKSHESFS	300
SKIKDCCEKP	IVERSQCIME	AEPDEKPADL	PSLVEKYIQD	KEVCKSFEAG	HDAFMSEFVY	360
EYSRRHPEFS	TQLILRIAKG	YETLLEKCK	TDNPAECYAN	AQDQLNEHIK	ETQDVVKTNC	420
DLTTHGEAD	FLKALLIRYT	KMPQVSTET	LLEIGKKMTG	VGTKCKCLPE	DRRLPCSEGY	480
LSMVIQDMCR	RQETSPINDN	VSHCCSDSYA	YRRPCFSAMG	VDTKYVPPPF	NPEMFNFDEK	540
LCSAPAEERA	AGQMKLLVNL	IKRKPMQTEE	QIKTIADGFT	AMVDKCKQKS	DIETCFGEEG	600
ANLIVQSRAT	LGIGA					615

SEQ ID NO: 201 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 201

MPFGNTHNNF	KLNFKVEEEY	PDLTKHNNHM	AKVLTKDMYA	KLRDKQTPSG	FTLDDVIQTG	60
VDNPGHPFIM	TVGCVAGDEE	SYEIFKDILL	PIISDRHSGY	KPTDKKHTDL	NFENLKGGDD	120
LDPNVVLSR	VRTGRSTKGY	TLPPHNSRGE	RRAVEKLSVE	ALNTLDGEFK	GKYYPNKM	180
DAEQEQLIAD	HFLFDKPVSP	LLLGAGMARD	WPDARGIWHN	DAKSFLVWVN	EEDHLRVISM	240
EKGGMNKEVF	RRFCVGLKRI	EETFKKHNHG	FMWNEHLGYV	LTCPSNLGTG	LRGGVHVKLP	300
KLSTHPKFEE	ILTRLRLQKR	GTGGVDTASV	GGVFDISNAD	RLGSSEVDQV	QMVVDGVKLM	360
VEMEKKLEKG	EAIDGMIPAQ	K				381

SEQ ID NO: 202 moltype = AA length = 895
 FEATURE Location/Qualifiers
 source 1..895
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 202

MTAETQVQY	GSYMTSEQVY	MTQEDDWDNR	LLLDPAWEKQ	QRKTFTAWCN	SHLRKAGTQI	60
ENIEEDFRNG	LKMLLLEVI	SGERLPKPKD	GKMRPHKIAN	VNKALDFICS	KGKVLVSIGA	120
EEIVDGNVIM	TLGMIWTIIL	RFAIQDISVE	ETSAKEGLLL	WCQRKTAPYR	NNVNQNFHIS	180
WKDGLALCAL	IHRHRPLDID	YSKLRKDDPM	GNLNTAFEVA	EKYLIDIPKML	DAEDIVNTPK	240
PDEKAIMTYV	SCFYHAFAGA	EQAETAANRI	CKVLAVNQEN	EKLMEYEKL	ASELLEWIRR	300
TIPWLENRVA	EQTMRAMQOK	LEDFRDYRRV	HKPPKVQKEC	QLEINFNTLQ	TKLRLSNRPA	360
FMPSEGMVVS	DIANAWKGLE	QVEKGYEEWL	LTEIRRLERL	DHLAEKFKQK	SSLHESWTS	420
KVELLSMKDY	ESASLMEIRA	LMRKHEAFES	DLAAHQDRVE	QIAAIAQELN	ELDYHDAVTI	480
NARCOGICDQ	WDNLGTLTQK	RRDSLEERVE	LWETIDQLYL	EFAKRAAPFN	NWMDGAMEDL	540
QDMFIVHSIE	BIQSLITAHN	QFKATLPEAD	KERVATMGIO	NEIVKIAQTY	GIKLSGVNPY	600
TNLSPOQDIT	KWDVAKHLVP	LRDQMLQEEV	ARQQSNERLR	RQFAAQANII	GPWQTKMEE	660
ISHVSVDIAG	SLEEQMSNLK	QYEQNIINYK	CNIDKLEGDH	QLSQESLIFD	NKHTNYTMEH	720
VRVGWEQLLT	TIARTINEVE	NQLLTRDAKG	ISQEQLENEFR	ASFNHFDRKR	NGMMDPDDFR	780
ACLISMGYDL	GEVEFARIMT	LVDSNNTGVV	TFQAFIDFMT	RETAETDTAD	QVMASPKILA	840
SDKTYITVEE	LRRELPEPQA	DYCISRMTSY	IGSGAPPGAL	DYISFSSALY	GESDL	895

SEQ ID NO: 203 moltype = AA length = 843
 FEATURE Location/Qualifiers
 source 1..843
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 203

MPKPLTDQDK	RKQISVRGLA	GVENVSDLKT	NFNRHLHFTL	VKDRNVATKR	DYYFALANTV	60
RDHLVGRWIR	TQQHYEYKDP	KRVYVLSLEF	YMGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELQDIEEDA	GLGNGGLGRL	AACFLDSMAS	LGLAAYGYGI	RYEFGIFNQK	IVNGWQVEEA	180
DDWLRYGPNW	EKARPEYMRP	VHFYGRVEHT	PEGVKWVDTO	VVLALPYDTP	VPGYRNNVVN	240
TMRLWSAKAP	CDPNLKDFNV	GGYIQAVLDR	NLAENISRVL	YPNDNFFEGK	ELRLKQEFV	300
VAAATLQDIVR	RFKSKKFGST	EVVRVDLSTL	PDKVAIQLND	THPAMAIPEL	MRILLDTEHQ	360
TWEKAWDICT	RTCAYTNHTV	LPEALERWPT	DLLQNLPRH	LEIIEINRR	HLEIRIAKLYP	420
GDHDLRRMS	LVEEGDQKKI	NMAHLCTVGS	HAVNGVARIH	SDILKATLTK	DFYEVDPHKF	480
QNKTNIGITR	RWLVMCNPGL	AEVIAERIGE	DYIRDLQDLK	KLLTFVDDDS	LIHDIKVKQ	540
ENKLKFSAYL	EEHYKVKINP	NSMFDVQVKR	IHEYKRQLLN	CLHIITLYNR	IKKEPNKNWT	600
PRTIMIGGKA	APGYHTAKLI	IRLITAIGEI	VNHDVVGDR	LKVIFLENYK	VTLAEKIIPA	660
SDLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	AEAEAGEENLF	IFGMRVDDVD	720
AMDKSGYDAM	DYYNRIPELK	QAMDQIAGGF	FSPDQHDLPK	DIVNMLLHHD	RFKVFADYEA	780
YIKCQEKVSA	LYKNPKWEWK	MVIHNIAGCG	KFSSDRTISQ	YAREIWGMPE	SLERIPAPDD	840
PRQ						843

SEQ ID NO: 204 moltype = AA length = 363
 FEATURE Location/Qualifiers
 source 1..363
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 204

MPHSFPFLST	BQKKELSDIA	QKIVAPGKGI	LAADSTGSGV	AKRFQSINAE	NTEENRRLRY	60
QLLFTADRV	PCIGGVIFFH	ETLYQKTDDG	KLFPQLIKER	DMVVGIVKDK	GVVPLAGTNG	120

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ETTTQGLDGL	YERCAQYKGD	GADFAKWRCV	LKITDTPPSE	LAIKENANVL	ARYASICQMH	180
GIVPIVEPEI	LPDGDHDLKR	CQVTEKVL	AVYKALSDHH	VYLEGTLKLP	NMVTGHS	240
QKYSNQEIAM	ATVTALRRTV	PPAVPGITFL	SGGQSEEEAS	INLNAMNQCP	LHRPWAITS	300
YGRALQASAL	KAWGGKPENG	KACQEEFIKR	ALNNSKACQG	KYVSSGDKGA	AGGESLYVAN	360
HAY						363
SEQ ID NO: 205	moltype = AA length = 193					
FEATURE	Location/Qualifiers					
source	1..193					
	mol_type = protein					
	organism = Salmo salar					
SEQUENCE: 205						
MAPKKDAKAP	AKKAEPAKAA	APAPAPEPDV	VAAPPPLDLS	TVKVEFTPDQ	QEDYKEAFGL	60
FDRVGDQKVA	YNQVADIMRA	LGQNPTNKEV	RKVLGNPSDE	DMAAKRLEFE	AFPLMLQHIV	120
NDPNKGTFFD	YVEGLRVFDK	EGNGTVMGAE	LRIVLGTLGE	KMSEAEIDAL	MQQGEDENG	180
INFEAFVKHI	MSI					193
SEQ ID NO: 206	moltype = AA length = 334					
FEATURE	Location/Qualifiers					
source	1..334					
	mol_type = protein					
	organism = Salmo salar					
SEQUENCE: 206						
MVKVGVNGFG	RIGRLVTRAA	FHSKKGVEIV	AINDPFIDLD	YMVYMFKYDS	THGRFPHGEVK	60
AEGGKLVIDG	HKITVFHERD	PANIKWGDAG	ATYVVESTGV	FTTIEKASTH	LKGGAKRVVI	120
SAPSADAPMF	VMGVNHEKYD	NSLKVVSNAS	CTTNCLAPLA	KVIHDNYHII	EGLMSTVHAV	180
TATQKTVDGP	SGKLWRDRGR	ASQNIIPAST	GAAKAVGKVI	PELNGKITGM	AFRVPTPNVS	240
VVDLTVRLRK	PASVDAKKV	VKAAADGPMK	GILGYTEQQV	VSSDFNGDTH	SSIFDAGAGI	300
ALNDHFPVKLV	TWYDNEFGYS	NRVIDLMAHM	ATKE			334
SEQ ID NO: 207	moltype = AA length = 284					
FEATURE	Location/Qualifiers					
source	1..284					
	mol_type = protein					
	organism = Salmo salar					
SEQUENCE: 207						
MDAIKKMQM	LKLDKENALD	RAEGAEGDKK	AAEDKSKQLE	DDLVALQKKL	KGTEDELDKY	60
SESLKDAQEK	LEVAEKTATD	AEADVASLNR	RIQLVEEELD	RAQERLATAL	TKLEEAKEAA	120
DESERGMKVI	ENRASKDEEK	MELQDIQLKE	AKHIAEEADR	KYEEVARKLV	IIESDLERTE	180
ERAELSSEGC	SELEEEELKT	TNNLKSLEAQ	AEKYSQKEDK	YEEIKVLTLD	KLKEAETRAE	240
FAERSVAKLE	KTIDDELEDEL	YAQKLKYKAI	SEELDNALND	MTSI		284
SEQ ID NO: 208	moltype = AA length = 1138					
FEATURE	Location/Qualifiers					
source	1..1138					
	mol_type = protein					
	organism = Salmo salar					
SEQUENCE: 208						
MPEAKKPAAK	TDKAAAAPAD	KEARSAIEEG	SECRDELPPD	ETVPGGQSEL	TGLFVERPES	60
TTVIKGNVT	FVAKVDSSDL	LKPNMKWLK	GKWLDLGSKA	GKHVQFKEAY	DRNSKVITYE	120
MSIIKVVEGD	AGGYRCEVTS	KDKCDSCTFE	VTVEAVQEEQ	PANILDAFKR	SGDAGEDAGD	180
LDFSALLKKR	EKKARDEPKE	DVDVWEILKD	AKPCDYEKIA	FDYGITDLRG	LLKRLKKMKK	240
VEPKKSDAPL	KKLEQCYSD	KGKRTQMHVE	LHDPNVQVKW	LKNGVEIKPS	AKYVFECVGN	300
KRTLITINKN	LSDDAAYECV	VGEESFTFEV	FVKEPPVTIT	KLDDVHVVV	GEKVEFECEV	360
SEEGANVKWM	KDGVELTKDG	KIRIKKDGKK	HTLVINEATI	EDIGMYVYVT	NGGESKGELE	420
VEAKELEVLQ	SIADLSVKAC	EQAMFKCEVS	DEKVVGWKFK	DGVEVKPGNR	IKMSHIGRIH	480
KLTIDDVKPQ	DEGNYTFVPE	GYALSLSAKL	NFIEIKIEYV	PRQDPPKIHL	DTSSTGSKNT	540
IVVAGNKLRL	LDVEITGEPV	PTVCWMKGDT	VISEAEGVRV	VETRTTLSSF	VIEGAERPDE	600
GQYSIIVTNP	AGEDRAELTV	KIVDVPNPPE	NVRCMGVGED	TATITWDPKP	FDGGAPIKGY	660
LMERKKQSS	RWTKLNFEV	ESTTYEAKKM	IEGVFYEMRV	FAVNGIGISQ	PSGNSKPFMP	720
IAPTSEPTRL	TVEDVTDSTC	ALKWRPPERV	GAGGVDGYII	EWCKEGEDNW	VAANKEPVDK	780
NTYRVKGLPT	GEKLLFRVVA	MNIAGRSPPC	TMKQSVTIRE	IMEYPKIRLP	RQLRTKFIRK	840
VGEKINLVIP	FQKPRPVVN	WLKDGEPLEN	KSVGIRTSDF	DTILFIRSAE	RDHSGTYTSL	900
VQIDNMQDKA	DIHIQIVDKP	GPPINVMVTD	VWGFNAALEW	KPKDGTGNTD	ITGYTIQKAD	960
KKTQGWFTVY	EHNRRPSCA	SDLVMGNEYS	FRVFSENICG	LSDEVAFSKN	TAIIGKTDLE	1020
YNKPAFKERD	MSSSPKFAP	LVDRVVVAGY	STAIKCAVRA	YPKAKIVWMK	NKMIIGEDPK	1080
FLMQNNQGV	TLNIRKPGQF	DGGKYSCKAI	NDLGEDEVCE	RLEVRVLKEK	KEGDEEKK	1138
SEQ ID NO: 209	moltype = AA length = 1502					
FEATURE	Location/Qualifiers					
source	1..1502					
	mol_type = protein					
	organism = Salmo salar					
SEQUENCE: 209						
MATKVMHFNQ	KKHVSHSVSH	SSHTTKHVVK	EQSSRKSSSK	SVNQVQHTHA	TEAMAEPAYT	60
VPAFRQRSAD	BIEEYQVRSS	HVGKGLASIQ	KELHRMRVVT	KTQVENIAIK	REVQEMMHKK	120

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MTLSTETDKA	PDFMVALRPH	TVWEKTPVKL	FCTVDGHRP	IVKWYKGGKP	VDPLSAPGKY	180
KIESKYGVHS	LIISRCMVSD	TAEYSAVATN	SHGSTTSKAS	VIVKRPAGGA	YGSQQLFGQV	240
PHLTVIHHSK	LEITMLDRFG	VSGTEGGSI	SLVCTMVVVP	DLPNVPPLAQ	WYRDDKLLKA	300
GKLAELIKVG	GAATLTPLHL	AKDDEGLYTL	RMWTKDGTE	HSAYLFVKDA	APSVAGAPGA	360
PISVKAFDIN	SDYVLVWAKP	PNTTNEAAIT	GYFVDKRESG	SATWSQCND	PVQICKYPVH	420
GLNVGHAYHF	RVRVNSAGI	SRPSRESKDV	TALDPAERER	LQVIKLDGKH	EVVIKDDDL	480
GVVTLPGPEP	DMHASEIFKS	YIVLSWKPPS	PRGRAPLWYI	IEKSVSGTGA	WQRINTAVKL	540
QSPRYPVFDL	EAGKKYQFRV	YSENLYGASE	ASKPTDPIEK	VDEHGVPSAP	GQVVATRNTK	600
SSVVFQWDDP	KHPNHLMGYY	IDASLVGSKT	WAPCNHKKPY	NTRFVVHGLT	PGETYVFRVQ	660
AVNVYGLSDE	SQESTPIAVE	PALTTPSAPH	GITMLSCDGA	SMIAWNSPK	HRGGSKINAY	720
YVDKRDADSL	AWKEVNSAPT	NTRTYTVDGL	TEGTFYEFKV	QAGNLAGVGV	PSAPSTPLKC	780
EAWTSVAVPG	AYDLAFREVR	GDSLVLILKA	PVYTGAASVT	GYIVEMAKKG	HHYHPLNEEA	840
IGHCYLQVGT	LEAGAETYFK	VKAVNAEGVG	KASQTSEPVF	AKALPGTQEI	QCGVDEDTGD	900
IFLSPEACEI	SETSNFAWSK	NYKEISDCTR	VSIETKGKHS	KLTFVNPVDS	DLGAFSVHVT	960
DTDGVSASHT	VTEDALNTML	ELSYNIRHPI	VPLKHQLNVE	VLEKGVHFRW	LQCLKMSPAV	1020
NYRFIVNDKE	VTSSDSHKIS	HDVNTGVIQM	TVDHFSKASE	ATYTVQIHG	RAKNQSSLVL	1080
VGDVFKAALK	EAEPQRSEHI	RKQGPHPFAEY	LSYHVDDCT	VLLVCKVANV	KKETVFWHFK	1140
DEEEIVPETP	PNVMSGVAL	PISLFSRKDQ	GHYKAKLSDD	RGKDTSVFDI	SGQVFDIIN	1200
ATAHIAAGSSA	SELVLQCTPE	GIRLQCYMKY	YTEEIRTAWL	HKDSKISSSE	KMRIAGTAEM	1260
AWMQIREPSE	KEKGHYSIQI	MDATKSHTRT	FDLGGQAYTD	AYEEYLRLKA	AAFAEKNRGR	1320
VVGGLPDVVT	IMEQKTLSLT	CTVWGDPSPE	VTFWKNENEV	VSTEHAKITL	EANKFASLTI	1380
TAVTSEDSKG	YSEINVRNKY	GEPVEITVSV	YKQGESPPPE	KMGGRGATPA	PLASKTPAKT	1440
PTPAQTPTPS	LKSPTPSLKS	PTPSLRSPAK	SPTPTPKSPT	PPRRVKSPSP	ARSLKSPTTP	1500
RK						1502

SEQ ID NO: 210 moltype = AA length = 434
 FEATURE Location/Qualifiers
 source 1..434
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 210

MSITKIHARE	ILDSRGNTV	EVDLYTAKGR	FRAAVPSGAS	TGVHEALELR	DGDKSRYLKG	60
GTVKAVDHVN	KDIAAKLIEK	KFSVVDQEKI	DHFMLELDT	ENKSKFGANA	ILGVSLAVCK	120
AGAAEKGVPL	YRHIADLAGH	KDVLPCPAF	NVINGGSHAG	NKLAMQEFMI	LPIGASNFHE	180
AMRIGAEVYH	NLKNVIKAKY	GKDATNVGDE	GGFAPNILEN	NEALELLKTA	IEKAGYPDKI	240
IIGMDVAASE	FYKAGKYDLD	FKSPDDPARY	ITGDQLGLDY	KSPFKGYPVQ	SIEDPFDQDD	300
WAAWTKFTAA	YDIQVVGDDL	TVTNPKRIQQ	AVEKKACNCL	LLKVNQIGSV	TESIKACKLA	360
QSNQWGVVMS	HRSGETEDTF	IADLVVGLCT	GQIKTGAPCR	SERLAKYNQL	MRIEELGAK	420
AKFAGKDYRH	PKIN					434

SEQ ID NO: 211 moltype = AA length = 170
 FEATURE Location/Qualifiers
 source 1..170
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 211

MAPKKAKRRG	AAAEAGSSNV	FSMFEQSQIQ	EYKEAFTIID	QNRDGIISKD	DLRDVLASMG	60
QLNVKNEELE	AMVKEASGPI	NFTVFLTMFG	EKLKGADPED	VIVSAFKVLD	PEATGFIKKE	120
FLQELLTTQC	DRFSAEEMKN	LWAAFPDPVA	GNVDYKQICY	VITHGEEKEE		170

SEQ ID NO: 212 moltype = AA length = 1979
 FEATURE Location/Qualifiers
 source 1..1979
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 212

MTTLGLTEGH	PAKYQHLTNL	VPAISYELGQ	KKERVIMSTD	AEMAAFGPAA	IYLRKPERER	60
IASQAAPFDA	KTAFVVEPK	EMYLKGVLOS	KEGGKATVKT	LCNKVLTVKE	DDIHPMNPPK	120
YDKIEDMAMM	THLNEATVLY	NLKERYAAWM	IYTYSGLFV	TVNPKWLPV	YDSVVVNAYR	180
GKKRIEAPPH	IFSISDNAYQ	FMLTDHENQS	ILITGESGAG	KTVNTRKVIQ	YFATIAVAGG	240
KKEQTAAATS	GKIKGSLEDQ	IIAANPLLEA	YGNAKTVRND	NSSRFKGFIR	IHFGTGKLA	300
SADIETYLLE	KSRVTFQLSA	ERSYHIFYQL	MTGHQPQLLE	ALLITTNPYD	YPIISHGEIA	360
VKSIDDTTEF	IATDTADIL	GFTAEEKIGI	YKLTGSVMHH	GAMFKFKQQR	EEQAEPDGTE	420
EADKISYLMG	LNSADLLKAL	CYPRVKVGNE	MVTKGQTVPQ	VNNSTMALCK	SVYEKMFLWM	480
VVRINEMLDT	KQSRQFFIGV	LDIAGFEIFD	YNSLEQLCIN	FTNEKLQOFF	NHHMFVLEQE	540
EYKKEGIEWA	FIDFGMDLAA	CIELIEKPMG	IFSILEEECM	FPKASDTTFK	SKLYDQHIGK	600
TKAFKPKPKP	KGKAEAHFAL	VHYAGTVDYN	VTGWLDKNKD	PLNDSVVQLY	QKSSVKLLAM	660
LYVGAASQA	DDAARGGGGK	KHKGGSFQTV	SALFRENLGK	LMTNLRSTHP	HFVRCILPNE	720
SKTPGLMENF	LVIHQRLCNG	VLEGIRICRK	GFPSRIQYGD	FKQRYRVLNA	SVIPDGQFID	780
NKKASEKLLG	SIDVDSQYQ	FGHTKVFFKA	GLLGLTEEMR	DEKLATLVTM	TQALCRGYVM	840
RKEFVKMMAR	REAYYSIQYN	IRSFNMNVKHW	PWMKLYFKIK	PLLSAETEK	EMAAMKENFE	900
KMKEDLAKAL	AKIKGLEEKI	VSLQENNDL	QLQIAAEENT	LCDAEERCEG	LIKSKIQMEA	960
KLKETFERLE	DEEBELNALT	AKKRKLEDEC	SELKKDIDDL	ELTLAKVEKE	KHATENKVKN	1020
LTEEMSSQDE	SLIKLTKEKK	ALQEVHQQVL	DDLQAEEDKV	NNLTAKATKL	EQQVDDLEGS	1080
LBEKKVVRMD	LERAKRLEGG	DLKLALLESML	DLENDKQQSD	EKKKKDFET	SQLLSKIEDE	1140
QALGAQLQKK	IKELQARIEE	LEEBIEAERA	ARAKVEKQRS	DLSRELEBIS	ERLEEAGGAT	1200

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SVQIEMSKKR	BAEFQKLRRD	LEESTLQHEA	TAAALRRKQA	DTVAELGEQI	DNLQRVKQKL	1260
EKEKSEYKME	IDDLSSNMEA	IAKAKGNLEK	MCRSLEDQLS	ELKTKNDEHV	RQLNDINVQR	1320
ARLLTENGEL	GRQMEEKESL	VSQLTRAKQT	FTQQIEDLKR	QIEEEVKAKN	ALAHGLQSAR	1380
HDCDLLREQY	EEEQEAKAEL	QRALSKANSE	VAQWRSRYET	DAIQRTTEELE	DAKKKLTQRL	1440
QDAQEAVEAT	NAKCASLEKT	KQRLLEGEVED	LMIDVERANA	SAAQLDKKQR	NFDKVLSEWK	1500
QKYEEGQABL	BGSKEARSLS	STELFKMKNS	YEEALDHLET	LKRENKNLQH	EISDLTELVG	1560
ESGKSIHELD	KAKKTIVENEK	AEIQAALAEA	EGTLEHEESK	ILRIQLELNQ	IKGEVDRKLA	1620
EKDEETEIQIK	RNSQRMDSM	QSTLDSEIRS	RNDAMRVKKK	MEGDLNEMEI	QLSHANRQAA	1680
ESQKQLRNVQ	GALKDAQQLQ	DDAIRVADDM	KEQVAMVERR	NNLMMAEIEE	LRAALEQTER	1740
GRKVAEQELV	DASERVGLLH	SQNTSLINTK	KKLEVDLTQV	QGEMEDTVQE	ARNAEEKAKK	1800
AITDAAMMAE	ELKKEQDTSS	HLERMKKNLE	VTVKDLQHRL	DEAENLAMKG	GKKQLQKLEA	1860
RVRELESEVD	AEQKRGVEAI	KGVRKYERRV	KELTYQTEED	NKNVSRLODL	VDRLQLKMKA	1920
YKRAAEDABE	QSNIHLSRYR	KVQHELEEAQ	ERADISESQV	NKLRAKSREI	GSKGNVAEE	1979

SEQ ID NO: 213 moltype = AA length = 5447

FEATURE Location/Qualifiers

source 1..5447

mol_type = protein

organism = Salmo salar

SEQUENCE: 213

MDNEEEFEEE	FEETVTSTT	SHQTTSRAAA	VTQPGVPVR	RVVRKKVKVD	60
TSKFMTPYLA	HSQKMLDLFS	YNKYRAYDK	AKGKPYAITA	DTPEMMRIKK	120
RMEGNKVRCT	SLYDGEAREI	AHVKHVSELI	SKVLYRAKWD	ETKDRYLLPP	180
NAANYSKKLY	TEQWDEDKTM	CYPYSDSPSL	RRVAKAQEVL	SDILYKKGHA	240
VPNDMELAKK	NFEHRSLLKY	HEDYNKNVKG	HWCETPYIEV	AVARVAMNKL	300
DIKDQINPMQ	TETPEYKQHK	KAGHAASSIK	YKKDYENSKS	VADYNVLPAT	360
AGTILSDKVY	KASYEKSRS	SINYCDTPKF	KMDSVVQQFS	DAHVKDKYEN	420
EDIYMQHCKK	MEETKSEQNY	KADYEDIKTR	CFYPQTMSP	YEAHKVQAQC	480
TVKFTQVTD	PVQLQAAINA	KQLSDLNHYA	AYELDKFKCA	LPPDYPPFIQ	540
SCYRYDWEKV	KAKQFEVKG	AISILAAARH	TNIASDVYK	KEPEKNKGKM	600
KILHSMHVAK	IQADPREYKLN	YEKTKTKYIT	TLDMMSTLA	KKSQGIASMT	660
LPHDAMSLDL	AKKANIQSD	NEYHSDYNNS	TKGTGWVAPG	SMEVEKAKKA	720
QHPDTPVPTS	IADHPVMMQA	QVNQIMRSDL	HYKKGLEEH	QKYSPLPHNDP	780
NISKNYKQGS	WEETIAKGYD	MRPDAISVIH	AKHGRHIASD	IQYKKDYVKG	840
QDDPLLHVHY	EVARLQSDKN	YKKDYHKSCL	KYHSPVDMMS	VVHAKHASTV	900
HHYSLPLGAM	NLELARTMNT	NASDVNYKAD	WENSVKGTGW	IPIGSIPVEL	960
HKYRQHPENF	KFSSKMNQKI	MELCKHNQKI	MNVQYKASG	EKFLHTYNLP	1020
CNAVNISKNY	YTYAHRQDIL	KGHQVKADAI	PILAAARACTN	IASNYKYKLA	1080
FRNVKDDPLL	VHYMEVAKMQ	SEQNYKIDYH	KSCLKYHSPV	DMMSLVHAKH	1140
RKSLHNYTLL	PMAPVYKLN	TMMEIQSNNA	YKSDYNNMCR	GSGWVPIGSM	1200
ARDEQGYRVH	PSTLKFTSLT	QGMNMLAMT	NTKLLDDASY	RASGEKFKHT	1260
LQAKCNAQTM	SESHYKHKWV	EDIAKGYDIR	PDIAISVIHAK	HGRHIASDIQ	1320
KQIGCRNLQD	DPLLVHYMEV	AKMQSDKNYK	KDYHKSCLKY	HSPVDMMSLE	1380
FAGYRKIQHH	YTLTADSMNL	ALARTMNASS	SDYEYKSDWN	NSVRGTGWIP	1440
VGGDIQSEQK	YRTHPSKFQF	KKLMDSDMDIV	LATDNNKIMN	KQYETKQWHN	1500
APEIVLAKAN	AINMSQKTYK	AGLAKMYKNG	YDLKVDAIAI	QAAKQGTAKA	1560
MNRGKQIGCR	NLQDDPLLHV	YMEVARLQSD	KNYKKDYHKS	KLKYHSPVDM	1620
TVQTFAGYRK	IQHYHTVLAD	AMNVELAKQM	QTIASDNLKY	SDYETSVKGT	1680
EKAKTAGLIG	SEKLYRQHPS	SFKFTKDMHS	MDVVLNANN	AIMNKQTYVD	1740
IMPDSMEVVL	ARANKINCSA	TLYKLANEQD	KKKGYDLRND	AIAILAAKQG	1800
AGYRKQVGHH	IGARCVEDPD	LLMLAINSAL	IASDALYKID	FHQSKTKFHL	1860
KKNQIQVNH	APVTRLHNWT	CLPDSDDVVH	ARHVDIASD	AVYKADLCKL	1920
LDVEKAKKAG	EVLSEKLYRQ	HPSNFKFTSS	VQDIGMVLAK	QNNDNANKQA	1980
KNHIMPDMAM	IVLANANKSN	YSWNQYRKAM	VEDKKKGYDM	RNDIAISIAA	2040
KYKSSYRKQV	GHHIGARCVE	DDPLLVLALN	SAKIASDALY	KKDFHQSKTK	2100
ELCKKNQIQV	NDFNYRTHLH	NWTCLPDS	VDVHARHVDI	ASEAVYRADM	2160
TGSLDVEKAK	KAGEILSEKK	YRQHPSTLDF	KSPLDGMDIV	LAKHNGQIMN	2220
DKIKIHVGNE	LPELVLAKAN	AITMSNKLKY	QGFDETIKKG	YVLPVDAISV	2280
SDYKYKAGYS	KQVGHGIGAR	CVEEDPLLML	AINSAKIASD	ALYKKDFHQS	2340
IAFELCKKNQ	IQVNDNFYRT	HLHNWTCPLD	STDVVHARRV	YDLHSDAVYR	2400
WVPIGSLDVL	KAKKAAEELS	DRLYKQKPD	LKYTTDMHSM	PMVLAKANAE	2460
WEADKIKIHV	GNELPELVLA	KANAINNSYK	LYKAELEELY	KKGYNLKSDC	2520
ETISDYKYKA	GSYEQVGHG	GARCVEEDPL	IMLALNSAKI	ASDALYKIDF	2580
VDMLTIELAK	KCQMQLVNDN	YRTRLHNWTC	LPDSSDVVQA	RKVYALHSDA	2640
GTGWVPIGSI	DVEKAKKAGE	ILSERKYREH	PSNFKFTATT	DDMPYALATA	2700
VQQWEADKIQ	IHVGNELPEL	VLAKANAITM	SNKLYREGFN	EMIGQGYHLD	2760
KSREIISDYK	YKAGFRNQCG	HHIGAHCVED	DPLLVLAAAN	ARISSDINYL	2820
SIPLDMMSFD	LAKKSQFQIQ	EFTYRTKLHN	WTCMPDNTDV	VRARKVYSLH	2880
WLRGCGWLQP	GSVDVIKAKN	AQTILNERLY	RQPPSSIKFT	SVVDSPEVVL	2940
RKYREVDWTE	KTNIHLNNDL	PSLVLSKANA	IAISNKLTYK	DWDADKAGY	3000
KARKSRDIIS	DYKHYEGYRK	QVGHGIGARC	VEEDPLIMLA	LNSAKIASDA	3060
TKFHLPLVDML	TIELAKKQCV	QVNDNFYRTH	LHNWTCPLDS	TDVVQARKTY	3120
DLEWLRGCGW	NPHESVEVLK	VKHAQAVLAD	RGYRVKLDQ	KFTAPVDRID	3180
LNEANYRALW	HTEKTKYTIT	DTPGLATARE	VAKNLHPKLY	TKDWKVKAT	3240
IKQCIHNNKV	QSDHKYKADY	RKQAGHHIGA	RCVEDDPLIM	LAINSAKIAS	3300
SKTKFHLPLVD	MLTIELAKKC	QVQVNDHHR	TRLHNWTCPL	DSTDVQTARK	3360

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KADLEWLRCG	GWVAADCVDH	VKVRNAQKIM	NERLYKKDAK	DNFAKYTHIV	DRPEILATI	3420
NAFNLSDLKY	KETNAEKGK	YIGSEDTPL	AHSREVAKI	SEKYYKLAWD	ESKATGYQLD	3480
AHYIPVVGAK	KTREIVSDVK	YHDTHEKAKG	HYLAGTLVDF	PEVMRCGEQE	KQKGLRDYHK	3540
EYNTNKAHNM	IPADIIPNMV	AKRCQDMLSD	VLRYRTLLHNM	TCHPDQQDII	RARKNNEILS	3600
DVYYKDDLNM	IKGIGCYAWD	TPELVRAKNA	HLLQSDLYK	EEGKKEFNMY	SIVTDTPVVY	3660
TALLGHTWAS	DLNRYEAYHK	DKHMYTTILD	TNYAHNATL	KDIYSTKTYT	SAWDKIKAKD	3720
YTLPSDSLAL	KHAKQQKVIL	SNVKYREDYE	KFKSLYTLPK	SLEDDPQTLR	CLKAGKMGLD	3780
RLYKADYEKT	KAKNHIPDDM	LDVLSARQTO	NYVSEIGYRK	HLHQWICMPD	MQVYVQARKV	3840
NEQLSDIFYK	DDLNLWRGIG	CYAWDTPEIL	RVKAADKLQS	DNKYRAKGIE	AFKDYSVVID	3900
TPTYETAKLS	QNLSDLHYR	NDYATNVKGT	NSSPAVTVDI	ERARLANFNQ	SDNLYKEANK	3960
NFMPTGYSLP	HDNPLMKQAK	QNSIVASSVK	YKNDYEKSKA	TNYTMDPDGV	SFVNIRNANK	4020
VTNPRLYREL	YEKGKGKIH	TYDTPEIRQV	KMTQEAUSD	CYKEKYFANR	GTMIIPMGVTP	4080
QMIHCNHVNE	ITSLDRYKED	LLWLRLGIGCF	LYDTPEMSTV	RNITKFRSNY	EVEAKKNHAN	4140
FTVVLDTPPE	QRTVELKTHM	SDLMYRAQGN	VEKTKCTTTL	DSLQVLRVVK	AQQLSNKYMY	4200
TDLASKERAH	FTPEKDTPI	SHARAMKVVF	SEIKYKQDYE	KMKHKYTSIH	DTPILVRRAK	4260
AYLQSSDLRY	KESPEKSKGH	YHVSVDALDI	VCHRRVTDDI	SEVKYREKYI	NSLGTWRSMP	4320
DRPEFFHSRI	IGENVSDIKY	KEDLDWLKGI	GCYVWDRPEM	VQAEKNKVL	SERHYKETFE	4380
KNRHFRFKYC	DTPPFQAACK	ASVLIINDRHY	RAAYEKTDDK	YSSVLDDPRN	LLAKENAKNS	4440
QWKYREQYER	AKNKFTSLLE	TPHEHSHKRS	KAIGDNVYKM	DYNKNKAKGY	TLGHDTPHST	4500
HMKKVKEITS	VLKYKELYEK	SKAQINMDPE	SPEIRAAKEA	YKNISNLDYR	KKYQATKNKW	4560
IWTVDRPDDI	QAAKANFQOS	DIEYTYDKEM	MKGCVISIVD	DKLAVLAKMN	ADLSNVKYRE	4620
KFEKAGQYK	TVLDTPTNLH	AKSVRLASE	SKYKLAGKME	QQTQGTTL	TTRDTIHAKE	4680
MGKLVSEKQY	KQKFEKEKKG	SKYNSMRVPP	DVQHAMDVAK	NQSNFAYRQE	AKAKLNYTSV	4740
AMRPDIMRAT	QASKLASNIG	YREKARQEEA	HGGSVLVHRPD	IATATAVSKL	NSQVQYKAKY	4800
DKEKGLRPKY	DMKESVYVKT	QKDAHQQLASE	VKYKGLDKKL	HKPVTDMAES	LSMQHNLSTS	4860
KLSSDYQYKK	KHEDSKGHYL	MIADSPQQLH	HKEASELQSH	VKYKEKYEKA	KGKAMLDPET	4920
PTYVTAKAQ	HMQSAKDYHK	EYBESIKGKN	LSGLEVTAPM	LHVRHATKIA	SERDYRRDLE	4980
EGVKGGLTV	FEETPELLRA	RNATAILSQK	EYKKTLETEI	KGKGLMALAT	DTPDFMRARN	5040
ATDILSQQKY	KQTAEADRAS	YTSVIDTPDI	LHAQNIRNMV	SQKKYREEAE	KCMSHYVPVL	5100
DTPPEMMVRE	NQKNFSTIKY	KDLLCRCTAI	SDLPEVKVR	EMQKNISSVL	YKDSAKGTP	5160
QVFTPEMERA	KRNQEQISSV	LYSDSFRKQV	QGKAAFVLDT	PEMRRVKETQ	RNISGVKYHQ	5220
DPEKSKGSFT	PTTSDPVMDR	VKKNTQEFSD	INRYGIQRRV	VEMEKRRVVE	HEQETVTDLR	5280
VVRTNPGSVF	DYDPAEDNIQ	SRLHMHMSVQ	AQRRSKEHSR	STSAMSGVGD	EKSEVEDSPA	5340
HVSLYSNGF	ATSSMGYTH	KTVELQQRSS	SVATQQTTSV	SVPSPSTGT	KTVRAMYDYS	5400
SADMDEVSPK	DGDVIVNVQS	IDEGWMYGTV	QRSGKTGMPL	ANYVEAI		5447

SEQ ID NO: 214 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 214

MCDDDETTAL	VCDNGLGLVK	AGFAGDDAPR	AVFPSIVGRP	RHQGVVMGGM	QKDSYVGDEA	60
QSKRGILTLL	YPIEHGIITN	WDDMEKIWHH	TFYNELRVAP	EEHPTLLTEA	PLNPKANREK	120
MTQIMFETFN	VPAMYVAIQ	VLSLYASGRT	TGIVLDSGDG	VTHNVPIYEG	YALPHAIMRL	180
DLAGRDLDY	LMKLLTERGY	SFVTTAEREI	VRDIKEKLCY	VALDFENEMA	TAASSSSLEK	240
SYELPDGQVI	TIGNERFRCP	ETLFPQPSFI	MESAGIHETA	YNSIMKCDID	IRKDLYANNV	300
LSGGTTMYPG	IADRMQKEIT	ALAPSTMKIK	ITAPPERKYS	VWIGGSILAS	LSTFQAMWIT	360
KQBYDEAGPS	IVHRKCF					377

SEQ ID NO: 215 moltype = AA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 215

MQWLSVCSLL	VLLSVLSRSQ	AQNQICTIFT	EAKEDGFKSL	ILVGLAQNL	DSTLGLDVLPL	60
IAEALAMGVK	CCSDTPPEDC	ERDVADLFQS	AVCSSETLVE	KNDLMCCEK	TAAERTHCFV	120
DHKAKIPRDL	SLKAELEPAD	QCEDFKKDHK	AFVGRFIFKF	SKSNPMLPPH	VVLAIAGYGY	180
EVLTTCCGEA	EAQTCFDTKK	ATFQHAIAKR	VAEKSLCIV	HKKYGDVVVK	AKKLVSQYSQ	240
MPQASFOEMA	GMVDKIVATV	APCCSGDMVT	CMKERKTLVD	EVCADSVLS	RAAGLSACCK	300
EDAVHRGSCV	EAMKPDPKPD	GLSEHYDVHA	DIAAVCQFT	KTPDVAMGKL	VYEISVRHPE	360
SSQQVILRFA	KEAEQALLQC	CDMEDHAEV	KTALAGSDID	KKITDTEYY	KMKCAAEAAV	420
SDDNFEKSM	VYVTRIMPQA	SFDQLHMHVC	TVHDLVHACC	KDEPGHFVLP	CAEEKLTDAL	480
DATCDDYDPS	SINPHIAHCC	NQYSYMRHRC	ILAIQPDTEF	TPPELDASSF	HMGPELCTKD	540
SKDILLSGKK	LLYGVVRHKT	TITFDHLKTI	STKYHTMKDK	CCAAEDQAAC	FTEEAPKLVS	600
ESAEVLVKV						608

SEQ ID NO: 216 moltype = AA length = 608
 FEATURE Location/Qualifiers
 source 1..608
 mol_type = protein
 organism = Salmo salar

SEQUENCE: 216

MQWLSVCSLL	VLLSVLSRSQ	AQNQICTIFT	EAKEDGFKSL	ILVGLAQNL	DSTLGLDVLPL	60
IAEALAMGVK	CCSDTPPEDC	ERDVADLFQS	AVCSSETLVE	KNDLMCCEK	TAAERTHCFV	120

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DHKAKIPRDL	SLKAELPAD	QCEDFKKDHK	AFVGRFIFKF	SKSNPMLPPH	VVLAIAGYV	180
EVLTTCCGEA	EAQTCFDTTK	ATFQHAVMKR	VAELRSLCIV	HKKYGDRVVK	AKKLVQYSQK	240
MPQASFQEMG	GMVDKIVATV	APCCSGDMVT	CMKERKTLVD	EVCADSVLS	RAAGLSACCK	300
EDAVHRGSCV	EAMKPDPKPD	GLSEHYDIHA	DIAAVCQFTT	KTPDVAMGKL	VYEISVRHPE	360
SSQQVILRFA	KEAEQALLQC	CDMEDHAECV	KTALAGSDID	KKITDETDDY	KMCAAEAAV	420
SDDSEKSM	YYYTRIMPQA	SFDQLHMVSE	TVHDLHACC	KDEQGHFVLP	CAEEKLTDI	480
DATCDDYDPS	SINPHIAHCC	NQSYSMRRHC	ILAIQPDTEF	TPPELDASSF	HMGPELCTKD	540
SKDLLLSGKK	LLYGVVRHKT	TITEDHLKTI	STKYHTMKEK	CCAAEDQAAC	FTEEAPKLVS	600
ESAEVLVKV						608

SEQ ID NO: 217 moltype = AA length = 380
 FEATURE Location/Qualifiers
 source 1..380
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 217

MAKNCHNDYK	MKFSGDEEFP	DLSLHNNHMA	KVLTKDIYNK	LRGKSTPSGF	TVDDIIQTGV	60
DNPGHPFIMT	VGCVAGDEES	YEVFKDLLDP	IISDRHSGYK	PTDKHRTDLN	FENLKGDDDL	120
DPAYVLSSRV	RTGRSLIKGFT	LPPHNSRGER	RTIEKLSIEA	LATLSGEFKG	KYYPLNGMTD	180
KEQDQLINDH	FLFDKPVSP	LTCAGMARDW	PDARGIWHND	AKTFLVWVNE	EDHLRVISMV	240
QGGNMKEVFR	RFCTGLLKIE	ETFKKHNGHF	MWNEHLGVVL	TCPSNLGTGL	RGGVHVKLPK	300
LSTHAKFEI	LTRLRLQKRG	TGGVDTASVG	GVFDISNADR	LGSSEVDQVQ	MVVDGVKLMV	360
AMEKKLEKSE	SIDDMIPAQK					380

SEQ ID NO: 218 moltype = AA length = 892
 FEATURE Location/Qualifiers
 source 1..892
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 218

PHKLLQTNKG	GMIMLAYMTQ	EEGWDRDLLL	DPAAWKQQRK	TFTAWCNSHL	RKAGTQIENI	60
EEDFRNGLKL	MLLLEVISGE	RLPKPKDGKM	RPHKIANVKN	ALDYICSKGV	KLVSIGAEI	120
VDGNAMKTLG	MIWTIILRFA	IQDISVEETS	AKEGELLWCQ	RKTAPYRNVN	VQNFHISWKD	180
GLALCALIHR	HRPDLIDYSK	LKKDDPIGNL	NTAF EVAEKF	LDIPKMLDAE	DIVNTPKPDE	240
KAIMTYVSCF	YHAFAGAEQA	ETAANRICKV	LAVNQENEKL	MEEYEKLASE	LLEWIRRTIP	300
WLENRAAEQT	MRAMQQLKED	FRDYRRVHKP	PRVQEKQCLE	INFNTLQTKL	RLSNRPAPMP	360
SEKMYSDIA	NAWKGLEQVE	KGYEEWLLTE	IRRLERLDHL	AEKFKQKCSL	HETWTSKGEE	420
LLSQKDFESA	SLMEIRALMR	KHEAFESDLA	AHQDRVEQIA	AIAQELNELD	YHDAVTNNAR	480
CGGICQDQWN	LGTLTKRRD	SLERVEKLWE	TIDQLYLEFA	KRAAPFNWMM	DGAMEDLQDM	540
FIVHSIEETQ	SLITAHDOFK	ATLPEADKER	MATMGIQSEI	LKIAQTYGIK	LSGINPYTNL	600
SPQDISNKWD	AVKHLVPLRD	QMLQEEVARQ	QANERLRRQF	AAQANIIGPW	IQTKMEEIGH	660
VSADITGSL	DQMNMLQYQE	QNIINYKSNI	DKLEGDHQLS	QESLIFDNKH	TNYTMEHIRV	720
GWEQLLTITA	RTINEVENQI	LTRDAKGISQ	EQLNEFRASF	NHFDKRNNGM	MDPDDFRACL	780
ISMGYDLGEV	EFARIMTLVD	ANNTGVVTFQ	AFIDFMTRET	AETDTAEQVM	ASPKILASDK	840
SVITYDELE	ELPLEQAEC	ISRMTRYVGA	DGTTGALDYI	SFSSALYGES	DL	892

SEQ ID NO: 219 moltype = AA length = 843
 FEATURE Location/Qualifiers
 source 1..843
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 219

MAKPLTDQEK	RKQISVRGLA	GVENVTDIKS	NFNRLHLYTL	VKDRNVATKR	DYYFALANTV	60
RDHLVGRWIR	TQGHYYEKDP	KRVYLSLEF	YMGRTLQNTM	VNLALENACD	EATYQLGLDM	120
EELQDIEEDA	GLGNGGLGRL	AACFLDSMAS	LGLAAYGYGI	RYEFGIFNQK	IVDQWQVEEA	180
DDWLRYGPNW	EKARPEYMRP	VHYGRVEHT	PGGVKWVDTQ	VVLALPYDTP	VPGYRNNVVN	240
TMRLWSAKAP	CDPNLKFNV	GGYIQAVLDR	NLAENISRVL	YPNDFPEFGK	ELRLKQEFV	300
VAATLQDIIR	RFKSKFGST	EFVRMDLSTL	PDKVAIQLND	THPALAIPEL	MRILMDLEEL	360
PWEKAWDIVV	RTCAVTNHTV	LPEALERWPI	DLFQNLPRH	LDIVYEINRR	HLERVAQQFP	420
GDEGRLLRMS	LIEEGDHKKI	NMAHLICIVG	HAVNGVARIH	SEIIKDTVFK	DFYEMDPEKF	480
QNKTNGITPR	RWLVMNSPGL	AEVIAERIGD	DYIRDLQLK	NLLEFVDDDS	LIRDVAVKVQ	540
ENKMKFSAYL	EEHYKVKINP	ESMFDVQVKR	IHEYKRQLLN	CLHIITFYNR	IKKQPNKAWT	600
PRTIMIGGKA	APGYHTAKMI	IKLITAIGDI	VNHDAVVGDR	LKVIFLENYK	VTLAEKIIPA	660
SDLSEQISTA	GTEASGTGNM	KFMLNGALTI	GTMDGANVEM	AEAEAGEENLF	IFGMRVDDVK	720
AMDQTYGDAK	SYDRIPELKL	QAMDQISGGF	FSPGQPDLPK	DIVNMLMNH	RFKVFADYEE	780
YIKCQDKVSA	LYKDPKEWTK	MVIRNIGSSG	KFSSDRTIAQ	YAREIWMNEP	SLEKIPAPDD	840
PRD						843

SEQ ID NO: 220 moltype = AA length = 367
 FEATURE Location/Qualifiers
 source 1..367
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 220

MPHAYPFLST	EQKKELSDIA	KKIVAPGKGI	LAADSTGSGV	AKRFQGINAE	NTEENRRLYR	60
QLLFTADDRV	APCIGGVIFV	HETLYQKTDD	GKLFPQLIRE	RGQVVGIKVD	KGVMPLAGTD	120

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GETTTQGLDG LYERCAQYKK DGADFAKWRC VLKITEPTPS ELAIMENANV LARYASICQM 180
VRHHGIVPIV EPEILPDGDH DLQRTQYVTE KVLAAVYKAL SDHHVYLEGT LLKPNMVTAG 240
HSCTRKYSAQ EVAMATVTAL RRTVPPAVPG VTFLSGGQSE EQASINLNAI NQCPLGRPWA 300
LTFSYGRALQ ASALKAWGGK KENGKACQEE YIKRALNNNL ASQGYVSSG DKGAAGGESL 360
YVANHAY 367

SEQ ID NO: 221      moltype = AA length = 194
FEATURE            Location/Qualifiers
source              1..194
                    mol_type = protein
                    organism = Gadus morhua

SEQUENCE: 221
MAPKKDVKAP AAAAKKAEPK KKAEPAPPEV AVPAKTVDL SAVKVEFTPD QMEDYREAFG 60
LFDRVGDNKV CYNQIADIMR ALGQNPTNKE VKAILGNPSD EDMNSKRVDV EGFLPMMQTI 120
VNSPNKGTLN DYVEGLRVFD KEGNGTVMGA ELRIVLSTLG EKMTEAEIDA LMGGQEDENG 180
CINYESFVKH IMSI 194

SEQ ID NO: 222      moltype = AA length = 333
FEATURE            Location/Qualifiers
source              1..333
                    mol_type = protein
                    organism = Gadus morhua

SEQUENCE: 222
MVKVGINGFG RIGRLVTRAG FQSKKVEIVA INDPFIDLEY MVYMFKYDST HGRFHGEVKI 60
EGDKLVIDGH KISVFHERDP ANIKWGDAGA QYVVESTGVF TTIEKASAHK KGGAKRVIIS 120
APSADAPMFV MGVNHEKYEN SLQVVSNSAC TTNCLAPLAK VIHDNFGIVE GLMSTVHAIT 180
ATQKTVDGSP GKLRWDGRGA SQNIIPASTG AAKAVGKVIP ELNGKITGMA FRVPTPNVSV 240
VDLTVRLEKP AKYDDIKKVI KAASEGPMKG ILGYTEDQVV STDPNGDIRS SIFDAGAGIA 300
LNDHFVKLVN WYDNEFGYSN RVIDLMAHMS CKE 333

SEQ ID NO: 223      moltype = AA length = 284
FEATURE            Location/Qualifiers
source              1..284
                    mol_type = protein
                    organism = Gadus morhua

SEQUENCE: 223
MDAIKKKMQM LKLDKENALD RAEQAEGDKK AAEDRSKQLE DDLVALQKKL KGTEDELDKY 60
SEALKDAQEK LELAEEKAAD AESDVASLNR RIQLVEEELD RAQERLATAL TKLEEAEEKAA 120
DESERGMKVI ENRAMDEEK MELQEIQLKE AKHIAEADR KYEEVARKLV IIESDLERTE 180
ERAELSSEK SELEELKTV TNNLKSLEAQ AEKYSQKEDK YEEIKVLTD KLKEAETRAE 240
FAERSVAKLE KTIDLEDEL YAKKLKYKAI SEELDHALND MTSI 284

SEQ ID NO: 224      moltype = AA length = 1119
FEATURE            Location/Qualifiers
source              1..1119
                    mol_type = protein
                    organism = Gadus morhua

SEQUENCE: 224
MPEAADAGEL PTAEEDSGDE EGQGTETLGL FVERPQENVV IIAGADVUIT AKVDSTTLTR 60
KPTVKWLKKG WLDLASKAGK HLQFKETYDR ETKIITYEMK VIKAKVGAAG AAGAYRCEVT 120
SKDKFDSASF EITVEAAQSS DDVDILSSFK RADAGEDDGN LDPSALLKAA KKKRPPEDDK 180
PVIDVWELLK SAHPSEYEKI AFEYGITDLR GMLKRLKKMK AEPKPSEAFK KRMESCYSVE 240
KGKKIVLTCE VFDPNVQVKW LKNGQEIKGS AKYIMETIGN IRKLTIMKTT LADDAAYECV 300
VGEDKCFTEV FVKEPPVITIT KLMDDYHVVV GERVEFEVEV SEEGANAMWF FEDIEIFKDT 360
NPAKYRIKID GRKHMLIILE AGLEDIGMYH CWTNGGHTKG ELEVEEKELL VLQDIADLTV 420
KATDQAIFKC EVSDEKVTGK WFKDGVVLP DERHKMTHIG RFHKLIDNV KPDAGDYTF 480
IPNGYALSLS AKLNFLIEMI DYVPRQDPPK IHLDTGTMV SQNTIIVVAG NKLRLDVEIT 540
GEPAPTIVWS KGEVITATE GRVRVDNRKT LSSFVIEGAE REDEANYTIC VTNPAGEDKA 600
KLFVKIVDVP DPPQNVKCLS VGEDSSSMVW EPPAFDGGAP LKGYLMERKK KGSSRWTKLN 660
FDVYEATTYE AKRMIEGVYV EMRVFAINSI GMSQPSITSK PFMPLAPTSE PTRLLVHDVT 720
DGCTCLKWMA PERIGAGGLD GYIIIEYCKEG ETEWVVANKE LCERQGVVVR GLPVGEKINF 780
RVFAKNLAGV SPPATLAQPV TVREIVLDPK IRLPRDLRTK YIRKVGDKIN LVIPFQGKPR 840
PVVKWYKGA PVDEKIVGVR NSTVDSILFI RSAERDHSK YELVVQIENM EDRAIDLTV 900
VERPGPPTKV KEYEVWGFNA ALEWEAPKDD GNCEITGYTI QKADKKTMEW YTIYEHNRRP 960
SCTSSDLIMG NYEMFRYSE NLVGPSEEPG QSKNTAVIGK TGLAYNPPEF KEKDVSCIPT 1020
FTQPLLDREV VAGYSTAMSC SVKGFPPKPKI VMMKNKMILG GPKYIMLNN QGVLTNLNIRK 1080
PSTFDGGRYS CMAVNDLGA LVECKLEIRI PTDPTAEVK 1119

SEQ ID NO: 225      moltype = AA length = 1413
FEATURE            Location/Qualifiers
source              1..1413
                    mol_type = protein
                    organism = Gadus morhua

SEQUENCE: 225
MAMRSGNYNH EYYHKQSKYV VKEYSSSESV VDDYQHKAAV RIQGVVRKKM VEKYVREEMP 60
IRGPKFVVRL RSHTVFENSP IKLFCTVEGF PVPIVTWYKD GVILEHGSAG RYIVDDNVGM 120

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RSLEIPRCGT	DDTAQYSAVA	SNIHGQASSQ	ASIIIVKRFRQ	EDESCLYAWL	PYTVPMPLPSI	180
QYTKINITFT	EKFGPVFASE	GGASASLTATM	HLEPNLANLQ	PDSQWYRDDT	RLVDSKWVKI	240
NTGRGTSTLT	IPNLFKDDDEG	LYTLRLVTKG	GTAEHSAPIS	VSDGPPPVVG	APGAPMDIRI	300
HDANRDYVIV	SWKPNNTTTE	GPILGYFVDK	CEVGTENWAQ	CNDHPKICK	YPVPGLFEGH	360
SYFFRVRAVN	ARGISKPSRM	SDPIAALDPT	EFERLHAKL	GGKLDVVTYD	DDMEAEKGPP	420
GVSSSGIYASE	TDRTYVVLWS	KAPAYSSKAP	MWYIEKCQV	GTAAWQVRNT	QVPIRSPRYA	480
VFDLAEAGKEY	QFRVLSANMY	GTSEPSEPSK	LIQTQALKGV	PSAPGQVIST	RETDTSLLIQ	540
WAPPKEPNNL	IGYYIDQCVK	GSKDWTSAH	KPHSSTKFVV	SGLTTGETYV	FRVQAINELG	600
LSDESQESSP	LSVKAALTSP	TAPYDVALLG	CDGQSMVLNW	KKPLHTGGTA	INAYFVDKRR	660
SGATMNRREVH	IPAVTERVYQ	VEGLTQGAII	QFRVYAANLA	GLGKSSKASA	DFSCEAWTMP	720
EPGPAYDLTA	SEVRDTSVL	EWKKPVYGS	GPITGYHVEY	AKKGSSDWTR	ANETAVSHRY	780
LKVEGLEEGA	KYVFRVRGVN	EAGVGMASLP	SDSVVAKALA	GAQEVSCSVD	EKSGDIVFAF	840
ESCQMNVGSE	FLWKKNYKEI	TEFSSSIVVH	TEGNLCKLMF	KNAVKEDFGT	YSVSVSHTDG	900
VSSSCSISAE	ELQKMLELSY	SIRHPIIPLK	SELAYKILER	GRMRFWLQAE	EISSKVITYKF	960
FANNKELSGA	DPNKMAHDS	TGVIEFILDH	FTEDSEGTYT	CQITDGGGKA	QSSSLVLIGDA	1020
FKAALAEADY	QRREYIRVKE	GPHFSEFLT	KVGEDCSVT	VCKVANLKE	SEFHWYKEDT	1080
EIIPDVKPD	GSQVCKLPIH	LFSRKTGTV	KATISDDRKG	DMSQVDISGK	VFDDVINKIN	1140
QLAGASAAEL	VINCTATGIQ	LQCHMKYITS	EMNIHWMHGE	SKVGHSEKCV	IGGTPAMATM	1200
EVIEPNDDKDK	GVYSIVITDL	ENTNKRITDL	SGDVYDKAYA	QFQKLKAEAY	AEMNRGKVMG	1260
GLPDVVTIME	KKTLSTLCTV	CGDPKPLVSW	LKNGNDVEPD	NQYVVSLLDQ	KFACLTINGI	1320
GMDDSGKYTM	IVQNKYGGEC	VDIVVSVYRH	GDTIPVAKPN	LTAKTIIPPK	MPIELPQPKT	1380
AAPSPAPPAS	PSPAPPKAVG	GRGMKSPTPT	RRK			1413

SEQ ID NO: 226 moltype = AA length = 457
 FEATURE Location/Qualifiers
 source 1..457
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 226

MKSNPLTSNT	GEHASYLLHV	NGVMSITKIH	AREILDSRGN	PTVEVDLWTA	KGLFRAAVPS	60
GASTGHEAL	ELRDGDKTRY	LKGKTQKAVD	HVNKDISPKL	IEKNFSVVDQ	EKIDKFMLEL	120
DGTENKSKFG	ANAILGVSLA	VCKAGAAEKG	VPLYRHIADL	AGNKDVILPV	PAFNIVINGGS	180
HAGNKLAMQE	FMILPVGAA	FHEAMRIGAE	VYHNLKNVIK	AKYKDATNV	GDEGGFAPNI	240
LENNEALELL	KSAIEKAGYP	DKIIIGMDVA	ASEFYRSKGK	DLDFKSPDDP	ARHISGKELG	300
DLYKTFIANY	PVQSIEDPFD	QDDWANWADF	TAAVDIQVVG	DDLTVTNPKR	IQQAVEKKAC	360
NCLLLKVNQI	GSVTESIQAC	KLAQSNWGW	MVSHRSGETE	DTFISDLVVG	LCTGQSKTGA	420
PCRSERLSKY	NQLMRIEEL	GDKAKFAGKD	YRHPKIN			457

SEQ ID NO: 227 moltype = AA length = 170
 FEATURE Location/Qualifiers
 source 1..170
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 227

MAPKKAKRRA	GGGDSGSSNV	FSMFEQSQIQ	EYKEAFTIID	QNRDGIISKD	DLRDVLASMG	60
QINVNKEELE	AMVKEASGPI	NFTVFLTMFG	EKLKGADPED	VIVAAPKILD	PEATGSIKKE	120
FLQELLTTQC	DRFSPPEIKN	MWSAFPDPVA	GNVDYKNICY	VITHGEEKEE		170

SEQ ID NO: 228 moltype = AA length = 1935
 FEATURE Location/Qualifiers
 source 1..1935
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 228

IMGDGMQCF	GEAAVYLRKP	ERERLEAQT	PFDAKTAFFV	SEPKEMYLKG	VLQSRDGGKA	60
TVKTLCDKTL	TVKEEIIYPM	NPPKYDKIED	MAMMTHLNPE	TVLYNLKERY	AAWMIITYSG	120
LPCVTVPYK	WLPVYDSSV	NAYRGKKRIE	APPHIFSISD	NAYQAMLTDR	ENQSILITGE	180
SGAGKTVNTK	RVIQYFATIA	VSGPKKTEQV	AGKMQGSLED	QIIAANPLLE	AYGNAKTVRN	240
DNSSRFQKFI	RIHFATSGKL	ASADIETYLL	EKSRTVFQLS	AERSYHIFYQ	LMTGHKPELI	300
DALLITKNPF	DYHMISQGEI	TVKSIDDVEE	FIATDDAIDI	LGFTADEKIG	IYKLTGAVLH	360
HGNMKFKQKQ	REEQAEPDGN	EEADKIGYLM	GLNSADVLKA	LCYPRVKVGN	EYVTKGQTC	420
QVHNSVMALC	KSVYEKMLFW	MVVRINEMLD	TKQPRQFFIG	VLDIAGFEIF	NYNSLEQLCI	480
NFTNEKLQQF	FNHMFVLEQ	EYKKEGIEW	EFIDFGMDLA	ACIELIEKPM	GIFSILEEEC	540
MFPKASDTTF	KNKLHDQHLG	KTKAFKPKP	AKGKAEAHFS	LVHYAGTVDY	NITGWLDKNK	600
DPLNDSVIQL	YQKSSVKLLA	LLYASHALPD	ADAASKKGAK	KKGGSFQTVS	ALFRENLGKL	660
MTNLRSTHPH	FVRCLIPNES	KTPGLMENFL	VIHQLRCNGV	LEGIRICRKG	FPSRVLYGDF	720
KORYKVLNAS	VIPEGQFIDN	KKASEKLLGS	IDVDSTQYKF	GHTKVFFKAG	LLGALEEMRD	780
EKLAALVTLT	QALCRGFNMR	KEFVKMMERR	EAVFTIQYNI	RSFMNVKTWP	WMKVYFKIKP	840
LLKSAETEKE	MAQMKETFGK	TKEDLAKALS	KKKELEEKMV	SLLQEKNDLL	LQVQSEAETL	900
SDSEERCEGL	IKSKIQLEAR	LKETSERLED	EEEINSELTA	KRKLEDECS	ELKKDIDDL	960
LTLAKVEKEK	HATENKVNKL	TEEMSSQDEI	IAKLTKEKKA	LQEAHQQTLD	DLQAEEDKVN	1020
TLKAKCKLE	QQVDDLEGLS	EQEKKLRMDL	ERAKRKLEGD	LKLAQESIMD	LENDKQQSDE	1080
KIKKRDFTS	QLLNKIEDEQ	TLSAQFQKKI	KELQARIEEL	EEEIEAERAA	RAKVEKQRAD	1140
LSRELEEISE	RLEEAGGATS	LQIEMNKKRE	AEFQKLRRDL	EESTLQHEAT	SAALRKKQAD	1200
SVAELGEQID	NLQRVQKLE	KEKSECKMEI	DDLSSNMEAV	AKSKANLEKL	ARTLEDQLSE	1260
LKSKNDHVR	QLNDSVQRA	RLQTENGEFG	RQQEEKEALI	SQLTRGQAY	TQQIEELKRH	1320

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IEEEVKAKNA	LAHSVQ SARH	DCDLLREQFE	EEQEA KAE LQ	RGMSKANSEV	AQWR SKYETD	1380
AIQRTEEELE	AKKKLAQRLQ	DABEEAVEATN	SKCASLEKTK	QRLLEGEVEDL	MIDVERANAL	1440
AANLDKKQRN	FDKVLAEWKQ	KYEE SQAELE	GAQKEARSLS	TEMPKLNKSY	EESLDQLETL	1500
KRENKNLQQE	ISDLSEQMGE	TGKTIHELEK	GKKTVEVEKS	ELQTSLEAE	ATLEHEESKI	1560
LRVQLELTQI	KGEVDRKMAE	KDEEMEQIKR	NSLRVIDSMQ	SNLDSEIRSR	NDALRIKKKM	1620
EGDLNEMETI	LSHANRQASE	AQKQLRNVQG	QLKDAQHLHD	DALRGQEDMK	EQVAMVERRN	1680
NMLLAEIEEL	RAALEQTERG	RKVAEQELVD	ASERLGLLHS	QNTSLINTKK	KLETDLIQIQ	1740
GEVEDTVQEA	RNAEEKAKKA	ITDAAMMAEE	LKKEQDTSSH	LERMKKNLEL	TVKDLQHRLD	1800
EAESLAMKGG	KKQLQKLEAR	VRELETECEA	EQRGSEAIK	GVRKYERKVK	ELSYQCDEDK	1860
KNICRLQDLV	DKLQLKMKAY	KRQAESESEQ	ANTHLSRYRK	VQHELEEAQE	RADVSESQVN	1920
KLRTKSRNGI	TKIIN					1935

SEQ ID NO: 229 moltype = AA length = 6118
FEATURE Location/Qualifiers
source 1..6118
 mol_type = protein
 organism = *Gadus morhua*

SEQUENCE: 229

MATAGMPTRK	VRKRVAVDTS	KFMTPLYLAHS	MKMLEQYSHN	MYRKAYDLRS	GTPPVHIADT	60
PEMVRIRKAQ	EQLSEVKYRM	EGNKARAGSL	YDGEAKEIAH	VKHMSSELLSK	VLYRQKWDET	120
KDRYLLPPDA	PELVLA VKNA	ANYSKKLYTE	DWDLEKASFL	PYSDSPELRR	VAKAQEALS	180
IQYKGYDQR	KLKPTSLADR	PDVELAKKVF	TQRSDLKYKE	DYNKNVKGQW	CETPYFDVAT	240
ARVAMENLSS	RRYTQDWEAA	KDQINFMQTE	TPVYDTNKA	GTSASSVKYR	KDYEKT KDKS	300
DYNTLPATEN	PGLRLRYAG	TILSDKVYKK	NYEKSRGFSI	NYCDTPKFKM	DSVLKFTDT	360
HYQDKYKNEV	QGHYIGSYED	IYMLHCQKVE	GPKSEQNYKA	EYEDIKTRCF	FPQTTIPEYE	420
VDKKLAQCKD	KVYRQHPDKV	KFTQVTDSPV	QVQAALNAIQ	LSDINYKAKY	EQTKTKNSLP	480
ADYPPFIQSR	VNAFNLSENC	YKYNWEKEKA	KRFEMKGDAI	SILAAARHTN	IASDVYKKE	540
YEKNRGRMVG	ALSIHDDPKI	LHSVHVAKIV	SNREYTKDYE	KVTKYHTSL	DMMSVTLAKK	600
SQAIASMTGY	KKISNRYLMP	YDAILVDLAK	KANSIQSDNE	YKSDYNQYTK	GSPWITFGSM	660
DVELAKKAGE	ILSEKKYRKH	PDTVPFTAID	DHPVMLQARV	NQLQRSDFY	KGLEEIHQK	720
YSLPIDMPEL	QLARANSYNI	SKNYKAAWE	ETISKGYDLK	LDAISIKAAK	AARHAASDIQ	780
YKAYEKARG	HHVGFRNIKD	DPLL VHMQV	AKMQSEKNYK	KDYHKSCLKY	HSPVDMMSLV	840
HAKQASKAGT	YAGYRKQHH	YSLMPDAMNL	VLARKMNTDA	SDCEYKSEWN	NYVRGTGWVP	900
IGSIGVETAK	IGGQIQSENK	YRTHPSTFKF	SKLMDSMDLE	LATQAYTSAW	EKDKRTIHVM	960
PDSPEIVLAK	ANAITMSDVR	EYYKHLHQNG	YHLKGDAIPI	VAAKLGTKIA	SNYKYKLDYE	1020
KARGHHVGRF	NIKDDPLLHV	YMQVAKMQSE	KNYKKDYEKA	KLKYHSPVDM	MSLVHAKES	1080
KAQTYAGYRK	IPHTYFLLPD	MMQLQLCRKM	QEIQSDNEYK	SEYNNLYKGT	GWVPIGSIDV	1140
ELAKTANSAL	REHGYRQHP	SFKFTSKTDS	MDMVLALANS	KIIDQAAYKA	SGEKVKHTYH	1200
LPADAPETIQ	AKFNAQMTSK	SHYKQKWVEH	IAKGYDMRPD	AISILHAKQG	RHIASDIQYK	1260
KAYEKARGHH	VGFRSIKDDP	LLVHYMQVAK	MQSEKNYKGD	YEKAKLYKHS	PVDMMSLVHA	1320
KEASKAQTYA	GYRKIQHHYN	LMPDAMNLVL	ARKMNTDASD	CEYKSEWNYY	VRGTGWVPIG	1380
STGVETAKIG	GEIQSEKNYK	THPSTFKFSK	LMDSMDLELA	TANNKIMNKD	KRTIHVMPDS	1440
PEIVLAKANA	ITMSDKLYKA	GVMEMAKKGY	HLKGDAIPIV	AAKLGTKIAS	NYKYKLDYEK	1500
ARGHHVGFPR	IKDDPLL VHY	MQVAKMQSER	NYTKDYQKSK	LKYHSPVDM	SLVHAKESK	1560
AQTYAGYRKI	EHVYNLMPDA	MNLVLARKMN	TDASDCEYKS	EWNNYVRGTG	WVPIGSIGVE	1620
TAKIGGEIQS	ENKYRTHPST	FKFSKLMDSM	DLELATANNK	IMNKQAYTSA	WEKDKRTIHV	1680
MPDSPEIVLA	KANAITMSDK	LYKAGVMEMA	KKGYHLKGDA	IPIVAAKLG	KIASNYKYKL	1740
DYEKARGHHV	GFRCIQDDPL	LVHYMQVAK	QSEKNYKDYK	EKAKLYHSP	VDMMSLVHAK	1800
DATKAQTYAG	YRKIQHHYSL	LPDAMNLVLA	RNMQVNI SDN	QYKSDYESYC	KGIGWPIGS	1860
LDVEKAKIAG	KIQSEKLYRK	HPSNLKFTKD	MFSMDLT LAT	ANNQIMNKVL	LALYCDVYTL	1920
MNILLSGNHL	TNNFVVSFSK	LYKLDNELSK	KKGHNVKDA	TAI LAAKNCT	TIASDYKYKA	1980
GYRKQVGHNI	GARVEDDPL	LVLALNSARI	ASDALYKKDF	NKSKTKFNLP	VDMMAFELAK	2040
KNQIQVNHAN	YITRLHNWTC	LPDSTDV VNA	RRAYEIQSDN	YKEDQKLMQ	GIGWVPIGSL	2100
DVEKAKKAGE	ILSEHLYRKH	PSNYKFTSKC	DAMPLVLAKA	NAQITNKKAY	HDWDKEKSN	2160
IHIMPDA MDV	VLAKANRANY	SIKRYKQANE	DAKKGYDLR	NDAIGIIAAR	AQRNIISDIK	2220
YKAGYRKQVG	HHIGARCVED	DPLLMLALNS	ARIASDYLYK	KDFNKSCTKF	NLPVDMMAFE	2280
LAKKNQIQVN	HANYITRLHN	WTCPLDSTDV	VNARRAYEIQ	SDNYKEDQK	LMQIGWVPI	2340
GSLDVEKAKK	AGEILSEHLY	RKHPSNYKFT	SKCDAMPLVL	AKANAQITNK	KAYHDAWDKE	2400
KSNIHIMPDA	MDVVLAKANR	ANYSIKGYKQ	ANEDAKKKGY	DLRNDAGII	AARAQRNIIS	2460
DIKYKAGYRK	QVGHHIGARC	VEDDPLLVL	LNSARIASDY	LYKKDFNKS	TKFNLVDM	2520
AFELAKKNQI	QVNHANYITR	LHNWTCPLDS	TDVNVARRAY	EIQSDNYKKE	DQKLMQIGW	2580
VPIGSLDVEK	AKKAGQILSE	HLRYKHPSNY	KFTSKCDAMP	LVLAKANAQI	TNKKAYHEAW	2640
DKEKSNIHIM	PDAMDVLVLA	ANRANYSIKK	YKEANEESKK	KGYDLRNDAI	SIIAARAQRN	2700
IVSDYKYKAG	YRKQTHGHY	ARCVEDDPLL	VLALNSARIA	SDALYKKDFN	KSKTKFNLPV	2760
DMMAFELAKK	NIQVNHANY	ITRLHNWTC	PDSTDV VHA	HVYEQSDAV	YKADLKWLGQ	2820
MGWVPIGSLD	VEKTKKAGEI	LSERKYRQPA	SSHKFTSPLD	AMEIVLAKNN	ALTMNQRLYT	2880
EAWEKDKTKL	HITSETPEII	LSQQNAINMS	NKKYKQGYED	TIKKGYLPG	DAISVKS AKH	2940
SRNIISDYKY	KAGYRTQVGH	HIGARCVEDD	PLLMLALNSA	RIASDYLYKK	DFNKSCTKFN	3000
LPVDMMAFEL	AKKNQIQVN	ANYITRLHNW	TCLPDSTDV	NARKAYDLQS	DAVYKADMEF	3060
MKGIGWVPIG	SLDVVAKAHA	GKILSDHLYK	QKTDKLYKTS	DTKAMPMLLA	KANA EVMNKK	3120
SYVAWEKDK	IKTHLIDPL	EILLSRSNAY	NLSNKMYKED	VVKMFRKG YH	LKGDAI AIVA	3180
AKKGREIISD	IKYKAGYRKQ	VGHHIGALSV	RDDPLIMLAL	NSARIASDAL	YKDFNQSKT	3240
KFHLPVDMLT	IELAKKQVQV	VNDFNYRTRL	HNWTCPLDST	DVVQARKAYD	LQSDAVYKAD	3300
MEWIKQGQWV	PIGSLDVEKA	KKAAEILSER	KYREHPSKFK	FTAKTDDMPY	ALAKANA EIM	3360
NKQSYVAWE	SDKTKLHIMP	DTPEIILGQQ	NKLNNSRKYY	REAYEETLKK	GYLPLKDAIS	3420
VLAARASRNI	ISDYKYKAGF	RKQCGHHIGA	LSVQDDPLLM	MAANA AKIFS	ANLYKQDPHK	3480

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TKTKYKLPVD	MVAPDLAKTA	QSHMKEVTYR	TRLHNWTCMP	DNTDVVHARK	VYDLHSDAVY	3540
RADMEWIRGC	GWSQDSVDV	VKARNAQKIL	NENLYRQAPS	TVKFTSVVDL	PAIVLSKQNA	3600
ENISEMKYTA	AWNKDCLTFH	LKPDIPSVQL	AKANAINISN	KLYTKAWDNQ	KAKAYHIKED	3660
TIAYLTKARAS	RDIISDYKYK	TGYRKQVGHG	IGALSVRDDP	LIMLALNSAR	IASDALYKKD	3720
FNQSKTKFHL	PVDMLTIELA	KKCQIQVNDP	NYRTRLHNWT	CLPDSSTDVQ	ARKAYDLQSD	3780
AVYKADMWFI	RGCGWVPHFC	VDVVVKVHAQ	KILADRGYRV	KLDTQKFTAP	VDRMDFVCAK	3840
NAAEVLNDAK	YREAWHTDKS	KFSLVDTPGL	ATSREVAKIM	HPKLYTKDWD	KVKATGYFMP	3900
GDAVPIKQCT	QTTKVQSSHK	YLEGYRKQVG	HHIGALSVPD	DPLIMLAMNS	AKIASEALYK	3960
KDFNKSCTKF	HLPVDMLTIE	LAKKCCQIQV	DFNYRTRLHN	WTCLPDSSTDV	VNARKAYDLQ	4020
SDFVYKADLE	WIRGCGWVPA	ESVHHVKVRK	AQEILNERLY	KKGAIENFSK	YTHVVDPEL	4080
ILARVAAANL	SDLKYKETFN	NEKGHYIGSE	DTPQLAHSRE	CAKNISEAKY	HESHEKAKGH	4140
YMANLTVDFP	EVVRCGCEQK	AKALRVYQKE	YHESKTNNHI	PSDIIANQVA	KRCQDMLSDV	4200
IYRTHLHQWT	CPPDQLDTPR	ARKGNEILSD	VFYKEDLTWM	KGMGCYVWDT	PEIVRAKKS	4260
ELQSELKYKE	AGKKEFNYS	IVTDTVPVYT	AILGSTWASE	LNRYEAYHKE	KHLYTTILT	4320
HDYARCHDFK	HFFSNKIYTA	AWDKIKAKGY	KIPHDSSHAM	HAKQKQVILS	GVKYKEDYEK	4380
FKSLYSLPKC	LQDDPLKARC	IKAGKLNLDL	LYKEDYEKTK	AKNHI PADML	DIVSGREVQR	4440
NVSEINYRKR	LHQVWCPLDM	QVVFQARKVN	DQLSDIFYKD	DLNYLRGIGC	YAWDTPPEILR	4500
VKTAGDQOSE	NKYKAKAVEA	YKDYTVVLDL	PTYETAKQNA	VNLSDLHYRL	DYNANVKGTN	4560
TAPAVTMEFE	NITFKFNYSQ	VEAKKNHSCF	SMATGYLMTN	DTPVVKQVKN	NSLVASSVKY	4620
KAAYEMMKAT	GTYLHPEGVT	FVNTRKVNQV	INPRLYREQY	EKDKDKIHST	LDTPDIRQVK	4680
MNQEHLSEL	YREKYNNTKG	QLINMPMTPO	LMHCQHVNQI	TSDLRYKEDL	LWLRGLGCFM	4740
YDTPEMVNVN	NITFKFNYSQ	VEAKKNHSCF	SVVLDTPPEY	RVTELKSHMS	MVSPTSNTSL	4800
NTLDSVDIQR	SKWAQQLSNK	FLYTDLASKE	RAHFSQEVET	PAMGLARTMK	VTNSYNKYKE	4860
QYKMKHRYT	SIADTPILIR	AKKAYLQSSD	LYKETTFERS	KGHYHTVKDA	LDILCHRRVT	4920
DDISEVKYRE	KYINSLGTWK	SIPDRPEFFF	NRIVGENVSS	VKYKEDLEWM	KGIGCFVWDT	4980
PEMVQADINK	TLYSGRLYKA	SFEKNRYNPK	YTCDDTPFFHA	AKNASSLINE	RSYRSSYEKN	5040
KDKFITITDD	PRYQLARENN	KMSQAKYKSR	AKELLKSGCN	ELHRPDILTA	LYQSVLGSKW	5100
KYREQYERVK	SKFTSILETP	EYESHKRSKK	IGDIVYRMEY	NKNKAKGYSL	PYDTPHQNMM	5160
KKVQALTSNL	KYREVYEKAK	SQINMDPDAP	EIRAAKEAYK	NTTNLDYKCK	YEATKNKWIW	5220
TVDRPDFISH	SRNTLQOSDV	EYKYDKEMMK	GCVIPIVDDK	ITLLAKKGCE	MASDIKYKEK	5280
YEKAKGHYNP	VADSPQILHA	KAVRTLASES	KYKEASKKEM	QSGSFTTLP	TRDTVHSKEV	5340
NKLISGKLYK	SKFEKEKGQS	KYNRMIVPPD	VQHAMDVAKS	QSSYAYKRDA	KDNLHYTTTA	5400
DRPDIRKATQ	AAKLISDVGY	REKARAEASR	GGSLTYRPDI	ALATEVSKLT	SLVDAKSALH	5460
GAASVDTLQM	RHSKMSALT	SNVYHEKFD	KEMKGGKKPY	DLKDSKIYQT	LKDATNLASE	5520
VKYKGDLLKL	HKPVTDMAES	LSMQHNLSTS	KLSSDYRYKK	KFKDSKGHYL	VIADTPPEQLH	5580
HKEATELQSS	VSGPSLGGVH	VCKDYKIDYE	EGMRGKNLCG	LEVTPAMIHV	RHATKIASER	5640
EYRKDLGEGV	KSGLSAPFEE	TPEHLRAKNA	THILSEREYK	KSLELEIKGK	GMLALATDTP	5700
DFMRARNATD	ILSQTQYKHT	AELDRASFTS	VIDTPDIIHA	QQMRNIISQN	KYKAEAGKTM	5760
SHYVPVFDTP	EMQVRVRENQ	NFSTLQYQAD	HKI IKGKVL	VKDTPPEMLR	KENTKNFSSV	5820
LYKDSKAKGT	PVVPTPEMER	VKNRQEHISS	VLYSDSFRKQ	VQKGAAPVLD	TPEMRRVKES	5880
QLNISGVRVH	QDHEKSKGSF	TAMASYPEME	RVKKNTOEFS	DINYRGIQRR	VVEMERRRVV	5940
EHEQETTTDL	RVMRTNPGSV	FDYDPAEDNI	QSRSLHMMSV	HAQRRSKEHS	RSTSAMS GPA	6000
DEKSEMSHEG	DHMMSTYSNG	FGASSVGYQT	TKTVELQQRS	SSVATQQTTV	SSIPSHSTPT	6060
GKTVRAMYDY	GAADNDEVTF	KDGDVIVNVQ	SIDEGWMYGT	VQRTGKTGML	PANYVEAV	6118

SEQ ID NO: 230 moltype = AA length = 1935
 FEATURE Location/Qualifiers
 source 1..1935
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 230

MSTDAEMAVY	GQAAIYLRKP	EKERLEAQSK	PFDAKSAAYV	TDVKELYLKC	TILKKEGGKV	60
TVKVHDETEE	KTVKDDVTP	MNPPKYDKIE	DMAMMTHLNE	ASVLYNLKER	YAAWMIYTYS	120
GLFCATVNPY	KWLPVYDQHV	VCAYRGKKRM	EAPPHIFSVS	DNAVQFMLTD	RENQSVLITG	180
ESGAGKTVNT	KRVIQYFATI	SVGGDKKKDI	GAGKIQGSLE	DQIIAANPLL	EAYGNAKTVR	240
NDNSSRFGKF	IRIHFGTSGK	LSSADIETYL	LEKSRVTFQL	PAERGYHIFY	QMMTNHKPEI	300
IEMTLITTNP	YDFPTCSMGQ	IAVASIDDKE	ELVATDTAID	ILGFNNDEKM	GIYKLTGAVL	360
HHGNMKFKQK	QREBQAEPEG	TEDADKVAYL	LGLNSADMLK	GLCYPRVKVG	NEYVTKGQTV	420
PQVLNSVSAL	AKSIYERMFL	WMVIRINQML	DTKQPRSSFI	GVLDIAGFEI	FDYNSMEQLC	480
INFNTNEKLQ	FFNHHMFVLE	QEEYKKEGII	WEFIDFGMDL	AACIELIEKP	MGIFSIILEEE	540
CMFPKASDVT	FKAKLYDQHL	GKTKAFKKPK	PAKGKAEAHF	SLVHYAGTVD	YNI TGWLDKN	600
KDPLNDSVVQ	LYQKSGVKLL	PVLYPPVASE	DSTKKGGKKK	GGSMQTVSSQ	FRENLGKLM	660
NLRSTHPHFV	RCLIPNESKT	PGLMENFLVI	HQLRCNGVLE	GIRICRGFPF	SRILYADFKQ	720
RYKVLNASVI	PEGQFIDNKK	ASEKLLGSID	VNHEEYKFGH	TKVFFKAGLL	GTLEEMRDEK	780
LATLVGMIQA	LCRGFQMRKE	FKKMMERRHA	VYTIQYNIRS	FMNVKTWPMW	KVYKIKPLLL	840
KSAETEKELS	AMKENYEMKM	KDLATALAKK	KELEEKMVSL	LQEKNDLSLQ	VASEGENLND	900
AEERCEGLIK	SKIQLEARLK	ETTERLEDEE	EMNADLTAKK	RKLEDECSSEL	KKDIDDEL	960
LAKVEKEKHA	TENKVKNLLE	EMASQDESVA	KLSEKKKALQ	EAHQQTLDL	QAEEDKVNTL	1020
SKAKTKLEQQ	VDDLEGSLEQ	EKKLRMDLER	SKRKLEGLDK	LAQESIMDL	NDKQQSDEKI	1080
KKKDFETSQ	LSKIDFQSL	AIQLQKKIKE	LQARIEELEE	EIEAERAARA	KVEKQADLS	1140
RELEIEISERL	EEAGGATSAQ	IEMNKKREAE	FQKLRRDLEE	STLQHEATAA	ALRKKQADSV	1200
AELGEQIDNL	QRVKQKLEKE	KSEYKMEIDD	LSSNMETVAK	AKGNLEKICR	TLEDQLSEIK	1260
AKSDENSRIQ	NDIGAQRARL	LTENGFEGRQ	LEEKEALVSQ	LTRGKQAYTQ	QIEELKRQIE	1320
EEVKAKNALA	HGVQSAHRDC	DLLEQFEEEE	QEAKAELQRC	MSKANGEVAQ	WRSKYETDAI	1380
QRTELEESK	KKLAQRQLEA	EEQIEAVNSK	CASLEKTKQR	LQGEVEDLMV	DVERANGLAA	1440
NLDKKQRNFD	KVLAEWKQKY	EEGQAELEGT	QKEARSLSTE	LFKMKNSYEE	ALDQLETLKR	1500

-continued

ENKNLQQEIS	DLSEQIGETG	KSIHELEKSK	KQVETEKTEI	QTALEEAEGT	LEHEESKILR	1560
VQLELNQIKG	EVDRKLAEKD	EEMEQIKRNS	LRITDSMQST	LDSEVRSRND	ALRIKKKMEG	1620
DLNEMEIQLS	HANRQAAEAQ	KQLRNVQGGQ	KDAQLHLDDA	VRGGEDLKEQ	AAMVDRRNGL	1680
MVAEIEELRA	ALEQTERGRK	VAEQELIDAS	ERVGLLHSQN	TSLNLTKKKL	ETDLVQVQGE	1740
VDDTVQEARN	AEEKAKKAIT	DAAMMAEELK	KEQDTSSHLE	RMKKNLETV	KDLQHRLEDA	1800
ENLAMKGGK	QLQKLESFVR	ELETEVEGEQ	RRGVDVAVGV	RKYERRVKEL	TYQTEEDKKN	1860
GARLQDLVDK	LQLKVKYKVR	QAEEAEEQAN	SYLSKCRKVQ	HELEEAEEERA	DIAETQVNKL	1920
RSKSRDGGKG	KDPAE					1935

SEQ ID NO: 231 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Gadus morhua

SEQUENCE: 231

MCDDDETTAL	VCDNGSGLVK	AGFAGDDAPR	AVFPSIVGRP	RHQGVVMVGMG	QKDSYVGDGA	60
QSKRGILTLK	YPIEHGIITN	WDDMEKIWHH	TFYNELRVAP	EEHPTLLTEA	PLNPKANREK	120
MTQIMFETFN	VPAMYVAIQ	VLSLYASGRT	TGIVLDSGDG	VTHNVPVYEG	YALPHAIMRL	180
DLAGRDLDY	LMKILTERGY	SFVTTAEREI	VRDIKEKLCY	VALDFENEMA	TAASSSSLEK	240
SYELPDGQVI	TIGNERFRCP	ETLFPQSPFI	MESAGIHETA	YNSIMKCDID	IRKDLYANNV	300
LSGGTTMYPG	IADRMQKEIT	ALAPSTMKIK	IIAPPERKYS	VWIGGSILAS	LSTFQQMWIS	360
KQEYDEAGPS	IVHRKCF					377

SEQ ID NO: 232 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Thunnus maccoyii

SEQUENCE: 232

MCDDDETTAL	VCDNGSGLVK	AGFAGDDAPR	AVFPSIVGRP	RHQGVVMVGMG	QKDSYVGDGA	60
QSKRGILTLK	YPIEHGIITN	WDDMEKIWHH	TFYNELRVAP	EEHPTLLTEA	PLNPKANREK	120
MTQIMFETFN	VPAMYVAIQ	VLSLYASGRT	TGIVLDAGDG	VTHNVPVYEG	YALPHAIMRL	180
DLAGRDLDY	LMKILTERGY	SFVTTAEREI	VRDIKEKLCY	VALDFENEMA	TAASSSSLEK	240
SYELPDGQVI	TIGNERFRCP	ETLFPQSPFI	MESAGIHETA	YNSIMKCDID	IRKDLYANNV	300
LSGGTTMYPG	IADRMQKEIT	ALAPSTMKIK	IIAPPERKYS	VWIGGSILAS	LSTFQQMWIS	360
KQEYDEAGPS	IVHRKCF					377

1.-41. (canceled)

42. A method for producing a pet food product, the method comprising:

- (a) contacting a plurality of host cells with a recombinant nucleic acid, wherein the recombinant nucleic acid encodes an albumin protein of an animal selected from the group consisting of a mammal, a bird, and a fish;
- (b) expressing the albumin protein encoded by the recombinant nucleic acid in the plurality of host cells; and
- (c) generating a pet food product comprising the albumin protein expressed in the plurality of host cells.

43. The method of claim 42, wherein the albumin protein is a rabbit albumin protein.

44. The method of claim 42, wherein host cells of the plurality of host cells comprise one or more members selected from the group consisting of *Pichia* cells, *Saccharomyces* cells, *Schizosaccharomyces* cells, *Trichoderma* cells, *Aspergillus* cells, *Escherichia* cells, *Bacillus* cells, and *Lactococcus* cells.

45. The method of claim 42, wherein the albumin protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 16, 78, 93, 108, 123, 138, 161, 174, 187, 200, 215, or 216.

46. The method of claim 45, wherein the amino acid sequence is at least 70% identical to SEQ ID NO 78.

47. The method of claim 45, wherein the amino acid sequence is at least 90% identical to any one of SEQ ID NOs: 16, 78, 123, 161, 174, 187, 200, 215, or 216.

48. The method of claim 45, wherein the amino acid sequence is at least 95% identical to any one of SEQ ID NOs: 16, 78, 108, 123, 161, 174, 187, 200, 215, or 216.

49. The method of claim 42, further comprising isolating the albumin protein from the plurality of host cells.

50. A method for producing a pet food product, the method comprising:

- (a) contacting a plurality of host cells with a first recombinant nucleic acid encoding an albumin protein and a second recombinant nucleic acid that encodes a protein different from the albumin protein;
- (b) expressing the albumin protein encoded by the first recombinant nucleic acid and the protein encoded by the second recombinant nucleic acid in the plurality of host cells; and
- (c) generating a pet food product comprising the albumin protein and the protein expressed in the plurality of host cells.

51. The method of claim 50, further comprising isolating the albumin protein and the protein from the plurality of host cells.

52. The method of claim 50, wherein the albumin protein or the protein is a rabbit protein.

53. The method of claim 52, wherein the albumin protein and the protein are rabbit proteins.

54. The method of claim 50, wherein the albumin protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 16, 78, 93, 108, 123, 138, 161, 174, 187, 200, 215, or 216.

55. The method of claim 54, wherein the amino acid sequence is at least 70% identical to SEQ ID NO 78.

56. The method of claim **50**, wherein the protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 1-15, 17-77, 79-92, 94-107, 109-122, 124-137, 139-160, 162-173, 175-186, 188-199, 201-204, or 217-232.

57. The method of claim **56**, wherein the amino acid sequence is at least 70% identical to any one of SEQ ID NOs: 4, 54, 57, 58, 59, 60, 63, 64, 67, 74, or 77.

58. A method for producing a pet food product, the method comprising:

- (a) contacting a plurality of host cells with a recombinant nucleic acid;
- (b) expressing a protein encoded by the recombinant nucleic acid; and
- (c) generating a pet food product comprising the protein expressed in the plurality of host cells,

wherein the protein comprises an amino acid sequence that is at least 70% identical to any one of SEQ ID NOs: 5, 8, 9, 11, 14, 16, 19-21, 23, 24, 26-29, 32-35, 37-41, 44-49, 51-54, 56, 57, 59, 65-79, 81, 82, 84, 93, 94, 96,

97, 99, 108, 109, 111, 112, 114, 117, 123, 124, 126, 127, 129, 138-141, 143, 147, 148, 150, 151, 153, 161, 162, 164, 165, 167, 169, 170, 172, 174, 175, 177, 178, 180, 184, 185, 187, 188, 190-192, 197, 200, 201, 203-206, 208, 209, 213, 215, 216, 217, 219-222, 224, 225, or 229.

59. The method of claim **58**, wherein the protein is a rabbit protein.

60. The method of claim **58**, wherein the protein is an albumin protein.

61. The method of claim **58**, wherein the amino acid sequence is any one of SEQ ID NOs: 5, 8, 9, 11, 14, 16, 19-21, 23, 24, 26-29, 32-35, 37-41, 44-49, 51-54, 56, 57, 59, 65-79, 81, 82, 84, 93, 94, 96, 97, 99, 108, 109, 111, 112, 114, 117, 123, 124, 126, 127, 129, 138-141, 143, 147, 148, 150, 151, 153, 161, 162, 164, 165, 167, 169, 170, 172, 174, 175, 177, 178, 180, 184, 185, 187, 188, 190-192, 197, 200, 201, 203-206, 208, 209, 213, 215, 216, 217, 219-222, 224, 225, or 229.

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